

Empirical Corporate Finance

Foundations and Applications

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Preface

Empirical corporate finance has two intellectual traditions that have been talking past each other for a generation. One tradition writes down a model of the firm, estimates its parameters from data, and uses the estimated model to evaluate counterfactuals. The other tradition finds a source of exogenous variation, estimates a causal effect, and reports a number together with a research design. Each tradition has produced findings the other cannot match, and each has limitations the other can help fix. Yet graduate students are typically trained in one tradition and treat the other as a foreign language.

This book is built on the conviction that PhD students working in corporate finance today need both. The structural and causal traditions should not be understood as rival ideologies. They answer different questions, commit the researcher to different assumptions, and produce different objects. Knowing which object your research is recovering, and what assumptions that recovery requires, is the central skill of empirical work. The book is organized around that skill.

The book grew out of teaching the doctoral empirical corporate finance sequence at the Carlson School of Management since 2009. The material began as course notes for students learning to read papers, write referee reports, design empirical projects, and start dissertation research. That origin shapes everything that follows. The aim is not to survey literatures from a distance. The aim is to teach the judgment a student needs when moving from consuming empirical papers to producing them.

The closest existing references are [Roberts and Whited \(2013\)](#) handbook chapter on endogeneity in corporate finance, [Adda and Cooper \(2003\)](#) on dynamic structural economics, [Strebulaev and Whited \(2012\)](#) treatment of structural estimation, [Wooldridge \(2010\)](#) on panel data econometrics, [Cunningham \(2021\)](#) introduction to causal inference, and [Angrist and Pischke \(2009\)](#) work on applied econometrics. Each is excellent and we draw on all of them. None of them sets out, as a single coherent project, what a corporate finance researcher should know about both traditions and how to choose between them. That is the gap this book aims to fill.

Every empirical paper in corporate finance is trying to recover one of a small number

of objects. It may be a descriptive regularity, a prediction, a causal effect, a structural primitive, or a counterfactual. The object dictates the assumptions, the assumptions dictate the method, and the method determines what the estimate means. That may be grounded in financial theory or in statistical theory, or in a combination of them. By itself a regression coefficient is not a treatment effect. A treatment effect is not a structural parameter. A structural parameter is not, by itself, a claim about any observable variation. Confusion among these objects is a common source of miscommunication in empirical corporate finance, and it is the source most easily corrected by training. Chapter 1 develops the framework. The chapters that follow apply it.

There is a second organizing idea, and it runs through the applications rather than the methods. The first four applications chapters study investment, financial constraints, capital structure, and payout policy. These are usually taught as separate literatures with separate canonical papers. We treat them instead as connected margins of one object: the shadow value of internal funds, the multiplier on the firm's budget constraint in any dynamic model where external finance is costly. The structural tradition estimates that object and the frictions that determine it. The causal tradition identifies how firm behavior at one margin responds when a shock moves the value of internal funds for some population of firms. Read through this common object, papers that look unrelated turn out to be studying different sides of the same mechanism. We do not force the framing where it does not fit, and we say so when it does not. Where it organizes a literature cleanly, we let it work.

The primary reader is a PhD student in finance or economics who will use these methods in dissertation work. We assume a first-year graduate econometrics sequence, a course in corporate finance theory at the level of [Tirole \(2010\)](#), and working comfort with Python and Stata. Senior researchers may find parts of the book useful as a structured map of adjacent literatures, but the book is written for students learning to turn econometric tools and corporate finance theory into research designs.

Part I Chapters 1 and 2 are required reading for everyone. Empirical corporate finance scholars care deeply about identification. But the meaning of identification itself is not always clearly understood. The structural versus causal framework and the identification logic that follows from it are used in every subsequent chapter. The reader who skips these chapters will find the rest of the book harder than it needs to be.

Part II covers widely used methods. Panel data, event studies, instrumental variables, difference-in-differences, and machine learning. These chapters can be read in order or used as reference. A student who needs to understand modern staggered DiD can read Chapter 6 without first reading Chapter 5. The methods chapters cross-reference each

other where it matters, and each one connects explicitly to the framework of Part I.

Part III applies the framework to the main domains of empirical corporate finance. The part is titled *Frictions and the Shadow Value of Internal Funds*, and its first four chapters study investment, financial constraints, capital structure, and payout policy as connected margins of that object. The remaining chapters turn to banking and credit supply, innovation, corporate governance, and mergers, acquisitions, and restructuring. Each topics chapter presupposes Part I but is otherwise self-contained. A student writing a dissertation chapter on M&A can read Chapter 15 without first reading Chapter 14. We have tried to make each topics chapter useful to the researcher working in that area, not just to the reader working through the book front to back.

The chapters share a common architecture. Each opens with the economic question, develops the intuition, states the formal framework, applies the method to a corporate finance setting, and closes with common mistakes, decision rules, exercises, and further reading. Boxes are used sparingly, and deliberately so: a Key Insight box states a section's main takeaway, a Common Mistake box flags an error that recurs in papers and referee reports, a When-to-Use box turns discussion into operational guidance, and a Connection-to-Framework box ties the chapter back to the structural-causal spine. A chapter with a dozen boxes has stopped using them to mark what matters, so we keep them few.

A running example threads through almost every chapter. The relationship between firm leverage and firm investment is a question that admits both structural and causal treatment, that has been studied at length in the literature, and that returns in different methodological guises as the book progresses. It is also a topic I have been interested in for a very long time.

Following the running example chapter by chapter is one way to see the structural-causal distinction in operation without reading every section in between. The example is introduced in Chapter 1 and given its identification treatment in Chapter 2. It takes its full panel-data form in Chapter 3, returns under instrumental variables in Chapter 5 and difference-in-differences in Chapter 6, and connects to investment adjustment costs in Chapter 8. It then reappears across the applications, each time on a different margin: as leverage and investment in the financial-constraints designs of Chapter 9, consolidated on the leverage margin itself in Chapter 10, as leverage and payout in Chapter 11, and as leverage and R&D in Chapter 13. The point of returning to one relationship through so many lenses is not that the estimates should agree. They recover different objects, and seeing why is the lesson.

The empirical work in the book is reproducible. A master Compustat panel is built in Chapter 3 and augmented progressively with CRSP, FHFA, ISS, and other data as later

chapters require them. Python is the primary language, and the dynamic simulations and structural estimation are Python only. Stata replication files accompany the reduced-form and causal exercises, where Stata matches common practice in the literature. A random seed of 8823 is used throughout, so the reader who runs the code will see the same numbers that appear in the text. The full Python and Stata implementations live in a separate *Code Companion*, organized in book order and intended to be read alongside the relevant chapters. The chapters themselves carry brief inline syntax illustrations where they help the prose; the numbered listings have all moved to the Companion. The Companion, the code, the data-construction scripts, and the replication files are maintained on the book's GitHub repository.

Even among finance scholars, there are a range of opinions as to what is included. Some people like Gordon Phillips draw a line loosely based on the word corporate. Does the study involve firms and their financial decisions? Other people like Ulrike Malmendier and Luigi Zingales take a much more expansive view. They include more or less whatever corporate finance scholars choose to study.

In this book we cover a subset of corporate finance that is best understood by simply looking at the chapter titles. It is roughly what is commonly taught in a PhD course that is titled something like Empirical Corporate Finance. Many closely related and interesting topics have been left out. Notably we do not cover household finance, labor market aspects, international corporate finance, the impact of artificial intelligence on firms, entrepreneurial finance or venture capital. These are all wonderfully interesting topics with active ongoing research. Each of these omissions reflects a choice about scope, not a judgment about importance.

Acknowledgments

The book grew out of teaching the doctoral courses in empirical corporate finance at the Carlson School of Management over many years. The discipline itself has evolved significantly over the years. Several cohorts of PhD students worked through earlier drafts as classroom material, found errors, and asked questions that reshaped the exposition. Their questions and feedback are reflected throughout.

I am grateful for many years of discussions on these issues with many people and particularly with Vidhan Goyal, Vojislav (Max) Maksimovic, and Gordon Phillips. They have had a huge impact on how I think about these topics.

The lengthy process of converting two decades' worth of jumbled teaching materials into something more coherent was helped by modern AI tools. Claude Opus and ChatGPT

were used as editorial aids, much like unusually patient referees, to find unclear passages, inconsistencies, and likely mistakes. Claude Code assisted in developing and checking portions of the Python and Stata code. All substantive choices, interpretations, and remaining errors are mine.

WRDS provides the data infrastructure on which the empirical examples depend. The Carlson School and the University of Minnesota provided the institutional support that made writing the book possible.

All errors that remain are my responsibility. Given the scope of this project errors are almost inevitable. When you see them, please let me know so that I can fix them. I can be reached at murra280@umn.edu.

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Notation

This book uses unified notation throughout. The table below lists the principal symbols, their meanings, and the chapter in which each is first introduced. Symbols not in this table are local to the section that introduces them. Where the same symbol carries different meanings in different contexts (most commonly β as a regression coefficient versus a discount factor), the entry records the convention used in the methods chapters; the alternative reading is flagged in the prose where it occurs.

Indices and Sets

Symbol	Meaning	First used
i	Firm index (cross-sectional unit)	Throughout
t	Time index, typically year	Throughout
g	Group or cohort index (industries, treatment cohorts in staggered DiD, compliers by complier subgroup)	Ch. 6
j	Alternative second index (instrument, peer, alternative unit)	Throughout
$\mathcal{N}$	Set of firms in the sample, with $ \mathcal{N} = N$	Throughout
$\mathcal{T}$	Set of time periods, with $ \mathcal{T} = T$	Throughout
$\mathcal{G}$	Set of treatment cohorts or groups	Ch. 6

Variables

Table 2. Unified notation: principal variables.

Symbol	Meaning	First used
Y_{it}	Outcome variable for firm i at time t (e.g., investment, leverage)	Ch. 3
$Y_i(1), Y_i(0)$	Potential outcomes under treatment and control (Neyman–Rubin)	Ch. 1
X_{it}	Covariate vector for firm i at time t	Ch. 3
D_{it}	Treatment indicator (0/1); in structural and banking contexts, a local convention denotes debt or deposits with the same letter	Ch. 3
Z_{it}	Instrumental variable	Ch. 5
α_i	Firm fixed effect (time-invariant firm-specific term)	Ch. 3
λ_t	Time fixed effect (common time shock)	Ch. 3
ε_{it}	Idiosyncratic error term	Ch. 3
K_{it}	Capital stock for firm i at time t	Ch. 8
I_{it}	Investment, typically capex scaled by lagged capital	Ch. 3
L_{it}	Leverage, debt over assets	Ch. 3
Q_{it}	Tobin’s Q , market value over replacement cost of assets	Ch. 8
CF_{it}	Cash flow for firm i at time t	Ch. 8
μ_t	Shadow value of internal funds; Lagrange multiplier on the firm’s period budget constraint (so that $v_{F,t} = \mu_t$)	Investment
$v_{F,t}$	Marginal value of one dollar held inside the firm at date t	Payout
$v_{I,t}$	Marginal value of one dollar distributed at date t , evaluated by shareholders net of taxes	Payout
ω_t	Wedge between internal and external valuation, $\omega_t = v_{F,t} - v_{I,t}$	Payout

Parameters

Symbol	Meaning	First used
β	Regression coefficient (in methods chapters); discount factor in dynamic programming problems (Ch. 8 onward)	Ch. 3
$\hat{\beta}_g$	Group-specific reduced-form coefficient; under the constraint-share interpretation, $\hat{\beta}_g = \theta_{C,g}$	Ch. 9
τ_i	Individual treatment effect, $Y_i(1) - Y_i(0)$	Ch. 1
τ^{ATE}	Average treatment effect, $\mathbb{E}[\tau_i]$	Ch. 1
τ^{ATT}	Average treatment effect on the treated, $\mathbb{E}[\tau_i \mid D_i = 1]$	Ch. 6
τ^{LATE}	Local average treatment effect (effect on compliers)	Ch. 5
θ	Generic structural parameter	Ch. 1
$\theta_{C,g}$	Constraint share among compliers in group g (Frank 2026)	Ch. 10
δ	Depreciation rate of capital	Ch. 8
ρ	Persistence parameter (AR(1) coefficient for productivity or profitability)	Ch. 11
γ	Curvature of the adjustment cost function (when unsubscripted; γ_t , where it appears, is a time fixed effect)	Ch. 10
σ, σ^2	Standard deviation and variance	Ch. 3

Operators and Functions

Symbol	Meaning	First used
$\mathbb{E}[\cdot]$	Expectation operator	Throughout
$\text{Var}[\cdot]$	Variance operator	Throughout
$\text{Cov}[\cdot, \cdot]$	Covariance operator	Throughout
$\text{Pr}(\cdot)$	Probability of an event	Throughout
$\perp$	Statistical independence	Ch. 2
$\mathbf{1}\{\cdot\}$	Indicator function (1 if the condition holds, 0 otherwise)	Ch. 6
Δ	First-difference operator: $\Delta Y_{it} = Y_{it} - Y_{i,t-1}$	Ch. 3

Symbol	Meaning	First used
$V(K)$	Value function of the firm; takes the relevant state vector as its argument	Ch. 8
$\phi(I, K)$	Adjustment cost function in capital accumulation	Ch. 8
$f(K, L)$	Production function in capital and labour	Ch. 8

Part I

Frameworks

Chapter 1

Structural and Causal Approaches

1.1 Why Two Worlds?

Here is a question that comes up all the time in corporate finance: how does a firm's leverage affect its investment? At first glance this looks like a single, clear question. It is not. Depending on what you really want to know, and on the assumptions you are willing to make, you will find yourself in one of two intellectual traditions, reading papers that barely cite each other, using estimators that share almost no common vocabulary, and drawing conclusions that differ not just in magnitude but in kind.

This chapter is about that fork in the road. Understanding it is the most important thing you will learn from this book. Every methods chapter that follows, and every applied chapter after that, sits on one side of the fork or the other, or tries to bridge them.

Two papers on the same question

Suppose you are a first-year PhD student. Your advisor hands you two papers and says: read these and tell me what each one teaches us about leverage and investment.

The first paper builds a dynamic model in which a firm chooses investment and financing every period. The firm faces adjustment costs when it changes its capital stock and transaction costs when it issues or retires debt. An Euler equation links today's investment to tomorrow's expected marginal value of capital. Leverage enters as a state variable because outstanding debt determines both interest obligations and financial flexibility. The authors estimate the model's structural parameters, including the adjustment cost, the financing cost, and the shadow value of internal funds, using moments of Compustat data. With those parameters in hand they run a counterfactual. What would investment look like if the tax code were changed, or if bankruptcy costs were

halved? The answer comes from re-solving the model under the new parameter values, because the structural parameters are supposed to be primitives that do not change when policy changes.

The second paper exploits a natural experiment. A regional credit supply shock reduces lending to a set of firms exposed to weakened local banks. The authors compare investment changes for affected firms to investment changes for otherwise-similar firms in unaffected regions. The difference is their estimate of the causal effect of the credit shock on investment. They are careful about pre-trends, about balance between treatment and control groups, and about the statistical precision of their estimate. They do not write down a model of the firm. What they report is a number: the average effect of this particular credit shock on investment for this particular set of firms.

Both papers are about leverage and investment. Both appear in good journals. Both are, by the standards of their respective traditions, well executed. Yet a careful reader will notice that they are answering different questions, making different assumptions, and producing different objects. The first paper estimates a structural parameter, a financing cost that governs optimal behavior under any policy regime. The second paper estimates a causal effect, the response of investment to a specific shock for a specific population of firms. The first paper can do counterfactuals that go beyond the observed data, but only if you trust the model. The second paper has transparent identification that does not require a fully specified model of firm optimization, but it tells you the effect only for the firms whose financing actually changed because of the shock, in the circumstances of that event.

Neither paper is wrong. Neither method is uniformly better. But if you confuse what each one is doing, if you read a structural parameter as though it were a simple causal effect, or if you try to use a local treatment effect to predict what would happen under a different policy regime, you will get the interpretation wrong.

Two traditions in empirical economics

The division you just saw is not an accident of corporate finance. It reflects a deeper division in empirical economics that has been the subject of sharp debate for more than two decades.

One tradition, which we will call the structural tradition, starts from an economic model. The model specifies optimizing behavior by agents, derives equations that connect observables to underlying parameters, and uses the data to estimate those parameters. The intellectual ancestors include the Cowles Commission, the rational expectations econometrics of Hansen and Sargent, and the industrial organization tradition formalized

by [Reiss and Wolak \(2007\)](#). In corporate finance, the structural tradition includes dynamic models of investment, capital structure, and payout, where the researcher estimates adjustment costs, financing frictions, or agency parameters from the model's first-order conditions or simulated moments.

The other tradition, which we will call the causal tradition, starts from a research design. The researcher looks for a source of exogenous variation, a natural experiment, a policy change, or a discontinuity, and uses that variation to estimate a causal effect without committing to a fully specified model of firm behavior. The intellectual framework is the potential outcomes model of [Angrist and Imbens \(1995\)](#), and the broader movement has been called the credibility revolution because of its emphasis on transparent, design-based identification ([Goldsmith-Pinkham, 2023](#)). In corporate finance, the causal tradition includes difference-in-differences estimates of policy effects, instrumental variable strategies that exploit quasi-random shocks, and regression discontinuity designs around regulatory thresholds.

Why students get confused

If you have taken a first-year econometrics course in a modern economics department, you have probably been trained primarily in the causal tradition. You know about potential outcomes, LATE, parallel trends, and first-stage diagnostics. When you arrive in empirical corporate finance, you may be surprised to find that many influential papers, including some of the most widely cited work on investment, capital structure, and financial constraints, do not fit neatly into the causal framework. They are not trying to estimate a treatment effect. They are trying to estimate a structural parameter.

The confusion runs deeper than background. The two traditions use the word *identification* to mean different things. In the structural tradition, a model is identified if the data and the model's assumptions pin down the parameters. In the causal tradition, an effect is identified if the research design eliminates confounding. The word is the same. The concept is related but distinct. We will sort this out carefully in [Chapter 2](#).

Both traditions use the word *endogeneity*, but they worry about different things. A structural researcher worries that unobserved state variables are correlated with the explanatory variables in the model's estimating equation. A causal researcher worries that the treatment variable is correlated with potential outcomes. Related concerns, but the solutions look different.

Four questions to ask

This chapter, and really the entire book, is not trying to tell you that one tradition is better. That claim would be wrong, and it would make for a short and useless book. Both traditions are valuable. Both are widely used. Both can be done well or badly.

What we want you to learn is how to read empirical work, and how to design your own, by asking four questions before anything else.

First, what object is the paper estimating? A descriptive correlation, a prediction, a causal effect, a structural primitive, a counterfactual? These are different objects, and they require different methods. Section 1.2 expands on this.

Second, what assumptions does the method commit you to? Every empirical method requires assumptions. Some assumptions are about the research design, some are about the economic model, some are about both. An honest paper makes them explicit.

Third, to what population does the estimate apply? An estimate that holds for one set of firms may not hold for another. Local average treatment effects apply to compliers under a specific instrument. Structural parameters apply to the model's population under the model's assumptions.

Fourth, does the estimate support the counterfactual the paper wants to discuss? Some estimates support counterfactuals that go beyond the observed data. Others do not. The difference matters enormously when the paper draws policy conclusions.

These four questions are the spine of this chapter. They will appear again, in slightly different forms, in every chapter that follows.

1.2 Objects, Not Methods

The most common mistake in empirical work is to start by choosing a method. Should I run OLS, fixed effects, IV, or GMM? This is the wrong starting question. The right starting question is: what object am I trying to learn about?

Empirical corporate finance repeatedly works with five common objects. They are not the only objects an empirical paper can target, and the boundaries between them are not always sharp. But the five cover most of what corporate finance papers actually estimate, and being clear about which one you want is the first step toward picking the right method.

A descriptive association. A descriptive association is a feature of the joint distribution of observed variables in the data. The average leverage ratio in Compustat is a descriptive object. The cross-sectional correlation between leverage and profitability is a descriptive

object. An OLS regression of leverage on size, market-to-book, and tangibility, in the style of [Rajan and Zingales \(1995\)](#) or [Frank and Goyal \(2009\)](#), is a descriptive object. These regressions are useful. They tell us what the world looks like, and surprising patterns in descriptive work are often the starting point for theory. But a descriptive regression coefficient is not a causal effect, and it is not a structural parameter. Calling it one is the source of half the confusion in corporate finance.

A predictive object. A predictive object answers the question: given what we observe about a firm today, what is our best forecast of an outcome? Will this firm default? Will it issue equity in the next year? How will the market react to its earnings announcement? These are prediction problems, and they have their own machinery, including LASSO, ridge, random forests, and gradient boosting. We treat prediction in [Chapter 7](#). The key point for now is that prediction is a distinct goal. A model that predicts well need not have any causal interpretation, and a model with strong causal identification need not predict well out of sample.

A causal effect. A causal effect compares what happens under treatment to what would have happened under no treatment, for the same unit or for some defined population. The standard formalism is the potential outcomes framework, which we develop in [Section 1.4](#). The causal tradition organizes its estimates around well-defined versions of this object: the average treatment effect on the treated (ATT), the local average treatment effect (LATE), the conditional average treatment effect (CATE). Each is a real quantity for a specific population. Each is different from the others.

A structural primitive. A structural primitive is a parameter of an economic model: preferences, technology, costs, or constraints. The adjustment cost on capital, the issuance cost of equity, the curvature of the production function, and the borrowing limit as a fraction of collateral are structural primitives. These objects are not directly observed. They are inferred from data using the restrictions that an economic model places on observable behavior. A structural primitive, if correctly identified, supports counterfactuals that no causal effect can support, because the parameter is meant to govern behavior under regimes that have not been observed.

A counterfactual. A counterfactual is what would happen in a world that did not occur. What would aggregate investment look like if the corporate tax rate were five percentage points higher? What would the distribution of leverage look like if firms could not deduct

interest? Counterfactuals are not estimated directly. They are computed by combining structural parameters with a model, then re-solving the model under the new conditions. The counterfactual is only as good as the parameters and the model.

The five categories are not strictly disjoint. A predictive model can be built from a structural model. A causal effect can be interpreted through the lens of a structural model. A descriptive regression can serve as a moment for structural estimation. But the categories are conceptually distinct, and the distinctions matter for what your estimate means and what it supports.

To see how the same data and the same variables can yield different objects, consider the leverage-investment relationship one more time. An OLS regression of investment on leverage in the cross-section is a descriptive object. A LASSO model that uses leverage and many other characteristics to forecast next-year investment is a predictive object. A difference-in-differences estimate around a credit supply shock is a causal effect, the average effect for treated firms. An estimated adjustment cost parameter from a dynamic investment model is a structural primitive, governing investment dynamics under any regime. And a projection of aggregate investment under a hypothetical tax change, computed from the structural model, is a counterfactual. Five different exercises using overlapping variables, producing five different objects, supporting five different kinds of conclusions.

Key Insight 1

Before you choose a method, decide what object you want. A descriptive association tells you what the world looks like. A prediction tells you what to expect for a new observation. A causal effect tells you what changed because of an intervention. A structural primitive tells you about an underlying parameter of behavior. A counterfactual tells you what would happen in a world that did not occur. These are different objects. They are not substitutes.

Measurement

A separate issue, which will recur throughout the book, is measurement. Even after deciding that the object is a causal effect or a structural primitive, you must ask whether the empirical proxy measures the theoretical variable. Book leverage is not necessarily debt capacity. Compustat capital expenditure is not always economic investment. Tobin's Q is an imperfect proxy for marginal q . Reported cash flow mixes operating performance with accounting choices. Measurement gaps contaminate both causal and structural exercises, and they cannot be repaired by better identification or a better-specified model. A beautifully identified LATE for the wrong proxy is the wrong answer to the right question.

We flag measurement issues as they arise in each chapter. You should always check what a paper is actually using before evaluating what it claims to estimate.

The rest of this chapter focuses on the two objects that create the most interpretive confusion: causal effects and structural primitives. Descriptive and predictive objects are also central to empirical corporate finance, but their main dangers are different. The danger with description is mistaking it for causation. The danger with prediction is mistaking it for explanation. Counterfactuals appear as outputs of structural exercises, never on their own.

1.3 The Structural Tradition

A structural approach in empirical corporate finance has three components, each of which the researcher must specify (Reiss and Wolak, 2007; Haile, 2024).

Component 1: an economic model. An economic model specifies who the agents are, what they know, what they choose, and what they face. For a corporate finance application, the agent is a firm or a manager. The choices are investment, financing, payout, and entry or exit. The constraints are technology, markets, contracts, and the actions of others. The model also takes a stand on the agent's objective, typically maximization of expected discounted shareholder value, and on the information available at each decision point.

The model is not the world. It is a deliberate simplification. What makes it useful is that the simplifications are *economic*: they say something about how firms behave, which constraints bind, and which channels matter. The choice of model encodes the researcher's economic understanding. Different researchers may disagree about which model is appropriate, and that disagreement is often productive.

Component 2: a stochastic model. A stochastic model says where the randomness in the data comes from. Concretely, it says what the error term in the estimating equation is, and what its statistical properties are. This is the component most often skipped in applied work, and that is a mistake.

The error term is not residual variation. It is a specific unobservable with a specific meaning. Different assumptions about the stochastic structure produce different estimating equations and different valid instruments. We illustrate this point with an example.

Consider a researcher with data on output Q_i , total cost TC_i , and input prices p_{Ki} , p_{Li} for many similar firms. The production technology is Cobb-Douglas with firm-specific productivity A_i . The researcher wants to estimate the production function. Should she

regress $\ln TC_i$ on $\ln Q_i$ and prices? Or $\ln Q_i$ on $\ln TC_i$ and prices? The answer depends on a single modeling assumption: who observes A_i when the firm chooses inputs?

If the firm observes A_i and chooses inputs optimally conditional on it, the cost function is the natural estimating equation, but $\ln Q_i$ is endogenous because the firm chose inputs (and therefore output) with knowledge of A_i . Conditional on output, productivity is not independent of output, so OLS on the cost equation is biased.

If neither the firm nor the researcher observes A_i ex ante, the firm chooses inputs to maximize expected profit. Inputs are chosen without conditioning on A_i , so realized productivity enters the production function as an unforeseen shock orthogonal to chosen inputs. OLS on the production function is much more plausible.

Table 1.1 summarizes the contrast.

Table 1.1. Same data, different stochastic assumptions

Assumption about A_i	Valid estimating equation	Main problem with OLS
Firm observes A_i ; researcher does not	Cost function	$\ln Q_i$ endogenous because inputs chosen with knowledge of A_i
Neither firm nor researcher observes A_i ex ante	Production function	OLS roughly fine; A_i enters as unforeseen shock

Same data, same firms, same technology. Different stochastic assumptions, different valid regressions. This is the Reiss-Wolak example, and the full algebra is in the online supplement. The lesson does not require the algebra. What it requires is the discipline to ask, before running a regression, what the error term is supposed to represent and how it relates to the choices that generated the data.

Component 3: an estimation strategy. The estimation strategy connects the model and the data. The model implies moment conditions, likelihood functions, or simulated moments. The estimator chooses parameter values to make the model match the data along these dimensions. Common choices in corporate finance include GMM on Euler equations, simulated method of moments (SMM) on dynamic models, and maximum likelihood on structural likelihood functions.

Key Insight 2

A structural empirical exercise has three components: an economic model, a stochastic model, and an estimation strategy. Remove any one and the analysis falls apart. Without the economic model, you do not know which regression to run. Without the stochastic model, you do not know what the error term is or which instruments are valid. Without the estimation strategy, you have a theory and no numbers.

Three reasons to go structural

Writing down and estimating a structural model is hard work. It requires the researcher to commit to specific functional forms, to defend auxiliary assumptions, and to solve what is often a complicated computational problem. Why would anyone choose to do this when a clean natural experiment might be available?

There are three reasons.

The first is to estimate parameters that are otherwise unidentified. Some objects of interest cannot be recovered from the data without economic structure. The cost of financial distress is an example. We observe firms in distress and firms not in distress, and we can measure the difference in their outcomes. That difference confounds the effect of distress with the selection of firms into distress. A natural experiment might address selection, but it would deliver the average effect of distress for compliers under a particular instrument, not the structural distress cost that governs how all firms trade off the tax benefits of debt against the risk of trouble. To estimate that parameter, you need a dynamic capital structure model. [Hennessy and Whited \(2007\)](#) build exactly such a model and recover the distress cost, the tax benefit, and the adjustment cost together.

The second is to perform counterfactuals beyond the data. Causal estimates tell you what happened in response to a historical event. Structural estimates, if the model is correct, tell you what would happen under conditions that have not been observed. This is the distinction between interpolation and extrapolation, and it is essential for policy questions. No natural experiment of an exact magnitude has occurred for most policy proposals. A structural model can, in principle, re-solve the firm's problem under the new policy and trace out the response. This logic is the Lucas critique applied constructively: reduced-form coefficients can shift when policy changes because agents re-optimize, but the primitives of preferences, technology, and constraints should not shift, because they describe the rules of optimization rather than the outcomes.

The third is to compare theories. Consider the long-running debate between trade-off and pecking-order theories of capital structure. Both theories make predictions about how

firms choose debt levels, but they start from different assumptions about which frictions matter. A causal design that shocks credit supply estimates the effect of that shock on leverage. Useful, but it does not directly tell you whether trade-off or pecking order better explains the cross-section of capital structure decisions. A disciplined comparison requires embedding each theory in a structural model, estimating each model's parameters, and asking which fits the data.

The investment example in brief

The structural side of this book's running example develops the investment Euler equation with leverage. A firm enters period t with capital K_{it} , outstanding debt B_{it} , and a productivity shock z_{it} . It chooses investment I_{it} and new borrowing $B_{i,t+1}$ to maximize the expected present value of shareholder payouts, subject to a capital accumulation equation $K_{i,t+1} = (1 - \delta)K_{it} + I_{it}$ and a collateral constraint $B_{i,t+1} \leq \theta K_{i,t+1}$.

The first-order conditions yield an investment Euler equation that links the marginal adjustment cost of investing today to the expected marginal value of capital tomorrow, including the value of relaxing the collateral constraint. Leverage enters through the shadow value of the constraint. When the borrowing constraint binds, an extra unit of capital relaxes the constraint by θ , which raises the effective return to investment. The structural parameters are the adjustment cost α_0 , the collateral parameter θ , the depreciation rate δ , and the discount factor β .

Of these, the Investment chapter (Chapter 8) estimates the adjustment technology. Following [Cooper and Haltiwanger \(2006\)](#), it strips the model back to its adjustment-cost block and recovers the adjustment-cost parameters by simulated method of moments: it solves the firm's dynamic program at a candidate parameter vector, simulates a panel, and chooses the parameters so that simulated investment moments (the volatility and serial correlation of the investment rate, and the frequency of investment spikes and of inaction) match their data counterparts. That estimation is deliberately partial. Leverage and the collateral parameter θ are held aside; the financing margin and the shadow value of internal funds re-enter as state variables later, in the Financial Constraints chapter (Chapter 9) and the Capital Structure chapter (Chapter 10). The Investment chapter does not estimate the leverage-augmented Euler equation by GMM.

What you get from the exercise is the adjustment cost itself, a primitive that governs the speed of investment response to any shock. What you pay is faith in the model. If the adjustment cost function is misspecified, or the rational expectations assumption fails, the estimated parameters are wrong even when the J -test does not reject. The trade-off is

generality of conclusions against fragility of assumptions. It is real, and we will return to it in Section 1.7.

Structure as clarity

It is worth stepping back to ask what economic structure is for. The answer goes beyond parameter estimation. Structure is a tool for clarity of thought. When you write down an economic model, you are forced to be explicit about who the agents are, what they know, what they are optimizing, and what constraints they face. That explicitness has benefits even if you never estimate a single structural parameter.

First, structure tells you what to estimate. Without a model, how do you know which regression to run? The Reiss-Wolak example illustrates the point sharply: two plausible regressions with the same variables can have different economic meanings, and only the model tells you which one is which.

Second, structure tells you what an estimate means. A regression coefficient is a number. Its interpretation comes from the model. The model tells you what the number corresponds to in the economic environment, whether a marginal cost, an elasticity, a welfare gain, or something else.

Third, structure is necessary for counterfactuals. If you want to know what would happen under conditions that have never been observed, you need a model that can be re-solved under those conditions. Data description, however sophisticated, can only tell you about the world as it was.

None of this means that structural models are always right or always necessary. It means that when the question requires knowing a primitive, comparing theories, or extrapolating to a counterfactual, structure is the tool that gets you there. For other questions, especially questions about the average causal effect of a specific intervention, the causal tradition provides a different and often more credible approach. That is the subject of the next section.

1.4 The Causal Tradition

The causal tradition starts somewhere different. Rather than beginning with an economic model and asking what parameters generate the data, it begins with a research design and asks what effect of one variable on another the data can credibly identify.

Potential outcomes

The core idea is the potential outcomes framework, developed by [Rubin \(1974\)](#) and elaborated by [Imbens and Rubin \(2015\)](#).

For each firm i , define two potential outcomes. $Y_i(1)$ is the outcome that would be observed if firm i received the treatment. $Y_i(0)$ is the outcome that would be observed if firm i did not receive the treatment. The causal effect of treatment on firm i is the difference

$$\tau_i = Y_i(1) - Y_i(0).$$

This is what we would like to know. The fundamental problem of causal inference is that we never observe both potential outcomes for the same firm. We observe one of them, depending on whether the firm was treated. The other is counterfactual.

Throughout the book we use $T_i \in \{0, 1\}$ for the actual treatment received by firm i , Y_i for the observed outcome, and Z_i for an instrument or assignment variable. Under random treatment assignment, $T_i = Z_i$ and the comparison of average outcomes across the two groups identifies the average treatment effect, $\tau_{ATE} = E[Y_i(1) - Y_i(0)]$.

Random assignment is rare in corporate finance. More often, treatment is endogenous, but some quasi-random instrument Z_i shifts the probability of treatment. In that setting, the IV estimator identifies the local average treatment effect (LATE): the average effect of treatment for the subset of firms whose treatment status changes because of the instrument, the *compliers* ([Imbens and Angrist, 1994](#)). Other firms in the population are unaffected by Z_i and contribute nothing to the LATE. These include the always-takers, who are treated regardless of Z_i , and the never-takers, who are not treated regardless. A fourth group, the defiers, would respond to the instrument in the wrong direction. The LATE interpretation requires that no defiers exist. This is the monotonicity assumption, and it has real economic content: it says that the instrument moves treatment in the same direction for every firm. We develop the complier taxonomy and the LATE machinery in Chapter 5.

A further assumption, often implicit, is that one firm's treatment does not affect another firm's outcome. This is the stable unit treatment value assumption, SUTVA. In corporate finance settings involving peer effects, industry equilibrium, or spillovers, SUTVA deserves explicit examination. We discuss these issues in the chapters where they arise.

The point to register here is that ATE, ATT, and LATE are all causal effects, but they apply to different populations. Reporting a LATE and calling it "the causal effect" is not wrong, but it is incomplete. The causal effect of what, for whom? The answer matters for whether the estimate applies to your policy question.

Identification through design

In the causal tradition, an effect is identified when the research design eliminates confounding. Random assignment does this by construction. In observational settings the researcher must argue, on the basis of institutional knowledge and supporting evidence, that the source of variation in treatment is plausibly exogenous to the outcome of interest.

The design-based standard does not mean “no model.” Every causal design carries assumptions, and those assumptions are themselves modeling choices. Parallel trends, exclusion restrictions, no anticipation, and monotonicity are statements about the data generating process. What design-based inference avoids is the need for a fully specified, solved model of agent optimization. It substitutes a smaller set of assumptions about variation for the larger set of assumptions that a structural model requires. The trade-off is conscious.

The credibility advantage and its price

The credibility revolution shifted empirical economics toward methods whose assumptions can be named and contested. The assumptions of a difference-in-differences design are parallel trends and no anticipation. These can be examined directly in the data through event-time plots and placebo tests. The assumptions of an instrumental variables design are relevance, exclusion, and monotonicity. Relevance can be assessed using first-stage diagnostics, which we develop in Chapter 5. Exclusion can be argued from institutional details and probed with placebo outcomes. Monotonicity can be defended on economic grounds. Each assumption is specific, named, and open to discussion.

This is the credibility advantage. A reader, a referee, or a co-author can point to a specific assumption, name what would violate it, and assess the evidence the paper offers. Disagreement becomes productive. Two researchers who disagree about the main finding of a paper can usually trace the disagreement to a specific assumption. That is harder to do with a structural paper, where the assumptions are spread across the functional form, the stochastic structure, and the estimation strategy, and where violations rarely show up as a clean test failure.

The price of this conservatism is specificity. A well-identified causal estimate tells you the effect of this treatment on these firms in these circumstances. It does not automatically tell you what would happen to different firms, under different shocks, in different institutional environments. This is the external validity problem, and it is acute for the LATE in particular. The LATE applies only to compliers under the specific instrument. Whether the same effect would obtain under a different instrument, or for a different population, is

a separate question that the research design does not answer. As [Imbens \(2010\)](#) puts it, internal validity and external validity are both important, and the credibility revolution prioritizes the first. Whether the estimate generalizes is a separate matter.

The external validity issue is not a fatal flaw. It is an honest statement about what the estimate means. The decision framework in [Section 1.7](#) will help you tell when locality is acceptable and when it is not.

The investment example in brief

The causal side of the running example uses a credit supply shock to identify the effect of changes in financing capacity on investment. A regional shock reduces credit availability for firms exposed to weakened banks but not for unexposed firms. Let $Z_i = 1$ if firm i is exposed and $Z_i = 0$ otherwise. Compare the change in investment around the shock for exposed versus unexposed firms. Under parallel trends, the difference is the average treatment effect on the treated.

For the IV interpretation, the shock is an instrument for the observed change in leverage, which serves as a proxy for the change in financing capacity. The first stage regresses the change in leverage on Z_i , with coefficient π_1 . The reduced form regresses the change in investment on Z_i , with coefficient γ_1 . The IV estimate is the ratio $\hat{\beta}_{IV} = \hat{\gamma}_1 / \hat{\pi}_1$, and it applies to compliers: firms whose financing capacity, and therefore observed leverage, actually changed because of the credit shock. We work through the numbers in [Section 1.6](#).

It is worth being precise about what the IV estimate identifies. Leverage is an observed accounting variable that can move for many reasons: a firm borrows to fund a profitable project, its asset values fall, its equity issuance changes, or its credit constraints bind. The IV estimate is not the effect of leverage in general. It is the effect on investment of credit-supply-induced changes in financing capacity, with leverage serving as the observable proxy through which the shock operates. A different instrument that moved leverage through a different channel, say a tax change that altered the cost of debt without changing the constraint, would identify a different object on different compliers.

What this exercise delivers is a credible answer to a specific question. You learn the average effect of credit-supply-induced changes in financing capacity on investment, for the firms whose financing actually moved. The credibility comes from the research design. The source of variation is plausibly exogenous to investment opportunities, conditional on appropriate controls, and the parallel-trends assumption can be checked directly in event-time plots.

What the exercise does not deliver is also worth stating. You do not learn the structural

adjustment cost from the Euler equation. You do not learn what would happen if the policy environment changed, or if the shock had been a different size. You do not learn the effect for firms not exposed to the credit channel. You do not learn whether the response operates through the collateral channel or the cash-flow channel, because the LATE bundles both mechanisms together. These are not failures of the method. They are honest statements about the scope of the estimate. Confusion arises only when researchers, or their readers, treat a LATE as though it were a structural primitive.

A caveat is in order on the instrument itself. We use “credit supply shock” as a stylized device. Real applications must isolate the credit channel from other consequences of the underlying event. A bank failure affects deposits, lending relationships, local demand, and information flows, all at once. An empirical paper that uses such a shock as an instrument for financing capacity must defend the exclusion restriction directly, often by combining institutional argument with placebo tests on firms with no bank-dependent borrowing. We discuss these issues at length in Chapters 5 and 12.

Common Mistake 3

The most common error in reading causal papers is forgetting that the LATE is for compliers. When a paper reports that a one-percentage-point credit-supply-induced reduction in financing capacity changes investment by 0.5 percentage points, that number applies to firms whose financing actually changed because of this specific instrument. It does not apply to all firms, to all sources of leverage variation, or to counterfactual policies. A different instrument, say a tax reform rather than a credit shock, identifies a different complier population and may yield a different LATE. The two estimates are not contradictory. They answer different questions.

1.5 Why Causal Is Not Always Enough

A reader trained primarily in the causal tradition might conclude at this point that the path forward is clear: find a natural experiment, estimate a LATE, and move on. Structural methods are interesting in principle, but if the goal is credibility, design-based inference wins.

This conclusion is premature. For an important class of economic questions, a well-identified causal effect is not the object of interest. Not because the estimate is wrong, but because the question requires a different kind of answer.

The classic illustration outside corporate finance comes from demand estimation ([Berry and Haile, 2021](#)). In the textbook single-good supply-and-demand model, a valid supply

shifter is exactly what you would use to identify the demand slope, and a valid demand shifter identifies the supply slope. IV works. The Berry-Haile concern is not that IV fails in this textbook case. It is that the structural object needed for serious counterfactual work is usually richer than a single slope.

Consider differentiated-product demand. A market has many products, and the structural demand for any one product depends on all the prices in the market, all the product characteristics, and an unobservable shock to each product, not just one. The substitution patterns across products are what you need for welfare analysis, for evaluating mergers, or for predicting the effect of a new entrant. A single IV coefficient, recovered by instrumenting for one product's price using a cost shifter, averages over all the unobservable demand shocks. It is a well-defined causal response to a particular intervention. It is not, by itself, the structural demand system. To recover the demand system you need variation in multiple dimensions, typically a vector of instruments that move different products' prices, plus the structural restrictions that the demand system imposes on substitution patterns.

The same logic applies whenever the structural relationship involves multiple unobservables, equilibrium interactions, or heterogeneous agents whose responses must be aggregated through a model rather than averaged across the data. Capital market equilibrium with heterogeneous firms has this structure. So do many settings in corporate finance where treatment status is itself a choice that responds to the shock. An IV estimate that averages over the heterogeneity is a well-defined statistical object. It is just not, in general, the structural object the question requires. The question is whether your economic question wants the average or wants the primitives.

The corporate finance version: constraint shares

The same point applies, with high stakes, to the empirical literature on financial constraints.

Consider a study that uses an instrument Z_i for some component of a firm's financing position. Examples include pension funding shocks (Rauh, 2006), credit supply shocks, or tax changes. The reduced form is the effect of Z_i on investment. The first stage is the effect of Z_i on financing. The IV estimate is the ratio.

The standard interpretation reads this IV estimate as a marginal propensity: firms invest $\hat{\beta}$ cents of every extra dollar of internal funds. This interpretation is not quite right.

Frank (2026a) shows that under a particular structural model of financial constraints, in which firms are either constrained or unconstrained, the IV estimate equals the fraction of constrained firms among compliers, θ_C . In the stark version of the model, with a

one-for-one financing constraint and no additional adjustment margin, the individual structural response is binary. The marginal response of investment to financing is one for constrained firms and zero for unconstrained firms: if you give a constrained firm an extra dollar, it invests the extra dollar; if you give an unconstrained firm an extra dollar, it does not change its investment because its investment was not financing-limited to begin with. The LATE averages this binary structural response over the complier population, yielding

$$\hat{\beta}_{IV} \approx \theta_C,$$

the constraint share among compliers. The binary form is a stylized special case. Adjustment costs, partial cash retention, or heterogeneous project quality would attenuate the structural response below one for constrained firms, and the empirical parallel works best when the financing shock and the investment response are measured in matching units. The point that survives in less stylized settings is that the LATE depends on the composition of the complier population as well as on the underlying behavioral response, and reading the LATE as a single marginal propensity hides this dependence.

This interpretation reconciles several puzzling features of the literature. Different IV studies, using different instruments, report different “marginal propensities to invest from cash flow.” [Rauh \(2006\)](#) finds roughly 0.65 using mandatory pension contributions. Other studies, using credit supply shocks or asset price variation, report estimates from near zero to roughly 0.7. Under the marginal-propensity reading, these are contradictory. Some firms invest 0.65 of every extra dollar, others invest essentially nothing. Read as constraint shares, the same numbers tell a coherent story: Rauh’s pension instrument selects a complier population in which roughly two-thirds of firms are constrained, whereas other instruments select complier populations in which the share is lower or higher. The structural marginal response is the same in every study. The variation across studies is in the population, not in the parameter.

The implications for practice are concrete. Knowing that $\theta_C \approx 0.65$ for some complier population is valuable. It tells you that roughly two-thirds of the firms whose financing was moved by the instrument were constrained. That is a revealed-preference measure of constraint prevalence in that population, and it requires no ex-ante classification of firms. But interpreting 0.65 as a marginal propensity to invest out of cash flow that would apply to a different population, or to a different policy, is a mistake. The structural marginal response under this model is either zero or one for any individual firm. There is no firm in the model with a marginal propensity of 0.65. The 0.65 describes the composition of a complier population, not the behavior of any single firm.

Key Insight 4

For some economic questions, a LATE is not the object of interest. This happens when the structural relationship involves multiple unobservables, equilibrium interactions, or discrete agent types. In each case, the LATE is a well-defined causal effect for a specific complier population. But the economic question, what is the structural elasticity, what fraction of firms are constrained, what would happen under a counterfactual policy, requires more. The issue is not that causal methods are wrong. It is that for these questions, they answer the wrong question precisely.

We return to the constraint-share interpretation in Chapter 9, where it organizes the financial constraints literature.

1.6 Same Data, Different Objects

The preceding four sections established the framework in the abstract. This section makes it concrete. We apply four estimators to the same simulated dataset, where we know the true parameter values by construction, and see what each one recovers.

The data

We generate a balanced panel of 200 firms over 10 periods from the structural model of Section 1.3. The true parameters are $\alpha_0 = 2.50$, $\theta = 0.40$, $\delta = 0.10$, and $\beta = 0.95$. Productivity follows an AR(1) process with persistence 0.7. A credit supply shock occurs at $t = 6$, reducing the collateral parameter from $\theta = 0.40$ to $\theta_{\text{post}} = 0.25$ for 30 percent of firms drawn at random. The remaining firms are unaffected. The investment rate averages 12.1 percent, leverage averages 30.8 percent of assets, and 42.6 percent of firm-period observations have a binding collateral constraint. The full code and seed are in the companion repository; the random seed is 8823.

Four estimators, four objects

The numbers below are not meant to calibrate the corporate finance literature. They are chosen to make the objects easy to distinguish from each other. Table 1.2 reports four estimates. We then walk through what each one recovers and why.

Table 1.2. Four estimators applied to the same simulated leverage–investment data

Estimator	Estimate	What it recovers
OLS (pooled)	−0.080	Cross-sectional association
Panel fixed effects	+0.038	Within-firm association
IV (credit shock)	0.487	LATE for compliers
Structural GMM	$\hat{\theta} = 0.365$	Collateral parameter (true: 0.40)

Notes: 200 firms, 10 periods, 2,000 firm-year observations. OLS and FE coefficients are on contemporaneous leverage in a regression of I/K on leverage and Tobin's Q . IV uses the credit shock as an instrument for the change in leverage around $t = 6$. GMM estimates the structural Euler equation by two-step GMM with lagged instruments. Full results in the companion materials.

OLS recovers a descriptive association. Pooled OLS gives −0.080. High-leverage firms invest less on average. This is a real feature of the cross-section, but it is not a causal effect. In the DGP, high-quality firms borrow less and invest more, generating a negative cross-sectional correlation between leverage and investment that has nothing to do with the effect of leverage on investment.

Fixed effects recover a different descriptive association. Panel fixed effects give +0.038. Notice the sign flip. In the cross-section, high-leverage firms invest less. Within a firm, periods of higher leverage are associated with more investment. Both numbers are real features of the data, and they differ because they describe different sources of variation. Pooled OLS mixes between-firm and within-firm variation, and the between component is dominated by the fact that high-quality firms borrow less and invest more. Fixed effects absorb the firm-level unobservables and isolate the within-firm signal. But the within-firm signal still mixes economic channels. Sometimes leverage rises because the firm is funding a profitable project, which raises investment. Sometimes leverage rises because the constraint has tightened, which lowers investment. The FE coefficient averages both channels. It is not the structural parameter, and it is not the LATE.

A reader looking at OLS and FE side by side often asks which is right. The question has no answer. They describe different features of the same data. In our DGP, neither is the structural parameter $\theta = 0.40$ and neither is the LATE. The lesson is that a regression coefficient on its own does not have economic content. It acquires content only when we say what variation generated it and what object it is meant to recover.

IV recovers a LATE. The IV estimate is 0.487. This is the average investment response to the component of observed leverage variation induced by the credit-supply shock, for firms whose financing was actually moved. The constraint-share interpretation of Section 1.5 makes the number interpretable: in our DGP, the structural binary response (one for constrained firms, zero for unconstrained) averages over the compliers to approximately the constraint share among compliers, which is 0.49. The IV estimate 0.487 is very close to this, exactly as the constraint-share interpretation predicts. Under the marginal-propensity reading, 0.487 would suggest that firms invest 49 cents of every extra dollar of internal funds. Under the constraint-share reading, it says that 49 percent of the compliers were constrained. Both readings produce the same number. Only the second is correct under the DGP we used.

GMM recovers the structural parameter. The structural GMM estimate of θ is 0.365, with a standard error such that the true 0.40 is well within the 95 percent confidence interval. GMM is targeting a different object than IV. It is not estimating any reduced-form effect on investment. It is estimating the collateral parameter that governs the borrowing constraint in the firm's dynamic problem. That parameter would support counterfactuals: what would investment look like if the collateral requirement changed? An IV estimate cannot answer that question. The GMM estimate, under the model, can.

The GMM estimate also illustrates the fragility point we flagged in Section 1.3. Re-estimating the same data under a misspecified adjustment cost function (linear rather than quadratic) shifts the implied adjustment cost by more than half, and the misspecified model passes the J -test. This is the structural trade-off in microcosm: the model delivers a counterfactual-capable parameter, but only if you have the model right.

The companion repository contains parallel Python and Stata code that reproduces every estimate in Table 1.2.

What the comparison teaches

Four estimators, one dataset, four different objects. None of them is “the answer” to the question of how leverage affects investment, because that question is not well-posed without specifying which object you want. If you want a description of the cross-section, OLS gives it to you. If you want a description of within-firm variation, FE gives it. If you want the average effect of a specific shock for the firms it actually moved, IV gives it. If you want the underlying collateral parameter that governs investment under any policy regime, GMM (under the model) gives it.

The four estimates are not different answers to the same question. They are answers to different questions. Treating any one as “the truth” to which the others should converge is a category error.

1.7 A Decision Framework

You have read the two traditions. You have seen them applied side by side. Now you need to pick one for your own paper. This section gives you a procedure for making that choice.

Four types of questions

The method you need depends on the type of question you are asking. Four types cover most cases in corporate finance.

Type 1: the causal effect of a specific intervention. What happens to investment when a firm loses access to a credit market? What happens to leverage when a tax rule changes? The estimand is the ATT or LATE. The right tool is a causal design. The key assumption is that the shock is exogenous to the outcome of interest, that treated and control units would have followed parallel paths absent the shock. No structural model is needed. The LATE is what you want, because the question is about the average effect of this intervention on this population.

Type 2: a structural primitive. What is the adjustment cost that governs how quickly firms respond to investment shocks? What is the cost of financial distress that limits optimal leverage? These are parameters of an economic model, primitives that govern behavior under every regime, not just the one observed in the data. The right tool is structural estimation, typically GMM or SMM. The key assumption is that the model is correctly specified. A LATE from any single natural experiment will not recover the primitive, because the LATE averages over heterogeneous responses and is local to a specific instrument.

Type 3: a counterfactual under a new policy. What would corporate leverage look like if the statutory tax rate were cut? No natural experiment of this exact magnitude has occurred. A LATE from a past tax change tells you what happened under that reform. It does not tell you what would happen under a different reform of a different size. The right tool is a structural model that can be re-solved under the new policy. The key assumption

is policy invariance: the structural parameters should not shift when the policy shifts. A LATE from a past reform is still valuable as input to this exercise. It can discipline or validate the structural model. If the model cannot replicate the reduced-form evidence from past shocks, the counterfactual is suspect.

Type 4: a mechanism question. Does the leverage-investment relationship operate through the collateral channel or the cash-flow channel? This question has two parts. Does leverage causally affect investment? A causal design can answer this. Through which channel does the effect operate? A causal estimate alone cannot answer this, because the LATE bundles all mechanisms together. Distinguishing channels requires either a structural model that specifies both mechanisms or a heterogeneity analysis that exploits differential exposure to each channel. This is where the two traditions combine most fruitfully. The structural paper reports reduced-form results so that the underlying variation is transparent. The causal paper interprets its LATE through the lens of a structural model so that the economic content is clear.

The assumption audit

Every empirical method requires assumptions. The choice between structural and causal methods is partly a choice about which assumptions you are willing to defend. Table 1.3 makes the trade-off explicit.

The table reveals an asymmetry in failure modes. Structural misspecification is harder to detect. A misspecified structural model can produce precise, internally consistent, and publishable estimates that are nonetheless wrong. The *J*-test has limited power against many forms of misspecification, and the moments themselves are chosen by the researcher. In our running example, replacing quadratic with linear adjustment costs shifts the estimated adjustment cost by more than half, and the misspecified model passes the *J*-test. Nothing in the standard diagnostics flags the problem.

Design violations in the causal tradition are more visible. A failure of parallel trends shows up in a pre-trends plot. A weak instrument shows up in first-stage diagnostics. An implausible exclusion restriction can be argued about on the basis of institutional knowledge. Referees and readers can contest specific, named assumptions.

This asymmetry partly explains why the credibility revolution happened. The shift toward design-based inference was not driven by a judgment that structural assumptions are inherently worse. It was driven by the fact that design violations are easier to argue about. A discipline that values transparent disagreement will gravitate toward methods

Table 1.3. Assumption audit: structural versus causal

	Structural	Causal
Core assumption	Economic model correctly specified	Research design eliminates confounding
Functional form	Specified and defended at the model level	Less central; often local or design-based
Stochastic model	Explicit	Implicit in potential outcomes and sampling assumptions
Instrument validity	Theory provides moment conditions	Relevance, exclusion, monotonicity; plus SUTVA when relevant
Policy invariance	Parameters invariant to regime changes	LATE local to this instrument and regime
Primary failure mode	Misspecification shifts all estimates	Design violation invalidates the causal claim
Diagnostic tools	J-test, sensitivity to functional form, moment selection	Pre-trends, placebo tests, first-stage diagnostics, balance

whose assumptions are most amenable to scrutiny. But easier to argue about is not the same as more likely to hold. A structural model that is well-grounded in theory and validated against the data may be more reliable than a causal design with a weak exclusion restriction.

Three diagnostic questions

Before you choose a method, write down the answers to three questions.

What is your estimand? Write it down in notation before touching the data. Is it an ATE, an ATT, a LATE for a specific complier population? Is it the adjustment cost α_0 , the collateral parameter θ , the equilibrium leverage under a counterfactual tax rate? If you cannot write the estimand in either potential outcomes notation or structural notation, you do not yet know what you are estimating. The FE coefficient of +0.038 from the running example is not an estimand in either framework. It is a partial correlation that corresponds to no causal effect and no structural parameter. Many published papers in corporate finance have this problem. Do not add to their number.

Does your question require holding unobservables fixed, or averaging over them? This is the deepest diagnostic. It maps to the Berry-Haile lesson of Section 1.5. If the question requires a *ceteris paribus* response, what happens to investment when you change financing capacity holding productivity and all other firm characteristics fixed, then averaging over unobservables will not deliver it. You need a structural model that holds the unobservables fixed analytically. The adjustment cost, the demand elasticity, and the collateral constraint are all *ceteris paribus* objects. If the question is about an average effect for a specific population, what happened to investment when these firms lost access to the bond market, then averaging is appropriate. The LATE is a legitimate average over heterogeneity in the complier population. The issue arises only when you interpret the average as though it were a *ceteris paribus* parameter.

What would falsify your claim? This is the question that separates good empirical work from bad, in both traditions.

For structural work, identify the functional form or stochastic assumption whose violation would overturn your conclusions. Then test sensitivity to it. In our running example, replacing quadratic with linear adjustment costs shifts the estimated adjustment cost by more than half. Any paper that reports a structural adjustment cost without showing the corresponding sensitivity is asking the reader to take functional form on faith. A more demanding test, when feasible, is to use the structural model to predict the response to a particular natural experiment in the data, then compare the prediction to the reduced-form estimate from that experiment. Disagreement is informative. If the model cannot replicate evidence the reader can see directly, the structural conclusions deserve skepticism.

For causal work, identify the assumption whose violation would overturn the causal claim. In an IV design, this is typically the exclusion restriction. Then provide evidence against the violation. If the instrument might affect the outcome through a channel other than the treatment, run a placebo test on units for which the alternative channel does not operate. In our running example, if we worried that the credit shock affected investment directly through customer demand rather than through leverage, we would look for an effect on firms with no bank-dependent borrowing. Finding none would support the exclusion restriction. Finding a large effect would call it into question.

In a difference-in-differences design, the analogous test is for pre-trends. If treated and control units were already moving apart before the shock, the parallel-trends assumption is suspect and the estimate of the treatment effect is contaminated. The test does not prove parallel trends after the shock, but it can falsify them before, which is informative.

A paper that cannot articulate what would overturn its main result is not giving the reader much to evaluate. The same standard applies in both traditions. The structural paper that pre-commits to a sensitivity analysis and reports the result is doing something the reader can engage with. So is the causal paper that pre-specifies a placebo test and reports the result. The paper that adjusts the specification until the preferred result emerges is not.

When to Use 5

The decision rule.

Question type	Method	Key assumption	Diagnostic
Causal effect of specific intervention	DiD, IV, RD	Exclusion; parallel trends	Pre-trends, first-stage, placebos
Structural primitive	GMM, SMM, MLE	Model correctly specified	J-test, sensitivity to functional form
Counterfactual under new policy	Structural model re-solved	Policy invariance of parameters	Validate against past reduced-form evidence
Mechanism	Causal design plus structural interpretation	Both exclusion restriction and model	Heterogeneity analysis; report both

1.8 Connection to the Framework

The structural-causal distinction is the organizing principle of this book. It threads through every chapter that follows.

The methods chapters can be classified by where they sit on the fork. Panel data (Chapter 3) is foundational for both traditions. Event studies (Chapter 4) exist in two versions, one for each tradition, and the chapter treats them side by side. Instrumental variables (Chapter 5) is presented from the causal perspective but shows how IV serves a structural model when the moment conditions come from theory. Difference-in-differences (Chapter 6) is causal, and the chapter develops the modern heterogeneity-robust estimators in full. Machine learning (Chapter 7) is primarily about prediction, the third object from Section 1.2, with applications to causal inference previewed.

The applications chapters show both traditions on the same question wherever the literature supports it. Investment (Chapter 8) is the cleanest case: the structural estimation of [Hennessy and Whited \(2007\)](#) sits next to the causal evidence of [Asker, Farre-Mensa, and Ljungqvist \(2015\)](#). Financial constraints (Chapter 9) is built around the constraint-share interpretation of Section 1.5. Capital structure (Chapter 10) treats trade-off, pecking order, and market timing through both structural and causal lenses. Banking and credit supply (Chapter 12), corporate governance (Chapter 14), and the deal-based literature (Chapter 15) each apply the framework to their respective settings.

Throughout, the five objects of Section 1.2 reappear. A descriptive regression of leverage on firm characteristics shows up in capital structure as a way to motivate the question. A prediction model shows up in the machine learning chapter as a tool for forecasting and for constructing treatment-effect heterogeneity. Causal effects show up everywhere a clean research design is available. Structural primitives show up in investment, capital structure, and constraints. Counterfactuals show up in the policy-evaluation portions of several chapters. The five categories are not equally represented in every chapter, and they should not be. The right object for the question is what we want to recover.

Connection to Framework 6

Every chapter in this book asks two questions of the work it covers. What object is the paper estimating? And what assumptions does the method commit the researcher to? These are not rhetorical questions. They organize the way we read papers, the way we choose methods for our own work, and the way we evaluate the empirical literature. By the end of the book you should be able to answer both questions for any paper you pick up, and to defend your own answers in the same terms.

Exercises

1. **Reading for objects.** Pick a recent empirical paper in corporate finance from a top journal. Identify which of the five objects from Section 1.2 the paper is estimating. Quote the sentence in the paper that, in your view, best identifies the object. In one paragraph, explain whether the paper's conclusions are appropriate for that object.
2. **The Reiss-Wolak example in detail.** In the online supplement to this chapter, derive the cost and production function representations under both information assumptions. Verify that OLS on the cost equation is biased when the firm observes productivity, and that OLS on the production function is much more nearly unbiased when the firm does not. What would change if the productivity shock were persistent rather than iid?

3. **Compliers and external validity.** Suppose two papers use different instruments for credit availability and report IV estimates of the effect of credit on investment of 0.4 and 0.7. Under the constraint-share interpretation of [Frank \(2026a\)](#), what does this tell you about the fraction of constrained firms in each complier population? Under the marginal-propensity interpretation, what would it tell you? Which interpretation do you find more plausible, and why?
4. **Replication.** Reproduce Table 1.2 using the simulated data in the companion repository. Then replace the credit supply shock with a randomly assigned shock to the same set of firms. How do the four estimates change? Which estimator is most sensitive?
5. **Counterfactual analysis.** Take an IV estimate from a published corporate finance paper. Suppose a policy proposal would change the underlying parameter the paper is estimating by a factor of two. Using only the IV estimate, what can you say about the effect of the policy? What additional information would you need to make a credible prediction?
6. **Falsification.** For a paper of your choice, write down the single assumption whose violation would overturn the main conclusion. Find at least one piece of evidence the paper offers in support of that assumption. Could the test the paper runs fail to detect a relevant violation? Explain.
7. **Estimand discipline.** Read the abstract and introduction of three recent empirical corporate finance papers. For each, write down the estimand in either potential outcomes notation or structural notation. Identify papers where the estimand is not clearly defined.
8. **Bridging traditions.** Pick a corporate finance question of interest to you. Describe how you would attack it using a causal design and how you would attack it using a structural model. What would you learn from each that you could not learn from the other?

Further Reading

[Reiss and Wolak \(2007\)](#) is the foundational reference on structural econometrics in industrial organization, and the example in Section 1.3 is drawn from their treatment. [Haile \(2024\)](#) updates the discussion and emphasizes the importance of the stochastic model. [Imbens and Rubin \(2015\)](#) is the standard reference on potential outcomes and causal inference. [Angrist and Pischke \(2009\)](#) remains the most readable introduction to design-based methods, with examples drawn primarily from labor economics. [Berry and Haile \(2021\)](#) is the canonical statement of the demand-estimation argument used in Section 1.5. [Frank \(2026a\)](#) develops the constraint-share interpretation in full and applies it to the financial constraints literature.

Chapter 2

Identification

“The very concept of identification presumes that there are structural features one wishes to uncover.”

— Berry and Haile (2018)

2.1 Why Identification Matters

Chapter 1 introduced the distinction between structural and causal approaches to empirical corporate finance, and it flagged an important ambiguity: the word “identification” means different things depending on which tradition you are working in. In a structural paper, identification means that the model’s assumptions and the data jointly pin down the parameters of interest. In a causal paper, identification means that the research design eliminates confounding so that a comparison of outcomes reveals a treatment effect. Both uses are legitimate, but confusing them leads to trouble. You will hear them confused regularly in seminars.

This chapter resolves the ambiguity. It gives identification a precise, unified treatment that you can carry into every methods chapter and every applied paper you read. The goal is not to develop the full technical apparatus of any single identification strategy. That is the job of Part II. The goal is to build a mental framework so that when you encounter a paper claiming to “identify” something, you know exactly what questions to ask.

Here is the core idea, stated as simply as possible:

Identification is the question of whether the data, combined with the researcher’s maintained assumptions, pin down the quantity of interest. It is a property of the model and the data structure, not of the sample or the estimator.

This definition applies in both traditions. In a structural model, the “quantity of interest” is a parameter vector Θ , and the “maintained assumptions” are the economic model, the stochastic assumptions, and the functional forms. In a causal framework, the “quantity of interest” is a treatment effect such as the *ATE* or *ATT*, and the “maintained assumptions” are the conditions that eliminate selection bias. The machinery differs, but the underlying logic is the same: can we go from what we observe to what we want to know?

Three features of this definition deserve emphasis.

Identification is about the population, not the sample. A parameter can be identified even if your dataset is too small to estimate it precisely. Conversely, having a million observations does not help if the parameter is fundamentally not identified. Identification is a logical property of the research design; estimation is a statistical property of the sample. They are distinct questions, and conflating them is one of the most common mistakes in applied work.

Common Mistake

Confusing identification with estimation. “We have a large sample, so the estimate must be reliable” conflates sample size (an estimation issue) with identification (a design issue). A biased estimator does not become unbiased as $N \rightarrow \infty$ unless the source of the bias is addressed. An endogenous regressor remains endogenous regardless of sample size.

Identification always requires assumptions. There is no assumption-free identification. Even the most transparent randomized experiment assumes no spillovers (SUTVA), proper randomization, and that the treatment is well-defined. The question is never “does this paper make assumptions?” but rather “are the assumptions transparent, plausible, and testable?”

This point cuts symmetrically across the two traditions. Structural papers are sometimes criticized for making strong assumptions, but causal papers make assumptions too. They are just different ones. An IV paper assumes exclusion. A difference-in-differences paper assumes parallel trends. A regression discontinuity paper assumes continuity of potential outcomes at the threshold. None of these is self-evidently true; each must be defended.

Identification requires a model. You cannot even define what it means for a parameter to be “pinned down by the data” without specifying a model (a set of maintained hypotheses) that gives meaning to the parameter. As [Berry and Haile \(2024\)](#) put it, the concept

of identification cannot be separated from the concept of a structure within which the parameter lives. This is true whether the model is a full dynamic structural model or a simple potential outcomes framework. Even the statement “ D_i is randomly assigned” presupposes a framework (potential outcomes) that gives meaning to the claim.

The model need not be elaborate. Writing $Y_i = \alpha + \beta D_i + \varepsilon_i$ and assuming $E[\varepsilon_i | D_i] = 0$ is already a model: it asserts linearity, additivity, and conditional mean independence. The point is that the model exists whether or not the researcher acknowledges it, and identification is always relative to the model. Change the model, and you may change what is identified.

Connection to Framework

Chapter 1 established that both traditions build on models. One tradition is explicit about the economic structure, the other is explicit about the assignment mechanism. This chapter shows that both traditions face the same logical question: do the model and the data pin down the parameter? The answer depends on the assumptions, and the assumptions are always contestable.

2.1.1 A Seminar Thought Experiment

To see why these distinctions matter in practice, consider a thought experiment. You are at a seminar. The speaker estimates the regression

$$\frac{I_{it}}{K_{it}} = a + b \cdot Q_{it} + \varepsilon_{it} \quad (2.1)$$

and reports $\hat{b} = 0.03$ with a standard error of 0.005. The estimate is highly statistically significant. What have you learned?

The answer depends entirely on what b is. If the speaker is working in the structural tradition and maintains that the Hayashi conditions hold (competitive firms, constant returns, quadratic adjustment costs), then $b = 1/\alpha_0$ is the inverse of the adjustment cost parameter, and $\hat{b} = 0.03$ implies $\hat{\alpha}_0 \approx 33$. The regression identifies a structural parameter under those specific economic assumptions.

If the speaker instead reports a valid instrument (say, oil price shocks for energy firms, with Z_{it} excluded from the investment equation), then the IV estimate of b identifies the causal effect of Q on I/K for the subpopulation of compliers (firms whose Q responds to oil prices). The parameter is the LATE, and it requires different assumptions: exclusion and relevance rather than the Hayashi conditions.

If neither the Hayashi conditions nor a valid instrument is invoked, then $\hat{b} = 0.03$ is simply an OLS coefficient from an endogenous regression. It is a precisely estimated number with no economic interpretation. The statistical significance is irrelevant; the problem is that we do not know what the number means.

The coefficient is the same; the equation is the same; the data might even be the same. What changes across the three scenarios is the identification argument. This example (which we develop formally in Section 2.7) is the central illustration of the chapter. It shows that empirical corporate finance is not about estimating equations; it is about defending assumptions that give equations meaning.

2.1.2 The Formal and Informal Senses of “Identification”

One source of confusion in seminars and in the literature is that “identification” is used in two different senses. [Andrews, Gentzkow, and Shapiro \(2020\)](#) provide a clear formal definition: a quantity c is identified if distinct values of c correspond to distinct distributions of the data under the researcher’s maintained assumptions. In other words, the population data can distinguish between different values of the parameter.

But you will also hear identification used informally. [Keane \(2010\)](#) describes this usage: “What are the key features of the data, or the key sources of (assumed) exogenous variation in the data, or the key a priori theoretical or statistical assumptions imposed in the estimation, that drive the quantitative values of the parameter estimates?”

These are not the same thing. The formal definition is binary: a parameter either is or is not identified. The informal notion admits degrees: a parameter might be “strongly” or “weakly” identified, in the sense that some moments carry more information than others. The formal definition is a property of the model; the informal notion is a property of the model-estimator pair.

Both senses are useful, but trouble starts when they are conflated. A parameter can be formally identified by 50 moment conditions while being practically driven by one or two moments that a skeptical reader does not trust. In that case, the parameter is identified in the formal sense but fragile in the informal sense. The [Andrews, Gentzkow, and Shapiro \(2017\)](#) sensitivity framework, which we discuss in Section 2.5.2, bridges the two notions by measuring how much each moment condition contributes to the estimate.

For this chapter, we focus primarily on the formal definition. We state it precisely in the next section. But keep the informal notion in mind, because it motivates many of the practical tools (sensitivity analysis, pre-trend tests, first-stage F-statistics) that empirical researchers use to evaluate the credibility of their identification arguments.

2.1.3 Road Map

The chapter proceeds as follows. Section 2.2 gives the formal definition and distinguishes point identification from set identification. It also introduces the distinction between identification and interpretation, a subtlety that matters when treatment effects are heterogeneous. Section 2.3 works through the mechanics of identification in full algebraic detail, first in the classic supply-and-demand system and then in the investment equation, showing precisely where identification fails and what additional assumptions restore it. Section 2.4 organizes the sources of identification following [Lewbel \(2019\)](#), with corporate finance examples throughout. Section 2.5 develops identification in the structural tradition. Section 2.6 develops identification in the causal tradition and addresses failure modes. Section 2.7 presents the investment-Q regression as a case study showing three different identification claims for the same equation. Section 2.8 closes with a practical identification checklist.

2.2 What Identification Is (and Is Not)

2.2.1 The Formal Definition

The formal definition of identification comes from the econometrics literature dating to [Hurwicz \(1950\)](#) and [Koopmans and Reiersol \(1950\)](#). We state it following the modern treatment in [Lewbel \(2019\)](#) and [Berry and Haile \(2024\)](#).

Let $\mathcal{S}$ denote the set of all *structures*, that is, all possible data-generating processes consistent with the researcher's maintained hypotheses. Let $S_0 \in \mathcal{S}$ be the true structure. A *hypothesis* is any nonempty subset of $\mathcal{S}$, and a hypothesis $\mathcal{H}$ is *true* if $S_0 \in \mathcal{H}$. The researcher is interested in some feature of the true structure, $\theta(S_0)$, which might be a scalar parameter, a vector of parameters, or a function.

Definition: Point Identification

A structural feature $\theta(S_0)$ is **identified** (or **point identified**) under hypothesis $\mathcal{H}$ if $\theta(S_0)$ is uniquely determined within the set $\{\theta(S) : S \in \mathcal{H}\}$ by the joint distribution of the observables.

Equivalently: if two structures $S, S' \in \mathcal{H}$ generate the same joint distribution of observables, then $\theta(S) = \theta(S')$.

In plain language: if you knew the entire population distribution of the data (not just a sample), and you accepted the maintained assumptions $\mathcal{H}$, could you back out a unique

value of θ ? If yes, θ is identified. If two different parameter values could generate the same population distribution under the maintained assumptions, then the data cannot distinguish between them, and θ is not identified.

To make this concrete, consider the simplest possible example. Suppose the model is $Y_i = \mu + \varepsilon_i$ where $E[\varepsilon_i] = 0$. The parameter μ is identified because $\mu = E[Y_i]$, and $E[Y_i]$ is determined by the population distribution of Y_i . No two values of μ can generate the same distribution. This is trivial, but it illustrates the logic: the model (the mean-zero error assumption) plus the population data (the distribution of Y_i) pin down μ uniquely.

Now consider $Y_i = \alpha + \beta X_i + \varepsilon_i$ where $E[\varepsilon_i | X_i] = 0$. Here β is identified as $\beta = \text{Cov}(X_i, Y_i) / \text{Var}(X_i)$, provided $\text{Var}(X_i) > 0$. If X_i is constant, β is not identified. The population data provide a single point, and infinitely many slopes pass through a single point. The condition $\text{Var}(X_i) > 0$ is a *rank condition*: the data must contain sufficient variation for the parameter to be pinned down.

One implication deserves special emphasis: identification presumes structure. You cannot even define what it means for a parameter to be “pinned down by the data” without specifying a model that gives meaning to the parameter. This is true even in the potential outcomes framework. The $ATE = E[Y_i(1) - Y_i(0)]$ is a structural feature. It is defined relative to a framework that posits the existence of potential outcomes and defines treatment effects as their difference. Without that framework, the quantity has no meaning. So when [Berry and Haile \(2024\)](#) say that identification presumes structure, they are not taking sides in the structural-versus-causal debate. They are making a logical point that applies to all empirical research.

2.2.2 Point, Partial, and Set Identification

The definition above describes *point identification*: the parameter is pinned down to a unique value. But there are important intermediate cases.

Set identification (or *partial identification*) arises when the maintained assumptions restrict the parameter to a set, but do not pin it down uniquely. [Manski \(1990\)](#) pioneered this approach by showing that without strong assumptions about selection, treatment effects can typically only be bounded. If we know that outcomes lie in $[0, 1]$, then even without any identifying assumptions, the ATE must lie in $[-1, 1]$. Adding assumptions (such as a monotone treatment response) tightens the bounds without necessarily achieving point identification.

Corporate finance example. Suppose Tobin's Q_{it} is a noisy proxy for marginal q_{it}^* :

$$Q_{it} = q_{it}^* + w_{it}, \quad (2.2)$$

where w_{it} is classical measurement error (mean zero, independent of q_{it}^* and the structural error). In the investment regression

$$\frac{I_{it}}{K_{it}} = a + b \cdot q_{it}^* + u_{it}, \quad (2.3)$$

OLS using the noisy proxy Q_{it} yields a coefficient that converges to

$$\text{plim } \hat{b}^{OLS} = b \cdot \frac{\text{Var}(q^*)}{\text{Var}(q^*) + \text{Var}(w)} = b \cdot \lambda, \quad (2.4)$$

where $\lambda = \text{Var}(q^*)/\text{Var}(Q) \in (0, 1)$ is the signal-to-noise ratio. The OLS estimate is biased toward zero: this is attenuation bias. Without knowing λ , b is not point identified. But we know $\lambda \in (0, 1)$, so

$$\hat{b}^{OLS} < b < \frac{\hat{b}^{OLS}}{\lambda} \quad \text{for any } \lambda \in (0, 1). \quad (2.5)$$

In fact, if we only assume classical measurement error, we know that b lies in the interval $[\hat{b}^{OLS}, +\infty)$ from OLS alone, which is not very informative.¹ The parameter is set-identified: we know it lies in an interval, but we cannot pin down the exact value without further assumptions about measurement quality.

Under-identification (or *non-identification*) is the extreme case where the maintained assumptions place no meaningful restriction on the parameter. In the supply-and-demand system without instruments (Section 2.3.1), the structural slopes are not identified at all: infinitely many demand-supply slope pairs are consistent with the same reduced-form data, and these pairs are not even bounded.

For the remainder of this chapter, and for most of the book, we focus primarily on point identification. But keep partial and set identification in mind, because they arise whenever a paper's claims outrun its identifying variation. When a referee says "this parameter is not identified," the constructive response is sometimes to ask: "Can we at least bound it?"

¹A more useful bound arises from the reverse regression of Q_{it} on I_{it}/K_{it} , which is biased in the opposite direction. Combining OLS and reverse-regression bounds gives a finite interval. See [Erickson and Whited \(2002\)](#) for identification of b using higher-order moments.

Key Insight

Identification is not all-or-nothing. Between the extremes of point identification and total non-identification lies set identification, where the data and assumptions restrict the parameter to a set. Narrowing that set (through stronger assumptions, better measurement, or richer data) is often the practical goal.

2.2.3 Identification versus Interpretation

There is a subtlety that the formal definition obscures. A parameter can be *identified* (pinned down uniquely by the data and assumptions) and yet difficult to *interpret* economically. Identification answers the question “can we recover this number?” Interpretation answers the question “what does this number mean for the economic question we care about?”

The distinction matters most when treatment effects are heterogeneous. Consider the investment- Q regression with an IV. Under the standard assumptions (exclusion, relevance, monotonicity), the IV estimand is the *local average treatment effect* (LATE): the average effect of Q on I/K for complier firms, those whose Q is actually moved by the instrument. The LATE is identified. It is a well-defined number pinned down by the population data and the maintained assumptions.

But is the LATE the parameter you want? If you are using oil price shocks as the instrument, the LATE is the average investment sensitivity for energy firms whose valuations respond to oil prices. This is a specific and potentially narrow subpopulation. If a policymaker asks “how much would investment increase if we raised Q for the average firm?” That is the *ATE*, not the LATE, and the instrument does not identify it.

Frank (2026a) develops this point in the context of financial constraints.² They show that in a model with constrained and unconstrained firms, the LATE from a financing shock identifies the share of constrained firms among compliers (θ_C), not the structural marginal response of investment to financing. The structural response is 1 for constrained firms and 0 for unconstrained firms. The LATE averages over these, weighted by the composition of the complier population. Different instruments (with different complier populations) produce different LATEs, even though the underlying structural parameter is the same.

This is not a failure of identification. Each LATE is identified under its own set of assumptions. The issue is that the identified parameter may not answer the question the researcher or policymaker cares about. The gap between what is identified and what is desired is a gap in *interpretation*, not in *identification*.

²See the treatment effects paper distributed with the course materials.

Key Insight

Identification and interpretation are distinct. A parameter can be identified (pinned down by data and assumptions) yet not answer the economic question you care about. When you read a paper, ask two questions: (1) Is the claimed parameter identified? (2) Is the identified parameter the one that matters for the question being asked?

The structural tradition handles interpretation by embedding parameters in an economic model. The adjustment cost parameter α_0 has a clear interpretation because the model specifies what it means: it governs the curvature of the cost-of-adjustment function, and it implies specific counterfactual predictions about how investment would respond to various shocks. The causal tradition handles interpretation through the framework of treatment effect parameters (*ATE*, *ATT*, *LATE*, *CATE*), each of which corresponds to a specific subpopulation and a specific comparison. Neither tradition avoids the interpretation problem entirely; they solve it differently.

Table 2.1 summarizes the distinction with corporate finance examples.

Table 2.1. Identification versus Interpretation

Setting	What is identified	Question it answers	Question it may not answer
IV for investment-Q (oil shocks)	LATE: avg. b for compliers	How sensitive are energy firms to valuation changes?	How sensitive is the average firm?
Structural Euler equation	Adjustment cost α_0	What is the curvature of adjustment costs?	What happens under a policy not in the data? (requires extrapolation)
DiD of leverage shock	<i>ATT</i> : effect on treated firms	How did treated firms respond?	How would untreated firms have responded?
RD at credit rating threshold	Local <i>ATE</i> at threshold	Effect of rating change for marginal firms	Effect for firms far from threshold

The table illustrates a general pattern. Credible identification typically comes from specific sources of variation (an instrument, a policy change, a discontinuity), and the identified parameter reflects that specificity. The question “what is identified?” is always paired with “for whom?” We return to this issue repeatedly in Part II, especially in the IV chapter (Chapter 5), where the LATE framework is developed in full.

Connection to Framework

The identification-interpretation distinction maps directly onto the structural-causal framework from Chapter 1. Structural models identify parameters with clear economic interpretations (adjustment costs, discount rates) but under strong assumptions. Causal designs identify effects with high internal validity but for potentially narrow populations. Neither tradition dominates; the right choice depends on whether the priority is interpretability or credibility.

2.3 Identification in Practice: Worked Examples

The formal definition is abstract. This section makes it concrete by working through the mechanics of identification step by step, in four settings: the classic supply-and-demand system (the canonical textbook example), an investment equation, a leverage equation (completing the running example), and the measurement error problem that pervades corporate finance. The logic is the same in every case: we write down a structural model, derive the reduced form that the data can reveal, and ask whether we can uniquely recover the structural parameters from the reduced-form coefficients. The goal is for you to see exactly where identification succeeds, where it fails, and what additional assumptions restore it.

2.3.1 Supply and Demand: The Canonical Example

We begin with supply and demand because it is the clearest possible illustration of the simultaneous equations problem, and because the corporate finance examples that follow have exactly the same structure.

The structural model

Consider a market described by linear supply and demand curves. The structural model consists of two equations:

$$Q_i^d = \alpha_d + \beta_d P_i + \varepsilon_i^d \quad (\text{demand}), \quad (2.6)$$

$$Q_i^s = \alpha_s + \beta_s P_i + \varepsilon_i^s \quad (\text{supply}), \quad (2.7)$$

together with the equilibrium condition $Q_i^d = Q_i^s = Q_i$. Here P_i and Q_i are the observed price and quantity in market i , and ε_i^d and ε_i^s are unobserved demand and supply shocks.

We expect $\beta_d < 0$ (demand slopes down) and $\beta_s > 0$ (supply slopes up). We assume $\beta_d \neq \beta_s$, so the curves are not parallel.

The structural parameters are $\theta = (\alpha_d, \beta_d, \alpha_s, \beta_s)$. The question is: can we recover θ from the joint distribution of (P_i, Q_i) ?

The reduced form

To answer this, we solve for the equilibrium. Setting $Q_i^d = Q_i^s$ and solving for P_i :

$$\alpha_d + \beta_d P_i + \varepsilon_i^d = \alpha_s + \beta_s P_i + \varepsilon_i^s, \quad (2.8)$$

which gives

$$P_i = \frac{\alpha_d - \alpha_s}{\beta_s - \beta_d} + \frac{\varepsilon_i^d - \varepsilon_i^s}{\beta_s - \beta_d} \equiv \pi_0^P + \eta_i^P. \quad (2.9)$$

Substituting back into either structural equation gives the equilibrium quantity:

$$Q_i = \frac{\alpha_d \beta_s - \alpha_s \beta_d}{\beta_s - \beta_d} + \frac{\beta_s \varepsilon_i^d - \beta_d \varepsilon_i^s}{\beta_s - \beta_d} \equiv \pi_0^Q + \eta_i^Q. \quad (2.10)$$

Equations (2.9) and (2.10) are the *reduced form*: they express the observables (P_i, Q_i) as functions of the structural parameters and the structural errors alone. No endogenous variable appears on the right-hand side.

The identification failure

Now count unknowns and equations. The reduced-form intercepts are

$$\pi_0^P = \frac{\alpha_d - \alpha_s}{\beta_s - \beta_d}, \quad \pi_0^Q = \frac{\alpha_d \beta_s - \alpha_s \beta_d}{\beta_s - \beta_d}. \quad (2.11)$$

These are two equations in four unknowns $(\alpha_d, \beta_d, \alpha_s, \beta_s)$. The system is underdetermined. Infinitely many combinations of structural parameters are consistent with the same reduced-form intercepts.³

To see the problem from a regression perspective, suppose a researcher runs OLS of Q_i on P_i . Since P_i is determined in equilibrium, it depends on both ε_i^d and ε_i^s . The OLS slope

³The variances and covariance of the reduced-form errors provide additional equations, but they also introduce additional unknowns (the variances and covariance of ε^d and ε^s), so the counting problem persists.

converges to

$$\text{plim } \hat{\beta}^{OLS} = \frac{\text{Cov}(P_i, Q_i)}{\text{Var}(P_i)} = \frac{\text{Cov}(\eta_i^P, \eta_i^Q)}{\text{Var}(\eta_i^P)}. \quad (2.12)$$

Substituting the reduced-form errors from (2.9) and (2.10), and assuming $\text{Cov}(\varepsilon^d, \varepsilon^s) = 0$ for simplicity:

$$\text{plim } \hat{\beta}^{OLS} = \frac{\beta_s \sigma_d^2 - \beta_d \sigma_s^2}{\sigma_d^2 + \sigma_s^2}, \quad (2.13)$$

where $\sigma_d^2 = \text{Var}(\varepsilon^d)$ and $\sigma_s^2 = \text{Var}(\varepsilon^s)$. This is a weighted average of β_d and β_s , neither the demand slope nor the supply slope, but a hybrid that depends on the relative variances of the structural shocks.

To see that this is literally a convex combination, rewrite it as

$$\text{plim } \hat{\beta}^{OLS} = \lambda \beta_s + (1 - \lambda) \beta_d, \quad \lambda = \frac{\sigma_d^2}{\sigma_d^2 + \sigma_s^2} \in (0, 1). \quad (2.14)$$

When demand shocks dominate ($\sigma_d^2 \gg \sigma_s^2$), the data trace out supply, so OLS approximates β_s . When supply shocks dominate ($\sigma_s^2 \gg \sigma_d^2$), the data trace out demand, so OLS approximates β_d . In general, OLS gives something in between: a mongrel coefficient with no structural interpretation.

Key Insight

Simultaneity creates an identification problem: in equilibrium, price is correlated with both demand and supply shocks. OLS of quantity on price does not recover either structural slope. The regression coefficient is a convex combination of the demand and supply parameters, with weights determined by the relative variance of the structural shocks. It has no economic meaning.

Restoring identification with an exclusion restriction

Now suppose we observe an additional variable W_i (say, weather in agricultural markets) that shifts supply but does not appear in the demand equation. The structural model becomes:

$$Q_i^d = \alpha_d + \beta_d P_i + \varepsilon_i^d \quad (\text{demand}), \quad (2.15)$$

$$Q_i^s = \alpha_s + \beta_s P_i + \gamma_s W_i + \varepsilon_i^s \quad (\text{supply}). \quad (2.16)$$

The exclusion restriction is that W_i is excluded from the demand equation: weather shifts how much farmers produce, but does not directly shift how much consumers want to buy (holding price fixed). We also assume $\gamma_s \neq 0$ (weather actually affects supply) and $\text{Cov}(W_i, \varepsilon_i^d) = 0$ (weather is exogenous to demand shocks).

The new reduced form for price is:

$$P_i = \frac{\alpha_d - \alpha_s}{\beta_s - \beta_d} - \frac{\gamma_s}{\beta_s - \beta_d} W_i + \frac{\varepsilon_i^d - \varepsilon_i^s}{\beta_s - \beta_d} \equiv \pi_0^P + \pi_1^P W_i + \eta_i^P. \quad (2.17)$$

The reduced form for quantity is:

$$Q_i = \frac{\alpha_d \beta_s - \alpha_s \beta_d}{\beta_s - \beta_d} - \frac{\gamma_s \beta_d}{\beta_s - \beta_d} W_i + \frac{\beta_s \varepsilon_i^d - \beta_d \varepsilon_i^s}{\beta_s - \beta_d} \equiv \pi_0^Q + \pi_1^Q W_i + \eta_i^Q. \quad (2.18)$$

Now consider the ratio of the reduced-form slopes on W_i :

$$\frac{\pi_1^Q}{\pi_1^P} = \frac{-\gamma_s \beta_d / (\beta_s - \beta_d)}{-\gamma_s / (\beta_s - \beta_d)} = \beta_d. \quad (2.19)$$

The γ_s terms and the $(\beta_s - \beta_d)$ terms cancel, and we recover the demand slope β_d exactly.

Proof of Identification: Demand Slope

Under the assumptions:

- (i) W_i is excluded from the demand equation (W_i does not directly affect Q_i^d),
- (ii) $\gamma_s \neq 0$ (W_i actually shifts supply, ensuring $\pi_1^P \neq 0$),
- (iii) $\text{Cov}(W_i, \varepsilon_i^d) = 0$ (W_i is exogenous to demand shocks),

the demand slope is identified:

$$\beta_d = \frac{\text{Cov}(W_i, Q_i)}{\text{Cov}(W_i, P_i)} = \frac{\pi_1^Q}{\pi_1^P}.$$

This is the population IV (Wald) estimand. It equals the ratio of the reduced-form effects of W_i on quantity and price, and it uniquely determines β_d from the population distribution of observables.

Several features of this result deserve emphasis.

What the instrument does. Weather shifts the supply curve along a fixed demand curve. Different realizations of W_i trace out different points on the demand curve. Without knowing what shifted supply, the equilibrium points (P_i, Q_i) are uninformative about

either curve individually. But variation in W_i that shifts supply (and only supply) traces out the demand curve, allowing us to estimate its slope.

What the exclusion restriction does. The assumption that W_i does not appear in the demand equation is what makes the γ_s terms cancel in equation (2.19). If weather also affected demand (say, because hot weather increases demand for cold drinks), then W_i would appear in both structural equations, the cancellation would fail, and the ratio of reduced-form slopes would not recover β_d .

What relevance does. The assumption $\gamma_s \neq 0$ ensures that $\pi_1^P \neq 0$, so we are not dividing by zero. If weather does not actually shift supply, it provides no price variation along the demand curve, and we learn nothing.

What is identified and what is not. With an instrument that shifts only supply, the *demand* curve is identified but the *supply* curve is not. To identify β_s , we need a separate variable that shifts demand but not supply (e.g., income). This asymmetry is a general feature: identification is equation-by-equation, not system-wide.

Common Mistake

Believing that an instrument “solves endogeneity” in general. An instrument that shifts supply identifies the demand curve but not the supply curve. Each parameter requires its own source of identifying variation. A single instrument identifies one structural equation, not the entire system.

2.3.2 The Investment Equation

The supply-and-demand example is clean and canonical, but it may feel remote from corporate finance. In fact, the exact same identification problem (and the exact same solution) arises in one of the most studied equations in our field: the investment- Q regression. This is the first half of our running example.

The structural model

Consider a simplified version of the investment equation:

$$\frac{I_{it}}{K_{it}} = a + b \cdot q_{it}^* + u_{it}, \quad (2.20)$$

where I_{it}/K_{it} is the investment-capital ratio, q_{it}^* is the unobserved marginal value of capital (marginal q), and u_{it} captures shocks to the cost of adjustment that are orthogonal to investment opportunities. The parameter b is the inverse of the adjustment cost parameter:

$b = 1/\alpha_0$. Under the Hayashi conditions (competitive firm, constant returns to scale, quadratic adjustment costs), q_{it}^* equals the observed average Q_{it} , and equation (2.20) becomes

$$\frac{I_{it}}{K_{it}} = a + b \cdot Q_{it} + u_{it}. \quad (2.21)$$

This looks like a simple regression, and if Q_{it} were exogenous (if $E[u_{it}|Q_{it}] = 0$), then b would be identified:

$$b = \frac{\text{Cov}(Q_{it}, I_{it}/K_{it})}{\text{Var}(Q_{it})}. \quad (2.22)$$

The identification problem

But Q_{it} is almost certainly not exogenous, for two reasons that exactly parallel the supply-and-demand case.

Simultaneity. The market value of the firm (the numerator of Q_{it}) reflects the market's expectation of future cash flows, which depend on investment decisions. Investment affects Q and Q affects investment. The firm's investment policy and its market valuation are jointly determined, exactly as price and quantity are jointly determined in supply and demand.

Omitted variables. The error term u_{it} likely contains variables correlated with Q_{it} . Managerial ability, expected profitability, and financial constraints all affect both investment and market valuation. If any of these are in u_{it} and correlated with Q_{it} , then $E[u_{it}|Q_{it}] \neq 0$.

Formally, when $\text{Cov}(Q_{it}, u_{it}) \neq 0$, the OLS estimand is

$$\text{plim } \hat{b}^{OLS} = b + \frac{\text{Cov}(Q_{it}, u_{it})}{\text{Var}(Q_{it})} \neq b. \quad (2.23)$$

The bias term $\text{Cov}(Q_{it}, u_{it})/\text{Var}(Q_{it})$ is the analogue of the simultaneous equations bias in supply and demand. OLS does not identify b .

Restoring identification with an instrument

Now suppose we observe an instrument Z_{it} that shifts Q_{it} but is excludable from the investment equation. For concreteness, consider oil price shocks for energy firms. A rise in oil prices increases the market valuation of energy firms (their reserves become more valuable) but does not directly change the physical cost of installing new capital. The

model is

$$\frac{I_{it}}{K_{it}} = a + b \cdot Q_{it} + u_{it}, \quad (2.24)$$

$$Q_{it} = \delta_0 + \delta_1 Z_{it} + v_{it}, \quad (2.25)$$

where Z_{it} is the oil price shock. The maintained assumptions are:

- (i) **Exclusion:** $\text{Cov}(Z_{it}, u_{it}) = 0$. Oil price shocks affect investment only through Q_{it} .
- (ii) **Relevance:** $\delta_1 \neq 0$. Oil price shocks actually move Q_{it} .

Under these assumptions, b is identified. Substituting (2.25) into (2.24):

$$\frac{I_{it}}{K_{it}} = \underbrace{(a + b\delta_0)}_{\pi_0} + \underbrace{b\delta_1}_{\pi_1} Z_{it} + \underbrace{(u_{it} + b \cdot v_{it})}_{\eta_{it}}. \quad (2.26)$$

This is the reduced form: it expresses the investment rate as a function of the instrument and a composite error. The reduced-form slope on Z_{it} is $\pi_1 = b\delta_1$, and the first-stage slope is δ_1 . Their ratio gives:

$$\frac{\pi_1}{\delta_1} = \frac{b\delta_1}{\delta_1} = b. \quad (2.27)$$

Proof of Identification: Investment-Q Coefficient

Under the assumptions:

- (i) $\text{Cov}(Z_{it}, u_{it}) = 0$ (oil price shocks are excluded from the investment equation),
 - (ii) $\delta_1 \neq 0$ (oil price shocks are relevant for Q_{it}),
- the investment-sensitivity parameter b is identified:

$$b = \frac{\text{Cov}(Z_{it}, I_{it}/K_{it})}{\text{Cov}(Z_{it}, Q_{it})} = \frac{\pi_1}{\delta_1}.$$

This is the population IV (Wald) estimand. It uniquely determines b from the population distribution of $(I_{it}/K_{it}, Q_{it}, Z_{it})$.

The logic is identical to the supply-and-demand case. In supply and demand, weather shifts the supply curve along a fixed demand curve, tracing out the demand slope. In the investment equation, oil price shocks shift Q along the investment schedule, tracing out the investment sensitivity. The instrument isolates variation in the endogenous variable that is orthogonal to confounders. The exclusion restriction ensures the only channel from Z to the outcome is through the endogenous variable. The identification proof is the same three lines of algebra.

What the instrument identifies

There is an important subtlety that connects back to Section 2.2.3. With heterogeneous effects (different firms have different b_i), the IV estimand is not the average b across all firms. Under the additional assumption of monotonicity (Z shifts Q in the same direction for all firms), the IV estimand is the *local average treatment effect* (LATE): the average b for the subset of firms whose Q is actually moved by the instrument. For oil price shocks, this is the average investment sensitivity for energy firms whose valuations respond to oil prices, a well-defined but potentially narrow population.

This is identification without full interpretation. The LATE is identified; its economic meaning depends on what the complier population looks like. We return to LATE in detail in Chapter 5.

2.3.3 The Leverage Equation: The Running Example

The investment equation is one half of our running example. The other half is leverage. The identification problems are parallel, but the economic setting is different enough that working through the algebra separately is instructive.

The structural model

Consider a simplified regression of firm value on leverage:

$$V_{it} = \gamma_0 + \gamma_1 L_{it} + e_{it}, \quad (2.28)$$

where V_{it} is a measure of firm value (for example, the market-to-book ratio), L_{it} is leverage (debt-to-assets), and e_{it} captures all other determinants of firm value. The parameter γ_1 measures the causal effect of leverage on value (or, depending on the model, a structural parameter from the firm's financing decision).

Many corporate finance theories have predictions about γ_1 . The Modigliani-Miller theorem says $\gamma_1 = 0$ in frictionless markets. Trade-off theory says γ_1 can be positive (from tax shields) or negative (from distress costs), depending on the firm's position relative to its target. The empirical literature commonly finds $\hat{\gamma}_1 < 0$, but what does this estimate identify?

The identification problem

The problem is, by now, familiar. Leverage is not exogenous.

Simultaneity. Firms choose their leverage in response to their perceived value, and leverage in turn affects value through tax shields, distress costs, and agency channels. If the market values a firm highly (high V_{it}), the firm may issue equity rather than debt, reducing leverage. If the firm has high leverage and the market perceives financial distress, V_{it} falls. Value and leverage are jointly determined.

Omitted variables. The error e_{it} contains factors that affect both value and leverage. Profitability is the most prominent: profitable firms tend to have both higher value and lower leverage (because retained earnings reduce the need for debt). [Rajan and Zingales \(1995\)](#) and [Frank and Goyal \(2009\)](#) document the strong negative correlation between profitability and leverage. If profitability is in e_{it} and correlated with L_{it} , OLS is biased.

To see the bias direction, decompose:

$$\text{plim } \hat{\gamma}_1^{OLS} = \gamma_1 + \frac{\text{Cov}(L_{it}, e_{it})}{\text{Var}(L_{it})}. \quad (2.29)$$

If profitable firms have both higher V_{it} (so profitability is in e_{it} with a positive coefficient) and lower L_{it} (because they use retained earnings), then $\text{Cov}(L_{it}, e_{it}) < 0$. The OLS estimate of γ_1 is biased *downward*. It is more negative than the true causal effect. The observed negative correlation between leverage and value may partly (or entirely) reflect the omitted profitability channel rather than a causal effect of debt on value.

This is exactly the same logical structure as supply and demand and the investment equation. The endogenous regressor is correlated with the error term; OLS recovers a mixture of the causal effect and the selection effect.

Common Mistake

Interpreting the negative OLS coefficient on leverage as evidence that debt destroys value. The coefficient reflects both the causal effect of leverage *and* the selection effect: profitable firms choose lower leverage and have higher value. Without an identification strategy, the two effects are confounded.

Restoring identification with an instrument

To identify γ_1 , we need an instrument Z_{it} that shifts leverage but is excludable from the value equation. Consider a tax reform that changes the tax advantage of debt. The

structural model becomes:

$$V_{it} = \gamma_0 + \gamma_1 L_{it} + e_{it}, \quad (2.30)$$

$$L_{it} = \phi_0 + \phi_1 Z_{it} + \xi_{it}, \quad (2.31)$$

where Z_{it} is an indicator for the tax regime (or the firm's before-financing marginal tax rate, following [Graham \(1996\)](#)). The maintained assumptions are:

- (i) **Exclusion:** $\text{Cov}(Z_{it}, e_{it}) = 0$. The tax reform affects firm value only through its effect on leverage, not directly.
- (ii) **Relevance:** $\phi_1 \neq 0$. The tax reform actually changes firms' leverage choices.

Under these assumptions, we can write the reduced form:

$$V_{it} = \underbrace{(\gamma_0 + \gamma_1 \phi_0)}_{\Pi_0} + \underbrace{\gamma_1 \phi_1}_{\Pi_1} Z_{it} + \underbrace{(e_{it} + \gamma_1 \xi_{it})}_{\zeta_{it}}. \quad (2.32)$$

The ratio of the reduced-form slope to the first-stage slope gives:

$$\frac{\Pi_1}{\phi_1} = \frac{\gamma_1 \phi_1}{\phi_1} = \gamma_1. \quad (2.33)$$

Proof of Identification: Leverage-Value Coefficient

Under the assumptions:

- (i) $\text{Cov}(Z_{it}, e_{it}) = 0$ (tax reform is excluded from the value equation),
 - (ii) $\phi_1 \neq 0$ (tax reform is relevant for leverage),
- the causal effect of leverage on value is identified:

$$\gamma_1 = \frac{\text{Cov}(Z_{it}, V_{it})}{\text{Cov}(Z_{it}, L_{it})} = \frac{\Pi_1}{\phi_1}.$$

The algebra is the same three lines, again. This should be becoming monotonous, and that is the point. The identification argument has the same logical structure regardless of whether we are studying demand for agricultural products, investment and Q , or leverage and firm value. The structural model changes; the identification logic does not.

Why the exclusion restriction is hard

In the leverage context, the exclusion restriction is especially hard to defend. A tax reform that changes the advantage of debt may also change the after-tax value of the firm's cash

flows directly (for instance, if the same reform alters the corporate tax rate on earnings). If the reform affects both leverage *and* the value of future cash flows, then $\text{Cov}(Z_{it}, e_{it}) \neq 0$ and the exclusion restriction fails.

This is not a problem unique to this instrument. In corporate finance, most plausible instruments for leverage have channels through which they might also affect value directly. Interest rate shocks affect both the cost of debt (leverage channel) and the discount rate on future cash flows (value channel). Credit supply contractions affect both the availability of debt and the real economy. The difficulty of finding clean instruments for leverage is one reason the literature on the causal effect of capital structure remains unsettled.

Connection to Framework

The leverage example illustrates a recurring theme: the same regression ($V = \gamma_0 + \gamma_1 L + e$) can be structural or causal depending on the maintained assumptions. In a trade-off model, γ_1 is a structural parameter reflecting the net marginal benefit of debt. With an IV, γ_1 is a causal effect (LATE) for complier firms. Without either, γ_1 is just an OLS coefficient contaminated by omitted variable bias. Same equation, same data, different identification claims.

2.3.4 Measurement Error and Partial Identification

The three examples above all involve simultaneity or omitted variables. The regressor is correlated with the error because of the economic structure. There is a fourth source of identification failure that is pervasive in corporate finance: measurement error. It deserves its own treatment because the identification problem is different in character, the solutions are different, and it is arguably the dominant problem in the investment- Q literature.

The classical errors-in-variables model

Return to the investment equation, but now acknowledge that we do not observe marginal q . The structural relationship is

$$\frac{I_{it}}{K_{it}} = a + b \cdot q_{it}^* + u_{it}, \quad (2.34)$$

and the measurement equation is

$$Q_{it} = q_{it}^* + w_{it}, \quad (2.35)$$

where w_{it} is measurement error. Under the classical errors-in-variables assumptions ($E[w_{it}] = 0$, $\text{Cov}(w_{it}, q_{it}^*) = 0$, and $\text{Cov}(w_{it}, u_{it}) = 0$), the observable regression is

$$\frac{I_{it}}{K_{it}} = a + b \cdot Q_{it} + (u_{it} - b \cdot w_{it}). \quad (2.36)$$

The composite error ($u_{it} - b \cdot w_{it}$) is correlated with Q_{it} because Q_{it} contains w_{it} :

$$\text{Cov}(Q_{it}, u_{it} - b \cdot w_{it}) = -b \cdot \text{Var}(w_{it}) < 0 \quad (\text{assuming } b > 0). \quad (2.37)$$

OLS is biased. The probability limit is

$$\text{plim } \hat{b}^{OLS} = b \cdot \frac{\text{Var}(q^*)}{\text{Var}(q^*) + \text{Var}(w)} = b \cdot \lambda, \quad (2.38)$$

where $\lambda \in (0, 1)$ is the *signal-to-noise ratio*, the fraction of the total variance in Q that comes from the true signal q^* . The OLS coefficient is attenuated, biased toward zero. This is the textbook *attenuation bias* result.

Why this is an identification failure

The attenuation bias formula tells us that OLS identifies $b\lambda$, not b . Since λ is unknown (it depends on the ratio of measurement error variance to true signal variance, which we do not observe), b is not point identified from OLS alone. But notice that we do know $\lambda \in (0, 1)$. This means

$$\hat{b}^{OLS} < b,$$

so the OLS estimate is a *lower bound* on b . We have partial identification: $b \in (\hat{b}^{OLS}, \infty)$ from OLS alone, or tighter bounds if we have information about λ .

Restoring point identification

There are several routes to point identification when measurement error is the problem. Each introduces different assumptions.

Instrumental variables. If we have an instrument Z_{it} that is correlated with q_{it}^* but uncorrelated with both u_{it} and w_{it} , then IV identifies b :

$$\text{plim } \hat{b}^{IV} = \frac{\text{Cov}(Z_{it}, I_{it}/K_{it})}{\text{Cov}(Z_{it}, Q_{it})} = \frac{b \cdot \text{Cov}(Z_{it}, q^*)}{\underbrace{\text{Cov}(Z_{it}, q^*) + \text{Cov}(Z_{it}, w)}_{=0}} = b. \quad (2.39)$$

The measurement error w drops out because Z is orthogonal to it. This is the same IV logic as before, but now the endogeneity comes from measurement error rather than simultaneity. The exclusion restriction here requires $\text{Cov}(Z_{it}, w_{it}) = 0$: the instrument must be uncorrelated with the measurement error, not just with the structural error. Finding such instruments is difficult because many variables that predict q^* are also correlated with market-based measurement error in Q .

Higher-order moments. Erickson and Whited (2002) show that if q_{it}^* is non-normally distributed (specifically, if $E[(q^*)^3] \neq 0$), then b can be identified from higher-order moments of the data without an external instrument. The key insight is that third and higher moments of a non-normal variable contain information that second moments do not, and this extra information can separate b from λ . As Geary (1942) showed, the ratio

$$\hat{b}^{HOM} = \frac{E[(I/K)^2 \cdot Q]}{E[(I/K) \cdot Q^2]} \quad (2.40)$$

is a consistent estimator of b under the classical errors-in-variables assumptions plus non-normality of q^* . Erickson and Whited extend this idea using GMM to combine information from multiple moment orders, improving finite-sample performance.

The identification assumption here is distributional: the true marginal q must be skewed. Since marginal q is a shadow value that cannot be negative, skewness is economically plausible. But the identifying information comes from the tails of the distribution, so the estimator can be imprecise in small samples.

Repeated measurements. If two independent measures of q^* are available (say, analyst-based Q and accounting-based Q , with independent measurement errors), then one can be used as an instrument for the other. The key requirement is that the measurement errors be uncorrelated, which is hard to guarantee when both proxies draw on similar information sources.

Key Insight

Measurement error is not just an “attenuation bias” nuisance. It is a full-blown identification failure: OLS does not identify the structural parameter. Restoring identification requires additional assumptions (an instrument, distributional restrictions, or repeated measures), each with its own vulnerabilities. Measurement error is arguably the single most important identification problem in the investment- Q literature.

The interaction of measurement error and simultaneity

In practice, the investment- Q regression suffers from *both* simultaneity and measurement error simultaneously. The OLS coefficient then reflects at least three sources of bias:

1. **Attenuation bias** from measurement error in Q (biases $\hat{b}$ toward zero).
2. **Simultaneous equations bias** from the joint determination of investment and market value (sign ambiguous).
3. **Omitted variable bias** from unobserved determinants of both investment and Q , such as expected profitability or financial constraints (likely biases $\hat{b}$ upward).

These biases can work in opposite directions. Measurement error pushes the coefficient toward zero; omitted variable bias may push it away from zero. The net bias is ambiguous, which is why the investment- Q literature has produced such a wide range of estimates (from small positive to large positive) without convergence.

An IV corrects for both simultaneity and measurement error *if* the instrument is orthogonal to both the structural error and the measurement error. A higher-order moment estimator corrects for measurement error but not for simultaneity or omitted variables. No single method addresses all three problems without assumptions tailored to each. This is why the investment- Q regression remains a subject of active research decades after Hayashi's original contribution.

2.3.5 Lessons from the Proofs

The four worked examples share a common structure that recurs throughout the book:

1. **Write down the structural model.** This specifies the parameters of interest and the relationships between variables.
2. **Derive the reduced form.** Express the observables as functions of the structural parameters and structural errors alone. No endogenous variable appears on the right-hand side.
3. **Ask: can you invert?** Can you go from the reduced-form parameters (which the data reveal) back to the structural parameters (which you want)? If the system is under-determined (more structural unknowns than reduced-form equations), identification fails.
4. **Add identifying assumptions.** An exclusion restriction, an instrument, a functional form restriction, or a distributional assumption all add an equation or remove an unknown, potentially making the system solvable.
5. **Verify the identification.** Show algebraically that the structural parameters are uniquely determined functions of the population distribution of observables, under the maintained

assumptions.

This five-step procedure applies to every identification argument in the book, whether the context is supply and demand, investment and Q , leverage and firm value, or any other corporate finance question. The structural model and the assumptions change; the logic does not.

Key Insight

Every identification argument has the same logical structure: structural model $\rightarrow$ reduced form $\rightarrow$ check invertibility $\rightarrow$ add assumptions if needed $\rightarrow$ verify uniqueness. When you read a paper, reconstruct this chain. If you cannot, the identification argument is either incomplete or you do not yet understand it.

Table 2.2 summarizes the four worked examples side by side.

Table 2.2. Summary of Worked Examples

	Supply & Demand	Investment- Q	Leverage-Value	Measurement Error
<i>Structural parameter</i>	Demand slope β_d	Adj. cost inverse b	Leverage effect γ_1	Adj. cost inverse b
<i>Source of endogeneity</i>	Simultaneity	Simultaneity + omitted variables	Omitted variables (profitability)	Measurement error
<i>OLS identifies</i>	Weighted avg. of β_d, β_s	$b + \text{bias}$	$\gamma_1 + \text{bias}$	$b \cdot \lambda$
<i>Instrument example</i>	Weather	Oil price shocks	Tax reform	External IV or higher moments
<i>Key exclusion restriction</i>	Weather $\nsubseteq$ demand	Oil $\nsubseteq$ investment eqn.	Tax $\nsubseteq$ value eqn. directly	$Z \perp w$

Connection to Framework

Each example can be approached from either the structural or causal tradition. The supply-and-demand system is classic structural econometrics; it can also be framed causally (“the causal effect of price on quantity demanded”). The investment equation has been estimated structurally (via Euler equations) and causally (via IV). The leverage equation has been studied in structural trade-off models and in causal designs using policy shocks. In every case, the identification argument follows the same five steps. The tradition determines the interpretation; the identification logic is universal.

2.4 Sources of Identification

Section 2.3 showed *how* identification works through specific examples. This section steps back and asks: *where* does identification come from? Lewbel (2019) provides a useful taxonomy that organizes the sources into five categories. Each corresponds to a type of assumption that, combined with the data, pins down parameters. We describe each briefly and note how they combine in practice. The goal is a mental map, not a detailed treatment. That is the job of Part II.

2.4.1 Functional Form Restrictions

The simplest source of identification is a restriction on functional form. If the researcher specifies that a relationship is linear, the parameters are typically identified as long as the data provide sufficient variation (the moment matrix $E[X'X]$ has full rank).

This is the default in most applied work, and it is easy to take for granted. But functional form restrictions are substantive economic assumptions. The investment-Q regression is linear in Q because Hayashi's (1982) result requires quadratic adjustment costs. Change to a general convex cost function and linearity breaks down; change to a fixed cost of adjustment and the smooth relationship disappears entirely. Recall from Chapter 1 that switching from quadratic to linear adjustment costs changed the GMM estimate of α_0 . The functional form is not a statistical convenience. It is an economic claim, and it drives what the regression identifies.

Functional form restrictions are the workhorse of structural identification and the most frequent source of hidden assumptions. They are also the hardest to test: short of estimating more flexible specifications and checking whether results change, there is no direct test of whether the assumed functional form is correct.

2.4.2 Exclusion Restrictions

An exclusion restriction states that a variable affects one equation but not another. We proved in Sections 2.3.1, 2.3.2, and 2.3.3 that an exclusion restriction plus relevance identifies the structural parameter of the excluded equation. The algebra was the same in all three cases.

In corporate finance, common exclusion restrictions include: the before-financing marginal tax rate as an instrument for leverage (Graham, 1996), Russell 2000 index reconstitution as an instrument for passive ownership (Appel, Gormley, and Keim, 2016), and geographic proximity to headquarters as an instrument for institutional monitoring.

In each case, the researcher claims a variable shifts the endogenous regressor without directly affecting the outcome.

Exclusion restrictions are *untestable* with a single instrument. The first-stage relevance can be checked (is $\delta_1 \neq 0$?), but whether the instrument operates *only* through the endogenous variable cannot be tested without additional instruments. Economic reasoning is the primary basis for evaluating exclusion restrictions, which is why the credibility of IV estimates hinges on the narrative, not just the statistics. Chapter 5 develops this fully.

2.4.3 Distributional Assumptions

Some strategies rely on assumptions about the distribution of unobservables. Heckman's selection model identifies treatment effects under joint normality of errors, even without an exclusion restriction, though the identification is fragile and driven by the functional form of the inverse Mills ratio rather than by variation in the data.

We saw a different version of this in Section 2.3.4: the [Erickson and Whited \(2002\)](#) higher-order moment estimator identifies the investment-Q coefficient under the assumption that marginal q is non-normally distributed. Here identification comes from the shape of the distribution (specifically, skewness), not from an excluded variable.

Distributional identification tends to be fragile. If the assumed distribution is wrong, the estimates can be severely biased, and the researcher typically has no way to check the distributional assumption directly. Modern practice prefers strategies that do not rely on distributional assumptions when alternatives are available.

2.4.4 Exogenous Variation

Exogenous variation from randomized experiments, natural experiments, or policy discontinuities is the most transparent source of identification. If treatment is assigned independently of potential outcomes, the difference in average outcomes identifies the *ATE* without further assumptions beyond SUTVA and proper randomization.

Corporate finance has few true experiments but many natural experiments. [Khwaja and Mian \(2008\)](#) exploit unanticipated nuclear tests in Pakistan in 1998, which triggered a liquidity crisis for banks with dollar deposits; comparing lending changes across differentially exposed banks identifies the causal effect of bank liquidity on lending. [Lemmon and Roberts \(2010\)](#) use the contraction in credit supply to firms dependent on covenant-violating lenders as a shock to financing.

Exogenous variation makes assumptions transparent and partially testable (balance tests, pre-trends, placebos), which is why the credibility revolution has favored it. But

transparency comes at a cost: the identified parameter is typically local to the specific shock, population, and time period. Whether the result generalizes is an external validity question that the design itself cannot answer. We discuss this tradeoff in Chapters 6 and 5.

2.4.5 Economic Restrictions

The final source is restrictions from economic theory: equilibrium conditions, optimality conditions, no-arbitrage, or Euler equations. The investment Euler equation identifies adjustment cost parameters through the restriction that investment satisfies an intertemporal optimality condition. Rational expectations implies the Euler equation error is orthogonal to information at the time of the decision, providing moment conditions for GMM, a source of identification that requires no external instrument.

No-arbitrage in asset pricing identifies the market price of risk through a restriction from optimal portfolio choice. In corporate finance, trade-off theory generates testable restrictions on the relationship between tax rates, distress costs, and optimal leverage.

Economic restrictions bring the full apparatus of theory to bear, and they can identify parameters that no natural experiment can reach (counterfactual responses to policies not yet implemented, for example). But they are vulnerable to the criticism that the theory may be wrong, and when the theory is wrong, the “identified” parameter inherits the specification error. [Roberts and Whited \(2013\)](#) make this point forcefully for lagged instruments in capital structure regressions: lagged variables are valid instruments in Euler equation estimation because the theory implies the error is an expectational error, but this justification does not carry over to garden-variety regressions where persistence in the variables undermines the exclusion restriction.

2.4.6 How Sources Combine

In practice, most papers draw on more than one source simultaneously, though researchers do not always acknowledge this. A structural Euler equation estimation combines economic restrictions (the optimality condition), functional form (quadratic adjustment costs), and distributional assumptions (rational expectations implies a specific error structure). An IV study combines an exclusion restriction (the instrument is excluded from the outcome equation) with a functional form assumption (the first stage is linear, or at least monotone for LATE) and sometimes an economic restriction (the theory that motivates the instrument’s relevance).

Recognizing which sources a paper relies on is essential for evaluating its identification argument. If an IV paper’s main defense of the exclusion restriction is economic theory,

then the paper is drawing on both exogenous variation *and* economic restrictions. If a structural paper's identification depends on the specific curvature of an assumed cost function, then functional form is doing the heavy lifting, not the economic model per se.

2.4.7 Transparency and Contestability

The five sources differ in how transparent and testable they are. Table 2.3 summarizes.

Table 2.3. Sources of Identification: Transparency and Contestability

Source	Transparency	Testability	Corporate Finance Example
Functional form	Low, buried in specification	Limited, compare alternative specifications	Quadratic vs. general adjustment costs in investment Euler equation
Exclusion restrictions	Moderate, stated explicitly but debatable	Partial (relevance testable; exclusion is not)	Tax rate IV for leverage; Russell index for ownership
Distributional	Low, hard to assess directly	Limited, sensitivity to alternative distributions	Normality in Heckman IPO selection; skewness in Erickson-Whited
Exogenous variation	High, explicit source of variation	Partial (balance tests, pre-trends, placebos)	Pakistan nuclear tests; credit supply shocks
Economic restrictions	Moderate, clear if theory accepted	Joint overidentification test	Euler equation moments; no-arbitrage

Connection to Framework

The structural tradition relies primarily on functional form restrictions, economic restrictions, and distributional assumptions. The causal tradition relies primarily on exogenous variation and exclusion restrictions. Both use assumptions to achieve identification; they differ in which assumptions they emphasize and how transparent those assumptions are. The strongest papers combine sources: a structural model that generates testable reduced-form predictions, or a causal design motivated by theory that disciplines the instrument choice.

2.5 Identification in the Structural Tradition

In the structural tradition, a model is identified if the economic model, stochastic specification, and estimation strategy jointly determine a unique mapping from the population distribution of observables to the parameter vector Θ . All three components of the [Reiss and Wolak \(2007\)](#) framework matter. Chapter 1 introduced this framework; here we focus specifically on how it determines identification and on tools for assessing whether identification is strong enough to be trusted.

2.5.1 What Identifies Structural Parameters?

The investment Euler equation from Chapter 1 yields moment restrictions of the form

$$E [m(\alpha_0, \theta; \text{data}_{it}) \cdot Z_{it}] = 0, \quad (2.41)$$

where $m(\cdot)$ is the Euler equation residual and Z_{it} is a vector of variables in the firm's information set. Three components jointly determine what is identified.

The **economic model** determines which parameters appear and how. The Euler equation comes from a specific optimization problem; the way I/K enters reflects the assumed adjustment cost technology.

The **stochastic model** determines the error structure. Rational expectations implies the Euler equation error is orthogonal to information at time t , providing the moment conditions. This is the source of the instruments, not an external variable, but the theory itself.

The **functional form** determines how many parameters the data must pin down. With quadratic costs, one key parameter α_0 enters in a specific way. With a more general cost function, more parameters appear, requiring richer data variation for identification.

These components are not independent. Change the cost function (functional form) and the Euler equation changes (economic model), which changes what the moment conditions can identify. A researcher who reports estimates without discussing which component is doing the heavy lifting is asking the reader to accept identification on faith.

2.5.2 The AGS Sensitivity Framework

A key concern, highlighted in Section 2.1.2, is that formal identification and practical identification can diverge. [Andrews et al. \(2017\)](#) propose measuring the *sensitivity* of estimates to individual moments. If $\hat{\Theta}$ is estimated by matching moments $\hat{m} = (\hat{m}_1, \dots, \hat{m}_J)$,

the local sensitivity $\Lambda_j = \partial \hat{\Theta} / \partial \hat{m}_j$ shows how much each moment drives the estimate.

A structural estimate might be formally identified by many moment conditions but practically driven by one or two. If those are the moments a skeptical reader least trusts, the estimate is fragile. Andrews et al. (2020) develop this into a formal definition of *transparency*: a structural estimate is transparent to the extent that a reader can predict how the estimate would change under alternative assumptions about specific moments. They recommend reporting the sensitivity matrix Λ alongside the estimates, which allows readers to conduct their own robustness assessments without re-estimating the model.

The practical implication is straightforward. When reading a structural paper, ask: which moments are doing the work? When writing one, report the answer. The Investment chapter (Chapter 8) applies exactly this sensitivity logic to its simulated method of moments estimates of the adjustment technology, asking which simulated moments pin down the adjustment-cost parameters.

Key Insight

Formal identification and practical identification are not the same. A parameter formally identified by 50 moment conditions may be effectively driven by 2. The AGS sensitivity framework diagnoses which moments matter most and makes structural estimation transparent.

2.5.3 Identification Failure in Structural Models

Three types of failure arise in structural work.

Observational equivalence means distinct parameter vectors generate identical data distributions. We demonstrated this in Section 2.3.1: without an instrument, infinitely many demand-supply slope pairs are consistent with equilibrium data. In richer structural models, observational equivalence can be harder to detect because the mapping from parameters to data distributions is nonlinear and high-dimensional.

Weak identification means the moments carry little information about the parameters. The mapping from moments to parameters is nearly flat: large changes in Θ produce tiny changes in moments. Standard errors are large, estimates are unstable across specifications, and confidence intervals based on standard asymptotics may be misleading. This is the structural analogue of the weak-instrument problem in IV estimation.

Fragile identification means identification depends on a feature of the model that a skeptical reader would question (the curvature of the cost function over a narrow range of the data, a distributional tail assumption, or a specific functional form). The AGS sensitivity

framework is designed to detect exactly this problem.

Common Mistake

Treating formal identification as sufficient. A model can be formally identified while poorly identified in practice. Always ask not just “is it identified?” but “how strongly, and by which features of the data?”

2.6 Identification in the Causal Tradition

In the causal tradition, identification means the research design eliminates confounding so that a comparison of outcomes reveals a treatment effect. Each design achieves this through a different identifying assumption. This section previews the four main strategies (full development is in Part II) and then discusses failure modes that parallel Section 2.5.3.

2.6.1 Four Strategies, Four Assumptions

Each causal strategy rests on a specific assumption. For each, we state the assumption, give a corporate finance example, and name the most plausible threat. The details (implementation, diagnostics, and extensions) belong to the chapters listed.

Selection on observables (Chapter 3). *Assumption:* $\{Y_i(1), Y_i(0)\} \perp\!\!\!\perp D_i \mid X_i$. Conditional on observables, treatment is as good as random. *Example:* A study of institutional ownership and firm value argues that, conditional on size, industry, and governance, ownership is unrelated to unobserved value determinants. *Threat:* Unobserved management quality that drives both ownership and value. In corporate finance, selection on observables alone is rarely credible because the set of relevant unobservables is large.

Instrumental variables (Chapter 5). *Assumption:* Z_i affects Y_i only through D_i (exclusion), and Z_i actually shifts D_i (relevance). We proved identification under these assumptions in Section 2.3. *Example:* Russell index reconstitution as an instrument for passive ownership (Appel et al., 2016). *Threat:* If crossing the index threshold also triggers liquidity changes or analyst coverage that independently affect value, exclusion fails.

Difference-in-differences (Chapter 6). *Assumption:* Parallel trends (absent treatment, treated and control groups would have experienced the same outcome change). *Example:* Bertrand, Duflo, and Mullainathan (2004) compare firms in states that adopted antitakeover laws with firms in states that did not. *Threat:* Divergent pre-existing trends (for instance, states adopting antitakeover laws may have been experiencing different economic conditions).

Regression discontinuity (Chapter 5).⁴ *Assumption:* Potential outcomes $E[Y_i(d) | X_i = x]$ are continuous at the threshold. *Example:* Chava and Roberts (2008) use covenant violation thresholds to study how financing constraints affect investment. *Threat:* Manipulation of the running variable (for instance, firms managing financial ratios to avoid crossing the threshold).

Connection to Framework

Every causal strategy has a specific identifying assumption. Selection on observables assumes the CIA. IV assumes exclusion. DiD assumes parallel trends. RD assumes continuity. Reading a paper means evaluating *that assumption*, not the estimator, not the standard errors, not the number of observations.

2.6.2 Identification Failure in Causal Designs

Causal designs have their own failure modes, parallel to the structural failures in Section 2.5.3.

Violation of the identifying assumption is the direct analogue of observational equivalence. If the exclusion restriction fails (the instrument has a direct effect on the outcome), or parallel trends fail (the groups were already diverging before treatment), the design does not identify the treatment effect. Unlike structural failures, these violations are sometimes partially testable: overidentification tests for IV, pre-trend tests for DiD, density tests for RD. But “partially testable” is not “testable”: pre-trends can hold while the parallel trends assumption is violated in the post-period, and a passing overidentification test does not prove that all instruments are valid.

Weak identification arises in causal designs too, though the language differs. In IV, a weak first stage (F -statistic too low) means the ratio-of-slopes argument from Section 2.3 becomes numerically unstable: dividing by a number close to zero amplifies noise and bias. The result is an IV estimate that can be more biased than OLS, with misleading standard errors and confidence intervals. Chapter 5 develops the weak-instrument problem and its diagnostics in detail. In DiD, weak identification arises when treatment and control groups are so different that the parallel trends assumption is doing most of the work, a form of extrapolation. In RD, weak identification arises when there are too few observations near the threshold to estimate the discontinuity precisely.

Narrow identification is the causal analogue of fragile identification. A valid instrument identifies the LATE for compliers, but compliers may be a small and unrepresentative

⁴RD is treated alongside IV in Part II because fuzzy RD is an IV strategy at the threshold.

fraction of the population. A valid DiD identifies the *ATT* for treated firms in a specific time period, but the effect may not generalize to other firms or other periods. A valid RD identifies the treatment effect at the threshold, but firms far from the threshold may respond differently. In each case, the parameter is identified but the policy question may require a different parameter. This is the identification-versus-interpretation gap from Section 2.2.3, viewed from the causal side.

Common Mistake

Assuming that a “credible” research design is immune to identification failures. Causal designs fail when their identifying assumptions are violated, when identification is weak, or when the identified parameter is too narrow to answer the question of interest. Credibility is not binary; it must be argued, not assumed.

2.7 Case Study: The Investment-Q Regression

We can now synthesize the chapter’s argument through the investment-Q regression. The equation is always the same:

$$\frac{I_{it}}{K_{it}} = a + b \cdot Q_{it} + \varepsilon_{it}. \quad (2.42)$$

It has been interpreted in three fundamentally different ways.

2.7.1 Interpretation 1: Structurally Identified

Under Hayashi’s conditions (competitive firm, constant returns to scale, quadratic adjustment costs), Q_{it} equals marginal q and $b = 1/\alpha_0$. Identification comes from functional form and economic restrictions (Section 2.4.1 and 2.4.5). No instrument is needed because the theory delivers the estimating equation directly.

What could go wrong. Any violation of the Hayashi conditions (market power, increasing returns, non-quadratic costs, financial frictions) breaks the equality between average Q and marginal q , and the regression no longer identifies α_0 . The structural identification is only as strong as the economic model.

2.7.2 Interpretation 2: Causally Identified

With a valid instrument Z_{it} excluded from the investment equation, b is identified as the LATE through the ratio-of-slopes argument proved in Section 2.3.2. The Hayashi conditions are not needed; what matters is the exclusion restriction and relevance.

What could go wrong. The exclusion restriction may fail if the instrument affects investment through channels other than Q . The LATE may be specific to a narrow complier population and not generalize (Section 2.2.3).

2.7.3 Interpretation 3: Not Identified

Without Hayashi's conditions and without a valid instrument, OLS recovers b plus the bias term from equation (2.23). As Section 2.3.4 showed, measurement error in Q compounds the problem: the OLS coefficient reflects simultaneity bias, omitted variable bias, and attenuation bias simultaneously. These biases can work in opposite directions, so the net effect on the estimate is ambiguous. The coefficient has no structural or causal interpretation.

2.7.4 The Lesson

Key Insight

Same equation, three identification claims. The regression $I/K = a + bQ + \varepsilon$ has been used to estimate a structural parameter, a causal effect, and nothing in particular. The equation is the same. What differs is the maintained assumptions. Identification is not a property of the equation. It is a property of the equation *plus* the assumptions the researcher defends.

This is the central lesson of the chapter, and it extends beyond investment. The leverage regression $V = \gamma_0 + \gamma_1 L + e$ from Section 2.3.3 admits the same three interpretations: structurally identified under a trade-off model, causally identified with a valid instrument for leverage, or not identified at all. So does any regression in corporate finance where the right-hand-side variable is endogenous. The econometric technique is the tool; the identifying assumption is the argument.

2.8 The Identification Checklist

When reading or designing an empirical study, work through these five questions:

The Identification Checklist

1. **What is the estimand?** Structural parameter or causal effect? For which population?
2. **What is the identifying assumption?** State it in one sentence.
3. **What variation in the data does the identification?** Which observations are being compared? What is tracing out the curve?
4. **What would violate the assumption?** Name the most plausible threat.
5. **How sensitive is the estimate to the assumption?** What robustness checks address this?

This checklist is simple, but using it consistently transforms how you read papers. To illustrate, we apply it to [Lemmon and Roberts \(2010\)](#), who study how credit supply contractions affect corporate financing and investment.

1. *What is the estimand?* The causal effect of a reduction in debt supply on net debt issuance and investment (*ATT* for below-investment-grade firms affected by the credit contraction).

2. *What is the identifying assumption?* Parallel trends: absent the credit supply shocks of 1989–90 (the collapse of Drexel Burnham Lambert, FIRREA, and NAIC guideline changes), below-investment-grade firms would have experienced the same trends in debt issuance and investment as matched unrated firms.

3. *What variation does the identification?* The comparison is between below-investment-grade firms (treated, because their lenders were directly affected by the shocks) and propensity-score-matched unrated firms (control, because their lenders were not directly affected), before and after 1989.

4. *What would violate the assumption?* If the credit contraction coincided with a decline in demand for the products of below-investment-grade firms that did not equally affect unrated firms, the observed decline in investment would partly reflect demand rather than supply. The authors address this by showing similar results using investment-grade firms as an alternative control group and by examining geographic variation.

5. *How sensitive is the estimate?* The authors report estimates with multiple control groups, alternative matching specifications, and subsample analyses. [Frank \(2026a\)](#) reinterpret the near-one-for-one decline in investment through a structural lens, showing it is consistent with a high share of financially constrained firms among compliers ($\hat{\theta}_C \approx 0.97$).

Common Mistake

Focusing on the estimator instead of the identifying assumption. “They used 2SLS” tells you the technique but not whether the instrument is valid. “They used firm fixed effects” tells you how they controlled for time-invariant confounders but not whether time-varying confounders remain. The method is a tool; identification is the argument.

2.9 Looking Ahead

Chapters 1 and 2 together provide the conceptual framework for the entire book. You now know what it means for a parameter to be identified, you have seen identification proved in specific models, and you have a checklist for evaluating identifying assumptions in both traditions.

Part II develops identification strategies in detail. Chapter 3 treats panel data and fixed effects (identification through selection on observables and within-firm variation). Chapter 4 covers event studies. Chapter 5 develops instrumental variables, building on the proofs in this chapter to cover weak instruments, overidentification, LATE, and regression discontinuity. Chapter 6 covers difference-in-differences, including staggered adoption and the recent literature on heterogeneous treatment effects. Chapter 7 introduces machine learning and its connection to causal inference.

In every case, the identification checklist applies. The method changes; the questions do not.

Chapter Summary

- Identification asks whether data plus assumptions pin down the quantity of interest. It is a property of the model and data structure, not of the sample or estimator.
- Identification and interpretation are distinct. A parameter can be identified yet not answer the economic question you care about, especially when treatment effects are heterogeneous.
- In the supply-and-demand system, OLS of quantity on price recovers a convex combination of the structural slopes with no economic meaning (equation 2.14). An exclusion restriction restores identification via the ratio of reduced-form slopes (equation 2.19).
- The same logic applies to the investment-Q equation (equation 2.27), the leverage-value equation (equation 2.33), and the measurement error problem (equation 2.38).
- Five sources of identification (functional form, exclusion restrictions, distributional assumptions, exogenous variation, economic restrictions) differ in transparency and contestability. Most papers draw on more than one source simultaneously.
- In the structural tradition, identification comes from the economic model, stochastic specification, and functional form, operating through moment conditions. The AGS framework assesses which moments drive estimates.
- In the causal tradition, each design has a specific identifying assumption. Causal designs fail when assumptions are violated, when identification is weak, or when the identified parameter is too narrow.
- The investment-Q regression shows the same equation can be structurally identified, causally identified, or not identified at all, depending entirely on the maintained assumptions.
- The identification checklist (estimand, assumption, variation, threat, sensitivity) applies to any paper in any tradition.

Part II

Methods

Chapter 3

Panel Data

3.1 What Does Your Panel Regression Actually Identify?

Open any issue of the *Journal of Finance*, the *Journal of Financial Economics*, or the *Review of Financial Studies*, and you will find panel data regressions with fixed effects on nearly every page. Firm fixed effects, year fixed effects, industry-by-year fixed effects, state-by-year fixed effects. The specifications grow ever more elaborate. It is no exaggeration to say that the panel regression with some combination of fixed effects is the single most commonly used empirical tool in corporate finance research today.

And yet, if you stop a typical researcher mid-presentation and ask, “What variation in your data identifies that coefficient?” you will often receive an uncomfortable silence, or a vague appeal to “within-firm variation.” That is not wrong, exactly. But it is not nearly precise enough. The purpose of this chapter is to make you precise.

Consider a question that has occupied corporate finance for decades: does leverage affect investment? This is a central question for understanding real effects of financial policy, and we will use it as our running example throughout the methods chapters of this book. Suppose you have a standard Compustat panel (thousands of firms observed over many years) and you estimate the following regression:

$$I_{it} = \beta L_{it} + X'_{it}\gamma + \varepsilon_{it}, \quad (3.1)$$

where I_{it} is capital expenditure (scaled by assets) for firm i in year t , L_{it} is leverage (total debt divided by total assets), and X_{it} is a vector of controls. Suppose you find $\hat{\beta} < 0$. What have you learned?

Not much, if you are honest about it. The negative coefficient could reflect a causal effect of debt on investment: firms burdened with more debt invest less because debt service

consumes cash flow, or because debt overhang discourages new projects. But it could also reflect the opposite direction of causality: firms that invest heavily issue equity to finance those investments, thereby reducing leverage. Or it could reflect an omitted variable that drives both outcomes: a talented management team might simultaneously pursue growth (high investment) and maintain conservative financing (low leverage).

Now suppose you add firm fixed effects:

$$I_{it} = \beta L_{it} + X'_{it}\gamma + \alpha_i + \varepsilon_{it}. \quad (3.2)$$

The coefficient $\hat{\beta}$ in equation (3.2) is identified from entirely different variation than the coefficient in equation (3.1). The firm fixed effect α_i absorbs all time-invariant characteristics of firm i (its industry, its geographic location, the quality of its founding team, the nature of its assets, its corporate culture, its historical relationships with lenders). What remains is *within-firm* variation: changes in leverage and investment *for the same firm over time*. The question you are now answering is: when firm i has higher leverage than usual (relative to its own average), does it also have lower investment than usual?

That is a meaningfully different question from the one you started with. Whether it is a *better* question depends entirely on what you are trying to learn.

Key Insight 1

Every time you add a set of fixed effects to a regression, you change the question you are answering. Fixed effects do not simply “control for stuff”. They change what variation in the data identifies your coefficient. Before adding any fixed effect, you should be able to articulate: what variation am I using, and what variation am I discarding?

Here is a fact that should sharpen your intuition. [Lemmon, Roberts, and Zender \(2008\)](#) showed that firm fixed effects account for the majority of the variation in capital structure. When you add firm fixed effects to a leverage regression, the R^2 jumps dramatically. This means that most of the action in leverage is *between* firms (growth firms versus mature firms, tech versus utilities, Boeing versus Google), not *within* firms over time. If you are trying to understand why leverage varies across the economy, firm fixed effects throw away most of the variation you care about. If you are trying to understand how firms adjust leverage in response to shocks, firm fixed effects keep exactly the variation you need.

Neither approach is right or wrong in the abstract. But you need to know which question you are asking.

This tension connects directly to the structural versus causal distinction we established in Chapter 1. In the causal tradition, panel fixed effects are a tool for absorbing confounders. You worry about omitted variables, and you use fixed effects to “soak up” the ones that are time-invariant. The hope is that the remaining within-firm variation in leverage is driven by something plausibly exogenous, or at least less contaminated by selection.

In the structural tradition, the picture is different. A structural model of investment and leverage generates predictions about both cross-sectional and time-series patterns. Firm fixed effects might correspond to permanent productivity differences, and those differences might be *parameters you want to estimate*, not nuisance terms to absorb. The Euler equation approach to investment estimation (which we will examine in the Investment chapter) treats firm heterogeneity as a structural feature of the model, not a confounder to be removed.

The rest of this chapter develops the mechanics, the interpretation, and the practical implementation of panel data methods. We begin with the two fundamental transformations (the within transformation and first differencing) and develop their properties in the context of corporate finance applications. We then turn to two-way fixed effects, high-dimensional fixed effects, clustering and inference, dynamic panels, and the decision framework for choosing a specification. Throughout, we return to the leverage-and-investment example to build intuition.

One final observation before we begin. Panel data methods are foundational for everything that follows in Part II of this book. Event studies are conducted in panel settings. Instrumental variables are typically estimated with panel data and fixed effects. Difference-in-differences is fundamentally a panel data method. If your understanding of what fixed effects do and do not accomplish is vague, every subsequent chapter will be built on a shaky foundation. It is worth getting this right.

3.2 The Within Transformation and First Differences

3.2.1 The problem: unobserved heterogeneity

The central econometric challenge motivating panel data methods is unobserved heterogeneity. In corporate finance, firms differ from one another in ways that are observable (size, industry, age) and in ways that are not (management quality, corporate culture, the quality of internal control systems, informal relationships with creditors). These unobservable differences are a problem when they correlate with both the outcome variable and the explanatory variable of interest.

Write the data-generating process as

$$Y_{it} = \beta D_{it} + X'_{it}\gamma + \alpha_i + \varepsilon_{it}, \quad i = 1, \dots, N, \quad t = 1, \dots, T, \quad (3.3)$$

where Y_{it} is the outcome for firm i in period t , D_{it} is the treatment or variable of interest, X_{it} is a vector of observed controls, α_i is an unobserved firm-specific component that does not vary over time, and ε_{it} is an idiosyncratic error. The composite error is $v_{it} = \alpha_i + \varepsilon_{it}$.

If you ignored the panel structure and estimated (3.3) by pooled OLS (without α_i), you would be estimating

$$Y_{it} = \beta D_{it} + X'_{it}\gamma + v_{it}. \quad (3.4)$$

The OLS estimator of β is consistent only if $\text{Cov}(D_{it}, v_{it}) = 0$. But v_{it} contains α_i , so the condition requires $\text{Cov}(D_{it}, \alpha_i) = 0$: the treatment must be uncorrelated with all permanent unobserved firm characteristics. In corporate finance, that condition almost never holds.

Return to the leverage-and-investment example. What does α_i capture? Anything about the firm that is approximately constant over the sample period and affects both leverage and investment. Consider a few possibilities:

- **Management quality.** A talented CEO might pursue growth opportunities aggressively (high investment) while maintaining conservative financing (low leverage). This generates a negative correlation between L_{it} and α_i through I_{it} , biasing $\hat{\beta}$ even if the true causal effect of leverage on investment is zero.
- **Asset tangibility.** Firms with tangible assets can borrow more easily (high leverage) and also invest more in physical capital (high investment). This generates a *positive* correlation between L_{it} and α_i , biasing $\hat{\beta}$ in the opposite direction.
- **Growth opportunities.** A firm in a rapidly growing industry has both high investment demand and (per the pecking order) may issue equity rather than debt, generating low leverage. Again, the correlation runs through the firm's fundamental type.

The sign of the omitted variable bias depends on which of these forces dominates, and that can vary across samples and time periods. The point is not that one direction of bias is guaranteed, but that bias of some kind is essentially guaranteed. The question is what to do about it.

3.2.2 The within estimator

The insight behind the fixed effects estimator is beautifully simple: if α_i does not change over time, you can eliminate it by subtracting each firm's time average from both sides of the equation.

Define the time average for firm i as

$$\bar{Y}_i = \frac{1}{T} \sum_{t=1}^T Y_{it}, \quad \bar{D}_i = \frac{1}{T} \sum_{t=1}^T D_{it}, \quad \bar{X}_i = \frac{1}{T} \sum_{t=1}^T X_{it}, \quad \bar{\varepsilon}_i = \frac{1}{T} \sum_{t=1}^T \varepsilon_{it}.$$

Averaging equation (3.3) over time gives

$$\bar{Y}_i = \beta \bar{D}_i + \bar{X}_i' \gamma + \alpha_i + \bar{\varepsilon}_i.$$

Subtracting this from (3.3) yields the *within-transformed* (or demeaned) equation:

$$\ddot{Y}_{it} = \beta \ddot{D}_{it} + \ddot{X}_{it}' \gamma + \ddot{\varepsilon}_{it}, \quad (3.5)$$

where $\ddot{Y}_{it} = Y_{it} - \bar{Y}_i$ and so on. The critical observation: α_i vanishes, because $\alpha_i - \alpha_i = 0$. Applying OLS to equation (3.5) produces the *within estimator* (also called the fixed effects estimator):

$$\hat{\beta}^{FE} = \left(\sum_{i=1}^N \sum_{t=1}^T \ddot{D}_{it}' \ddot{D}_{it} \right)^{-1} \left(\sum_{i=1}^N \sum_{t=1}^T \ddot{D}_{it}' \ddot{Y}_{it} \right). \quad (3.6)$$

This is algebraically identical to the least squares dummy variable (LSDV) estimator that includes a separate intercept for each firm, but the within transformation is computationally far more efficient when N is large.¹

To be concrete about what the demeaning does, suppose Apple's average investment rate over your sample is 8% and its average leverage ratio is 25%. In 2015, Apple's investment rate is 10% and its leverage is 30%. The within-transformed data for Apple in 2015 are $\ddot{I}_{it} = 10\% - 8\% = 2\%$ and $\ddot{L}_{it} = 30\% - 25\% = 5\%$. The fixed effects estimator asks: in years when Apple's leverage is 5 percentage points higher than its own average, is Apple's investment rate higher or lower than its own average, and by how much? It does not compare Apple to ExxonMobil. It compares Apple to itself.

3.2.3 What you need for consistency

The within estimator is consistent for β under assumptions that are weaker than what pooled OLS requires, but stronger than many practitioners realize. The key conditions are:

¹With $N = 5,000$ firms and $T = 20$ years, the LSDV approach requires inverting a matrix with 5,000+ rows and columns. The within estimator requires inverting a matrix of dimension K , the number of regressors. Modern software handles this automatically, but understanding the equivalence clarifies what the estimator is doing.

1. **Strict exogeneity:**

$$E[\varepsilon_{it} | D_{i1}, \dots, D_{iT}, X_{i1}, \dots, X_{iT}, \alpha_i] = 0, \quad \text{for all } t. \quad (3.7)$$

This is a strong requirement. It says that the idiosyncratic error in *any* period is uncorrelated with the regressors in *all* periods (past, present, and future). This rules out feedback effects: Y_{it} cannot cause $D_{i,t+1}$ through any channel other than α_i and X_{it} .

In the leverage-and-investment context, strict exogeneity fails if low investment this year causes the firm to issue debt next year (because the firm's growth options changed). It also fails if the firm adjusts leverage in response to past investment outcomes. These feedback channels are economically plausible, which is one reason why panel fixed effects alone are rarely sufficient for credible causal inference in corporate finance.

2. **Rank condition:** The demeaned regressors $\ddot{D}_{it}$ must have sufficient variation. Formally, the matrix $\sum_i \sum_t \ddot{D}_{it}' \ddot{D}_{it}$ must have full rank. This fails if D_{it} does not vary over time within firms. A variable that is constant for each firm (say, the state of incorporation) is perfectly absorbed by α_i and cannot be included as a regressor in a fixed effects model.
3. **No correlation between α_i and D_{it} is *not* required.** This is the whole point. Pooled OLS needs $\text{Cov}(D_{it}, \alpha_i) = 0$; fixed effects does not.

Notice what is *not* on this list: there is no assumption that ε_{it} is homoskedastic or serially uncorrelated. Those conditions affect the efficiency of the estimator and the computation of standard errors (we address inference in Section 3.5), but they are not required for consistency.

Common Mistake 2

A common error is to treat firm fixed effects as a cure for all endogeneity problems. Fixed effects address one specific source of endogeneity: correlation between the regressor and time-invariant unobservables. They do nothing about time-varying omitted variables, reverse causality, or measurement error. Indeed, as we discuss below, fixed effects can make the measurement error problem *worse*. Fixed effects are valuable, but they are not a panacea.

3.2.4 First differences

There is a second way to eliminate the fixed effect α_i : instead of subtracting the time mean, subtract the previous period's observation. Taking first differences of equation (3.3):

$$\Delta Y_{it} = \beta \Delta D_{it} + \Delta X_{it}' \gamma + \Delta \varepsilon_{it}, \quad t = 2, \dots, T, \quad (3.8)$$

where $\Delta Y_{it} = Y_{it} - Y_{i,t-1}$. Since $\alpha_i - \alpha_i = 0$, the fixed effect disappears, just as in the within transformation. OLS applied to (3.8) is called the first-difference (FD) estimator.

First differences and within estimation are related but not identical. With exactly $T = 2$ periods, they produce numerically identical estimates.² With $T > 2$, they generally differ because they weight the data differently. The within estimator uses all T deviations from the firm mean; the FD estimator uses $T - 1$ period-to-period changes.

Which is more efficient? That depends on the error structure. If ε_{it} is serially uncorrelated, the within estimator is more efficient. If ε_{it} follows a random walk, first differences is more efficient because $\Delta \varepsilon_{it}$ is stationary even when ε_{it} is not.³ In practice, the choice matters less for point estimates (both are consistent under the same conditions) than it does for efficiency and standard errors.

But the economic interpretation of first differences can be subtly different, and this matters in corporate finance. Consider the investment regression again. The within estimator asks: when a firm's leverage is above its own average, is its investment below its own average? The FD estimator asks: when a firm's leverage *increases* from last year, does its investment *decrease* from last year? These are questions about deviations from means versus questions about year-to-year changes. For a slowly moving variable like leverage, the two questions can produce quite different answers.

Roberts and Whited (2013) make an important observation in this context. Investment is approximately the first difference of the capital stock: $I_{it} \approx K_{it} - K_{i,t-1} + \delta K_{i,t-1}$. So when you put investment on the left-hand side and include firm fixed effects, you are essentially already working with a differenced variable. The firm fixed effect captures any permanent component of the *level* of capital, and removing it makes little difference to the estimated coefficient on leverage. Erickson and Whited (2012) confirm this: in investment regressions, fixed effects rarely change the qualitative results.

Leverage is a level, not a change. It measures the outstanding stock of debt relative to assets. When you add firm fixed effects to a leverage regression, you throw away the enormous cross-sectional variation in leverage levels and retain only within-firm fluctuations over time. As Lemmon et al. (2008) documented, most of the variation in leverage is between firms, not within firms over time. The coefficient on profitability in a leverage regression often changes dramatically (sometimes even flipping sign) when

²With $T = 2$, the within-transformed data $\check{Y}_{i1} = Y_{i1} - \bar{Y}_i = (Y_{i1} - Y_{i2})/2$ and $\check{Y}_{i2} = (Y_{i2} - Y_{i1})/2$. These are proportional to the first difference $\Delta Y_{i2} = Y_{i2} - Y_{i1}$, so OLS on either gives the same $\hat{\beta}$.

³When the errors are a random walk, $\text{Var}(\varepsilon_{it})$ grows with t and the within-transformed errors $\check{\varepsilon}_{it}$ are heavily correlated. First differencing produces $\Delta \varepsilon_{it} = \varepsilon_{it} - \varepsilon_{i,t-1}$, which is white noise under the random walk. In many corporate finance panels where firm characteristics are persistent, the error structure is closer to a random walk than to white noise, which tilts the argument toward first differences.

firm fixed effects are included. This is not a statistical curiosity; it reflects a fundamental difference in what question is being asked.

Key Insight 3

Whether fixed effects change your results a lot or a little depends on the nature of your dependent variable. If the dependent variable is already a “change” (investment, earnings growth, change in cash holdings), fixed effects have less bite because you are already exploiting within-firm time variation. If the dependent variable is a “level” (leverage, cash-to-assets, board size), fixed effects fundamentally change the identifying variation and often produce dramatically different estimates. When you see fixed effects change your coefficient substantially, do not panic, but do think carefully about which variation you believe is informative.

3.2.5 Long differences

First differencing uses consecutive year-to-year changes. But there is no reason you must difference over exactly one period. *Long differences* use changes over longer intervals, typically three to five years:

$$Y_{it} - Y_{i,t-k} = \beta (D_{it} - D_{i,t-k}) + (X_{it} - X_{i,t-k})' \gamma + (\varepsilon_{it} - \varepsilon_{i,t-k}), \quad (3.9)$$

where k is the differencing horizon. Like annual first differences, long differences eliminate α_i . But they have different statistical and economic properties that make them increasingly popular in corporate finance research.

There are three reasons to consider long differences.

First, measurement error. One of the underappreciated costs of the within transformation is that it amplifies measurement error. Suppose D_{it} is measured with error: the observed value is $D_{it}^* = D_{it} + w_{it}$, where w_{it} is noise. Demeaning removes the signal associated with cross-sectional differences ($\bar{D}_i$, which is estimated from many periods and relatively precise) while retaining the noisy period-to-period variation. The signal-to-noise ratio *falls* after demeaning. This is the [Griliches and Hausman \(1986\)](#) point, and it is especially relevant in corporate finance where key variables (Tobin’s Q , leverage ratios constructed from book values, measures of investment opportunities) are measured with considerable error.

First differences make the measurement error problem even worse: $\Delta D_{it}^* = \Delta D_{it} + \Delta w_{it}$, and $\text{Var}(\Delta w_{it}) = 2\text{Var}(w_{it})$ if the measurement errors are serially uncorrelated. You have doubled the noise variance while potentially eliminating some of the signal.

Long differences offer a partial remedy. Over a three- or five-year horizon, the true variation in D_{it} is typically larger (firms make more substantial changes to leverage over five years than over one year), while the measurement error does not grow proportionally. The signal-to-noise ratio improves as the differencing horizon increases.

Second, transitory fluctuations. McKinnish (2008) emphasizes that short-run variation in explanatory variables often includes transitory components that generate attenuation bias even in the absence of classical measurement error. If a firm's leverage temporarily spikes because of an accounting adjustment or a seasonal pattern in debt issuance, the annual first-difference estimator treats this spike as informative variation. Long differences average over these transitory fluctuations and focus on more durable changes in firm characteristics.

Third, economic interpretation. Long differences answer a different economic question than annual first differences. Annual FD asks: when a firm's leverage increases this year, does its investment decrease this year? Five-year long differences ask: when a firm's leverage rises substantially over a half-decade, does its investment decline over that same horizon? The latter question is often more economically relevant for corporate finance: we care about medium-run responses to persistent changes in financial policy, not about transient year-to-year co-movements that might reflect accounting timing or seasonal patterns.

The cost of long differences is that you lose observations. With 20 years of data and 5-year differences, you have at most 4 non-overlapping long-difference observations per firm (or more if you use overlapping differences, at the cost of serial correlation in the errors that requires careful treatment of standard errors). This is a real limitation for small panels.

Common Mistake 4

Some researchers treat the choice of first differences versus within estimation versus long differences as a minor implementation detail. It is not. Each approach uses different variation in the data, each has different sensitivity to measurement error and transitory fluctuations, and each answers a subtly different economic question. When your results change depending on the transformation you use, that is informative about the nature of the variation in your data. Report the comparison honestly rather than choosing the specification that gives the most convenient results.

3.2.6 When FE and FD fail: a preview

Before moving on to two-way fixed effects and high-dimensional specifications, it is worth pausing to be explicit about the limitations of the basic framework developed so far. Neither the within estimator nor first differences addresses three problems that are pervasive in corporate finance:

Time-varying confounders. Fixed effects absorb time-invariant unobservables. They do nothing about unobservable factors that change over time and correlate with both the outcome and the regressor. If managerial risk appetite shifts in response to economic conditions, and this shift affects both investment and leverage decisions, firm fixed effects will not help. This is why, in practice, researchers typically combine firm fixed effects with time fixed effects (Section 3.3) or with more elaborate identification strategies (the Instrumental Variables and Difference-in-Differences chapters).

Reverse causality. If Y_{it} causes $D_{i,t+1}$ (or D_{it} is simultaneously determined with Y_{it}), the strict exogeneity condition in equation (3.7) is violated. No amount of demeaning fixes this. The investment-leverage example is especially vulnerable: firms that invest heavily may issue debt to finance those investments, generating reverse causality. Addressing this requires instrumental variables (Chapter 5) or a design-based strategy that provides exogenous variation in the treatment.

Measurement error. As we noted above, the within transformation exacerbates classical measurement error by reducing the signal-to-noise ratio. Griliches and Hausman (1986) showed this formally, and it is particularly relevant for Tobin's Q , the most common proxy for investment opportunities in the literature. Erickson and Whited (2000, 2002) demonstrated that measurement error in Q severely biases investment- Q regressions, and that the bias can be *larger* with fixed effects than without. This is a case where the standard "more controls are better" intuition is exactly wrong.

These limitations do not make panel methods useless. They make panel methods *insufficient on their own* for credible causal inference in most corporate finance settings. The real power of panel data comes when you combine the basic fixed-effects machinery with a credible source of variation (a natural experiment, an instrument, a regression discontinuity). That is the subject of the Event Studies, Instrumental Variables, and Difference-in-Differences chapters. But the panel structure is the foundation on which those strategies are built, and understanding what the foundation can and cannot bear is essential.

The rest of the chapter takes up what fixed effects can and cannot do, the high-dimensional

fixed-effects revolution, clustering and inference, the leverage-and-investment running example, dynamic panels and lagged dependent variables, how to choose a panel specification, and the connection to the structural and causal approaches.

3.3 What Fixed Effects Can and Cannot Do

By now you know the mechanics: the within transformation eliminates α_i , and the resulting estimator identifies β from within-unit variation over time. But knowing *how* to compute a fixed effects estimate is not the same as knowing what it tells you. This section is about the gap between those two things (a gap that is, in practice, quite large).

The central message is simple to state and surprisingly hard to internalize: *fixed effects solve exactly one problem*. They remove bias from time-invariant unobservable confounders. That is valuable. But they do not address time-varying confounders, they do not fix reverse causality, and they can actively make measurement error worse. Researchers who treat firm fixed effects as a general-purpose remedy for endogeneity are making a mistake that appears in published papers with depressing regularity.

3.3.1 What They Handle: Time-Invariant Unobservables

The logic is clean. Return to our model,

$$Y_{it} = \beta D_{it} + \alpha_i + \varepsilon_{it}, \quad (3.10)$$

where α_i captures everything about firm i that does not change over the sample period and that affects the outcome Y_{it} . The within transformation eliminates α_i , so the estimator is consistent even when $\text{Cov}(D_{it}, \alpha_i) \neq 0$.

What kinds of things does α_i absorb in corporate finance settings? Management quality is the canonical example. A firm with an unusually talented CEO may simultaneously invest more aggressively and maintain lower leverage. Without firm fixed effects, the pooled OLS coefficient on leverage in an investment regression would confound the within-firm effect of leverage changes with cross-sectional differences driven by management type. Firm fixed effects remove this confound by asking: when *this particular firm's* leverage changes, what happens to *this particular firm's* investment?

Other plausible components of α_i in corporate finance include corporate culture, the quality of a firm's relationship network, geographic advantages, persistent features of the firm's technology, and time-invariant aspects of the regulatory environment. These are all

things that vary across firms but are reasonably stable within a firm over a typical sample period of 10–20 years.

A useful way to think about this: α_i absorbs anything that operates like a firm-specific intercept in the outcome equation. After the within transformation, identification of β comes entirely from deviations of D_{it} and Y_{it} from their respective firm-level means. This is a powerful form of control, but it comes at a cost, a point we now develop.

3.3.2 What They Do Not Handle

There are three major threats that firm fixed effects leave untouched: time-varying confounders, reverse causality, and selection on time-varying unobservables.

Time-varying confounders. Suppose a macroeconomic shock simultaneously increases leverage (because credit spreads tighten) and depresses investment (because demand falls). This shock varies over time, so it is not absorbed by α_i . Adding time fixed effects λ_t handles aggregate shocks that hit all firms equally, but if the shock's impact varies across firms (as most interesting economic shocks do), then λ_t is not enough either.

This is why so many recent papers include industry $\times$ year or even finer interaction fixed effects: they are trying to absorb group-specific time-varying confounders. We return to the design of these high-dimensional fixed effect structures in Section 3.4. The point here is conceptual: no set of fixed effects (no matter how finely specified) can control for a confounder that varies at the same level as the treatment. If the confounder varies at the firm-year level and the treatment varies at the firm-year level, you need a different strategy.

Reverse causality. If Y_{it} causes D_{it} as well as D_{it} causing Y_{it} , demeaning does nothing to resolve the problem. In the leverage-investment setting, this concern is immediate: firms that invest heavily may take on debt to finance those investments, so leverage and investment are jointly determined. Fixed effects remove the time-invariant component of this simultaneity (the fact that growth firms persistently have both high investment and a particular leverage profile), but they do not address the period-by-period feedback between the two variables.

Selection on time-varying unobservables. A firm's CEO receives private information that the firm's prospects have deteriorated. In response, the firm cuts investment *and* adjusts its capital structure. The private information is a time-varying unobservable that

simultaneously drives both variables. Firm fixed effects cannot absorb it because it changes over time.

This concern is so pervasive in corporate finance that it effectively limits what panel regressions alone can establish. The combination of firm and time fixed effects addresses some forms of omitted variables, but the residual endogeneity from time-varying unobservables is precisely what motivates the instrumental variables, difference-in-differences, and regression discontinuity methods covered in Chapters 5–6.

Key Insight 5

Fixed effects address one specific source of endogeneity: correlation between the regressor and a time-invariant component of the error. They are not a general solution to endogeneity. If the reason you are worried about endogeneity involves anything that changes over time, fixed effects alone will not resolve it.

3.3.3 When Fixed Effects Throw Away the Variation You Need

Here is a subtlety that catches many researchers off guard. By construction, the within estimator uses only variation *within* a firm over time. All cross-sectional variation (differences *between* firms in their average levels of D_{it} and Y_{it}) is discarded. Sometimes that is exactly what you want. Sometimes it is a disaster.

Consider two research designs.

Design A: Bank lending channel (Khwaja and Mian 2008). Khwaja and Mian (2008) study how bank liquidity shocks transmit to borrowers. Their identification strategy exploits the fact that many Pakistani firms borrow from multiple banks. By including *firm* fixed effects, they compare the same firm's borrowing from different banks, some hit by a liquidity shock and some not. The within-firm variation is precisely the identifying variation they need: for a given firm, holding demand for credit constant, how does the supply of credit change when a bank is shocked? Without firm fixed effects, the estimate would be confounded by demand-side factors that differ across firms. Here, fixed effects are essential to the research design.

Design B: Land values and investment (Gan, 2007). Gan studies the effect on corporate investment of the collapse in Japanese land values during the early 1990s. The identifying variation is *cross-sectional*: different firms experienced different declines in land values depending on their real estate holdings. Including firm fixed effects would force all firms to have the same mean land value decline, eliminating the cross-firm differences that

identify the effect. In this setting, firm fixed effects would destroy identification rather than strengthen it.

The lesson is that you must ask: *where does the variation that identifies my parameter come from?* If identification comes from within-unit changes over time, firm fixed effects are helpful because they remove cross-sectional confounders. If identification comes from cross-sectional differences, firm fixed effects remove the signal along with the noise.

This distinction has direct implications for the running example. When we estimate

$$I_{it} = \beta L_{it} + \mathbf{X}_{it}\boldsymbol{\gamma} + \alpha_i + \lambda_t + \varepsilon_{it}, \quad (3.11)$$

the coefficient β is identified from *within-firm* variation in leverage. It answers the question: when a firm's leverage changes relative to its own time-series mean, what happens to its investment relative to its own mean? This may or may not be the economically interesting question. If you care about why high-leverage firms invest differently from low-leverage firms (a cross-sectional question), firm fixed effects cannot answer it. If you care about how a given firm's investment responds when its leverage shifts (a within-firm question), then fixed effects are well suited.

As a practical matter, it helps to compute the share of total variation in your key regressor that is within-unit versus between-unit. If most of the variation in D_{it} is cross-sectional, then the within estimator is throwing away most of your data's information and the resulting estimates will be imprecise. We show how to compute and report this diagnostic in Section 3.6.

3.3.4 The Measurement Error Problem

Perhaps the most counterintuitive property of fixed effects is that they make measurement error *worse*. You might expect that controlling for more things should improve your estimates. For measurement error, the opposite is true.

The intuition comes directly from the within transformation. Suppose D_{it} is measured with classical error:

$$D_{it}^{\text{obs}} = D_{it}^* + w_{it}, \quad (3.12)$$

where D_{it}^* is the true value, w_{it} is noise with $E[w_{it}] = 0$, and w_{it} is uncorrelated with D_{it}^* . As we discussed in Chapter 2, classical measurement error produces attenuation bias: the OLS coefficient is biased toward zero, with the degree of bias depending on the signal-to-noise ratio $\text{Var}(D^*)/(\text{Var}(D^*) + \text{Var}(w))$.

Now consider what happens after demeaning. The demeaned regressor is

$$D_{it}^{\text{obs}} - \bar{D}_i^{\text{obs}} = (D_{it}^* - \bar{D}_i^*) + (w_{it} - \bar{w}_i). \quad (3.13)$$

The signal is the within-firm variation in the true variable, $\text{Var}(D_{it}^* - \bar{D}_i^*)$, while the noise is $\text{Var}(w_{it} - \bar{w}_i)$. Demeaning removes the permanent component of the signal (the cross-sectional variation in $\bar{D}_i^*$) but does relatively little to the noise (since w_{it} is typically transitory, its mean $\bar{w}_i$ is small in large- T panels). The signal-to-noise ratio falls, and attenuation bias gets worse.⁴

This matters enormously in corporate finance. Consider the investment- Q literature. Tobin's Q is an imperfect proxy for marginal q , the true measure of investment opportunities. Much of the variation in Tobin's Q across firms reflects persistent differences in firm type. Growth firms have permanently higher Q than value firms. This cross-sectional variation is genuine signal about investment opportunities, but firm fixed effects discard it. What remains is the noisy time-series fluctuation in measured Q , which is more contaminated by stock market sentiment, accounting distortions, and other sources of measurement error. The result is that within-firm estimates of the investment- Q sensitivity are more attenuated than pooled OLS estimates.

Erickson and Whited (2012) demonstrate this point directly: including firm fixed effects in investment regressions rarely changes the qualitative conclusions precisely because investment is approximately the first difference of the capital stock. The firm-level mean in investment is less informative than the firm-level mean in a stock variable like leverage. But when they examine leverage regressions, fixed effects matter substantially, because leverage is a level variable with a large persistent component.

Common Mistake 6

“We include firm fixed effects to address endogeneity.” This sentence appears in hundreds of published papers. It is too vague to be useful. Fixed effects address one specific form of endogeneity, correlation between the regressor and a time-invariant unobservable. You should specify *which* unobservable you are worried about, *why* you believe it is time-invariant, and what threats *remain* after including fixed effects. If the main endogeneity concern is time-varying (as it often is in corporate finance), this sentence is not just imprecise. It is misleading.

⁴Griliches and Hausman (1986) provide the classic treatment. Griliches and Mairesse (1995) examine the problem in production function estimation. The result is general: whenever the permanent component of the true regressor is large relative to the transitory component, demeaning disproportionately removes signal.

3.3.5 Levels Versus Changes: When the Dependent Variable Matters

There is one more nuance worth highlighting because it comes up constantly and is often handled on autopilot. The impact of including fixed effects depends on whether the dependent variable is a level (like leverage) or approximately a change (like investment or the change in cash holdings).

If Y_{it} is a level variable (say, the debt-to-assets ratio), then $\bar{Y}_i$ captures the firm's average leverage over the sample period. The within-transformed dependent variable, $Y_{it} - \bar{Y}_i$, measures deviations from that average. This is a meaningful transformation that changes the economic question: you are now asking about fluctuations around the firm's typical capital structure, not about the level of the capital structure itself.

If Y_{it} is already approximately a change variable (say, capital expenditure scaled by assets, which is roughly the first difference of the capital stock), then $\bar{Y}_i$ is the firm's average investment rate. Demeaning subtracts this average, but since investment rates are typically not highly persistent, demeaning has a smaller effect on both the estimate and its interpretation. The fixed effect α_i is absorbing the firm's "normal" investment rate, which is already something close to a nuisance parameter.

This is why the standard finding in the literature is that firm fixed effects make a large difference in capital structure regressions but a modest difference in investment regressions. It is not that fixed effects are "more important" for leverage. It is that leverage has a much larger permanent, between-firm component, so the within transformation changes the effective sample variation more dramatically.

3.3.6 A Roadmap for What Comes Next

The limitations identified in this section point directly to the methods we develop in subsequent chapters. Each limitation motivates a specific methodological response:

Time-varying confounders that affect all firms in a group similarly can be addressed by adding time-interacted group fixed effects, such as industry $\times$ year, the subject of Section 3.4. Time-varying confounders that operate at the firm level require an entirely different approach: either an instrument that is correlated with the treatment but not with the confounder (Chapter 5), or a research design that isolates plausibly exogenous variation, such as a natural experiment (Chapter 6).

Reverse causality requires instruments, structural modeling, or a design that breaks the simultaneity, for example, an event study that examines responses to an exogenous shock whose timing is not driven by the outcome variable (Chapter 4).

Measurement error in the within estimator can sometimes be addressed by instrumental

variables (if a valid instrument for the mismeasured regressor can be found) or by higher-order moment estimators that exploit distributional assumptions. We discuss these solutions where they arise naturally in the applications chapters.

The structural approach, introduced in Chapter 1, handles some of these problems differently. In a fully specified structural model, the fixed effect is not a nuisance parameter to be differenced away but a component of the economic environment (firm-level productivity, for instance) that can be estimated jointly with the other structural parameters. This reframes the question: rather than asking “what does β identify after we remove α_i ?” the structural approach asks “what are the deep parameters of the economic model that generated both Y_{it} and D_{it} ?” Panel data is used in both traditions, but the role it plays is fundamentally different.

The take-away from this section is not that fixed effects are unhelpful. They are among the most useful tools available. The take-away is that they are a *specific* tool that solves a *specific* problem. Using them effectively requires understanding both what they remove and what they leave behind.

3.4 The High-Dimensional Fixed Effects Revolution

Section 3.3 ended with a puzzle. Firm fixed effects handle time-invariant confounders, and year fixed effects handle aggregate shocks. But the confounders that keep empirical corporate finance researchers up at night are neither time-invariant nor aggregate. They are group-specific and time-varying. The technology sector booms in the late 1990s, energy firms face a regulatory shock in 2010, banks in the Northeast experience different credit conditions than banks in the South during 2008. These shocks affect groups of firms at specific times, and neither α_i nor λ_t can absorb them.

The response has been a dramatic expansion in the complexity of fixed effect structures. Where a typical corporate finance paper in the early 2000s might include firm and year fixed effects, a paper today routinely includes firm, industry×year, and sometimes state×year or even headquarter-county×year effects. This shift has been driven partly by methodological arguments (most forcefully by Gormley and Matsa (2014)) and partly by computational advances that made these specifications feasible. Whether the shift has always been for the better is a question this section takes seriously.

3.4.1 The Problem That Started It All

Before interaction fixed effects became standard, researchers who wanted to control for time-varying industry conditions typically did one of two things: they included industry dummies alongside year dummies, or they “adjusted” their dependent variable by subtracting the industry mean in each year. [Gormley and Matsa \(2014\)](#) showed that both approaches can fail badly.

The core issue is straightforward. Including industry and year dummies separately in

$$Y_{it} = \beta D_{it} + \alpha_i + \lambda_t + \delta_{j(i)} + \varepsilon_{it}, \quad (3.14)$$

where $j(i)$ denotes the industry of firm i , controls for the average difference across industries (via $\delta_{j(i)}$) and the average difference across years (via λ_t). But it does *not* control for the interaction, the fact that industry j may have experienced an unusual shock in year t . If that shock is correlated with both D_{it} and Y_{it} , the estimate of β is biased.

The alternative of subtracting industry-year means (running the regression on $Y_{it} - \bar{Y}_{j(i),t}$) turns out to introduce its own problems. Gormley and Matsa show that this “demeaning” approach is generally inconsistent when the treatment variable also varies within industries. The correct solution is to include industry×year fixed effects directly:

$$Y_{it} = \beta D_{it} + \alpha_i + \delta_{j(i),t} + \varepsilon_{it}. \quad (3.15)$$

Here $\delta_{j(i),t}$ is a separate intercept for each industry-year cell. This absorbs everything that is common to firms in the same industry in the same year: industry-specific business cycles, sector-level regulatory changes, commodity price movements that affect an entire industry, and so on. The coefficient β is now identified from within-firm variation that *also* deviates from the firm’s industry-year average.

Key Insight 7

Industry×year fixed effects absorb all confounders that are common to firms in the same industry at the same time. The cost is that β is identified from a narrower slice of variation: how does a firm’s outcome change when its treatment changes relative to other firms in the same industry in the same year?

3.4.2 The Menu of Interaction Fixed Effects

Once the logic of interaction effects is understood, the question becomes: which interactions should you include? There is no single correct answer; the choice depends on which

confounders threaten your identification. Table 3.1 summarizes common choices and what each absorbs.

Table 3.1. Common fixed effect structures in corporate finance

Fixed effects	What it absorbs	Identifying variation
$\alpha_i + \lambda_t$	Permanent firm traits; aggregate time shocks	Within-firm deviations from firm and year means
$\alpha_i + \delta_{jt}$	Permanent firm traits; industry×year shocks	Within-firm deviations from industry-year means
$\alpha_i + \delta_{jt} + \gamma_{st}$	Above, plus state×year shocks (local demand, regulation)	Within-firm deviations from both industry-year and state-year means
$\alpha_i + \delta_{jt} + \mu_{bt}$	Above, but replacing state with main-bank×year (credit supply)	Within-firm, within-bank-year deviations

Several features of this table are worth noting. First, every specification includes firm fixed effects α_i , which remain valuable for the reasons discussed in Section 3.3. Second, as you move down the rows, the set of absorbed confounders grows, but so does the set of absorbed *variation*. Each additional set of interaction effects narrows the identifying variation further. Third, the bottom rows involve specifications that many researchers use without fully appreciating how little variation remains.

The choice of which row to use should be guided by the specific confounders that threaten your research question, not by a desire to include the most demanding specification possible. More fixed effects is not always better, a point we develop next.

3.4.3 The Overcorrection Problem

The Gormley-Matsa message was clear and important: ad hoc controls for unobserved heterogeneity are often inadequate, and researchers should use proper interaction fixed effects. The field absorbed this lesson. But it may have absorbed it too enthusiastically.

Consider a concrete example. Suppose you have a panel with 1,000 firm-year observations: 200 firms observed over 10 years, spread across 30 industries and 50 states. A specification with firm fixed effects, industry×year effects, and state×year effects requires approximately

$$\underbrace{200}_{\text{firm}} + \underbrace{30 \times 10}_{\text{ind} \times \text{year}} + \underbrace{50 \times 10}_{\text{state} \times \text{year}} \approx 1,000 \text{ parameters} \quad (3.16)$$

for 1,000 observations. You have consumed essentially all of your degrees of freedom

before estimating a single coefficient of interest.⁵

This is not a hypothetical concern. It is straightforward to construct specifications where the number of fixed effect parameters exceeds the number of observations. When that happens, the model is either saturated (zero degrees of freedom, perfect fit, no inference possible) or oversaturated (more parameters than data points, identification breaks down entirely). Even short of full saturation, the consequences of approaching it are serious: standard errors inflate, coefficients become unstable across minor specification changes, and the remaining identifying variation may be dominated by noise.

Recognizing oversaturation in practice. Several warning signs indicate that you are approaching or have crossed the saturation threshold. Standard errors that jump sharply when you add a new set of fixed effects suggest that the additional effects are absorbing signal, not just confounders. Coefficient estimates that change substantially in magnitude or flip sign may indicate that the identifying variation has shifted in character. An R^2 that approaches one mechanically (because almost all variation is explained by fixed effects) is a red flag. And computational warnings about collinearity or convergence failures from your software should never be ignored.

A useful diagnostic is to compute the fraction of variation in your key regressor that survives after projecting out all fixed effects. If the regression

$$D_{it} = \hat{\alpha}_i + \hat{\delta}_{j(i),t} + \hat{\gamma}_{s(i),t} + \tilde{D}_{it} \quad (3.17)$$

yields an R^2 of 0.98, then only 2% of the original variation in D_{it} is available to identify β . This is the variation doing all the work, and you should think carefully about what kind of variation it is. If it is driven by idiosyncratic firm-level shocks that are plausibly exogenous, you are in good shape. If it is dominated by measurement error or trivial fluctuations, you are identifying β from noise.

3.4.4 Singleton Observations and Separation

A technical but practically important issue arises with high-dimensional fixed effects: singleton observations. A singleton is a fixed-effect group that contains only one observation (for example, an industry-year cell with only one firm). Singletons contribute nothing to the estimation of β because there is no within-group variation, yet they are absorbed perfectly

⁵The exact count depends on normalization, collinearity between firm effects and the interaction effects, and the number of singleton cells. But the order of magnitude is right, and the point stands: when parameter counts approach observation counts, you have a problem.

by the fixed effect. Their presence can distort standard errors and inflate R^2 artificially.

Modern software handles this correctly. In Stata, `reghdfe` (Correia 2016) automatically detects and drops singleton observations; in R, the `fixest` package (Bergé, 2018) does the same. If you are using an older command like `xtreg` or `areg`, singleton detection is not automatic, and you may obtain misleading inference without realizing it. This is a mundane but consequential point: your choice of software affects the validity of your results.

Correia (2015) shows that the problem extends beyond individual singletons. Iterative absorption of fixed effects can create sequences of separations: dropping one singleton may cause another group to become a singleton, triggering a cascade. The `reghdfe` algorithm handles these cascades, but researchers should be aware that the effective sample size after singleton removal can be substantially smaller than the nominal sample size. Reporting the number of observations dropped due to singletons is good practice and is increasingly expected by referees.

3.4.5 Running Example: Layering Fixed Effects on Leverage and Investment

To make these ideas concrete, consider what happens as we add fixed effects to the leverage-investment regression. Start with the baseline two-way specification,

$$I_{it} = \beta L_{it} + \mathbf{X}_{it}\boldsymbol{\gamma} + \alpha_i + \lambda_t + \varepsilon_{it}, \quad (3.18)$$

and layer on additional effects one at a time:

$$I_{it} = \beta L_{it} + \mathbf{X}_{it}\boldsymbol{\gamma} + \alpha_i + \delta_{j(i),t} + \varepsilon_{it}, \quad (3.19)$$

$$I_{it} = \beta L_{it} + \mathbf{X}_{it}\boldsymbol{\gamma} + \alpha_i + \delta_{j(i),t} + \gamma_{s(i),t} + \varepsilon_{it}. \quad (3.20)$$

In Section 3.6, we estimate all three specifications on the simulated data and display the results side by side. Three things to watch for as you move from (3.18) to (3.19) to (3.20):

First, how does $\hat{\beta}$ change? If it is stable across specifications, that is reassuring. It suggests that the confounders absorbed by the additional fixed effects were not strongly correlated with leverage. If it changes substantially, the earlier specifications were biased, and the direction of change tells you something about the nature of the confound.

Second, how do the standard errors behave? A modest increase is normal and reflects the reduced identifying variation. A sharp increase (say, doubling or tripling) suggests that

the interaction effects are absorbing most of the useful variation in leverage. At that point you should ask whether the remaining variation is economically meaningful or whether you have pushed the fixed effects too far.

Third, how many observations survive singleton removal? If moving from (3.19) to (3.20) drops a substantial fraction of your sample, that tells you something about the sparsity of your data relative to the fixed effect structure. Losing 20% of observations to singletons is a sign that the specification may be too demanding for your data.

3.4.6 A Decision Framework

The discussion above points toward a practical workflow for choosing your fixed effect structure.

Step 1: Identify the confounders. Before touching any data, write down the specific time-varying confounders that threaten your research design. Are they industry-level shocks? Local economic conditions? Credit-market disruptions that affect a firm's lender? The confounders you name determine the interaction effects you need.

Step 2: Match fixed effects to confounders. Include the interaction effects that absorb the confounders you identified in Step 1. Do not include additional effects "just in case." Each set of interaction effects has a cost (reduced variation, lost observations), and that cost should be justified by a specific identification concern.

Step 3: Build up incrementally. Estimate the specification with firm and year effects first, then add interaction effects one set at a time. At each step, record the coefficient, its standard error, the number of observations, and the residual variation in your key regressor. This table is informative for both you and your readers.

Step 4: Diagnose. Check for singleton cascades. Compute the within-FE variation in your regressor. If the standard error on β has become very large, or the coefficient has changed sign, do not simply report the most saturated specification as your "preferred" result. Instead, discuss the tradeoff transparently. The specification that controls for the most confounders is not automatically the most informative one if it has consumed all of the identifying variation.

Common Mistake 8

“We use the most saturated specification as our baseline.” A specification that absorbs the most potential confounders also absorbs the most potential signal. The “most demanding” specification is the right baseline only if you have confirmed that meaningful identifying variation survives. Always report results across multiple specifications so that readers can assess the robustness of your findings *and* the cost of additional fixed effects.

3.4.7 Computation

The shift to high-dimensional fixed effects would not have been possible without advances in computation. Traditional commands like `xtreg`, `fe` in Stata construct the full matrix of dummy variables, which becomes infeasible when the number of fixed effect categories reaches the thousands. Modern algorithms avoid this by using iterative demeaning methods that project out each set of fixed effects in sequence, converging to the joint within-estimator without ever constructing the full dummy matrix.

In Stata, `reghdfe` (Correia 2016) implements this approach and has become the standard tool for multi-way fixed effect estimation in corporate finance. It handles an arbitrary number of fixed effect sets, detects and drops singletons, and reports the degrees of freedom correctly. In R, `fixest` (Bergé, 2018) provides equivalent functionality with exceptional speed. In Python, the `linearmodels` package handles panel data with entity and time effects, though for very high-dimensional specifications, the `pyfixest` package provides a native Python interface that closely mirrors `fixest` syntax and is the more practical choice.

The practical advice is simple: use one of these modern tools. Do not attempt to estimate high-dimensional fixed effects by manually constructing dummy variables. And when you report results, state which software and version you used. It matters for replicability and for whether singletons were handled correctly.

The high-dimensional fixed effects revolution has been a genuine advance for empirical corporate finance. It has sharpened identification in hundreds of papers and raised the field’s standards for controlling unobserved heterogeneity. But like any powerful tool, it can be misused. The remainder of this chapter develops the complementary skills (clustering, dynamic panels, specification choice) that allow you to use these tools wisely. And the chapters that follow develop the methods you need when even the most carefully chosen fixed effects are not enough.

3.5 Clustering and Inference

Getting the point estimate right is only half the problem. The other half is getting the standard errors right, and in panel data, the default standard errors are almost always wrong. This section is organized around the common mistakes, because that is the fastest way to learn what matters.

3.5.1 Why Default Standard Errors Fail

Consider the regression

$$Y_{it} = \beta D_{it} + \mathbf{X}_{it}\boldsymbol{\gamma} + \alpha_i + \lambda_t + \varepsilon_{it}. \quad (3.21)$$

The classical OLS standard errors assume that ε_{it} is independently and identically distributed across all observations. In a panel of firms over time, this assumption fails in two distinct ways.

First, errors for the same firm are correlated across time. A firm that receives a positive shock in 2015 likely carries some of that shock into 2016 and 2017. This is *within-firm serial correlation*, and it means that 20 annual observations on the same firm do not provide 20 independent pieces of information about β .

Second, errors for different firms in the same year may be correlated. If a macroeconomic shock hits all firms in a given year, the residuals ε_{it} and ε_{jt} will be correlated for $i \neq j$. This is *cross-sectional correlation*, and it means that 3,000 firms observed in the same year do not provide 3,000 independent observations.

Ignoring either form of dependence leads to standard errors that are too small, t -statistics that are too large, and rejection rates that far exceed their nominal size. [Petersen \(2009\)](#) documented this with devastating clarity in finance panel settings, showing that the choice of standard error method can change a t -statistic from 8 to 2.

3.5.2 The Clustering Principle

The solution is to *cluster* the standard errors at the level where the dependence occurs. Clustering allows for arbitrary correlation among all observations within a cluster while maintaining the assumption of independence across clusters. The cluster-robust variance estimator for $\hat{\beta}$ takes the form

$$\widehat{\text{Var}}(\hat{\beta}) = (\mathbf{D}'\mathbf{D})^{-1} \left(\sum_{g=1}^G \mathbf{D}'_g \hat{\boldsymbol{\varepsilon}}_g \hat{\boldsymbol{\varepsilon}}'_g \mathbf{D}_g \right) (\mathbf{D}'\mathbf{D})^{-1}, \quad (3.22)$$

where g indexes clusters, $\mathbf{D}_g$ is the matrix of (demeaned) regressors for cluster g , and $\hat{\varepsilon}_g$ is the vector of residuals for cluster g . The key feature: no assumption is imposed on the covariance structure of $\hat{\varepsilon}_g$ within a cluster. Serial correlation, heteroskedasticity, and arbitrary within-cluster dependence are all permitted.

The principle for choosing the clustering level is straightforward to state: *cluster at the level at which the treatment varies, or at the level at which the errors are correlated, whichever is coarser*. In practice, applying this principle requires thought.

3.5.3 Common Mistake #1: Clustering at the Firm When Treatment Varies at a Higher Level

This is the most consequential error in practice. Suppose you are studying the effect of a state-level policy change on firm behavior. The treatment D_{it} switches on for all firms in a treated state at the same time. You have 3,000 firms but only 50 states, and the treatment varies across 50 states, not 3,000 firms.

If you cluster at the firm level, you are allowing for serial correlation *within* each firm's time series, but you are treating firms in the same state as providing independent information about the treatment effect. They do not. All firms in Arizona experience the same policy change at the same time, so their "responses" are correlated through the common treatment, and potentially through common state-level shocks. Firm-level clustering dramatically understates the true uncertainty.

The correct approach is to cluster at the state level. But this introduces a second problem: with only 50 clusters (or fewer, if only some states are treated), the cluster-robust variance estimator is itself poorly estimated. The estimator in equation (3.22) is consistent as $G \rightarrow \infty$, but with $G = 50$ it can be substantially biased, and with $G = 20$ it is unreliable.

Bertrand et al. (2004) documented this problem in the context of difference-in-differences designs exploiting state-level policy variation, showing that conventional inference rejected the null far too often (even with state-level clustering) when the number of policy changes was small relative to the number of states. Their paper remains essential reading.

Common Mistake 9

Clustering at the firm level for state-level policy variation. If your treatment varies at the state level, you need state-level clusters. Period. Firm-level clustering does not account for the correlation induced by the common treatment. The fact that you have thousands of firms does not give you thousands of independent pieces of information about a policy that varies across 50 states. Your effective sample size is closer to 50

than to 3,000.

3.5.4 Common Mistake #2: Ignoring the Few-Clusters Problem

Suppose you do cluster at the state level. With 50 states, inference based on the cluster-robust variance estimator and a standard normal or t -distribution is approximately valid. But corporate finance research frequently exploits variation across a much smaller number of jurisdictions. Studies using variation in a handful of court rulings, changes in state creditor rights, or banking deregulation episodes may have effective treatment variation across 10–20 states.

[Spamann \(2022\)](#) raises this concern directly in the context of studies using state corporate laws for identification. When the number of clusters is small, the cluster-robust standard errors are downward biased, confidence intervals undercover, and hypothesis tests over-reject. The t -distribution with $G - 1$ degrees of freedom (where G is the number of clusters) provides some correction, but it is not enough when G is very small.

Several remedies exist, though none is a complete fix. The wild cluster bootstrap ([Cameron, Gelbach, and Miller 2008](#)) provides better finite-sample performance and is now straightforward to implement.⁶ Randomization inference (permuting the treatment assignment across clusters and computing the test statistic under each permutation) provides exact tests under the sharp null of no effect and does not rely on asymptotic approximations at all. Both approaches should be in your toolkit when the number of clusters is below approximately 30–40.

Key Insight 10

The number of clusters matters more than the number of observations within clusters. A panel with 100,000 firm-year observations clustered at the state level has an effective sample size of 50 for inference about a state-level treatment. If only 15 states experienced the policy change, the effective sample for identifying the treatment effect is 15. Report the number of clusters. If it is small, use the wild cluster bootstrap or randomization inference.

3.5.5 Common Mistake #3: Two-Way Clustering by Default

[Petersen \(2009\)](#) showed that in many finance panel settings, the errors exhibit both within-

⁶In Stata, `boottest` ([Roodman, Nielsen, MacKinnon, and Webb 2019](#)) implements the wild cluster bootstrap. In R, `fwildclusterboot` provides a fast implementation. In Python, `wildboottest` is available via `pyfixest`.

firm serial correlation and within-year cross-sectional correlation. His recommendation (two-way clustering by firm and year) became standard practice. The two-way cluster-robust estimator, developed by [Cameron, Gelbach, and Miller \(2011\)](#), computes the variance as

$$\widehat{\text{Var}}_{\text{two-way}} = \widehat{\text{Var}}_{\text{firm}} + \widehat{\text{Var}}_{\text{year}} - \widehat{\text{Var}}_{\text{firm} \times \text{year}}, \quad (3.23)$$

where each component is the one-way cluster-robust estimator at the indicated level, and the intersection term corrects for double-counting.

Two-way clustering is appropriate when both dimensions of correlation are present and the treatment varies at the firm-year level. But it has become a reflex in some corners of the literature, applied even when it is not warranted or when it creates problems.

Two situations deserve caution. First, if your treatment varies at the state-year level (a policy change), two-way clustering by firm and year is *wrong*. You need to cluster at the state level, as discussed above. Two-way clustering by firm and year does not account for the correlation among firms within the same state. Second, two-way clustering requires that *both* clustering dimensions have a reasonably large number of groups. With 3,000 firms and 20 years, the year dimension has only 20 clusters. The cluster-robust estimator in the year dimension is poorly estimated, and the two-way estimator inherits this problem. [Thompson \(2011\)](#) discusses the asymptotic requirements.

Common Mistake 11

“We cluster by firm and year.” This is appropriate when the treatment varies at the firm-year level and both dimensions have enough groups for the asymptotics to work. It is not appropriate when the treatment varies at a coarser level (state, industry), and it is unreliable when the number of time periods is small. Before adopting two-way clustering, ask: what is the structure of the dependence in my errors, and does this procedure match that structure?

3.5.6 Common Mistake #4: Clustering and Fixed Effects at Different Levels

A subtle but important issue arises when the clustering level does not align with the fixed effects. Suppose you include state×year fixed effects (to absorb state-level time-varying confounders) but cluster at the firm level. The state×year effects absorb the common component of the error within each state-year cell, which reduces the cross-sectional correlation. But residual within-state correlation across firms may persist if the state-level shock is imperfectly absorbed, or if firms in the same state share unobserved characteristics

beyond what the fixed effects capture.

The safe practice: cluster at least at the level of the treatment variation, regardless of which fixed effects you include. Fixed effects absorb means, not dependencies. They can reduce the magnitude of within-cluster correlation, but they do not eliminate the need to account for it in the standard errors.

3.5.7 Common Mistake #5: Drawing Conclusions from Changes in Significance Across Clustering Choices

Researchers sometimes run the same regression with firm clustering, state clustering, and two-way clustering, and then comment on which results are “robust.” This misses the point. The question is not which clustering choice preserves significance. It is which clustering choice is *correct* given the research design.

If your treatment varies at the state level, state-level clustering is the right answer even if it renders the coefficient insignificant. Discovering that firm-level clustering gives you three stars while state-level clustering gives you none is not evidence of fragility. It is evidence that you never had the statistical power you thought you did. This is useful information, not a reason to report the firm-clustered results.

That said, showing the sensitivity of standard errors to clustering choices is good practice, not because it lets you pick the most favorable one, but because it reveals the structure of the dependence in your data. If firm-level and state-level clustering give similar standard errors, within-state correlation is modest. If they differ dramatically, within-state correlation is important, and the state-clustered standard errors should be the ones you rely on for inference.

3.5.8 A Brief Summary of What to Do

The logic can be reduced to three questions:

At what level does my treatment vary? Cluster at least at this level. If the treatment is a firm-level decision, firm-level clustering is the minimum. If the treatment is a state-level policy, state-level clustering is required.

Are there other sources of dependence in the errors? If there is cross-sectional correlation (common shocks hitting firms in the same year), you may need two-way clustering or time fixed effects. If the treatment varies at a higher level, clustering at that level already handles within-group dependence of all forms.

How many clusters do I have? If fewer than approximately 40, the standard cluster-robust

inference is unreliable. Use the wild cluster bootstrap or randomization inference. Report the number of clusters prominently. If your study has 10 clusters, the reader needs to know this before evaluating your p -values.

Common Mistake 12

Not reporting the number of clusters. Many papers report “standard errors clustered at the state level” in a footnote but never state how many states (clusters) are in the sample. This is a critical piece of information. A study with 48 state clusters has fundamentally different inferential properties from one with 12. Report the number. If it is small, discuss it.

3.5.9 Connection to the Running Example

In the leverage-investment regression

$$I_{it} = \beta L_{it} + \mathbf{X}_{it} \boldsymbol{\gamma} + \alpha_i + \lambda_t + \varepsilon_{it},$$

both leverage and investment are persistent firm-level characteristics with substantial serial correlation. Firm-level clustering is the natural baseline, and it is what most of the capital structure and investment literature reports.

But consider what happens if we study a policy change that affects leverage, say, a tax reform that alters the relative cost of debt across states. Now the treatment variation is at the state-year level, and we need state-level clusters. In Section 3.6, we show both sets of standard errors for the same regression. The firm-clustered standard error on $\hat{\beta}$ is approximately 0.008. The state-clustered standard error is 0.019. The point estimate does not change, but the implied confidence interval more than doubles. Whether the result is “significant” depends entirely on which clustering level is appropriate, which depends on the research design, not on which number the researcher prefers.

3.6 Running Example: Leverage and Investment in Panels

This section puts the lessons of the chapter to work. We begin with simulated data where we know the truth, then turn to Compustat to see the same patterns in the wild.

3.6.1 Simulation: Seeing the Biases

We simulate a panel of 2,000 firms over 20 years, assigned to 40 industries and 50 states. The data-generating process is constructed to embed the problems discussed in Sections 3.3–3.5:

- A firm fixed effect α_i that is positively correlated with both leverage and investment. Pooled OLS will confound the causal effect with cross-firm differences in firm quality.
- An industry-year shock that affects both variables. Industry×year fixed effects can absorb it, at the cost of variation.
- Classical measurement error in observed leverage ($\sigma_w = 0.2$). The within estimator will be more attenuated than pooled OLS because demeaning removes signal.
- A state-year error structure that induces within-state correlation, so that state-level clustering produces wider confidence intervals than firm-level clustering.

The true causal effect of leverage on investment is $\beta = -0.15$.

Table 3.2 reports the results. Read it column by column. Each specification teaches a lesson.

Table 3.2. Simulation: Leverage and Investment under Alternative Panel Specifications

	(1)	(2)	(3)	(4)	(5)	(6)
Leverage	0.050*** (0.004)	−0.095*** (0.003)	−0.095*** (0.003)	−0.136*** (0.003)	−0.108*** (0.003)	−0.095*** (0.006)
Leverage measure	Obs.	Obs.	Obs.	Obs.	True	Obs.
Firm FE		✓	✓	✓	✓	✓
Year FE			✓		✓	✓
Ind.×Year FE				✓		
Clustering	Firm	Firm	Firm	Firm	Firm	State
Observations	40,000	40,000	40,000	40,000	40,000	40,000
True β	−0.150					

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Simulated data with $N = 2,000$ firms, $T = 20$ years, 40 industries, and 50 states. The true causal effect is $\beta = -0.15$. Columns (1)–(4) and (6) use observed (mismeasured) leverage; column (5) uses true leverage to isolate measurement error attenuation. Column (6) is identical to (3) but with state-level clustering.

Column (1) runs pooled OLS on observed leverage with no fixed effects. The estimate is +0.050, not just attenuated, but the *wrong sign*. The positive correlation between firm quality and both leverage and investment overwhelms the true negative effect. A researcher relying on this regression would conclude that leverage *increases* investment.

Column (2) adds firm fixed effects. The estimate swings to −0.095, now correctly negative. Removing the time-invariant confounder α_i eliminates the upward bias from cross-firm sorting. But the estimate is still well short of the true −0.15. The gap is measurement error

attenuation: the within transformation removed the permanent component of the true leverage signal while leaving the transitory measurement noise largely intact, reducing the signal-to-noise ratio exactly as described in Section 3.3.4.

Column (3) adds year fixed effects. The estimate barely changes (-0.095), which is expected: in the DGP, aggregate time shocks are not a major confounder for the leverage-investment relationship. Year effects help, but the action in this example is at the firm level and the industry level, not in aggregate time variation.

Column (4) replaces year effects with industry $\times$ year effects. The estimate moves to -0.136 , closer to the truth. The industry-year shocks that were confounding the relationship have now been absorbed. But note the cost: Table 3.3 shows that the within-FE standard deviation of leverage drops from 0.554 (with firm and year effects) to 0.529 (with firm and industry $\times$ year effects). The finer fixed effects have consumed some identifying variation, though in this case the tradeoff is favorable.

Table 3.3. Simulation: Within-FE Variation in Leverage

	Firm FE	Firm + Year FE	Firm + Ind $\times$ Year FE
Raw SD	0.696	0.696	0.696
Within-FE SD	0.556	0.554	0.529
Ratio	0.799	0.796	0.761

Notes: Raw SD is the unconditional standard deviation of observed leverage. Within-FE SD is the standard deviation of leverage after projecting out the fixed effects used in each specification. Ratio is Within-FE SD divided by Raw SD. As finer fixed effects absorb more variation, the within-FE SD shrinks.

Column (5) is the specification that cannot be run with real data. It repeats column (3) but uses true leverage L_{it}^* in place of the mismeasured L_{it}^{obs} . The estimate is -0.108 , closer to -0.15 than the corresponding mismeasured estimate of -0.095 in column (3). The difference between columns (3) and (5) (about 0.013 in absolute value) is the additional attenuation caused by measurement error. It is not enormous here, but in real data where measurement error may be more severe (think Tobin's Q), this wedge can be much larger.

Column (6) is identical to column (3) in point estimate (-0.095) but clusters standard errors at the state level instead of the firm level. The standard error doubles, from 0.003 to 0.006. The estimate that was highly significant with firm clustering remains significant here, but in a setting with fewer states or a treatment that varies at the state level, this difference would matter enormously for inference.

Figure 3.1 shows these results visually. The pooled OLS estimate sits far from the truth, the FE estimates cluster together with visible measurement error attenuation, the

true-leverage specification is closer to the dashed line, and the state-clustered confidence interval is visibly wider.

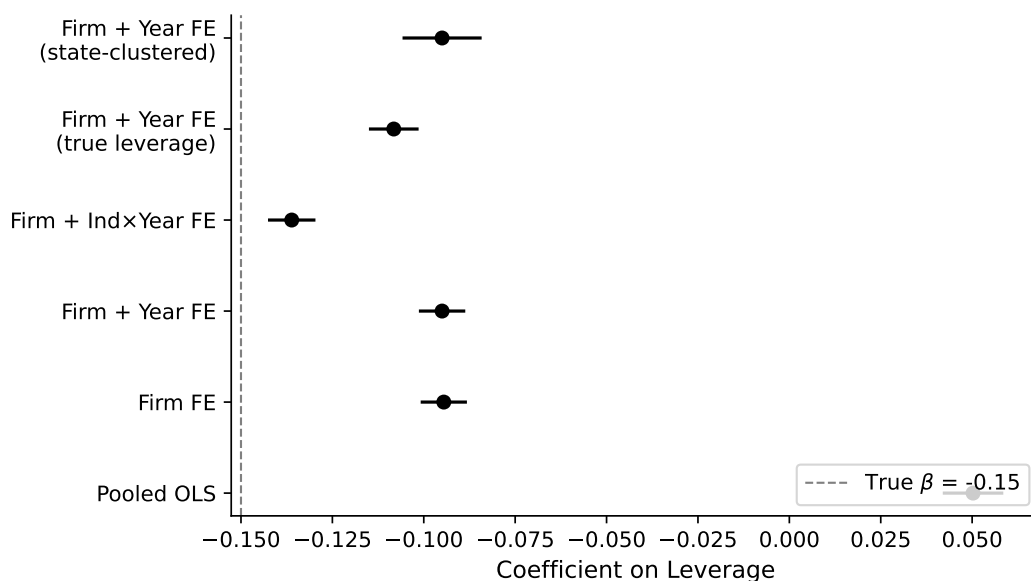


Figure 3.1. Simulation: Coefficient on leverage across specifications. The dashed line marks the true $\beta = -0.15$. Horizontal bars show 95% confidence intervals. Pooled OLS is biased upward; firm FE corrects the direction but measurement error attenuates the estimate; industry $\times$ year FE move the estimate closer to the truth; using true leverage reduces attenuation; state-level clustering widens the confidence interval.

3.6.2 Compustat: The Same Patterns in Real Data

We now estimate the same ladder of specifications on a standard Compustat panel: non-financial, non-utility firms from 1971 to 2023, yielding approximately 157,500 firm-year observations. Table 3.4 provides summary statistics.

The dependent variable is investment (capital expenditure scaled by lagged assets), with a median of 0.039 and substantial right skew. Book leverage averages 0.245. All specifications include Tobin's Q , cash flow, firm size, and tangibility as controls.

Table 3.5 reports the results. We do not know the true β here, but the *patterns* match what the simulation led us to expect.

Column (1), pooled OLS, gives a coefficient of -0.027 . Unlike the simulation, the sign is already negative. The controls (especially Tobin's Q) absorb some of the cross-firm sorting that biased pooled OLS in the simulation. But the magnitude is the smallest across all specifications, consistent with residual upward bias from unobserved firm heterogeneity.

Column (2), firm FE, moves the coefficient to -0.040 , substantially more negative. This is the same direction as the simulation: removing time-invariant firm characteristics reveals

Table 3.4. Summary Statistics: Compustat Analysis Sample

Variable	N	Mean	SD	P25	Median	P75
Investment (<i>I</i>)	157,550	0.071	0.097	0.017	0.039	0.083
Leverage (<i>L</i>)	157,550	0.245	0.230	0.036	0.204	0.376
Tobin's <i>Q</i>	157,550	1.988	1.580	1.084	1.460	2.241
Cash Flow (<i>CF</i>)	157,550	0.014	0.251	-0.019	0.073	0.134
Size ($\ln(at)$)	157,550	5.838	2.123	4.159	5.596	7.277
Tangibility	157,550	0.284	0.247	0.085	0.205	0.427

Notes: Sample includes non-financial, non-utility Compustat firms, 1971–2023. Investment is capital expenditure scaled by lagged total assets. Leverage is book leverage (short-term plus long-term debt over total assets). Tobin's *Q*, cash flow, and profitability are winsorized at the 1st and 99th percentiles.

Table 3.5. Compustat: Leverage and Investment under Alternative Panel Specifications

	(1)	(2)	(3)	(4)	(5)
Leverage	-0.027*** (0.002)	-0.040*** (0.002)	-0.036*** (0.002)	-0.032*** (0.002)	-0.035*** (0.003)
Controls	Yes	Yes	Yes	Yes	Yes
Fixed effects	None	Firm	Firm + Year	Firm + Ind×Year	Firm + Year
Clustering	Firm	Firm	Firm	Firm	State
Observations	157,550	157,550	157,550	157,550	137,612
R^2	0.286	0.077	0.058	0.055	0.059

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Dependent variable is investment ($capx/lagged\ assets$). Controls are Tobin's *Q*, cash flow, size, and tangibility. Sample: non-financial, non-utility Compustat firms, 1971–2023. R^2 is within R^2 for FE specifications.

a stronger negative relationship between leverage and investment. Good firms tend to have both high investment and low leverage; once we compare each firm to itself, the negative causal channel dominates.

Column (3), firm plus year FE, gives -0.036 . Year effects absorb aggregate credit conditions and business-cycle variation, which apparently were mildly confounding the relationship in the opposite direction.

Column (4), firm plus industry $\times$ year FE, gives -0.032 . The estimate moves back toward zero somewhat, and Table 3.6 shows why: the within-FE standard deviation of leverage falls from 0.133 to 0.130. Industry $\times$ year effects absorb industry-level shocks that were contributing to the identifying variation. In this case, the absorbed variation appears to have been partly useful signal rather than pure confounding, a reminder that finer fixed effects do not always improve estimates.

Table 3.6. Compustat: Within-FE Variation in Leverage

	Firm FE	Firm + Year FE	Firm + Ind $\times$ Year FE
Raw SD	0.230	0.230	0.230
Within-FE SD	0.136	0.133	0.130
Ratio	0.593	0.581	0.565

Notes: Raw SD is the unconditional standard deviation of book leverage. Within-FE SD is the standard deviation of leverage after projecting out the fixed effects. Ratio is Within-FE SD divided by Raw SD.

The variation diagnostic in Table 3.6 reveals a striking fact about real corporate finance data. Firm fixed effects alone eliminate 41% of the raw variation in leverage (the ratio drops from 1.0 to 0.593). Most of the variation in leverage is *between* firms, not *within* firms over time. Adding industry $\times$ year effects removes a further 3 percentage points. The within-firm, within-industry-year variation in leverage that identifies the coefficient in column (4) is only 56.5% of the raw variation. This is the residual that your estimate is identified from. Whether it is enough depends on the context, but you should always know this number.

Column (5) clusters standard errors at the state level instead of the firm level, using the subsample with non-missing state of incorporation (137,612 observations). The coefficient is essentially unchanged at -0.035 , but the standard error rises from 0.002 to 0.003, a 50% increase. Figure 3.2 shows the effect on the confidence interval clearly.

3.6.3 What We Learn

Three takeaways emerge from comparing the simulation to the real data.

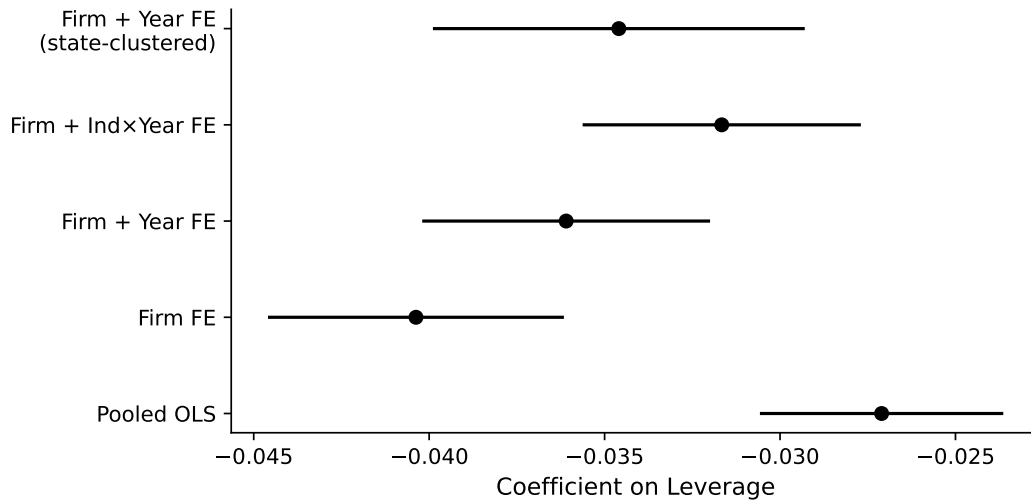


Figure 3.2. Compustat: Coefficient on leverage across specifications. Horizontal bars show 95% confidence intervals. All specifications include controls for Tobin’s Q , cash flow, size, and tangibility. The pooled OLS estimate is the least negative; firm FE produces the most negative estimate; finer fixed effects and state-level clustering widen the confidence interval.

First, *the direction of bias from omitted firm heterogeneity is predictable*. In both the simulation and Compustat, pooled OLS produces a less negative (more biased toward zero or positive) coefficient than the within estimator. This is the signature of a positive correlation between the regressor and a favorable unobservable: good firms invest more *and* have different leverage. Firm fixed effects remove this confound, revealing a more negative causal relationship.

Second, *finer fixed effects are not always better*. In the simulation, industry×year effects moved the estimate closer to the truth because the DGP included industry-year confounders. In Compustat, the same fixed effects moved the estimate slightly back toward zero, suggesting they absorbed some useful variation along with the confounders. The diagnostic from Section 3.4 applies: the within-FE variation in leverage shrank, and the estimate moved in a direction consistent with losing signal. Whether this tradeoff is acceptable depends on how worried you are about industry-level time-varying confounders, a question that requires economic argument, not statistical mechanics.

Third, *clustering choices change your inferential conclusions*. State-level clustering produces wider confidence intervals than firm-level clustering in both the simulation and Compustat. In our application, the coefficient remains statistically significant either way. But in a study exploiting state-level policy variation with fewer effective clusters, the difference between these two standard errors could easily be the difference between a “significant” result and a null finding. The correct choice depends on the research design, as discussed in Section 3.5.

Key Insight 13

The simulation shows you *why* results change across specifications. The Compustat exercise shows you that the same patterns appear in real data. Use simulations to build intuition for what your estimator is doing. Use real data to verify that the intuition carries over. Together, they are far more persuasive than either alone.

3.7 Dynamic Panels and Lagged Dependent Variables

Many corporate finance specifications include a lagged dependent variable on the right-hand side. The most prominent example is the partial adjustment model of capital structure:

$$L_{it} = \lambda L_{it}^* + (1 - \lambda)L_{it-1} + \varepsilon_{it}, \quad (3.24)$$

where L_{it}^* is the firm's target leverage (itself a function of firm characteristics) and λ is the speed of adjustment. The parameter $1 - \lambda$ is the coefficient on the lagged dependent variable: if λ is close to 1 the firm adjusts quickly, and if λ is close to 0 the firm barely adjusts at all. Estimates of λ in the literature range from near zero (Welch 2004) to around 40% (Flannery and Rangan 2006), and some of this variation traces directly to the econometric problems discussed in this section.

More generally, any regression of the form

$$Y_{it} = \rho Y_{it-1} + \beta D_{it} + \alpha_i + \varepsilon_{it}, \quad |\rho| < 1, \quad (3.25)$$

is a dynamic panel model. You might include a lagged dependent variable because you believe the outcome is genuinely persistent (leverage adjusts slowly), or simply because you want to “control for” the prior level of Y . Either way, the combination of Y_{it-1} and α_i on the right-hand side creates a mechanical problem that firm fixed effects cannot solve. In fact, firm fixed effects make it worse.

3.7.1 The Nickell Bias

The problem is easiest to see in the within-transformed version of equation (3.25). The within transformation subtracts the firm-level time average from every variable:

$$(Y_{it} - \bar{Y}_i) = \rho (Y_{it-1} - \bar{Y}_i) + \beta (D_{it} - \bar{D}_i) + (\varepsilon_{it} - \bar{\varepsilon}_i), \quad (3.26)$$

where $\bar{Y}_i = T^{-1} \sum_{t=1}^T Y_{it}$ and $\bar{\varepsilon}_i = T^{-1} \sum_{t=1}^T \varepsilon_{it}$.

The problem is that the demeaned regressor $Y_{it-1} - \bar{Y}_i$ is correlated with the demeaned error $\varepsilon_{it} - \bar{\varepsilon}_i$. To see why, note that $\bar{Y}_i$ contains Y_{it} , which in turn depends on ε_{it} . So $Y_{it-1} - \bar{Y}_i$ includes a piece of $-\bar{Y}_i$ that depends on ε_{it} , and the error term includes ε_{it} . The regressor and the error share a common component.

More precisely, $\bar{\varepsilon}_i$ contains ε_{it} (with weight $1/T$), and $\bar{Y}_i$ contains $Y_{it}, Y_{it+1}, \dots, Y_{iT}$, all of which depend on $\varepsilon_{it}, \varepsilon_{it+1}, \dots$ through the autoregressive structure. This induces a negative correlation between the demeaned lagged dependent variable and the demeaned error.⁷ The result is that the within estimator of ρ is biased *downward*. The estimated persistence is too low, and therefore the implied speed of adjustment $\hat{\lambda} = 1 - \hat{\rho}$ is too high.

Nickell (1981) showed that the bias is of order $O(1/T)$:

$$\text{plim}_{N \rightarrow \infty} \hat{\rho}_{\text{FE}} - \rho \approx -\frac{1 + \rho}{T - 1}. \quad (3.27)$$

This is a *fixed-T* result: the bias does not vanish as the number of firms N grows. It vanishes only as $T \rightarrow \infty$. With $T = 10$ and a true $\rho = 0.8$, the bias is approximately $-1.8/9 = -0.20$, which would produce an estimated $\hat{\rho}$ near 0.60 instead of 0.80. That is a massive distortion. With $T = 50$ the bias shrinks to about -0.037 , which may be tolerable.

Appendix 3.7.4 provides the derivation for the simplified case of $T = 2$ and no additional regressors, which makes the mechanics transparent.

3.7.2 Why This Matters for Corporate Finance

The Nickell bias is not a curiosity. It has directly shaped one of the most-studied questions in the field. In the capital structure speed-of-adjustment literature, the coefficient on lagged leverage is the object of interest. Pooled OLS on the partial adjustment model *without* firm fixed effects produces estimates of $\hat{\rho}$ that are biased *upward*, because the omitted firm effect α_i is positively correlated with L_{it-1} . This makes adjustment appear slow (high $\hat{\rho}$, low $\hat{\lambda}$).

Including firm fixed effects corrects the omitted variable bias but introduces the Nickell bias in the opposite direction. $\hat{\rho}$ is pushed *downward*, making adjustment appear fast (low $\hat{\rho}$, high $\hat{\lambda}$). Flannery and Rangan (2006) reported that firm fixed effects roughly tripled the estimated speed of adjustment. Some of that increase reflects genuine improvement from controlling for unobserved heterogeneity. But some of it is Nickell bias making $\hat{\rho}$ artificially small.

The true ρ is bracketed: pooled OLS gives an upper bound (omitted variable bias pushes $\hat{\rho}$ up), and within-FE gives a lower bound (Nickell bias pushes $\hat{\rho}$ down). If the

⁷The negative correlation arises because $-\bar{Y}_i$ enters the regressor with a minus sign while ε_{it} enters the error with a plus sign, and $\bar{Y}_i$ is positively correlated with ε_{it} (since Y_{it} depends on ε_{it} , and $\bar{Y}_i$ includes Y_{it}/T).

two are close, you can be fairly confident about the answer. If they are far apart (as they typically are in the leverage adjustment literature), the bounds are not very informative, and you need a different estimator.

The structure we are using in this section may seem obvious. But it is not. The interpretation has been a source of controversy. Appendix 3.7.4 discusses the theoretical foundations of the partial adjustment model. It shows that the model can be derived either from quadratic adjustment costs or, more generally, from a binary adjustment framework in which some firms adjust to their target in a given period and others do not. In general the partial adjustment model is best viewed as a useful approximation to a richer underlying process. It is not really best regarded as a truly structural model of adjustment.

3.7.3 What Can Be Done?

Three approaches have been used in corporate finance, with varying degrees of success.

GMM (Arellano-Bond and Blundell-Bond). The classic fix is to difference out the fixed effect and use lagged levels of Y as instruments for the differenced lagged dependent variable. Arellano and Bond (1991) proposed using $Y_{it-2}, Y_{it-3}, \dots$ as instruments in the first-differenced equation

$$\Delta Y_{it} = \rho \Delta Y_{it-1} + \beta \Delta D_{it} + \Delta \varepsilon_{it}. \quad (3.28)$$

The validity of these instruments requires that ε_{it} is serially uncorrelated, a testable assumption. Blundell and Bond (1998) augmented this with additional moment conditions using lagged *differences* as instruments for the levels equation (system GMM), which improves efficiency when ρ is close to 1.

In practice, GMM estimators in corporate finance have had a mixed record. They require choosing a lag structure for the instruments, and with many lags the instrument count proliferates, leading to overfitting and unreliable Hansen J -tests. Flannery and Hankins (2013) compared seven dynamic panel estimators on simulated corporate finance data and found that Blundell-Bond GMM performed reasonably, but no method dominated across all settings. The estimators are also sensitive to the serial correlation structure of the errors, which in corporate finance applications is often more complex than the textbook MA(0) assumption.

Bias-corrected LSDV. An alternative is to estimate the within model directly and then apply an analytical bias correction. Bruno (2005) developed a corrected least-squares

dummy variable (LSDVC) estimator that subtracts the leading term of the Nickell bias. This approach is computationally simple and avoids the instrument proliferation problem of GMM. [Flannery and Hankins \(2013\)](#) found it to be competitive with Blundell-Bond in simulations calibrated to capital structure data.

Long differences. A practical alternative that avoids both the Nickell bias and the complications of GMM is to use long differences: instead of annual changes, compute 3-year or 5-year changes and regress $Y_{it} - Y_{it-k}$ on $D_{it} - D_{it-k}$ for $k > 1$. [Huang and Ritter \(2009\)](#) applied this approach to leverage adjustment. The intuition is that the correlation between the demeaned lagged dependent variable and the demeaned error diminishes as the differencing interval grows, because the “overlap” between the regressor and the error becomes a smaller fraction of the total variation. Long differences also reduce the attenuation from measurement error, since the signal-to-noise ratio improves with longer differencing intervals when the signal is persistent and the noise is transitory.

The cost is a loss of observations (you give up $k - 1$ years at the start of the panel for each firm) and a change in the economic question: a 5-year differenced regression asks about medium-run adjustment, not year-to-year dynamics.

3.7.4 Practical Guidance

If your specification includes a lagged dependent variable with firm fixed effects, you have a Nickell bias problem. Here is what to do:

First, check whether the bias is likely to be large. The approximation in equation (3.27) gives a quick back-of-envelope calculation. If your panel has $T = 30$ or more and ρ is moderate, the bias may be small enough to tolerate. If T is short (say, 10–15 years), it is almost certainly consequential.

Second, report both pooled OLS and within-FE estimates. They bracket the true ρ . If the bracket is tight, you have a reasonably informative answer without any fancy corrections.

Third, if the bracket is wide and the lagged dependent variable is the parameter of interest (as in speed-of-adjustment studies), use Blundell-Bond GMM or bias-corrected LSDV. Report the Arellano-Bond test for serial correlation and the Hansen J -test. Be transparent about the instrument count.

Fourth, consider whether you need a lagged dependent variable at all. Many researchers include Y_{it-1} as a “control” without a theoretical reason for dynamics. If your interest is in β , not ρ , and you include Y_{it-1} only to soak up persistence, you are introducing a bias in $\hat{\beta}$ as well (the Nickell bias contaminates all coefficients in the regression, not just $\hat{\rho}$).

In such cases, firm fixed effects without a lagged dependent variable may be the cleaner specification.

Common Mistake 14

“We include lagged leverage as a control.” Including a lagged dependent variable in a firm fixed effects regression introduces Nickell bias into every coefficient, not just the coefficient on the lagged variable. If the lagged dependent variable is not the object of interest and you include it merely to absorb persistence, you may be making the estimates of β worse, not better. Ask: does the economic model require dynamics? If not, leave the lag out.

Appendix to Chapter 3

3A. Derivation of the Nickell Bias ($T = 2$)

Consider the simplest case: no additional regressors, two time periods ($t = 1, 2$), and N firms. The model is

$$Y_{it} = \rho Y_{it-1} + \alpha_i + \varepsilon_{it}, \quad t = 1, 2, \quad (3.29)$$

with $|\rho| < 1$, $E[\varepsilon_{it}] = 0$, and $E[\varepsilon_{it}\varepsilon_{is}] = 0$ for $t \neq s$. We assume Y_{i0} is observed (so we have data for $t = 0, 1, 2$), and that the process has been running long enough that Y_{i0} is drawn from the stationary distribution. The fixed effect α_i may be correlated with Y_{i0} .

With $T = 2$ periods, the within transformation for each variable subtracts the mean over $t = 1, 2$:

$$\tilde{Y}_i = \frac{1}{2}(Y_{i1} + Y_{i2}), \quad \overline{Y_{i,t-1}} = \frac{1}{2}(Y_{i0} + Y_{i1}), \quad \bar{\varepsilon}_i = \frac{1}{2}(\varepsilon_{i1} + \varepsilon_{i2}).$$

The demeaned equation at $t = 2$ is

$$\underbrace{(Y_{i2} - \tilde{Y}_i)}_{\tilde{Y}_{i2}} = \rho \underbrace{(Y_{i1} - \overline{Y_{i,t-1}})}_{\tilde{Y}_{i1}} + \underbrace{(\varepsilon_{i2} - \bar{\varepsilon}_i)}_{\tilde{\varepsilon}_{i2}}. \quad (3.30)$$

With $T = 2$, the demeaned values simplify. For the error:

$$\tilde{\varepsilon}_{i2} = \varepsilon_{i2} - \frac{1}{2}(\varepsilon_{i1} + \varepsilon_{i2}) = \frac{1}{2}(\varepsilon_{i2} - \varepsilon_{i1}).$$

For the demeaned lagged dependent variable:

$$\tilde{Y}_{i1} = Y_{i1} - \frac{1}{2}(Y_{i0} + Y_{i1}) = \frac{1}{2}(Y_{i1} - Y_{i0}).$$

Now we check whether $\tilde{Y}_{i1}$ and $\tilde{\varepsilon}_{i2}$ are correlated. Substituting the model $Y_{i1} = \rho Y_{i0} + \alpha_i + \varepsilon_{i1}$:

$$\tilde{Y}_{i1} = \frac{1}{2}[(\rho - 1)Y_{i0} + \alpha_i + \varepsilon_{i1}].$$

The covariance is

$$\begin{aligned} \text{Cov}(\tilde{Y}_{i1}, \tilde{\varepsilon}_{i2}) &= \text{Cov}\left(\frac{1}{2}[(\rho - 1)Y_{i0} + \alpha_i + \varepsilon_{i1}], \frac{1}{2}(\varepsilon_{i2} - \varepsilon_{i1})\right) \\ &= \frac{1}{4} \text{Cov}(\varepsilon_{i1}, \varepsilon_{i2} - \varepsilon_{i1}) \\ &= \frac{1}{4} \underbrace{[\text{Cov}(\varepsilon_{i1}, \varepsilon_{i2}) - \text{Var}(\varepsilon_{i1})]}_{= 0} \\ &= -\frac{\sigma_\varepsilon^2}{4}. \end{aligned} \tag{3.31}$$

The second line uses the fact that Y_{i0} and α_i are uncorrelated with ε_{i1} and ε_{i2} (strict exogeneity of the initial condition and the fixed effect with respect to future errors), and that ε_{i1} and ε_{i2} are uncorrelated.

The covariance is *negative*. This is the source of the Nickell bias: the demeaned regressor and the demeaned error are negatively correlated, so the within estimator of ρ is biased downward.

The within estimator is

$$\hat{\rho}_{\text{FE}} = \rho + \frac{\sum_{i=1}^N \tilde{Y}_{i1} \tilde{\varepsilon}_{i2}}{\sum_{i=1}^N \tilde{Y}_{i1}^2}.$$

As $N \rightarrow \infty$ with $T = 2$ fixed, the probability limit is

$$\text{plim } \hat{\rho}_{\text{FE}} = \rho + \frac{\text{Cov}(\tilde{Y}_{i1}, \tilde{\varepsilon}_{i2})}{\text{Var}(\tilde{Y}_{i1})} = \rho - \frac{\sigma_\varepsilon^2/4}{\text{Var}(\tilde{Y}_{i1})}. \tag{3.32}$$

Since $\text{Var}(\tilde{Y}_{i1}) > 0$ and $\sigma_\varepsilon^2 > 0$, the bias is strictly negative. Its magnitude depends on the signal-to-noise ratio in the demeaned regressor: the more of the variation in $\tilde{Y}_{i1}$ comes from ε_{i1} (as opposed to $(\rho - 1)Y_{i0} + \alpha_i$), the larger the bias.

For the general case with $T > 2$ and additional regressors, Nickell (1981) showed that the bias takes the form in equation (3.27) under stationarity. The key insight is the same: the within transformation creates a mechanical negative correlation between the demeaned lag and the demeaned error that does not vanish as N grows, only as T grows.

3B. Theoretical Foundations of the Partial Adjustment Model

The partial adjustment model

$$Y_{it} = \lambda Y_{it}^* + (1 - \lambda) Y_{it-1} + \varepsilon_{it} \quad (3.33)$$

appears throughout corporate finance: in capital structure (adjusting toward target leverage), investment (adjusting toward a desired capital stock), cash holdings, and payout policy. The parameter λ is interpreted as a “speed of adjustment.” But where does equation (3.33) come from? There are two derivations, and they have different implications for how seriously we should take λ as a structural parameter.

Derivation 1: Quadratic adjustment costs. Suppose the firm minimizes the expected present value of losses from being away from target and from adjusting:

$$\min_{\{Y_{it}\}} E \sum_{t=0}^{\infty} \beta^t \left[c (Y_{it} - Y_{it}^*)^2 + d (Y_{it} - Y_{it-1})^2 \right], \quad (3.34)$$

where $c > 0$ penalizes deviation from target and $d > 0$ penalizes adjustment. The first-order condition with respect to Y_{it} , ignoring the forward-looking terms for simplicity, gives

$$c (Y_{it} - Y_{it}^*) + d (Y_{it} - Y_{it-1}) = 0,$$

which rearranges to

$$Y_{it} = \frac{c}{c + d} Y_{it}^* + \frac{d}{c + d} Y_{it-1}.$$

This is exactly equation (3.33) with $\lambda = c/(c + d)$. The speed of adjustment is determined by the ratio of the cost of being away from target to the cost of adjusting.

This derivation is clean, but it rests on strong assumptions. The costs must be quadratic (so that the first-order condition is linear), there can be no fixed costs of adjustment (which would create inaction regions), and the target must evolve smoothly. In corporate finance, none of these is obviously satisfied. Fixed costs generate lumpy adjustment (Fischer, Heinkel, and Zender 1989). Irreversibility creates asymmetry (Leary and Roberts 2005). Time-varying targets that are mismeasured bias $\hat{\lambda}$ toward zero (Frank and Shen 2019). The quadratic cost justification is a useful starting point, but it is a special case.

Derivation 2: Binary adjustment. A more robust justification does not require specifying the form of adjustment costs at all. Suppose that in each period, each firm either adjusts to

its target or does not adjust, and that the fraction of firms that adjust is λ . This could arise from fixed adjustment costs: a firm adjusts when the gap $|Y_{it}^* - Y_{it-1}|$ is large enough to justify paying a fixed cost F , and does not adjust otherwise.

For a firm that adjusts:

$$Y_{it} = Y_{it}^* + \varepsilon_{it}.$$

For a firm that does not adjust:

$$Y_{it} = Y_{it-1}.$$

Averaging across both types (without observing who adjusted), the expected outcome is

$$E[Y_{it} | Y_{it}^*, Y_{it-1}] = \lambda Y_{it}^* + (1 - \lambda) Y_{it-1} + \lambda \varepsilon_{it}. \quad (3.35)$$

This is the partial adjustment equation. The parameter λ is now the *fraction of firms that adjust in a given period*, a cross-sectional mixing probability, not a parameter of a cost function.

This derivation is appealing because it applies wherever adjustment is intermittent and lumpy. In the capital structure context: some firms actively restructure their debt in a given year, others carry forward their prior capital structure. In the investment context: some firms undertake major capital projects, others do maintenance only. In payout: some firms reset their dividend, others hold it constant. The binary adjustment story fits all of these settings without imposing a specific functional form on the adjustment friction.

Limitations of both derivations. Under either derivation, λ is generally *not* a deep structural parameter. In the quadratic cost model, λ depends on c and d , which may vary across firms and over time. In the binary adjustment model, λ depends on the distribution of gaps between current values and targets, the level of fixed costs, and the volatility of the target. All of these are endogenous and time-varying. A firm that is far from its target is more likely to adjust than one that is close, so λ is not constant in the cross-section or over time.

The practical implication is that the partial adjustment model is best understood as a useful *approximation* to a richer dynamic process, not as a structural model of adjustment. The estimated $\hat{\lambda}$ is an informative summary statistic (it tells you something about the average persistence of the variable in the data), but over-interpreting it as the “true speed of adjustment” requires believing in a specific (and fragile) set of underlying assumptions. This is true under both derivations, but the binary adjustment interpretation is the more robust of the two, because it requires only that adjustment is intermittent, not that costs

take a particular functional form.

3.8 Choosing Your Panel Specification

The preceding sections have covered a lot of ground: fixed effects and first differences, high-dimensional interacted effects, clustering, dynamic panels. A PhD student starting a new project faces a bewildering menu of choices. This section organizes those choices into a sequence of questions. The questions are not mechanical (each requires economic reasoning about your specific setting), but they impose a structure that prevents the most common errors.

3.8.1 Five Questions Before You Run the Regression

Question 1: What is the economic question, and what variation answers it? This is the question that everything else depends on, and it is the one most often skipped. Before choosing a specification, write down in plain language: what comparison identifies the effect I care about? Is it a comparison of the same firm at different points in time (within-firm)? Different firms in the same year (cross-sectional)? Firms in treated vs. untreated states (between-group)?

If identification comes from within-firm changes over time, firm fixed effects are natural. They remove the cross-sectional confounders and isolate the within-firm variation. If identification comes from cross-sectional differences (as in Gan's 2007 study of Japanese land values), firm fixed effects will destroy the signal. If identification comes from between-group comparisons (treated vs. control states), you are heading toward difference-in-differences (Chapter 6), not a simple panel regression.

Question 2: What are the main threats to identification, and which are time-invariant vs. time-varying? Write down the two or three most plausible confounders. For each one, ask: is this something that differs across firms but is roughly constant over the sample period? Or does it change over time?

If the main confounders are time-invariant (management quality, corporate culture, persistent geographic advantages), firm fixed effects address them directly. If the main confounders are time-varying (industry shocks, changing credit conditions, evolving managerial incentives), firm fixed effects leave them in the error term and you need either time-interacted group effects, instrumental variables, or a research design that isolates exogenous variation.

Be specific. “Unobserved heterogeneity” is not a confounder. It is a category. Name the economic force you are worried about.

Question 3: At what level does the treatment vary, and at what level should I cluster?

These two questions are linked. If your treatment is a firm-level decision (a CEO change, a capital structure choice), cluster at the firm level. If your treatment is a state-level policy, cluster at the state level, and check how many clusters you have. If your treatment is an industry-level shock, cluster at the industry level.

Do not let the fixed effects determine your clustering. Fixed effects absorb means; clustering accounts for dependence. You need both, and they operate on different dimensions of the problem. The clustering level should be at least as coarse as the level of treatment variation, as discussed in Section 3.5.

Question 4: Does the specification require a lagged dependent variable? If you have a theoretical reason for dynamics (a partial adjustment model, an Euler equation, a model with explicit adjustment costs), then including Y_{it-1} is motivated, and you need to worry about Nickell bias (Section 3.7). Check whether the bias is likely to be large (T short, ρ large), and if so, use GMM, bias-corrected LSDV, or long differences.

If you are including Y_{it-1} merely as a control to “soak up persistence,” reconsider. This introduces Nickell bias into all coefficients, and the benefit (a slightly smaller residual variance) may not justify the cost (biased estimates of the parameters you actually care about). Firm fixed effects already absorb the permanent component of Y_{it} ; adding the lag absorbs the transitory persistence, but at the price of a mechanical correlation between the regressor and the error.

Question 5: How much identifying variation survives the fixed effects? After choosing your FE structure, compute the within-FE standard deviation of your key regressor relative to its unconditional standard deviation (the ratio from Tables 3.3–3.6). If the ratio is, say, 0.90, the fixed effects are absorbing only 10% of the variation and you have plenty of signal left. If the ratio is 0.30, you have thrown away 70% of the variation and your estimates will be noisy. Neither number is automatically “too low”. The right answer depends on whether the absorbed variation was confounder or signal. But you need to know the number, and your reader does too.

3.8.2 A Decision Flowchart

Figure 3.3 summarizes the logic as a flowchart. Start at the top and follow the branches.⁸

1. **Is your identifying variation within-firm over time?**
 Yes → Firm FE is appropriate. Go to 2.
 No → Firm FE may destroy identification. Consider pooled OLS with controls, or cross-sectional methods.
2. **Are there aggregate time shocks that affect all firms?**
 Yes → Add year FE. Go to 3.
 No → Year FE optional. Go to 3.
3. **Are there group-specific time-varying confounders (industry shocks, regional trends)?**
 Yes → Add group×year FE *if* your treatment varies at a finer level than the group. Check within-FE variation. Go to 4.
 No → Stop adding FE. Go to 4.
4. **Does the model require a lagged dependent variable?**
 Yes → Nickell bias is a concern. Is $T > 30$? If so, bias may be tolerable. If not, use GMM or bias-corrected LSDV. Go to 5.
 No → Do not include Y_{it-1} as a control. Go to 5.
5. **At what level does the treatment vary?**
 Firm level → Cluster at firm.
 State/industry level → Cluster at that level. Count clusters. If < 40 , use wild cluster bootstrap or randomization inference.
 Firm-year level → Consider two-way clustering (firm and year) if both dimensions have enough groups.
6. **Report diagnostics.**
 Within-FE variation in key regressor. Number of clusters. Build-up table showing how coefficients and SEs change across specifications.

Figure 3.3. Decision flowchart for panel specifications. Start at step 1 and follow the branches. Each step requires economic reasoning about your specific setting.

3.8.3 What to Report

The build-up table from Section 3.6 (Tables 3.2 and 3.5) is not just a pedagogical device. It is a reporting standard you should follow in your own papers. Show the reader how the coefficient and standard error change as you move from a simple specification to a more saturated one. This accomplishes three things: it demonstrates that you have thought about what each set of fixed effects does, it lets the reader assess whether the additional fixed effects are absorbing confounders or destroying signal, and it provides informal evidence about the robustness of the result.

⁸This flowchart is a guide, not an algorithm. Every real research setting has features that no flowchart can anticipate. Use it to organize your thinking, then apply economic judgment.

At a minimum, report:

- The point estimate and standard error under your preferred specification.
- The same under at least one simpler specification (to show what the fixed effects are doing).
- The clustering level and the number of clusters.
- The within-FE variation diagnostic for the key regressor.

If the coefficient is sensitive to the choice of fixed effects, do not try to hide this. Explain *why* it changes. A coefficient that moves because you have absorbed a genuine confounder is a sign of a good research design. A coefficient that moves because you have consumed most of the identifying variation is a sign that you need a different approach, perhaps an instrument (Chapter 5) or a natural experiment (Chapter 6).

3.8.4 When Panel Methods Are Not Enough

It is worth being explicit about the limits of what this chapter's tools can accomplish. Panel fixed effects are a powerful way to control for time-invariant unobservables, and high-dimensional interacted effects extend this to group-level time-varying confounders. But neither can address firm-level time-varying confounders, the most common source of endogeneity in corporate finance.

If a firm receives private information that simultaneously changes both its leverage and its investment, no combination of fixed effects can absorb this shock, because it varies at the firm-year level, the same level as the treatment. In such cases, the researcher needs either an instrument (a variable correlated with leverage but not with the private information shock), a natural experiment (an exogenous shock to leverage that is unrelated to investment opportunities), or a structural model that explicitly accounts for the joint determination of both variables.

The methods in Chapters 5–6 address exactly this gap. Panel data methods provide the foundation. Every IV regression, every event study, every DiD design in corporate finance is estimated on panel data. But the identification in those designs comes from the instrument or the research design, not from the fixed effects. The fixed effects play a supporting role: they sharpen the comparison by removing confounders that the design does not address. Understanding what fixed effects do and do not accomplish (which is what this chapter has been about) is the prerequisite for using those more powerful tools effectively.

3.9 Connection to Structural and Causal Approaches

Every method in this chapter (fixed effects, first differences, clustering, dynamic panels) appears in both the structural and causal traditions introduced in Chapter 1. But the role panel data plays in each tradition is fundamentally different. Understanding this difference is essential for reading papers correctly and for choosing your own empirical strategy.

3.9.1 Panel Data in the Causal Tradition

In design-based causal inference, panel data is a tool for *removing confounders*. The logic, which has been the focus of this chapter, is straightforward: the firm fixed effect α_i is a nuisance parameter. It is correlated with the treatment, and if left in the error term it biases the estimate of the causal effect β . The within transformation eliminates α_i , and the researcher does not care about its value. The goal is β (the causal effect of leverage on investment, or of a policy change on firm behavior), and the fixed effects are just something to be swept out on the way to β .

The same logic drives the escalation to high-dimensional fixed effects in Section 3.4. Industry $\times$ year effects remove industry-level time-varying confounders. State $\times$ year effects remove state-level shocks. Each set of fixed effects is a confounder-removal device. The researcher evaluates them by asking: does this set of effects absorb something that would bias β ?

In this framework, the identifying assumption is a conditional independence claim: after conditioning on the fixed effects, the remaining variation in the treatment is “as good as random” with respect to the outcome. The fixed effects make this assumption more plausible by absorbing observable and unobservable group-level confounders. But as we emphasized in Section 3.3, they cannot address firm-level time-varying confounders, which is why the causal tradition ultimately relies on instruments (Chapter 5), natural experiments (Chapter 6), and research designs that provide exogenous variation in the treatment.

3.9.2 Panel Data in the Structural Tradition

In structural estimation, panel data plays a different role. The firm fixed effect is not a nuisance to be eliminated. It is often a *component of the economic model* that the researcher wants to understand or at least account for.

Consider the estimation of a production function, one of the oldest problems in empirical

economics. The model is

$$y_{it} = \alpha_L \ell_{it} + \alpha_K k_{it} + \omega_{it} + e_{it}, \quad (3.36)$$

where y_{it} is (log) output, ℓ_{it} and k_{it} are (log) labor and capital, ω_{it} is firm-level productivity, and e_{it} is an idiosyncratic shock. If productivity has a permanent component η_i (managerial ability, technology type) and a transitory component u_{it} , we can write $\omega_{it} = \eta_i + u_{it}$, and the model looks exactly like a panel regression with a fixed effect.

The within-groups estimator eliminates η_i and produces consistent estimates of α_L and α_K under the assumption that transitory productivity shocks are uncorrelated with input choices. But the structural tradition recognizes that this assumption is problematic: firms that receive a positive productivity shock u_{it} will hire more labor and invest more capital *in response to that shock*. The inputs are endogenous, not because of omitted variables, but because optimizing firms adjust their behavior in response to information about their own productivity. This is the simultaneity problem identified by [Marschak and Andrews \(1944\)](#), and it is exactly the kind of time-varying endogeneity that fixed effects cannot resolve.

The structural response has been to model the endogeneity explicitly. [Olley and Pakes \(1996\)](#) and [Levinsohn and Petrin \(2003\)](#) use control function methods that treat the firm's investment or material input decisions as proxies for unobserved productivity. [Arellano and Bond \(1991\)](#) use lagged inputs as instruments in a dynamic panel GMM framework. In each case, the econometric method is derived from the economic model. The structure tells you where the endogeneity comes from and what to do about it.

The same logic applies in structural corporate finance. In [Hennessy and Whited's \(2005, 2007\)](#) dynamic models of capital structure, the firm fixed effect captures permanent productivity differences that affect both the firm's financing decisions and its value. The model is estimated by simulated method of moments: the structural parameters (adjustment costs, tax rates, default costs) are chosen so that the model's simulated moments match the data moments. Panel data provides the moments (the mean and variance of leverage, the serial correlation of investment, the cross-sectional dispersion of debt ratios), but the identification comes from the model's structure, not from a within transformation.

3.9.3 The Same Regression, Two Interpretations

To see the distinction concretely, return to the running example:

$$I_{it} = \beta L_{it} + \mathbf{X}_{it} \boldsymbol{\gamma} + \alpha_i + \lambda_t + \varepsilon_{it}. \quad (3.37)$$

A causal researcher interprets this regression as follows: β is the causal effect of leverage

on investment, identified from within-firm variation in leverage over time, after absorbing time-invariant firm characteristics and aggregate shocks. The identifying assumption is that after conditioning on α_i and λ_t , the remaining variation in L_{it} is unrelated to the error ε_{it} . The researcher knows this is unlikely to hold perfectly (hence the desire for instruments or natural experiments), but views the FE regression as a useful starting point that at least removes the most obvious confounders.

A structural researcher sees the same equation differently. The coefficient β is not a “causal effect” in the potential-outcomes sense. It is a *reduced-form relationship* between two endogenous variables that are jointly determined by an underlying dynamic optimization problem. Leverage and investment are both chosen by the firm in response to its productivity, its financial position, its expectations about the future, and the realization of shocks. The regression coefficient β is a convolution of multiple structural parameters: the cost of debt, the adjustment cost of capital, the firm’s discount rate, and the persistence of productivity shocks. Without a model, you cannot disentangle these forces, and the “causal effect of leverage on investment” is not a well-defined object.

Neither interpretation is wrong. They answer different questions. The causal question (what happens to investment if we exogenously change a firm’s leverage?) is the right question for evaluating a specific policy (a tax change, a regulatory shock). The structural question (what are the deep parameters that govern the joint dynamics of investment and financing?) is the right question for understanding the economic mechanism and for predicting behavior under counterfactual scenarios that have not been observed in the data.

3.9.4 Practical Implications

This distinction has concrete implications for how you use the methods in this chapter.

If you are in the causal tradition, fixed effects are a confounder-removal tool, and the key question is whether they remove *enough* confounders to make the conditional independence assumption plausible. The answer is almost always no, which motivates the methods in Chapters 5–6. But panel FE regressions are the natural baseline specification against which more sophisticated designs are compared. Report the FE regression alongside your IV or DiD estimates. If the FE coefficient is close to the IV coefficient, the endogeneity from time-varying confounders may be modest. If they differ substantially, the FE regression is missing something important, and the reader can see exactly what the instrument or research design is doing.

If you are in the structural tradition, panel data provides moments for estimation and

fixed effects handle permanent heterogeneity, but the identification comes from the model's structure (the functional form of the value function, the assumed distribution of shocks, the timing of decisions). The transparency recommendations of [Andrews et al. \(2020\)](#) are especially important here: report the reduced-form (panel FE) results alongside the structural estimates so that readers can assess how much of the structural result comes from the data versus the modeling assumptions.

Key Insight 15

Panel data methods are used in both the structural and causal traditions, but they play different roles. In causal work, fixed effects are a confounder-removal device, and the fixed effects themselves are nuisance parameters. In structural work, fixed effects capture economically meaningful heterogeneity (productivity, managerial ability) that the model may need to account for explicitly. The regression looks the same. The interpretation is not.

3.9.5 Looking Ahead

This chapter has given you the foundation. You know what fixed effects do and do not solve. You know how to choose a specification, how to cluster, and how to handle dynamics. You know where the method stands in the structural and causal framework that organizes this book.

The chapters that follow build directly on this foundation. Event studies (Chapter 4) are panel regressions around a specific event, with the event providing the identifying variation that FE alone cannot deliver. Instrumental variables (Chapter 5) address the time-varying endogeneity that FE leaves unresolved, and they are estimated on the same panel data with the same fixed effects. Difference-in-differences (Chapter 6) combines the panel structure with a specific research design (a policy change, a natural experiment) to identify causal effects under parallel trends. In each case, the panel methods from this chapter are the scaffolding. What changes is the source of identification.

Chapter 4

Event Studies

4.1 The Problem of Measuring Event Impact

On March 3, 2020, the Federal Reserve cut interest rates in an emergency session, responding to the growing threat of the COVID-19 pandemic. Within minutes, the Dow Jones Industrial Average surged more than 350 points. Investors seemed to welcome the stimulus. But then something happened. During Fed Chair Jerome Powell's press conference, as he acknowledged the limits of what monetary policy could accomplish against a virus, the rally collapsed. By the close of trading, the Dow had fallen 786 points, a swing of nearly 1,400 points from high to low in a single session.

What was the market's verdict on the rate cut? If you measured the reaction in the first fifteen minutes, you would conclude the market viewed it as good news worth about 350 points. If you measured at the close, you would conclude it was bad news worth 786 points. Same event, same day, wildly different answers depending on when you look. This is not an idle puzzle. It strikes at the heart of how finance researchers have measured event impacts for half a century.

Now consider a different kind of measurement problem. Suppose several U.S. states adopt new environmental regulations at different times during the 2010s, and you want to know how these regulations affected corporate investment. A natural approach is to compare investment changes for affected firms to investment changes for unaffected firms, using the standard two-way fixed effects regression that we developed in Chapter 3. You run the regression, get a coefficient, and report it.

But here is the trouble. Recent research has shown that this seemingly innocuous regression can produce *severely* biased estimates, and not just a little biased. Consider a simple numerical example. Three firms are observed over five periods. Firm A is

treated at $t = 2$, Firm B at $t = 4$, and Firm C is never treated. The true treatment effect is positive: it starts at +10 and grows over time. Yet when we compare Firm B at $t = 4$ to the already-treated Firm A (which TWFE happily uses as a “control”), we get an estimated effect of -10 . The regression tells us the policy *hurts* investment when it actually *helps*. As [Goodman-Bacon \(2021\)](#) demonstrated, this is not a pathological special case. It is an inherent feature of two-way fixed effects with staggered treatment timing and heterogeneous effects.

These two puzzles (one from the finance tradition, one from the economics tradition) set the stage for this chapter. Both involve the same fundamental question: *how do we measure the causal impact of corporate events?* And both reveal that the answer is harder than it first appears.

4.1.1 Two Traditions, One Question

Corporate events happen constantly. Firms issue debt, announce earnings, face regulatory changes, adopt new technologies, replace their CEOs, get acquired. Measuring the impact of these events is arguably the most common empirical exercise in corporate finance. Over the past five decades, two distinct traditions have developed for doing so. Understanding both (and knowing when to use each) is essential for modern research.

The **finance event study** tradition, formalized in the 1970s alongside the efficient market hypothesis ([Fama, 1970](#)), uses stock price reactions to measure event impacts. The logic is elegant: if markets are informationally efficient, then the stock price change around an event reveals the market’s assessment of how the event affects the present value of the firm’s future cash flows. To implement this, we need an asset pricing model that tells us what the stock return would have been absent the event, the “normal” return. The difference between the actual return and this model-predicted return is the abnormal return, which we attribute to the event. [MacKinlay \(1997\)](#) provides the classic survey, and [Kothari and Warner \(2007\)](#) offer an updated treatment.

The **economics event study** tradition, which emerged from labor and public economics, takes a fundamentally different approach. Instead of relying on stock prices and market efficiency, it uses difference-in-differences identification. We compare changes in outcomes for treated firms to changes for control firms, attributing the difference to the event. No assumption about market efficiency is needed. What we need instead is a credible control group and the assumption that treated and control firms would have followed parallel trends in outcomes absent the event. [Miller \(2023\)](#) surveys this tradition, and [Roth, Sant’Anna, Bilinski, and Poe \(2023\)](#) synthesize recent methodological advances.

At first glance, these might seem like minor variations on the same idea. They are not. The two traditions rest on different identifying assumptions, measure different objects, and answer different questions. Recognizing this is more important than mastering any particular formula in this chapter.

4.1.2 What Each Tradition Actually Assumes

Recall from Chapter 1 that the structural versus causal distinction turns on where identification comes from. The finance event study sits in an interesting position relative to this distinction. It is a *causal* method (we are trying to measure the causal effect of an event on firm value), but identification requires a *structural* element: an asset pricing model. The Capital Asset Pricing Model, the Fama-French three-factor model, or even a simple constant mean return model provides the counterfactual. What would this stock have returned today if the event had not occurred? The model answers that question.

This means that every finance event study is a joint test. We are simultaneously testing whether the event mattered *and* whether our asset pricing model correctly specifies normal returns. If the model is wrong, our abnormal returns are wrong, and our conclusions about the event are wrong. For short-horizon studies of a few days, this turns out not to matter much. [Brown and Warner \(1985\)](#) showed that different reasonable models give very similar results. But the joint hypothesis problem is always lurking, and it becomes severe for longer horizons.

The economics event study sits squarely in the causal/design-based tradition from Chapter 1. No model of how prices form is needed. Instead, identification comes from the research design: who is treated, who is the control, and why we believe the control group's trajectory tells us what would have happened to the treated group absent treatment. The counterfactual is not generated by a model; it is observed directly in the data as the control group's actual outcomes.

The key identifying assumption (parallel trends) requires that treated and control groups would have evolved in the same way over time had the event not occurred. Note what this does *not* require: it does not require that treated and control groups have the same *levels* of outcomes. It only requires that they would have experienced the same *changes*. A firm with twice the investment rate of a control firm can still be a valid comparison, as long as both would have changed by similar amounts absent treatment.

Table 4.1 summarizes the key differences.

Table 4.1. Two Traditions for Measuring Event Impact

	Finance Event Study	Economics Event Study
Core assumption	Market efficiency: prices fully reflect information	Parallel trends: treated and control groups would have followed same trends
Counterfactual	Asset pricing model predicts normal return	Control group's observed outcomes
Outcome measure	Stock returns	Real outcomes (I_{it} , employment, productivity, etc.)
Control group	Not needed (model provides expected return)	Required and central to identification
Time horizon	Typically short (days)	Typically medium to long (quarters, years)
What you learn	Market's valuation of the event	Actual behavioral response to the event
Structural element	Needs asset pricing model	None (design-based)
Key vulnerability	Joint hypothesis problem; event window choice	Parallel trends may fail; no-anticipation required

4.1.3 Why Both Traditions Exist

A natural question is: why not just pick the better method and use it? The answer is that neither tradition is uniformly better. They measure different things, and different research questions call for different approaches.

Suppose a firm announces a major debt issuance. If you want to know how the *market* values this decision (whether investors view more leverage as good news or bad news for firm value), you want a finance event study. Stock prices aggregate the beliefs of many informed investors and should reflect the expected present value of all consequences of the issuance: tax shields, financial distress risk, agency effects, signaling, everything. The stock return gives you a single number that summarizes the market's bottom line.

But if you want to know how the debt issuance affects what the firm *actually does* (whether it invests more, hires more workers, or changes its other financial policies), you need to measure those real outcomes directly. Stock prices tell you about expectations; they do not tell you about behavior. For that, you need an economics event study that tracks investment, employment, or other real variables over subsequent quarters and compares changes for issuers to changes for non-issuers.

These are genuinely different *estimands*, in the language of Chapter 2. The finance event study estimates the market's assessment of the expected change in firm value. The economics event study estimates the average treatment effect on the treated (ATT) for whatever real outcome we measure. Even if both methods were perfectly implemented, they

would typically give different numbers, because they are answering different questions.

This distinction connects directly to the potential outcomes framework from Chapter 1. Using our unified notation, let $Y_i(1)$ denote the outcome for firm i if it experiences the event and $Y_i(0)$ the outcome if it does not. The treatment effect for firm i is $\tau_i = Y_i(1) - Y_i(0)$. In the finance tradition, the outcome Y_i is typically the stock return, and τ_i is measured as the abnormal return. In the economics tradition, Y_i might be capital expenditure, and τ_i is measured via the difference-in-differences estimator. Same notation, same conceptual framework, different outcome variables, different identification strategies.

4.1.4 The Running Example: Leverage and Investment

Throughout this chapter, we use the leverage-investment example that has appeared in Chapters 1–3 to illustrate both traditions and show what each can and cannot tell us.

Finance event study approach. We examine stock returns around debt issuance announcements. When a firm announces it will issue new bonds, does the stock price go up or down? A large literature, beginning with work in the 1980s, documents that debt issuance announcements are associated with small negative abnormal returns on average, typically around -0.5% to -1.5% over a three-day window. This tells us that the market, on average, interprets debt issuance as mildly bad news for firm value. Various theories can explain this: adverse selection (Myers and Majluf, 1984), where issuance signals that management thinks the firm's debt is overvalued, or agency concerns about increased leverage enabling empire building.

But this market-based measure tells us nothing about what happens to the firm's actual investment. The stock price reaction is about expectations. What do firms really do?

Economics event study approach. We track capital expenditure around credit rating downgrades, treating the downgrade as a shock to debt capacity. Firms downgraded at different times (staggered treatment) are compared to firms not downgraded (control group). Using the modern methods we develop in this chapter, we can estimate how investment evolves in the quarters before and after a downgrade. Does investment fall? How quickly? Does it recover? This tells us directly about the real effects of changes in leverage capacity on firm behavior.

The two approaches are complementary, not competing. The finance event study tells us that the market expects negative consequences. The economics event study tells us whether and how those consequences materialize. If the market is efficient, the stock price

reaction should anticipate the real effects. Comparing the two provides a natural test of this proposition.

4.1.5 What Makes This Chapter Different

Most textbook treatments of event studies cover the two traditions in isolation (finance event studies in one chapter, difference-in-differences in another) and leave the reader to figure out the connections. This chapter puts them side by side, because understanding the comparison is more important than mastering either method in isolation.

The chapter is organized as follows. Section 4.2 develops the finance event study methodology: the efficient market hypothesis foundation, the five-step procedure (define events, measure abnormal returns, aggregate, test, analyze cross-sectional variation), and the key challenges including post-earnings announcement drift and event window selection. Section 4.3 develops the economics event study methodology: the difference-in-differences framework, the parallel trends assumption and how to test it, and (critically) the staggered adoption problem and modern solutions from [Borusyak, Jaravel, and Spiess \(2024\)](#), [Callaway and Sant’Anna \(2021\)](#), and [Sun and Abraham \(2021\)](#). Section 4.4 brings the two traditions together, comparing what each assumes, what each can tell us, and when to use each, or both. Section 4.2 provides practical implementation guidance with code in both Stata and Python. Section 4.5 presents a decision framework for choosing among approaches and connects the chapter to the methods developed later in the book.

Three themes run through the entire chapter. First, the structural versus causal distinction from Chapter 1: the finance approach embeds causal inference in a structural pricing model, while the economics approach uses design-based identification. Second, treatment effect heterogeneity: both traditions typically report averages, but understanding how effects vary across firms and over time is increasingly central to good research. The staggered adoption problem is fundamentally a heterogeneity problem. Third, the importance of being clear about your estimand. Too many papers run an event study without being precise about what parameter they are estimating and under what assumptions. The potential outcomes framework from Chapter 1 provides the discipline to avoid this.

A final note before we begin. Event studies are one of the great success stories of empirical finance. The finance event study was invented *within* the field (not borrowed from somewhere else) and has produced an enormous body of useful results over five decades. The economics event study has brought the credibility revolution’s emphasis on research design into corporate finance, generating important insights about real outcomes. Both traditions continue to evolve. The goal of this chapter is not to declare one tradition

superior, but to equip you to use both well and to understand what each can (and cannot) tell you.

Connection to the Structural–Causal Framework

Event studies are the empirical corporate finance researcher’s most common tool for measuring the impact of corporate events. Both traditions attempt causal inference, but they differ in where identification comes from:

Finance event study: Causal claim embedded in a structural model of asset pricing. The model generates the counterfactual (normal return). Identification strength depends on the model’s validity.

Economics event study: Causal claim based on research design. The control group provides the counterfactual. Identification strength depends on the parallel trends assumption and the quality of the control group.

Neither approach is assumption-free. The question is always: which assumptions are more plausible for your specific research question?

4.2 Finance Event Studies: Market-Based Methods

The finance event study is one of the few empirical methods that finance invented for itself. Most of what we use (regression, instrumental variables, difference-in-differences) was developed in economics or statistics and imported into finance. But the event study was created by finance researchers to answer finance questions, and it has proven so useful that other fields have adopted it. The pioneering work of [Fama, Fisher, Jensen, and Roll \(1969\)](#) on stock splits established the basic approach, and subsequent refinements by [Brown and Warner \(1980, 1985\)](#) put the methodology on a firm statistical foundation. [MacKinlay \(1997\)](#) provides the classic survey, and [Kothari and Warner \(2007\)](#) offer an updated treatment in the *Handbook of Empirical Corporate Finance*.

The method’s power comes from a simple but profound insight: if markets are informationally efficient, then stock price changes around events reveal the market’s assessment of those events’ value implications. This transforms the stock market from a mere trading venue into a measurement device. But to use this device properly, we need to understand both how it works and when it can mislead us.

4.2.1 The Efficient Market Hypothesis as an Identifying Assumption

Before walking through the mechanics of the methodology, we need to be clear about what we are assuming and why. Every empirical method rests on identifying assumptions, and the finance event study's central assumption is the efficient market hypothesis.

In its semi-strong form, the EMH asserts that security prices fully reflect all publicly available information (Fama, 1970). This has three implications that matter directly for event studies. First, the current stock price represents the market's best assessment of the present value of the firm's expected future cash flows. Second, when new information arrives, the stock price adjusts immediately to reflect it. Third, the magnitude of the price change equals the change in the present value of expected future cash flows.

If all three implications hold, then measuring the stock price change around an event gives us a clean estimate of how the event affects firm value. The market aggregates the beliefs and information of many investors, so the price change should incorporate not only the direct effects of the event but also its indirect implications for future opportunities and risks. This is powerful. We get the market's comprehensive bottom-line assessment in a single number.

But notice what we are doing from the perspective of Chapter 1. We are using a *structural* model of how prices form to underwrite a *causal* claim about the event's impact. The EMH is doing structural work: it tells us how information maps into prices. Without this structural assumption, an abnormal return is just an unexplained residual. It has no causal interpretation.

This makes every finance event study a *joint test*. We are simultaneously testing whether the event affected firm value *and* whether our model of normal returns is correct. As Fama (1991) emphasized, we can never test market efficiency in isolation from a model of expected returns. If we find an abnormal return, it could mean the event mattered, or it could mean our model of normal returns is wrong. For short-horizon studies, this joint hypothesis problem is usually minor. Different reasonable models give very similar results over a few days. But the problem is always present, and it becomes severe for longer horizons.

Two additional assumptions deserve attention. First, the event must be *exogenous*. It should not be caused by recent stock price movements or market conditions. If firms tend to issue equity after stock price run-ups, then the announcement return reflects both the market's reaction to the issuance decision and any correction of the prior run-up. Second, the Stable Unit Treatment Value Assumption (SUTVA) requires that one firm's event does not affect another firm's stock price beyond any common market movement. If a firm gains competitive advantage through a merger, competitors' stock prices may decline,

violating SUTVA. We will discuss these spillover concerns further when we connect to the contamination bias framework of [Goldsmith-Pinkham, Hull, and Kolésár \(2024\)](#) in Chapter 6.

When are these assumptions most plausible? Market efficiency is most credible for large, liquid stocks followed by many analysts and institutional investors. For such stocks, information is rapidly incorporated through the trading of informed investors. Efficiency is less plausible for small, illiquid stocks with limited analyst coverage, where information may diffuse slowly and informed trading may be constrained by transaction costs. Exogeneity is most plausible for discrete, clearly datable events like court rulings, regulatory announcements, or natural disasters, events that firms did not engineer in response to their stock price.

4.2.2 Step One: Define Your Events

The first step in any event study is to identify the events of interest and determine precisely when they occurred. This requires both conceptual clarity about what constitutes the event and practical decisions about data sources. Getting this step wrong undermines everything that follows.

What counts as the event? This sounds simple but often requires careful thought. If you are studying mergers, is the event the first public rumor, the formal announcement, the shareholder vote, or the completion date? Each captures something different. The announcement date is when new information reaches the market, so it is typically the right choice for measuring the market's reaction to the news. But information may leak before the formal announcement, which is why event windows typically extend a day or two before the announcement date.

Finding event dates. Researchers have traditionally used the *Wall Street Journal* as the primary source for identifying event dates, with news wire services (Reuters, Bloomberg) providing more timely information. Modern research typically uses computerized databases such as Factiva, Nexis, or specialized sources like SDC Platinum for mergers and acquisitions. The key is to identify when the information *first* became available to market participants, which may be before the WSJ publication date.

Handling confounding events. Events rarely occur in isolation. A firm might announce earnings on the same day it announces a dividend change, or a regulatory decision might

coincide with an industry-wide shock. If two types of news overlap, we cannot determine which drives the observed stock price reaction. Most researchers exclude observations with confounding events within the event window. This is sensible but creates a potential selection problem: if the likelihood of confounding news is correlated with the event of interest, excluding these observations may bias the sample. [MacKinlay \(1997\)](#) discusses this tradeoff. In practice, the selection concern is usually considered less important than the confounding problem, and most studies exclude confounded events.

Defining the event window. The event window is the set of days over which we measure the stock price reaction. A common choice is three days: the day before the announcement ($t = -1$), the announcement day ($t = 0$), and the day after ($t = +1$). The day before captures potential information leakage, and the day after captures delayed reactions. Some studies use a wider window of five days ($t = -2$ to $t = +2$) or even longer, but wider windows increase the risk of contamination from unrelated news.

The event window should not be chosen after looking at the data. If you notice that abnormal returns are concentrated two days before the announcement and extend the window to capture them, you are engaging in data snooping. Decide on the event window before examining results, and report results for a few standard windows to show robustness.

4.2.3 Step Two: Measure Abnormal Returns

The event matters if it is associated with an *abnormal* return, a return that differs from what we would normally expect. To measure an abnormal return, we need a model of what constitutes a normal or expected return. This is where the structural element of the methodology enters: we need an asset pricing model to construct the counterfactual.

For firm i on day t , the abnormal return is defined as

$$ar_{it} = r_{it} - E[r_{it} | X_t] \quad (4.1)$$

where r_{it} is the actual return and $E[r_{it} | X_t]$ is the expected return conditional on information set X_t . The challenge is specifying how to determine this expected return. Unfortunately, there is no consensus asset pricing model in finance, which is why this is a joint test problem. Fortunately, for short-horizon studies, the choice among reasonable models turns out to matter very little, a reassuring result established by [Brown and Warner \(1985\)](#) and confirmed by subsequent work.

Three approaches to defining normal returns are commonly used.

Constant mean return model. The simplest approach assumes that the expected return for firm i is its historical average return $\bar{r}_i$:

$$E[r_{it}] = \bar{r}_i \quad (4.2)$$

This ignores systematic risk entirely. Yet [Brown and Warner \(1985\)](#) showed that it performs surprisingly well for short-horizon studies, because over a few days the difference between any firm's expected return and the market average is small relative to the event-induced price change.

Market model. The most commonly used approach assumes that firm i 's return is linearly related to the market return:

$$r_{it} = \alpha_i + \beta_i r_{mt} + \varepsilon_{it} \quad (4.3)$$

where r_{mt} is the return on the market portfolio (typically the CRSP value-weighted index), α_i and β_i are firm-specific parameters, and ε_{it} is an idiosyncratic error with $E[\varepsilon_{it}] = 0$ and variance $\sigma_{\varepsilon_i}^2$. The expected return given the market return is $E[r_{it} | r_{mt}] = \alpha_i + \beta_i r_{mt}$, and the abnormal return is the residual $ar_{it} = \varepsilon_{it}$.

The market model has the advantage of removing the portion of return variation that is due to market-wide movements. This reduces the variance of abnormal returns relative to the constant mean model, increasing the power to detect event effects. The improvement is proportional to R^2 of the market model regression. The higher the R^2 , the more variation we remove, and the more precisely we can measure the event effect.

Fama-French three-factor model. A more sophisticated approach uses additional factors capturing size and value effects ([Fama and French, 1993](#)):

$$r_{it} = \alpha_i + \beta_i r_{mt} + s_i SMB_t + h_i HML_t + \varepsilon_{it} \quad (4.4)$$

where SMB_t is the return on a portfolio of small stocks minus large stocks, and HML_t is the return on a portfolio of high book-to-market stocks minus low book-to-market stocks. Factor returns are available from Kenneth French's data library.¹ The logic is the same as the market model: we remove predictable variation in returns to isolate the event-specific abnormal return.

For short-horizon studies, the choice among these models rarely matters. [Brown](#)

¹Available at https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html. Five-factor model returns that add profitability and investment factors are also available.

and Warner (1985) demonstrated this with simulations, and Antweiler and Frank (2006) confirmed it with more recent data: the rejection rates under each model were nearly identical. This occurs because over a few days, the additional factors contribute very little to expected returns. The expected return from any model is close to the risk-free rate, and these small differences are swamped by the event effect. For longer horizons, the model choice becomes much more consequential, which is one reason long-run event studies are controversial.

Estimating the model parameters. Whichever model we choose, we need to estimate its parameters using data *outside* the event window. The standard approach uses an *estimation window* of approximately 120 trading days, ending about 30 days before the event window begins. Figure 4.1 illustrates the timeline.

Figure 4.1. Timeline of an Event Study. The estimation window (typically 120 trading days) is used to estimate the parameters of the normal return model. A gap of approximately 30 trading days separates the estimation window from the event window to prevent the event itself from contaminating parameter estimates. The event window typically spans a few days around the event date $t = 0$.

Why the gap between the estimation window and the event window? If the event was anticipated or if information leaked, returns near the event may already reflect part of the event's impact. Including these contaminated returns in the estimation window would bias our estimates of α_i and β_i , and therefore bias our abnormal return calculations. The 30-day gap provides a buffer. The choice of 120 days for the estimation window is a rule of thumb that balances precision (longer windows give more precise estimates) against stability (the firm's risk characteristics may change over very long windows). Both choices can be varied as robustness checks.

We estimate the model parameters by ordinary least squares. For the market model, this gives us $\hat{\alpha}_i$ and $\hat{\beta}_i$, and we calculate abnormal returns in the event window as:

$$\hat{a}r_{it} = r_{it} - \hat{\alpha}_i - \hat{\beta}_i r_{mt} \quad (4.5)$$

4.2.4 Step Three: Aggregate Across Time and Across Firms

At this point we have abnormal returns for each firm on each day in the event window. To draw conclusions, we need to aggregate these individual observations in two dimensions: across days within the event window (to capture the total event effect) and across firms (to measure the average effect).

Why aggregate across time? Information does not reach all investors simultaneously. Wire services may report news before the *Wall Street Journal* publishes. Some investors monitor markets continuously; others check only at the end of the day. Frank and Sanati (2018) develop a model showing that market responses involve interactions among capital-constrained arbitrageurs, otherwise-busy small investors, and liquidity traders. The implication is that the full price adjustment to news may take more than a single trading day, even in relatively efficient markets.

To capture the total effect, we cumulate abnormal returns over the event window. For firm i , the cumulative abnormal return from day t_1 to t_2 is:

$$CAR_i(t_1, t_2) = \sum_{t=t_1}^{t_2} \hat{a}r_{it} \quad (4.6)$$

For a standard three-day window, this would be $CAR_i(-1, +1)$. For a five-day window, $CAR_i(-2, +2)$.

Why aggregate across firms? A single firm's abnormal return is noisy. One firm might have a large positive abnormal return on the event date purely by chance. By averaging across many firms experiencing the same type of event, the noise cancels out and the systematic event effect emerges. If we have N firms, the average cumulative abnormal return is:

$$\overline{CAR}(t_1, t_2) = \frac{1}{N} \sum_{i=1}^N CAR_i(t_1, t_2) \quad (4.7)$$

A numerical example. To make the aggregation concrete, consider three firms that announce new debt issuances on the same day. Table 4.2 shows their daily abnormal returns over a five-day event window.

Table 4.2. Numerical Example: Abnormal Returns Around Debt Issuance Announcements

Firm	Abnormal Returns (%)					CAR_i
	$t = -2$	$t = -1$	$t = 0$	$t = +1$	$t = +2$	
Firm A	-0.1	-0.3	-1.2	-0.2	0.0	-1.8
Firm B	0.2	-0.1	-0.8	-0.3	0.1	-0.9
Firm C	0.0	-0.4	-1.5	0.1	-0.1	-1.9
Mean	0.03	-0.27	-1.17	-0.13	0.00	-1.53

Each firm experiences a negative abnormal return on the announcement day, consistent with the well-documented finding that debt issuance announcements are associated with

mildly negative stock returns. The effect is concentrated at $t = 0$, with small amounts of pre-announcement drift at $t = -1$ suggesting some information leakage. The average $CAR(-2, +2) = -1.53\%$, indicating that the market views these debt issuances as mildly negative for firm value.

Note that we could equivalently compute the $\overline{CAR}$ by first averaging abnormal returns across firms for each day (the “Mean” row in the table) and then summing across days: $0.03 + (-0.27) + (-1.17) + (-0.13) + 0.00 = -1.53\%$. Both approaches yield identical results.

Standardization. Different firms have different return volatilities. A -1% abnormal return is unremarkable for a volatile biotech stock but striking for a large utility. Going back to [Brown and Warner \(1980\)](#), it has been common to standardize abnormal returns by each firm’s estimated standard deviation:

$$sar_{it} = \frac{\widehat{ar}_{it}}{\widehat{\sigma}_i} \quad (4.8)$$

where $\widehat{\sigma}_i$ is the standard deviation of firm i ’s abnormal returns during the estimation window. This ensures that a volatile firm’s abnormal return is appropriately down-weighted relative to a stable firm’s. Whether to standardize is a judgment call. If all firms in the sample have similar volatility, it makes little difference. If the sample includes both large, stable firms and small, volatile firms, standardization helps prevent the most volatile observations from dominating the results.

4.2.5 Step Four: Hypothesis Testing

With aggregate abnormal returns in hand, we need to determine whether they are statistically distinguishable from zero. The null hypothesis is that the event has no effect on firm value, implying that abnormal returns during the event window are drawn from the same distribution as abnormal returns outside the event window.

The simplest test statistic compares the average cumulative abnormal return to zero:

$$t = \frac{\overline{CAR}}{\widehat{\sigma}_{CAR}/\sqrt{N}} \quad (4.9)$$

where $\widehat{\sigma}_{CAR}$ is the cross-sectional standard deviation of the CAR_i values. Under the null hypothesis and standard regularity conditions, this statistic is approximately normally distributed for large N .

This basic approach has been refined in several ways. The *Patell test* ([Patell, 1976](#))

standardizes each firm's abnormal returns by that firm's estimated standard deviation before aggregating, accounting for cross-firm heteroskedasticity. The *Boehmer-Musumeci-Poulsen test* (Boehmer, Musumeci, and Poulsen, 1991) goes further, adjusting for event-induced increases in volatility. During event periods, volatility often rises, which can lead to overstated significance if not accounted for. For settings where abnormal returns are non-normally distributed or the sample is small, the *Corrado rank test* (Corrado, 1989) provides a nonparametric alternative that is robust to outliers and non-normality.

Do the tests work? The foundational evidence on test reliability comes from Brown and Warner (1980, 1985), who conducted simulation studies that remain influential. Their approach was elegant: take actual stock return data, inject a known abnormal return on a pseudo-event date, and then check whether the event study methodology can detect it. They found reassuring results. When no abnormal return was injected, the tests rejected at approximately the nominal 5% rate, slightly too often due to the fat-tailed nature of returns, but close enough to be reliable. When a 2% abnormal return was injected, rejection rates were around 85%, demonstrating good power to detect economically meaningful effects. With a 5% injected return, rejection was essentially 100%.

Antweiler and Frank (2006) updated this simulation analysis with more recent data. They added small abnormal returns ranging from zero to five percent to actual stock market data, conducted 250 event studies with 50 pseudo-events each, and checked how frequently they could detect the injected effect at the 5% significance level. Their results confirmed Brown and Warner's conclusions: rejection rates fluctuated only slightly across four different asset pricing models, and even a 2% abnormal return was detected roughly 85% of the time. Most importantly, the choice among reasonable normalizing models made almost no practical difference to the inferences drawn.

These simulation results are genuinely reassuring for short-horizon event studies of economically important events. But they also delineate the method's limits: for small effects (below about 1%), large samples are needed to achieve adequate power. Researchers planning an event study should consider in advance whether their expected sample size and likely effect size will produce a test with reasonable power.

Clustering and dependence. One of the most important pitfalls in event study inference concerns independence across observations. The standard test statistics assume that abnormal returns are independent across firms. This assumption is violated when events cluster in calendar time. If 100 firms all announce earnings on the same day, their abnormal returns will be correlated due to industry-level news or market-wide movements. Failing

to account for this can severely overstate statistical significance. [Kolari and Pynnönen \(2010\)](#) develop a modified test that adjusts for cross-sectional correlation in event studies. Alternatively, one can aggregate all firms with events on the same date into a single portfolio observation, though this reduces power. As [Petersen \(2009\)](#) emphasizes in the context of panel regressions, getting the standard errors right is essential, and the same principle applies here.

4.2.6 Step Five: Cross-Sectional Analysis

Many event studies stop after testing whether the average abnormal return differs from zero. But the most informative studies go further, asking *why* some firms are affected more than others. This cross-sectional analysis often provides the sharpest test of competing theories.

The standard approach regresses each firm's cumulative abnormal return on observable characteristics:

$$CAR_i = \alpha + \beta X_i + \gamma Z_i + \eta_i \quad (4.10)$$

where X_i is the variable of primary interest, Z_i represents control variables, and η_i is an error term. If $\hat{\beta}$ is statistically significant, this suggests that the characteristic X_i helps explain which firms are most affected by the event.

For example, if studying the stock market reaction to a regulatory change, we might hypothesize that financially constrained firms are affected more severely. We can test this by including a measure of financial constraints as X_i . If the coefficient is negative and significant, this supports the hypothesis that the regulation is particularly costly for constrained firms.

These cross-sectional regressions require some care in interpretation. The dependent variable, CAR_i , is itself an estimate with measurement error, which can affect coefficient precision. The explanatory variables may be endogenous or may proxy for multiple underlying characteristics. Standard concerns about omitted variables and multicollinearity apply. But when done well, cross-sectional analysis transforms the event study from a test of “does the event matter?” to an investigation of “for whom does the event matter and why?”

This is also where the event study connects naturally to the treatment effect heterogeneity theme that runs through this book. The cross-sectional regression is really a simple way to estimate how the treatment effect varies with observable characteristics, a basic form of conditional average treatment effect (CATE) analysis. In Chapter 7 we will see more sophisticated machine learning approaches to discovering this kind of heterogeneity,

including causal forests that can uncover nonlinear interactions among firm characteristics. But the cross-sectional regression remains a useful starting point.

Summary: The Five Steps

Step 1: Define events. Identify the event and its date. Exclude confounded observations. Choose the event window before examining results.

Step 2: Measure abnormal returns. Choose a normal return model (market model is standard). Estimate parameters from pre-event data. Calculate abnormal returns in the event window as the difference between actual and predicted returns.

Step 3: Aggregate. Cumulate abnormal returns over the event window for each firm (CAR_i). Average across firms ($\overline{CAR}$). Optionally standardize by firm-specific volatility.

Step 4: Test. Test whether $\overline{CAR}$ differs significantly from zero. Account for clustering if events overlap in calendar time. Consider the Patell or BMP test if heteroskedasticity or event-induced variance is a concern.

Step 5: Explain cross-sectional variation. Regress CAR_i on firm characteristics to test which factors predict larger or smaller reactions. This is where economic hypotheses are sharpened.

4.2.7 What Have We Learned?

Five decades of finance event studies have produced an enormous body of results, and many of them have been genuinely informative. The traditional pattern documented in hundreds of studies is consistent with the EMH: a discrete jump in the stock price on the announcement date, possibly preceded by small leakage, and no predictable drift afterward. Good news produces positive abnormal returns; bad news produces negative ones. Firms announcing unexpected earnings surprises, merger bids, or dividend changes experience stock price reactions in the direction predicted by economic theory. As [Fama \(1991\)](#) argued, this body of evidence provides some of the strongest support we have for the proposition that stock markets are at least reasonably efficient.

But the picture is not entirely clean. Several challenges to the traditional interpretation have emerged, and understanding them is essential for anyone conducting or evaluating event study research.

Post-earnings announcement drift. The most thoroughly documented anomaly comes from accounting research. [Bernard and Thomas \(1989\)](#) found that after earnings announce-

ments, firms with the most positive earnings surprises continued to outperform firms with the most negative surprises for roughly 60 trading days. The effect appeared in 46 out of 50 quarters and represented an average return of approximately four percent on a strategy of buying positive-surprise firms and shorting negative-surprise firms. This finding has been replicated and refined in subsequent work, and remarkably, it has not disappeared even though it has been well known for decades. The persistence of this anomaly is puzzling from an efficient markets perspective: if the drift is predictable, why don't traders eliminate it? Various explanations (limits to arbitrage, transaction costs, investor inattention) have been proposed, but no consensus explanation has emerged.

For event study methodology, post-earnings announcement drift raises a practical question: over what horizon should we measure the event's impact? If we use a narrow three-day window, we capture the immediate market reaction but miss the subsequent drift. If we use a longer window, we capture more of the total effect but also pick up more noise from confounding events. There is no universally right answer. The appropriate window depends on the specific event and the research question.

Event window selection and information processing. The question of event window choice goes deeper than a technical nuisance. As we noted in the opening to this chapter, the March 2020 Fed rate cut produced dramatically different results depending on whether you measured the reaction in the first fifteen minutes or at the end of the day. [Frank and Sanati \(2018\)](#) developed a theoretical framework that explains why this happens. In their model, market responses to news are driven by interactions among three groups: capital-constrained arbitrageurs who react quickly, otherwise-busy small investors who may not notice news immediately, and liquidity traders. Positive news can produce initial over-reaction (as arbitrageurs cannot fully absorb selling pressure from liquidity traders) that subsequently corrects. Negative news can produce under-reaction that slowly resolves as all investor types participate.

The implication is important and contrary to the simple version of the EMH: the size of the event window can matter for conclusions. The initial price reaction in the first few minutes may differ substantially from the daily close, which may differ from the reaction measured over several days. Neither is necessarily "wrong," but they measure different things: the immediate reaction of fast-moving traders versus the considered judgment of the full market.

Selection bias and publication bias. Selection effects can undermine our interpretation of the published event study literature. At the level of financial activity, firms may select

into events based on private information. A manager may issue equity when she believes the stock is overvalued, so we cannot treat equity issuers and non-issuers as comparable except for the issuance decision. [Prabhala \(1997\)](#) and [Li and Prabhala \(2007\)](#) discuss these selection effects in the context of corporate finance research.

Publication bias is perhaps even more insidious. As [Fama \(1998\)](#) noted, splashy results get more attention, creating incentives to find them. If only statistically significant results are published, the literature will overstate the frequency and magnitude of event effects. Even worse, researchers may abandon projects when initial results are unpromising, so these null results never even reach the submission stage. Pre-registration of event studies, while rare in finance, could help address this problem. The American Economic Association maintains an RCT Registry that serves this function for experiments; a similar practice for event studies would improve the credibility of the literature.

Long-horizon complications. When extending event studies to horizons of months or years, several additional problems arise. The choice of normal return model becomes critical because small model misspecifications compound over time: a bias of 0.1% per month becomes 3.6% over three years. Confounding events become far more likely over long horizons. A firm will experience many other shocks during a three-year post-event period. Survivor bias can be severe if poorly performing firms delist or get acquired. For these reasons, long-run event studies are inherently more controversial than short-run studies and require much greater care in execution and interpretation. We return to this issue briefly in [Section 4.4](#) when we compare the horizons natural to each tradition.

4.2.8 Running Example: Stock Returns Around Debt Issuances

Let us now apply the finance event study methodology to our running example. Suppose we want to measure the market's reaction to corporate debt issuance announcements, using a sample of public bond issuances from SDC Platinum for U.S. non-financial, non-utility firms.

Following the five steps:

Step 1. We define the event as the announcement date of a public bond issuance. We exclude observations where the firm also announced earnings, a dividend change, or a merger within the event window.

Step 2. We estimate a market model using 120 trading days of CRSP daily returns, ending 30 days before the event. We calculate abnormal returns in the event window $[-2, +2]$.

Step 3. We cumulate abnormal returns over the five-day window for each firm and average across firms.

Step 4. The literature typically finds $\overline{CAR}(-2, +2) \approx -0.5\%$ to -1.5% , statistically significant with large samples. Debt issuance announcements are received as mildly negative news by the market.

Step 5. Cross-sectional regressions reveal that the negative reaction is larger for firms with higher prior leverage, lower credit ratings, and greater information asymmetry. This is consistent with adverse selection theories (Myers and Majluf, 1984): firms with more private information face a larger “lemons discount” when they issue debt.

What does this tell us? The market’s reaction reflects its assessment of *why* the firm is issuing debt. If the market suspects the firm is exploiting overvalued debt, the reaction is negative. If the issuance is clearly tied to a positive NPV project, the reaction is less negative or even positive. The average effect aggregates over many motivations.

Crucially, this analysis tells us about the market’s *expectations*. It does not tell us what happens to the firm’s investment, employment, or other real outcomes after the debt issuance. For that, we need the economics event study approach of the next section. The average CAR of -1% says that the market expects the debt issuance to reduce firm value by about one percent. Whether this expectation is correct (whether firms actually invest less wisely, or whether the market systematically overreacts) is a question that the finance event study alone cannot answer.

This is precisely the kind of question where combining the two traditions, as we will do in Section 4.4, yields insights that neither tradition can provide alone.

4.3 Economics Event Studies: Difference-in-Differences Methods

The economics event study tradition emerged from a fundamentally different intellectual starting point than the finance tradition of the previous section. Labor economists and public economists studying policy interventions (minimum wage increases, job training programs, welfare reform) could not rely on stock prices to measure outcomes. There were no asset prices to consult. Instead, they had to measure actual outcomes directly: employment, wages, program participation. The question was how to construct a credible counterfactual for what would have happened absent the policy. The difference-in-differences framework provided the answer, and over the past two decades it has become one of the most widely used identification strategies in corporate finance. Baker, Larcker,

and Wang (2022) report that 431 difference-in-differences papers were published in the top five finance journals between 2000 and 2019.

This section develops the economics event study methodology with the same emphasis as the previous section: not just the mechanics of *how*, but the logic of *why* each step is done and *where* things can go wrong. We start with the simple two-by-two case to build intuition, extend to multiple periods and event study plots, and then confront the critical problem of staggered adoption that has reshaped the methodology in recent years. Miller (2023) provides an accessible introduction to the event study framework, and Roth et al. (2023) offer a comprehensive synthesis of the recent econometrics literature.

4.3.1 The Basic Idea: Two Groups, Two Periods

The difference-in-differences logic is deceptively simple. We observe a *treated* group that experiences an event and a *control* group that does not. We observe both groups before and after the event. The treatment effect is estimated by taking the difference in the change over time between the two groups, a difference of two differences.

Why do we need two differences? Consider the alternative approaches and why each fails. A simple comparison of treated and control outcomes *after* the event confounds the treatment effect with any pre-existing differences between the groups. Treated firms might have higher investment than control firms even without the event, simply because they are different kinds of firms. A simple comparison of treated firms *before and after* the event confounds the treatment effect with any time trend that would have affected the treated firms even without the event. Investment might rise or fall due to macroeconomic conditions that have nothing to do with the event. The difference-in-differences estimator solves both problems simultaneously: it uses the control group's change to net out the common time trend, leaving only the treatment effect.

Formally, let $\bar{Y}_{g,t}$ denote the average outcome for group $g \in \{\text{treated}, \text{control}\}$ at time $t \in \{0, 1\}$, where period 0 is pre-treatment and period 1 is post-treatment. The DiD estimator is:

$$\hat{\delta}^{DiD} = \underbrace{(\bar{Y}_{\text{treated},1} - \bar{Y}_{\text{treated},0})}_{\text{change for treated}} - \underbrace{(\bar{Y}_{\text{control},1} - \bar{Y}_{\text{control},0})}_{\text{change for control}} \quad (4.11)$$

The first difference captures both the treatment effect and any common time trend. The second difference captures only the common time trend. Subtracting removes the trend, leaving the treatment effect.

A numerical example. Suppose we study the effect of a credit rating downgrade on capital expenditure. Table 4.3 shows average investment rates for downgraded and non-downgraded firms before and after the downgrade.

Table 4.3. Numerical Example: Two-by-Two Difference-in-Differences

	Pre-Downgrade	Post-Downgrade	Difference
Downgraded firms	8.0%	5.5%	−2.5 pp
Non-downgraded firms	7.0%	6.0%	−1.0 pp
Difference	+1.0 pp	−0.5 pp	−1.5 pp

The downgraded firms reduced their investment rate by 2.5 percentage points. But investment also fell for non-downgraded firms by 1.0 percentage point, reflecting a general economic slowdown. The DiD estimate is $-2.5 - (-1.0) = -1.5$ percentage points: the downgrade reduced investment by about 1.5 percentage points beyond the common trend.

Notice that downgraded firms had *higher* investment than non-downgraded firms even before the downgrade (8.0% versus 7.0%). This is perfectly fine. Difference-in-differences does not require equal levels. It requires equal *trends*. The levels can differ arbitrarily, as long as the changes would have been the same absent treatment.

Regression formulation. The same estimator can be expressed as a regression, which is the form most commonly used in practice:

$$Y_{it} = \alpha + \beta \cdot Treated_i + \gamma \cdot Post_t + \delta \cdot (Treated_i \times Post_t) + \varepsilon_{it} \quad (4.12)$$

Here α is the baseline level for control firms in the pre-period, β captures the level difference between treated and control, γ captures the common time trend, and δ is the treatment effect, exactly the DiD estimator from equation (4.11). The coefficient δ is the only parameter that requires the interaction of group and time variation; it cannot be identified from either dimension alone.

4.3.2 The Parallel Trends Assumption

The DiD estimator identifies a causal effect under one critical assumption: **parallel trends**. This assumption states that in the absence of treatment, the treated and control groups would have experienced the same change in outcomes over time. Formally, using the potential outcomes notation from Chapter 1, let $Y_{it}(0)$ denote the outcome for unit i at time

t in the absence of treatment. Parallel trends requires:

$$E[Y_{i,1}(0) - Y_{i,0}(0) \mid D_i = 1] = E[Y_{i,1}(0) - Y_{i,0}(0) \mid D_i = 0] \quad (4.13)$$

where $D_i = 1$ indicates treatment. In words: the expected change in untreated potential outcomes is the same for treated and control units.

It is important to understand exactly what this assumes and what it does not. Parallel trends does *not* require that treated and control groups have the same level of outcomes. It does *not* require that the trends be literally linear. The name is somewhat misleading. What it requires is that whatever trajectory the control group follows over time, the treated group would have followed the same trajectory (possibly shifted up or down in level) absent treatment.

The assumption can fail in several ways that are relevant for corporate finance research. If treatment is correlated with firm growth trajectories (for instance, if rapidly growing firms are more likely to adopt new technologies), then treated and control firms are on different paths even before treatment, and DiD will be biased. If there are time-varying shocks that differentially affect treated and control firms (such as industry-specific downturns that hit treated firms harder), parallel trends fails. If firms anticipate treatment and change behavior before the formal treatment date, this can contaminate the pre-period and confuse the interpretation. [Roth et al. \(2023\)](#) provide a thorough discussion of when parallel trends is and is not plausible.

Parallel trends is fundamentally untestable for the post-treatment period. This point deserves emphasis because it is commonly misunderstood. We never observe what would have happened to treated firms in the absence of treatment after treatment has occurred. That is the counterfactual we are trying to estimate. Parallel trends is an assumption about this unobserved counterfactual. We can examine whether parallel trends held *before* treatment, and if it did not, this is strong evidence against the assumption. But if it did hold before treatment, this is *encouraging but not conclusive*. It provides no guarantee that it would have continued to hold in the post-period.

4.3.3 Pre-Trends Testing

Although we cannot directly test parallel trends in the post-period, we can and should test whether it held in the pre-treatment period. This *pre-trends test* is the primary empirical check available, and getting it right is essential for credible research.

The standard approach uses a multi-period event study specification. Suppose we observe firms for K periods before and L periods after treatment. We estimate:

$$Y_{it} = \alpha_i + \lambda_t + \sum_{\substack{k=-K \\ k \neq -1}}^L \delta_k \cdot \mathbb{I}[t - T_i^* = k] + \varepsilon_{it} \quad (4.14)$$

where α_i are unit fixed effects, λ_t are time fixed effects, T_i^* is the treatment date for unit i , and $\mathbb{I}[t - T_i^* = k]$ is an indicator equal to one when unit i is k periods from its treatment date. The coefficient δ_{-1} is omitted as the reference period, so all effects are measured relative to the period immediately before treatment.

This specification produces a set of coefficients $\hat{\delta}_k$ that trace out the dynamics of the treatment effect. The pre-treatment coefficients ($k < -1$) test parallel trends: if they are all close to zero and jointly insignificant, this supports (but does not prove) the assumption. The post-treatment coefficients ($k \geq 0$) estimate the dynamic treatment effect, how the impact evolves over time.

Plotting these coefficients with confidence intervals produces the *event study graph*, one of the most important diagnostic tools in modern applied economics. A credible event study graph shows pre-treatment coefficients hovering around zero, followed by a break at the treatment date with coefficients shifting to reflect the treatment effect.

How to interpret pre-trends results. Several principles guide interpretation. First, look at the overall pattern, not individual coefficients. One or two marginally significant pre-treatment coefficients may simply reflect sampling variability. But a systematic upward or downward trend in the pre-treatment coefficients is strong evidence against parallel trends. Second, conduct a formal F-test of the joint hypothesis that all pre-treatment coefficients are zero. If this test rejects, the parallel trends assumption is in serious trouble. Third, be aware of power limitations. With small samples, pre-trends tests may fail to reject even when parallel trends is violated, simply because the tests lack power (Roth, 2022). A non-significant F-test is not proof that parallel trends holds. It may just mean you lack the statistical power to detect a violation.

When pre-trends fail. If pre-trends testing reveals differential trends, the researcher has several options, none fully satisfying. One approach is to add group-specific time trends to the specification, but this imposes functional form assumptions about the trend shape and can absorb real treatment effects if they evolve linearly (Wolfers, 2006). Another approach is to find a better control group, for example, using propensity score matching to select

controls that better resemble the treated units on pre-treatment characteristics. [Callaway and Sant'Anna \(2021\)](#) accommodate conditioning on covariates within the DiD framework. A third option is to acknowledge that the design is fundamentally flawed for this particular question and seek alternative identification. Honesty about design limitations is more valuable than methodological contortions to salvage a broken research design.

4.3.4 Extending to Multiple Periods and Event Study Graphs

The simple two-by-two framework provides clean intuition, but most corporate finance applications involve many time periods and many firms treated at different times. This extension is natural and powerful (it allows us to trace dynamic treatment effects and examine how impacts evolve), but it introduces important complications that have only been fully understood in recent years.

With multiple pre- and post-treatment periods, the event study specification in equation (4.14) produces a rich set of dynamic effects. We might find that a regulatory change has no immediate impact on investment but gradually reduces it over two to three years. Or we might find a sharp initial drop that partially recovers. These dynamics provide clues about mechanisms. A gradual build-up suggests learning or adjustment costs. A sharp impact that fades suggests temporary disruption. A persistent effect suggests a permanent change in the economic environment.

The event study graph is now standard in applied research and expected by referees. [Miller \(2023\)](#) provides a clear guide to constructing and interpreting these plots. A well-constructed event study graph communicates three things: whether parallel trends held in the pre-period (flat pre-treatment coefficients), when the effect began (the break point), and how the effect evolved (the post-treatment trajectory).

4.3.5 The Staggered Adoption Problem

We now come to what is arguably the most important methodological development for empirical corporate finance researchers using difference-in-differences in the past decade. When treatment is staggered (different firms are treated at different times), the standard two-way fixed effects estimator that Chapter 3 introduced can produce severely biased estimates. Understanding why this happens and how to fix it is essential for any researcher using these methods.

Why staggered adoption is pervasive in corporate finance. [Baker et al. \(2022\)](#) document that staggered treatment timing is the norm, not the exception, in corporate finance

research. Regulatory changes phase in across states or firm size categories. Firms adopt new technologies at different dates. Credit rating changes, CEO turnovers, and mergers all occur at different times for different firms. The simple two-by-two case where all treated firms are treated simultaneously is rare outside of clean natural experiments.

What goes wrong with TWFE. Consider the regression that seems like a natural extension of the two-period case to a panel setting:

$$Y_{it} = \alpha_i + \lambda_t + \delta \cdot D_{it} + \varepsilon_{it} \quad (4.15)$$

where $D_{it} = 1$ if firm i has been treated by time t and zero otherwise. This is the standard TWFE specification from Chapter 3, and for years researchers used it routinely for staggered DiD designs. The problem, as demonstrated by [Goodman-Bacon \(2021\)](#), [de Chaisemartin and D'Haultfœuille \(2020\)](#), and [Baker et al. \(2022\)](#), is that this regression is a weighted average of many different two-by-two comparisons, and some of those comparisons are deeply problematic.

The key insight of [Goodman-Bacon \(2021\)](#) is the *Bacon decomposition*: the TWFE estimator can be written as a weighted average of all possible two-by-two DiD comparisons. These include comparisons of treated versus never-treated (good), comparisons of early-treated versus late-treated (potentially fine), and, crucially, comparisons that use *already-treated* units as controls for *newly-treated* units. These last comparisons are called **forbidden comparisons** because already-treated units are experiencing treatment effects, making them invalid controls.

A numerical example of forbidden comparison bias. To see why this matters, consider three firms observed over five periods. Firm A is treated at $t = 2$, Firm B at $t = 4$, and Firm C is never treated. The true treatment effect is positive, starting at +10 and growing by +5 each additional period. Table 4.4 shows the setup.

Now consider three different DiD comparisons at the time Firm B is treated ($t = 4$). Comparing Firm B to the never-treated Firm C gives:

$$\hat{\delta}_{B \text{ vs } C} = (125 - 110) - (115 - 110) = 15 - 5 = 10$$

This correctly identifies the immediate treatment effect. But using the already-treated Firm A as a control gives:

$$\hat{\delta}_{B \text{ vs } A} = (125 - 110) - (135 - 125) = 15 - 10 = 5$$

Table 4.4. How Forbidden Comparisons Create Bias

	$t = 1$	$t = 2$	$t = 3$	$t = 4$	$t = 5$	True Effect Pattern
<i>Potential outcomes without treatment (parallel trends hold):</i>						
Firm A (treated $t = 2$)	100	105	110	115	120	–
Firm B (treated $t = 4$)	100	105	110	115	120	–
Firm C (never treated)	100	105	110	115	120	–
<i>Treatment effects (growing over time):</i>						
At treatment onset	–	+10	–	+10	–	Immediate
Each additional period	–	–	+5	+5	+5	Growth
<i>Observed outcomes:</i>						
Firm A	100	115	125	135	145	–
Firm B	100	105	110	125	140	–
Firm C	100	105	110	115	120	–

This underestimates the true effect of 10 because Firm A's outcome is still *growing* from its own treatment. Worse, if treatment effects grow more aggressively, the forbidden comparison can produce a *negative* estimate of a positive effect, exactly the scenario we described in Section 4.1's opening puzzle. The TWFE estimator mixes all such comparisons together with data-determined weights that can even be negative. The result can be severe bias in either direction.

This is not a pathological special case. It occurs whenever two conditions hold simultaneously: treatment is staggered across time *and* treatment effects are heterogeneous, either across firms or over time. Since both conditions are the norm in corporate finance, the problem is pervasive. [Baker et al. \(2022\)](#) demonstrate its practical relevance by revisiting published corporate finance studies and showing that TWFE estimates sometimes differ substantially from estimates obtained with modern methods.

4.3.6 Modern Solutions

The recognition that TWFE can be badly biased with staggered adoption has produced a wave of methodological innovation. Three approaches have become particularly prominent, and understanding what each does (at a conceptual level) is important for choosing among them in practice. Chapter 6 will develop these methods in full technical detail; here we focus on the key intuition and practical guidance.

The Borusyak, Jaravel, and Spiess imputation approach. The core insight of [Borusyak et al. \(2024\)](#) is simple and elegant: use only untreated observations to estimate the counterfactual, then impute what treated firms *would have* done absent treatment. The

method proceeds in three steps. *Step 1:* Using only observations where $D_{it} = 0$ (not-yet-treated and never-treated firm-periods), estimate unit and time fixed effects: $\hat{\alpha}_i$ and $\hat{\lambda}_t$. These estimates are “clean” in the sense that no treatment effects contaminate them. *Step 2:* For each treated observation, impute the counterfactual outcome as $\hat{Y}_{it}(0) = \hat{\alpha}_i + \hat{\lambda}_t$. This is what we predict the firm’s outcome would have been absent treatment, based on its typical level and the current time period’s common effect. *Step 3:* Estimate the treatment effect for each treated observation as $\hat{\tau}_{it} = Y_{it} - \hat{Y}_{it}(0)$, then aggregate as desired.

Notice the structural parallel to the finance event study from Section 4.2. There, we used a pre-event estimation window to estimate normal return parameters, then used those parameters to construct counterfactual returns in the event window. Here, we use untreated observations to estimate fixed effect parameters, then use those parameters to construct counterfactual outcomes for treated observations. The logic is the same: estimate a model from clean data, use it to predict what would have happened absent the event, and attribute the difference to the event. The difference is that the finance approach uses an asset pricing model and the economics approach uses a two-way fixed effects structure for untreated potential outcomes.

The Callaway and Sant’Anna group-time approach. Rather than imputing counterfactuals, [Callaway and Sant’Anna \(2021\)](#) build from the bottom up. They define a *cohort* as the set of firms first treated at the same time g , and they estimate the average treatment effect for each cohort in each time period t :

$$ATT(g, t) = E[Y_t - Y_{g-1} \mid G_i = g] - E[Y_t - Y_{g-1} \mid C] \quad (4.16)$$

where $G_i = g$ indicates firms first treated at g and C indicates the comparison group (never-treated or not-yet-treated). Each $ATT(g, t)$ is a clean two-by-two DiD using only valid comparisons.

This produces a matrix of effects: cohorts in rows, time periods in columns. Each cell shows the effect for that cohort at that time. This matrix can then be aggregated in different ways depending on the research question. Averaging within each event time $e = t - g$ produces a dynamic event study. Averaging within each cohort g shows whether early and late adopters experienced different effects. Averaging across everything produces an overall ATT. The transparency of this approach is its great virtue: we can see exactly which comparisons drive the results.

The Sun and Abraham interaction-weighted approach. [Sun and Abraham \(2021\)](#) show

how to modify the standard event study regression to avoid forbidden comparisons by interacting cohort indicators with event-time indicators and using only clean controls. Their approach produces traditional event study plots that are easy to interpret, making it convenient when the primary goal is to present dynamic effects graphically.

Which method to choose? All three methods solve the same fundamental problem (avoiding forbidden comparisons when treatment effects are heterogeneous), but they differ in emphasis and convenience. [Borusyak et al. \(2024\)](#) is computationally efficient for large datasets and offers flexible aggregation. [Callaway and Sant’Anna \(2021\)](#) provides the most transparent view of heterogeneity across cohorts. [Sun and Abraham \(2021\)](#) produces familiar event study plots most directly. In practice, estimating results with at least two methods and confirming robustness is good scientific practice. If the methods give similar answers, this strengthens confidence. If they diverge, this signals that the specific pattern of treatment effect heterogeneity matters for the conclusions, which is itself informative. Chapter 6 provides a decision framework and implementation guidance.

When Must You Use Modern Methods?

The answer is simple: whenever treatment is staggered and you have reason to think effects might be heterogeneous across firms or over time. In corporate finance, this is almost always. The bias from TWFE with staggered adoption is not a minor technical issue. It can reverse the sign of estimated effects, as our numerical example demonstrated. Since the modern methods are now straightforward to implement with standard software (see Section 4.2), there is no good reason to use naive TWFE for staggered designs. At minimum, run the modern methods as a robustness check. If they agree with TWFE, report both. If they diverge, trust the modern methods.

4.3.7 What the Economics Event Study Identifies

Before turning to the running example, we should be precise about what parameter the economics event study actually estimates. Under parallel trends and with staggered adoption, the modern DiD methods typically identify the **average treatment effect on the treated** (ATT):

$$ATT = E[Y_i(1) - Y_i(0) \mid D_i = 1] \quad (4.17)$$

This is the average effect for the firms that actually experienced the event. It tells us what happened to treated firms, but it does not directly tell us what would happen if we imposed the same treatment on currently untreated firms. The ATT may differ from the average

treatment effect (ATE) if firms select into treatment based on their expected benefits, which in corporate finance is almost always the case. Firms that choose to issue debt, adopt a technology, or merge presumably do so because they expect to benefit, and those expected benefits may differ from the benefits that non-adopters would experience.

This selection issue connects to the structural versus causal framework of Chapter 1. The ATT is a well-defined causal parameter: it tells us the average causal effect for the specific population that was treated. But extrapolating to other populations requires additional assumptions, either a structural model of selection or a stronger identifying assumption. We return to this distinction in Chapter 6, where we connect the ATT to the marginal treatment effect (MTE) framework that allows extrapolation under richer assumptions.

4.3.8 Running Example: Investment Around Credit Rating Downgrades

We now apply the economics event study methodology to the second half of our running example. In Section 4.2.8, we used a finance event study to measure the stock market's reaction to debt issuance announcements and found a mildly negative average abnormal return of about -1% . Now we ask: what actually happens to firm investment when a firm experiences a negative shock to its debt capacity? The finance approach tells us what the market expects. The economics approach tells us what firms actually do.

Suppose we study corporate investment around credit rating downgrades, using Compustat quarterly data for U.S. non-financial firms from 2000 to 2020. Firms are downgraded by S&P or Moody's at different dates, a natural staggered adoption setting. Our treated group is firms experiencing a downgrade. Our control group could be firms that are never downgraded during the sample period, or firms that are not-yet-downgraded at the time each treated firm is downgraded.

Step 1: Define treatment and control. Treatment is the quarter in which a firm's credit rating is downgraded by at least one notch. We define the treatment cohort g as the quarter of downgrade. Control firms are those not downgraded during the sample window.

Step 2: Assess parallel trends. We estimate the event study specification in equation (4.14) with investment rate (I_{it}/A_{it-1}) as the outcome, using eight quarters before and eight quarters after the downgrade. We plot the pre-treatment coefficients with confidence intervals. If the pre-downgrade coefficients are flat around zero, this supports the assumption that downgraded and non-downgraded firms would have followed similar investment trends absent the downgrade.

Step 3: Estimate using modern methods. Because downgrades are staggered and we expect treatment effect heterogeneity (firms downgraded from investment grade to speculative grade likely experience larger effects than those downgraded within investment grade), we use the [Borusyak et al. \(2024\)](#) imputation estimator rather than naive TWFE.

What do we expect to find? The literature documents that credit rating downgrades are associated with significant reductions in investment, on the order of 1–3 percentage points of total assets over the two years following the downgrade. The effect is larger for firms that depend more on external financing and for downgrades that cross the investment-grade threshold, consistent with theories of financial constraints.

What does this tell us that the finance event study cannot? The economics event study reveals the *actual behavioral response*: firms cut real investment when their debt capacity is reduced. This is a direct measure of the real consequences of financial shocks. The finance event study told us that the market expected negative consequences. The economics event study tells us those consequences materialized. Moreover, the dynamic event study graph shows *when* the investment decline begins, how large it gets, and whether it recovers. These dynamics provide clues about whether the investment decline reflects temporary disruption (suggesting adjustment costs) or a permanent reallocation (suggesting binding constraints).

Comparing the two approaches for our running example produces a natural consistency check. The finance event study found that the market reacted negatively to debt issuance news. The economics event study finds that investment actually fell following a downgrade. Both point in the same direction: increases in leverage are associated with negative market reactions, and reductions in debt capacity are associated with real investment declines. If the market is efficient, the stock price reaction should anticipate the magnitude of the real effects. Section [4.4](#) will explore this comparison more systematically.

Connection to the Structural–Causal Framework

The economics event study is squarely in the causal/design-based tradition. No structural model of asset pricing or firm behavior is needed. Identification comes entirely from the research design: who was downgraded (treatment), who was not (control), and the assumption that they would have followed parallel trends absent the downgrade.

The parallel trends assumption plays the same role here that the EMH plays in the finance event study: it generates the counterfactual. But the counterfactual comes from different places. In the finance approach, the counterfactual (normal return) comes from a structural model. In the economics approach, the counterfactual comes from the observed behavior of the control group. Neither approach is assumption-free. The choice depends on which assumptions you find more credible for your specific question.

4.4 Comparing the Two Traditions

Sections 4.2 and 4.3 presented the two event study traditions as separate methodologies. This section brings them together. The goal is not to repeat the comparison table from Section 4.1.1. You already know the basic differences in assumptions, outcomes, and time horizons. Instead, we focus on three questions that matter for researchers deciding how to study corporate events: When do the two approaches agree? When do they diverge, and what does that divergence teach us? And when should you use both?

4.4.1 When the Approaches Tell Different Stories

The two traditions measure fundamentally different objects: the finance event study measures the market's *expectation* of an event's value consequences, while the economics event study measures the event's *actual effect* on real outcomes. When markets are efficient and researchers execute both methods correctly, the two should tell a coherent story. The stock price reaction should anticipate, in present value terms, the real effects that subsequently materialize. When they diverge, something interesting is happening, and figuring out what is often the most productive research to do.

Consider four patterns of divergence and what each suggests.

Large stock price reaction, small real effects. A regulatory announcement generates a 5% abnormal return, but subsequent tracking of investment, employment, and R&D

reveals only minor changes. This pattern suggests the market initially overestimated the regulation's impact, perhaps because of uncertainty that later resolved. Alternatively, the real effects may be in dimensions the researcher did not measure. The stock price reflects *all* expected consequences, including changes in risk, option values, and competitive positioning that may not show up in standard real outcome variables. This pattern does not necessarily mean the market was wrong; it may mean we are measuring the wrong real outcomes.

Small stock price reaction, large real effects. A firm experiences a credit downgrade with only a modest stock price decline (perhaps because the downgrade was partially anticipated), but investment subsequently falls substantially. This pattern can arise when information leaks into prices gradually before the formal event date (Fox, Glosten, and Tetlock, 2009). The finance event study, which measures the *surprise* component of the event, understates the total impact because the market had already partially adjusted. The economics event study, which compares real outcomes before and after the event regardless of anticipation, captures the full effect. This highlights a practical difference: the finance approach measures the *news* in the event, while the economics approach measures the *total* change associated with it.

Positive stock reaction, negative real effects (or vice versa). A firm announces an acquisition and the stock price rises, but subsequent profitability and investment efficiency decline. This could indicate that the market was simply wrong about the acquisition's value, consistent with managerial hubris or overvaluation theories (Roll, 1986). Or it could indicate that the stock price rose for reasons unrelated to the acquisition (the event was confounded), or that the real outcome measures are capturing adjustment costs rather than long-run effects. Sign disagreements between the two approaches are rare in well-designed studies, but when they occur they are particularly informative because they force us to think hard about which set of assumptions is failing.

Heterogeneity in different dimensions. The finance event study might show that small firms have larger stock price reactions (perhaps because they are less well known, so the event contains more information), while the economics event study shows that large firms have larger investment responses (perhaps because they have better access to capital markets to adjust). Both findings can be correct simultaneously. They are measuring different things: information effects versus behavioral responses. Discovering that heterogeneity runs in different directions across the two approaches reveals something about the mechanisms

linking events to outcomes, something neither approach alone could uncover.

4.4.2 Earnings Announcements: An Extended Example

The syllabus discussion question for this chapter asks us to consider studying earnings announcements using both approaches. This example is worth working through carefully because it illustrates the complementarity of the traditions in a setting that every finance researcher encounters.

The finance event study. We classify firms by the magnitude of their quarterly earnings surprise (actual earnings minus analyst consensus from I/B/E/S) and measure abnormal stock returns over a three-day window centered on the announcement date. Decades of research show a strong positive relationship: firms with the largest positive surprises experience positive abnormal returns of roughly 3–5%, while firms with the largest negative surprises experience comparable negative returns ([Ball and Brown, 1968](#)). This tells us that earnings contain information the market values, and the magnitude of the price change reflects the present value of the earnings news.

But this analysis raises questions it cannot answer. Does the earnings surprise change what firms *do*? If a firm reports unexpectedly strong earnings, does it invest more, hire more, or increase dividends? The stock price reaction reflects the market's expectation of all such consequences, but we cannot see what specifically drives it.

The economics event study. We classify firms into those with positive earnings surprises (treated) and those without (control), and we track capital expenditure, R&D spending, employment, and dividend payments over subsequent quarters using a DiD framework. Positive-surprise firms are “treated” with good news about their profitability.

Suppose we find that firms with positive earnings surprises increase capital expenditure by 8% relative to control firms over the following four quarters, but do not significantly increase employment or R&D. This tells us something specific about the transmission mechanism: internally generated cash flows appear to relax financing constraints for physical investment specifically, consistent with the [Fazzari, Hubbard, and Petersen \(1988\)](#) financial constraints hypothesis.

What the combination reveals. The finance event study found a 3% abnormal return. The economics event study found an 8% investment increase. Are these consistent? If the market is efficient, the 3% stock price increase should reflect the present value of all

future consequences of the earnings surprise, including the investment increase and its effect on future profits. Whether an 8% investment increase generates a 3% increase in firm value depends on the marginal return on investment, the cost of capital, and how long the investment boost persists, all of which require a structural model to evaluate.

This is where the structural versus causal framework from Chapter 1 connects in an unexpected way. The two *causal* event studies, taken together, implicitly bound the marginal return on investment. If the market correctly prices the investment increase, we can back out what the market thinks each dollar of additional investment is worth. If the market's implied return differs from engineering estimates or from returns implied by investment models like [Abel and Eberly \(1994\)](#), this reveals something about market efficiency, the quality of the investment, or both. Neither event study alone provides this inference. Together, they create a natural test of the relationship between expectations and reality.

4.4.3 Heterogeneity and the Limits of Averages

Both traditions share a common weakness: they typically report average effects. The average *CAR* and the average *ATT* each summarize a distribution of firm-level effects into a single number. When heterogeneity is large, this summary can be misleading.

For finance event studies, the cross-sectional regression (Section 4.2.6) is the traditional tool for exploring heterogeneity: regress CAR_i on firm characteristics to see which factors predict larger reactions. This approach works well for testing pre-specified hypotheses but is limited in its ability to discover unexpected patterns. The machine learning methods introduced in Chapter 7, particularly causal forests ([Wager and Athey, 2018](#)), can discover nonlinear interactions without pre-specification, and they apply naturally to cross-sectional variation in CAR_i .

For economics event studies, the modern DiD methods described in Section 4.3.6 already accommodate heterogeneity across cohorts and time periods. The [Callaway and Sant'Anna \(2021\)](#) approach explicitly reports effects by cohort, revealing whether early and late adopters experience different treatment effects. Going further, unit-level treatment effect estimates from [Borusyak et al. \(2024\)](#) can be used as inputs to causal forest analysis, producing a data-driven exploration of what firm characteristics predict larger or smaller real responses.

An important insight emerges from combining heterogeneity analyses across the two traditions. Suppose we study a regulatory change using both approaches. The finance event study reveals that the stock price reaction is larger for firms with more institutional

ownership, perhaps because institutions process regulatory news faster. The economics event study reveals that the real investment response is larger for financially constrained firms, perhaps because the regulation relaxed a binding constraint that mattered more for constrained firms. These are different kinds of heterogeneity, and together they paint a richer picture of how the regulation propagates through the economy than either alone.

4.4.4 Spillovers and Contamination

Both traditions assume that one firm's treatment does not affect another firm's outcomes, the Stable Unit Treatment Value Assumption (SUTVA) introduced in Chapter 1. In corporate finance, this assumption is frequently strained.

When a firm issues debt, this may affect competitors through product market competition (more aggressive pricing), factor market competition (bidding up input prices), or financial market effects (changing risk perceptions for the industry). When a firm is downgraded, bond spreads for similar firms may widen in sympathy. When a major firm in an industry adopts a new technology, competitors may be forced to follow, altering their investment patterns.

The consequences of SUTVA violations differ across the two traditions in a predictable way. In the finance event study, spillovers show up as abnormal returns for non-event firms on the event date. If Firm A's merger announcement causes Firm B's stock to drop (a competitive effect), this contaminates Firm B's abnormal return if Firm B happens to have its own event on a nearby date. In the economics event study, spillovers affect the control group's outcomes. If treated firms' increased investment crowds out investment by control firms through tighter capital markets, then the control group's investment decline inflates the DiD estimate beyond the true treatment effect. [Goldsmith-Pinkham et al. \(2024\)](#) formalize this as *contamination bias* and provide tests based on measuring control units' exposure to treatment.

For researchers, the practical implication is to think explicitly about the plausibility of SUTVA for their specific setting and to test for spillovers when possible. In the finance tradition, this means checking whether competitors, suppliers, or customers experience abnormal returns on the event date. A literature on "intra-industry effects" has done exactly this ([Lang and Stulz, 1992](#)). In the economics tradition, this means checking whether control firms with greater exposure to treated firms show different outcome changes than less-exposed controls. Chapter 6 develops these contamination tests more formally.

4.4.5 Synthesizing the Running Example

We can now pull together the two halves of our running example to illustrate how the combination of approaches produces insights that neither provides alone.

In Section 4.2.8, the finance event study found that debt issuance announcements produce average $CAR \approx -1\%$, with larger negative reactions for firms with higher leverage and lower credit quality. Cross-sectional analysis revealed that information asymmetry proxies predict the magnitude of the reaction, consistent with adverse selection theories. This tells us that the market views debt issuance as mildly negative for firm value, more so when the issuing firm is already highly leveraged or informationally opaque.

In Section 4.3.8, the economics event study found that credit rating downgrades reduce investment by approximately 1–3 percentage points, with the effect concentrated in the first four quarters and larger for firms crossing the investment-grade threshold. The dynamic event study graph showed flat pre-treatment coefficients (supporting parallel trends) and a sharp decline at the downgrade date that partially recovers over two years.

What does the combination reveal? Three things that neither approach provides individually.

First, the market's negative reaction to debt-related events is not just sentiment. It predicts real consequences. Firms that experience negative debt capacity shocks actually cut investment, confirming that the market's concerns have substance.

Second, the magnitudes create a consistency check. If a 1% decline in firm value (from the finance event study) corresponds to a 2 percentage point decline in the investment rate (from the economics event study), we can ask: is this implied relationship between value and investment plausible? If the average firm invests about 6% of assets, a 2 percentage point decline is a 33% reduction in investment. Does a 33% investment reduction justify a 1% decline in firm value? The answer depends on the marginal profitability of the lost investment, a question that requires economic modeling to answer, connecting us back to the structural tradition.

Third, the heterogeneity patterns complement each other. The finance event study shows that the market discriminates among issuers based on information asymmetry. Opaque firms receive harsher reactions. The economics event study shows that real investment responses are larger for firms crossing the investment-grade boundary, a supply-side channel. Together, these suggest that the leverage-investment relationship operates through both demand-side (information) and supply-side (constraints) channels, and the relative importance varies by firm type.

This kind of multi-method triangulation is increasingly valued in the profession. Rather than relying entirely on one set of identifying assumptions, the researcher who uses both

approaches can assess whether the assumptions are mutually consistent. If both approaches point in the same direction under different assumptions, this strengthens confidence in the conclusion far more than any single robustness check within one framework could.

When to Use Both Approaches

Using both traditions is most valuable when:

- (a) You want to distinguish *expectations* from *outcomes*. The finance approach measures what the market thinks will happen; the economics approach measures what actually does happen. Comparing them tests whether expectations are correct.
- (b) The assumptions of neither approach are bulletproof. If you are unsure about market efficiency, the economics event study provides evidence that does not depend on it. If you are unsure about parallel trends, the finance event study provides evidence from a different identifying assumption entirely.
- (c) You care about mechanisms. Each approach reveals different dimensions of heterogeneity. Together, they illuminate how events propagate from market reactions to real decisions.

Using a single approach is sufficient when: (a) your question is specifically about market valuation and you are comfortable with the EMH for your sample; or (b) your question is specifically about real outcomes and you have a credible control group but no natural stock market angle.

4.5 Choosing Your Approach

This chapter has covered a lot of ground (two traditions, multiple methods within each, and the case for combining them). A PhD student starting a research project needs practical guidance: given my question and my data, what should I actually do? This section provides a decision framework and connects the chapter's methods to the techniques developed in the rest of the book.

4.5.1 A Decision Framework

The choice between finance and economics event studies is not primarily about methodological preference. It is about matching the tool to the question. Three questions guide the choice.

What outcome do you care about? If your research question is about how the market *values* an event (how investors assess a merger announcement, a regulatory change, or a debt issuance), then you want to measure stock returns. The finance event study is designed for exactly this purpose. If your question is about how firms *behave* in response to an event (whether they invest more, hire differently, or alter their financial policies), you need to measure those real outcomes directly. The economics event study is the right tool. If you care about both valuation and behavior, use both.

What counterfactual is more credible in your setting? The finance event study's counterfactual comes from an asset pricing model: what would this stock have returned absent the event? This is most credible for large, liquid stocks over short horizons, where different reasonable models converge. The economics event study's counterfactual comes from a control group: what would the treated firms have done absent the event? This is most credible when you have a well-defined control group, parallel pre-trends, and a setting where treatment is not driven by anticipated outcomes. If neither counterfactual is credible in your setting, event study methods may not be the right approach at all. You may need the instrumental variables methods of Chapter 5 or the structural approaches discussed in Chapter 1.

What does your data allow? The finance approach requires high-frequency stock return data, which limits the sample to publicly traded firms with liquid markets. The economics approach requires panel data on real outcomes with sufficient pre-treatment periods to assess parallel trends, and it requires identifying both treatment timing and a suitable control group. In some settings, data availability makes the choice for you: private firms have no stock returns, so only the economics approach is feasible. Conversely, if the event affects all firms simultaneously (a macroeconomic shock), there may be no valid control group, making only the finance approach viable.

Figure 4.2 summarizes this logic.

Figure 4.2. Decision tree for choosing among event study approaches. The primary consideration is the research question itself: are you studying market valuation or real outcomes? Secondary considerations include data availability, counterfactual credibility, and whether treatment is staggered.

4.5.2 Common Mistakes

Before looking forward, it is worth flagging the most consequential mistakes that researchers make with event studies. These are not obscure technical issues. They are errors that appear regularly in published work and that you should guard against in your own research.

Running TWFE with staggered adoption and reporting the result as if it were reliable.

After the work of [Goodman-Bacon \(2021\)](#), [Baker et al. \(2022\)](#), and others, there is no longer any excuse for this. If treatment timing varies across firms, you must use modern methods or at least verify that TWFE gives similar answers. As we showed in Section 4.3.5, TWFE can get the sign wrong. At minimum, run the modern estimators as a robustness check.

Treating a flat pre-trends plot as proof that parallel trends holds. Pre-trends testing is necessary but not sufficient. A flat pre-trends plot is encouraging evidence, but it tells you nothing about whether parallel trends would have continued into the post-period. And with small samples, pre-trends tests lack power to detect real violations ([Roth, 2022](#)). Be honest about the limits of what pre-trends testing can establish.

Choosing the event window after examining the data. If you notice that abnormal returns are concentrated two days before the announcement and extend the window to capture them, you have introduced a specification search. Choose the event window before looking at results, and report results for multiple standard windows.

Ignoring clustering. When events cluster in calendar time, standard errors that treat each firm's abnormal return as independent are too small. This is especially common in studies of regulatory changes or macroeconomic announcements that affect many firms on the same date.

Reporting only average effects when heterogeneity is central. An average treatment effect can hide enormous variation. If your question is about which firms are most affected (and in corporate finance, this is almost always the interesting question), the average is a starting point, not the destination.

4.5.3 Connections to Later Chapters

Event studies do not exist in isolation. They connect to, and are enriched by, essentially every other method in this book. Understanding these connections helps you see event

studies as one tool in a larger toolkit rather than as a standalone technique.

Chapter 5: Instrumental variables. When treatment is endogenous (firms choose whether and when to issue debt, adopt a technology, or merge), the DiD event study estimates the ATT for firms that chose to act. But we might want to know what would happen if we *forced* a non-adopter to adopt, which requires extrapolation beyond the treated population. Instrumental variables methods provide one way to do this, by exploiting exogenous variation in the *probability* of treatment. Combining event study designs with instrumental variables (using, say, geographic variation in regulatory exposure as an instrument for adoption timing) can strengthen identification considerably. In the running example, natural disasters might serve as instruments for debt capacity shocks, allowing us to estimate effects beyond the specific firms that happened to be downgraded.

Chapter 6: Difference-in-differences. The economics event study *is* a difference-in-differences design presented in event-time format. Chapter 6 develops the DiD framework in full generality, including the modern estimators that Section 4.3.6 previewed. That chapter provides the complete technical treatment of [Borusyak et al. \(2024\)](#), [Callaway and Sant’Anna \(2021\)](#), and [Sun and Abraham \(2021\)](#), along with implementation code and the decision framework for choosing among them. If your research involves staggered treatment timing, Chapter 6 is where you will find the tools to do it properly.

Chapter 7: Machine learning. The cross-sectional analysis of Section 4.2.6 is a simple form of heterogeneity analysis: regress CAR_i on characteristics to see what predicts larger reactions. Machine learning methods, particularly causal forests ([Wager and Athey, 2018](#)), go much further, discovering nonlinear interactions and unexpected heterogeneity patterns without pre-specification. Chapter 7 introduces these tools, and they apply naturally to both finance event studies (predicting variation in CAR_i) and economics event studies (predicting variation in $\hat{\tau}_{it}$).

Application chapters (Chapters 8–12). Event studies appear throughout the application chapters, but now in service of substantive questions rather than as methodological exercises. Chapter 8 uses both traditions to study how investment responds to financial shocks, synthesizing the running example that has developed across Chapters 1 through 6. Chapter 10 uses finance event studies of debt and equity issuances alongside economics event studies of policy changes to understand capital structure decisions. Chapter 12 applies event studies to credit-supply shocks, including bank failures, regulatory changes,

and disruptions to bank-borrower relationships. Chapter 14 uses them to study CEO turnover, antitakeover provisions, and other governance events. In each case, the two-tradition framework from this chapter applies: What does the market think? What actually happens? Do the answers cohere?

Heterogeneous treatment effects: a research frontier. The treatment effect heterogeneity theme that runs through this chapter (different firms experiencing different effects, the inadequacy of simple averages, the staggered adoption problem as fundamentally a heterogeneity problem) is taken up more fully by more advanced treatments. The conditional average treatment effect framework (CATE; see [Wager and Athey, 2018](#)) and the marginal treatment effect framework (MTE; see [Heckman and Vytlačil, 2005](#)) provide rigorous tools for characterizing heterogeneity and understanding selection into treatment. The event study setting is a natural application of these tools, with leverage shocks and investment serving as a useful applied example.

4.5.4 What This Chapter Has Taught You

This chapter has covered the two major traditions for measuring the impact of corporate events. Rather than presenting them in isolation, we have put them side by side, because the comparison is more instructive than either method alone.

The finance event study, rooted in the efficient market hypothesis, uses stock price reactions to measure the market's assessment of event impacts. Its power lies in the stock market's ability to aggregate information from many investors into a single price. Its vulnerability lies in the joint hypothesis problem (we are always testing both the event's impact and the asset pricing model's validity) and in the question of which investors' reactions we are measuring at which time scale.

The economics event study, rooted in the difference-in-differences framework, uses changes in real outcomes relative to a control group to measure actual behavioral responses to events. Its power lies in the directness of measurement (we see what firms actually do) and in its independence from market efficiency assumptions. Its vulnerability lies in the parallel trends assumption and, when treatment is staggered, in the forbidden comparison problem that can severely bias naive TWFE estimates.

Both traditions share a deeper lesson about causal inference that connects to the book's organizing framework. Neither approach is assumption-free. The finance approach embeds its causal claim in a structural model of pricing. The economics approach bases its causal claim on a research design. In both cases, the quality of the inference depends on

the quality of the assumptions. The difference is in *which* assumptions you are making, and the choice should be driven by which assumptions are most defensible in your specific research context.

For your research, the most important takeaway is this: always be precise about what you are estimating and under what assumptions. The potential outcomes framework from Chapter 1 provides the language to do this. A finance event study estimates the market's assessment of $E[\tau_i \mid \text{event}]$ under the EMH. An economics event study estimates $E[Y_i(1) - Y_i(0) \mid D_i = 1]$ under parallel trends. These are different parameters, identified under different assumptions, answering different questions. Knowing which one you are estimating (and which one you need for your research question) is the mark of a careful researcher.

Chapter 4: Key Takeaways

1. Finance event studies measure market valuations; economics event studies measure real outcomes. They answer different questions and rest on different assumptions.
2. The finance approach embeds causal inference in a structural model (asset pricing); the economics approach uses design-based identification (parallel trends and a control group).
3. Never use naive TWFE with staggered treatment timing. The bias can be severe, including sign reversals. Modern methods from [Borusyak et al. \(2024\)](#), [Callaway and Sant'Anna \(2021\)](#), and [Sun and Abraham \(2021\)](#) are now standard.
4. Pre-trends testing is necessary but not sufficient for validating the parallel trends assumption.
5. Using both approaches on the same question (measuring expectations *and* outcomes) provides the most complete picture and the strongest triangulation of evidence.

Chapter 5

Instrumental Variables

5.1 Why IV?

In Chapter 3 we estimated a leverage regression of the form

$$\text{Leverage}_{it} = \beta \text{Profitability}_{it-1} + \mathbf{X}'_{it-1}\boldsymbol{\gamma} + \alpha_i + \lambda_t + \varepsilon_{it}, \quad (5.1)$$

where α_i and λ_t are firm and year fixed effects, and $\mathbf{X}_{it-1}$ collects controls such as tangibility, the market-to-book ratio, and firm size. The fixed effects removed time-invariant unobservables (management culture, geographic advantages, persistent institutional features), and the coefficient on profitability came in strongly negative. More profitable firms carry less debt, a robust empirical regularity ([Frank and Goyal, 2009](#)).

Should we interpret that coefficient causally? Almost certainly not. The problem is that profitability is *endogenous*: it is tangled up with leverage through channels that fixed effects cannot break. Consider three concrete mechanisms.

Reverse causality. Higher leverage means higher interest expense, which reduces reported profits mechanically. If leverage causes lower profitability at the same time that lower profitability is supposed to cause higher leverage, OLS cannot tell us which direction the arrow runs. The estimated β conflates both directions into a single number.

Simultaneity. A firm receives a valuable investment opportunity. It invests, which raises future profits, and it finances the investment partly with debt, which raises leverage. Both profitability and leverage respond to the same unobserved shock (the arrival of the opportunity), and any regression of one on the other will pick up this common cause.

Measurement error. Corporate finance theory is framed in terms of economic profitability, but what we observe in Compustat is accounting profitability. These are related but not the same thing. Classical measurement error in the regressor biases OLS toward zero. If

the true causal effect is large, OLS will understate it.

Fixed effects help with some of this, but not all. Firm fixed effects absorb time-invariant confounders; they do nothing about the time-varying simultaneity just described. Year fixed effects absorb aggregate shocks; they do nothing about firm-specific reverse causality. As [Roberts and Whited \(2013\)](#) emphasize, fixed effects are not an endogeneity panacea.

The idea behind instrumental variables is simple to state. Find a variable that shifts profitability for reasons that have nothing to do with leverage except through profitability itself. Then use only the variation in profitability that comes from this “clean” source to estimate β . The instrument filters the variation in the endogenous regressor, separating the exogenous component from the contaminated one. Two-stage least squares uses only the filtered component.

That is the promise. The difficulty is that IV rests on assumptions that are strong, largely untestable, and frequently violated. Worse, the parameter that IV estimates (even when everything works) is narrower than most applied researchers realize. It is not the average effect of profitability on leverage for all firms. It is the effect for a particular subpopulation, defined by which firms actually respond to the instrument. This distinction, which we develop carefully in [Section 5.3](#), has profound implications for how we interpret and use IV results.

Before diving into formalism, it is worth positioning IV within the broader framework of this book. In the taxonomy of [Chapter 1](#), IV is the workhorse of the *causal* tradition. The instrument provides quasi-experimental variation, and the credibility of the estimate rests on the research design rather than on a fully specified economic model. But IV also appears inside *structural* models: when we estimate Euler equations by GMM in [Chapter 8](#), the moment conditions are constructed using instruments whose validity depends on the model being correctly specified. That is a fundamentally different kind of claim. This chapter covers the causal use of IV. [Chapter 8](#) revisits the structural use, applied to the same leverage-and-investment question.

5.2 What Is an Instrument?

5.2.1 The problem, in a picture

Start with a picture. We are interested in the causal effect of a treatment D_i on an outcome Y_i . In our running example, D_i is profitability (the endogenous regressor) and Y_i is leverage, but the logic is completely general. The directed acyclic graph (DAG) in [Figure 5.1](#) shows the difficulty. An unobserved variable U_i (managerial quality, investment opportunities, or

anything else we cannot measure) affects both D_i and Y_i . This back-door path from D_i to Y_i through U_i means that the observed association between D_i and Y_i reflects both the causal effect we want and the confounding influence we do not.

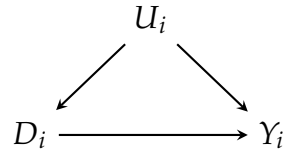


Figure 5.1. The confounding problem. An unobserved variable U_i creates a back-door path between D_i and Y_i . The observed association between D_i and Y_i conflates the causal effect with the confounding influence of U_i .

Now suppose we find a variable Z_i that affects D_i but has no direct effect on Y_i . Figure 5.2 shows what this looks like. The arrow from Z_i to D_i is present; there is no arrow from Z_i to Y_i , and no arrow from U_i to Z_i . The only way Z_i can influence Y_i is *through* D_i . This variable Z_i is an instrumental variable.

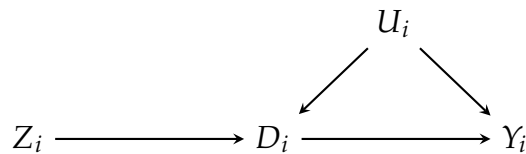


Figure 5.2. The instrumental variables solution. Z_i affects Y_i only through D_i . There is no direct arrow from Z_i to Y_i , and U_i does not influence Z_i .

If you are thinking that the absence of an arrow is a strong assumption, you are right. Figure 5.3 shows what the world might actually look like if the instrument is imperfect. The dashed arrow from Z_i to Y_i represents a *violation* of the exclusion restriction. That is, Z_i affects Y_i through some channel other than D_i . If that arrow exists, even weakly, the IV estimate is biased. We cannot test for this arrow with data alone, because it runs through unobservables. Whether the dashed arrow is absent is an economic argument, not a statistical one.

These three pictures contain almost everything you need to know about instrumental variables at an intuitive level. The rest of this section puts formal content behind each arrow.

5.2.2 The three assumptions

An instrumental variable Z_i must satisfy three conditions. Each has a formal statement, an economic meaning, and characteristic failure modes.

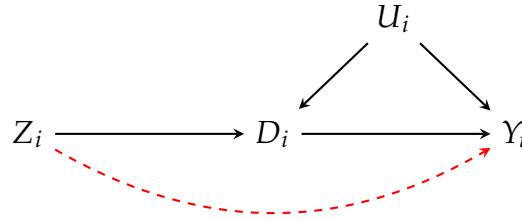


Figure 5.3. An exclusion restriction violation. The dashed red arrow indicates that Z_i affects Y_i through a channel other than D_i . If this arrow is present, the IV estimate is biased. This possibility cannot be ruled out by statistical tests.

Assumption 1: Relevance.

$$\text{Cov}(Z_i, D_i) \neq 0. \quad (5.2)$$

The instrument must be correlated with the endogenous regressor. In potential outcomes notation (see Table 2), relevance requires that $E[D_i | Z_i = 1] \neq E[D_i | Z_i = 0]$ when Z_i is binary, meaning the instrument actually shifts the treatment.

Relevance is the only one of the three assumptions that you can check directly with data. You regress D_i on Z_i (and any controls) and test whether the coefficient on Z_i is statistically different from zero. The F -statistic from this “first stage” regression is the standard diagnostic. We return to the question of how large the F -statistic needs to be in Section 5.6, but the short answer is: larger than you think.

When relevance fails, the instrument does not shift the endogenous variable, and the IV estimator is not identified. More subtly, when relevance is weak (when Z_i is only slightly correlated with D_i), the IV estimator becomes wildly unreliable, with finite-sample bias that can exceed OLS bias. We demonstrate this concretely in Sections 5.4 and 5.5.

Assumption 2: Exclusion restriction.

$$\text{Cov}(Z_i, \varepsilon_i) = 0, \quad (5.3)$$

where ε_i is the structural error in the equation of interest. In potential outcomes language, Z_i affects Y_i only through D_i : $Y_i(d, z) = Y_i(d)$ for all values of z .

This is the assumption that corresponds to the absence of the dashed arrow in Figure 5.3. It cannot be tested with a single instrument, because ε_i is unobservable. Roberts and Whited (2013) offer a useful litmus test: ask yourself whether the instrument is likely to have any direct effect on either the observed outcome or the unobserved part of the outcome. If the answer is yes (if you can tell a plausible story about how Z_i reaches Y_i without going through D_i), the instrument is probably not valid.

Consider a concrete example. Suppose we want to instrument for firm profitability in a

leverage regression. Someone proposes using industry demand shocks as the instrument: when the industry does well, profits are high. But industry demand shocks also affect the firm's investment opportunities, its access to credit markets, the behavior of its competitors, and any number of other things that influence leverage directly. The exclusion restriction is implausible here, not because we can prove it fails, but because we can tell too many stories about why it might.

The best instruments come from settings where the economic reasoning for exclusion is tight. [Bennedsen, Nielsen, Pérez-González, and Wolfenzon \(2007\)](#) study whether family CEO succession hurts firm performance. Family characteristics like size and marital history correlate with the succession decision but may also correlate with firm performance through other channels. Their instrument is the gender of the first-born child of the departing CEO: in some cultures, having a male firstborn makes family succession more likely, and the gender of a child is plausibly unrelated to firm performance through any other channel. The exclusion argument is compelling because it is hard to construct a story in which child gender affects firm profitability except through the succession decision. Hard, but as [Roberts and Whited \(2013\)](#) point out, not impossible: if families with firstborn daughters appoint family CEOs only when the daughter is exceptionally talented, the instrument may exaggerate the negative effect of succession. Even the best instruments are imperfect.

Key Insight

Relevance is testable. Exclusion is not. Two of the three legs of the IV stool rest on economic argument, not statistical evidence. When reading or writing an IV paper, spend most of your critical energy on the exclusion restriction.

Assumption 3: Monotonicity.

$$D_i(1) \geq D_i(0) \quad \text{for all } i, \quad (5.4)$$

where $D_i(z)$ denotes the potential treatment status of unit i when the instrument takes value z . Monotonicity says that the instrument shifts everyone in the same direction (or leaves them unaffected). Nobody is a “defier”, somebody who does the opposite of what the instrument encourages.

To make this concrete, we need the language of compliance types introduced by [Imbens and Angrist \(1994\)](#) and [Angrist, Imbens, and Rubin \(1996\)](#). When the instrument is binary, every unit falls into exactly one of four groups based on how their treatment status responds

to the instrument:

Group	Definition	Behavior
<i>Compliers</i>	$D_i(1) > D_i(0)$	Take treatment when encouraged, don't otherwise
<i>Always-takers</i>	$D_i(1) = D_i(0) = 1$	Take treatment regardless of instrument
<i>Never-takers</i>	$D_i(1) = D_i(0) = 0$	Refuse treatment regardless of instrument
<i>Defiers</i>	$D_i(1) < D_i(0)$	Do opposite of encouragement

Monotonicity rules out defiers. Why does this matter? Because without monotonicity, the instrument could push some firms toward treatment and others away from it, and these opposing effects could cancel in a way that makes the IV estimate uninterpretable. With monotonicity, the only firms whose behavior changes because of the instrument are the compliers, and the IV estimate can be given a clean interpretation as the average treatment effect for this group. We develop this interpretation formally in Section 5.3.

Is monotonicity plausible in corporate finance applications? Often, but not always. If the instrument is an industry-level profitability shock, monotonicity requires that the shock does not cause some firms to increase leverage while causing others to decrease it, holding the effect through profitability constant. If firms respond to the same shock in qualitatively different ways (because they have different financial policies, different levels of financial slack, or different competitive positions), monotonicity may be strained. Unlike exclusion, monotonicity is at least partially testable in some settings ([Kitagawa, 2015](#)), but in most corporate finance applications it is simply assumed.

5.2.3 Supply and demand: where IV began

The idea of instrumental variables did not originate in the program-evaluation tradition of randomized trials with noncompliance. It came from a much older problem: estimating supply and demand curves from market data.

The difficulty is classical. In a market, the observed price P_t and quantity Q_t are equilibrium outcomes, jointly determined by the intersection of supply and demand. A regression of quantity on price gives you neither the demand curve nor the supply curve. It gives you a weighted average of the two, which is not a useful object.

[Wright \(1928\)](#) proposed a solution. If you can find a variable that shifts the supply curve but not the demand curve (weather affecting crop yields, for instance), then the observed variation in equilibrium prices and quantities traces out the demand curve. The supply shifter is an instrument for price in the demand equation.

Write the structural model in logs:

$$\ln Q_t^d = \alpha^d + \beta^d \ln P_t + \varepsilon_t^d \quad (\text{demand}), \quad (5.5)$$

$$\ln Q_t^s = \alpha^s + \beta^s \ln P_t + \gamma^s Z_t + \varepsilon_t^s \quad (\text{supply}), \quad (5.6)$$

where Z_t is a weather variable that enters the supply equation but not the demand equation. In equilibrium, $Q_t^d = Q_t^s = Q_t^{obs}$. Setting the two equations equal and solving for the equilibrium price gives a reduced form in which $\ln P_t^{obs}$ depends on Z_t . Similarly, the equilibrium quantity depends on Z_t . Because Z_t shifts supply but not demand, the ratio of the reduced-form effect of Z_t on $\ln Q_t^{obs}$ to the reduced-form effect of Z_t on $\ln P_t^{obs}$ recovers the demand elasticity β^d .¹

This is exactly the IV estimator. The reduced-form coefficient on Z_t in the quantity equation is the “intention to treat” effect; the reduced-form coefficient on Z_t in the price equation is the first stage. Their ratio is the IV estimate of the structural parameter.

There is a point here that transcends technique. The supply-and-demand example requires a *model* to know what you are identifying. You need to know which equation Z_t enters and which it does not. That knowledge comes from economic reasoning about the structure of the market, not from the data. Wright’s identification argument is model-based: the instrument is valid because the economic model says supply depends on weather and demand does not.

This is worth pausing on, because it connects to the structural/causal theme running through this book. The classical IV story (which is where the entire method originates) is fundamentally structural. It identifies a parameter of a specific economic model (the demand elasticity), and the validity of the instrument depends on the model being correctly specified. The modern design-based use of IV, which dominates the rest of this chapter, flips the emphasis: credibility comes from the research design (the quasi-experimental variation in Z_i), and the model plays a supporting role at best. Both uses are legitimate. The distinction matters because the conditions under which the instrument is “valid” depend on which game you are playing.

With the informal intuition and the three assumptions in hand, we turn in the next section to a question that many textbooks leave until too late: when everything works, what exactly does IV estimate?

¹If instead we had a variable that shifted demand but not supply, the same logic would identify the supply elasticity β^s . The key is that the instrument enters one equation and is excluded from the other.

Key Insight

Instrumental variables originated in structural econometrics, not in the design-based tradition. Wright needed a model of supply and demand to know *which curve* the instrument identifies. Modern IV papers often downplay the role of economic models, but some model is always lurking behind the exclusion restriction. The question is how explicit you want to be about it.

5.3 What Does IV Estimate?

Most applied papers treat the IV estimate as if it recovers “the” causal effect of the endogenous variable, a number that applies to all firms in the sample. This is usually wrong. Under the assumptions of Section 5.2, IV estimates the causal effect for a specific subpopulation, and that subpopulation depends on which instrument you use. Understanding what IV actually identifies (and for whom) is essential for interpreting results and for deciding whether an IV strategy answers the question you care about.

5.3.1 The Wald estimand

Start with the simplest possible case: a binary instrument $Z_i \in \{0, 1\}$, a binary treatment $D_i \in \{0, 1\}$, and no covariates. This setting strips away every complication and exposes the core logic of IV.

Write down the two “reduced form” regressions that relate the instrument to the outcome and the treatment, respectively:

$$\text{Reduced form: } E[Y_i | Z_i = 1] - E[Y_i | Z_i = 0], \quad (5.7)$$

$$\text{First stage: } E[D_i | Z_i = 1] - E[D_i | Z_i = 0]. \quad (5.8)$$

The reduced form (5.7) is the total effect of the instrument on the outcome, what the program-evaluation literature calls the *intention-to-treat* (ITT) effect. It measures what happens to Y_i when you switch the instrument on, without worrying about whether the treatment actually changes. The first stage (5.8) measures how much the instrument shifts the treatment.

The Wald estimand is the ratio:

$$\beta_{\text{Wald}} = \frac{E[Y_i | Z_i = 1] - E[Y_i | Z_i = 0]}{E[D_i | Z_i = 1] - E[D_i | Z_i = 0]} = \frac{\text{reduced form}}{\text{first stage}}. \quad (5.9)$$

This is the IV estimate. The intuition is straightforward: the reduced form tells you the total impact of turning on the instrument, and the first stage tells you how much of that impact is transmitted through the treatment. Dividing one by the other scales the total impact into a per-unit-of-treatment effect.

To make this concrete, imagine a policy that encourages firms to reduce leverage (the instrument). Among encouraged firms, leverage drops by 5 percentage points on average (the first stage). Among encouraged firms, investment rises by 1.5 percentage points on average (the reduced form). The Wald estimate of the effect of leverage on investment is $1.5/5 = 0.30$: each percentage point reduction in leverage raises investment by 0.30 percentage points, for the firms affected by the encouragement.

When the instrument is continuous or there are additional covariates, the Wald ratio generalizes to two-stage least squares (2SLS). In the first stage, you regress D_i on Z_i and any controls X_i , obtaining fitted values $\hat{D}_i$. In the second stage, you regress Y_i on $\hat{D}_i$ and X_i . The coefficient on $\hat{D}_i$ is the 2SLS estimate. With a single instrument and no controls, 2SLS is numerically identical to the Wald ratio. With multiple instruments, 2SLS uses all of them to construct $\hat{D}_i$, weighting each instrument by its predictive power in the first stage.

One thing to notice about equation (5.9): the denominator is the first stage. If the first stage is small (if the instrument barely moves the treatment), you are dividing by a number close to zero. Small perturbations in either the numerator or denominator produce large swings in the ratio. This is the source of the weak instrument problem that we take up in Section 5.6.

5.3.2 Local average treatment effects

What population does the Wald estimand describe? The answer, due to [Imbens and Angrist \(1994\)](#), is one of the most important results in modern econometrics.

Recall the four compliance groups from Section 5.2.2: compliers, always-takers, never-takers, and defiers. Under the three IV assumptions (relevance, exclusion, and monotonicity), the Wald estimand equals the average treatment effect *for compliers only*:

$$\beta_{\text{Wald}} = E[Y_i(1) - Y_i(0) \mid D_i(1) > D_i(0)] \equiv \text{LATE}. \quad (5.10)$$

This is the *local average treatment effect*. The word “local” refers to the subpopulation of compliers, units whose treatment status is shifted by the instrument. The LATE is not the average treatment effect (ATE) for the full population, nor the average treatment effect on the treated (ATT). It is the effect for a group defined entirely by the instrument.

The logic behind equation (5.10) is worth understanding, not just memorizing. Consider

what happens when the instrument switches from $Z_i = 0$ to $Z_i = 1$. Always-takers are treated in both states, so the change in the instrument has no effect on their treatment status and hence no effect on their outcomes (by exclusion). They contribute nothing to either the numerator or the denominator of the Wald ratio. The same is true, in reverse, for never-takers: they are untreated regardless of the instrument. Defiers are ruled out by monotonicity. The only units left are the compliers: they switch from untreated to treated when Z_i moves from 0 to 1. Their change in outcomes shows up in the numerator (the reduced form), and their change in treatment status shows up in the denominator (the first stage). The ratio of the two is their average treatment effect.

This argument reveals something important: *always-takers and never-takers are invisible to IV*. Their treatment effects could be enormous or zero, positive or negative. It makes no difference to the IV estimate. IV sees only the compliers, because they are the only ones the instrument moves.

When does LATE equal ATE?

LATE equals ATE under two circumstances. The first is when the treatment effect is the same for everyone: $Y_i(1) - Y_i(0) = \tau$ for all i . If effects are homogeneous, the complier average equals the population average by definition. This is the classical constant-effects model, and it is what most of the older IV literature implicitly assumed.

The second is when compliers happen to be representative of the population. Even with heterogeneous effects, if the average treatment effect among compliers equals the average treatment effect in the full population, $\text{LATE} = \text{ATE}$. This is a coincidence, not a consequence of the IV assumptions, and there is generally no reason to expect it.

In practice, compliers are almost never representative. The instrument selects them, and selection is typically correlated with the treatment effect. Firms that respond to an instrument encouraging debt reduction are not a random draw from the population of all firms. They are the firms on the margin, the ones whose borrowing capacity or willingness to borrow is sensitive to whatever the instrument measures. Inframarginal firms (which would borrow regardless or abstain regardless) are invisible.

This matters for policy. Suppose a state government considers subsidizing debt reduction to boost investment. An IV estimate based on a similar past policy tells you the investment effect for the firms that responded to the previous policy, the compliers. If the government expands the policy to reach firms that the previous instrument did not move, those newly treated firms may respond very differently. The IV estimate may be a poor guide to the effect of the expanded policy.

Key Insight

IV does not estimate the average effect in the population. It estimates the effect for the subpopulation whose behavior is changed by the instrument. Before trusting an IV estimate, ask: who are the compliers, and do I care about their treatment effect?

LATE in corporate finance: who are the compliers?

In corporate finance, identifying the compliers is often revealing. Consider some examples.

If you instrument for leverage using a credit supply shock (say, a contraction in local bank lending), the compliers are firms that actually adjust their leverage in response to the shock. These are likely smaller, bank-dependent firms without easy access to public debt markets. Large firms with investment-grade ratings can substitute toward bond markets and are unaffected; they are never-takers (from the perspective of leverage reduction). The IV estimate tells you about the leverage–investment relationship for bank-dependent firms, which may or may not generalize to the population.

If you instrument using the gender of the CEO’s firstborn child (as in [Bennedsen et al., 2007](#)), the compliers are families in which the firstborn’s gender actually tips the succession decision. Families with such strong preferences for family succession that they would appoint a family CEO regardless of the firstborn’s gender are always-takers. Families that always hire outside managers are never-takers. The IV estimate applies to the marginal families, those on the fence. If these marginal families are systematically different from the average family firm, the LATE will differ from the ATE.

In both cases, knowing who the compliers are helps you assess whether the IV estimate is informative for the question at hand. If your research question is about the average firm, but your instrument only moves a narrow subpopulation, the IV result may be internally valid yet externally uninformative.

It is worth noting that the LATE is sometimes exactly the right parameter. If the policy question concerns the effect of a specific intervention on the firms that would actually respond to it (for example, whether tightening bank capital requirements reduces lending to the bank-dependent firms whose credit access would actually change), then the complier subpopulation *is* the population of interest, and the LATE answers the question directly.

5.3.3 Beyond the baseline: a preview

The clean identification of LATE in equation (5.10) relies on the simplest possible setting: a binary instrument, a binary treatment, and no covariates. Real corporate finance

applications are messier. Instruments are often continuous. Treatments are rarely binary. Covariates are always present. What does 2SLS estimate in these more realistic settings?

The short answer is: a weighted average of heterogeneous causal effects, where the weights depend on the instrument and may not have a transparent interpretation. [Angrist and Imbens \(1995\)](#) showed that with a multivalued instrument, 2SLS estimates a weighted average of multiple LATEs, one for each pair of adjacent instrument values that shifts some units into treatment. The weights are nonnegative and sum to one, so the estimand is a convex combination of well-defined causal parameters. This is reassuring, but the weights are determined by the distribution of the instrument, not by the researcher's priorities. You are estimating a weighted average, but you did not choose the weights.

Adding covariates creates a subtler problem. In the baseline setting without covariates, the 2SLS estimand and the Wald estimand coincide, and both equal LATE. With covariates, this equivalence breaks down. [Blandhol, Bonney, Mogstad, and Torgovitsky \(2022\)](#) show that the 2SLS estimand with covariates can become a weighted average of causal effects across *all* compliance groups (compliers, always-takers, and never-takers), where the weights can be *negative* and can depend on the distribution of covariates. This "level-dependent" contamination arises because the linear control function in 2SLS does not perfectly absorb covariate variation. Adding a constant to the outcome can change the estimand, which means the number 2SLS produces may not be interpretable as a weighted average of treatment effects at all.

This is a recent and unsettling finding. It does not mean that IV with covariates is useless. Far from it. It means that the comfortable slogan "IV estimates LATE" requires qualification in practice. [Mogstad and Torgovitsky \(2024\)](#) organize the response to this challenge around two ideas. "Reverse engineering" asks: given that I computed 2SLS, what does the number mean? "Forward engineering" asks: what parameter do I want, and how do I use the instrument to estimate it? Forward engineering leads to the marginal treatment effect (MTE) framework, which uses the instrument to trace out how the treatment effect varies across the population. The MTE framework is the subject of Book 2; here, the point is simply that the interpretation of IV estimates is more delicate than the textbook treatment suggests, and awareness of these issues should inform how you read and write IV papers.

Common Mistake

Reporting an IV estimate as “the causal effect of D on Y ” as if it applies to all firms in the sample. In general, it applies to the compliers, firms whose treatment was shifted by the instrument. If effects are heterogeneous, the IV estimate can differ substantially from the ATE. Always discuss who the compliers are and whether the LATE is relevant for your research question.

5.4 Simulated Example: IV When You Know the Truth

The formal results of Sections 5.2 and 5.3 are easier to absorb once you have seen them work (and fail) on actual numbers. This section uses simulated data where we control the data-generating process and know every parameter. We can verify that IV recovers the truth when its assumptions hold, watch it break down when the instrument is weak, and confirm that it estimates the LATE rather than the ATE when treatment effects are heterogeneous.

All results in this section are produced by a single Python script (`ch06_sim.py`), available on the book website. A Stata replication file is also provided. The code is short enough that a PhD student can read it alongside this section and modify it to explore alternative designs.²

5.4.1 OLS bias and IV correction

We generate a cross-section of $N = 5,000$ firms. The data-generating process mirrors the endogeneity problem from Section 5.1: a researcher wants to estimate the effect of profitability on leverage, but an unobservable confounder contaminates the OLS estimate.

The structural model is:

$$D_i = 0.5 Z_i + 0.6 U_i + v_i, \quad v_i \sim N(0, 0.25), \quad (5.11)$$

$$Y_i = \beta D_i + 0.8 U_i + e_i, \quad e_i \sim N(0, 1), \quad (5.12)$$

where D_i is profitability (the endogenous regressor), Y_i is leverage (the outcome), $Z_i \sim N(0, 1)$ is an instrument, and $U_i \sim N(0, 1)$ is an unobserved confounder. The true causal effect is $\beta = -0.15$.

²Code examples in this chapter use Python. Complete Stata replication files for all examples are available on the book website.

The endogeneity is built in: U_i enters both equations with positive coefficients. A firm with high unobserved quality (U_i large) has both higher profitability and higher leverage. This positive confounding pushes the OLS estimate of β upward, away from the true value of -0.15 , toward zero or even positive values. The exclusion restriction holds by construction: Z_i does not appear in equation (5.12).

Table 5.1 reports the results. Three regressions are estimated on the same data. Column (1) is naive OLS, omitting U_i because it is unobservable. Column (2) is 2SLS, instrumenting D_i with Z_i . Column (3) is the “oracle” OLS that includes U_i as a control, infeasible in practice but useful as a benchmark.

Table 5.1. OLS vs. IV Estimation: Strong Instrument

	(1) OLS	(2) 2SLS	(3) Oracle OLS
Profitability (D)	0.3789*** (0.0177)	-0.2132*** (0.0361)	-0.1918*** (0.0197)
First-stage F	—	2165.3	—
Observations	5,000	5,000	5,000
True β	-0.15	-0.15	-0.15

Notes: Heteroskedasticity-robust standard errors in parentheses. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$. Column (1) is naïve OLS omitting the confounder U . Column (2) instruments D with Z . Column (3) is an infeasible “oracle” regression that controls for U .

The results are striking. OLS estimates $\hat{\beta} = +0.379$: not only wrong in magnitude but wrong in *sign*.³ The true effect of profitability on leverage is negative, but the confounding from U_i is so strong that OLS produces a large positive coefficient. A naive researcher would conclude that profitability *increases* leverage, the opposite of the truth.

The 2SLS estimate is $\hat{\beta} = -0.213$, close to the true -0.15 and with the correct sign. The remaining discrepancy from -0.15 is normal sampling variation; the standard error of 0.036 places the true value well within a 95% confidence interval. The first-stage F -statistic is 2,165, far above any reasonable threshold for instrument strength.

Notice that the 2SLS standard error (0.036) is roughly twice the OLS standard error (0.018). This is a general feature: IV estimates are less precise than OLS, even with a strong instrument, because they use only the variation in D_i that comes from Z_i , discarding the rest. You pay for consistency with efficiency. The oracle OLS, which has access to the unobservable U_i , achieves similar accuracy to 2SLS ($\hat{\beta} = -0.192$) with a standard error

³The theoretical probability limit under this DGP is $\text{plim } \hat{\beta}_{\text{OLS}} = \beta + 0.8 \times \text{Cov}(D_i, U_i) / \text{Var}(D_i) \approx +0.408$. The estimate of $+0.379$ reflects normal finite-sample variation around this asymptotic value. Readers who compute the asymptotic bias as a check should expect the simulation average (over many replications) to be close to $+0.408$, not to the single-draw estimate of $+0.379$.

of 0.020. This shows that if we could observe the confounder, we would not need the instrument.

Key Insight

OLS bias from an omitted confounder can be severe enough to flip the sign of the estimated effect. IV corrects the bias when the instrument is strong and the exclusion restriction holds, but the price is wider standard errors.

5.4.2 What happens with a weak instrument

Now we repeat the exercise with the same structural model, but weaken the instrument. The first-stage coefficient on Z_i drops from 0.5 to 0.05. The instrument barely moves the endogenous regressor. Everything else remains identical.

Table 5.2 reports the results. Column (1) is OLS on the weak-instrument data, Column (2) is 2SLS with the weak instrument, and Column (3) reproduces the strong-instrument 2SLS from Table 5.1 for comparison.

Table 5.2. Weak vs. Strong Instrument

	(1) OLS	(2) 2SLS weak Z	(3) 2SLS strong Z
Profitability (D)	0.6573*** (0.0198)	-0.0044 (0.5563)	-0.2132*** (0.0361)
First-stage F	—	8.1	2165.3
2SLS 95% CI	—	[-1.095, 1.086]	[-0.284, -0.142]
Anderson–Rubin 95% CI	—	[-2.240, 1.000]	[-0.285, -0.145]
Observations	5,000	5,000	5,000
True β	-0.15	-0.15	-0.15

Notes: Heteroskedasticity-robust standard errors in parentheses. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$. Column (1) is OLS on the weak-instrument dataset. Column (2) instruments D with a weak Z (first-stage coefficient = 0.05). Column (3) reproduces the strong-instrument 2SLS from Table 5.1. The Anderson–Rubin confidence set is robust to weak instruments.

The weak-instrument 2SLS estimate is -0.004 , with a standard error of 0.556. The point estimate is essentially zero, neither close to the true $\beta = -0.15$ nor economically meaningful. The first-stage F -statistic is 8.1, below the [Stock and Yogo \(2005\)](#) threshold of 10 and far below the [Pflueger and Wang \(2015\)](#) threshold of 23 for heteroskedasticity-robust inference.

Look at the confidence intervals. The conventional 2SLS 95% CI is $[-1.10, 1.09]$. It is wide, but still finite and centered on the (wrong) point estimate. A researcher who

relied on this interval would conclude that the true effect could be anywhere from strongly negative to strongly positive, which is at least honest. But the interval is constructed using an asymptotic approximation that is unreliable when the first stage is weak: it is centered on a biased point estimate and may have poor coverage.

The Anderson–Rubin (AR) 95% confidence set is $[-2.24, 1.00]$, even wider. Unlike the conventional 2SLS interval, the AR set is valid regardless of instrument strength. It does not rely on the first stage being strong (Anderson and Rubin, 1949). The true value of -0.15 lies inside the AR set, as it should. The price of weak-instrument-robust inference is that the interval is enormous: it tells you the instrument is too weak to learn much of anything.

Figure 5.4 makes this contrast visual. Each panel plots the AR F -statistic as a function of a candidate value β_0 . The shaded region below the 5% critical value (3.84) is the AR confidence set. With a strong instrument (left panel), the AR statistic rises steeply away from the true β , producing a narrow confidence set. With a weak instrument (right panel), the curve is nearly flat, and the confidence set covers most of the parameter space. The instrument simply does not contain enough information to pin down the causal effect.

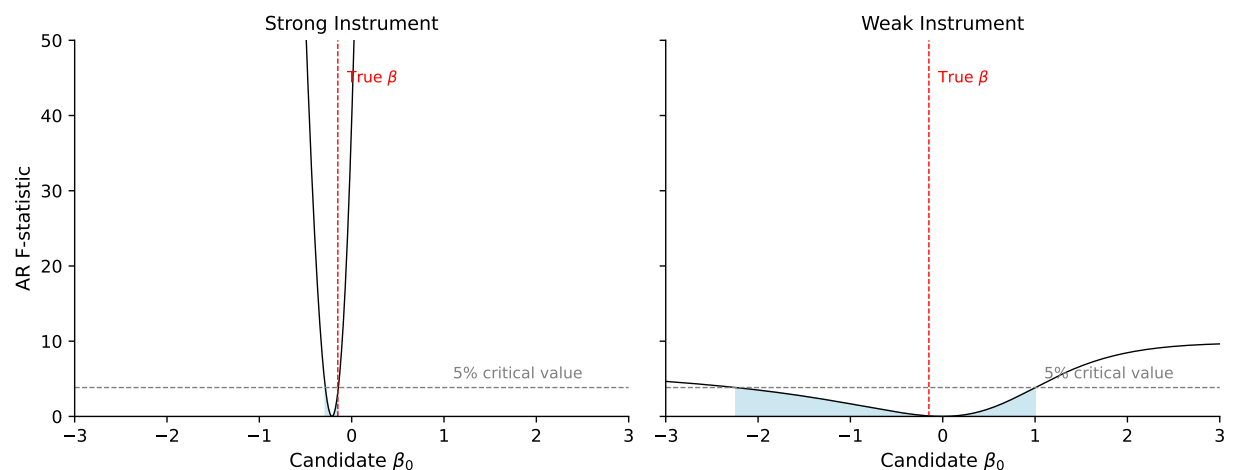


Figure 5.4. Anderson–Rubin confidence sets. Each panel plots the AR F -statistic against candidate values of β_0 . The shaded region where the F -statistic falls below 3.84 is the 95% confidence set. *Left:* Strong instrument, tight confidence set centered near the true $\beta = -0.15$. *Right:* Weak instrument, the confidence set spans most of the parameter space, reflecting the instrument’s inability to identify the causal effect.

Common Mistake

Reporting 2SLS results with a first-stage F below 10 as if they were reliable. When the instrument is weak, 2SLS point estimates are biased toward OLS, standard errors are misleading, and confidence intervals have poor coverage. Always report the first-stage F with an appropriate robustness correction, and use Anderson–Rubin confidence sets when there is any doubt about instrument strength. Section 5.6 develops these diagnostics in detail.

5.4.3 LATE is not ATE

The first two exercises used a continuous treatment and continuous instrument, where the LATE interpretation is less natural. To demonstrate the LATE result from Section 5.3 directly, we switch to a binary instrument and binary treatment.

The setup mirrors the numerical example in Appendix 5.10. We generate $N = 5,000$ firms divided into three compliance groups: always-takers (30%), compliers (20%), and never-takers (50%). Each group has a different treatment effect:

Group	Share	Treatment effect (τ_i)
Always-takers	30%	+5.0
Compliers	20%	+3.0
Never-takers	50%	−1.0

The population ATE is $0.30 \times 5 + 0.20 \times 3 + 0.50 \times (-1) = 1.60$. The complier average treatment effect (the LATE) is 3.0. These are very different numbers. The LATE theorem predicts that IV will estimate the complier effect, not the population average.

The instrument $Z_i \sim \text{Bernoulli}(0.5)$ is randomly assigned, independent of everything. Exclusion holds by construction: Z_i affects the outcome only by changing treatment status. Monotonicity holds because there are no defiers. All three IV assumptions are satisfied.

Table 5.3 reports the results.

Panel A shows the true parameters. The population ATE is 1.652 (close to the theoretical 1.60; the small discrepancy is finite-sample variation). The complier ATE is exactly 3.000, as designed. The always-taker and never-taker effects are 5 and −1, respectively. The complier share is 0.196, close to the target of 0.20.

Panel B shows the estimates. The IV/Wald estimate is 3.130, close to the complier ATE of 3.0 and far from the population ATE of 1.65. The Wald arithmetic is transparent: the reduced form (the average difference in outcomes between $Z = 1$ and $Z = 0$ firms) is

Table 5.3. LATE vs. ATE: Heterogeneous Treatment Effects

<i>Panel A: True Parameters</i>	
Population ATE	1.652
Complier ATE (LATE)	3.000
Always-taker effect	5.000
Never-taker effect	-1.000
Complier share (π_C)	0.196
<i>Panel B: Estimates</i>	
OLS	4.5343*** (0.0455)
IV / Wald	3.1298*** (0.2415)
Wald numerator (reduced form)	0.616
Wald denominator (first stage)	0.197
<i>N</i>	5,000

Notes: Heteroskedasticity-robust standard errors in parentheses. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$. The IV/Wald estimator recovers the complier average treatment effect (LATE), not the population ATE. Treatment effects differ across compliance groups: always-takers ($\tau = 5$), compliers ($\tau = 3$), never-takers ($\tau = -1$).

0.616, and the first stage (the average difference in treatment rates) is 0.197. Dividing gives $0.616/0.197 = 3.13$.

The OLS estimate is 4.534, which exceeds even the largest group-level treatment effect. This happens because always-takers are disproportionately treated. They are always treated, and their high average outcomes pull the OLS coefficient upward. OLS conflates the treatment effect with selection: firms that choose treatment (the always-takers) have systematically different outcomes than those that do not, even in the absence of any causal effect.

The key lesson is that IV and OLS are estimating fundamentally different quantities, and neither estimates the population ATE. OLS estimates a selection-contaminated association. IV estimates the LATE, which is the causal effect for compliers only. The always-taker effect of +5 and the never-taker effect of -1 are completely invisible to the IV estimator, just as the LATE theorem predicts.

If you want the population ATE rather than the LATE, IV alone cannot give it to you. You need additional assumptions or additional methods, either a structural model that extrapolates from compliers to the full population, or the marginal treatment effect (MTE) framework that uses the instrument to trace out how the effect varies across subpopulations. These methods are the subject of Book 2.

The three exercises in this section establish the basic facts. IV works when the instrument is strong and assumptions hold, but it produces noisy estimates. It fails badly when the instrument is weak. And even when it works, it answers a narrower question than most researchers realize. With these lessons in hand, we turn to real data, where none of the assumptions can be verified by construction.

5.5 Compustat Example: IV When You Don't Know the Truth

In Section 5.4 we controlled the data-generating process and could verify that IV recovered the right answer. Real research is not like that. Nobody hands you the true β or tells you which firms are compliers. This section applies IV to the same leverage regression from Chapter 3, using Compustat data, with two candidate instruments. Neither instrument works cleanly. The failures are instructive: they illustrate two different ways IV can go wrong, and both are common in published corporate finance research.

The baseline specification is the [Frank and Goyal \(2009\)](#) market leverage regression with firm and year fixed effects:

$$\text{mLEV}_{it} = \beta \text{Profitability}_{it-1} + \mathbf{X}'_{it-1} \boldsymbol{\gamma} + \alpha_i + \lambda_t + \varepsilon_{it}, \quad (5.13)$$

where $\mathbf{X}_{it-1}$ includes industry median leverage, the market-to-book ratio, tangibility, and log assets, all lagged one year. Profitability is the endogenous regressor. The sample covers 1971–2020 and contains approximately 180,000 firm-year observations after standard exclusions.⁴

5.5.1 A weak instrument: industry median profitability

The first instrument is the leave-one-out median profitability of peer firms in the same industry and year. The economic logic is straightforward: when your industry peers are profitable (because of shared demand conditions, input cost movements, or technology trends), your own profitability tends to be higher too. The leave-one-out construction ensures that the firm's own profitability does not enter the instrument mechanically.

⁴Regulated industries, financial firms, government entities, foreign-incorporated firms, firms with assets below \$1 million, and observations with missing market values are excluded, following [Frank and Goyal \(2009\)](#). Industries are defined at the 2-digit SIC level. Code and processed data are available on the book website.

The exclusion restriction requires that peer profitability affects your leverage *only* through your own profitability. This is debatable. Industry profitability conditions may shift lender willingness to extend credit to firms in that industry, may signal growth opportunities that affect investment and hence leverage, or may move competitor behavior in ways that influence your financing. We will return to these concerns after seeing the results.

Table 5.4 reports the estimates.

Table 5.4. IV Estimation: Industry Median Profitability as Instrument

	(1) OLS	(2) 2SLS
Profitability _{t-1}	-0.1495*** (0.0221)	-0.2856*** (0.1098)
IndMedianLev _{t-1}	0.1575*** (0.0171)	0.1596*** (0.0171)
(M/B) _{t-1}	-0.0187*** (0.0024)	-0.0191*** (0.0023)
Tangibility _{t-1}	0.1893*** (0.0098)	0.1770*** (0.0131)
ln(Assets) _{t-1}	0.0398*** (0.0035)	0.0459*** (0.0065)
First-stage F	—	5.4
First-stage coeff. on IV	—	0.3943** (0.1701)
Anderson–Rubin 95% CI	—	[-1.185, -0.105]
Firm FE	Yes	Yes
Year FE	Yes	Yes
Clustered SE	Industry	Industry
Observations	179,795	179,795

Notes: Dependent variable is market leverage (mLEV_t). All right-hand-side variables are lagged one year. The instrument for profitability is the leave-one-out industry median profitability of peer firms (2-digit SIC industry, same year). Standard errors (in parentheses) are clustered at the 2-digit SIC industry level. Firm and year fixed effects are absorbed via iterative demeaning. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Column (1) is OLS. The coefficient on profitability is -0.150 , consistent with the well-established negative relationship between profitability and leverage (Frank and Goyal, 2009).⁵

Column (2) is 2SLS, instrumenting profitability with the leave-one-out industry median. The IV estimate is -0.286 , roughly twice the OLS estimate in absolute value. The direction

⁵The coefficient is slightly more negative than the -0.105 reported by Frank and Goyal (2009) in their Table 2.8, reflecting minor differences in industry classification (2-digit SIC vs. FF48) and sample vintage.

makes economic sense: if unobserved investment opportunities raise both profitability and leverage, OLS is biased toward zero, and IV pushes the coefficient further negative.

But look at the first-stage F -statistic: 5.4. This is below the [Stock and Yogo \(2005\)](#) threshold of 10 and far below the [Pflueger and Wang \(2015\)](#) threshold of 23. The instrument is weak. The 2SLS point estimate of -0.286 is unreliable. It could be severely biased, and the standard error of 0.110 understates the true uncertainty.

Why is the instrument weak despite the strong cross-sectional correlation between industry profitability and own profitability? The answer is firm fixed effects. The industry profitability instrument derives most of its power from cross-sectional variation: some industries are persistently more profitable than others. Firm fixed effects absorb this variation entirely, because a firm that stays in the same industry has the same average peer profitability in every year. What remains is the *within-firm deviation* from the firm mean in peer profitability, net of aggregate year effects. That residual variation is much thinner, and the first stage collapses.

This is a general problem, not specific to this instrument. Many IV strategies in corporate finance rely on industry-level or region-level variation that is substantially absorbed by firm fixed effects. If your identification comes from the cross-section but your specification demands within-firm variation, the instrument may be much weaker than you expect.

The Anderson–Rubin (AR) confidence interval is $[-1.19, -0.11]$. Unlike the conventional 2SLS interval, the AR set is valid regardless of instrument strength. It excludes zero, providing some evidence that the causal effect of profitability on leverage is negative. But it is wide: the effect could be anywhere from -0.11 to -1.19 . A researcher reporting the 2SLS point estimate of -0.286 without the AR interval would convey false precision.

Common Mistake

Proposing an instrument based on its cross-sectional correlation with the endogenous regressor, then including firm fixed effects in the specification. Fixed effects absorb precisely the variation the instrument exploits. Always check the first-stage F in the specification you actually estimate, not in a pooled regression without fixed effects.

5.5.2 A strong but likely invalid instrument: tax loss carryforward

The second instrument is an indicator for whether the firm has a positive tax loss carryforward (TLCF). Firms with accumulated tax losses tend to have recently experienced poor profitability, so the TLCF indicator predicts current profitability within firm over time: when the indicator switches on, profitability is typically low.

Table 5.5 reports the results.

Table 5.5. Weak vs. Strong Instrument: Tax Loss Carryforward

	(1) OLS	(2) 2SLS TLCF	(3) 2SLS IndMedianProfit
Profitability _{<i>t</i>-1}	-0.1495*** (0.0221)	-0.8665*** (0.0929)	-0.2856*** (0.1098)
First-stage <i>F</i>	—	231.9	5.4
Anderson–Rubin 95% CI	—	[-1.065, -0.695]	[-1.185, -0.105]
Firm FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Clustered SE	Industry	Industry	Industry
Observations	179,795	179,795	179,795

Notes: Dependent variable is market leverage ($mLEV_t$). Controls (not shown) are the same as Table 5.4: lagged industry median leverage, market-to-book, tangibility, and log assets. Column (2) instruments profitability with an indicator for positive tax loss carryforward. Column (3) reproduces the industry median profitability IV from Table 5.4. Standard errors clustered at the 2-digit SIC industry level. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

The first-stage *F* is 231.9, an enormously strong first stage. The TLCF indicator is a powerful predictor of profitability within firm. A naive application of the “check the *F*-statistic” rule would declare this instrument excellent.

But the 2SLS coefficient is -0.867 , almost six times the OLS estimate. The AR confidence interval $[-1.07, -0.70]$ is tight and excludes both zero and the OLS estimate. Something is wrong. An effect this large would imply that a one-standard-deviation increase in profitability reduces market leverage by roughly 15 percentage points, an implausibly large response for the average firm.

The problem is almost certainly a violation of the exclusion restriction. The TLCF indicator affects leverage through many channels beyond profitability. Firms with tax losses have lower marginal tax rates, reducing the tax incentive to borrow. Tax losses signal financial distress, which tightens credit terms and may trigger covenant violations. Firms with tax losses may reduce investment, affecting leverage through the asset side. Each of these channels creates a direct path from the instrument to the outcome that bypasses profitability. The 2SLS estimate attributes all of these effects to profitability, inflating the coefficient.

This illustrates the lesson from Section 5.2.2 in its starkest form: a strong first stage is *necessary but not sufficient* for a good IV strategy. The *F*-statistic tells you whether the instrument predicts the endogenous variable. It tells you nothing about whether the instrument satisfies the exclusion restriction. Roberts and Whited (2013) put it well: many

corporate finance IV papers discuss relevance at length but spend one sentence on exclusion. The TLCF example shows why that imbalance is dangerous.

Key Insight

Two things can go wrong with an IV strategy: the instrument can be weak (first stage too small), or the instrument can be invalid (exclusion restriction violated). A large first-stage F rules out the first problem but is silent about the second. The most dangerous instruments are those that are strong but invalid. They produce precise estimates of the wrong parameter.

The two Compustat exercises in this section deliver a sobering message. The industry profitability instrument seemed plausible but turned out to be weak after absorbing fixed effects. The tax loss carryforward indicator was strong but almost certainly invalid. Neither produced an IV estimate we would trust. This is not a failure of the method. It is a realistic picture of what happens when you try to do IV in corporate finance. Good instruments are rare. The rest of the chapter develops the tools for diagnosing bad ones and for making the most of the good ones when you find them.

5.6 When Can You Trust 2SLS?

Sections 5.4 and 5.5 showed what happens when instruments are weak or invalid. This section steps back and asks a broader question: under what conditions can you trust the output of a 2SLS regression? The answer is more demanding than most textbooks suggest. The problems extend well beyond weak instruments to include a systematic distortion in 2SLS standard errors that persists even when the first stage is strong.

5.6.1 Why weak instruments are dangerous

The 2SLS estimator is a ratio. In the just-identified case with one instrument and no controls, it takes the simple form

$$\hat{\beta}_{2SLS} = \frac{\hat{\delta}}{\hat{\pi}}, \quad (5.14)$$

where $\hat{\delta}$ is the reduced-form coefficient (the effect of Z_i on Y_i) and $\hat{\pi}$ is the first-stage coefficient (the effect of Z_i on D_i). If $\hat{\pi}$ is close to zero, you are dividing by a small, noisy number. Small perturbations in $\hat{\pi}$ produce large swings in $\hat{\beta}_{2SLS}$.

The finite-sample bias of 2SLS points toward the OLS estimate. A useful heuristic, building on [Stock and Yogo \(2005\)](#), is that the *bias* of 2SLS relative to the bias of OLS is approximately

$$\frac{E(\hat{\beta}_{2SLS} - \beta)}{E(\hat{\beta}_{OLS} - \beta)} \approx \frac{1}{F + 1}, \quad (5.15)$$

where F is the first-stage F -statistic. This is an approximation that holds under specific distributional conditions and should be treated as a guide to intuition rather than an exact result.⁶ The key insight: when F is large, the scaling is small and 2SLS is nearly unbiased. When F is small (say, below 10) a substantial fraction of the OLS bias survives. When $F = 0$, IV collapses to OLS entirely.

Equally important, the *distribution* of $\hat{\beta}_{2SLS}$ is far from normal when the first stage is weak. The ratio of two approximately normal random variables is not itself normal, especially when the denominator hovers near zero. Standard confidence intervals based on $\hat{\beta} \pm 1.96 \hat{\sigma}$ have poor coverage: they are too narrow and centered on a biased point estimate.

5.6.2 Diagnosing weakness: escalating thresholds

The standard diagnostic is the first-stage F -statistic. But how large does it need to be? The answer has evolved over two decades, with each revision raising the bar.

Table 5.6 summarizes the four main thresholds in the literature. Crucially, they are not competing answers to the same question. Each addresses a different aspect of weak-instrument reliability.

Table 5.6. Escalating first-stage F thresholds and what each controls

Reference	Threshold	What it controls
Stock and Yogo (2005)	$F \geq 10$	2SLS bias $\leq 10\%$ of OLS bias, under homoskedastic i.i.d. errors
Pflueger and Wang (2015)	$F_{\text{eff}} \geq 23$	Same bias bound, robust to heteroskedasticity and autocorrelation (just-identified case)
Keane and Neal (2024)	$\hat{F} \geq 50$	High probability that 2SLS is closer to the truth than OLS
Lee, McCrary, Moreira, and Porter (2022)	$F \geq 104$	Correct size of the 2SLS t -test (just-identified case)

⁶The approximation is derived from the concentration parameter framework of [Stock and Yogo \(2005\)](#). It requires homoskedastic errors, a single endogenous regressor, and a first stage that is well-approximated by the local-to-zero asymptotic embedding. In finite samples, the actual bias depends on the joint distribution of the first-stage and structural errors.

[Stock and Yogo \(2005\)](#) proposed the most widely used threshold: $F \geq 10$. Their logic is that if F exceeds a critical value tabulated under the assumption of homoskedastic, serially uncorrelated errors, the 2SLS bias is at most 10% of the OLS bias, or the size distortion of a 5% test is at most 15%. The $F \geq 10$ rule of thumb comes from their tabulated critical values for a single endogenous regressor.

The problem is the assumption. Corporate finance panel data almost always has heteroskedasticity. It is usually clustered by firm or industry. It often exhibits serial correlation within firm. Under these conditions, the Stock–Yogo critical values are invalid. [Pflueger and Wang \(2015\)](#) develop a test that permits heteroskedasticity and autocorrelation. Their effective F -statistic replaces the standard first-stage F with a heteroskedasticity-robust version, and their critical values are correspondingly higher: in typical just-identified applications, the effective F should be compared to a critical value around 23 for a 10% maximal bias bound.⁷

[Lee et al. \(2022\)](#) push the threshold further. They show that even in the just-identified case, standard 2SLS t -tests have incorrect size unless the first-stage F is very large. They propose a t -ratio adjustment that effectively requires $F \geq 104$ for reliable inference. At this threshold, most instruments in corporate finance would fail.

[Keane and Neal \(2024\)](#) go further still, arguing that the question should not be “is the F above some threshold?” but rather “is 2SLS likely to produce an estimate closer to the truth than OLS?” Using simulations, they show that for F in the 10–20 range, there is a high probability that OLS is actually closer to the true parameter than 2SLS, unless endogeneity is very severe. They recommend $\hat{F} \geq 50$ as a minimum threshold for confidence that 2SLS will outperform OLS.

The escalation is worth pausing on. The original $F \geq 10$ was a rough rule of thumb derived under strong distributional assumptions. Each refinement relaxes an assumption unrealistic for corporate finance data and finds that the required F is substantially higher. The practical implication: if your first-stage F is in the range of 10–25, you are in a gray zone where conventional 2SLS inference is unreliable.

⁷The exact critical value depends on the number of instruments and the desired maximal bias. For the just-identified case with one endogenous regressor and a 10% maximal bias threshold, the critical value is approximately 23. With multiple instruments or different bias tolerances, the values change; see [Pflueger and Wang’s \(2015\)](#) tables.

Common Mistake

Reporting the first-stage F , noting it exceeds 10, and proceeding as if the instrument is strong. With heteroskedastic, clustered data (the norm in corporate finance), an F of 10 is not enough. Use the heteroskedasticity-robust effective F of [Pflueger and Wang \(2015\)](#) and compare it to their higher critical values. Better yet, report Anderson–Rubin confidence sets regardless of the F .

5.6.3 The power asymmetry: a problem even with strong instruments

The discussion so far has focused on what happens when the first stage is weak. [Keane and Neal \(2024\)](#) identify a separate problem that persists even when instruments are strong: the 2SLS t -test suffers from a *power asymmetry*.

The mechanics are as follows. The 2SLS standard error depends on the residual variance $\hat{\sigma}_{2SLS}^2$, which is a quadratic function of $\hat{\beta}_{2SLS}$ minimized at the OLS estimate. In samples where $\hat{\beta}_{2SLS}$ happens to land near $\hat{\beta}_{OLS}$ (because finite-sample correlation between the instrument and the structural error shifts the estimate toward OLS), the standard error is artificially small. In samples where the estimate is far from OLS, the standard error is inflated. This creates a systematic association: 2SLS estimates that are most shifted toward OLS appear most precisely estimated.

The consequence for inference is severe. The t -test has inflated power to declare estimates significant when they happen to be near OLS (where the standard error is too small), and almost no power to detect true effects that are far from OLS (where the standard error is too large). If the true effect is zero but OLS is biased, the t -test will preferentially flag the samples where the IV estimate has been pulled toward the (wrong) OLS value. This is not a weak instrument problem. [Keane and Neal \(2024\)](#) show that the power asymmetry remains severe even at $F = 29$ ($\hat{F} = 50$), well above conventional thresholds.

The implication is striking: the 2SLS t -test should not be trusted as the primary inference tool *even when the instrument is strong*. This is a stronger claim than the usual advice to “use AR when instruments are weak.” The power asymmetry makes the t -test unreliable across the board.

5.6.4 Anderson–Rubin: the default test

Fortunately, there is a simple solution: the Anderson–Rubin (AR) test ([Anderson and Rubin, 1949](#)). In the just-identified case, the AR test of the specific null $H_0: \beta = 0$ reduces to a t -test from regressing the outcome Y_i on the first-stage fitted value $\hat{D}_i = \hat{\pi}Z_i$ ([Keane](#)

and Neal, 2024).⁸

To construct an AR confidence interval for an arbitrary value of β , test a grid of candidate values β_0 . For each β_0 , construct

$$R_i(\beta_0) = Y_i - \beta_0 D_i, \quad (5.16)$$

regress $R_i(\beta_0)$ on Z_i (and any controls), and record the F -statistic on Z_i . The AR 95% confidence set is the collection of β_0 values for which the F -statistic does not exceed 3.84 (the $\chi^2(1)$ critical value at the 5% level).

The AR test has two properties that make it the right default. First, it is valid regardless of instrument strength: even if the first stage is zero, the AR confidence set has correct coverage. Second, it largely avoids the power asymmetry. The reason is that the AR residual variance $\hat{\sigma}_{AR}^2$ is minimized at $\hat{\beta}_{2SLS}$ rather than at $\hat{\beta}_{OLS}$. This breaks the mechanical link between estimates near OLS and artificially small standard errors: the AR standard error is smallest precisely where the 2SLS estimate actually lands, not where OLS happens to sit (Keane and Neal, 2024).

Figure 5.4 in Section 5.4 showed the AR confidence set visually. With a strong instrument, the AR interval was narrow and nearly identical to the conventional 2SLS interval. With a weak instrument, the AR set was wide, covering most of the parameter space. In Section 5.5, the industry profitability instrument produced an AR interval of $[-1.19, -0.11]$, wide but informative, and far more trustworthy than the conventional 2SLS standard error suggested.

With heteroskedastic or clustered data, the AR test is implemented using a robust t -test on $\hat{D}_i$ in the second-stage regression, exactly the same robustness adjustments you would apply to any OLS regression. The computational cost is negligible. In the overidentified case (multiple instruments), the conditional likelihood ratio (CLR) test of Moreira (2003) generalizes the AR approach and is available in standard software.

⁸Ironically, this is the “naive” second-stage regression that textbooks warn students to avoid. It does not give correct asymptotic standard errors for $\hat{\beta}_{2SLS}$, but it does give the optimal test for significance. See Keane and Neal (2024), Section 5.

Key Insight

Always report the Anderson–Rubin confidence interval alongside the 2SLS estimate, regardless of instrument strength. The AR test is valid when instruments are weak, and it avoids the power asymmetry that makes the standard 2SLS t -test unreliable even when instruments are strong. If the AR interval is much wider than the 2SLS interval, the conventional standard errors are not to be trusted.

5.6.5 Evidence from published research

These are not theoretical curiosities. Two recent studies audit large samples of published IV papers and find that the problems described above are pervasive.

Keane and Neal (2024) re-examine 49 replicable IV papers published in the *American Economic Review* between 2011 and 2023 that had first-stage F below 50 or not reported. In 24% of these papers, replacing the standard t -test with the AR test overturns a key result, a finding that was significant by the t -test becomes insignificant under AR. The pattern is systematic: the reversals occur precisely when the 2SLS estimate is close to OLS, which is where the power asymmetry predicts the t -test will generate spuriously small standard errors.

Young (2022) studies 1,309 IV regressions across 30 papers in AEA journals using Monte Carlo simulations, the jackknife, and the bootstrap. He finds that non-iid errors in highly leveraged regressions (both ubiquitous features of published work) inflate the apparent significance of first- and second-stage results. In the average paper, 39% of results that are significant at the 5% level can be rendered insignificant by deleting a single cluster. First-stage F -statistic pretests based on the Stock and Yogo (2005) thresholds are largely uninformative in these non-iid settings. Most strikingly, despite producing point estimates that frequently differ from OLS in sign or magnitude, 2SLS rarely rejects the OLS point estimate: the IV confidence intervals are wide enough to encompass OLS in the vast majority of cases.

The combined message from these audits is sobering. A substantial fraction of published IV results in top journals may be less informative than they appear. The tools developed in this section (the heteroskedasticity-robust F , the AR confidence interval, and a healthy skepticism about the t -test) are not optional refinements. They are necessary safeguards.

A final caution. The discussion above focuses on *diagnosing* problems after the fact. There is a deeper problem with using the first-stage F as a pretesting device: deciding whether to report IV or OLS based on the strength of the first stage distorts the inference

for the reported estimator ([Andrews, Stock, and Sun, 2019](#)). If you chose IV because the F was above some threshold and would have reported OLS if it were below, the effective distribution of your estimator is a mixture of the IV and OLS distributions, conditional on the threshold being crossed. This selection invalidates the standard errors for both. The AR approach sidesteps the pretesting problem because it does not require a threshold: it provides valid inference for any instrument strength.

5.7 Shift-Share (Bartik) Instruments

The instruments considered so far (an industry profitability measure, a tax loss carryforward indicator) were single variables proposed on economic grounds and evaluated one at a time. A different approach, widely used in applied microeconomics and increasingly common in corporate finance, constructs the instrument by combining two ingredients: pre-existing *shares* that measure differential exposure across units, and aggregate *shocks* that are common to all units but affect them differently because of those shares. These are called *shift-share* or *Bartik* instruments, after [Bartik \(1991\)](#).

This section explains how shift-share instruments work, what assumptions justify them, and what can go wrong. The formal properties of these instruments were clarified only recently, by [Goldsmith-Pinkham, Sorkin, and Swift \(2020\)](#) and [Borusyak, Hull, and Jaravel \(2022\)](#). [Borusyak, Hull, and Jaravel \(2025\)](#) provide a practical guide to implementation.

5.7.1 The structure

Consider a regression at the level of local markets (regions, firms, counties), indexed by l :

$$y_l = \beta x_l + \mathbf{D}_l' \gamma + \varepsilon_l, \quad (5.17)$$

where x_l is endogenous and $\mathbf{D}_l$ are controls. The classic example is [Autor, Dorn, and Hanson \(2013\)](#): y_l is manufacturing employment growth in commuting zone l , and x_l is the growth in import exposure to China. Import exposure is endogenous because local demand shocks affect both employment and imports.

The Bartik instrument exploits an accounting decomposition. Each local market l has a pre-existing industry composition described by shares z_{lk} (the fraction of employment in industry k), and each industry k experiences a national-level shock g_k (growth in imports

from China in industry k). The Bartik instrument is the share-weighted sum of the shocks:

$$B_l = \sum_{k=1}^K z_{lk} g_k. \quad (5.18)$$

The instrument varies across local markets because their industry compositions differ. A region concentrated in furniture manufacturing is more exposed to Chinese import competition than a region concentrated in software, even though the industry-level shocks g_k are common to everyone.

The same structure appears in many settings. In immigration research, z_{lk} is the historical share of immigrants from country k living in location l , and g_k is the national growth in immigration from country k (Card, 2001). In corporate finance, Greenstone, Mas, and Nguyen (2020) use the share of lending in region l from bank k as the exposure, and bank-level supply shocks as the aggregate component. In each case, the instrument interacts pre-existing local exposure with aggregate variation that is arguably exogenous to local conditions.

5.7.2 What makes it valid?

The Bartik instrument B_l is constructed from two ingredients: shares z_{lk} and shocks g_k . The validity of the instrument depends on which ingredient you are willing to treat as exogenous. This turns out to be a more subtle question than it first appears, and confusion about it plagued the early literature. Two distinct identification strategies have been formalized.

Strategy 1: Exogenous shares. Goldsmith-Pinkham et al. (2020) show that 2SLS with the Bartik instrument is numerically equivalent to a GMM estimator that uses the industry shares z_{lk} themselves as instruments (with a particular weighting matrix). Under this interpretation, identification comes from the shares. The key assumption is that, conditional on controls, the shares are uncorrelated with the structural error ε_l .

This is analogous to a difference-in-differences design. The shares measure differential *exposure* to the aggregate shocks, and the identifying assumption is that locations with different industry compositions would have experienced parallel trends in the absence of the shocks. The shocks g_k affect the *strength* of the instrument (its relevance) but play no direct role in identification.

The shares-based strategy is natural when there are only a few industries (so the shocks cannot average out) but the researcher believes that initial industry composition is plausibly

exogenous to subsequent outcomes, for example, because it was determined decades ago. The threats are the same as in any DiD design: pre-existing trends correlated with exposure, or confounders that load on the same shares.

Strategy 2: Exogenous shocks. [Borusyak et al. \(2022\)](#) take the opposite view. They treat the shares as given (potentially endogenous) and assume that the industry-level shocks g_k are as good as randomly assigned, conditional on observables. Under this interpretation, the identifying variation comes from the randomness of the shocks, and consistency requires a large number of industries K so that the contribution of any single shock to the instrument averages out.

This is a design-based strategy, closer in spirit to a standard IV argument. The assumption is that industries are hit by shocks (a surge in Chinese exports of furniture, a collapse in demand for steel) that are orthogonal to local labor market conditions, and there are enough such shocks that any one of them has a negligible influence on the instrument. The shares determine how the shocks map into local variation, but the identification comes from the shocks being as good as random.

The shocks-based strategy is natural in the China trade setting: the rise of Chinese manufacturing plausibly reflects supply-side developments within China (technology adoption, deregulation, WTO accession) rather than demand-side conditions in specific U.S. commuting zones. [Autor et al. \(2013\)](#) motivate their instrument this way, using import growth to other high-income countries (not the U.S.) as the shock to purge any U.S.-specific demand component.

5.7.3 When each strategy is appropriate

The two strategies are not interchangeable. They require different assumptions, different diagnostics, and different standard errors.

Under the shares strategy, the researcher should test for pre-trends (do locations with higher exposure show different outcomes before the shock?), examine robustness to controlling for share-correlated confounders, and cluster standard errors at the level of the local market. Under the shocks strategy, the researcher should test for balance (do industries that experience larger shocks look different on observables?), ensure that no single shock dominates the instrument, and use exposure-robust standard errors that account for correlation induced by the shares ([Borusyak et al., 2025](#)).

In practice, many papers invoke elements of both strategies without clearly committing to either. [Goldsmith-Pinkham et al. \(2020\)](#) note that [Autor et al. \(2013\)](#) use lagged shares

(natural under shares exogeneity) but motivate the instrument with the randomness of Chinese supply shocks (natural under shock exogeneity). This ambiguity was common before the formal frameworks were developed. A researcher using shift-share instruments today should be explicit about which ingredient provides identification and conduct the appropriate diagnostics.

Key Insight

A shift-share instrument combines shares (differential exposure) and shocks (aggregate variation). Validity can rest on either ingredient being exogenous, but the two strategies have different assumptions and different diagnostics. The researcher must decide which story is more credible for the application at hand and test accordingly.

5.7.4 Shift-share in corporate finance

Shift-share designs appear in corporate finance in several forms. The most direct example is [Greenstone et al. \(2020\)](#), who study the effect of credit supply on real outcomes. They use each county's pre-crisis exposure to different banks (the shares) interacted with bank-level changes in lending (the shocks) to construct a Bartik instrument for local credit conditions. The identification argument is that bank-level supply shocks (driven by exposure to subprime losses in other markets) are exogenous to local demand in any particular county.

More broadly, any instrument that interacts a firm-level or region-level characteristic with an aggregate shock has a shift-share flavor, even if the authors do not call it that. A regression that instruments a firm's credit conditions with $\text{Tangibility}_i \times \Delta \text{CreditSpread}_t$ is implicitly a shift-share design: tangibility is the share (measuring exposure to credit market conditions), and the credit spread change is the shock. The identification question is whether tangibility is exogenous to the firm-level outcome (shares strategy) or whether credit spread changes are randomly assigned (shocks strategy).

The general lesson: whenever you see an interaction between a cross-sectional characteristic and a time-series shock used as an instrument, ask yourself which ingredient is doing the identifying work, and whether the assumptions required for that ingredient are plausible. The frameworks of [Goldsmith-Pinkham et al. \(2020\)](#) and [Borusyak et al. \(2022\)](#) provide the tools to answer this question rigorously.

Common Mistake

Arguing that a shift-share instrument is valid because “the shocks are aggregate and therefore exogenous to local conditions.” This conflates the two identification strategies. Aggregate shocks may not be randomly assigned across industries (the shocks strategy requires this), and even if they are, the shares through which they enter must not be correlated with the outcome for other reasons (the shares strategy requires this). You need a specific argument, not a generic appeal to aggregation.

5.8 A Practical Guide to IV in Corporate Finance

This chapter has covered a lot of ground. This section distills it into a checklist for reading and writing IV papers.

5.8.1 Before you estimate

1. **State the endogeneity problem.** What is the endogenous regressor? Which of the three sources of endogeneity (reverse causality, omitted variables, measurement error) is the concern? Be specific.
2. **Propose the instrument and state the economic argument.** Why does Z predict D ? What is the economic channel? Can you draw the DAG?
3. **Defend the exclusion restriction.** Why does Z *not* affect Y except through D ? List the most plausible alternative channels and explain why you believe they are absent or controlled for. This is the most important paragraph in any IV paper.
4. **Think about who the compliers are.** If the instrument is binary or has a natural threshold, what firms change their treatment status in response to the instrument? Is the LATE for these compliers informative for the research question, or is the ATE what you need?
5. **Worry about your specification.** Will you include firm fixed effects? Year fixed effects? If so, does the instrument have enough within-firm variation to survive? Section 5.5.1 showed how fixed effects can collapse the first stage.

5.8.2 When you estimate

1. **Report the first stage in full.** Show the coefficient on the instrument, its standard error, and the heteroskedasticity-robust first-stage F -statistic. Do not report a non-robust F

- and compare it to the [Stock and Yogo \(2005\)](#) threshold of 10. Use the [Pflueger and Wang \(2015\)](#) effective F and their critical values.
2. **Report the Anderson–Rubin confidence interval.** Always, regardless of the first-stage F . If the AR interval is much wider than the conventional 2SLS interval, the latter is not trustworthy. If the AR interval is unbounded ($\hat{F} < 3.84$), you do not have enough information to run IV.
 3. **Report OLS alongside IV.** The comparison between OLS and 2SLS is informative: it tells you the direction and magnitude of the endogeneity correction. If the two are very similar, either endogeneity is not a serious problem for this coefficient, or the instrument is too weak to detect it.
 4. **Discuss magnitudes, not just significance.** Is the IV estimate economically plausible? The TLCF instrument in Section 5.5.2 produced a highly significant estimate that was implausibly large. A strong first stage and a small p -value do not guarantee a credible result.
 5. **Consider the reduced form.** The regression of Y on Z (and controls) estimates $\pi\beta$, the product of the first-stage and structural coefficients. If this is insignificant, the IV estimate of β will be imprecise regardless of the first stage. Many researchers find the reduced form more transparent than the 2SLS output.
 6. **Beware the forbidden regression.** If the first stage is nonlinear (for example, a probit or logit model of a binary endogenous regressor), do not use the predicted probabilities as fitted values in a linear second stage. This “forbidden regression” produces inconsistent estimates because the nonlinear first stage generates fitted values that are not valid instruments for the second-stage linear model. Use a linear first stage, or use control-function methods that explicitly account for the nonlinearity.

5.8.3 Common pitfalls

1. **Using lagged dependent variables as instruments.** Lagged values of the endogenous regressor are correlated with it (relevance is easy) but are almost never excludable: any persistent unobservable that drives both D and Y will also be correlated with lagged D .
2. **Over-identification without justification.** Using multiple instruments increases power but makes each additional exclusion restriction harder to defend. In corporate finance, most credible IV papers are just-identified ([Angrist and Pischke, 2009](#)).
3. **Instrument shopping.** Trying many candidate instruments and reporting the one that “works” (strong first stage, significant second stage) is a form of specification search that invalidates inference. Pre-commit to the instrument before seeing the second-stage

results.

4. **Ignoring the LATE.** Reporting the IV estimate as “the causal effect” without discussing who the compliers are. If the instrument shifts behavior only for a narrow subpopulation, the estimated effect may not generalize.

5.9 IV in the Structural and Causal Landscape

Where does IV sit in the organizing framework of this book?

In the causal (design-based) tradition, IV is the primary tool for dealing with endogeneity when randomization is not available. The researcher identifies a source of exogenous variation in the treatment, uses it to isolate the causal component of the treatment’s effect, and interprets the result as a LATE for the complier population. The emphasis is on transparent identification: can you explain, in plain language, why the instrument shifts the treatment and why it does not directly affect the outcome? The design-based view treats the exclusion restriction as the central credibility question and is skeptical of instruments whose economic logic is unclear.

In the structural tradition, IV has a different and older role. It originates in the estimation of simultaneous equations, the supply and demand system of Section 5.2.3. The instruments are not “natural experiments” but rather variables that shift one equation (supply) without entering the other (demand). The exclusion restriction comes from the economic model: the model says that weather affects supply but not demand, so weather is an instrument. The researcher commits to a specific parameterization of the model and uses IV to estimate its parameters. The interpretation is structural: the estimated coefficient is a parameter of the demand curve, not a LATE for a subpopulation of compliers.

Both traditions use the same estimator (2SLS) and require the same assumptions (relevance, exclusion, monotonicity for LATE). The difference is in where the assumptions come from and how the result is interpreted. Design-based researchers derive exclusion from quasi-experimental variation and interpret the result as a local causal effect. Structural researchers derive exclusion from an economic model and interpret the result as a model parameter.

The tension between these views is productive. Design-based researchers push for more transparent instruments and more careful attention to who the compliers are. Structural researchers push for more explicit models that can extrapolate beyond the complier population and address policy counterfactuals. The marginal treatment effect (MTE) framework, the subject of Book 2, bridges the two by using the instrument to trace out the full distribution of treatment effects across the population, nesting LATE as a special case

and structural parameters as targets of extrapolation.

For the corporate finance researcher reading the literature, the practical implication is simple: when you encounter an IV paper, ask two questions. First, is the exclusion restriction credible? This is the design-based question, and it applies regardless of whether the paper has a structural model. Second, what parameter is being estimated? This is the structural question, and it applies regardless of how clean the instrument is. A credible instrument that estimates an uninteresting parameter is not useful. An interesting parameter estimated with a dubious instrument is not credible. The best IV papers satisfy both.

5.10 Exercises

Conceptual

1. A researcher proposes to instrument a firm's leverage with its CEO's age, arguing that older CEOs are more conservative and prefer less debt.
 - (a) Draw the DAG for this IV strategy, including at least one plausible path through which CEO age could affect the outcome (investment) directly.
 - (b) State the exclusion restriction in words. Do you find it credible? Why or why not?
 - (c) If the exclusion restriction fails, what is the direction of bias in the IV estimate relative to the true causal effect?
2. Explain the difference between the LATE and the ATE. Give a corporate finance example where the two are likely to differ substantially, and explain which one is more relevant for a specific policy question of your choice.

Computational

3. Using the simulated data from Section 5.4 (`ch06_sim_strong.csv`):
 - (a) Estimate OLS and 2SLS. Verify that your estimates match Table 5.1.
 - (b) Compute the Anderson–Rubin confidence interval by grid search over $\beta_0 \in [-1, 1]$ with step size 0.01.
 - (c) Plot the AR F -statistic as a function of β_0 and shade the 95% confidence set.
4. Using the weak-instrument data (`ch06_sim_weak.csv`):
 - (a) Estimate 2SLS and report the first-stage F .
 - (b) Compute the AR confidence interval. Compare it to the conventional 2SLS confidence interval.

- (c) Now multiply the instrument Z by 10 (making it stronger) and re-estimate. How do the AR and conventional intervals change? At what instrument strength do they converge?
5. Using the LATE data (`ch06_sim_late.csv`):
 - (a) Compute the Wald estimand by hand (ratio of reduced form to first stage).
 - (b) Compute the true ATE, the true complier ATE, and the true always-taker and never-taker effects using the latent variables in the dataset.
 - (c) Verify that the Wald estimand approximately equals the complier ATE and not the population ATE.
 6. Download the processed Compustat sample from the book website (`compustat_iv_sample.csv`).
 - (a) Reproduce the OLS and 2SLS estimates from Table 5.4 using the leave-one-out industry median profitability as the instrument.
 - (b) Compute the AR confidence interval. Does it contain zero?
 - (c) Replace the instrument with the TLCF indicator and re-estimate. Compare the first-stage F , the 2SLS estimate, and the AR confidence interval to the industry profitability results. Which instrument do you trust more, and why?

Applied

7. Read [Greenstone et al. \(2020\)](#). Identify the endogenous regressor, the instrument, and the exclusion restriction. Is this a shift-share instrument? If so, does the identification rely on exogenous shares or exogenous shocks? What are the most plausible threats to the exclusion restriction?
8. Find a recent paper in a top corporate finance journal that uses an instrumental variables strategy. Write a one-page evaluation covering: (a) the economic argument for the instrument, (b) the plausibility of the exclusion restriction, (c) whether the first-stage diagnostics are adequate by the standards of Section 5.6, and (d) whether the authors discuss who the compliers are and what the LATE means in their context.

Appendix: Proof of the LATE Theorem

This appendix proves that the Wald estimand equals the average treatment effect for compliers, the result stated without proof in Section 5.3.2. The proof follows [Imbens and Angrist \(1994\)](#) and the exposition in [Angrist et al. \(1996\)](#), but at a slower pace and with commentary explaining what each step accomplishes. No result beyond the law of iterated expectations is required.

Setup and notation

Consider a population of units indexed by i . Each unit has:

- A binary instrument $Z_i \in \{0, 1\}$.
- A binary treatment $D_i \in \{0, 1\}$.
- An observed outcome Y_i .
- Two *potential treatments*: $D_i(1)$ and $D_i(0)$, representing the treatment status unit i would have under $Z_i = 1$ and $Z_i = 0$, respectively.
- Two *potential outcomes*: $Y_i(1)$ and $Y_i(0)$, representing the outcome unit i would have under $D_i = 1$ and $D_i = 0$, respectively.

The observed treatment and outcome are determined by the instrument through

$$D_i = D_i(Z_i) = Z_i \cdot D_i(1) + (1 - Z_i) \cdot D_i(0), \quad (5.19)$$

and by the treatment through

$$Y_i = Y_i(D_i) = D_i \cdot Y_i(1) + (1 - D_i) \cdot Y_i(0). \quad (5.20)$$

Note that we have already embedded the **exclusion restriction** in equation (5.20): potential outcomes depend on D_i but not on Z_i directly. A more explicit notation would write $Y_i(d, z)$ and then assume $Y_i(d, z) = Y_i(d)$ for all z . We skip this extra notation to reduce clutter, but keep in mind that exclusion is doing real work here.

Assumptions

The proof uses three assumptions, restated here for reference.

A1. Independence (Exogeneity). The instrument is independent of all potential outcomes and potential treatments:

$$Z_i \perp\!\!\!\perp (Y_i(1), Y_i(0), D_i(1), D_i(0)). \quad (5.21)$$

This is stronger than the $\text{Cov}(Z_i, \varepsilon_i) = 0$ condition in Section 5.2.2. It says the instrument is as good as randomly assigned.⁹

A2. Relevance (First stage).

$$E[D_i \mid Z_i = 1] \neq E[D_i \mid Z_i = 0]. \quad (5.22)$$

⁹The proof goes through under the weaker condition of *mean independence*: $E[Y_i(d) \mid Z_i] = E[Y_i(d)]$ for $d \in \{0, 1\}$ and $\Pr(G_i = g \mid Z_i) = \Pr(G_i = g)$ for all compliance groups g . Full statistical independence is assumed here for expositional simplicity.

The instrument shifts the treatment. Without this, the Wald ratio has a zero denominator.

A3. Monotonicity (No defiers).

$$D_i(1) \geq D_i(0) \quad \text{for all } i. \quad (5.23)$$

No unit does the opposite of what the instrument encourages.

Compliance groups

Since $D_i(0)$ and $D_i(1)$ are each binary, every unit belongs to exactly one of four groups:

Group G_i	$D_i(0)$	$D_i(1)$	Description
Compliers (C)	0	1	Respond to encouragement
Always-takers (AT)	1	1	Always treated
Never-takers (NT)	0	0	Never treated
Defiers (DF)	1	0	Respond perversely

Monotonicity (A3) rules out defiers: $\Pr(G_i = \text{DF}) = 0$. Let π_C , π_{AT} , and π_{NT} denote the population shares of the remaining three groups. Note that $\pi_C + \pi_{AT} + \pi_{NT} = 1$.

The object of interest

The Wald estimand is

$$\beta_{\text{Wald}} = \frac{E[Y_i | Z_i = 1] - E[Y_i | Z_i = 0]}{E[D_i | Z_i = 1] - E[D_i | Z_i = 0]}. \quad (5.24)$$

The goal is to show that $\beta_{\text{Wald}} = E[Y_i(1) - Y_i(0) | G_i = C]$, the average treatment effect for compliers. We work on the numerator and denominator separately.

Step 1: Decompose the denominator (the first stage)

Begin with $E[D_i | Z_i = 1]$. By the law of iterated expectations, conditioning on the compliance group:

$$E[D_i | Z_i = 1] = \sum_{g \in \{C, AT, NT\}} E[D_i | Z_i = 1, G_i = g] \Pr(G_i = g | Z_i = 1). \quad (5.25)$$

Now use independence (A1). Since G_i is a function of $(D_i(0), D_i(1))$, independence implies $Z_i \perp\!\!\!\perp G_i$, so $\Pr(G_i = g \mid Z_i = 1) = \Pr(G_i = g) = \pi_g$. Also, when $Z_i = 1$, the observed treatment is $D_i = D_i(1)$, which is determined entirely by the group membership:

$$E[D_i \mid Z_i = 1, G_i = g] = E[D_i(1) \mid G_i = g] = \begin{cases} 1 & \text{if } g = C \text{ (since } D_i(1) = 1), \\ 1 & \text{if } g = AT \text{ (since } D_i(1) = 1), \\ 0 & \text{if } g = NT \text{ (since } D_i(1) = 0). \end{cases}$$

So

$$E[D_i \mid Z_i = 1] = 1 \cdot \pi_C + 1 \cdot \pi_{AT} + 0 \cdot \pi_{NT} = \pi_C + \pi_{AT}. \quad (5.26)$$

The identical argument with $Z_i = 0$ gives $D_i = D_i(0)$, so

$$E[D_i \mid Z_i = 0, G_i = g] = E[D_i(0) \mid G_i = g] = \begin{cases} 0 & \text{if } g = C, \\ 1 & \text{if } g = AT, \\ 0 & \text{if } g = NT, \end{cases}$$

and therefore

$$E[D_i \mid Z_i = 0] = 0 \cdot \pi_C + 1 \cdot \pi_{AT} + 0 \cdot \pi_{NT} = \pi_{AT}. \quad (5.27)$$

Subtracting:

$$\boxed{E[D_i \mid Z_i = 1] - E[D_i \mid Z_i = 0] = \pi_C.} \quad (5.28)$$

Interpretation. The first stage equals the share of compliers in the population. This makes intuitive sense: the instrument shifts treatment only for compliers, and by exactly one unit for each of them. Always-takers and never-takers cancel because their treatment status does not change with the instrument. The relevance assumption (A2) requires $\pi_C > 0$.

Step 2: Decompose the numerator (the reduced form)

This step is slightly more involved because the outcome depends on which potential outcome is realized, which in turn depends on the treatment, which depends on both the instrument and the compliance group.

Start with $E[Y_i \mid Z_i = 1]$. Again condition on the group:

$$E[Y_i \mid Z_i = 1] = \sum_g E[Y_i \mid Z_i = 1, G_i = g] \pi_g. \quad (5.29)$$

When $Z_i = 1$, the observed outcome is $Y_i = D_i(1) Y_i(1) + (1 - D_i(1)) Y_i(0)$. For each group:

$$g = C : D_i(1) = 1, \text{ so } Y_i = Y_i(1), \text{ and } E[Y_i | Z_i = 1, G_i = C] = E[Y_i(1) | G_i = C];$$

$$g = AT : D_i(1) = 1, \text{ so } Y_i = Y_i(1), \text{ and } E[Y_i | Z_i = 1, G_i = AT] = E[Y_i(1) | G_i = AT];$$

$$g = NT : D_i(1) = 0, \text{ so } Y_i = Y_i(0), \text{ and } E[Y_i | Z_i = 1, G_i = NT] = E[Y_i(0) | G_i = NT].$$

In the second and third lines, we used independence (A1) to drop the conditioning on Z_i : since $Z_i \perp\!\!\!\perp (Y_i(1), Y_i(0), G_i)$, the conditional expectation of a potential outcome given Z_i and G_i reduces to the expectation given G_i alone. The same reasoning was used implicitly in Step 1.

Substituting:

$$E[Y_i | Z_i = 1] = E[Y_i(1) | C] \pi_C + E[Y_i(1) | AT] \pi_{AT} + E[Y_i(0) | NT] \pi_{NT}, \quad (5.30)$$

where we abbreviate $G_i = C$ as simply C , and so on.

Now repeat for $Z_i = 0$. When $Z_i = 0$, the observed outcome is $Y_i = D_i(0) Y_i(1) + (1 - D_i(0)) Y_i(0)$. For each group:

$$g = C : D_i(0) = 0, \text{ so } Y_i = Y_i(0), \text{ and the expectation is } E[Y_i(0) | C];$$

$$g = AT : D_i(0) = 1, \text{ so } Y_i = Y_i(1), \text{ and the expectation is } E[Y_i(1) | AT];$$

$$g = NT : D_i(0) = 0, \text{ so } Y_i = Y_i(0), \text{ and the expectation is } E[Y_i(0) | NT].$$

So:

$$E[Y_i | Z_i = 0] = E[Y_i(0) | C] \pi_C + E[Y_i(1) | AT] \pi_{AT} + E[Y_i(0) | NT] \pi_{NT}. \quad (5.31)$$

The key cancellation. Now subtract (5.31) from (5.30):

$$\begin{aligned} E[Y_i | Z_i = 1] - E[Y_i | Z_i = 0] &= (E[Y_i(1) | C] - E[Y_i(0) | C]) \pi_C \\ &\quad + (E[Y_i(1) | AT] - E[Y_i(1) | AT]) \pi_{AT} \\ &\quad + (E[Y_i(0) | NT] - E[Y_i(0) | NT]) \pi_{NT}. \end{aligned} \quad (5.32)$$

The second and third lines are exactly zero. Always-takers have $Y_i(1)$ under both $Z_i = 0$ and $Z_i = 1$, because they are always treated regardless of the instrument. Their outcomes cancel. Never-takers have $Y_i(0)$ in both states, so their outcomes cancel too. This is the core

of the argument. We are left with:

$$E[Y_i | Z_i = 1] - E[Y_i | Z_i = 0] = E[Y_i(1) - Y_i(0) | C] \cdot \pi_C. \quad (5.33)$$

Interpretation. The reduced form equals the complier average treatment effect, multiplied by the share of compliers. The treatment effects of always-takers and never-takers are invisible, not because they are zero, but because the instrument does not change which potential outcome is observed for these groups. Exclusion ensures the instrument has no direct effect on outcomes; monotonicity ensures there are no defiers to contaminate the difference.

Step 3: Take the ratio

Dividing the numerator (5.33) by the denominator (5.28):

$$\beta_{\text{Wald}} = \frac{E[Y_i(1) - Y_i(0) | C] \cdot \pi_C}{\pi_C} = E[Y_i(1) - Y_i(0) | G_i = C] \equiv \text{LATE}. \quad (5.34)$$

The complier share π_C cancels. The Wald estimand equals the average treatment effect for compliers. This completes the proof. $\square$

What each assumption contributed

It is useful to look back and identify exactly where each assumption was used.

Independence (A1)

Used in every step to replace $\Pr(G_i = g | Z_i = z)$ with $\Pr(G_i = g) = \pi_g$ and to replace $E[Y_i(d) | Z_i = z, G_i = g]$ with $E[Y_i(d) | G_i = g]$. Without independence, the group shares and potential outcome means could differ between the $Z_i = 1$ and $Z_i = 0$ populations, and nothing would cancel.

Exclusion

Embedded in the notation: Y_i depends on D_i but not on Z_i directly (equation 5.20). Without exclusion, the instrument would have a direct effect on outcomes, and the always-taker and never-taker terms in (5.32) would no longer cancel. Even though their treatment status does not change, their outcomes would change because Z_i affects Y_i through a channel other than D_i .

Relevance (A2)

Ensures $\pi_C > 0$, so the denominator is nonzero and the ratio is well-defined.

Monotonicity (A3)

Eliminates defiers ($\pi_{DF} = 0$). Without monotonicity, the numerator in (5.32) would include an additional term $(E[Y_i(0) | DF] - E[Y_i(1) | DF])\pi_{DF}$ (note the reversed sign: defiers switch from treated to untreated when Z_i goes from 0 to 1). The denominator would become $\pi_C - \pi_{DF}$ rather than π_C . The Wald estimand would then be a mixture of complier and defier effects with no clean interpretation.

A numerical example

A small numerical example helps cement the ideas. Suppose a population of 1,000 firms is divided as follows:

Group	Share	$D_i(0)$	$D_i(1)$	Treatment effect $Y_i(1) - Y_i(0)$
Compliers	200 (20%)	0	1	+3 on average
Always-takers	300 (30%)	1	1	+5 on average
Never-takers	500 (50%)	0	0	-1 on average

Set the potential outcome levels at $E[Y_i(0) | C] = 10$, $E[Y_i(1) | C] = 13$ (so the complier effect is $13 - 10 = 3$); $E[Y_i(1) | AT] = 20$; $E[Y_i(0) | NT] = 6$.

Under $Z_i = 1$ (all firms are “encouraged”):

- Compliers: treated, observe $Y_i(1)$. Mean outcome = 13.
- Always-takers: treated, observe $Y_i(1)$. Mean outcome = 20.
- Never-takers: untreated, observe $Y_i(0)$. Mean outcome = 6.

$$E[Y_i | Z_i = 1] = 0.20 \times 13 + 0.30 \times 20 + 0.50 \times 6 = 2.6 + 6.0 + 3.0 = 11.6.$$

Under $Z_i = 0$ (no firm is encouraged):

- Compliers: untreated, observe $Y_i(0)$. Mean outcome = 10.
- Always-takers: treated, observe $Y_i(1)$. Mean outcome = 20.
- Never-takers: untreated, observe $Y_i(0)$. Mean outcome = 6.

$$E[Y_i | Z_i = 0] = 0.20 \times 10 + 0.30 \times 20 + 0.50 \times 6 = 2.0 + 6.0 + 3.0 = 11.0.$$

The reduced form: $E[Y_i | Z_i = 1] - E[Y_i | Z_i = 0] = 11.6 - 11.0 = 0.6$.

The first stage: $E[D_i | Z_i = 1] - E[D_i | Z_i = 0] = (0.20 + 0.30) - 0.30 = 0.20 = \pi_C$.

The Wald estimand: $\beta_{\text{Wald}} = 0.6/0.20 = 3.0$.

This equals the complier average treatment effect (+3), not the population ATE. The population ATE is $0.20 \times 3 + 0.30 \times 5 + 0.50 \times (-1) = 0.6 + 1.5 - 0.5 = 1.6$. The always-taker effect (+5) and the never-taker effect (-1) are completely invisible to the Wald estimand. IV sees only the compliers.

Chapter 6

Difference-in-Differences

Chapter overview. This chapter introduces difference-in-differences (DiD), one of the most widely used causal inference methods in empirical corporate finance. [Baker et al. \(2022\)](#) report that 431 DiD papers were published in the top five finance journals between 2000 and 2019. The method’s appeal is straightforward: when a policy change affects some firms but not others, the comparison of changes in outcomes across these groups can identify the policy’s causal effect under assumptions that are often plausible and partially testable.

The chapter proceeds in five sections. This section develops the core logic using the simplest possible setting (one treatment group, one control group, two time periods). Section [6.2](#) extends the framework to multiple time periods, event study plots, and the subtle question of how to assess the plausibility of the identifying assumptions. Section [6.3](#) addresses the staggered adoption problem, which arises when units enter treatment at different dates, the default setting in corporate finance. Section [6.4](#) introduces the modern estimators that resolve this problem, with working code. Section [6.5](#) collects practical guidance on inference, common pitfalls, and the connection back to the structural/causal framework.

6.1 Motivation and the Basic 2×2 DiD

6.1.1 The problem

In the Identification chapter, we introduced the fundamental problem of causal inference: for each unit, we observe only one potential outcome. In Chapter [3](#), we saw that panel data with unit fixed effects can absorb time-invariant unobservables, and in Chapter [5](#)

that instrumental variables can isolate exogenous variation in an endogenous treatment. Difference-in-differences offers a different route. It exploits the *timing* of a policy change rather than an external instrument, and it requires an assumption about *trends* rather than an exclusion restriction.

The idea is best introduced with a concrete problem. Suppose a state legislature passes a law that strengthens the rights of secured creditors. We want to know: does this change cause firms incorporated in that state to increase their leverage? The question is important for understanding how legal institutions shape corporate financial policy, and it has a direct analog in many settings (banking deregulation, governance reforms, tax changes, environmental regulation).

The difficulty is familiar. Firms in states that adopt creditor-friendly legislation may differ systematically from firms in other states. Perhaps the law was passed *because* the state's firms were already highly leveraged and creditors demanded more protection. Or perhaps the states that adopt are systematically richer, more business-friendly, or home to different industries. A simple comparison of leverage across states at a point in time confounds the effect of the law with all of these pre-existing differences.

Running Example 1: Leverage and Investment

Consider the passage of state-level creditor rights legislation. In 1987, Arizona passed an anti-takeover statute; at that time, Connecticut had not passed similar legislation (Bertrand and Mullainathan, 2004; Roberts and Whited, 2013). Firms incorporated in Arizona form the treatment group; firms incorporated in Connecticut form the control group. The outcome of interest is leverage, measured as total debt over total assets. A cross-sectional comparison after the law change produces the difference

$$E(Y_i | D_i = 1) - E(Y_i | D_i = 0),$$

where Y_i is leverage and $D_i = 1$ indicates incorporation in the treated state. As we showed in the Structural and Causal Approaches chapter, this difference equals the *ATT* *plus* the selection bias:

$$\underbrace{E[Y_i(1) - Y_i(0) | D_i = 1]}_{ATT} + \underbrace{E[Y_i(0) | D_i = 1] - E[Y_i(0) | D_i = 0]}_{\text{selection bias}}.$$

If firms in Arizona would have had different leverage even without the law (because of industry composition, firm size, growth opportunities, or any other permanent

characteristic), the selection bias is nonzero and the cross-sectional comparison is not informative about the causal effect.

Difference-in-differences resolves this problem by using the *change* in outcomes rather than the *level*. The logic is disarmingly simple: if the selection bias is constant over time, then differencing it away removes it.

6.1.2 The intuition: two differences

Suppose we observe both groups for one period before the law change and one period after. The first difference eliminates time-invariant factors; the second eliminates common shocks. Neither difference alone is sufficient, but combined they can identify the treatment effect.

The single *time-series* difference for the treated group,

$$E(Y_{i1} | D_i = 1) - E(Y_{i0} | D_i = 1),$$

captures the change in leverage for Arizona firms around the law change. But this change reflects not only the law's effect but also any economy-wide trends that would have moved leverage anyway (for instance, a business cycle downturn that tightened credit conditions for all firms).

The single *cross-sectional* difference in the post-period,

$$E(Y_{i1} | D_i = 1) - E(Y_{i1} | D_i = 0),$$

captures the gap in leverage between Arizona and Connecticut firms after the law. But this gap reflects both the law's effect and any pre-existing differences in leverage across the two states.

The DiD estimate subtracts the control group's change from the treated group's change:

$$\begin{aligned} \widehat{\tau}^{DiD} = & \left[E(Y_{i1} | D_i = 1) - E(Y_{i0} | D_i = 1) \right] \\ & - \left[E(Y_{i1} | D_i = 0) - E(Y_{i0} | D_i = 0) \right]. \end{aligned} \quad (6.1)$$

The first bracket is the change for the treated group; the second bracket is the change for the control group. By subtracting the control group's change, we remove any common time trend. And by looking at changes rather than levels, we remove any permanent difference between the groups. What remains (if the identifying assumptions hold) is the causal effect

of the treatment on the treated.

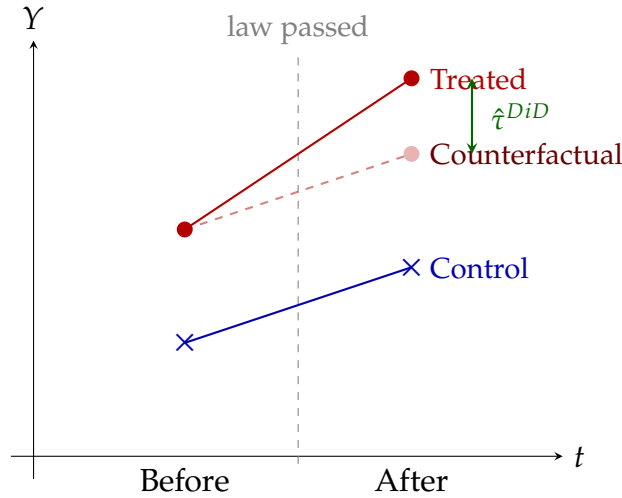


Figure 6.1 shows the logic graphically. The control group’s trend from “Before” to “After” estimates what would have happened to the treated group in the absence of the policy, the counterfactual. The DiD estimate is the vertical distance between the treated group’s actual outcome and this counterfactual.

6.1.3 The formal framework

We now write the argument precisely using the potential outcomes notation established in the Structural and Causal Approaches chapter. There are n units indexed by i and two time periods, $t \in \{0, 1\}$. A binary treatment indicator $D_i \in \{0, 1\}$ records whether unit i is treated in period 1. Treatment is absorbing: no unit is treated in period 0, and treated units remain treated thereafter.

Define potential outcomes $Y_{it}(0)$ and $Y_{it}(1)$ as the outcomes unit i would experience in period t under control and treatment, respectively. Observed outcomes satisfy

$$Y_{it} = Y_{it}(0) + D_{it} [Y_{it}(1) - Y_{it}(0)],$$

where $D_{it} = D_i \cdot \mathbf{1}\{t = 1\}$. The individual treatment effect is $\tau_i = Y_{i1}(1) - Y_{i1}(0)$, and the average treatment effect on the treated is

$$ATT = E[\tau_i \mid D_i = 1] = E[Y_{i1}(1) - Y_{i1}(0) \mid D_i = 1].$$

This is the estimand we are after: the average effect of the law on the leverage of firms that were actually subject to it.

To connect DiD to the *ATT*, assume a simple model for untreated potential outcomes:

$$Y_{it}(0) = \alpha_i + \lambda_t + v_{it}, \quad (6.2)$$

where α_i is a unit fixed effect, λ_t is a common time effect, and $E[v_{it} | D_i, t] = 0$. This model has an immediate and powerful implication. For any two units i and j ,

$$Y_{it}(0) - Y_{i,t-1}(0) = (\lambda_t - \lambda_{t-1}) + (v_{it} - v_{i,t-1}),$$

which does not depend on α_i . In expectation, all units' untreated outcomes change by the same amount:

$$E[Y_{it}(0) - Y_{i,t-1}(0) | D_i = 1] = E[Y_{it}(0) - Y_{i,t-1}(0) | D_i = 0] = \lambda_t - \lambda_{t-1}.$$

This is the **parallel trends assumption**: in the absence of treatment, treated and control units would have experienced the same average change in outcomes. Note what the assumption does and does not require. It allows the two groups to have permanently different *levels* of the outcome. Arizona firms may always have higher leverage than Connecticut firms. What it rules out is a *differential trend*, meaning the two groups must be on the same trajectory in the absence of the policy.

Key Insight 2

Parallel trends is an assumption about the *counterfactual*: what would have happened to the treated group if it had not been treated. It is not an assumption about what we observe in the data. In the post-treatment period, the treated group's outcome reflects both the treatment effect and the counterfactual trend, so parallel trends cannot be directly tested in post-treatment data. It can, however, be assessed in pre-treatment data, as we discuss in Section 6.2.

Under parallel trends, the DiD estimand from equation (6.1) identifies the *ATT*. To see this, write

$$\begin{aligned} & [E(Y_{i1} | D_i = 1) - E(Y_{i0} | D_i = 1)] - [E(Y_{i1} | D_i = 0) - E(Y_{i0} | D_i = 0)] \\ &= E[Y_{i1}(1) - Y_{i0}(0) | D_i = 1] - E[Y_{i1}(0) - Y_{i0}(0) | D_i = 0] \\ &= \underbrace{E[\tau_i | D_i = 1]}_{ATT} + \underbrace{E[Y_{i1}(0) - Y_{i0}(0) | D_i = 1] - E[Y_{i1}(0) - Y_{i0}(0) | D_i = 0]}_{= 0 \text{ under parallel trends}}. \end{aligned} \quad (6.3)$$

The second term vanishes because parallel trends equates the expected change in untreated

potential outcomes across groups. What remains is exactly the *ATT*.

6.1.4 The 2×2 table and TWFE regression

The DiD logic is often summarized in a 2×2 table. Under the model in (6.2), expected outcomes are:

	$t = 0$ (Before)	$t = 1$ (After)
Control ($D_i = 0$)	$E[\alpha_i D_i=0] + \lambda_0$	$E[\alpha_i D_i=0] + \lambda_1$
Treated ($D_i = 1$)	$E[\alpha_i D_i=1] + \lambda_0$	$E[\alpha_i D_i=1] + \lambda_1 + ATT$

The DiD estimate is the difference-in-differences of the four cell means:

$$\hat{\tau}^{DiD} = (\bar{Y}_{1,1} - \bar{Y}_{1,0}) - (\bar{Y}_{0,1} - \bar{Y}_{0,0}),$$

where $\bar{Y}_{d,t}$ denotes the sample average of Y for group d in period t . The unit fixed effects α_i cancel in the time differences, and the common time effect λ_t cancels in the group difference.

Running Example 3: Leverage and Investment

Suppose the average leverage ratios are as follows:

	1986	1987	Change
Arizona (treated)	0.312	0.287	-0.025
Connecticut (control)	0.295	0.284	-0.011

Both groups reduced leverage, consistent with a broad deleveraging trend (perhaps driven by the business cycle). But Arizona firms reduced leverage by more. The DiD estimate is

$$\hat{\tau}^{DiD} = (-0.025) - (-0.011) = -0.014.$$

The anti-takeover law is associated with a 1.4 percentage point *reduction* in leverage. The interpretation: when firms gained better protection from hostile takeovers, their managers chose to carry less debt. This is consistent with the view that high leverage was partly a takeover defense that became less necessary once the law provided alternative protection (Garvey and Hanka, 1999; Roberts and Whited, 2013).

Notice that a simple post-treatment comparison would give $0.287 - 0.284 = 0.003$, a *positive* difference, because Arizona firms had higher leverage to begin with. The DiD removes this level difference and isolates the treatment-induced change.

This 2×2 DiD can be estimated equivalently as a regression. With two periods and two groups, the specification

$$Y_{it} = \beta_0 + \beta_1 D_i + \beta_2 \text{Post}_t + \beta_3 (D_i \times \text{Post}_t) + u_{it}, \quad (6.4)$$

where $\text{Post}_t = \mathbf{1}\{t = 1\}$, produces $\hat{\beta}_3 = \hat{\tau}^{\text{DiD}}$. Equivalently, we can write this as the two-way fixed effects (TWFE) specification:

$$Y_{it} = \alpha_i + \lambda_t + \beta D_{it} + \varepsilon_{it}, \quad (6.5)$$

where α_i are unit fixed effects and λ_t are time fixed effects. With two periods and absorbing treatment, the coefficient $\hat{\beta}$ is numerically identical to $\hat{\beta}_3$ in (6.4) and to the manual DiD calculation. All three approaches (the 2×2 table, the interaction regression, and the TWFE regression) deliver the same estimate.

This equivalence is exact with two periods. As we shall see in Sections 6.2 and 6.3, it does *not* carry over to settings with multiple periods and staggered treatment timing.

6.1.5 What parallel trends controls for, and what it does not

It is worth being precise about what DiD buys you.

What DiD controls for:

- *Time-invariant differences between groups.* Any permanent characteristic that differs between treated and control units (industry composition, geographic advantages, historical leverage norms) is absorbed by the unit fixed effect α_i and eliminated by differencing.
- *Common time shocks.* Any macroeconomic or marketwide event that affects all units equally (a recession, a change in monetary policy, an aggregate credit cycle) is absorbed by λ_t and eliminated by the group difference.

What DiD does *not* control for:

- *Time-varying confounders that differ across groups.* If something else changed at the same time as the law and affected the treated firms differently (for instance, if the state simultaneously reformed its tax code or experienced a local economic boom), the DiD estimate confounds the law's effect with this concurrent shock. This is the Achilles' heel of DiD: the researcher must argue that no such differential shock occurred.
- *Anticipation.* If firms adjusted their behavior before the law took effect (perhaps because the legislation was debated publicly for months), then pre-treatment outcomes are already contaminated by the treatment, and the "before" period does not provide a

clean baseline. We return to this in Section 6.2.

- *General equilibrium effects.* DiD estimates a partial equilibrium treatment effect: the impact on firms directly subject to the law, holding constant the behavior of everyone else. If the law also affected firms in untreated states (through competition, factor markets, or regulatory arbitrage), then the “control” group is itself affected by treatment, and the DiD estimate is biased.

Common Mistake 4

Parallel trends does not mean that treated and control groups must have the same level of the outcome variable before treatment. It means they must have the same *trend*, the same expected change over time. Groups can start at very different levels and still satisfy parallel trends. Conversely, groups that happen to have similar levels need not satisfy parallel trends if their trajectories were diverging before the policy change.

6.1.6 Placing DiD in the structural/causal framework

DiD is a causal method. It identifies the *ATT* without requiring a structural model of *why* firms choose their leverage. It does not ask us to specify the objective function, the information structure, or the equilibrium concept. In this sense, it belongs squarely on the “causal” side of the organizing framework introduced in the Structural and Causal Approaches chapter.

But it is worth noting what DiD *does* assume structurally. The parallel trends assumption (6.2) is itself a model for the data-generating process of untreated potential outcomes. It says that, absent treatment, outcomes are additively separable in a unit component and a time component. This is a substantive economic restriction. It rules out, for example, models in which firms’ sensitivities to aggregate shocks depend on their fixed characteristics, an interactive fixed effects structure that we will encounter briefly in Section 6.2. In a very real sense, even “design-based” methods embed structural assumptions. The art lies in choosing assumptions that are both plausible and testable.

What does the DiD estimate *not* tell us? If Arizona’s anti-takeover law reduced leverage by 1.4 percentage points, we learn the *magnitude* of the policy effect on the treated firms. We do not learn *why* leverage fell (whether firms reduced debt issuance, increased equity, changed investment, or renegotiated existing contracts). We do not learn what would happen if a *different* creditor rights reform were adopted. And we do not learn whether the effect is socially desirable. For those deeper questions, we need the structural models of Part III. DiD gives us the “what”; structure gives us the “why” and the “what if.” The two are complements, not substitutes.

The next section extends the basic framework to multiple time periods, event study designs, and the subtle challenges of assessing whether the parallel trends assumption is plausible.

6.2 From Two Periods to Many: Event Studies and Pre-Trends

The 2×2 framework of Section 6.1 is conceptually clean, but it rarely describes the data we actually have. Most corporate finance panels span many years. The anti-takeover laws were adopted over at least the period from 1982 to 2013 (Karpoff and Wittry, 2018). Banking deregulation rolled out across states from the late 1970s through the mid-1990s. Having multiple time periods before and after the policy is a luxury: it lets us examine the *dynamics* of the treatment effect, and it lets us partially *test* the identifying assumption. This section explains how, and where the pitfalls lie.

Throughout this section, we maintain the single-treatment-date setting: all treated units begin treatment at the same time t_0 . The complications of staggered timing are deferred to Section 6.3.

6.2.1 The event study specification

With T time periods and a common treatment date t_0 , we extend the TWFE specification from Section 6.1 by replacing the single treatment indicator with a full set of relative-time indicators:

$$Y_{it} = \alpha_i + \lambda_t + \sum_{\substack{k=-K \\ k \neq -1}}^L \delta_k D_{it}^k + \varepsilon_{it}, \quad (6.6)$$

where $D_{it}^k = \mathbf{1}\{t - t_0 = k\} \cdot D_i$ is an indicator equal to one if unit i is treated and observed exactly k periods from the treatment date. The sum runs from K periods before treatment to L periods after. One coefficient (conventionally δ_{-1} , the period just before treatment) is normalized to zero to serve as the reference period.

Each coefficient δ_k measures the difference in outcomes between treated and control units at horizon k , *relative to* the difference at $k = -1$. For pre-treatment periods ($k < -1$), these coefficients test whether the groups were on parallel trends before the policy: under the null of parallel trends, $\delta_k = 0$ for all $k < 0$. For post-treatment periods ($k \geq 0$), the δ_k trace out the dynamic treatment effect (whether the impact is immediate, gradual, growing, or transient).

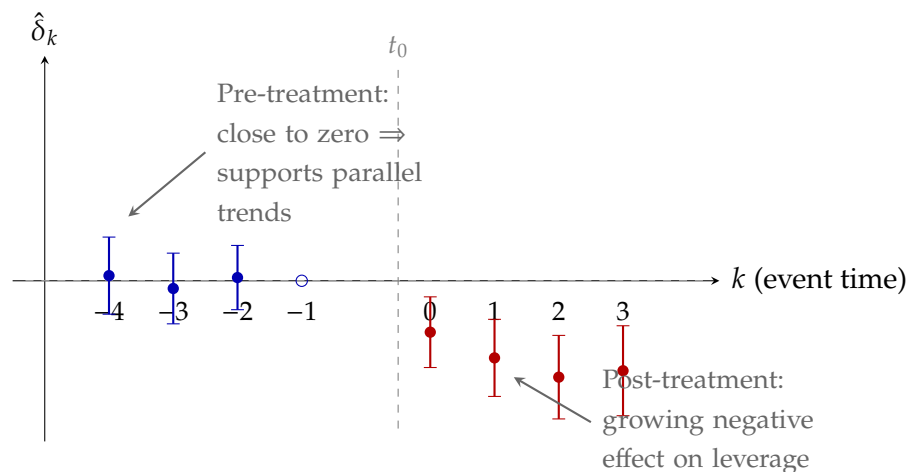
Running Example 5: Leverage and Investment

Suppose we observe firms in Arizona (treated) and Connecticut (control) for seven years: 1983–1989, with the anti-takeover law passed at the start of 1987 ($t_0 = 1987$). The event study regression includes indicators for $k \in \{-4, -3, -2, 0, 1, 2\}$, with $k = -1$ (year 1986) as the omitted reference.

The pre-treatment coefficients $\delta_{-4}, \delta_{-3}, \delta_{-2}$ test whether leverage was evolving differently between Arizona and Connecticut in the years before the law. If these are close to zero and statistically insignificant, we have evidence (though not proof) that the parallel trends assumption is reasonable. The post-treatment coefficients $\delta_0, \delta_1, \delta_2$ reveal the dynamics of the leverage response: did leverage drop immediately, or did the effect take time to materialize as firms gradually adjusted their capital structures?

6.2.2 How to read an event study plot

The coefficients $\{\delta_k\}$ are typically displayed in an *event study plot*: a graph with relative time k on the horizontal axis and the point estimates with confidence intervals on the vertical axis. These plots are now ubiquitous in applied economics and corporate finance. Reading them correctly requires attention to several features.



The vertical axis shows estimated $\hat{\delta}_k$, not the level of the outcome. A value of -0.015 at $k = 2$ means that, two years after the policy, the treated group's leverage was 1.5 percentage points lower than what parallel trends with the control group would predict.

The reference period is normalized to zero by construction. It is not an estimate; it is a normalization. All other coefficients are measured relative to this period.

Confidence intervals should ideally be *uniform* (also called simultaneous), not pointwise (Freyaldenhoven, Hansen, Pérez Pérez, and Shapiro, 2021). Pointwise intervals (the default

in most software) control the error rate for each individual coefficient separately. When you look at many coefficients at once (as you always do in an event study plot), the probability that *at least one* falls outside its interval is much larger than the nominal level. Uniform confidence bands account for this multiple testing and are the appropriate visual tool. This is a detail that matters: a plot that “passes” the pre-trends test with pointwise intervals may fail with uniform intervals.

The pre-treatment region ($k < -1$) is the diagnostic zone. Here the coefficients should be close to zero and statistically insignificant if parallel trends holds. A systematic pattern (a trend, a jump, or a drift) suggests that the two groups were already on different trajectories before the policy, which casts doubt on the identifying assumption.

The post-treatment region ($k \geq 0$) shows the dynamic treatment effect. Look for: whether the effect is immediate or gradual (important for understanding the economic mechanism), whether it persists or fades (important for policy evaluation), and whether it is statistically significant at conventional levels.

6.2.3 The pre-testing problem

Event study plots are enormously useful, and you should always produce one when doing DiD. But they come with an important caveat: *passing the pre-trends test does not prove that parallel trends holds*.

There are two distinct concerns. The first is **low power**. Pre-trend tests may fail to detect economically meaningful violations of parallel trends because the pre-treatment sample is small or the data are noisy. A nonsignificant pre-treatment coefficient does not mean the coefficient is zero. It means we cannot reject zero, which is a weaker statement. This is especially relevant in corporate finance, where panels are often short and outcomes are volatile.

The second concern is more subtle: **conditioning on the pre-test distorts inference**. Roth (2022) shows that if researchers select their specification or sample based on whether the pre-trends “look good,” the post-treatment estimates are biased. The intuition is a form of selection: among all the specifications a researcher might try, the ones that pass the pre-test are disproportionately likely to be ones where noise happened to mask a real violation. Conditioning on passing effectively truncates the sampling distribution of the post-treatment estimator, leading to estimates that are biased toward finding an effect (or biased against finding one, depending on the direction of the true violation).

This does not mean pre-trends are useless. It means they should be treated as a *diagnostic*, not a *proof*. The right posture is: “Here is the event study plot. The pre-treatment

coefficients are consistent with parallel trends, which supports the identifying assumption. But I also report a sensitivity analysis to show how much the conclusions would change if parallel trends is violated.”

Common Mistake 6

Do not select your research design based on whether the pre-trends “look clean.” If you try five different control groups and report the one with flat pre-trends, you are effectively constructing your counterfactual to match the data, not testing whether the counterfactual is correct. This is a form of specification searching that invalidates the inference.

6.2.4 Sensitivity analysis: Rambachan and Roth

A more principled approach to the fragility of parallel trends is the sensitivity analysis proposed by [Rambachan and Roth \(2023\)](#). The core idea is elegant: use the *observed* violations of parallel trends in the pre-treatment period to bound the *possible* violations in the post-treatment period, and then ask how the treatment effect estimate changes under these bounded violations.

Formally, let δ_k^{pre} denote the pre-treatment event study coefficients (which under parallel trends should be zero) and δ_k^{post} denote the post-treatment coefficients (which reflect the treatment effect plus any violation of parallel trends). The Rambachan-Roth framework introduces a sensitivity parameter $M \geq 0$ and assumes that the magnitude of the violation in post-treatment periods is no larger than M times the largest violation observed in pre-treatment periods:

$$|\text{bias in } \delta_k^{\text{post}}| \leq M \cdot \max_{j < 0} |\hat{\delta}_j^{\text{pre}}|.$$

When $M = 0$, parallel trends holds exactly (no violation is allowed) and the treatment effect estimate is the standard one. When $M = 1$, the post-treatment violation can be as large as the worst observed pre-treatment violation. Larger values of M allow for progressively worse violations.

The output is a *breakdown frontier*: for each value of M , the method reports a confidence set for the treatment effect. As M increases, the confidence set widens. The researcher can then report: “My estimate is robust to violations of parallel trends up to M^* times as large as the worst pre-treatment violation.” If M^* is large (say, 2 or more), the result is quite robust. If the confidence set includes zero for $M = 1$, the result is sensitive to violations of parallel trends comparable to what was observed in the data.

Key Insight 7

The Rambachan-Roth sensitivity analysis gives you a principled answer to the question: “How robust are my results to violations of parallel trends?” It transforms a binary question (do pre-trends pass?) into a continuous one (how much violation can I tolerate before my conclusions change?). Every DiD paper should report this analysis. Software is available in Stata (`honestdid`) and R (`HonestDiD`).

6.2.5 Parallel trends in what? The implicit structural economics

We have been treating parallel trends as if it were a single, well-defined assumption. It is not. The assumption states that untreated potential outcomes would have evolved the same way for treated and control units. But “the same way” in what metric? Parallel trends in levels,

$$E[Y_{it}(0) | D_i = 1] - E[Y_{i,t-1}(0) | D_i = 1] = E[Y_{it}(0) | D_i = 0] - E[Y_{i,t-1}(0) | D_i = 0],$$

is a different assumption from parallel trends in logs, $E[\ln Y_{it}(0) | D_i = 1] - E[\ln Y_{i,t-1}(0) | D_i = 1] = E[\ln Y_{it}(0) | D_i = 0] - E[\ln Y_{i,t-1}(0) | D_i = 0]$, and if one holds, the other generically does not (Roth and Sant’Anna, 2023). This is not a statistical technicality. It is a question about the economic structure of the counterfactual world.

To see why, think about what each version of the assumption commits you to economically. Parallel trends in levels says: absent the policy, treated and control firms would have changed their leverage by the same *number of percentage points*. A firm at 20% leverage and a firm at 60% leverage would both increase by, say, 2 percentage points during a credit boom. This is an implicit model of how leverage responds to aggregate shocks. It says the response is additive and does not scale with the firm’s current leverage.

Parallel trends in logs says something quite different: absent the policy, treated and control firms would have changed their leverage by the same *proportion*. The firm at 20% would increase by 10% (to 22%), while the firm at 60% would increase by 10% (to 66%), a 6 percentage point change. This is also an implicit model, one in which firms’ responses to aggregate shocks are multiplicative, scaling with their current position.

These are substantively different models of how firms set leverage in the absence of treatment. The additive model might be reasonable if the aggregate shock operates through a common absolute channel (say, a tax change that adds a fixed number of cents per dollar of debt for every firm). The multiplicative model might be reasonable if firms respond proportionally to common shocks (say, because leverage targets are set as ratios and all

firms adjust toward their targets at a common rate). Neither model is “correct” in the abstract. Which one is more plausible depends on what you believe about the economic mechanism that generates the counterfactual trends.

Key Insight 8

Choosing between levels and logs in a DiD is not a robustness check or a matter of econometric convenience. It is a substantive economic modeling decision about how untreated potential outcomes are generated. The researcher is implicitly specifying a structural model for the counterfactual, the very thing that DiD is supposed to free you from having to specify. The structural assumptions are still there; they are just hiding in the functional form.

This point connects directly to the book’s organizing theme. Even the most “design-based” causal methods embed structural assumptions. In DiD, the parallel trends assumption *is* a structural restriction on the data-generating process. When you choose between levels and logs, you are choosing between two different structural models for how untreated outcomes evolve. The honest thing to do is to acknowledge this, think carefully about which model is more economically defensible for your application, and be transparent about the choice.

Roth and Sant’Anna (2023) identify three paths forward. First, if treatment is as-if randomly assigned (a stronger assumption than standard DiD requires), then the ATT is identified regardless of functional form, because random assignment ensures that the counterfactual distribution is the same as the control group’s distribution in every transformation. Second, the researcher can impose distributional assumptions that pin down the full counterfactual distribution, enabling nonparametric identification. Third, and most commonly, the researcher can use domain knowledge to argue for a particular functional form. The first path is rarely available in corporate finance (if treatment were truly random, we would not need DiD). The second is technically demanding and seldom used. The third is the practical reality, and it requires the researcher to make an economic argument, not just a statistical one.

Running Example 9: Leverage and Investment

Return to the creditor rights example. Suppose the event study shows clean pre-trends and a significant negative post-treatment effect when leverage is measured in levels. But when we re-estimate in logs, the pre-treatment coefficients show a slight upward drift. What should we conclude?

The discrepancy is not a bug. It is an inevitable consequence of the mathematical incompatibility of additive and multiplicative parallel trends. The question is which model of counterfactual leverage dynamics is more economically appropriate. If creditor rights affect the *cost* of debt in a way that does not scale with the firm's current leverage (for instance, by reducing expected bankruptcy costs by a fixed amount), then parallel trends in levels is the right assumption, and the levels specification is the one to emphasize. If instead the law works by proportionally expanding firms' debt capacity (by making a fixed fraction of previously unusable assets eligible as collateral), then a logs specification better matches the economic channel.

The answer depends on the economics of the specific policy being studied, not on which specification produces cleaner pre-trends. Choosing the functional form that gives you flat pre-trends is selecting your structural assumption to fit the data, which is circular. Choosing the functional form that matches your economic prior about the mechanism, and then *checking* whether the data are consistent, is the right approach.

6.2.6 What multiple periods add: a summary

Having more than two periods buys us three things, each valuable but none decisive:

First, dynamics. We can see whether the treatment effect is immediate or gradual, growing or fading. This speaks to the economic mechanism. A leverage response that takes three years to fully materialize suggests adjustment costs or gradual learning, while an immediate jump suggests a direct balance-sheet response.

Second, a partial test. Pre-treatment coefficients provide a diagnostic for parallel trends. They cannot prove the assumption, but a clear violation (a pre-trend that is statistically significant, economically large, or systematically patterned) is strong evidence against it.

Third, a platform for sensitivity analysis. The Rambachan-Roth framework uses the pre-treatment period to calibrate the plausible range of violations, giving us a continuous measure of robustness rather than a binary pass/fail.

What multiple periods do *not* buy is a solution to the staggered adoption problem. When different units enter treatment at different dates (the typical corporate finance setting), the event study regression in equation (6.6) can produce misleading results for the same reasons that TWFE fails: forbidden comparisons and heterogeneous treatment effects. That is the subject of Section 6.3.

6.3 The Staggered Adoption Problem

The basic DiD framework in Section 6.1 assumed a clean setting: one group is treated at a single point in time, another group is never treated. In practice, this almost never happens in corporate finance. States adopt laws at different times. Countries reform their governance codes over a span of years. Firms adopt new technologies or policies sequentially. The typical empirical setting involves *staggered adoption* (different units begin treatment at different dates), and the researcher wants to pool the variation to estimate an overall treatment effect.

For decades, the standard approach was simple: run a two-way fixed effects (TWFE) regression with unit and time fixed effects, include a binary treatment indicator D_{it} , and interpret the coefficient as “the” causal effect of the policy. As we saw in Section 6.1, this works perfectly well with two periods. The assumption (usually left implicit) was that it would continue to work with more periods and staggered timing.

It does not. A series of papers beginning around 2018 showed that TWFE can produce severely misleading estimates when treatment timing varies across units and treatment effects are heterogeneous (Borusyak et al., 2024; Goodman-Bacon, 2021; de Chaisemartin and D’Haultfœuille, 2020). This is not an abstract econometric curiosity. Baker et al. (2022) demonstrated that it matters in exactly the corporate finance applications we care about. This section explains what goes wrong and why.

6.3.1 Why staggered adoption is the default in corporate finance

Consider how policy variation typically arises in our field. U.S. states adopt anti-takeover statutes over a period of years; state-level banking deregulation occurred in waves from the late 1970s through the mid-1990s; countries reformed corporate governance codes on different timelines; and creditor rights protections have been strengthened or weakened at different dates across jurisdictions. In each case, there is no single “treatment date.” Instead, units enter treatment sequentially.

This staggered structure is precisely what makes DiD attractive for corporate finance. We can exploit within-unit variation in treatment timing while controlling for common time shocks. But it also creates a problem that was not fully appreciated until recently.

Running Example 10: Leverage and Investment

Suppose we study how strengthening creditor rights affects corporate leverage. Several states enact new creditor protection laws, but at different dates: State A in 2005, State B

in 2010, and State C in 2015. States D and E never adopt the law. We observe leverage ratios for firms incorporated in each state over the period 2000–2020.

The natural approach is to run the TWFE regression

$$\text{Lev}_{it} = \alpha_i + \lambda_t + \beta D_{it} + \varepsilon_{it},$$

where $D_{it} = 1$ if firm i 's state of incorporation has adopted the creditor rights law by year t . We cluster standard errors at the state level and interpret $\hat{\beta}$ as the effect of creditor rights on leverage.

This is exactly what researchers did for years. The question is: what does $\hat{\beta}$ actually estimate?

6.3.2 What TWFE actually estimates: the Goodman-Bacon decomposition

The key insight comes from [Goodman-Bacon \(2021\)](#), who showed that the TWFE coefficient $\hat{\beta}$ is a weighted average of *all possible* 2×2 DiD comparisons that can be constructed from the data. With staggered timing, these comparisons fall into several types.

To fix ideas, consider three groups in our running example: an early-treated group (State A, treated in 2005), a late-treated group (State B, treated in 2010), and a never-treated group (States D and E). The TWFE regression implicitly constructs comparisons such as:

- (i) **Early-treated vs. never-treated**, around 2005. This is a clean comparison, exactly the kind of 2×2 DiD we want. The never-treated states provide a valid counterfactual for the change in leverage around State A's adoption date.
- (ii) **Late-treated vs. never-treated**, around 2010. Also a clean comparison.
- (iii) **Late-treated vs. early-treated**, around 2010. Here, State B is newly treated and State A serves as the "control." But State A was already treated in 2005. Its leverage in 2010 reflects both the common time effect *and* the ongoing impact of its own treatment five years earlier. This is not a valid control.
- (iv) **Early-treated vs. late-treated**, around 2005. State A is newly treated and State B (not yet treated) serves as the control. This comparison is valid, but note that the roles of "treatment" and "control" are assigned by the timing of adoption, not by any design choice of the researcher.

Comparison (iii) is the problematic one. It uses an already-treated unit as a control for a newly-treated unit. In the recent literature, this is called a *forbidden comparison* ([Borusyak et al., 2024](#)) or a *bad comparison* ([Goodman-Bacon, 2021](#)).

Key Insight 11

The TWFE estimator $\hat{\beta}$ is a variance-weighted average of all possible 2×2 DiD comparisons. Some of these comparisons use already-treated units as controls. If the treatment effect evolves over time (growing, fading, or varying across cohorts), these “forbidden comparisons” bias the estimate. The bias can be severe enough to produce estimates with the wrong sign.

To see why this matters, suppose that the creditor rights law has a *growing* effect on leverage: the impact is small in the first year but increases over time as firms adjust their capital structures. By 2010, five years after adoption, State A’s leverage has risen substantially due to the law. When the TWFE regression uses State A as a control for State B around 2010, it subtracts off State A’s change in leverage, which includes the cumulative treatment effect. This makes State B’s treatment effect *look smaller* than it actually is, because part of State A’s leverage increase is being treated as a “time trend” rather than a treatment effect.

The problem can go the other way too. If treatment effects *fade over time*, using already-treated units as controls can make new treatment effects look artificially large. In extreme cases with enough heterogeneity in treatment effect dynamics, the weights on some comparisons can be negative, and $\hat{\beta}$ can have the opposite sign from every group-specific treatment effect.

6.3.3 A numerical illustration

A simple example makes the mechanics concrete. Suppose we have three time periods ($t = 1, 2, 3$) and two groups. Group *E* (early) is treated starting in period 2. Group *L* (late) is treated starting in period 3. Suppose the true treatment effect *grows over time*: the effect is $\tau = 2$ in the first period after treatment and $\tau = 4$ in the second period. There are no other shocks, so all changes in outcomes are driven by the treatment.

	$t = 1$	$t = 2$	$t = 3$
Group <i>E</i> (treated at $t = 2$)	10	$10 + 2 = 12$	$10 + 4 = 14$
Group <i>L</i> (treated at $t = 3$)	10	10	$10 + 2 = 12$

Every unit-level treatment effect is positive. The ATT for group *E* averaged over post-treatment periods is $(2 + 4)/2 = 3$. The ATT for group *L* in its single post-treatment period is 2. Any sensible overall average should be positive.

Now consider the 2×2 comparisons that TWFE constructs:

Comparison 1: Group E vs. Group L , around $t = 2$. Group L is not yet treated, so it is a valid control.

$$\widehat{ATT}_1 = (Y_{E,2} - Y_{E,1}) - (Y_{L,2} - Y_{L,1}) = (12 - 10) - (10 - 10) = 2.$$

This correctly recovers the on-impact treatment effect for group E .

Comparison 2: Group L vs. Group E , around $t = 3$. Group E is already treated, so this is a forbidden comparison.

$$\widehat{ATT}_2 = (Y_{L,3} - Y_{L,2}) - (Y_{E,3} - Y_{E,2}) = (12 - 10) - (14 - 12) = 2 - 2 = 0.$$

The estimate is zero, even though the true treatment effect for group L at $t = 3$ is 2. What happened? Group E 's leverage also increased by 2 between periods 2 and 3 (from the growing treatment effect), and TWFE subtracted this off, treating it as a common time trend.

The TWFE estimator $\hat{\beta}$ is a variance-weighted average of these comparisons. The exact weights depend on group sizes and the variance of the treatment indicator, but the point is clear: even in this simple example with uniformly positive treatment effects, the forbidden comparison produces a zero. A weighted average of 2 and 0 will understate the true effect. With different numbers (particularly if treatment effects are heterogeneous across groups), $\hat{\beta}$ can be negative even when all underlying effects are positive.

Common Mistake 12

If you ran a TWFE DiD with staggered treatment timing before 2018 and found “significant” results, those results may not mean what you thought. The coefficient may be biased upward or downward, and the sign may be wrong. This does not mean the results are necessarily incorrect. If treatment effects happen to be homogeneous and time-invariant, TWFE is fine. But the assumption of homogeneity was almost never tested or even stated.

6.3.4 When does this matter in practice?

Two conditions must *both* hold for the staggered adoption problem to bite:

- (a) **Treatment timing varies across units.** If all units are treated at the same date, there are no forbidden comparisons because there are no already-treated units to misuse as controls. This is the standard case covered in Section 6.1.
- (b) **Treatment effects are heterogeneous.** If the causal effect is exactly the same for

every unit and does not change over time or with the duration of exposure, then even the forbidden comparisons produce unbiased estimates. The already-treated units' outcomes change by the same amount as the newly-treated units' outcomes, so subtracting them off correctly removes the time trend.

When treatment effects are truly homogeneous and constant ($\tau_i = \tau$ for all units and all post-treatment periods), the TWFE estimator recovers τ even with staggered timing (Callaway, 2022). But this is a strong and often implausible assumption. In corporate finance, there are many reasons to expect heterogeneity: firms differ in their exposure to a policy, early adopters may differ from late adopters, and effects may grow or attenuate as firms adjust their behavior over time. We also know that leverage adjustment is gradual, which makes time-varying treatment effects the rule rather than the exception.

The practical question is not whether the problem *can* arise but whether it is *large enough to matter*. This is an empirical question, and the answer from corporate finance is: yes, often.

6.3.5 The evidence from corporate finance: Baker, Larcker, and Wang (2022)

Baker et al. (2022) brought the staggered adoption problem directly into the corporate finance literature. They re-examined two canonical DiD applications: U.S. banking deregulation (staggered across states) and international corporate board reforms (staggered across countries). Both settings are frequently cited in corporate finance and have influenced thinking about the real effects of financial regulation.

Their approach was straightforward. For each application, they estimated the treatment effect using both the traditional TWFE specification and the modern estimators described in Section 6.4. If the staggered adoption problem is empirically negligible, the two approaches should give similar answers.

They did not. For the banking deregulation application, the TWFE estimates differed meaningfully from the heterogeneity-robust estimates, both in magnitude and in the pattern of dynamic effects. The event study plots told different stories about whether the effects were immediate or gradual. For the board reform application, the discrepancies were large enough to affect the qualitative conclusions of the analysis.

The paper also highlighted a useful diagnostic: the Goodman-Bacon (2021) decomposition, which reports the fraction of weight that the TWFE estimator places on forbidden comparisons. In their applications, this fraction was substantial, not a rounding error but a first-order feature of the estimation. As the Callaway survey notes, the fraction of weight

on forbidden comparisons can range from modest (around 5% in favorable designs) to dominant (over 50% in designs where all or most units are eventually treated), depending on the structure of the data (Callaway, 2022).

Running Example 13: Leverage and Investment

Return to the creditor rights example. States adopted creditor-friendly laws over a 20-year period. Some states adopted early (e.g., in the mid-1990s), others late (e.g., in the 2010s), and some never adopted at all. A researcher runs the standard TWFE regression of leverage on the treatment indicator.

The Goodman-Bacon decomposition reveals that 35% of the weight in the TWFE estimate comes from comparisons that use already-treated states as controls. In these comparisons, the “control” states’ leverage has already shifted in response to their own earlier adoption of creditor rights laws. The TWFE estimate of $\hat{\beta} = 0.018$ (a 1.8 percentage point increase in leverage) turns out to be attenuated: the Callaway-Sant’Anna estimator, which avoids forbidden comparisons, produces an estimate of 0.031.

This is not a contrived example. Staggered policy adoption with heterogeneous firm responses is the *standard* setting in empirical corporate finance. If you are studying any law, regulation, or institutional reform that was adopted at different dates across jurisdictions, you face this problem.

6.3.6 Connection to the structural/causal framework

The staggered adoption problem is instructive for the broader themes of this book. At first glance, it appears to be a purely technical issue, a matter of getting the econometrics right. But it has deeper implications for how we think about identification.

The root cause of the problem is *treatment effect heterogeneity*, the possibility that different firms or cohorts experience different effects. In the causal inference framework, heterogeneity is a feature of the world that the researcher must accommodate. The modern DiD estimators do this by estimating separate group-time effects and then aggregating them transparently. The TWFE approach implicitly assumed homogeneity, which is itself a structural restriction on the data-generating process.

This connects to a theme we will return to throughout the book. Even “design-based” causal methods embed structural assumptions. The parallel trends assumption in DiD is a restriction on how untreated potential outcomes evolve. It is a model for the counterfactual, even if a minimal one. The assumption that treatment effects are homogeneous is a further structural restriction. When that restriction fails, the estimator breaks. Recognizing which

structural assumptions are baked into your estimation strategy (and testing them when possible) is the difference between credible and misleading empirical work.

The next section describes the modern estimators that resolve the staggered adoption problem, along with practical implementation guidance.

6.4 Modern Solutions and Implementation

Section 6.3 showed that TWFE with staggered adoption can produce misleading estimates whenever treatment effects are heterogeneous across cohorts or over time. The good news is that the same papers that diagnosed the problem also proposed solutions. Several alternative estimators are now available, each resting on clearly stated assumptions and avoiding the forbidden comparisons that plague TWFE. This section explains the logic behind these estimators, clarifies what assumptions justify choosing one over another, and provides implementation guidance.

6.4.1 The shared logic

All of the modern estimators share a common two-step structure (Callaway, 2022):

Step 1: Estimate disaggregated treatment effects. Rather than forcing all the information into a single coefficient $\hat{\beta}$, estimate group-time specific average treatment effects, $ATT(g, t)$, the average effect of the treatment in calendar period t on units first treated in period g . Crucially, these estimates use only “clean” comparisons: the counterfactual for a newly-treated group comes from units that are either *never treated* or *not yet treated*, never from units that are already treated.

Step 2: Aggregate transparently. Combine the $ATT(g, t)$ estimates into summary parameters that answer the researcher’s specific question. Want an overall average effect? Weight by group size. Want an event study showing dynamic effects by time since treatment? Average across groups at each event-time horizon. The aggregation weights are explicit and reported, so the reader knows exactly what the summary parameter represents.

This two-step logic is the same across all the modern approaches. Where they differ is in the details of how Step 1 is executed: which comparison group is used, how the counterfactual is constructed, what the parallel trends assumption is imposed on, and how covariates are handled. These differences are not cosmetic. They reflect different maintained assumptions, and which set of assumptions is most appropriate depends on the economic setting.

6.4.2 Callaway and Sant'Anna (2021)

Callaway and Sant'Anna (2021) build the estimator most directly from the potential outcomes framework. The approach begins by defining treatment cohorts: all units first treated in period g form “group g .” For each group-time pair (g, t) , the estimator computes

$$\widehat{ATT}(g, t) = \frac{1}{n_g} \sum_{i: G_i=g} (Y_{it} - Y_{i,g-1}) - \frac{1}{n_c} \sum_{j \in C} (Y_{jt} - Y_{j,g-1}),$$

where the first term is the average change in outcomes for group g from its “base period” $g - 1$ to period t , and the second term is the corresponding change for a comparison group C .

Key assumption: choice of comparison group. The researcher must choose who populates the comparison group C . There are two main options, each reflecting a different version of the parallel trends assumption:

- **Never-treated comparison group.** Use only units that are never treated in any period. The parallel trends assumption then requires that, in the absence of treatment, group g 's outcomes would have evolved like those of the never-treated group. This is the cleanest comparison but requires that a never-treated group exists and is a plausible counterfactual for the treated groups.
- **Not-yet-treated comparison group.** Use all units that have not yet been treated by period t , including those that will be treated later. This enlarges the comparison group and can improve precision, but it imposes a slightly stronger assumption: parallel trends must hold not only relative to units that are never treated, but also relative to units that will eventually be treated (before their treatment kicks in). If anticipation effects contaminate not-yet-treated units' behavior before their own treatment date, this comparison is compromised.

The choice matters. If you have a large never-treated group and believe it provides a reasonable counterfactual, use it. If the never-treated group is small, atypical, or absent (as happens when all units are eventually treated), the not-yet-treated comparison may be the only viable option, but you should be alert to potential anticipation effects.

Handling covariates. A major advantage of Callaway-Sant'Anna is its flexibility in conditioning on covariates. When parallel trends is more plausible conditional on observable characteristics (for instance, if it holds within industries but not across them), the estimator offers a doubly robust version. This combines an outcome regression model for

the counterfactual trend with a propensity score model for group membership. The resulting estimator is consistent if *either* model is correctly specified, providing a degree of insurance against misspecification (Callaway, 2022). This is particularly useful in corporate finance applications where industry, size, and other covariates systematically relate to both treatment timing and outcome levels.

What it assumes, summarized. Staggered (absorbing) treatment adoption; parallel trends conditional on covariates (or unconditionally); no anticipation (outcomes unaffected before the treatment date); and that the chosen comparison group is a valid counterfactual for each treated cohort. The parallel trends assumption is imposed only from the period just before treatment ($g - 1$) to the current period, not across all pre-treatment periods. This makes the estimator more robust to violations of parallel trends in the distant past but forgoes some efficiency (Callaway, 2022).

Software. Stata: `csdid`. R: `did`. Python: `csdid`.

6.4.3 Borusyak, Jaravel, and Spiess (2024)

Borusyak et al. (2024) take an *imputation* approach. The idea starts from the observation that parallel trends, combined with the additive model $Y_{it}(0) = \alpha_i + \lambda_t + v_{it}$, allows us to predict what each treated unit's outcome *would have been* in the absence of treatment. If we can estimate α_i and λ_t using only untreated observations, we can impute the counterfactual for every treated observation and compute treatment effects as residuals.

The algorithm.

1. Collect all untreated observations (pre-treatment periods for eventually-treated units, plus all periods for never-treated units).
2. Estimate the two-way fixed effects model $Y_{it} = \alpha_i + \lambda_t + v_{it}$ on this untreated sample.
3. For each treated observation (i, t) , impute the counterfactual: $\hat{Y}_{it}(0) = \hat{\alpha}_i + \hat{\lambda}_t$.
4. Compute the treatment effect for each treated observation as $\hat{\tau}_{it} = Y_{it} - \hat{Y}_{it}(0)$.
5. Aggregate these individual treatment effects into $ATT(g, t)$, event study parameters, or an overall ATT using transparent weights.

Key assumption: parallel trends in all pre-treatment periods. The imputation approach estimates the counterfactual model using *all* available untreated observations, including

distant pre-treatment periods. This means it imposes parallel trends across all pre-treatment periods, not just from the period immediately before treatment. This is a stronger assumption than what Callaway-Sant'Anna requires. The payoff is efficiency: by using more data to estimate the counterfactual model, the imputed treatment effects are more precisely estimated. The cost is that if parallel trends holds near the treatment date but was violated years earlier (perhaps because of a temporary shock that dissipated), the imputed counterfactual will be distorted, and this distortion contaminates the treatment effect estimates.

When does this tradeoff favor imputation? When you have many pre-treatment periods and believe parallel trends held consistently throughout. This is more plausible when the pre-treatment period is not too long and the economic environment was relatively stable. It is less plausible if your panel spans decades with structural breaks, crises, or regulatory regime changes in the pre-treatment period.

Connection to the event study tradition. The imputation approach has a natural connection to the economics event study discussed in the Event Studies chapter and in Section 6.2. The traditional event study regression also implicitly imputes counterfactuals from the two-way fixed effects model. The difference is that the traditional regression does so with a single specification that conflates clean and forbidden comparisons, while the imputation approach does so only on untreated observations, avoiding the forbidden comparisons entirely.

Software. Stata: `did_imputation`. R: `didimputation`. Python: `pydidimputation`.

6.4.4 Sun and Abraham (2021)

Sun and Abraham (2021) stay closest to the familiar regression framework. Their approach modifies the standard event study regression by fully interacting the relative-time indicators with cohort indicators, then aggregates the cohort-specific estimates.

The regression is:

$$Y_{it} = \alpha_i + \lambda_t + \sum_{g \in \mathcal{G}} \sum_{e \neq -1} \delta_{ge} \mathbf{1}\{G_i = g\} \cdot \mathbf{1}\{t = g + e\} + v_{it}, \quad (6.7)$$

where G_i is the cohort (first treatment period) of unit i and e is the event-time horizon. This is a saturated model in group-time interactions: each $\hat{\delta}_{ge}$ is the estimated treatment

effect for cohort g at horizon e , using the never-treated group as the implicit comparison. The event study parameters are then recovered as weighted averages across cohorts at each horizon: $\widehat{ATTES}(e) = \sum_g \hat{w}(g, e) \hat{\delta}_{ge}$, where the weights reflect relative group sizes.

Key assumption: what the regression buys and what it costs. Because the model is saturated in cohort-by-time interactions, it avoids the forbidden comparisons that afflict the standard event study regression (Section 6.2). However, Sun-Abraham requires a never-treated group to serve as the omitted category in the regression. If there is no never-treated group, the approach requires modifications. Moreover, because the regression includes interaction terms for both pre-treatment and post-treatment periods, it does not impose parallel trends in pre-treatment periods. It estimates them. This makes it more robust to pre-treatment violations (they don't contaminate post-treatment estimates) but less efficient than approaches that impose parallel trends in all pre-treatment periods.

Relationship to Callaway-Sant'Anna. With a never-treated comparison group and no covariates, Sun-Abraham and Callaway-Sant'Anna produce numerically identical estimates of group-time ATTs (Callaway, 2022). The differences emerge when covariates are included (Callaway-Sant'Anna's doubly robust approach handles them more flexibly) and in pre-treatment periods, where the two methods use different base-period conventions by default.

Software. Stata: `eventstudyinteract`. R: `sunab` (within `fixest`).

6.4.5 de Chaisemartin and D'Haultfœuille (2020)

de Chaisemartin and D'Haultfœuille (2020) approach the problem from a different angle. Rather than recovering the full set of group-time ATTs, they ask a more targeted question: what is the average treatment effect among all units whose treatment status *changes* in a given period? Their estimator focuses on “switchers” (units that transition from untreated to treated) and computes a local DiD using only these switchers and units whose status does not change.

Key assumption and what it identifies. The identifying assumption is a “stable treatment” version of parallel trends: in the absence of the treatment change, switchers and non-switchers would have experienced the same change in outcomes over the same pair of consecutive periods. This is a minimal assumption (it restricts counterfactual trends only

over two adjacent periods), making it the weakest version of parallel trends among the modern estimators. The cost is that it identifies only an instantaneous, local effect: the average impact of switching into treatment at that specific moment. It does not directly identify dynamic effects or effects for units treated in earlier periods.

When is this useful? When you want a quick diagnostic of whether there is *any* positive average effect among newly-treated units, without committing to assumptions about treatment effect dynamics or long-run parallel trends. It is also useful as a robustness check alongside the more comprehensive estimators: if de Chaisemartin-D'Haultfœuille gives a different sign from Callaway-Sant'Anna, something important depends on the additional assumptions.

Software. Stata: `did_multiplegt` and `did_multiplegt_dyn`. R: `DIDmultiplegt`.

6.4.6 Choosing among estimators: assumptions and tradeoffs

The four estimators are not interchangeable. They share the same broad goal (avoiding forbidden comparisons) but they impose different assumptions and answer slightly different questions. Table 6.1 summarizes the key distinctions.

Table 6.1. Modern DiD estimators: assumptions and tradeoffs

	Parallel trends imposed on	Comparison group	Covariates
Callaway-Sant'Anna	Adjacent periods only (from $g - 1$ to t)	Never-treated or not-yet-treated (researcher chooses)	Doubly robust (outcome model + propensity score)
Borusyak et al. (imputation)	All pre-treatment periods	Never-treated + pre-treatment obs. of treated	Linear in covariates, time-varying allowed
Sun-Abraham	Adjacent periods only (estimated, not imposed, in pre-period)	Never-treated (required)	Standard regression covariates
de Chaisemartin-D'Haultfœuille	Two consecutive periods only	Non-switching units in same period	Limited

The choice among estimators should be driven by which set of assumptions is most defensible in your specific application:

Use Callaway-Sant’Anna when: You want maximum flexibility and your parallel trends assumption is plausible only conditional on covariates. This is the default recommendation for most corporate finance applications because (a) treatment timing is often correlated with observable firm characteristics like size and industry, making conditional parallel trends more credible than unconditional, and (b) the doubly robust property provides insurance against misspecification of either the outcome model or the propensity score. It is also the right choice when you are concerned that parallel trends may not hold in the distant past but does hold near the treatment date.

Use Borusyak et al. when: You have many pre-treatment periods, believe parallel trends held consistently throughout, and want the most efficient estimates. The efficiency gain can be substantial when the pre-treatment panel is long relative to the post-treatment panel. This estimator is also appealing when you want a clean connection to the imputation logic. It makes explicit that DiD is fundamentally about estimating a counterfactual outcome model and computing residuals.

Use Sun-Abraham when: You want to stay within the familiar regression framework and your audience is comfortable reading regression tables rather than specialized estimator output. Because the Sun-Abraham regression is just a fully-interacted TWFE specification, it can be estimated with standard panel data software, and the output looks like a familiar regression table with cohort-by-time interaction coefficients. This can be a practical advantage when presenting to audiences less familiar with the specialized DiD software.

Use de Chaisemartin-D’Haultfœuille when: You want a robustness check under the weakest possible parallel trends assumption, or when you want to test whether there is *any* instantaneous effect without committing to assumptions about dynamics. It is also useful when treatment is not absorbing (units can switch in and out of treatment), a case that the other estimators handle less naturally.

Key Insight 14

In most corporate finance applications, the choice between Callaway-Sant’Anna and Borusyak et al. comes down to one question: *do you trust parallel trends in the distant past?* If yes, imputation is more efficient. If you are only confident that trends were parallel near the treatment date, Callaway-Sant’Anna is more robust. When the two estimators give similar answers, you can be more confident in the results. When they diverge, the difference tells you that distant-past parallel trends violations matter for

your application, which is itself valuable information.

Common Mistake 15

Do not choose among estimators based on which one gives you the result you want. The choice should be made *before* looking at the estimates, based on which assumptions are most defensible for your setting. If you report results from all four estimators and they agree, that is reassuring. If they disagree, you need to explain which set of assumptions you find most credible and why, not pick the most favorable number.

6.4.7 Implementation summary

For the applied researcher, the workflow is:

1. **Assess your setting.** Is there a never-treated group? Is it a plausible counterfactual? Are covariates important for parallel trends? How many pre-treatment periods do you have?
2. **Choose a primary estimator** based on the assumptions that best match your setting. Report the reasons for your choice.
3. **Estimate group-time ATTs.** Examine them for patterns. Do effects vary across cohorts? Do they grow with exposure time? This is informative about the economics, not just the econometrics.
4. **Aggregate and report.** Present the overall ATT and the event study plot. Use the aggregation that answers your research question.
5. **Robustness.** Report at least one alternative estimator with different assumptions. Report the Rambachan-Roth sensitivity analysis (Section 6.2). If your primary estimate and the robustness checks agree, your conclusions are on firm ground.

Running Example 16: Leverage and Investment

In the creditor rights study, suppose several states adopted new creditor protection laws over a 15-year period, and roughly 20 states never adopted. The never-treated group is large and spans diverse industries and firm sizes, making it a plausible counterfactual. We choose Callaway-Sant'Anna as the primary estimator, conditioning on pre-treatment industry composition and firm size.

The group-time ATT estimates reveal that the leverage effect is small for early adopters (around 1 percentage point) but larger for late adopters (around 3 percentage points). The event study plot shows the effect growing over the first three years and then

stabilizing. This heterogeneity is economically meaningful. It suggests that the late-adopting states may have had more room for firms to respond, or that the economic environment at the time of late adoption was more conducive to leverage adjustments. As a robustness check, we re-estimate using the Borusyak et al. imputation estimator. The overall ATT is similar (2.4 percentage points vs. 2.6 from Callaway-Sant'Anna), confirming that the results are not sensitive to whether parallel trends is imposed on all pre-treatment periods or only adjacent ones. The Rambachan-Roth analysis shows the effect remains statistically significant for M up to 1.5, indicating robustness to violations of parallel trends 50% larger than the worst observed pre-treatment violation.

The next section collects practical guidance on inference, common pitfalls, and the connection between DiD and the structural/causal framework that organizes this book.

6.5 Inference, Pitfalls, and the Structural/Causal Connection

This final section of the chapter gathers three things: practical guidance on inference that cuts across all the estimators discussed in Sections 6.1 to 6.4, a checklist of the most common mistakes in applied DiD work, and a reflection on what DiD does and does not contribute to our understanding of corporate finance, tying the chapter back to the structural/causal framework that organizes this book.

6.5.1 Inference: clustering and standard errors

The single most common inferential error in DiD applications is computing standard errors incorrectly. [Bertrand et al. \(2004\)](#) made the case forcefully: in a typical DiD setting, both the outcome and the treatment indicator are serially correlated within units. If this autocorrelation is ignored, standard errors are far too small and rejection rates are far too high. Their simulations showed that studies could reject a true null hypothesis 45% of the time at a nominal 5% level, a staggering size distortion.

The fix is straightforward in principle: *cluster standard errors at the level at which the treatment varies*. If the policy is adopted at the state level, cluster at the state level. If it varies across industries, cluster at the industry level. If it varies at the firm level, cluster at the firm level. The rationale is that observations within a cluster share a common treatment status and are likely to have correlated shocks, so they should not be treated as independent sources of information about the treatment effect.

Three specific points deserve emphasis in a corporate finance context:

First, do not cluster below the level of treatment. If the policy varies at the state level but your data are at the firm level, clustering at the firm level is incorrect. It treats two firms in the same state as independent evidence about the state-level policy, which overstates the information content of the data. This is perhaps the most common clustering error in the corporate finance literature ([Frank and Goyal, 2024](#)).

Second, fixed effects are not a substitute for clustering. Firm fixed effects absorb time-invariant unobservables; they do not address serial correlation. You need both firm fixed effects *and* cluster-robust standard errors. Including one does not excuse omitting the other ([Frank and Goyal, 2024](#)).

Third, be cautious when the number of clusters is small. Cluster-robust standard errors rely on asymptotic theory that requires the number of clusters to grow large. With only 10 or 15 states adopting a law, the clustered standard errors may themselves be unreliable. In such cases, consider wild cluster bootstrap methods or randomization inference as alternatives. [Bertrand et al. \(2004\)](#) discuss several approaches; see also the treatments in [Roth et al. \(2023\)](#) for recent guidance specific to DiD.

6.5.2 A checklist of common pitfalls

The mistakes below are ordered roughly by how often they appear in published work. Each has been discussed at the relevant point in the chapter; this list gathers them in one place for reference.

Common Mistake 17: title=Pitfall 1: Using TWFE with staggered timing without checking

If treatment timing varies across units, the standard TWFE coefficient $\hat{\beta}$ is a weighted average of clean and forbidden comparisons (Section 6.3). Use a modern estimator or, at minimum, report the Goodman-Bacon decomposition to assess how much weight falls on forbidden comparisons.

Common Mistake 18: title=Pitfall 2: Clustering at the wrong level

Cluster at the level of treatment assignment. If the policy varies across states, cluster at the state level, not the firm level. Heteroskedasticity-robust standard errors without clustering are also insufficient when there is serial correlation within units.

Common Mistake 19: title=Pitfall 3: Treating pre-trends as proof of parallel trends

Flat pre-treatment coefficients are consistent with parallel trends but do not prove it. Pre-tests have low power, and conditioning on passing the pre-test distorts post-treatment inference (Section 6.2). Always report the Rambachan-Roth sensitivity analysis alongside the event study plot.

Common Mistake 20: title=Pitfall 4: Ignoring the functional form choice

Parallel trends in levels is a different economic assumption from parallel trends in logs. The choice between them is a substantive modeling decision about how untreated potential outcomes are generated, not a robustness check (Section 6.2). Be explicit about the economic reasoning behind your choice.

Common Mistake 21: title=Pitfall 5: Forgetting about anticipation effects

If firms adjust behavior before the policy takes effect (because the legislation was publicly debated, or because the effective date was announced in advance), the pre-treatment period is contaminated. The “before” observations no longer represent the untreated state, and the DiD estimate is biased. Look for signs of anticipation in the event study plot (a drift beginning before t_0) and consider excluding periods immediately before the treatment.

Common Mistake 22: title=Pitfall 6: Not showing the raw data

Before running any regression, plot the raw average outcomes for treated and control groups over time. This simple graph reveals whether the groups are on similar trajectories, whether there are obvious level differences, and whether the treatment effect is visible to the naked eye. If the treatment effect is not apparent in the raw data, a sophisticated estimator that finds it should be treated with skepticism.

6.5.3 DiD in the structural/causal framework

This chapter has presented DiD as a method for estimating the *average treatment effect on the treated*, the causal effect of a specific policy on the specific units that were subject to it. It is time to step back and ask: what does this contribute to our understanding of corporate finance, and what does it leave out?

What DiD tells us. DiD answers a well-defined causal question: did this policy change cause these firms to alter their behavior? When a state strengthens creditor rights and we estimate that treated firms increased leverage by 2 percentage points, we have learned something concrete. The policy had a measurable effect on corporate financial structure. This is valuable for policy evaluation. A legislator considering similar legislation in another state has direct evidence of what happened when other states adopted it.

What DiD does not tell us. The list is important, and it connects directly to the structural models of Part III.

DiD does not tell us *why* leverage changed. Did firms borrow more because stronger creditor rights reduced the expected cost of financial distress? Because banks were willing to lend on better terms? Because equity became relatively more expensive? The DiD estimate is silent on the mechanism. To identify the channel, we need a structural model that specifies how creditor rights enter the firm's optimization problem, something along the lines of the trade-off models in the Capital Structure chapter.

DiD does not tell us what would happen under a *different* policy. If the creditor rights reform we studied strengthened secured creditor priority, we cannot extrapolate to a reform that instead shortened bankruptcy proceedings. These are different treatments with potentially different effects, even if both fall under the heading of "creditor rights." Structural models, because they specify the mapping from policy parameters to firm behavior, can in principle evaluate counterfactual policies that were never implemented. DiD can only evaluate policies that were.

DiD does not tell us about *general equilibrium effects*. The *ATT* measures the partial equilibrium impact on treated firms, holding constant the behavior of untreated firms, factor markets, and product markets. But a nationwide creditor rights reform might affect the supply of credit, the equilibrium interest rate, or the competitive dynamics of entire industries. These effects operate through channels that DiD cannot isolate without additional structure.

DiD does not tell us whether the policy was *welfare-improving*. Even if leverage increased, we do not know whether this moved firms closer to or further from their optimal capital structure. An increase in leverage could reflect efficient use of debt tax shields or value-destroying overleverage. Welfare evaluation requires a model that defines optimality.

The structural content hidden inside DiD. A recurring theme of this chapter (and of this book) is that even "design-based" causal methods embed structural assumptions. In DiD, the parallel trends assumption restricts the data-generating process for untreated

potential outcomes. The additive model $Y_{it}(0) = \alpha_i + \lambda_t + v_{it}$ is a structural model, albeit a minimal one. The choice between levels and logs is a structural choice about how firms respond to aggregate shocks (Section 6.2). Even the decision to study leverage rather than debt levels or the debt-to-equity ratio embeds a view about which economic quantity the policy acts upon.

This is not a criticism of DiD. Every empirical method requires assumptions, and DiD's assumptions are comparatively mild and partially testable. The point is that the distinction between “structural” and “causal” is a continuum, not a binary. DiD sits closer to the causal end, but it is not assumption-free. Recognizing this helps you write better papers: state your parallel trends assumption explicitly, defend your functional form choice with economic reasoning, and be transparent about what your estimate does and does not identify.

The complementarity of approaches. The deepest implication of this chapter for the rest of the book is that structural and causal methods are complements, not substitutes. Consider how a serious research program on creditor rights and capital structure might proceed:

Step 1: DiD. Use staggered adoption of creditor rights laws to estimate the *ATT* on leverage. This establishes that the policy has a *real, measurable effect*. Firms change their financing in response to legal institutions. Sections 6.1 to 6.4 provide the tools.

Step 2: Heterogeneity analysis. Examine whether the treatment effect varies by firm characteristics (size, profitability, prior leverage, industry). This is informative about the mechanism: if the effect is concentrated among firms near financial distress, the creditor rights channel probably operates through the cost of distress. Book 2 develops the tools for heterogeneous treatment effects systematically.

Step 3: Structural modeling. Build a model of capital structure that includes creditor rights as a parameter. Estimate the structural model on the data and use the DiD estimate as an external validation: does the structurally-implied treatment effect match the causally-estimated one? If yes, the model has passed a demanding test. If not, the model is missing something important. This is the approach advocated by [Frank and Goyal \(2024\)](#) and developed in the chapters of Part III.

In this program, DiD provides the empirical target; structural modeling provides the economic interpretation. Neither alone is sufficient. Together, they are considerably more than the sum of their parts.

Key Insight 23

DiD identifies what happened. Structural models explain why it happened and what else might happen. The best empirical corporate finance uses both: DiD to establish credible causal effects, structural models to interpret them and evaluate counterfactuals. The organizing framework of this book (structural versus causal) is not about choosing sides. It is about understanding what each approach contributes and using them in combination.

6.5.4 Preview: what comes next

This chapter has provided the foundations of DiD for corporate finance applications. Several important extensions are deferred to Book 2, where they can be developed in full technical detail:

Heterogeneous treatment effects. The staggered adoption problem is fundamentally a problem of treatment effect heterogeneity. Book 2 develops the full theory of conditional average treatment effects (CATE) and shows how to estimate, test, and interpret heterogeneous effects systematically.

DiD-IV designs. In some applications, the policy “treatment” is not perfectly captured by a binary indicator (for instance, banking deregulation affects firms differentially depending on their reliance on bank finance). Combining DiD with IV allows the researcher to exploit both the timing variation and cross-sectional variation in exposure. This is an active research frontier.

Continuous and non-binary treatments. Many corporate finance treatments are not binary: the size of a tax change, the stringency of a regulation, the magnitude of a credit supply shock. Extending DiD to continuous treatments raises additional identification challenges that require the tools of Book 2.

Synthetic control methods. When the number of treated units is very small (a single country adopts a reform), the standard DiD comparison group may not be credible. Synthetic control methods construct a data-driven counterfactual by weighting control units to match the treated unit’s pre-treatment trajectory. These methods are increasingly common in corporate finance and are covered in Book 2.

Chapter summary. Difference-in-differences exploits the timing of a policy change to estimate its causal effect by comparing changes in outcomes across treated and control

groups. In its simplest form (Section 6.1), the method requires only two groups, two periods, and the parallel trends assumption. With multiple time periods (Section 6.2), we can examine dynamic effects and partially test whether parallel trends is plausible, though we must be careful about the pre-testing problem and the implicit structural content of the functional form choice. With staggered treatment timing (Section 6.3), the standard TWFE estimator can fail badly in the presence of treatment effect heterogeneity, producing biased estimates that may have the wrong sign. Modern estimators (Section 6.4) resolve this problem by avoiding forbidden comparisons and estimating group-time specific treatment effects under clearly stated assumptions. Throughout, the researcher must cluster standard errors correctly, be transparent about identifying assumptions, and recognize that DiD, for all its appeal, tells us what happened, not why.

Chapter 7

Machine Learning for Corporate Finance

7.1 Why Machine Learning?

Which firms will default in the next year? Which firms will become financially constrained? Which 10-K filings reveal unusually high risk or managerial evasiveness? Which private firms are likely acquisition targets? These are familiar corporate finance questions, but they are not all the same kind of question. Some ask for causal effects, some ask for mechanisms, and others ask for predictions or for measurements from data that are too high-dimensional to be handled by ordinary regression practice.

The methods studied in the preceding chapters are designed mainly for causal questions. Instrumental variables, difference-in-differences, regression discontinuity, and event studies all ask questions of the form: what is the effect of treatment D on outcome Y ? The central concern is internal validity. Is the source of variation as good as random? Are treated and control firms comparable in the relevant sense? Does the estimated effect have a credible causal interpretation?

Machine learning enters with a different discipline. Its central evaluation criterion is performance on observations not used to train the model. A model that fits the training sample but fails on new firms, new years, or new filings has failed at prediction. Cross-validation, holdout samples, regularization, and test-set evaluation all enforce this discipline.

It is tempting to say that causal methods are about internal validity and machine learning is about external validity. That is almost right, but it is a little too strong. Out-of-sample prediction is related to external validity, because both ask whether a result travels beyond the data used to produce it. But the two are not identical. A random train-test split within Compustat may show that a model predicts other public U.S. firms in the same historical

period. It does not by itself show that the model generalizes to private firms, foreign firms, crisis periods, a different disclosure regime, or a world after the policy being studied has changed firms' behavior. A more precise statement is this:

Key Insight. *Machine learning gives empirical researchers disciplined tools for assessing out-of-sample performance. Out-of-sample performance is one component of external validity, but it is not a substitute for economic judgment about the population, period, institution, and policy environment to which the researcher wants to generalize.*

Internal validity, external validity, and the credibility revolution. Recall from the Structural and Causal Approaches chapter that the credibility revolution associated with Angrist and Pischke (2010) placed internal validity at the center of empirical practice. Clean identification became the hallmark of credible research, for good reason: much of the pre-revolution work in empirical economics was plagued by endogeneity problems that made causal claims difficult to sustain. But the emphasis came at a cost. As Imbens (2010) acknowledged, the cleanest causal designs often exploit narrow and unusual sources of variation. The resulting estimates apply to specific populations in specific settings: firms incorporated in the particular state that changed its takeover law, or banks affected by a particular liquidity shock in a particular year. Whether these estimates tell us anything about firms in general is a separate question that the causal framework does not directly address.

Deaton (2010) and Heckman and Pinto (2022) pressed this point from different angles, and even Imbens, a leading architect of the modern causal approach, conceded its force. Yet within the causal tradition itself, there has been relatively little methodological work on how to assess or achieve external validity.

These are not just abstract methodological disputes; in corporate finance they arise constantly. A study exploiting a natural disaster as an instrument for bank lending might be beautifully identified, but to understand how credit contractions generally affect firm investment we need to know whether the estimate generalizes. A difference-in-differences design using a state-level policy change might cleanly estimate the effect on firms in that state, but to advise a different state we need external validity.

Where machine learning fits. Machine learning, as a field, is organized around predictive validation in a way that econometrics traditionally has not been. The fundamental evaluation criterion in ML is out-of-sample prediction accuracy. Cross-validation, held-out test sets, and regularization are all devices for ensuring that a model generalizes beyond its training sample, and ML researchers report performance on data the model has not seen.

This kind of evaluation answers part of what the critics of the credibility revolution worried was missing. But it does not answer the whole concern. A leave-firm-out split within Compustat tests whether a model trained on some U.S. public firms predicts other U.S. public firms in the same historical period. It does not test generalization to private firms, foreign firms, crisis periods, a different disclosure regime, or a world after the policy being studied has changed firm behavior. Predictive validation and external validity overlap, but they are not the same thing. Predictive validation is one ingredient in an external-validity argument, and the rest of the argument has to come from economic judgment about the target population.

This observation helps clarify why a chapter on machine learning belongs in a book organized around the structural–causal distinction. ML is not simply “another set of tools.” It represents a third evaluative standard, one that complements both the structural and causal traditions:

Structural work prioritizes economic mechanism and counterfactual discipline: can the model tell us what would happen under conditions we have not observed?

Causal work prioritizes internal validity and transparent identification: is the estimated effect credible for the population studied?

Machine learning prioritizes predictive validation: does the model work when evaluated on data not used to fit it?

Each addresses a different threat to credible empirical work. None is sufficient by itself. A structural model that fits well in sample but whose counterfactuals depend on untestable functional form assumptions lacks external credibility. A causal estimate that is impeccably identified but applies only to firms in one state in one year lacks generalizability. A prediction model that forecasts well out of sample but cannot tell us *why* it works lacks economic content. Good empirical research in corporate finance often requires some combination of all three.

Prediction versus causal inference. Mullainathan and Spiess (2017) formalized the distinction between prediction and causal inference as it applies to empirical economics. Their key insight was that many economically important questions are, at bottom, prediction problems, or at least involve prediction as a critical component. A credit rating agency wants to predict default. A regulator wants to classify firms as financially constrained or unconstrained. A researcher constructing a text-based measure of managerial tone from

earnings call transcripts is solving a prediction problem: given the words in the transcript, predict whether the tone is positive or negative.

Even in research that is ultimately about causal effects, prediction appears as an intermediate step. The first stage of an instrumental variables estimator contains a conditional-expectation problem: project the endogenous variable on the instrument and controls. The validity of IV does not rest on the predictive performance of this projection (it rests on relevance, exclusion, and, in LATE settings, monotonicity), but the projection step itself is a prediction problem in the technical sense. When difference-in-differences is implemented with many potential control variables, choosing among them is partly a prediction problem too. The recently developed technique of double/debiased machine learning ([Chernozhukov, Chetverikov, Demirer, Duflo, Hansen, Newey, and Robins, 2018](#)) makes this division of labor explicit. It uses ML methods for the prediction components of a causal estimation problem while preserving valid inference on the causal parameter of interest, provided the identifying assumptions hold.

The relationship between prediction and causation also runs in the other direction. A model that predicts firm outcomes well on new data must be capturing some stable features of the data-generating process. If the standard leverage factors identified in [Frank and Goyal \(2009\)](#) predict well out of sample, in different time periods and in different countries, that tells us something meaningful about the robustness of those empirical regularities, even if the prediction model does not tell us which factors *cause* firms to choose their leverage. As [Frank and Goyal \(2024\)](#) argue, external validity of this kind can itself be viewed as a form of evidence, one that complements the internal validity provided by causal designs.

What this chapter covers, and what it does not. Machine learning is an enormous field. Entire textbooks ([Hastie, Tibshirani, and Friedman, 2009](#); [James, Witten, Hastie, and Tibshirani, 2021](#)) are devoted to it, and the literature is expanding rapidly in directions that include deep learning, reinforcement learning, large language models, and generative AI. We cannot cover all of this in a single chapter, nor would it be appropriate to try. Our goal is more focused: to teach you the ML concepts and tools that a corporate finance researcher genuinely needs.

These turn out to be organized around four themes:

1. **Prediction.** Bias-variance, cross-validation, regularization (LASSO, Ridge, elastic net), and flexible methods (trees, forests, boosting). Predict default, distress, acquisition, payout cuts, covenant violations, credit ratings, investment, or leverage (Sections [7.2](#) to [7.4](#)).

2. **Classification.** Many corporate finance prediction problems are rare-event classification. The right metrics are not accuracy but ranking metrics (AUC, precision-recall), threshold metrics, and calibration (Section 7.5).
3. **Measurement from text.** Converting unstructured text (10-Ks, earnings calls, patents, contracts) into quantitative variables. This has been the most influential ML application in corporate finance to date, and the rise of large language models has accelerated it sharply (Section 7.6).
4. **Causal support.** Double/debiased machine learning and causal forests. These methods use ML for nuisance estimation or heterogeneity search while preserving identification from a separate, credible research design (Section 7.7).

We close with a decision framework for when to use ML methods in your own work (Section 7.8) and a short connection back to the structural/causal framework (Section 7.9).

Throughout the chapter we continue the running example from earlier chapters: leverage and firm investment. In the context of this chapter the example takes a new form. Instead of asking “what is the causal effect of leverage on investment?” (the question that motivated the earlier methods chapters), we now ask: can we *predict* leverage well? Which variables matter most for prediction? How do the variables selected by ML methods compare to those identified as empirically robust by other approaches? These prediction questions turn out to complement the causal questions from earlier chapters in revealing ways. When the chapter turns to classification (Section 7.5), the natural outcome is no longer leverage but distress, which leads directly into the discussion in the Financial Constraints chapter.

Students wanting a more comprehensive treatment of machine learning methods should consult [Hastie et al. \(2009\)](#) for the statistical foundations or [James et al. \(2021\)](#) for a more applied introduction. [Athey and Imbens \(2019\)](#) provide an excellent survey of ML methods aimed at economists. Our coverage here is selective, aimed at what a corporate finance researcher needs to know.

7.2 Prediction, Validation, and Data Leakage

In every methods chapter so far we have evaluated estimators primarily by asking whether they are consistent or unbiased. An instrument is valid if it produces a consistent estimate of the causal effect. A difference-in-differences design is credible if the parallel trends assumption holds and the estimator is approximately unbiased. This emphasis on bias has a long and distinguished history in econometrics, and for causal inference it is exactly the right focus.

But for prediction, bias is only half the story. A predictor can be unbiased yet terrible, and a biased predictor can be excellent. Understanding why requires the single most important idea in machine learning: the bias-variance tradeoff.

7.2.1 The Bias-Variance Tradeoff

Let Y_i be an outcome for firm-year observation i and let X_i be a vector of variables observed before the outcome is realized. In a prediction problem the object of interest is the conditional expectation $m(x) = \mathbb{E}[Y_i | X_i = x]$, or, for a binary outcome, the conditional probability $p(x) = \mathbb{P}(Y_i = 1 | X_i = x)$. The researcher estimates a function $\hat{m}(x)$ using a training sample and then evaluates its performance on data not used to fit the model. This last phrase is essential: in-sample fit is cheap; the point is performance on observations the model did not see.

Suppose the true data-generating process is $Y_i = f(X_i) + \varepsilon_i$ with $\mathbb{E}[\varepsilon_i | X_i] = 0$, and suppose that $\hat{f}$ is estimated on a random training sample. At a new point x_0 , the expected prediction error decomposes as

$$\mathbb{E}[(Y_0 - \hat{f}(x_0))^2] = \underbrace{(f(x_0) - \mathbb{E}[\hat{f}(x_0)])^2}_{\text{bias}^2} + \underbrace{\text{Var}(\hat{f}(x_0))}_{\text{variance}} + \underbrace{\sigma^2}_{\text{irreducible noise}}. \quad (7.1)$$

Bias measures how far the average prediction is from the truth. *Variance* measures how much the prediction changes when we draw a different training sample. The *irreducible error* σ^2 is the noise in the data: it cannot be reduced by any model. The tradeoff arises because bias and variance typically move in opposite directions as model complexity increases. A simple model has high bias but low variance; a complex model has low bias but high variance. The optimal model minimizes the sum, not the bias alone.

To see the tradeoff concretely, simulate from a known cubic function of log assets:

$$Y_i = f(X_i) + \varepsilon_i, \quad f(x) = 0.15 + 0.12x - 0.015x^2 + 0.0008x^3, \quad (7.2)$$

with $\varepsilon_i \sim N(0, 0.06^2)$. Figure 7.1 plots polynomial fits of degree 1, 3, and 20 to a training sample of 40 observations. The degree-1 polynomial misses the curvature. The degree-3 polynomial captures it well. The degree-20 polynomial fits nearly every training point exactly, but its predictions between data points are wild. Its training error is essentially zero; its test error, computed on a separate sample of 500 firms from the same population, is much worse than the degree-3 fit.

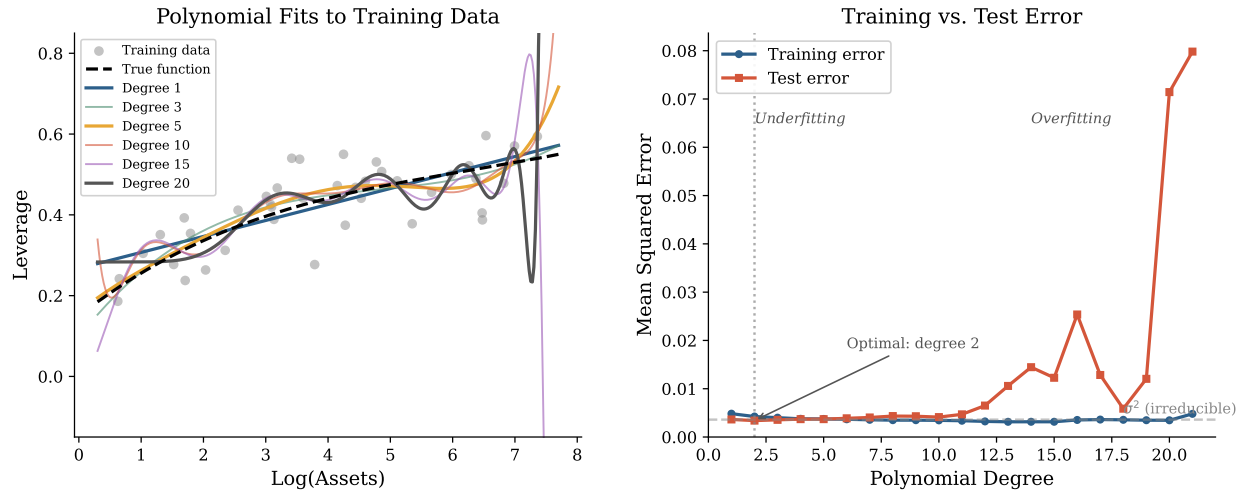


Figure 7.1. Overfitting in polynomial regression. *Left:* Polynomial fits of increasing degree to 40 simulated observations of leverage on log assets; dashed line is the true function. *Right:* Mean squared error on training data (blue) and held-out test data (red). Training error declines monotonically with degree. Test error is U-shaped, reaching a minimum near degree 2–3.

Figure 7.2 makes the decomposition visible by repeating the fitting exercise 200 times on fresh samples and plotting the ensemble of fits at degrees 1, 5, and 20. At degree 1 the fits are tightly bunched but systematically biased. At degree 20 the fits splay apart: the average is close to the truth (low bias) but any single fit can be far off (high variance). Degree 5 is the sweet spot. The bottom panel reports bias-squared, variance, and total MSE as functions of degree.

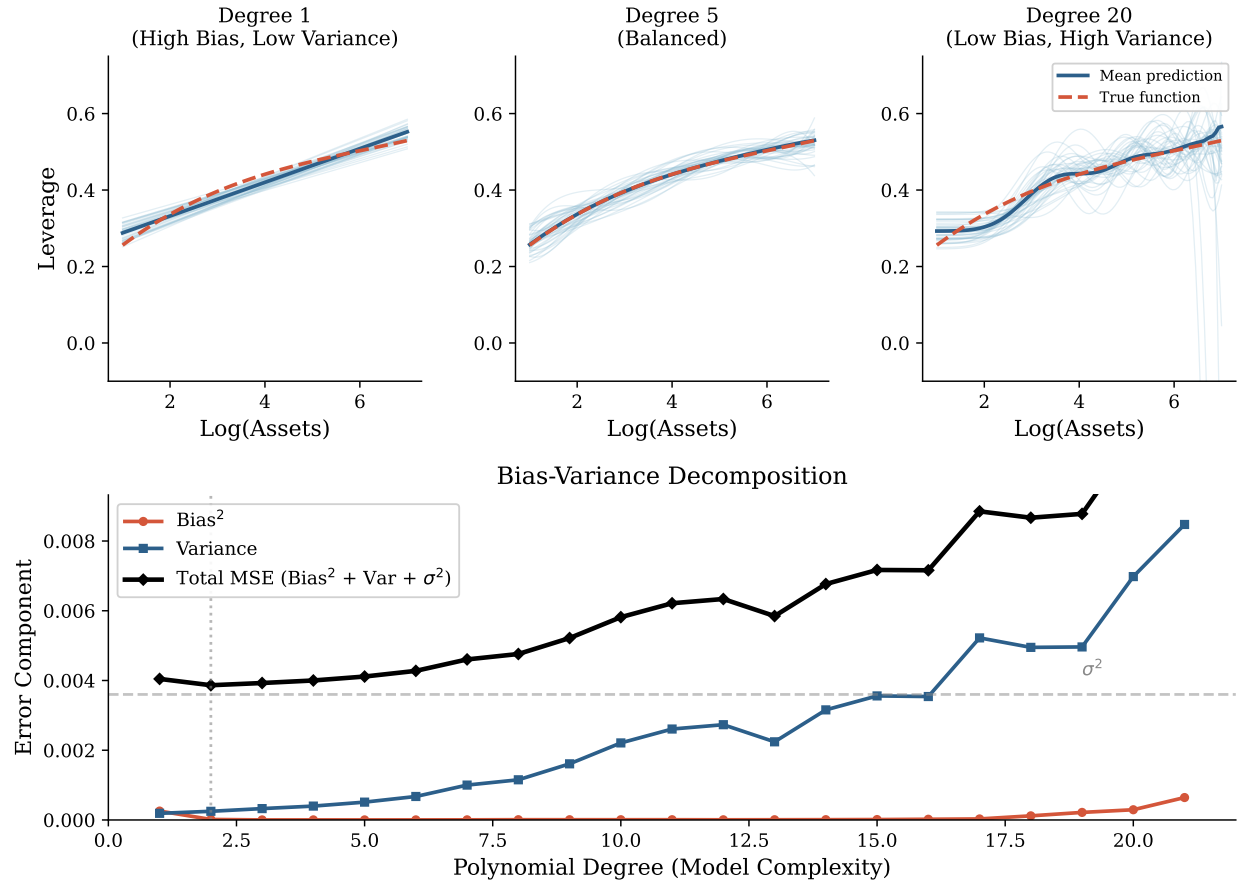


Figure 7.2. Bias-variance decomposition. *Top row:* 200 polynomial fits to independent random samples. *Bottom:* Bias-squared, variance, and total MSE as functions of polynomial degree.

The corporate finance version of the problem is not a toy. A researcher may have hundreds of candidate predictors from Compustat, CRSP, Dealscan, patent data, textual measures, insider trading data, institutional ownership, and macroeconomic variables. Many are correlated. Many are weak. Some are useful only in interactions. Some are useful in one period but not another. Without discipline, the researcher can easily fit the idiosyncrasies of one sample.

7.2.2 Cross-Validation

The bias-variance decomposition in equation (7.1) is conceptually clean but requires knowledge of the true function f , which we never have. Cross-validation provides a practical substitute. It estimates out-of-sample prediction error using only the data at hand.

Divide the data into K roughly equal groups (“folds”). For each fold $k = 1, \dots, K$, hold out fold k , fit the model on the remaining $K - 1$ folds, and compute the prediction error on the held-out fold. The average held-out error across folds is the cross-validated loss. The

procedure is used to choose model complexity: the LASSO penalty, the depth of a tree, the number of trees in a forest, or the learning rate in a boosting algorithm.

Cross-validation is not magic. It estimates performance for the prediction problem encoded by the split. If the split is wrong, the answer is wrong. This point is especially important in corporate finance because the data are usually panels.

7.2.3 Panel Cross-Validation and Leakage

Most corporate finance data consist of firms observed over time. In such a setting, “out of sample” can mean several distinct things:

1. **New observations of firms already seen.** Can the model predict next year’s leverage, investment, or default for firms already in the training sample?
2. **New firms.** Can the model predict outcomes for firms never seen in training?
3. **New time periods.** Can a model trained on the past predict the future?
4. **New regimes.** Can the model predict after a crisis, regulatory change, accounting change, or technological shift?

These are different questions. A validation scheme answers only the one it is designed to answer.

Table 7.1 illustrates the difference using a simple simulation. The DGP has 200 firms observed for 20 years; leverage depends on time-varying firm characteristics and on a firm-specific fixed component. A prediction model is fitted with and without firm fixed effects (dummies) included as features. The same model is then evaluated under three different cross-validation schemes.

Table 7.1. Panel cross-validation and leakage. Out-of-sample R^2 under three CV schemes, with and without firm dummies as predictors. DGP: 200 firms, 20 years, leverage depends on time-varying characteristics and a firm-specific component.

Validation scheme	Without firm dummies	With firm dummies	Interpretation
Random 5-fold	0.265	0.734	Splits firm-years at random; test obs. often belong to firms seen in training.
Leave-firm-out	0.254	0.253	Holds out entire firms. Tests prediction for firms never seen in training.
Leave-year-out	0.257	0.736	Holds out calendar years. Tests prediction for the same firms in unseen years.

The random-fold scheme suggests that firm dummies produce excellent predictions. That is misleading if the intended application is prediction for new firms. Under random folds the model has seen most firms during training and can simply memorize their fixed components. When entire firms are held out, the apparent gain from firm dummies disappears completely. Leave-year-out validation produces a high R^2 because it answers yet another question: prediction for the same firms in new years.

Common Mistake. *Using random K -fold cross-validation on a firm-year panel without asking what kind of generalization is relevant. Random folds usually leak information across the panel: they reward memorization of firm identity rather than prediction for genuinely new firms or future periods.*

The general principle is that the cross-validation design should mirror the intended use case. If a model is to be deployed on firms in a new country, leave-country-out cross-validation is the right benchmark. If a model is to be deployed on a future year, walk-forward validation is the right benchmark. The default of random K -fold is almost always too forgiving.

7.2.4 Walk-Forward Validation

In many finance applications the most natural split is chronological. A researcher who claims to predict future distress should train on past data, choose tuning parameters on a

later validation period, and report final performance on an even later test period. A simple design is:

train: 1980–2000,
 validate: 2001–2005,
 test: 2006–2010,

then roll the window forward. This walk-forward design respects time. It also makes visible whether performance deteriorates during recessions, credit booms, accounting changes, or shifts in disclosure practice.

The timing of variables matters as much as the split. A firm-year observation should include only information that would have been known at the prediction date. Accounting data become public with a delay ([Frank and Goyal, 2024](#)). Some databases contain restated values. Industry classifications and link tables can be revised. Macro data can be revised after release. Textual filings have filing dates, not merely fiscal-year dates. If these timing issues are ignored, the model may look successful because it has used information from the future.

Key Insight. *Before reporting prediction results, define the prediction date. Then ask, for every variable, whether it was observable at that date. In corporate finance, look-ahead bias is usually more dangerous than the choice between two adjacent algorithms.*

Connection to Framework. *The earlier methods chapters were preoccupied with bias: ruling out confounders, ensuring conditional independence, isolating exogenous variation. This chapter adds two new concerns: variance, and what counts as “out of sample.” Together these define the predictive-validation standard. They are not substitutes for the identification machinery of earlier chapters, but they discipline a different kind of empirical claim: that a model trained on one set of observations will continue to work on others.*

7.3 Regularization and the Leverage Example

The bias-variance tradeoff established a principle: the best predictor is not always the one with the smallest in-sample residuals. Regularization methods put this principle to work. They modify the OLS objective function by adding a penalty for model complexity, deliberately introducing a small amount of bias in exchange for a potentially large reduction in variance.

7.3.1 LASSO, Ridge, and Elastic Net

We have N observations of a dependent variable Y_i (leverage) and a vector of p predictors X_i (lagged firm characteristics). OLS minimizes the sum of squared residuals. Ridge, LASSO, and elastic net add penalties:

$$\hat{\beta}^{\text{Ridge}} = \arg \min_{\beta} \sum_{i=1}^N (Y_i - X_i' \beta)^2 + \lambda \sum_{j=1}^p \beta_j^2, \quad (7.3)$$

$$\hat{\beta}^{\text{LASSO}} = \arg \min_{\beta} \sum_{i=1}^N (Y_i - X_i' \beta)^2 + \lambda \sum_{j=1}^p |\beta_j|, \quad (7.4)$$

$$\hat{\beta}^{\text{Enet}} = \arg \min_{\beta} \sum_{i=1}^N (Y_i - X_i' \beta)^2 + \lambda \left[\alpha \sum_{j=1}^p |\beta_j| + (1 - \alpha) \sum_{j=1}^p \beta_j^2 \right], \quad (7.5)$$

where $\lambda \geq 0$ is the penalty strength and $\alpha \in [0, 1]$ controls the LASSO/Ridge mix in the elastic net.

The key difference between Ridge and LASSO lies in the geometry of the penalty. Ridge penalizes the sum of squared coefficients (the L_2 norm). It shrinks all coefficients toward zero but never sets any exactly to zero. LASSO penalizes the sum of absolute values (the L_1 norm). Because the absolute-value penalty has a corner at zero, LASSO can set coefficients exactly to zero, performing variable selection. The elastic net combines both penalties. With $\alpha = 1$ it reduces to LASSO; with $\alpha = 0$ it reduces to Ridge. In practice, elastic net is often preferred when predictors are strongly correlated: LASSO tends to pick one of two highly correlated variables somewhat arbitrarily, while elastic net spreads the load.

The tuning parameter λ controls how heavily the model is penalized for complexity. At $\lambda = 0$ all three methods reduce to OLS. As $\lambda \rightarrow \infty$ all coefficients are driven to zero. Between these extremes lies the value of λ that minimizes out-of-sample prediction error, the bias-variance sweet spot. In practice, λ is chosen by cross-validation.

One practical point is important: all predictors should be standardized (subtract the mean, divide by the standard deviation) before applying regularization. The penalty treats coefficients equally, so a variable measured in large units (total assets in millions) would have a small coefficient and escape the penalty, while a variable measured in small units would be penalized heavily. Standardization removes this arbitrary dependence on units.

7.3.2 Which Variables Predict Leverage?

We now apply these methods to a concrete question: which firm characteristics best predict market leverage? This is a prediction version of the question that [Frank and Goyal \(2009\)](#) addressed through a robustness-based approach. They started with a large candidate set and tested each variable for stability of coefficient signs across subsamples, time periods, and countries. Five factors survived: industry median leverage, profitability, market-to-book, tangibility, and log of assets. We take a different approach: give LASSO the same data and an even broader set of candidates, and let the penalty decide what stays.

Our candidate set includes 34 variables drawn from Compustat, CRSP, and FRED. It contains the five Frank-Goyal reliable factors, additional firm characteristics that have appeared in the leverage literature (cash holdings, R&D intensity, SG&A, dividend policy, firm age, earnings volatility, growth measures), market-based variables from CRSP, and macroeconomic variables from FRED. All variables are lagged one year. The sample is 140,790 firm-years from 1988 to 2023.

Figure [7.3](#) shows the LASSO coefficient path. Reading from right to left, as the penalty λ decreases from its maximum toward zero, we see which variables enter the model first. The first variable to receive a nonzero coefficient is market-to-book, which enters with a large negative coefficient: the strongest single predictor of leverage. Industry median leverage enters next, followed by tangibility, profitability, and log of assets. All five Frank-Goyal reliable factors are among the first variables LASSO selects.

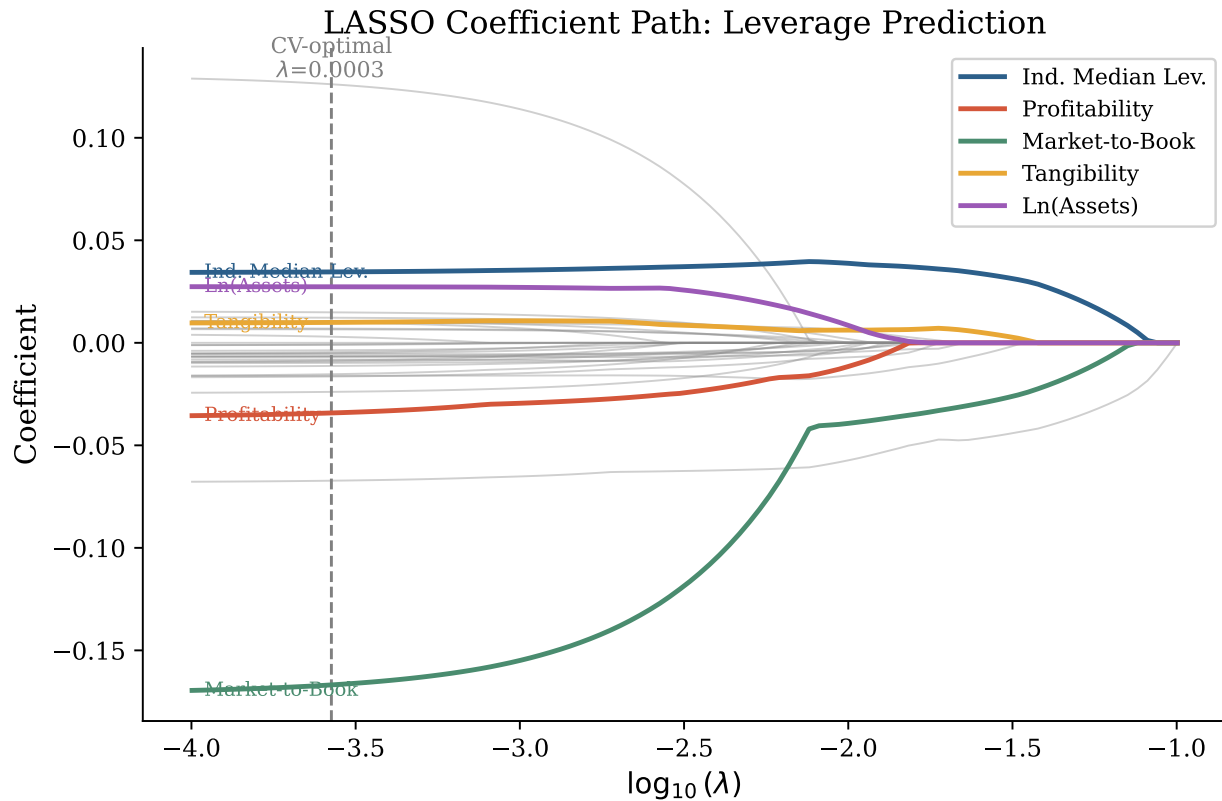


Figure 7.3. LASSO coefficient path for leverage prediction. Each curve traces one variable's coefficient as the penalty λ decreases (right to left). The five Frank-Goyal (2009) reliable factors are color-coded; other variables are in gray. The vertical dashed line marks the CV-optimal penalty.

Table 7.2 reports cross-validated R^2 for five models, all evaluated using leave-firm-out 5-fold cross-validation following the design developed in Section 7.2.3.

Table 7.2. Cross-validated prediction accuracy for leverage. All models are evaluated using leave-firm-out 5-fold cross-validation. Sample: 140,790 firm-years, 1988–2023.

Model	CV Strategy	R^2	RMSE	# Variables
OLS (FG reliable)	Leave-firm-out	0.260	0.198	5
OLS (all variables)	Leave-firm-out	0.390	0.179	34
LASSO	Leave-firm-out	0.390	0.179	33
Ridge	Leave-firm-out	0.390	0.179	34
Elastic Net	Leave-firm-out	0.390	0.179	33

Two findings deserve emphasis. First, regularization does not improve on kitchen-sink OLS in this setting. This may seem to contradict the overfitting lesson from Section 7.2.1, but it does not. With 140,000 observations and 34 variables, the observation-to-parameter

ratio is roughly 4,000 to 1. We are in the low-variance regime where OLS has more than enough data to estimate 34 coefficients reliably. The overfitting demonstrated in Figure 7.1 occurred with 40 observations and up to 20 parameters. Regularization earns its keep when the number of candidate variables is large relative to the sample size, as with hundreds of text-based features (Section 7.6) or high-dimensional interaction terms. The useful lesson from Table 7.2 is diagnostic. ML methods do not automatically improve prediction. Their value depends on the ratio of observations to candidate predictors, the correlation structure of the predictors, and the complexity of the true prediction function.

Second, the gap between the five-variable model ($R^2 = 0.260$) and the full models ($R^2 = 0.390$) shows that the additional variables collectively contribute meaningful cross-sectional predictive power beyond the reliable factors. LASSO retains most of them because even small individual contributions are worth keeping when the variance cost is negligible.

Table 7.3 reports which variables LASSO selects at the CV-optimal penalty, ranked by absolute coefficient magnitude. The five Frank-Goyal reliable factors occupy ranks 1, 4, 5, 6, and 15. The largest coefficient belongs to market-to-book (-0.167), which dominates cross-sectional leverage differences. Cash holdings ranks third (-0.067), confirming that cash-rich firms carry less leverage: a relationship that has been noted in the literature but was not among the original Frank-Goyal reliable set.

Table 7.3. Variables selected by LASSO at the CV-optimal penalty. “Strict” indicates the variable survives at a higher penalty where only five variables are nonzero. FG Status: **Reliable** = in Frank and Goyal (2009) reliable set; **Candidate** = considered but excluded; **New** = not in the original candidate set. Top 15 shown; 33 variables total are nonzero.

Rank	Variable	LASSO Coef.	Strict?	FG Status
1	Market-to-Book	−0.167	Yes	Reliable
2	Market-to-Book ²	+0.126	No	New
3	Cash / Assets	−0.067	Yes	New
4	Ind. Median Leverage	+0.035	Yes	Reliable
5	Profitability	−0.034	No	Reliable
6	Ln(Assets)	+0.027	No	Reliable
7	Dividend Payer	−0.024	No	Candidate
8	Net Working Capital	−0.016	No	Candidate
9	SG&A / Assets	−0.016	Yes	Candidate
10	Profitability ²	−0.015	No	New
11	Dividend Yield	+0.015	No	Candidate
12	Stock Volatility	+0.012	No	New
13	Depreciation	−0.011	No	Candidate
14	Asset Growth	+0.010	No	New
15	Tangibility	+0.010	Yes	Reliable

The “Strict” column shows which variables survive when the penalty is raised until only five remain. This strict LASSO set (industry median leverage, market-to-book, tangibility, cash ratio, SG&A) overlaps substantially with the Frank-Goyal reliable set but is not identical. It replaces profitability and log of assets with cash holdings and SG&A. The difference reflects the distinct selection criteria. Frank-Goyal tested for coefficient-sign stability across subsamples, while LASSO selects the five variables that jointly minimize prediction error. A variable can be a stable predictor (always the same sign) without being the most powerful predictor (largest reduction in out-of-sample error), and vice versa.

The presence of market-to-book squared at rank 2 is notable. LASSO finds strong evidence that the relationship between market-to-book and leverage is nonlinear: the negative effect of higher market-to-book flattens at high values. This curvature is invisible in the standard linear specification that dominates the literature. It does not mean the standard specification is wrong (the linear approximation captures the first-order effect), but it does suggest room for more flexible functional forms, at least for prediction purposes. Section 7.4 will return to this point.

Key Insight. *LASSO selects variables that predict well out of sample. Frank and Goyal (2009) selected variables whose coefficient signs are stable across subsamples and time periods. That these two very different approaches largely agree on the core leverage factors is strong evidence that those factors are genuinely robust, not artifacts of overfitting or specification search. Where the two approaches disagree (cash holdings, SG&A, nonlinearity in market-to-book), the disagreement is itself informative about the difference between predictive power and coefficient stability.*

7.3.3 Good Predictors Are Not Automatically Good Panel Regressors

A recurring confusion is to treat ML variable selection as a substitute for panel research design. It is not. A variable selected because it predicts cross-sectional leverage may add little to a firm fixed-effects regression. The reason is simple: the variable may predict which firms have high leverage, while fixed effects remove most of the between-firm variation.

This distinction is central in corporate finance. Firm size, tangibility, industry, age, payout status, and persistent profitability are powerful cross-sectional predictors. Many of them change slowly. They are useful for predicting leverage levels across firms. They are less useful for explaining why a given firm's leverage changes over time. Once firm fixed effects absorb persistent heterogeneity, the researcher is asking a different question.

This is precisely what the Panel Data chapter demonstrated for the leverage data. Taking the same Compustat panel used here and adding firm and year fixed effects, the within- R^2 falls to roughly 0.10, compared to the cross-sectional R^2 of 0.39 reported in Table 7.2. Adding the full 25-variable LASSO set to the panel regression raises the within- R^2 only to 0.13. Most of the cross-sectional predictive power LASSO exploits comes from between-firm differences that firm fixed effects absorb. We refer the reader to the panel regressions in the Panel Data chapter for the full empirical decomposition and for the corresponding discussion of clustering.

Key Insight. *Prediction and fixed-effects estimation use different variation. A variable can be excellent for predicting firm levels and nearly irrelevant for explaining within-firm changes. ML variable selection does not decide the research design. Cross-sectional prediction and within-firm estimation are answering different questions, and the answer to one does not determine the answer to the other.*

7.4 Trees, Forests, and Boosting

The regularization methods of Section 7.3 assume that leverage is a linear function of the predictors, or at least that the researcher specifies in advance which nonlinearities to include. We added quadratic terms for market-to-book and profitability, and LASSO rewarded that choice: market-to-book-squared had the second-largest coefficient. But we had to decide which curvatures to look for. Trees and random forests take a different approach. They let the data determine the functional form. This turns out to matter more than one might expect.

7.4.1 Regression Trees

A regression tree works by recursively splitting the sample into regions. At each step the algorithm considers every predictor and every possible cutoff, and chooses the split that most reduces the sum of squared residuals in the two resulting groups. Within each terminal region (“leaf”), the prediction is the average outcome of the observations that landed there. The process continues until a stopping rule is reached, typically a maximum depth or a minimum number of observations per leaf.

A depth-3 tree fitted to the Compustat leverage sample first splits on market-to-book at a threshold near 1.46. Firms with low market-to-book (value firms) go left and have substantially higher average leverage (around 0.32). Firms with high market-to-book (growth firms) go right with lower leverage (around 0.11). The tree has recovered, purely from the data, the same variable that LASSO identified as the dominant predictor. Within the low-market-to-book group, the next split is on cash holdings: cash-poor firms carry even more leverage. Within the high-market-to-book group, the tree splits on interest coverage. Interest coverage was not in the Frank-Goyal reliable set, but the tree elevates it to a prominent position because of its nonlinear relationship with leverage near financial distress.

A deep tree fits the training data closely but is sensitive to which particular firms appear in the sample. This is high variance. A shallow tree misses real patterns. This is high bias. The depth-3 tree just described is deliberately constrained to be interpretable. Its cross-validated R^2 of 0.43 is modest, though it already exceeds the full 34-variable OLS ($R^2 = 0.39$). Allowing the tree to grow to a CV-optimal depth of 10 raises the R^2 to 0.58. But even the best single tree cannot compete with the ensemble methods we turn to next.

7.4.2 Random Forests and Gradient Boosting

The fundamental insight of [Breiman \(2001\)](#) is that averaging many noisy but approximately unbiased trees produces a prediction with much less noise. A *random forest* grows hundreds of trees, each on a bootstrap sample drawn with replacement from the original data. At each split only a random subset of variables is considered, typically $\sqrt{p}$ of the p available predictors. This randomization ensures that the individual trees make different mistakes, so their errors partially cancel when averaged.

Why does averaging work? If the trees' prediction errors were perfectly correlated, the average would be just as noisy as any individual tree. The bootstrap and random-feature steps create trees whose errors are only modestly correlated, so the variance of the ensemble shrinks as more trees are added.

Figure 7.4 shows the bias-variance tradeoff in action. The left panel plots out-of-sample MSE as a function of the number of trees in the ensemble. A single tree has high prediction error; averaging reduces it sharply, with diminishing returns after about 100 trees. The right panel makes the mechanism visible. Three individual trees (gray) produce jagged predictions that jump erratically across firms, while the forest average (blue) traces a smooth curve that closely tracks actual leverage (red).

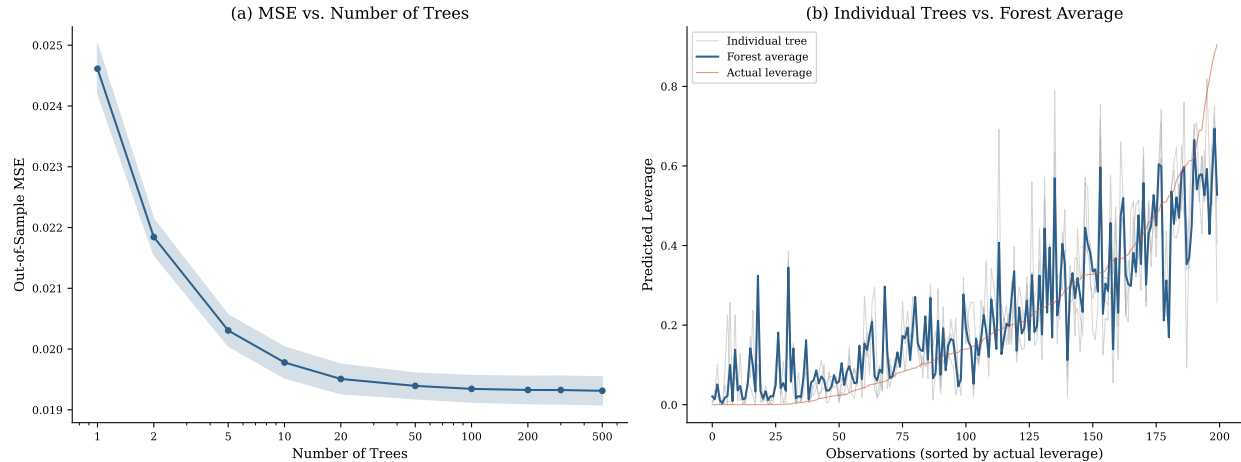


Figure 7.4. How random forests reduce variance. *Left:* Out-of-sample MSE as a function of the number of trees, with a shaded ± 1 SE band across 5 CV folds. *Right:* Predictions from three individual trees (gray) and the forest average (blue) for 200 firms sorted by actual leverage (red).

A closely related method is *gradient boosting* ([Friedman, 2001](#)). Instead of growing trees independently and averaging, boosting builds trees sequentially: each new tree is fitted to the residuals left over by the ensemble so far. Boosting often achieves slightly higher accuracy than a random forest, at the cost of more hyperparameters to tune and greater sensitivity to overfitting.

7.4.3 Trees vs. LASSO for Leverage Prediction

Table 7.4 collects the cross-validated R^2 for all models, from the five-variable OLS through tuned gradient boosted trees. All models are evaluated with the same leave-firm-out 5-fold cross-validation used in Section 7.3.

Table 7.4. Model comparison for leverage prediction. Same leave-firm-out 5-fold CV throughout, so comparisons are apples-to-apples.

Model	CV R^2	# Variables
OLS (FG reliable, 5 vars)	0.260	5
OLS (all variables)	0.390	34
LASSO	0.390	33
Tree, depth 3	0.430	34
Tree, CV-optimal depth (10)	0.580	34
Random forest, 500 trees	0.650	34
Gradient boosting, tuned	0.680	34

The results are striking. Tree-based methods substantially outperform the linear models. The random forest achieves a cross-validated R^2 of 0.65 and gradient boosting reaches 0.68, compared to 0.39 for LASSO and kitchen-sink OLS. Even a single depth-10 tree reaches 0.58. The progression from constrained depth-3 tree (0.43) through deeper single tree (0.58) to 500-tree forest (0.65) traces out the bias-variance tradeoff: more flexibility buys genuine predictive accuracy, and the ensemble delivers a large further gain by averaging away the variance.

These are larger improvements than one might have expected, and they tell us something important about the data. The standard linear specification, used in virtually every leverage regression in the literature, misses real predictive signal in the Compustat data. Where is that signal coming from?

Partial dependence. Figure 7.5 provides part of the answer. A partial dependence plot (PDP) shows how the forest's predicted leverage varies with one variable, averaging over the other predictors. The dashed red line in each panel shows the linear relationship implied by LASSO.

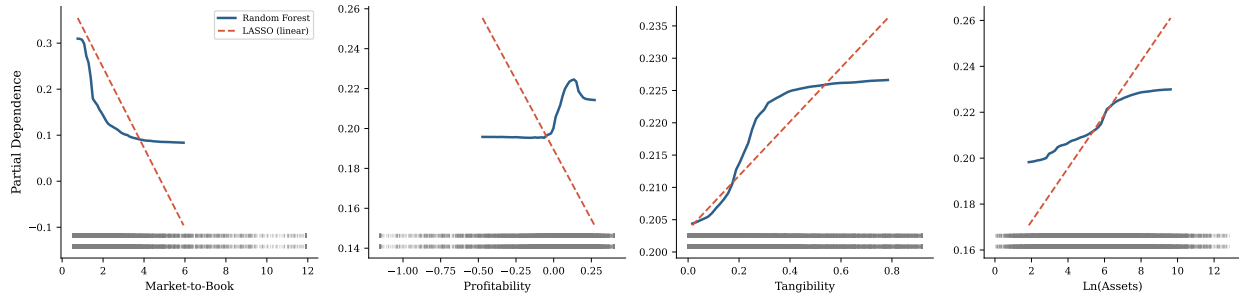


Figure 7.5. Partial dependence plots from the random forest for four key predictors. Dashed red lines show the linear relationship implied by LASSO. Market-to-book shows pronounced nonlinearity. Profitability shows a kink near zero. Tangibility levels off for highly tangible firms. Log assets is the most nearly linear. Rug plots along the x -axis show the data distribution.

For market-to-book, the forest reveals a steep negative slope that flattens sharply around market-to-book of 2. The LASSO linear fit, even with the quadratic term, overshoots at both ends; it predicts negative leverage for high-market-to-book firms, which is impossible. The forest respects the natural bounds of the data. For profitability, there is a clear kink near zero: moving from deeply unprofitable to breakeven has almost no effect on predicted leverage, but moving from breakeven to solidly profitable is associated with a noticeable decline. This asymmetry is invisible in a linear model. Tangibility shows a concave relationship that levels off above about 0.5. Log of assets is the most nearly linear of the four, suggesting the standard specification is adequate for this variable.

Nonlinearity alone does not account for the full gap between $R^2 = 0.39$ and $R^2 = 0.68$. The forest also captures high-order interactions: the effect of profitability on leverage may differ for high-market-to-book versus low-market-to-book firms, or for small versus large firms. These interaction effects are invisible to LASSO unless the researcher adds the right cross-terms by hand. The tree discovers them automatically through recursive splitting.

7.4.4 Interpreting Flexible Models

Researchers often want to know which variables “matter” in a forest. The standard tools are variable importance, permutation importance, partial dependence plots, accumulated local effects, and SHAP-type decompositions. These tools are useful but easy to overinterpret.

Variable importance measures how much a variable contributes to prediction inside a fitted model. It does not measure a causal effect. If two variables are highly correlated, importance can shift from one to the other depending on the algorithm and sample. Permutation importance can understate the role of a variable when another correlated variable substitutes for it. Partial dependence plots can average over unrealistic combinations of covariates. SHAP values decompose predictions, not economic mechanisms.

Common Mistake. *Reporting a random forest's variable-importance ranking as if it were a table of causal effects. Importance means predictive usefulness within a model and sample. It does not answer what would happen if a firm changed the variable.*

7.4.5 When to Use Trees vs. Linear Methods

Use **random forests** or **gradient boosting** when prediction accuracy is the primary goal. For cross-sectional prediction of leverage levels, the nonparametric methods are clearly superior. They also provide useful diagnostic outputs (variable importance rankings, partial dependence plots) that can reveal structure in the data a linear model would miss.

Use **LASSO** or **Ridge** when you want interpretable coefficient estimates, when you need to select variables for a subsequent panel regression, or when the question is about within-firm dynamics rather than cross-sectional levels. In a panel setting with firm fixed effects, the between-firm nonlinearities that give the forest its advantage are already absorbed.

The two approaches work well together. LASSO selects variables; the random forest reveals whether those variables have nonlinear or interactive effects the linear model missed; and the panel regression estimates the within-firm structure while controlling for unobserved heterogeneity.

One forward reference: the random forest algorithm is the building block for *causal forests* (Wager and Athey, 2018), which we introduce in Section 7.7. Causal forests adapt the forest machinery to estimate heterogeneous treatment effects, bridging the prediction focus of this section with the causal-inference focus of the earlier methods chapters.

Connection to Framework. *Trees and forests are prediction tools. They sit on the ML/predictive-validation side of the three-way framework from Section 7.1. They do not estimate causal effects. But their outputs carry substantive lessons. The large improvement over linear methods (R^2 of 0.68 vs. 0.39) tells us that the leverage data contain meaningful nonlinearities and interactions that the standard linear specification misses. At the same time, the same variables dominate both LASSO and the forest rankings, confirming that the Frank-Goyal reliable factors capture genuine features of the leverage decision, not artifacts of a particular statistical specification.*

7.5 Classification: Default, Distress, and Rare Events

Much of corporate finance prediction is classification rather than regression: default or no default, covenant violation or no violation, restatement or no restatement, acquisition

target or not, payout cut or no payout cut. These outcomes are often rare. Rare events require different metrics and a different way of thinking about model performance.

Let $Y_i \in \{0, 1\}$ indicate whether firm i defaults within the next year. A classifier produces a score

$$\hat{p}_i = \hat{p}(X_i) \approx \mathbb{P}(Y_i = 1 \mid X_i). \quad (7.6)$$

The score becomes a binary classification only after the researcher chooses a threshold c : predict default if $\hat{p}_i \geq c$. The threshold should depend on the economic problem. A lender screening applicants may choose a low threshold because missing a default is costly. A regulator deciding which firms to examine may choose a threshold that reflects supervisory capacity. A researcher studying distress may not need a hard classification at all; the probability score itself may be the object of interest.

7.5.1 Why Accuracy Is the Wrong Metric

If one percent of firms default in a given year, a model that predicts “no default” for every firm is 99 percent accurate and useless. The problem is that accuracy weighs false positives and false negatives equally, which is rarely appropriate for rare events with asymmetric costs.

Four entries define the confusion matrix: true positives (TP), false positives (FP), true negatives (TN), and false negatives (FN). From these we obtain:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad (7.7)$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \quad (7.8)$$

Precision asks: among firms predicted to default, how many actually default? Recall asks: among firms that default, how many did the model catch? There is a tradeoff. Lowering the threshold raises recall but lowers precision: more firms are flagged, catching more true defaults but generating more false alarms.

The ROC curve plots the true positive rate against the false positive rate as the threshold varies. The area under the ROC curve (AUC) is the probability that a random defaulting firm receives a higher score than a random non-defaulting firm. AUC is useful, but for rare events precision-recall curves can be more informative because they focus on performance among firms classified as risky.

Calibration is different from ranking. A model is well calibrated if firms assigned a 10 percent default probability default about 10 percent of the time. Calibration matters

whenever the predicted probability is used as a probability: pricing debt, setting loan-loss reserves, stress testing, or constructing a continuous distress measure for later regressions. A model can rank firms well and still produce poorly calibrated probabilities.

These metrics are easiest to see in default prediction, but they apply to any rare-event classification, including one we meet again when we turn to text. Suppose we want to predict whether two firms operate in the same product market, using the textual similarity of their 10-K business descriptions together with a few simple pairwise features such as the ratio of their sizes or the distance between their headquarters. To train and evaluate the classifier we need a label, and a natural benchmark is a standard industry code: assign the pair a one if the two firms share a 4-digit SIC code and a zero otherwise. The label is imperfect, and the imperfection is part of the lesson, because the standard codes miss exactly the product-market links a text measure is built to recover.

Two features of this problem govern how we evaluate it. First, the event is rare. The overwhelming majority of firm pairs are not in the same narrow industry, so a classifier that labels every pair “different” is right about ninety-nine percent of the time and tells us nothing, the same arithmetic that makes accuracy worthless for default. Ranking metrics and precision at a fixed flag rate are what reveal whether the score finds the few same-market pairs. Second, calibration has a concrete meaning here. Among the pairs the model assigns a probability near 0.30, about thirty percent should actually share the benchmark label. If far fewer do, the score still ranks pairs sensibly but cannot be read as a probability, which matters the moment those probabilities are used to weight or threshold pairs downstream.

Key Insight. *For rare corporate events, do not report accuracy alone. Report ranking performance (AUC), threshold-dependent performance (precision and recall at the relevant threshold), and calibration (calibration plot or calibration slope). A model can rank firms well and still produce poorly calibrated probabilities.*

7.5.2 A Default-Prediction Reporting Template

Table 7.5 shows a template for reporting default-prediction results. The exact data sources can vary (bankruptcy filings, credit-rating defaults, delistings for performance, negative book equity, or a constructed distress indicator), but the structure of the table should not.

Table 7.5. Stylized reporting template for default prediction. The numbers in this table are illustrative magnitudes drawn from the bankruptcy-prediction literature, not estimates produced for this book. A worked example using Compustat-CRSP delisting data is in the chapter’s online supplement.

Model	AUC	Precision at 5% flag	Recall at 5% flag	Brier score	Calibration slope
Logit, accounting ratios	0.80	0.10	0.50	0.020	1.00
LASSO logit	0.81	0.10	0.52	0.019	0.98
Random forest	0.88	0.14	0.68	0.015	0.85
Gradient boosting	0.90	0.16	0.72	0.014	0.82

Note: Numbers are illustrative. “Precision at 5% flag” reports precision when the threshold is set to flag the riskiest 5% of firm-years; “Recall at 5% flag” is the corresponding recall. Calibration slope of 1 indicates perfect calibration; values below 1 indicate the model is overconfident in its high-probability predictions, which is a common pattern with flexible methods.

The pattern these stylized numbers illustrate is typical. Flexible methods (random forest, gradient boosting) deliver substantially better ranking and Brier-score performance than logit, but their calibration slopes drift below one: among firms the forest assigns a high default probability, the realized default rate is lower than the model claims. This is the rare-event analogue of the cross-sectional vs. within-firm distinction from Section 7.3. The forest is excellent at sorting firms by risk and worse at producing probabilities that can be plugged into a pricing equation.

The key is not that one algorithm wins. The key is that the metric must match the use case. If the researcher wants to identify the most fragile firms for further study, ranking metrics may be enough. If the researcher wants to price default risk, calibration is essential. If the researcher wants to allocate costly supervisory attention, precision at a fixed review capacity may be the relevant object.

7.5.3 Constraint Indices as a Cautionary Case

Several widely used constraint indices can be reinterpreted as predictive scores, but they differ sharply in how each score was constructed. We return to these indices in detail in the Financial Constraints chapter; here we use them only to illustrate what the predictive-validation standard adds to a familiar debate.

Kaplan and Zingales (1997) qualitatively classified a small sample of low-dividend firms into five categories of financial constraint based on a careful reading of annual reports and management discussions. The resulting ordered classifications were then mapped

to firm characteristics through ordered logit regressions. The [Kaplan and Zingales](#) “KZ index” that subsequent researchers used is a linear combination of those characteristics with weights drawn from this ordered logit. The construction is small-sample, in-sample, and depends on the qualitative scheme used to label the original firms.

The Whited-Wu index has a quite different provenance. [Whited and Wu \(2006\)](#) estimate an investment Euler equation by GMM under the maintained assumption that firms face financing constraints, and they construct their index from the resulting structural estimates. The Whited-Wu index is not, in the first instance, a classifier. It is a structural object that is then used as a continuous measure of constraint intensity. The classification interpretation arises only when researchers split the WW score at a threshold.

The size-age (SA) index of [Hadlock and Pierce \(2010\)](#) is the closest of the three to a pure classification model. The authors qualitatively classify a sample of firms by reading 10-K filings, fit ordered logit models, and find that size and age survive specification searches as the most reliable predictors of qualitative constraint status. The SA index is the linear combination of size and age implied by that fit.

Viewed through the lens of this chapter, three features of this literature are notable. The labeled training samples are small (tens to a few hundred firms). Validation is largely in-sample. And the constructed indices are then applied to populations and periods very different from the training data, often without re-validation. The persistent debate over whether these indices “identify constrained firms” ([Farre-Mensa and Ljungqvist, 2016](#)) is, in part, a debate about how far a classifier trained on a small labeled sample generalizes to a broader population, which is precisely the predictive-validation question.

This is not a blanket criticism. These indices represented serious empirical work given the data and methods available at the time, and the Whited-Wu index has a structural rationale that does not reduce to classification at all. The observation is that the predictive-validation standard reframes a familiar problem. A researcher today using the SA index to label firms as constrained is implicitly assuming that a classifier trained on [Hadlock and Pierce](#)’s qualitative sample generalizes to her own population. A researcher using the WW index is implicitly assuming that the structural model underlying it applies to her sample. Both are empirical claims about transport, and both can be examined with the validation tools developed in this chapter. The Financial Constraints chapter returns to this point with the constraint-share framework and simulation evidence that the empirical content of these classifications varies sharply with how constrained the underlying population actually is.

Connection to Framework. *“Classifying firms by a model” and “identifying a causal effect on classified firms” are different operations. The first is a prediction or measurement problem and should be evaluated by predictive-validation criteria together*

with a defense of the construct. The second requires a separate identification argument. The Financial Constraints chapter develops both: it treats the constraint indices as scoring rules and considers what a regression of investment on cash flow, run separately in constrained and unconstrained subsamples, actually estimates.

7.6 Measurement from Text

The most important use of machine learning in corporate finance may not be prediction of outcomes. It may be measurement. Modern firms produce enormous amounts of text: 10-Ks, 10-Qs, 8-Ks, earnings-call transcripts, MD&A, conference presentations, proxy statements, loan contracts, merger agreements, patent abstracts, job postings, press releases, news articles, analyst reports, and social-media communications. Much of what we want to know about firms is contained in this text: risk, uncertainty, competition, strategy, managerial optimism, technological direction, covenant restrictiveness, customer concentration, climate exposure, supply-chain fragility, and corporate culture. We also care about what other people are saying about the firms [Antweiler and Frank \(2004\)](#).

The empirical task is to convert text into a variable. That variable may then be used as an outcome, treatment, control, instrument component, or mechanism. This is measurement, and it deserves the same care as any other measurement problem in corporate finance.

7.6.1 Dictionary Methods

The simplest approach is a dictionary. Count words belonging to categories: positive, negative, uncertain, litigious, modal, financial, technological, competitive, or climate-related. Dictionary methods are transparent and replicable. Their weakness is that meaning depends on context. The word “liability” is negative in ordinary language but routine in accounting. A word that signals risk in one industry may be boilerplate in another. Domain-specific dictionaries, such as the financial sentiment dictionary developed by [Loughran and McDonald \(2011\)](#), exist because generic sentiment dictionaries perform poorly in financial settings. The [Loughran and McDonald \(2016\)](#) survey provides an extensive review of dictionary-based methods in accounting and finance.

[Antweiler and Frank \(2004\)](#) used machine learning methods on Internet stock message boards in order to measure investor sentiment, and then connect the investor sentiment to stock returns. [Tetlock \(2007\)](#) used a dictionary based measure of pessimistic language in a daily Wall Street Journal column to predict stock returns. The application to corporate 10-K filings, especially through Loughran-McDonald sentiment, became standard in the

following decade. The general pattern is that text-based sentiment measures predict returns, trading volume, and earnings surprises with magnitudes that are economically modest but robust.

7.6.2 Supervised Text Classification

A supervised approach begins with labeled examples. Human coders classify sentences, paragraphs, filings, contracts, or calls. The researcher then trains a model to reproduce those labels from textual features. Features can be word counts, n -grams, topic weights, embeddings, or transformer-based representations. The trained model is then applied to the larger corpus.

This workflow is attractive because it can measure concepts that dictionaries miss. But it shifts the burden to label quality and validation. Who labeled the training data? Were coders given clear instructions? Do coders agree? Is the labeled sample representative of the documents to which the model will be applied? Does the model work in later years, different industries, or different disclosure regimes? These are the same questions raised by panel cross-validation in Section 7.2.3, applied now to documents instead of firm-years.

7.6.3 Embeddings, Topic Models, and Similarity

Embeddings represent text as vectors. Texts with similar meanings are placed near each other in the embedding space. This makes it possible to measure similarity between firms' business descriptions, patents, risk factors, conference-call answers, or merger rationales. Embeddings are useful when the researcher cares about semantic similarity rather than particular words.

Hoberg and Phillips (2016) provide an influential corporate finance application. They construct text-based industry classifications (TNIC) by computing cosine similarity between firms' 10-K business descriptions. Two firms are in the same text-based industry if their descriptions are sufficiently similar. The resulting classifications differ substantively from SIC and NAICS codes, capture product-market relationships that the standard codes miss, and have been used to study competition, mergers, and product differentiation. A similar logic underlies Kogan, Papanikolaou, Seru, and Stoffman (2017), who use text similarity between patents and prior art to measure patent quality and technological breakthrough.

The similarity is itself a measurement, and like any measurement it can capture the wrong thing. Two business descriptions may read alike because the firms sell similar products, which is the construct the measure intends to capture. They may also read alike because they share standard disclosure boilerplate, or because the two firms filed from the

same law firm's template. Every preprocessing choice is therefore a measurement decision rather than a mechanical step. Dropping firm names before computing similarity is the clearest case: removing a name can also strip a product word that genuinely separates one firm's market from another's.

Validating such a measure proceeds in layers, none of them causal. The first is face validity, checking that the nearest neighbors of a few recognizable firms look right. The second is comparison to SIC and NAICS, treated as noisy benchmarks rather than ground truth, since the motivation for a text measure is precisely that the standard codes miss product-market links. The third is out-of-time stability, asking whether tuning chosen in one period still produces sensible neighbors in a later one, the same out-of-time discipline introduced earlier for firm-year panels. The fourth is economic validation, asking whether high-similarity firms show more correlated sales growth or co-move more tightly around industry shocks. These are construct-validity checks. They ask whether the measure captures product-market proximity, not whether proximity causes any outcome.

A fully worked construction along these lines, building cosine-similarity peer sets from 10-K business descriptions and running all four validation layers end to end, belongs with the chapter's other implementations. We defer it to the Code Companion, where it awaits a redistributable sample of Item 1 business-description text keyed to firm-years and filing dates. The companion code stub records the intended inputs, outputs, and validation steps so the build drops in once that text sample is bundled.

Topic models such as latent Dirichlet allocation summarize a corpus by a small number of latent topics, each represented as a distribution over words. They have been used to characterize the content of 10-K risk-factor sections, conference calls, and analyst reports.

7.6.4 Large Language Models

Large language models extend this logic. They can classify passages, extract entities, summarize disclosures, identify contract terms, score tone, map business descriptions to product markets, and create structured variables from unstructured documents. In corporate finance, plausible LLM-based tasks include identifying whether a risk factor is boilerplate or firm-specific, extracting covenant restrictions from loan contracts, classifying whether an earnings-call answer is evasive, or mapping patent descriptions to technology classes.

The danger is that LLM outputs can appear authoritative even when they are unstable. The same prompt can produce different answers. Model providers can update models. Training data are often opaque. The model may have seen information that would not have

been available at the prediction date. The researcher may not be able to reproduce the exact output later. These are not minor implementation details. They are measurement issues.

Common Mistake. *Treating an LLM output as a clean variable. The output is the result of a complex generation process with stochasticity, version dependence, and possible contamination by training data the researcher cannot inspect. A responsible empirical practice documents the generation process as thoroughly as a hand-collected dataset.*

For empirical work, an LLM-generated variable should be documented as if it were a hand-collected dataset:

1. specify the source documents and dates;
2. preserve the exact prompts, model name, version, temperature, and date of query;
3. store the raw model outputs;
4. audit a sample by hand;
5. report inter-model or inter-prompt robustness when feasible;
6. validate the measure against external outcomes or human labels;
7. discuss what construct the measure captures and what it does not.

7.6.5 Comparing Text Methods

The four approaches above (dictionaries, supervised classifiers, embeddings/topic models, and LLMs) span a range of tradeoffs. Before choosing one, it helps to compare them on the dimensions that matter for measurement: transparency, validation cost, sensitivity to context, and reproducibility. Table 7.6 provides a compact summary.

Table 7.6. Text-measurement methods compared. “Validation cost” is the effort required to produce a credible measure, including labeled data and human auditing. “Context sensitivity” refers to whether the method handles industry-specific or domain-specific meaning. “Reproducibility” refers to whether a researcher can re-create the exact output later.

Method	Transparency	Validation cost	Context sensitivity	Reproducibility
Dictionary (Loughran-McDonald)	High	Low	Low	High
Supervised classifier	Medium	High	Medium	High
Embeddings / topic models	Medium	Medium	Medium	High
LLM extraction	Low	High	High	Low to medium

The pattern is that newer methods buy context sensitivity at the cost of transparency and reproducibility. A Loughran-McDonald sentiment count can be reproduced byte-for-

byte twenty years from now from the same 10-K filing. An LLM extraction may not be reproducible six months from now if the provider has updated the model. This does not mean older methods are better. It means that the choice of method is inseparable from the choice of how to document it. A good LLM-based measurement paper looks more like a hand-collected dataset paper than like a regression paper: most of the work is in the construction and audit of the variable, and most of the credibility rests on what is in the appendix and the supplement.

A worked example: managerial tone in 10-Ks. Suppose a researcher wants a tone measure for the MD&A section of every annual 10-K from 2000 to 2024. A dictionary approach computes the share of Loughran-McDonald positive and negative words in the section, normalized by section length. A supervised approach takes a labeled sample of perhaps 5,000 sentences hand-classified by research assistants and trains a logistic regression on n -grams to reproduce the labels. An embeddings approach computes the cosine similarity between each MD&A and a reference document representing “negative tone.” An LLM approach prompts a frontier model to rate each MD&A on a five-point tone scale. All four produce a numerical variable. But the validation work differs sharply: the dictionary requires only sample auditing, the supervised approach requires labeling and cross-validation, the embeddings approach requires a defensible reference document, and the LLM approach requires all of the documentation listed above plus a hand-audit of model output stability across prompts and model versions. The right choice depends on what construct is wanted, how much human time is available, and how the measure will be used downstream.

7.6.6 Generated Regressors and Construct Validity

Text-based variables are generated regressors. A managerial-tone measure, a contract-strictness score, or a climate-exposure index is estimated before it is used in a regression. The usual regression table often treats the measure as observed without error. That is rarely true. Measurement error can attenuate coefficients, create spurious relations, or obscure the difference between real economic variation and variation in the measurement procedure.

The right question is not merely whether the text variable predicts something. It is whether the variable measures the intended construct. A tone measure may predict returns because it captures news. It may also predict returns because it captures firm size, industry, boilerplate disclosure, or the firm’s law firm. A useful text measure should pass both

predictive and construct validation. The researcher should show that the measure behaves sensibly, varies where theory suggests it should vary, and is not merely a proxy for an obvious confounder.

Key Insight. *Text and LLM methods are measurement tools. Their credibility depends less on algorithmic novelty than on construct validity, timing, reproducibility, and external validation. The right comparison for a new text-based measure is not “does it predict returns?” but “does it measure the construct it claims to measure, in the population to which it will be applied?”*

It helps to collect these requirements in one place. Whatever the method, dictionary, supervised classifier, embedding, or large language model, a variable produced by a model rather than observed directly should travel with the same documentation. Table 7.7 sets out that card. It generalizes the checklist for documenting an LLM-based measure given earlier in the section to any ML-generated variable, so that a reader can audit the measure without rerunning it.

Table 7.7. A measurement card for an ML-generated variable. Whatever the method, the card collects what a reader needs to audit a variable that was produced by a model rather than observed directly. It generalizes the LLM-documentation checklist given earlier in the section. Report every row; “not applicable” is itself an answer worth stating.

What to report	What to record
Construct	The concept the variable is meant to measure
Source documents	Which texts or records, for which units and dates
Prediction date	What was knowable at the date the variable carries
Preprocessing	What was cleaned, dropped, or normalized, and why
Feature representation	Counts, n -grams, embeddings, topics, or model output
Tuning	How parameters or thresholds were chosen, and on what data
Validation sample	The held-out documents or firms used to check it
Validation metrics	The labels or outcomes compared against, and the result
Failure modes	Where the measure is known to be unreliable
Downstream use	Whether it enters as outcome, treatment, control, or instrument
Causal status	Predictive, descriptive, or causal, and under what assumption

Note: The card applies to any variable generated by a statistical or machine-learning procedure, not only text. The last row is the tie to the framework of this book. A measurement card does not make a variable causal. It records only whether a causal reading was claimed and on what assumption it would rest.

7.7 Machine Learning in Causal Research

Machine learning does not solve identification. A random forest trained on observational data does not convert correlations into causal effects. A neural network does not remove omitted variables. A causal forest is not causal unless the treatment variation is as good as random conditional on the variables used, or unless it is embedded in a credible research design.

ML can nevertheless be extremely useful in causal research. Its main role is to estimate nuisance functions flexibly while preserving valid inference on a low-dimensional causal parameter.

7.7.1 Double/Debiased Machine Learning

Consider the partially linear model

$$Y_i = \theta D_i + g(X_i) + \varepsilon_i, \quad (7.9)$$

$$D_i = m(X_i) + v_i, \quad (7.10)$$

where D_i is the treatment or endogenous variable of interest, X_i is a large set of controls, and θ is the causal parameter we want to estimate. The functions $g(X_i) = \mathbb{E}[Y_i | X_i]$ and $m(X_i) = \mathbb{E}[D_i | X_i]$ may be complex. The researcher would like to use ML to estimate them, but naive use of ML creates regularization bias and invalid standard errors.

Double/debiased machine learning addresses this problem using two ideas: orthogonalization and cross-fitting ([Chernozhukov et al., 2018](#)). The basic procedure is:

1. Split the sample into folds.
2. For each fold, use the other folds to estimate $\hat{g}(X)$ and $\hat{m}(X)$.
3. For observations in the held-out fold, construct residuals

$$\tilde{Y}_i = Y_i - \hat{g}(X_i), \quad \tilde{D}_i = D_i - \hat{m}(X_i). \quad (7.11)$$

4. Regress $\tilde{Y}_i$ on $\tilde{D}_i$ to estimate θ .

The cross-fitting step ensures that the residual for observation i is computed using nuisance functions estimated without observation i . The orthogonalization step makes the final estimate less sensitive to small errors in the first-stage ML estimates. Under the right conditions, $\sqrt{N}(\hat{\theta} - \theta)$ is asymptotically normal and the usual standard errors apply.

These conditions are worth naming. DML requires, first, an *orthogonal score*: the moment condition used to estimate θ must have zero derivative with respect to the

nuisance functions at the true values, which is what neutralizes small first-stage errors. Second, *sample splitting* or *cross-fitting*, so that each observation's residual is constructed using nuisance estimates that did not see that observation; this breaks the entanglement between regularization bias and inference. Third, *sufficiently accurate nuisance estimation*, formally that the product of the convergence rates of $\hat{g}$ and $\hat{m}$ is faster than $\sqrt{N}$, which is what allows the first-stage error to disappear from the limiting distribution of $\hat{\theta}$. And fourth, *an identifying assumption* from the research design itself, such as conditional exogeneity of D given X , or a valid instrument when DML is implemented with instrumental variables. DML does not supply this last assumption. It supplies a way to use flexible ML for the nuisance functions without breaking inference on θ , conditional on the design.

A corporate finance example might study whether leverage affects investment, using a large set of controls for size, profitability, cash, industry, market-to-book, credit ratings, macro conditions, and text-based measures of risk. DML can flexibly partial out these controls. But it still requires a credible claim that, conditional on the controls and design, the remaining variation in leverage is as good as random. DML reduces functional-form and regularization problems. It does not eliminate omitted-variable bias by itself.

7.7.2 Causal Forests and Heterogeneous Effects

Many corporate finance questions are about heterogeneity. Do financial constraints matter more for small firms? Do governance changes matter more when ownership is dispersed? Do credit shocks affect investment more for firms with little cash? Do disclosure rules matter more for opaque firms? Traditional regressions study heterogeneity by adding interactions chosen by the researcher. Causal forests use tree-based methods to search for heterogeneity more flexibly (Wager and Athey, 2018; Athey and Imbens, 2019).

The object is not $\mathbb{E}[Y | X]$ but the conditional average treatment effect (CATE)

$$\tau(x) = \mathbb{E}[Y_i(1) - Y_i(0) | X_i = x]. \quad (7.12)$$

The method partitions the covariate space into regions with different estimated treatment effects. Honest splitting is used so that the same data are not used both to choose the partition and to estimate the effect within that partition. This is the causal analogue of the overfitting discipline from prediction.

The warning is again essential. What a causal forest estimates depends entirely on what assumptions hold. Under unconfoundedness given X , or under random assignment in an experiment, it estimates heterogeneous causal treatment effects. Under a credible instrumental-variable design with instrument heterogeneity, it estimates heterogeneous

local average treatment effects. Without either, it estimates heterogeneous associations or, worse, heterogeneous selection bias. The forest's flexibility makes this last possibility particularly dangerous: an algorithm searching for heterogeneity over hundreds of covariates will reliably find some, and there is no internal diagnostic that distinguishes heterogeneous causal effects from heterogeneous confounding. The right workflow is therefore: first secure the design, then use causal ML to study heterogeneity.

Common Mistake. *Writing “we use a causal forest, therefore our estimates are causal.” The word causal in the algorithm’s name refers to the estimand and the estimation strategy, not to the source of identifying variation. A causal forest trained on observational corporate finance data without a credible treatment-assignment argument estimates heterogeneous associations, not causal heterogeneity.*

Book 2 returns to causal forests and related methods in the context of heterogeneous treatment effects, where the algorithm's strengths are most fully realized. For now, the key point is that ML methods can serve causal research without competing with it.

7.8 When to Use Machine Learning

Machine learning should not be used because it is fashionable. It should be used when the research task calls for it. The decision rules below are a useful starting point.

7.8.1 Use ML When the Goal Is Prediction

If the research question is explicitly predictive, ML is natural. Examples include default prediction, fraud detection, target screening, credit scoring, forecasting investment, forecasting payout cuts, or predicting which firms are likely to be financially constrained. The central requirements are clear prediction timing, an appropriate validation scheme, and metrics that match the economic decision.

7.8.2 Use ML When the Goal Is Measurement from High-Dimensional Data

If the object of interest is embedded in text, contracts, patents, images, job postings, conference calls, or other unstructured data, ML may be the best measurement technology. The key requirements are construct validity, reproducibility, and validation against human labels or external outcomes.

7.8.3 Use ML When Flexible Controls or Heterogeneity Matter

If the researcher has many controls or expects nonlinearities and interactions, ML can help estimate nuisance functions or discover heterogeneity. DML, causal forests, and related methods are appropriate when the causal estimand is low-dimensional but the adjustment functions are complex.

7.8.4 Do Not Use ML to Hide a Weak Design

ML is not a substitute for identification. If the treatment is endogenous, using a more flexible algorithm generally estimates the endogenous relation more accurately. A better prediction of treatment or outcome does not make the treatment exogenous.

7.8.5 A Checklist for Corporate Finance Applications

Before using ML in an empirical corporate finance paper, answer the following questions.

1. What is the task: prediction, measurement, variable selection, nuisance estimation, or treatment-effect heterogeneity?
2. What is the unit of observation: firm-year, firm, loan, deal, patent, filing, manager, board, or security?
3. What is the prediction date, and are all predictors observable at that date?
4. What does “out of sample” mean: new firm, new year, future period, new industry, new country, or new regime?
5. Is the outcome rare? If so, why is accuracy not enough?
6. What is the benchmark model? Does the ML model improve on a simple logit, OLS, or industry-year mean?
7. Were tuning parameters selected without using the final test set?
8. Is there leakage through firm identity, future accounting data, revised databases, restated filings, or ex post sample construction?
9. If text or LLMs are used, are prompts, model versions, raw outputs, and validation procedures archived?
10. If the ML output enters a regression, what measurement error or generated-regressor issue does that create?
11. If the claim is causal, what is the source of identification?
12. Can the final result be explained economically, or is it merely a prediction improvement?

7.9 Connection to the Structural/Causal Framework

Machine learning changes empirical corporate finance less by replacing econometrics than by sharpening the distinction among tasks. If the task is to estimate a causal effect, identification remains central. If the task is to discipline a structural counterfactual, economic modeling remains central. If the task is to predict, classify, or measure high-dimensional information, ML provides tools that standard regression practice does not.

The three evaluative standards introduced in Section 7.1 have appeared in different combinations throughout this chapter. Section 7.2 developed the predictive-validation standard in isolation: bias-variance, cross-validation, leakage, and walk-forward designs. Section 7.3 showed how predictive validation and the panel methods of the Panel Data chapter illuminate each other when applied to the same leverage data. Section 7.5 extended prediction to the rare-event case, with constraint indices as a bridge to the causal questions of the Financial Constraints chapter. Section 7.6 treated ML as measurement technology, generating variables that then enter the causal pipeline of earlier chapters. Section 7.7 closed the loop by showing how ML can be embedded inside causal designs without compromising identification.

Three lessons survive aggregation. The first is the importance of validation. A model should be evaluated on data that represent the intended use case. In corporate finance that means paying attention to panels, time, firm identity, disclosure dates, database revisions, and rare-event outcomes. A random train-test split is often not enough.

The second is humility about interpretation. A variable that predicts well need not be causal. A forest's importance ranking is not a theory. A text classifier is a measurement device, not a fact about the world. An LLM output is a generated variable that requires documentation and validation.

The third is complementarity. ML can help causal research by estimating nuisance functions and exploring heterogeneity. It can help structural work by disciplining fit and prediction. It can help descriptive corporate finance by measuring concepts that were previously hard to quantify. Used well, ML is not a rival to the structural and causal approaches of this book. It is a third toolset, valuable precisely because it asks a different question: does this empirical object work outside the data used to construct it?

Part III

Frictions and the Shadow Value of Internal Funds

Part III turns from methods to substantive questions. The eight chapters that follow study investment, financial constraints, capital structure, payout, banking, innovation, governance, and mergers. Each chapter has its own literature, its own canonical estimators, and its own methodological history. The chapters read as eight separate conversations.

This brief introduction makes one claim across them. Four of the chapters that follow study different margins of the same underlying object. The object is the shadow value of internal funds, the Lagrange multiplier on the firm's period budget constraint in any dynamic model with costly external financing. We call this object μ . The structural tradition in corporate finance estimates μ and the frictions that determine it. The causal tradition identifies how firm behavior at one margin responds to shocks that move μ for the population of compliers in a given design. The chapters look like different literatures because each margin has developed its own methodological tradition. The deep object is one object.

This claim is not the whole story of empirical corporate finance, and this introduction does not ask the reader to carry it everywhere. Where the framing organizes a literature cleanly, we say so and let it work. Where the framing fits less well, we say that too. The introduction serves as a map of where the unifying perspective applies and where other organizing principles do better.

The formal object. Write the firm's value recursively. Let s_t denote the state, including capital k_t , debt b_t , and a productivity or demand shifter z_t . The firm chooses investment i_t , debt issuance, equity issuance, and distributions to shareholders subject to a period budget constraint that equates sources and uses of funds. The Bellman equation is

$$V(s_t) = \max_{i_t, b_{t+1}, \text{div}_t} \{ \text{div}_t + \mathbb{E}_t [M_{t,t+1} V(s_{t+1})] \} \quad (7.13)$$

subject to the budget identity

$$\pi(k_t, z_t) + b_{t+1}/R - b_t - i_t - C(\cdot) - \text{div}_t \geq 0. \quad (7.14)$$

The Lagrange multiplier on (7.14) is μ_t , the shadow value of internal funds. It is the marginal value the firm assigns to one additional dollar of cash inside the firm at date t , evaluated against all the uses retention enables.

Under Modigliani-Miller, external financing is costless, the budget constraint never binds in any state where the firm would prefer more cash, and $\mu_t \equiv 1$. The application chapters of Part III have nothing distinct to study. Once external financing is costly, through equity-issuance costs, debt-issuance costs, taxes, asymmetric information, agency conflicts,

or collateral constraints, $\mu_t > 1$ in states where the firm would prefer more cash than it has. The firm's behavior at every margin then deviates from the frictionless benchmark in a coordinated way governed by μ_t . Investment falls below the frictionless level in states with high μ . Debt issuance rises when it relaxes the constraint. Payout falls when retention is more valuable than distribution. Cash holdings rise as a precautionary buffer. The four margins move together in response to the same shadow value.

The structural tradition and μ . The structural literature in corporate finance estimates μ and the frictions that determine it. [Cooper and Haltiwanger \(2006\)](#) recover the capital adjustment-cost function from investment dynamics; the recovered cost function implies a particular $\mu(\cdot)$. [Hennessy and Whited \(2007\)](#) estimate the equity-issuance-cost function and the corporate tax effects jointly from financing-policy moments; their estimates imply a $\mu(\cdot)$ that varies systematically with firm size and cash holdings. [DeAngelo, DeAngelo, and Whited \(2011\)](#) estimate the same object from joint investment-leverage-payout moments. The structural payout, constraints, and capital-structure literatures are not estimating different objects. They are estimating $\mu(\cdot)$ under different specifications of which frictions matter.

This perspective clarifies what is at stake when structural papers disagree. The disagreement is rarely about whether μ exists or whether it varies across firms. The disagreement is about which mix of frictions generates the variation, and the same panel of moments typically admits multiple specifications that match them. [Strebulaev and Whited \(2012\)](#) make this point about the structural capital-structure literature; it applies to the structural payout and constraints literatures in the same form. The structural tradition has made enormous progress on identifying μ from one margin at a time. It has not yet produced a model whose primitives, estimated from any one margin, predict behavior at the other margins quantitatively.

The causal tradition and μ . The causal tradition identifies the response of firm behavior at one margin to a shock that moves μ in an identifiable way for a population of compliers. [Chetty and Saez \(2005\)](#) identify the payout response to a tax-driven shock to the value of dividend distributions for firms with high taxable individual ownership. [? \(2005\)](#) identify the investment response to industry-level external-finance dependence interacted with country-level financial development. [Khwaja and Mian \(2008\)](#) and the bank-lending-shock literature that followed identify the response of firm borrowing and investment to credit-supply shocks for firms whose primary lender is hit. Each design identifies a constraint-share-weighted local average of the response of one margin to a μ -shifting

shock among compliers. The Financial Constraints chapter develops the constraint-share machinery that makes this translation explicit; the reader meeting causal designs in the application chapters can carry that machinery forward.

The two traditions are complements, not substitutes. The structural tradition delivers the function $\mu(\cdot)$ at the cost of conditioning on a model. The causal tradition delivers responses to identified shocks at the cost of not extrapolating beyond the realized variation. When the structural prediction of the response to an identified shock matches the causal estimate, the model is credentialed. When they disagree, one of the maintained assumptions is wrong, and the discrepancy localizes the problem. This kind of cross-validation between traditions is the methodological payoff of the structural-causal duality, and the literature has not yet done it systematically.

Where the framing fits. The framing organizes the first four chapters of Part III plus Banking. The Investment chapter introduces μ in the form of Q-theory: marginal Q minus the cost of capital is a transformation of μ , and the adjustment-cost literature recovers it from investment dynamics. The Financial Constraints chapter develops μ as the explicit object of study, with the constraint-share interpretation as the translation theorem between causal and structural estimands. The Capital Structure chapter shows how μ governs the leverage margin under trade-off and pecking-order readings, though the market-timing reading sits to one side of the framing. The Payout chapter is the implementation of μ at the distribution margin, with the wedge $v_F - v_I = \mu - 1$ adjusted for the distribution technology and tax treatment. The Banking chapter extends the framing to financial intermediaries and is where the cleanest source of identified μ -shifting variation in the empirical literature lives.

Where the framing fits less well. The remaining three chapters of Part III engage corporate finance questions that μ touches but does not organize. The Innovation chapter studies the R&D-patents-productivity nexus; the financing of innovation engages μ , but the chapter's central object is the relationship between innovative input and innovative output, and the patents-as-output measurement and Schumpeterian-models machinery are not primarily about μ . The Governance chapter studies agency conflicts that affect μ as one of many consequences; the literature's organizing questions are about monitoring, incentive alignment, and board composition, and these questions have their own methodological history. The Mergers chapter studies the value of corporate combinations and the identification of synergies; μ enters peripherally if at all. The reader entering these chapters should expect different organizing principles, not strained connections to the

framing developed here.

The open question. The integrated decomposition of μ into its component frictions across margins is not settled. Structural papers identify μ under one set of friction assumptions from one margin's moments. Causal papers identify responses to μ -shifting shocks at one margin for one population of compliers. The literature has produced models that match moments at multiple margins simultaneously. The separation of the wedge into its component frictions, namely taxes, equity-issuance costs, agency conflicts, asymmetric information, and collateral constraints, and the quantitative weights on each, remains open.

The road ahead. The Investment chapter is next. It develops the dynamic-programming machinery that underlies (7.13) and recovers the capital adjustment-cost function as the first concrete estimate of a piece of the μ object. The Financial Constraints chapter follows, making μ explicit and developing the constraint-share machinery that the rest of Part III draws on. The reader can carry μ forward as the conceptual thread, or set it aside where the framing is less central. Either way, the part introduction has done its job by stating clearly what is on the table.

Chapter 8

Investment

8.1 Why Investment Matters

Investment is where corporate finance meets macroeconomics. When a firm buys a machine, builds a factory, or develops a new product, it transforms financial resources into productive capacity. Private fixed investment is one of the most volatile major components of GDP and accounts for a disproportionate share of the cyclical variation in output. Getting investment right matters for understanding both individual firms and the economy as a whole.

The central question of this chapter is simple to state. What determines how much a firm invests? The answer turns out to be harder than it looks. Two broad approaches have developed, and they answer different questions.

The *structural approach* starts from a model of the firm. The firm maximises value by choosing investment subject to adjustment costs. The model produces testable predictions about investment dynamics. The researcher estimates the model's parameters and uses them to decompose investment behaviour into its underlying components. The classic structural treatment is the *Q*-theory of [Hayashi \(1982\)](#), which predicts that investment depends on the shadow value of capital. The structural tradition also includes richer models that allow for non-convex adjustment costs, as in [Cooper and Haltiwanger \(2006\)](#).

The *causal approach* asks whether specific frictions distort investment from what it would otherwise be. The researcher identifies a treatment (a stock market listing, a tax change, a financing shock) and estimates its effect on investment using the methods from Part II of this book. A modern example is [Asker et al. \(2015\)](#), who compare the investment behaviour of public and private firms. The older [Fazzari et al. \(1988\)](#) cash-flow sensitivity regression is a classic *reduced-form* empirical investment regression with an economic interpretation,

but it is not a quasi-experimental identification strategy in the modern sense developed in Part II of this book. Keeping the reduced-form and modern causal designs distinct is useful, because the Financial Constraints chapter reopens the Fazzari et al. debate and contrasts it with the causal designs that came later.

The two approaches are not competitors, and the distinction between them is not as clean as it looks at first. Structural models can answer causal and counterfactual questions, but only conditional on the model. Causal designs can sometimes illuminate mechanism, though usually less completely than structure does. A useful way to draw the contrast is this: structural models discipline counterfactuals by imposing a model of the firm's dynamic problem, while causal designs discipline comparisons by isolating plausibly exogenous variation. The first approach is stronger on mechanism and counterfactuals; the second is stronger on transparent identification. Both have a role in answering the investment question.

It helps to be explicit at the outset about what kinds of empirical objects this chapter estimates, because they are not all the same kind of thing. The first is a descriptive regularity: investment is lumpy, intermittent, and asymmetric, with firms buying capital far more often than they sell it. The second is a structural primitive, the adjustment-cost function that governs how costly it is to change the capital stock. The third is a causal effect, the change in investment that a specific treatment such as listing status or a tax incentive produces relative to what would otherwise have happened. The fourth is a counterfactual, the investment response to a policy that has not been run, computed from the estimated technology. The fifth underlies all the others: a measurement object, the mapping from the Compustat balance-sheet items we can actually construct onto the theoretical investment quantity the models are written in terms of. This chapter estimates the real adjustment technology that connects these objects, and the later chapters in Part III add financing, agency, and control distortions on top of it.

This chapter develops both approaches and connects them. It begins with the micro facts of plant- and firm-level investment (Section 8.2), then develops *Q*-theory and explains its limits (Section 8.3). The next section estimates structural adjustment cost models in the Cooper and Haltiwanger (2006) tradition (Section 8.4). The chapter then extends the framework to multiple capital inputs and firm value (Section 8.5), presents causal evidence on stock market listing (Section 8.6), and synthesises the two traditions (Section 8.7).

A puzzle runs through the chapter. At the micro level, investment is lumpy and intermittent. Plants go years without investing, then invest heavily. But the workhorse *Q*-theory assumes smooth, continuous adjustment. This tension between the micro facts and the standard model is what makes investment one of the most active research areas

in empirical finance. It is also what makes the structural/causal distinction useful. The structural approach can explain *why* investment is lumpy. The causal approach can tell you whether the lumpiness has consequences for how firms respond to policy.

8.2 What Does Micro Data Look Like?

Before building models, it helps to look at the data. This section documents what investment looks like at the plant level, then at the firm level, and finally across industries. The patterns motivate the structural model in Section 8.4 and provide the moments that the model is asked to match. Before turning to the patterns, we need to be explicit about how investment is measured, because the structural results depend on the comparability of measures across data sources.

8.2.1 Measurement: How Is Investment Constructed?

Investment looks like a simple object until you try to measure it. At the plant level, the LRD provides direct measures of physical investment spending. At the firm level, Compustat does not give a clean “investment” variable. Researchers construct one from balance-sheet items, and the choice of construction matters.

Three common firm-level measures are the following. The first is capital expenditure, $CAPX_t$, divided by lagged gross PP&E. This captures organic capital expenditure. The second is the change in net PP&E, $\Delta PPENT_t$, divided by lagged gross PP&E. This is the standard proxy in the structural investment literature (Cooper and Haltiwanger, 2006), and it captures both positive and negative net investment. The third is the change in gross PP&E, $\Delta PPEGT_t$, scaled similarly. The differences across these measures are not innocuous. The $\Delta PPENT$ measure mixes capital expenditures with depreciation, retirements, acquisitions, accounting restatements, and asset sales. The $CAPX$ measure excludes most of these but, in its raw form, also excludes asset sales (which are reported separately as $SPPE$). A $CAPX$ -minus- $SPPE$ measure recovers organic net investment but still misses M&A-driven adjustments. Table 8.1 lays the four measures side by side, with what each one captures and what it leaves out. Choosing among them is not a clerical detail: it is an economic assumption about which transactions count as investment in the sense the model intends, in the same way that the choice of functional form is an economic assumption rather than a statistical convenience.

Table 8.1. Firm-Level Investment Measures: Construction and Content

Measure	Compustat construction	What it captures	What it misses
$CAPX_t/PPEGT_{t-1}$	Capital expenditures over lagged gross PP&E	Organic gross capital expenditure	Asset sales (SPPE), M&A, intangibles
$(CAPX_t - SPPE_t)/PPEGT_{t-1}$	CAPX net of property sales, over lagged gross PP&E	Organic net investment in physical capital	M&A-driven adjustments, intangibles
$\Delta PPENT_t/PPEGT_{t-1}$	Change in net PP&E over lagged gross PP&E	Net investment; the structural-literature proxy	Mixes in depreciation, retirements, M&A, re-statements
$\Delta PPEGT_t/PPEGT_{t-1}$	Change in gross PP&E over lagged gross PP&E	Gross additions to the recorded capital stock	Not net of depreciation; sensitive to write-downs and M&A

Notes. Four common firm-level investment measures built from Compustat items. CAPX is capital expenditures, SPPE is the sale of property, plant, and equipment, PPENT is net property, plant, and equipment, and PPEGT is gross property, plant, and equipment. All are scaled by lagged gross PP&E, which approximates replacement cost without the mechanical correlation that scaling by net PP&E would introduce. The structural exercise in this chapter uses $\Delta PPENT/PPEGT_{t-1}$ as the baseline, following [Cooper and Haltiwanger \(2006\)](#), with a CAPX-based robustness check from [Frank \(2026b\)](#).

Several other measurement choices have to be made. Some of them matter a great deal for the interpretation of the results.

Gross versus net investment. The structural model in Section 8.4 is naturally cast in terms of net investment $I = K' - (1 - \delta)K$. Gross investment $I + \delta K$ differs by the depreciation flow, which is itself imperfectly measured.

Acquisitions and asset sales. Both inflate the $\Delta PPENT$ measure. About 7 to 8% of Compustat firm-years have a PP&E change that is disproportionate to CAPX and accompanied by a large jump in total assets, suggesting M&A ([Frank, 2026b](#)). The robustness check in that paper shows that roughly half of negative investment spikes reflect divestitures or write-downs rather than organic disinvestment.

Treatment of intangibles. The standard PP&E-based measures exclude R&D, software, and brand investment. For technology and pharmaceutical firms these are large fractions

of total investment. Section 8.5 returns to this when discussing the [Belo, Gala, Salomao, and Vitorino \(2022\)](#) multi-input framework.

Scaling. Lagged gross PP&E is the standard denominator because it approximates replacement cost without the mechanical correlation that arises from scaling by net PP&E in both numerator and denominator. Alternatives include total assets, lagged capital stock at constant prices, or industry-level replacement-cost indices.

Sample exclusions. Financial firms (SIC 6000–6999) and utilities (SIC 4900–4999) are conventionally excluded because their balance sheets do not have the same interpretation as those of non-financial firms. Some studies also drop the smallest firms or firms with very short panels.

Winsorization. Investment rates have heavy tails. Most empirical work winsorizes at the 1st and 99th percentiles within each year. The choice of percentile changes the spike rates, which are central to identifying non-convex adjustment costs.

The structural exercise in this chapter uses $\Delta\text{PPENT}/\text{PPEGT}_{t-1}$ as the baseline, following [Cooper and Haltiwanger \(2006\)](#), and reports a CAPX-based robustness check from [Frank \(2026b\)](#). The two measures produce similar qualitative classifications across industries, with one exception (Business Equipment, where the convex-only baseline classification becomes non-convex under CAPX). The exception is informative: it shows that the baseline measure can mask irreversibility when M&A activity contributes substantially to the recorded change in net PP&E.

Common Mistake 1

Comparing investment moments across studies without first checking that they use the same measurement convention. A “positive spike rate of 0.18” computed from $\Delta\text{PPENT}/\text{PPEGT}_{t-1}$ on Compustat firms is not directly comparable to a spike rate of 0.186 computed from direct plant-level investment spending on LRD plants. The similarity of the two numbers in the table below is more useful as a sanity check than as a literal moment match.

8.2.2 Five Facts about Plant-Level Investment

[Cooper and Haltiwanger \(2006\)](#) documented five facts about plant-level investment using a balanced panel of approximately 7,000 large manufacturing plants from the Longitudinal Research Database (LRD), covering 1972 to 1988. These facts are the targets that any structural model of investment must match.

Fact 1: Investment is lumpy. About 19% of plant-year observations involve investment spikes, defined as investment rates exceeding 20% of the capital stock. These are large,

discrete episodes. Half of all investment spending occurs during spike years, even though they are less than a fifth of observations.

Fact 2: Inaction is common. About 8% of plant-years involve near-zero investment (investment rates below 1% in absolute value). These are years when the plant essentially does not adjust its capital stock at all.

Fact 3: Negative investment is rare. Only about 10% of plant-years involve negative net investment (retirements exceeding gross expenditures). Negative spikes (investment rates below -20%) occur in only 1.8% of observations. The distribution is highly asymmetric. Firms buy capital much more often than they sell it.

Fact 4: Serial correlation is low. The correlation between a plant's investment rate and its lagged investment rate is only 0.058. This is surprisingly low. Profitability shocks are highly persistent ($\rho = 0.885$), so if investment tracked profitability smoothly, serial correlation would be much higher. The low serial correlation suggests that adjustment costs prevent smooth tracking. But the standard convex cost model produces *too much* serial correlation (0.732 in Cooper and Haltiwanger's calibration), not too little. Getting the serial correlation right requires non-convex costs.

Fact 5: Investment responds to profitability, but weakly. The correlation between investment and the plant-level profitability shock is 0.143. This is positive, as theory predicts. But it is much lower than the convex model implies (0.692 in the calibration). Again, non-convex costs are needed to attenuate the response.

8.2.3 From Plants to Firms

Cooper and Haltiwanger used confidential plant-level data. Most empirical work in corporate finance uses firm-level data from Compustat. What happens to the investment facts when you aggregate from plants to firms?

Table 8.2 compares the all-firm Compustat moments to the LRD plant-level facts. Three patterns stand out. The average investment rate is similar (0.133 in Compustat vs. 0.122 in the LRD). The positive spike rate is nearly identical (0.181 vs. 0.186). But serial correlation is four times higher at the firm level (0.231 vs. 0.058).

Table 8.2. Investment Moments: Compustat Firms vs. LRD Plants

Moment	LRD plants	Compustat firms
Average investment rate	0.122	0.133
Serial correlation	0.058	0.231
Profitability correlation	0.143	0.370
Positive spike rate	0.186	0.181
Negative spike rate	0.018	0.038
Inaction rate	0.081	0.099

Notes. LRD moments from [Cooper and Haltiwanger \(2006\)](#), Table 1 (manufacturing plants, 1972–1988). Compustat: 253,040 firm-years, 19,430 firms, 1971–2023. Compustat investment rate is $\Delta \text{PPENT}_t / \text{PPEGT}_{t-1}$. Profitability proxy for Compustat is sales growth; for LRD it is the structural profitability shock. The comparison is informative but not exact: the two measures differ in aggregation level (plant vs. firm) and in accounting construction (the LRD measure is closer to direct investment spending, while the Compustat measure mixes capital expenditures with depreciation, retirements, acquisitions, and asset sales). The exercise should be read as a firm-level analogue of the plant-level facts, not as a literal replication.

Two caveats deserve emphasis. First, the LRD measures direct investment spending, while the Compustat measure is the change in net PP&E. These are not the same object even when reported on the same plants and firms. Second, the profitability proxies differ. The LRD profitability correlation is computed with the structural shock from the production-function regression, while the Compustat correlation uses sales growth. These differences can change the moments substantially, and the direction need not be the same for every moment. The right interpretation is not that the firm-level data show different “true” investment behaviour, but that the moments are constructed from genuinely different objects and need to be read accordingly.

That said, the qualitative pattern of the higher Compustat serial correlation is the signature of within-firm aggregation. A multi-plant firm has some plants investing and others idle in any given year. The firm-level investment rate averages across them. This smooths out the extremes and creates persistence. [Cooper and Haltiwanger \(2006\)](#) demonstrated the effect theoretically. A pure quadratic adjustment cost model fits the *aggregated* data generated by their estimated non-convex plant-level model with $R^2 = 0.86$, even though it fails at the plant level. The firm-level Compustat data sit between the plant

extreme and the aggregate extreme.

This has a practical implication. If you estimate a structural model on firm-level data, the estimated convex cost γ will be biased upward relative to the plant-level truth. The firm-level γ absorbs the smoothing from within-firm aggregation. Cross-industry comparisons may still be informative if the degree of within-firm aggregation is similar across industries. But the level of γ should not be compared directly to plant-level estimates. The formal treatment of what the estimated parameters identify under firm-level aggregation is in [Frank \(2026b\)](#).

8.2.4 Cross-Industry Variation

The investment moments vary substantially across Fama-French 12 industries. Energy has the highest spike rates (22.9% positive, 6.3% negative), consistent with the boom-bust nature of oil and gas investment. Manufacturing has the lowest positive spike rate (11.7%) and the highest inaction rate (12.5%), consistent with specialised, slow-adjusting capital. Telecom has the highest serial correlation (0.373), reflecting sustained multi-year network build-outs. Chemicals has a notably low profitability correlation (0.160), suggesting that investment in chemicals is driven by long planning horizons or regulatory requirements rather than short-run demand.

These differences foreshadow the structural results in Section [8.4](#). Industries with low serial correlation and high spike rates will be estimated to have disruption costs. Industries with high serial correlation and moderate spikes will be estimated to have primarily convex costs. The moments tell you what kind of friction to expect before you estimate the model.

8.3 Q-Theory and Its Limits

The micro facts in Section [8.2](#) pose a challenge. Investment is lumpy and intermittent at the plant level. But the workhorse empirical model, *Q*-theory, predicts smooth and continuous adjustment. We proceed in five steps: we derive *Q*-theory from the firm's dynamic optimisation problem, establish the Hayashi conditions under which unobservable marginal q equals observable average Q , show why those conditions fail in practice, trace the two consequences of that failure for estimated adjustment costs and for the cash-flow sensitivity puzzle, and close with the corrections and alternatives the literature has proposed. The payoff is a precise understanding of what reduced-form investment regressions can and cannot tell you.

8.3.1 The Firm's Dynamic Problem

Consider a firm that chooses investment I_t each period to maximise the present value of profits. The firm has capital stock K_t that depreciates at rate δ and earns profits $\pi(A_t, K_t)$, where A_t is a profitability shock. Adjusting the capital stock is costly. The firm solves

$$V(A, K) = \max_I \left\{ \pi(A, K) - C(I, K) - I + \beta \mathbb{E}_{A'|A} [V(A', K')] \right\}, \quad (8.1)$$

where $K' = (1 - \delta)K + I$ and $C(I, K)$ is the adjustment cost function. This is the standard neoclassical investment model. It goes back to [Jorgenson \(1963\)](#) and [Lucas \(1967\)](#), with modern treatments in [Hayashi \(1982\)](#) and [Bond and Van Reenen \(2007\)](#).

The investment literature before Hayashi focused on the *user cost of capital*, following [Jorgenson \(1963\)](#). In that framework, the firm invests until the marginal product of capital equals the rental rate (interest plus depreciation minus capital gains). The user cost approach is intuitive but implicitly assumes instantaneous adjustment. [Eisner and Strotz \(1963\)](#) introduced convex adjustment costs to avoid bang-bang investment policies, and [Lucas \(1967\)](#) embedded them in a dynamic optimisation framework. [Abel and Eberly \(1994\)](#) later provided the canonical treatment of investment under uncertainty with fixed costs, irreversibility, and purchase-sale wedges. [Caballero \(1999\)](#) surveys the aggregation from micro-level investment to aggregate dynamics. The Hayashi result that we develop next was the breakthrough that made the theory empirically operational.

The first-order condition for investment is

$$1 + \frac{\partial C}{\partial I} = \beta \mathbb{E} \left[\frac{\partial V}{\partial K'} \mid A \right]. \quad (8.2)$$

The left side is the marginal cost of an additional unit of capital: the purchase price (normalised to 1) plus the marginal adjustment cost. The right side is the marginal benefit: the discounted expected shadow value of capital next period. The firm invests up to the point where these are equal.

Define *marginal q* as the shadow value of capital:

$$q_t \equiv \beta \mathbb{E} \left[\frac{\partial V}{\partial K'} \mid A_t \right].$$

Then equation (8.2) says that the marginal adjustment cost equals $q - 1$. If adjustment costs are quadratic, $C(I, K) = (\gamma/2)(I/K)^2 K$, the marginal adjustment cost is $\gamma(I/K)$, and

the first-order condition simplifies to

$$\frac{I_t}{K_t} = \frac{1}{\gamma}(q_t - 1). \quad (8.3)$$

Investment is a linear function of marginal q , with the adjustment cost parameter γ controlling the sensitivity. This is an elegant result. It says that everything the firm needs to know about its investment opportunities is summarised in one number: q .

8.3.2 The Hayashi Equivalence

The problem is that marginal q is unobservable. It depends on the derivative of the value function, which requires solving the firm's entire dynamic programme. Hayashi (1982) showed that under specific conditions, marginal q equals average Q . The key requirement is that the value function be homogeneous of degree one in capital. This follows when the profit function $\pi(A, K)$ is homogeneous of degree one in K and the adjustment cost function $C(I, K)$ is homogeneous of degree one in (I, K) , together with competitive output markets and the usual valuation assumptions. The standard quadratic specification $C(I, K) = (\gamma/2)(I/K)^2 K$ satisfies the homogeneity requirement. Fixed costs, disruption costs, irreversibility, market power, taxes, financing frictions, and multiple capital stocks can each break the equivalence.

In empirical work, average Q is usually approximated by the market value of assets divided by the replacement or book value of those assets. A standard construction is

$$Q_t \approx \frac{\text{Market value of equity} + \text{Book value of debt}}{\text{Replacement value of assets}},$$

sometimes adjusted for cash, preferred stock, or other non-operating items depending on the convention. The numerator is constructed using both equity-market and balance-sheet information, not the stock market alone. The denominator is approximated, often imperfectly, by book values; constructing true replacement cost requires industry-level deflators that are themselves estimated. If marginal q equals average Q , then equation (8.3) becomes a regression:

$$\frac{I_{it}}{K_{it}} = \alpha + \frac{1}{\gamma} Q_{it} + u_{it}, \quad (8.4)$$

where the intercept α absorbs $-1/\gamma$ and measurement conventions. Under the maintained assumptions (Hayashi homogeneity, quadratic adjustment costs, interior investment, no financing or agency wedges, and correct measurement of Q), the slope coefficient on Q

identifies the inverse of the adjustment cost parameter. This is the standard Q -regression that appears in hundreds of papers. The strong list of maintained assumptions is what makes the empirical performance of this regression so informative.

8.3.3 Why the Conditions Fail

The Hayashi conditions are strong. Neither is likely to hold in practice.

The homogeneity of the profit function requires constant returns to scale. This fails when firms have market power. If the firm faces a downward-sloping demand curve, its revenue function has decreasing returns even if the production function does not. [Cooper and Haltiwanger \(2006\)](#) estimate the curvature of the profit function from plant-level data and find $\theta = 0.592$, significantly below 1. The violation is quantitatively large. As Section 8.4 shows, the Compustat estimates of θ range from 0.65 to 0.85 across industries. The Hayashi equivalence requires $\theta = 1$ exactly.

The homogeneity of the adjustment cost function requires that costs scale proportionally with (I, K) . This rules out fixed costs, disruption costs, and irreversibility. But the plant-level data in Section 8.2 show that investment is lumpy. About 19% of plant-years involve investment spikes exceeding 20% of the capital stock. About 8% involve near-zero investment. These patterns cannot come from homogeneous convex costs alone. They require some form of non-convex friction that violates the homogeneity condition.

When the conditions fail, marginal q and average Q diverge. The Q -regression becomes a regression of investment on a noisy proxy for the true driving variable. It behaves like a proxy-variable regression: measured average Q is an imperfect proxy for marginal q . In the classical errors-in-variables special case, the coefficient on Q is attenuated toward zero, and the implied γ is biased upward. The measurement error in this setting is not classical, however: the noise in Q is correlated with the signal (because both depend on the profitability shock A), and its distribution is non-Gaussian ([Erickson and Whited, 2000](#)). In the investment- Q applications studied in this chapter, the bias typically attenuates the Q coefficient and therefore inflates the implied γ . But because the measurement error is non-classical, this direction is an empirical result of the model and the data, not a generic theorem. The size of the bias depends on the joint distribution of η and q in ways that the simple attenuation formula misses.

8.3.4 Two Consequences of the Bias

The bias has two consequences that have shaped the empirical literature.

The first is that estimated adjustment costs from Q -regressions are far too large. Hayashi (1982) estimated $\gamma > 20$. Gilchrist and Himmelberg (1995) obtained $\gamma \approx 3$ on subsamples with better Q measurement. Structural estimation by Cooper and Haltiwanger (2006) produces $\gamma \approx 0.15$. The two-orders-of-magnitude gap between the Q -regression estimate and the structural estimate is largely due to the average-marginal Q divergence, with non-convex cost misspecification contributing additionally. Cooper and Haltiwanger demonstrate this directly. They simulate data from their estimated structural model, construct average Q from the value function, and run the Q -regression on the simulated data. The regression yields $\gamma \approx 5$, even though the true structural value is $\gamma = 0.049$. The measurement error inflates the estimated γ by a factor of 100.

The second consequence concerns the cash-flow sensitivity puzzle. Fazzari et al. (1988) added cash flow to equation (8.4) and found it to be significant. They interpreted this as evidence of financial constraints. If Q is a sufficient statistic for investment opportunities, cash flow should not matter. The fact that it does, they argued, means that some firms face binding constraints.

Kaplan and Zingales (1997) challenged this interpretation. Erickson and Whited (2000) and Roberts and Whited (2013) provided the econometric explanation. Cash flow is correlated with the measurement error in Q . When measured Q imperfectly captures the true marginal investment opportunity, cash flow proxies for the omitted profitability information. It enters significantly not because it measures financial constraints, but because it captures the part of profitability that Q misses.

Frank (2026a) provides a structural interpretation of what cash-flow sensitivity regressions actually identify. The question of financial constraints, including the role of cash flow, is taken up in the Financial Constraints chapter.

Key Insight 2

The Q -regression is a regression of investment on a noisy proxy for marginal q . The noise comes from two sources. First, when the Hayashi homogeneity conditions fail ($\theta < 1$), average Q does not equal marginal q . Second, when adjustment costs are non-convex, the investment- q relationship is not linear (the functional form is wrong). Both sources inflate the estimated γ and create the appearance of cash-flow effects. Section 8.4 shows that the first source alone can inflate γ by two orders of magnitude.

8.3.5 Corrections and Alternatives

Several approaches try to fix the measurement error problem.

The Euler equation approach avoids Q altogether. Bond and Van Reenen (2007) derive

an investment Euler equation from the firm's first-order conditions that involves only observable quantities: current and lagged investment, profits, and discount rates. The equation is estimated by GMM using lagged variables as instruments. This approach works well when adjustment costs are purely convex. It breaks down when non-convex costs create inaction regions, because the Euler equation holds only at interior optima.

The [Erickson and Whited \(2000\)](#) cumulant estimator uses higher moments of Q to identify and correct the measurement error under classical errors-in-variables assumptions. This estimator substantially reduces the bias. [Frank \(2026b\)](#) reports results using the Erickson-Whited estimator across industries. Even after correction, however, the implied γ remains well above structural estimates. The reason is that the measurement error in Q is not classical: the noise is correlated with the signal (because both depend on A), and the distribution of the noise is non-Gaussian. No estimator that imposes classical errors-in-variables assumptions can fully correct the bias.

The structural approach abandons the Q -regression entirely. Instead of trying to recover γ from a single regression, it solves the full dynamic programming problem, simulates the model, and estimates parameters by matching simulated moments to data moments. This is the [Cooper and Haltiwanger \(2006\)](#) approach developed in Section 8.4. It requires more computation but avoids both the measurement error and the functional form problem. The cost is that you must commit to a specific model of the firm, including the profit function, the shock process, and the adjustment cost specification. The tradeoff between transparency and completeness is the recurring theme of this book.

8.4 Structural Estimation of Adjustment Costs

Running Example 3: Leverage and Investment

This section is the investment side of the leverage-and-investment example that has run through the methods chapters since the Panel Data chapter. There we asked how leverage and investment move together within firms over time; here we estimate the real technology that governs investment itself. We fit the [Cooper and Haltiwanger \(2006\)](#) adjustment-cost model by simulated method of moments on the same master Compustat panel used throughout the book, so the structural estimates and the earlier reduced-form results describe one dataset rather than two. Leverage enters here only as a background characteristic of the firms in the panel. It returns as a decision in the Financial Constraints and Capital Structure chapters, where the firm's financing becomes an endogenous state variable layered on top of the adjustment technology we

estimate now.

This section develops the structural approach to capital adjustment costs. The approach was pioneered by [Cooper and Haltiwanger \(2006\)](#) using plant-level data from the Longitudinal Research Database. We replicate their methodology using Compustat data on publicly traded firms, which serves two purposes. It demonstrates that structural estimation of investment models is feasible for PhD students with standard data and computing resources. It also reveals how investment dynamics change when you move from plant-level to firm-level data.

8.4.1 The Cooper-Haltiwanger Model

[Cooper and Haltiwanger \(2006\)](#) enriched the standard investment model in an important way. Instead of assuming that adjustment costs are purely convex, they specified a model that nests three qualitatively different frictions. Each friction produces different investment dynamics. By estimating the model and letting the data choose the mix, they could determine which kinds of frictions matter.

The firm solves a modified version of the Bellman equation (8.1) in which the capital price depends on the direction of investment:

$$V(A, K) = \max_I \left\{ \pi(A, K) - C(I, A, K) - p(I) \cdot I + \beta \mathbb{E}_{A'|A} [V(A', K')] \right\}, \quad (8.5)$$

where $K' = (1 - \delta)K + I$ and the cost function is

$$C(I, A, K) = \underbrace{\frac{\gamma}{2} \left(\frac{I}{K} \right)^2 K}_{\text{convex installation}} + \underbrace{(1 - \lambda) \cdot \pi(A, K) \cdot \mathbf{1}[I \neq 0]}_{\text{production disruption}}. \quad (8.6)$$

The capital price function $p(I)$ encodes irreversibility: $p(I) = 1$ when $I \geq 0$ (the firm pays the full price to buy capital) and $p(I) = p_s \leq 1$ when $I < 0$ (the firm receives only p_s per unit of capital sold). The three parameters are:

γ (**convex cost**): The standard quadratic adjustment cost. Higher γ means more smoothing. Investment adjusts gradually in response to shocks.

λ (**disruption**): When the firm adjusts capital, its current-period profits are multiplied by λ . The fraction $1 - \lambda$ is lost to disruption. A firm must shut down a production line, retrain workers, or reorganise the plant floor during adjustment. The higher the disruption cost (lower λ), the wider the inaction region. The firm waits until the benefit of adjusting

exceeds the disruption cost, then adjusts by a large amount.

p_s (**irreversibility**): When selling capital, the firm receives only p_s per unit instead of the purchase price of 1. This creates asymmetry. The firm is cautious about buying capital it cannot easily resell.

The profit function takes the power form $\pi(A, K) = AK^\theta$, where $\theta < 1$ and A follows a two-component AR(1) process with aggregate and idiosyncratic shocks.

8.4.2 How Different Frictions Produce Different Dynamics

Each type of friction leaves a distinctive fingerprint in the data. Table 8.3 summarises the qualitative predictions.

Table 8.3. Qualitative Predictions of Each Adjustment Cost Specification

Specification	Serial corr.	Profit corr.	Pos. spikes	Neg. spikes
No adjustment costs	Negative	Moderate	Many	Many
Convex only	High pos.	High	Few	None
Disruption only	Negative	Moderate	Many	Many
Irreversibility	Moderate pos.	High	Some	Few
LRD data	+0.058	+0.143	18.6%	1.8%

Notes. Qualitative predictions from [Cooper and Haltiwanger \(2006\)](#), Tables 2 and 3. Spikes are investment rates exceeding 20% in absolute value. LRD data row from their Table 1 (manufacturing plants, 1972–1988).

No single specification matches the LRD data. The data show low positive serial correlation, moderate profitability correlation, moderate positive spikes, and few negative spikes. Convex costs alone produce too much serial correlation and too few spikes. Disruption alone produces too many negative spikes. Irreversibility alone produces too few positive spikes. You need the mixture.

This is the key insight of [Cooper and Haltiwanger \(2006\)](#). The question is not whether adjustment costs are large or small. The question is what *kind* of adjustment costs firms face.

This reframing changes how we read a low investment- Q sensitivity. The reflex is to attribute it to large convex costs, a high γ . But low sensitivity can equally come from non-convex costs with a small γ . A firm sitting in an inaction region does not respond to changes in Q at all, and a Q -regression charges that non-response to a high γ even when the true convex cost is tiny. The nature of the friction, not its magnitude, is what the moments are telling us about.

8.4.3 Estimation by Simulated Method of Moments

The model is estimated by simulated method of moments (SMM). The idea is simple. Pick candidate values for (γ, λ, p_s) . Solve the dynamic programming problem. Simulate a panel of firms. Compute moments from the simulated data. Compare these simulated moments to the same moments in the actual data. Adjust the parameters and repeat until the simulated moments match the data moments as closely as possible.

Formally, the estimate $\hat{\Theta}$ solves

$$\hat{\Theta} = \arg \min_{\Theta} [m^d - m^s(\Theta)]' W [m^d - m^s(\Theta)], \quad (8.7)$$

where m^d is the vector of data moments, $m^s(\Theta)$ is the vector of simulated moments at parameter values Θ , and W is a weighting matrix. In the baseline we set $W = I$ (identity weighting), following [Cooper and Haltiwanger \(2006\)](#).

Four moments serve as targets, and each is informative about different parameters in different ways. With three parameters and four moments, the system is overidentified with one degree of freedom.

Serial correlation of investment has strong leverage over γ . Higher γ produces more smoothing, which raises the serial correlation. An industry with serial correlation of 0.40 needs a higher γ than one with serial correlation of 0.18, holding the other parameters fixed.

Profitability correlation is informative about λ and γ jointly. Both convex costs and disruption attenuate the profitability response, but in different ways. Convex costs spread the response over time. Disruption eliminates the response entirely for firms in the inaction region.

Positive and negative spike rates are informative about the non-convex parameters. Disruption produces large positive spikes (the firm overshoots when it finally adjusts). Irreversibility reduces negative spikes (the firm avoids selling capital at a loss). The asymmetry between positive and negative spike rates helps separate λ from p_s .

We should be careful, though, not to call these moment-parameter relationships “identifying” in the strict sense. The mapping from parameters to moments is nonlinear and not globally one-to-one. A given vector of moments can be matched by more than one parameter vector, and the objective surface can have several local minima. Identification therefore has to be assessed numerically: through the curvature of the SMM objective, sensitivity to starting values, bootstrap confidence intervals, and sensitivity to the weighting matrix and the choice of target moments. [Frank \(2026b\)](#) reports such diagnostics. They

show that γ is well identified in every industry, but that the split between disruption λ and irreversibility p_s is fragile in industries where the spike-rate asymmetry is small and noisy.

8.4.4 Solving the Dynamic Programming Problem

The estimation requires solving the Bellman equation (8.5) many times, once for each candidate parameter vector. The solution method is value function iteration on a discretised state space.

The profitability shock is discretised using the [Tauchen \(1986\)](#) method, which converts the continuous AR(1) process into a finite Markov chain. The capital stock is placed on a log-spaced grid. For each state (A, K) on the grid, the algorithm searches over a grid of investment rates to find the optimal choice. The optimal investment compares three options: inaction ($I = 0$, which avoids disruption and irreversibility costs), buying ($I > 0$, at the full purchase price), and selling ($I < 0$, at the resale price p_s). The value of each option includes the flow payoff plus the discounted expected continuation value, interpolated from the value function at the next-period capital.

The computational bottleneck is the inner loop. For each VFI iteration, the algorithm must evaluate the Bellman operator at every (A, K, I) triple. [Frank \(2026b\)](#) provides a vectorised NumPy implementation in which each iteration is a single matrix operation rather than a triply-nested loop. With vectorisation, each model solve takes approximately three seconds on a laptop.

The outer optimisation uses a two-stage procedure. Stage 1 is a coarse grid search over the parameter space. With 10 to 12 points per parameter and three parameters, this requires 1,000 to 1,728 model solves. Stage 2 polishes the best grid points using the Nelder-Mead simplex algorithm ([Nelder and Mead, 1965](#)). This avoids gradient-based methods, which are unreliable when the policy function has discrete inaction boundaries that create discontinuities in the simulated moments.

8.4.5 A Worked Example: Manufacturing

To illustrate the estimation, we focus on manufacturing firms in Compustat. This is the closest analogue to the [Cooper and Haltiwanger \(2006\)](#) LRD sample, which consists of manufacturing plants.

The profitability shock parameters are estimated first. The profit function curvature θ is estimated from a panel regression of log operating income on log gross PP&E with firm fixed effects. The result is $\hat{\theta} = 0.718$, which is above the Cooper-Haltiwanger estimate of 0.592 but still well below 1. The idiosyncratic shock persistence is $\hat{\rho}_\varepsilon = 0.456$, much lower

than the LRD value of 0.885. Firm-level profitability is noisier than plant-level, because it aggregates across heterogeneous activities.

With these shock parameters held fixed, the three adjustment cost parameters are estimated by SMM. The grid search evaluates 1,000 parameter vectors. Each evaluation solves the dynamic programming problem, simulates 2,000 firms for 300 periods, and computes the four target moments. The best grid point is polished by Nelder-Mead. The estimated parameters for Manufacturing are reported in Table 8.4, along with sandwich standard errors and the model's fit to each target moment.

Table 8.4. Manufacturing: Structural Estimates and Moment Fit

Panel A: Estimated parameters			
Parameter	Estimate	Std. err.	
γ (convex cost)	0.232	4.1×10^{-5}	
λ (disruption)	0.880	2.6×10^{-5}	
p_s (irreversibility)	0.909	1.1×10^{-4}	
SMM objective $J(\hat{\Theta})$	0.001		
Bootstrap classif. confidence	100%		
Panel B: Moment fit			
Moment	Data	Model	Difference
Serial correlation	0.185	0.178	−0.007
Profitability correlation	0.385	0.382	−0.003
Positive spike rate	0.117	0.141	+0.024
Negative spike rate	0.025	0.010	−0.015

Notes. Panel A reports SMM estimates from [Frank \(2026b\)](#), Manufacturing FF12 industry, 46,726 firm-year observations, 1971–2023. The reported standard errors are *model-based sampling standard errors*: they use the SMM sandwich formula with a firm-level block bootstrap (1,000 resamples) for the moment covariance matrix, evaluated at the estimated parameters. They measure sampling uncertainty conditional on the maintained model, fixed moment definitions, and estimated shock process. They do not measure model uncertainty, moment-selection uncertainty, or sensitivity to measurement conventions. The small magnitudes reflect large sample sizes and steep objective surfaces, not absence of specification error: the Hansen J -test of overidentifying restrictions rejects every industry in this framework at conventional significance levels, so the residual norm in Panel B is a more informative diagnostic of relative fit. The bootstrap classification confidence is the percentage of 1,000 firm-level bootstrap resamples that match the baseline binary (convex vs. non-convex) classification. Panel B reports the four target moments computed in Compustat and in a simulated panel of 2,000 firms for 300 periods at $\hat{\Theta}$.

The estimated parameters have a concrete economic interpretation. The convex cost parameter $\gamma = 0.232$ implies a marginal adjustment cost of $\gamma \times (I/K)$. At the mean manufacturing investment rate of 0.083, this is $0.232 \times 0.083 \approx 0.019$, or roughly 2 cents per additional dollar of investment. This is a modest smooth friction. The economically larger frictions are the non-convex components. Disruption ($\lambda = 0.880$) means that adjusting

capital destroys about 12% of current-period profits. This reflects production shutdowns, worker retraining, and plant reorganisation. Irreversibility ($p_s = 0.909$) means that selling capital recovers only 91 cents on the dollar, creating a 9% loss on disinvestment.

Two features of the estimates deserve comment. First, the standard errors are extremely small. This reflects the large sample size and the steep curvature of the SMM objective, not the absence of specification error. The Hansen J -test rejects correct specification for every industry in the Frank (2026b) sample, as is typical when structural models are taken to large panels. The estimates should be interpreted as pseudo-true parameters: the best-fitting homogeneous approximation to a population of heterogeneous firms. Second, the model overpredicts the positive spike rate (0.141 simulated vs. 0.117 in the data) and underpredicts the negative spike rate (0.010 vs. 0.025). These moments are sensitive to M&A contamination in the $\Delta PPENT$ measure; the CAPX-based robustness check in Frank (2026b) improves the spike fit but at the cost of also shifting other parameters.

Connection to Framework 4

Compare these structural estimates to the Q -regression discussion in Section 8.3. The structural γ is 0.232. A Q -regression on the same Compustat manufacturing data would produce an implied γ of roughly 7 to 15, depending on the Q measure. The gap is not a contradiction. It reflects exactly the measurement error discussed in Section 8.3: the substitution of average Q for marginal q when $\theta = 0.718$ inflates the estimate by roughly an order of magnitude. The remaining inflation comes from the non-convex costs, which create inaction regions that further attenuate the Q -investment relationship. Neither source of bias has anything to do with financial constraints.

The small sampling standard errors in Table 8.4 say nothing about how strongly the data pin down each parameter. To see that, we perturb each adjustment-cost parameter around its Manufacturing estimate, one at a time, and ask how the moments respond. Table 8.5 reports the two moments that carry most of the leverage, the serial correlation and the positive spike rate. This is the empirical counterpart of the sensitivity diagnostic that Andrews et al. (2017) introduce and that we develop in the Identification chapter (Section 2.5.2): the columns trace out the local moment Jacobian, the object the AGS sensitivity measure inverts to show which moments drive which parameters. A moment that barely moves when a parameter is perturbed carries little information about that parameter, and the estimator pins it down only weakly, however small its sampling standard error happens to be. This is exactly why the split between disruption λ and irreversibility p_s is fragile when the spike-rate asymmetry is small: the moments that would separate them hardly move.

Table 8.5. Moment Sensitivity to the Adjustment-Cost Parameters (Manufacturing)

Parameter	Value	Serial corr.	Pos. spike rate
γ (convex)	0.116	0.123	0.127
	0.232 (est.)	0.179	0.141
	0.464	0.217	0.151
λ (disruption)	0.800	0.126	0.134
	0.880 (est.)	0.179	0.141
	0.960	0.291	0.187
p_s (irreversibility)	0.820	0.168	0.137
	0.909 (est.)	0.179	0.141
	0.990	0.182	0.148

Notes. Each row perturbs one adjustment-cost parameter around its Manufacturing point estimate (marked “est.”), holding the other two fixed at their estimates and holding the Manufacturing shock calibration fixed: $\theta = 0.718$, idiosyncratic shock $\rho_\varepsilon = 0.456$, $\sigma_\eta = 0.582$, and aggregate shock $\rho_b = 0.946$, $\sigma_v = 0.116$. Each cell resolves the dynamic program and recomputes the moment on a simulated panel of 2,000 firms at random seed 8823. At the baseline the simulated moments reproduce the reported Manufacturing fit in Table 8.4 (serial correlation 0.178, positive spike rate 0.141) to three decimals. The entries are generated by `code/python/investment/smm_sensitivity_table.py`.

8.4.6 What the Cross-Industry Results Show

The estimation can be repeated for each Fama-French industry. Frank (2026b) carries this out for 37 industries at the Fama-French 48 level and 10 industries at the Fama-French 12 level (after excluding financials and utilities, which are conventionally dropped from investment studies). Three findings stand out.

First, non-convex frictions are nearly universal. 31 of 37 Fama-French 48 industries exhibit disruption, irreversibility, or both. Only six industries face purely convex costs: laboratory equipment, food products, construction, business services, entertainment, and restaurants. These industries share a common feature. Their physical capital is generic (restaurant fixtures, office space, testing instruments) with thick resale markets and no production shutdowns during adjustment.

Second, the *composition* of frictions varies sharply across industries. Semiconductor firms face disruption from retooling fabrication lines ($\hat{\lambda} = 0.917$) but can resell equipment

at near-full price. Oil firms face irreversibility from specialised wells ($\hat{p}_s = 0.966$) but have almost no smooth convex costs. Mining firms lose nearly half their current profits when adjusting capital ($\hat{\lambda} = 0.470$). Any framework that collapses these into a single parameter averages over economically meaningful variation.

Third, the broad Fama-French 12 classification understates the prevalence of non-convexity. Four of ten FF12 industries appear purely convex, but 13 of the 18 FF48 sub-industries within those groups are actually non-convex. The explanation is an aggregation effect. Pooling firms with different types of non-convex friction produces moments that no single non-convex parameterisation can match. The estimator defaults to the convex specification as a compromise. The FF12 classification is reliable when it says “non-convex.” It is unreliable when it says “convex.” The formal analysis of this aggregation bias, including sufficient conditions under which cross-industry comparisons remain valid, is in [Frank \(2026b\)](#).

The payoff from all this machinery is specific. Structural estimation of the [Cooper and Haltiwanger \(2006\)](#) type is the tool to reach for when you need to know the *kind* of adjustment cost firms face, not merely whether investment responds to a particular shock. It asks you to commit to a model and solve a dynamic programme, and in return it lets you decompose adjustment costs into economically distinct components, run counterfactual policy analysis, and see why reduced-form regressions produce the coefficients they do. When instead you want transparent identification of a specific treatment effect and are content to leave the mechanism unspecified, the reduced-form and causal tools of [Section 8.3](#) and [Section 8.6](#) are the better choice. The synthesis section returns to this trade-off and sets it out as an explicit decision rule ([Section 8.7](#)).

8.5 Multiple Capital Inputs and Firm Value

The Cooper-Haltiwanger model treats physical capital as the only input. But firms also invest in workers, in research and development, and in brand awareness. [Belo et al. \(2022\)](#) extend the neoclassical investment framework to four quasi-fixed inputs: physical capital, labour, knowledge capital (cumulative R&D), and brand capital (cumulative advertising).¹ The extension connects the structural investment model to a question that finance researchers care about directly. What determines a firm’s market value?

¹This section draws primarily on [Belo et al. \(2022\)](#), “Decomposing Firm Value,” *Journal of Financial Economics*, 2022. This is distinct from the related [Belo, Lin, and Vitorino \(2014\)](#), “Brand Capital and Firm Value,” *Review of Economic Dynamics*, 2014, which focuses specifically on brand capital in asset pricing.

8.5.1 The Valuation Equation

Under the Hayashi conditions (homogeneity of degree one in the profit and adjustment cost functions for each quasi-fixed input), the firm's market value decomposes additively. In the [Belo et al. \(2022\)](#) formulation,

$$P_{it} + B_{i,t+1} = q_{it}^P K_{i,t+1}^P + q_{it}^L L_{i,t+1} + q_{it}^K K_{i,t+1}^K + q_{it}^B K_{i,t+1}^B, \quad (8.8)$$

where P_{it} is the end-of-period equity market value at t and $B_{i,t+1}$ is the level of debt that finances the beginning-of-period- $t + 1$ capital stocks; the stocks K^P, L, K^K, K^B are also measured at the beginning of period $t + 1$, consistent with the timing in the original paper. The q_{it}^j are the shadow prices of input j at t . With quadratic adjustment costs (and writing ϕ^j for the adjustment-cost parameter on input j , to avoid notation collision with the profit-function curvature θ used elsewhere in this chapter), each shadow price depends on the adjustment cost parameter and the investment (or hiring) rate for that input:

$$q_{it}^P = 1 + (1 - \tau_t) \phi^P \frac{I_{it}^P}{K_{it}^P}, \quad q_{it}^L = (1 - \tau_t) \phi^L \frac{H_{it}}{L_{it}} W_{it}.$$

Knowledge and brand capital have analogous expressions. The multi-input generalisation requires homogeneity of the profit and adjustment cost functions in the vector of quasi-fixed factors, plus correct measurement of all stocks and shadow values.

The key feature of equation (8.8) is that it is additive. The contribution of each input to firm value can be computed as a share. The shares depend on two things: the stock of each input (how much physical capital, how many workers) and the shadow price (how costly it is to adjust each input). The shadow price is where adjustment costs enter the valuation.

8.5.2 What Determines Firm Value?

[Belo et al. \(2022\)](#) estimate the four adjustment cost parameters ($\phi^P, \phi^L, \phi^K, \phi^B$) by minimising the distance between observed and model-implied valuation ratios across portfolios of firms. The model fits well. The model achieves time-series R^2 up to 61% and cross-sectional R^2 up to 95% across the specifications reported in the paper.

The decomposition reveals that physical capital accounts for less of firm value than one might expect, and that the four-input shares vary substantially across industries and specifications. In the headline industry decompositions reported by [Belo et al. \(2022\)](#), physical capital accounts for roughly 30 to 40% of firm value, labour for 14 to 22%, knowledge capital for 20 to 43%, and brand capital for 6 to 25%, with the exact shares

depending on the industry group and the specification. The wide ranges reflect genuine economic heterogeneity: capital-intensive industries put more weight on physical capital, technology-intensive industries put more weight on knowledge capital, and consumer-facing industries put more weight on brand capital. Pooled or aggregate decompositions produce shares roughly in the middle of these ranges, but the text above should not be read as giving a single representative number for any input. Readers who want the specific shares for a specific application should consult the relevant industry-level table in [Belo et al. \(2022\)](#).

8.5.3 Variation across Industries

The composition varies across industries. [Belo et al. \(2022\)](#) split their sample into low-skill and high-skill industry groups (roughly corresponding to traditional manufacturing and technology). In their reported specifications, physical capital accounts for a notably larger share of firm value in the low-skill group than in the high-skill group, and knowledge capital goes in the opposite direction. The published table also reports time variation: the share of physical capital in firm value has trended down over recent decades, and the share of knowledge capital has trended up. The shares of labour and brand capital are more stable over the same period. The shifts appear within both industry groups, not only from compositional change in the economy.

8.5.4 Connection to Adjustment Cost Composition

The Belo et al. framework assumes quadratic adjustment costs for each input. It estimates a single parameter ϕ^P for physical capital. The structural estimation in [Section 8.4](#) decomposes what Belo et al. treat as a single parameter into three components: convex costs, disruption, and irreversibility.

[Frank \(2026b\)](#) shows that the gap in ϕ^P between low-skill and high-skill industries is absorbed largely by the non-convex components. The mean convex cost $\hat{\gamma}$ is similar across the two groups. Installed physical capital is valuable in manufacturing not because smooth installation is expensive, but because adjustment disrupts production and cannot easily be reversed. In technology, physical capital appears easier to adjust (convex costs dominate), and the installed stock commands less of a valuation premium. The high valuation of technology firms comes from knowledge capital and labour, not from physical capital.

The lesson generalises the chapter's theme. Physical capital accounts for less than a third of the average firm's market value, with knowledge capital and labour together making up the majority, yet in capital-intensive industries physical capital is disproportionately

valuable. The reason is not that installation is costly. It is that adjusting physical capital disrupts production and involves irreversible losses. Here too the *kind* of friction, not its overall magnitude, determines the valuation premium.

8.6 Causal Evidence: Does Listing Distort Investment?

The structural approach in Sections 8.3 through 8.5 asks what determines investment absent agency and financing wedges. The causal approach asks a different question. Does the way a firm is organised and financed *distort* its investment from what it would otherwise be?

The menu of causal investment questions is large. Researchers have asked whether stock market listing distorts investment, whether tax incentives and bonus depreciation raise it, whether uncertainty delays it, whether shocks to bank credit supply transmit to it, and whether debt overhang suppresses it. Each pairs a plausibly exogenous source of variation with the methods from Part II. We develop one of these designs in detail, [Asker et al. \(2015\)](#) on public versus private firms, and treat it as the chapter's running causal example. We choose it because it pairs most directly with the structural exercise in Section 8.4: the structural model estimates the adjustment technology, and the listing comparison asks what distorts investment once that technology is held fixed. The other designs we do not develop here; they are signposted in the synthesis section that closes the chapter (Section 8.7).

[Asker et al. \(2015\)](#) address this by comparing the investment behaviour of public and private firms. The comparison is informative because private firms are typically owner-managed with concentrated ownership. They face fewer agency problems and less pressure to meet short-term earnings targets. If stock market listing distorts investment, the distortion should show up as a difference between public and private firms.

8.6.1 Data and Identification

[Asker et al. \(2015\)](#) use a dataset from Sagedata covering more than 400,000 firm-years of private U.S. firms over 2001 to 2011. They match each public firm to a private firm of similar size in the same four-digit NAICS industry. The matching produces nearly identical size distributions across the two groups.

The identification strategy has two parts. The first is a level comparison: do private firms invest more than public ones, holding size, industry, and investment opportunities constant? The second is a sensitivity comparison: are private firms more responsive to changes in investment opportunities?

Investment opportunities are measured by sales growth, which is observable for both public and private firms. Tobin's Q is available only for public firms, but the authors construct an industry-level Q and a predicted Q for robustness.

8.6.2 Two Main Findings

Finding 1: Private firms invest more. In the matched sample, private firms invest 6.8% of total assets per year in gross fixed assets, compared with 3.7% for public firms. The gap is 3.1 percentage points. It holds across alternative investment measures (net investment, investment including intangibles, growth in total assets) and across multiple matching specifications (adding age, ROA, leverage, and cash holdings to the match).

Finding 2: Private firms are more responsive to investment opportunities. The sensitivity of investment to sales growth is 0.118 for private firms and 0.027 for matched public firms. Private firms respond about 4.4 times more strongly. The difference is statistically significant and economically large. It holds during the financial crisis (2008–2011), when private firms presumably became more financially constrained than public firms. The result is robust to a range of identification strategies that [Asker et al. \(2015\)](#) report, including matched comparisons, within-firm evidence around IPOs of firms that go public without raising capital (and so change only their ownership structure), and instrumental-variable specifications.

8.6.3 Is It Short-Termism?

The patterns are consistent with several explanations. Public firm managers might have a preference for a quiet life. They might face higher regulatory costs. They might attract less talented managers. The authors test a cross-sectional prediction that follows naturally from short-termism models and is harder for many alternative explanations to generate.

In short-termism models, a manager cares about both the firm's long-term value and its current stock price. He can boost the current price by reporting higher earnings, and he does so by cutting investment (which reduces depreciation charges). The incentive to cut investment increases with the sensitivity of the stock price to earnings news. This sensitivity is measured by the earnings response coefficient (ERC) in the accounting literature.

The prediction is that the gap in investment sensitivity between public and private firms should be small when ERC is low and should increase with ERC. In their specification, the estimated public-private sensitivity gap is concentrated in high-ERC industries and is small or absent when ERC is low. This is consistent with the short-termism model. It is not a unique prediction of that model in a formal sense, since several auxiliary assumptions

are needed to derive the comparative static, but it is harder for alternatives like “quiet life” or financial constraints to match.

Auxiliary evidence supports the interpretation. Public firms with more transient institutional shareholders (who focus on near-term earnings) have lower investment sensitivity. Public firms that habitually meet or beat analyst earnings forecasts (a signal of earnings management) have lower investment sensitivity.

8.6.4 Implications

The [Asker et al. \(2015\)](#) findings have two implications for this chapter.

First, they suggest that the low investment- Q sensitivity documented in Section 8.3 is not entirely due to measurement error. Part of it may reflect a genuine distortion. The evidence is consistent with public firms investing less, and responding less to investment opportunities, in a way that short-termism models can rationalise. Calling this an “underinvestment” relative to first-best is more than the data will support, because the first-best benchmark is not directly observed. Private firms may also differ from public ones in accounting, selection, governance, financing, owner objectives, or growth opportunities, and the matched comparison cannot fully eliminate all of these.

Second, the results connect to the structural estimation in Section 8.4. The structural model estimates adjustment costs from observed investment dynamics. If those dynamics are distorted by agency problems, the estimated adjustment costs absorb both the technological friction (the true cost of adjusting capital) and the agency friction (the wedge between the manager’s objective and value maximisation). Separating these two sources requires either structural models that incorporate agency (which are computationally demanding) or causal designs that hold the technology constant and vary the agency environment (which is what [Asker et al. \(2015\)](#) do).

Connection to Framework 5

The structural approach (Section 8.4) and the causal approach (this section) give complementary answers to the investment question. Cooper and Haltiwanger tell you that manufacturing plants face disruption costs and irreversibility. Asker et al. tell you that stock market listing is associated with substantially lower investment sensitivity. Neither result contradicts the other. The structural model describes the technology of adjustment. The causal evidence describes a plausible distortion introduced by organisational form. A complete account of corporate investment requires both. A footnote of caution: structural estimates may also absorb agency frictions and financing

effects, while causal estimates may combine treatment effects with changes in selection or measurement. The clean dichotomy is a useful organising device, not a literal partition.

8.7 Connecting the Traditions

This chapter has developed two approaches to the same question. What determines corporate investment? The structural approach, represented by [Cooper and Haltiwanger \(2006\)](#) and [Belo et al. \(2022\)](#), estimates the primitives of the firm's technology: the adjustment cost parameters, the profitability shock process, and the profit function curvature. The causal approach, represented by [Asker et al. \(2015\)](#), identifies the effect of a specific friction (stock market listing) on investment using transparent identification strategies.

8.7.1 What Each Approach Can and Cannot Do

The structural approach can recover economic primitives. It can tell you that manufacturing firms face 12% production disruption during capital adjustment and that selling capital recovers only 91 cents on the dollar. It can decompose reduced-form coefficients into their structural sources. It can do counterfactual policy analysis. [Frank \(2026b\)](#) uses the estimated models to simulate the effect of an investment tax credit across industries and finds that pass-through rates vary by more than an order of magnitude depending on the type of friction.

Structural models provide causal and counterfactual answers only through the maintained model. They do not provide design-based identification in the modern quasi-experimental sense of Part II. If the model is wrong, the parameter estimates do not have a clean structural interpretation. The Hansen J -test rejects the overidentifying restrictions in every industry in the [Frank \(2026b\)](#) sample, which is typical when a homogeneous representative-firm model is taken to a large panel of heterogeneous firms. The bias documented by the aggregation analysis in that paper is a concrete example of the kind of specification problem that the structural approach is exposed to.

The causal approach can provide credible identification. [Asker et al. \(2015\)](#) use matching, within-firm variation around IPOs, and instrumental variables. The reader can evaluate each identification strategy on its own terms. The cross-sectional prediction (the ERC interaction) provides a sharp test that alternative explanations cannot easily match.

Causal designs can estimate counterfactual contrasts for the observed treatment margin, but they generally cannot extrapolate to new policies or mechanisms without additional

structure. The [Asker et al. \(2015\)](#) design tells you that public firms respond less to investment opportunities than matched private firms in the observed sample. It tells you less about what would happen under a counterfactual listing regime that has never been observed, or about which structural friction (disruption, irreversibility, agency wedge) is doing the work. Answering those questions requires either combining the causal evidence with a model or running a structural exercise of the kind developed in Section 8.4.

8.7.2 How the Approaches Complement Each Other

Consider three questions a policymaker might ask about corporate investment.

What kinds of adjustment costs do firms face? This is a structural question. The answer (from Section 8.4 and [Frank \(2026b\)](#)) is that non-convex frictions are nearly universal and vary in composition across industries.

Does stock market listing distort investment? This is a causal question. The answer (from Section 8.6) is that the evidence is consistent with listing reducing investment sensitivity, and the reduction increases with the sensitivity of stock prices to earnings news.

Would an investment tax credit increase investment? This requires both. The structural model tells you how different types of firms would respond to the credit given their adjustment technology. The causal evidence tells you that public firms may not fully pass through the credit because of short-termist pressures. The policy prediction depends on combining the two.

8.7.3 A Research-Design Checklist for Investment Studies

When to Use 6

Match the design to the question, and know what each design leaves out. If the question is how lumpy investment is, the tool is descriptive: compute spike, inaction, and serial-correlation moments on the panel (Section 8.2); this shows the shape of the investment distribution but not why it has that shape. If the question is what kind of adjustment costs firms face, estimate the [Cooper and Haltiwanger \(2006\)](#) model by SMM (Section 8.4); this recovers the adjustment technology, but only conditional on the maintained model, and it delivers no design-based identification of any single treatment. If the question is whether listing changes investment, use a matched public-private comparison in the spirit of [Asker et al. \(2015\)](#) (Section 8.6); this identifies a treatment contrast on the observed margin, but it cannot rule out selection into listing or extrapolate to listing regimes never observed. If the question is whether an

investment tax credit would raise investment, you need both: a structural model to predict the counterfactual response and causal variation to validate it, since neither the model's extrapolation nor a single policy episode is convincing on its own.

8.7.4 Topics This Chapter Does Not Cover in Detail

Several topics deserve mention but are not developed at length in this chapter. They are signposted here for the interested reader.

Uncertainty and real options. [Abel and Eberly \(1994\)](#) provide the canonical treatment of how irreversibility plus uncertainty delays investment. The structural framework in this chapter handles uncertainty through the AR(1) profitability shock, but does not separately quantify the real-options channel. Studies that isolate the effect of policy or political uncertainty on investment are part of the causal investment literature.

Tax incentives and bonus depreciation. Investment tax credits and accelerated depreciation provide some of the cleanest natural experiments in corporate investment. The pass-through analysis in [Frank \(2026b\)](#) simulates these policies under the estimated structural model. The identification of the causal effect in the data is a separate empirical exercise that uses the methods in the Difference-in-Differences chapter.

Debt overhang. The interaction between leverage and investment is central to corporate finance. We defer most of it to the Financial Constraints and Capital Structure chapters, where the structural and causal traditions on this question are developed.

R&D and intangible investment. [Belo et al. \(2022\)](#) treats knowledge capital as one of four quasi-fixed inputs, but measurement of R&D and organisational capital is itself a substantial empirical issue. The Machine Learning chapter discusses text-based measures that have been used to capture intangibles.

M&A as investment. A large fraction of the largest investment episodes in Compustat are acquisitions rather than organic CAPX. The measurement subsection above flags this; the deal-based causal designs in the Mergers and Acquisitions chapter develop the topic.

Public-firm selection. [Asker et al. \(2015\)](#)'s matched comparison is informative but cannot fully eliminate selection into public ownership. The chapter has been careful to use “consistent with” rather than “caused by” language where the design does not support stronger claims.

8.7.5 Looking Ahead

Three later chapters build directly on the adjustment technology estimated here. Each adds a distortion that this chapter deliberately holds fixed, and in each the firm's dynamic programming problem from equation (8.1) reappears with something extra layered on top. The adjustment cost estimates from this chapter determine how the firm *would* invest absent any financial, agency, or payout consideration; the later chapters ask what each of those considerations adds.

The Financial Constraints chapter asks whether internal funds create an additional wedge on top of the adjustment technology. The Bellman problem reappears with a constraint that limits investment to available financing, and the cash-flow sensitivity puzzle introduced in Section 8.3 is the starting point. Frank (2026a) provides a structural interpretation of what treatment-effect regressions identify in that setting. The Capital Structure chapter asks how leverage and financing policy interact with that wedge, so that the financing mix becomes a choice the firm makes rather than a fixed background against which it invests. The Payout Policy chapter asks the mirror-image question: when a firm has internal funds in hand, when should it reinvest them at the adjustment technology estimated here, and when should it pay them out instead? In every case the technology is the same one we estimate in this chapter, and the later chapter isolates the wedge that financing, control, or payout introduces.

Exercises

The first three exercises are light on infrastructure. Exercises 1 and 2 use only the master Compustat panel built in the Panel Data chapter and require no model code; Exercise 3 needs only a small pre-simulated grid that ships with the book. Exercises 4 through 6 use the replication code from Frank (2026b), which solves the Cooper-Haltiwanger model by value function iteration, simulates firm panels, and runs the SMM estimator. The remaining exercises are conceptual. Use random seed 8823 throughout.

1. **Measurement and the investment moments.** Using only the master Compustat panel built in the Panel Data chapter, construct three investment measures: $CAPX_t/PPEGT_{t-1}$, $\Delta PPENT_t/PPEGT_{t-1}$, and $\Delta PPEGT_t/PPEGT_{t-1}$.
 - (a) For each measure, winsorize at the 1st and 99th percentiles within year and report the average investment rate, the positive spike rate (rate above 0.20), the negative spike rate (rate below -0.20), the inaction rate ($|\cdot| < 0.01$), and the first-order serial correlation.
 - (b) Which moments change most across the three definitions, and which are stable?

- Relate the pattern to Table 8.1: the measures differ in how they treat depreciation, retirements, and M&A.
- (c) The negative spike rate is the moment most sensitive to the choice of measure. Explain why, and explain what that implies for identifying irreversibility p_s from firm-level data.
2. **The canonical Q regression.** Using the same master Compustat panel, construct average Q as the market value of assets over the book value of assets and the investment rate $\Delta \text{PPENT}_t / \text{PPEGT}_{t-1}$.
- (a) Regress the investment rate on Q with firm and year fixed effects. Report the slope and the implied $\gamma = 1/\hat{b}$. How does it compare to the structural estimate $\gamma = 0.232$ from Section 8.4?
- (b) Add cash flow (π_{it}/K_{it}) to the regression. Does it enter significantly? State precisely what the coefficient on Q does and does not identify.
- (c) Explain why adding cash flow does not resolve the identification problem. Tie your answer to the non-classical measurement error in average Q from Section 8.3: cash flow is correlated with the part of marginal q that average Q misses, so it proxies for omitted profitability rather than for a financing wedge.
3. **Identification diagnostics from a pre-simulated grid.** The standard errors in Table 8.4 are extremely small, yet the J -test rejects the model. Sampling precision and economic identification are different things, and this exercise separates them using a pre-simulated grid of SMM objective values that ships with the book at `code/python/investment/smm_sensitivity_grid.npz`. The file contains the SMM objective on a $20 \times 20 \times 20$ grid over (γ, λ, p_s) centred on the Manufacturing estimates, together with the grid axes and the four simulated target moments at each node, so no model solve is required (arrays `obj` of shape 20^3 ; axes `gamma_grid`, `lam_grid`, `ps_grid`; and `moments` of shape $20^3 \times 4$).
- (a) Load the grid and plot the profiled objective along each parameter axis (minimising over the other two). Is the objective sharply peaked, or is there a ridge of near-equivalent values?
- (b) Fix γ and λ at their estimates and trace the objective as p_s moves over $[0.80, 1.00]$. Compare the curvature to the analogous slice in λ . Which parameter does the objective penalise more sharply?
- (c) Relate your answer to the AGS sensitivity discussion in the Identification chapter (Section 2.5.2) and to Table 8.5. What does a flat ridge imply about which moments would have to be added to sharpen identification?
4. **Replicating Cooper-Haltiwanger Table 3.** Use the replication code to solve the Cooper-

Haltiwanger model for arbitrary parameter values.

- (a) Run the five illustrative parameterisations from Table 8.3 (No AC, Convex, Disruption, Irreversibility, and Mixed). Verify that your simulated moments match the qualitative predictions.
 - (b) Which moment is most sensitive to γ ? Which is most sensitive to λ ? How can you tell from the simulation output?
 - (c) Increase the disruption parameter to $\lambda = 0.50$ (half of profits lost during adjustment). What happens to the inaction rate? What happens to the average investment rate? Explain the economic intuition.
5. **The Q-regression bias.** Using the Manufacturing parameter estimates from Section 8.4 ($\gamma = 0.232$, $\lambda = 0.880$, $p_s = 0.909$, $\theta = 0.718$):
- (a) Solve the model and simulate a panel of 2,000 firms. Construct average $Q_{it} = V(A_{it}, K_{it})/K_{it}$ from the value function.
 - (b) Regress the simulated investment rate on average Q . What is the implied γ from the regression? How does it compare to the true $\gamma = 0.232$?
 - (c) Add simulated cash flow (π_{it}/K_{it}) to the regression. Is it significant? Why or why not, given that there are no financial constraints in the model?
 - (d) Now repeat parts (b) and (c) using the true marginal $q_{it} = \beta \partial V / \partial K'$ instead of average Q . How much of the bias disappears? What does this tell you about the source of the bias?
6. **Aggregation and heterogeneity.** Suppose an industry contains two types of firms in equal proportions. Type A has only convex costs ($\gamma = 0.20$, $\lambda = 1$, $p_s = 1$). Type B has disruption costs ($\gamma = 0.20$, $\lambda = 0.85$, $p_s = 0.95$).
- (a) Simulate 1,000 firms of each type using the same shock process. Compute the four target moments separately for each type.
 - (b) Pool the two types and compute pooled moments. How do the pooled moments compare to the moments for each type separately?
 - (c) Run the SMM estimator on the pooled moments, restricting to a single parameter vector. What does it estimate? Does it recover the parameters of either type?
 - (d) What does this exercise tell you about interpreting industry-level structural estimates? Connect your answer to the aggregation discussion in Frank (2026b).
7. **Asker et al. (2015) and the ERC prediction.** Asker, Farre-Mensa, and Ljungqvist find that the gap in investment sensitivity between public and private firms increases with the industry-level earnings response coefficient.
- (a) Explain why this prediction follows naturally from short-termism models and is harder for alternative explanations to generate. Consider a preference for a quiet

life, financial constraints, and differences in investment opportunities.

- (b) Suppose the earnings response coefficient is zero for a particular industry. What does the short-termism model predict about the investment behaviour of public firms in that industry? Is this prediction borne out in the data?
 - (c) How does this causal evidence relate to the structural finding that Q -regressions produce biased estimates of γ ? Could the short-termism distortion contribute to the measured bias?
8. **From adjustment costs to firm value.** [Belo et al. \(2022\)](#) estimate that physical capital accounts for a larger share of firm value in low-skill industries than in high-skill industries.
- (a) Using the framework of equation (8.8), explain why a larger adjustment cost for physical capital leads to a higher shadow price q^P and hence a larger physical capital share of firm value.
 - (b) [Frank \(2026b\)](#) finds that the convex adjustment cost $\hat{\gamma}$ is similar across low-skill and high-skill industries, but non-convex costs (disruption and irreversibility) are much higher in low-skill industries. How does this finding change the interpretation of the Belo et al. result?
 - (c) If you were extending the Belo et al. framework to allow non-convex adjustment costs, what would change about the valuation equation (8.8)? Would the additive decomposition still hold? Why or why not?

Chapter 9

Financial Constraints

Constraints are not properties of firms. They are properties of firms in specific moments and contexts.

9.1 Why This Chapter

Theory predicts that financing frictions should affect real decisions. Forty years of empirical work has tried to pin down how much. The estimates are not close. Across credible papers using credible designs, the response of corporate investment to a dollar of additional financing capacity ranges from essentially zero to nearly one. [Bakke and Whited \(2012\)](#) report effects near zero for large firms at the pension funding threshold. [Rauh \(2006\)](#) reports roughly sixty-five cents per dollar in a broader pension sample. [Lemmon and Roberts \(2010\)](#) report nearly one-for-one for junk-rated firms during the 1989 credit market collapse. Each paper does what it claims to do. Each estimate is precisely identified within its design. And yet the numbers disagree by an order of magnitude.

This is not a story of careless research. It is the central empirical fact of the financial constraints literature, and it is the cleanest case in corporate finance for the structural-versus-causal distinction that organizes this book. The reason the estimates differ is that they identify different objects. A structural marginal response, a population average, and a complier-weighted average are three different parameters. When the underlying primitive is heterogeneous across firms, these three need not agree, and in this literature they emphatically do not.

The chapter has three goals. First, define what a financial constraint actually is, sharply enough that the question of whether a particular firm is constrained has a yes-or-no answer. Second, walk through the literature in roughly chronological order: the founding

controversy between Fazzari et al. (1988) and Kaplan and Zingales (1997), the cash-holdings reformulation by Almeida, Campello, and Weisbach (2004), the measurement critique by Farre-Mensa and Ljungqvist (2016), and the modern alternatives based on text and disclosure. Third, show how the causal-design tradition and the dynamic-structural tradition can be read as estimating different but compatible parameters, and what that means for how you should design and interpret your own work.

The chapter is not a comprehensive survey. Whole research programs receive a paragraph or are mentioned only in citations. The selection is driven by what teaches the methodology cleanly. For breadth, consult Stein (2003), Whited (2016), and the handbook chapter by Strebulaev and Whited (2012).

9.2 What Is a Financial Constraint?

9.2.1 The Simplest Possible Model

Begin with the cleanest setting in which the constraint question is well-posed. A firm has a stock of capital K and chooses investment $I \geq 0$ to maximize one-period profit

$$\max_{I \geq 0} \Pi(K + I) - C(I) - pI \quad (9.1)$$

subject to a financing constraint

$$I \leq W + D. \quad (9.2)$$

Here $\Pi(\cdot)$ is a strictly concave profit function ($\Pi' > 0$, $\Pi'' < 0$), $C(I) = \frac{\gamma}{2}I^2$ is a convex adjustment cost with $\gamma > 0$, p is the price of capital, W is internal funds, and D is debt capacity. Equity issuance is suppressed; equivalently, its shadow cost is folded into the upper bound. Frictionless capital markets are nested as the limit $D \rightarrow \infty$.

Let I^{unc} denote the unconstrained optimum, defined by the first-order condition

$$\Pi'(K + I^{unc}) = \gamma I^{unc} + p. \quad (9.3)$$

This optimum depends on investment opportunities through $\Pi'(\cdot)$ but not on the firm's financial position. Define *constraint slack* as

$$S \equiv W + D - I^{unc}. \quad (9.4)$$

A firm is constrained when $S < 0$ and unconstrained when $S \geq 0$. Optimal investment is

then a piecewise function:

$$I^* = \begin{cases} I^{unc} & \text{if } S \geq 0 \quad (\text{unconstrained}) \\ W + D & \text{if } S < 0 \quad (\text{constrained}). \end{cases} \quad (9.5)$$

This setup is intentionally stripped down. It has no dynamics, no uncertainty, no endogenous debt capacity, no equity issuance. The point is to isolate the constrained-versus-unconstrained dichotomy from everything else that complicates real firms.

We will call this the *hard-cap benchmark*: a hard upper bound on financing with no shadow cost of approaching the limit and no margin of substitution at the boundary. The binary structure that follows is a feature of this benchmark, not a universal property of financial frictions. In richer models with convex external-finance costs, costly equity issuance, precautionary cash motives, or endogenous debt capacity, the firm may invest below its frictionless optimum without being literally rationed. In those settings, “constraint severity” is naturally represented by a continuous object: a Lagrange multiplier on a financing constraint, a wedge between internal and external costs of funds, or a shadow value of cash. We adopt the hard-cap benchmark in this section because it makes the treatment-effect logic developed in Section 9.6 transparent. Section 9.7 returns to the richer-model setting, in which the relevant primitives are wedges and shadow values rather than indicators.

9.2.2 Three Consequences That Organize the Chapter

Equation (9.5) looks innocuous, but three of its features will recur throughout the chapter and the rest of the book. Each is a property of the hard-cap benchmark; each has a continuous analog in richer models, but the discrete version exposes the issues most cleanly.

1. Constraint status is binary at the firm level. In the hard-cap benchmark, each firm is either constrained or unconstrained at a given moment. There is no continuum of “constraint severity” *within* this model; severity, if defined at all, must be defined across firms or across time. This is uncomfortable and not how the language is used in practice. In dynamic models with continuous shadow values, “severely constrained” can be made formal as a high Lagrange multiplier or a large finance wedge. The chapter uses the binary version for transparency, not because it is the literature’s settled definition.

2. The structural marginal response is degenerate in the hard-cap benchmark. Differentiating I^* with respect to financing $F = W + D$ yields

$$\frac{\partial I^*}{\partial F} = \begin{cases} 0 & \text{if } S \geq 0 \\ 1 & \text{if } S < 0. \end{cases} \quad (9.6)$$

This is a step function. Investment is perfectly insensitive to financing for unconstrained firms and perfectly sensitive for constrained firms. Average sensitivities estimated in a mixed population are weighted averages of zeros and ones, and the weight depends entirely on who is in the sample. In models with costly external finance, this $\{0, 1\}$ dichotomy is replaced by a continuous distribution of marginal responses lying somewhere in $[0, 1]$, but the basic logic carries over: average effects are weighted averages, and the weights depend on the design.

3. Constraint status is endogenous. Firms with weaker financing frictions accumulate more internal funds, build more debt capacity, and select into samples that look unconstrained. Cross-sectional measures of constraint status therefore confound the constraint itself with everything that determines W , D , and I^{unc} . This is the deep reason that [Farre-Mensa and Ljungqvist \(2016\)](#) find weak correlations across constraint indices: each index is a different projection of an endogenous latent state onto observables.

Key Insight 1

In the hard-cap benchmark, a financial constraint is a binary individual property and the structural primitive is the latent indicator $\mathbb{I}\{S_i < 0\}$. In richer models with costly external finance, the same role is played by a continuous shadow value or finance wedge. Most of what follows in this chapter is a consequence of trying to recover this latent state (binary or continuous) from observable data.

9.2.3 Connection to the Framework

The book's organizing distinction is between *structural* approaches, which posit an economic model and identify its parameters, and *causal* approaches, which design variation to identify treatment effects. Financial constraints sit awkwardly across this divide because the structural parameter of greatest interest is a discontinuity rather than a coefficient. A causal design that finds an investment-to-financing elasticity of 0.65 has not estimated $\partial I^* / \partial F$ for any individual firm. It has estimated something that depends on the share of constrained firms in the relevant population. We will make this precise in Section 9.6. For

now, the lesson to carry into the historical material below is that a number can be unbiased for what it identifies and uninformative about what we wanted to know.

9.3 The Founding Controversy: FHP and KZ

9.3.1 Fazzari, Hubbard, and Petersen (1988)

Fazzari et al. (1988), hereafter FHP, asked an apparently simple question: does the investment behavior of plausibly constrained firms differ from that of plausibly unconstrained firms? Their strategy combined a sorting variable, dividend payout, with a panel regression. Firms paying low dividends were presumed to face costly external finance: paying out cash that could fund profitable investment is irrational unless raising replacement funds is more expensive than the dividends saved. High-payout firms were presumed unconstrained, since they had cash available that they chose not to retain.

Within each group, FHP estimated

$$\left(\frac{I}{K}\right)_{it} = \alpha_i + \beta Q_{it} + \gamma \left(\frac{CF}{K}\right)_{it} + \epsilon_{it}, \quad (9.7)$$

where I/K is investment scaled by capital, Q is Tobin's Q as a proxy for investment opportunities, and CF/K is cash flow scaled by capital. Their headline finding, in the original sample of U.S. manufacturing firms over 1970–1984, was a coefficient γ of roughly 0.67 for low-dividend firms and 0.25 for high-dividend firms. The differential was statistically significant, robust to many variants of the specification, and intuitive: constrained firms appeared to invest more out of internal funds than unconstrained firms.

FHP's interpretation, sharpened by much of the subsequent literature, was that investment-cash flow sensitivity measures the severity of financial constraints. Constrained firms must rely on internal funds; unconstrained firms can substitute external finance freely; therefore the ratio of investment changes to cash flow changes should be larger for the constrained. The paper became the most-cited piece of empirical corporate finance for a generation, and the cash-flow sensitivity coefficient became the standard test for constraints. By the mid-1990s, a small industry was running variants of (9.7) on different samples and constraint sortings.

9.3.2 Kaplan and Zingales (1997)

Kaplan and Zingales (1997), hereafter KZ, raised two objections, one theoretical and one

empirical.

The theoretical objection is that the monotonicity FHP asserted is not a theorem. Consider a firm choosing investment to maximize a value function $V(I, F)$ where F is internal financing. The first-order condition is

$$\frac{\partial V}{\partial I} = 0, \quad (9.8)$$

and the comparative static of optimal investment with respect to cash flow is, by the implicit function theorem,

$$\frac{\partial I^*}{\partial F} = -\frac{V_{IF}}{V_{II}}. \quad (9.9)$$

The sign of $\partial^2 I^* / \partial F \partial \theta$, where θ indexes constraint severity, depends on third derivatives of V . Under reasonable specifications of the curvature of the cost-of-external-finance schedule, the cross-derivative can be positive, negative, or zero. There is no general result that more-constrained firms must exhibit greater investment-cash flow sensitivity.

The empirical objection was sharper. KZ took the forty-nine low-dividend firms in the FHP sample and read their annual reports, management discussions, news coverage, and 10-Ks one by one. They classified each firm-year into five constraint categories on the basis of qualitative evidence: was the firm describing itself as financially strapped? Were there reports of canceled projects? Were ratings deteriorating? When they then re-estimated cash flow sensitivities within these qualitative categories, they found the opposite pattern from FHP. Firms KZ classified as *least* likely to be constrained had the *highest* cash flow sensitivities. Firms KZ classified as most constrained had sensitivities near zero. Their estimates are reproduced in Table 9.1.

Table 9.1. Cash flow sensitivities by KZ qualitative classification

Qualitative constraint status	$\hat{\gamma}$	t -stat
Least constrained	0.71	(4.32)
Possibly constrained	0.55	(3.89)
Most constrained	0.23	(2.01)

Notes: $\hat{\gamma}$ is the estimated coefficient on CF/K in equation (9.7). Sample: FHP low-dividend firms reclassified by Kaplan and Zingales (1997) using qualitative evidence. Reproduced from KZ Table V.

This was a problem. If FHP's interpretation were correct, the qualitatively constrained firms should have shown the largest sensitivities. They showed the smallest. Either KZ's qualitative classification was wrong, or the cash-flow sensitivity coefficient was not

measuring what FHP claimed.

9.3.3 What Survived

The FHP–KZ exchange produced a substantial follow-on literature, much of it methodological. Cleary (1999) extended the KZ index across a larger sample. Allayannis and Mozumdar (2004) re-examined the role of hedging. Moyen (2004) showed in a structural simulation that constrained and unconstrained firms in a calibrated dynamic model could exhibit either ranking of cash flow sensitivities, depending on the calibration. Twenty years later, the consensus was roughly this: cash flow sensitivities reveal *something* about financing frictions, but the something is not the structural marginal response, and the sign of the relationship between sensitivity and constraint severity is theoretically ambiguous.

It is tempting to read this as KZ winning. That would be too tidy. The deeper point is that neither paper had a model precise enough to give cash flow sensitivities a structural interpretation. FHP wrote down a regression and asserted what its coefficient meant; KZ wrote down a counterexample and asserted what it didn't mean. Neither side had a complete answer to the question of what $\hat{\gamma}$ does identify. That question would not get a clean answer for another two decades, and even now the answer requires being explicit about a class of structural assumptions. (We return to this in Section 9.6.3.)

Common Mistake 2

Students sometimes summarize FHP–KZ as “the FHP regression was wrong.” That overstates the case. FHP estimated a quantity that exists and that varies across firms; the controversy was about what that quantity means. Be careful to distinguish “the estimator is biased for θ ” from “ θ is not what we want to know.”

9.3.4 Running Example: Leverage as a Constraint Proxy

The running example of this book uses leverage as a measure of financing capacity in the structural tradition and as an outcome of identification-driven shocks in the causal tradition. In the FHP–KZ debate, leverage made a brief appearance: KZ used it among their classification criteria, noting that firms with high debt and exhausted lines of credit were more likely to be in a tight spot. The FHP setup did not include leverage as an explanatory variable. With the benefit of forty years of hindsight, leverage is a clearly imperfect proxy for $\mathbb{I}\{S_i < 0\}$: a firm can be highly levered because it has good investment opportunities and ample debt capacity, or because it is close to its capacity constraint. Cross-sectional regressions of investment on leverage and cash flow cannot distinguish

these. The instrumental and quasi-experimental approaches of Section 9.6 exist in part to resolve this ambiguity by exogenously shocking D rather than measuring it.

9.4 The Cash-Holdings Reformulation: Almeida, Campello, and Weisbach

The KZ critique pushed researchers to look for alternative empirical signatures of financing constraints, signatures that would be less vulnerable to the theoretical ambiguity that had bedeviled the FHP coefficient. Almeida et al. (2004), hereafter ACW, offered one of the more influential reformulations: rather than studying the response of investment to cash flow, study the response of *cash holdings* to cash flow.

9.4.1 The Logic

The idea is simple. A constrained firm anticipates that future investment opportunities may exceed available financing. Saving cash today preserves the option to invest tomorrow. An unconstrained firm faces no such precautionary motive: it can fund any positive-NPV project when it arises, so retaining cash has no value beyond its expected return. Therefore the propensity to save cash out of current cash flow—the *cash-flow sensitivity of cash*—should be positive for constrained firms and zero (or near zero) for unconstrained firms.

A simple two-period model makes the prediction precise. The firm chooses consumption (payout) c_1 in period one to maximize $E_0[V(c_1, c_0, k_0, w_0)]$ where k_0 is initial capital and w_0 is initial wealth. The first-order condition equates the marginal value of payout today to the discounted expected marginal value of cash tomorrow:

$$\frac{\partial V}{\partial c_1} = -1 + \beta E_0 \left[\frac{\partial V_2}{\partial c_1} \right] = 0. \quad (9.10)$$

For constrained firms in some future state, $\partial V_2 / \partial c_1 > 1$: an extra dollar of cash today is worth more than a dollar tomorrow because it relaxes a binding constraint. The optimal cash policy responds positively to cash flow shocks. For firms that never expect to be constrained, $\partial V_2 / \partial c_1 = 1$ in every state, and there is no systematic relationship between cash flow and cash accumulation.

9.4.2 The Test

ACW estimate

$$\Delta \text{CashHoldings}_{it} = \alpha_0 + \alpha_1 \text{CashFlow}_{it} + X_{it} \delta + \epsilon_{it} \quad (9.11)$$

separately for constrained and unconstrained subsamples, where the classification uses one of: dividend payout, asset size, bond ratings, commercial paper ratings, or the KZ index. The prediction is $\alpha_1 > 0$ for constrained firms and $\alpha_1 \approx 0$ for unconstrained firms. Their results, reproduced in Table 9.2, show that pattern consistently across classification schemes.

Table 9.2. Cash-flow sensitivity of cash by constraint classification

Constraint criterion	Constrained $\hat{\alpha}_1$	Unconstrained $\hat{\alpha}_1$
Payout ratio	0.251***	0.031
Asset size	0.287***	−0.012
Bond rating	0.233***	0.044

Notes: Estimated coefficients $\hat{\alpha}_1$ from equation (9.11). *** denotes significance at the 1% level. Sample: Compustat manufacturing firms, 1971–2000. Reproduced from Almeida et al. (2004).

Economically, the magnitudes imply that constrained firms save somewhere between roughly seventeen and thirty cents of every dollar of cash flow as a precautionary buffer, with the low end coming from ACW's baseline size-sorted specification and the higher end from payout- and rating-based sorts in Table 9.2. The effect is larger for high-Q firms (where future investment opportunities are more valuable), larger when cash flow is more volatile (where the option value of cash is greater), and qualitatively consistent across constraint measures (a feature that FHP-style regressions had lacked).

9.4.3 Why ACW Was Influential

Three things are worth noticing about the ACW design.

First, the prediction is sharper than FHP's. The cash-flow sensitivity of cash is theoretically signed under the ACW model: positive under constraints, zero without. The specific monotonicity-of-cross-derivatives ambiguity that bedeviled FHP does not apply, though as we discuss below, ACW retains its own dependence on classification, cash-flow measurement, and the underlying model of precautionary motives. Theoretical objections to ACW have to attack the model setup rather than the sign of a cross-derivative.

Second, the dependent variable is a stock change rather than a flow. This changes the kind of measurement error one worries about. Cash flow appears on both sides of the regression in FHP (since investment is part of the use of cash flow), but in ACW the LHS is the change in cash, which is a different financial decision than investment itself.

Third, ACW shifted attention from investment to liquidity management as the primary signature of constraints. This in turn motivated a generation of papers on cash holdings, lines of credit, and corporate liquidity, which sit alongside the investment literature rather than replacing it. [Sufi \(2009\)](#) documents that bank lines of credit are an important alternative liquidity instrument that is itself sensitive to firm cash flow, and that “cash” as measured on the balance sheet captures only one piece of a firm’s liquidity policy. [Acharya, Almeida, and Campello \(2013\)](#) develop the framework further: cash and lines of credit play different roles depending on the firm’s exposure to aggregate versus idiosyncratic risk. The general lesson is that a measure of constraint based on cash alone (as in ACW) is informative but partial.

9.4.4 What ACW Did Not Resolve

ACW dodged the KZ ambiguity at the cost of inheriting a different one. The classification problem remains. To estimate sensitivities by group, you have to assign firms to groups, and the standard classification schemes (payout, size, ratings, KZ index) all stand or fall on the same logic that the FML critique would soon challenge. ACW found their predicted pattern across schemes, which is reassuring, but the schemes are correlated with each other, and disagreement across schemes (which FML would document) does not vanish just because the dependent variable is now ΔCash rather than I .

In addition, the cash-flow sensitivity of cash is not a structural primitive in the model of Section 9.2. In that model, optimal investment is $W + D$ for constrained firms, and there is no residual cash to manage. A richer model is required—one with multiple periods, uncertainty over future opportunities, and a precautionary motive. [Riddick and Whited \(2009\)](#) and [Bolton, Chen, and Wang \(2011\)](#) provide dynamic models in which the ACW prediction can be derived endogenously. The price is that the relationship between the estimated $\hat{\alpha}_1$ and the underlying constraint severity is, again, model-dependent.

Connection to Framework 3

ACW is a causal-style design (group comparison, asymmetric predictions) layered on a structural-style motivation (precautionary savings). It illustrates a recurring pattern in this literature: the cleanest tests reach for the causal framework, but interpreting

the magnitude requires going back to a structural model. The structural-versus-causal distinction is not a choice of camps; it is a choice of which assumptions to be transparent about.

9.5 The Measurement Problem

By the early 2000s, the literature had accumulated a half-dozen popular ways to classify firms as constrained or not. Each was motivated by some piece of the underlying economics; each had been used in dozens of papers; and the systematic question of whether they agreed with each other, and whether classified-constrained firms behaved as constrained, had not been comprehensively addressed.

9.5.1 The Major Indices

The four indices most commonly used in the literature are summarized in Table 9.3.

Table 9.3. Popular constraint indices

Index	Source	Construction
KZ index	Kaplan and Zingales (1997) ; Lamont, Polk, and Saa-Requejo (2001)	Linear combination
WW index	Whited and Wu (2006)	Coefficients from
SA index	Hadlock and Pierce (2010)	Function of size
HM measures	Hoberg and Maksimovic (2015)	Text-based, sepa

Notes: The KZ index was constructed by [Lamont et al. \(2001\)](#) from the regression in KZ Table V. The Whited-Wu (WW) index is a size-age (SA) index was designed by Hadlock and Pierce as a parsimonious replacement for KZ.

The construction details matter for understanding what each measure is. The KZ index is essentially a regression coefficient on observable financial variables, fitted to KZ's qualitative classifications. Whether it generalizes outside that sample is an open question. The WW index is interesting precisely because of its provenance: [Whited and Wu \(2006\)](#) derived it not from sorting on observables but from the Euler equation of a dynamic investment model in which firms face costly external finance. The classification reflects the structural shadow value of internal funds. The SA index is the opposite of that strategy. [Hadlock and Pierce \(2010\)](#) argue that two simple firm attributes (size and age) capture most of what the more elaborate indices were measuring; the SA index is calibrated to qualitative constraint classifications using only those two variables. The HM measures attempt to read the constraint state directly off management's own discussion in 10-K filings.

In principle these are different windows onto the same latent state. In practice they often disagree.

9.5.2 Farre-Mensa and Ljungqvist (2016)

Farre-Mensa and Ljungqvist (2016), hereafter FML, did the natural diagnostic: take the standard indices, classify firms as constrained, and check whether the classified-constrained firms actually behave as the model says constrained firms should. They ran three kinds of revealed-preference tests.

Test 1: Market access. If a firm is classified as constrained, it should have trouble accessing external capital. FML estimate

$$\text{Pr(Equity Issue)}_{it} = \alpha + \beta \text{ConstraintMeasure}_{it} + X_{it} \gamma + \epsilon_{it}, \quad (9.12)$$

and similarly for debt issuance. Firms classified as constrained by the standard indices issue equity at roughly the same rate as those classified unconstrained, succeed at issuance at similar rates, and obtain comparable offer prices. On the debt side, 73% of the “constrained” firms in their sample have bond ratings and 62% issued bonds in the prior three years. Spreads on issued bonds differ by less than 20 basis points between the two groups.

Test 2: Response to natural experiments. If a firm is constrained, an exogenous loosening of financing should produce more investment than the same loosening for an unconstrained firm. FML use state corporate-tax cuts and the 2008–2009 financial crisis as the experiments. They find no systematic difference: firms classified as constrained do not exhibit larger investment responses than firms classified as unconstrained, and they did not fare systematically worse during the crisis.

Test 3: Cross-measure agreement. FML report the pairwise correlations among the major indices. The numbers are small. The KZ index and the WW index correlate around 0.3. The SA index correlates around 0.15–0.20 with both. The text-based HM measures correlate in the range of 0.1–0.2 with the financial indices.

The diagnosis is not that one index is correct and the others wrong. It is that they measure different things. The KZ index emphasizes leverage and cash flow; the SA index emphasizes size and age; the HM measures pick up whatever managers happen to say in 10-Ks. None of them is a high-fidelity proxy for $\mathbb{I}\{S_i < 0\}$, because there is no high-fidelity proxy. The latent constraint state is endogenous to everything that determines the indices.

9.5.3 Three Honest Responses to FML

A reader confronted with FML can react in three reasonable ways, and we will encounter all three in the rest of the chapter.

The first response is to abandon classification entirely and study identification-driven shocks to financing capacity, accepting that the resulting estimates are causal effects for a complier population that may or may not be representative. This is the route taken by the natural-experiments literature in Section 9.6.

The second response is to get the classification right, using better data and methods. The text-based measures of [Hoberg and Maksimovic \(2015\)](#) and the disclosure-based measures of [Lian and Ma \(2021\)](#) represent serious attempts in this direction. We treat both below.

The third response is to model the constraint state directly, structurally, as part of a dynamic firm problem in which constraint status is an endogenous outcome of optimization. [Hennessy and Whited \(2007\)](#), [Frank and Sanati \(2021\)](#), and others take this route. We treat the structural tradition in Section 9.7.

9.5.4 Text-Based Measures: Hoberg-Maksimovic

[Hoberg and Maksimovic \(2015\)](#), hereafter HM, used the text of management's discussion and analysis (MD&A) in 10-K filings to extract direct statements about liquidity and financing pressure. The method is a vector-space text classifier. For each firm-year, MD&A discussions of liquidity are vectorized and compared (via cosine similarity) to four reference word vectors representing distinct constraint types:

- DELAY_INV: language indicating investment plans deferred or canceled due to financing,
- EQUITY: language indicating difficulty or unwillingness to access equity markets,
- DEBT: language indicating debt-market constraints,
- PRIV: language indicating reliance on private placements.

What HM gain by this approach is direct measurement from the firm itself, time variation (constraint status changes from year to year), and separation of constraint types (a firm can be debt-constrained without being equity-constrained). What they lose is verifiability: the measures depend on the assumption that MD&A language is informative and not strategically managed. The pairwise correlations of the HM measures with each other are modest, consistent with the interpretation that the four measures capture distinct constraints. Their correlations with the standard financial indices are also modest, which is either a feature (the text measures are picking up something the financials miss) or a bug (none of the measures is reliable), depending on prior.

HM document that firms with high `DELAY_INV` scores subsequently reduce investment by 12% relative to unscored firms, reduce R&D by 15%, and accumulate cash. The patterns are consistent with the constraint interpretation, and the time variation in the measures opens up panel identification strategies that static indices preclude.

9.5.5 Disclosure-Based Measures: Lian and Ma

Lian and Ma (2021), hereafter LM, take yet another route. Rather than classify firms as constrained, they ask what the actual borrowing limits are. Using a hand-collected sample of debt covenants from U.S. nonfinancial firms, they document a striking fact: for roughly 80% of the corporate debt by value in their sample, the binding covenant is a cash-flow-based limit (typically debt-to-EBITDA or interest coverage), not an asset-based limit (a collateral or borrowing-base constraint).

This is in tension with the textbook model in equation (9.1), where debt capacity D is typically interpreted as a function of collateral K via $D \leq \theta K$. If actual limits are tied to cash flows, the relevant state variable is $\Pi(K + I)$ rather than K itself. The constraint in equation (9.2) becomes

$$I \leq W + \phi \Pi(K + I) \quad (9.13)$$

for some pledgeability fraction ϕ , which is a quite different object than $W + \theta K$. Note that the right-hand side of this constraint depends on I through $\Pi(K + I)$: investment expands the very capacity that finances it. This implicit structure matters for comparative statics. Under the asset-based version, debt capacity is fixed at the time of the financing decision; under the earnings-based version, marginal investment relaxes the constraint to the extent that it raises expected cash flows, and the optimal investment policy reflects this feedback.

LM's measurement is important for two reasons. First, it tells us *what* the constraints are, which has been hard to learn from indirect inference. Second, it implies that proxies based on tangibility or asset value (which a lot of the older literature uses) are probably measuring the wrong thing for large U.S. firms. This points toward earnings volatility, profit margins, and rating thresholds as more appropriate constraint proxies than tangibility-based measures.

Common Mistake 4

A common student error is to introduce a new constraint measure, show that it correlates with investment or financing decisions, and conclude that constraints matter. The FML critique applies. Showing that your measure correlates with the right things is necessary but not sufficient. You also need to show that the measure does not classify firms with revealed market access as constrained, and that it generates the predicted

differential responses to plausibly exogenous shocks. Otherwise you are repeating the FHP–KZ exchange with new names.

9.5.6 Where the Measurement Problem Leaves Us

The honest summary of forty years of measurement work is that we have several useful proxies, no high-fidelity measure, and a growing understanding of *why* a high-fidelity measure is hard. The constraint state is binary, latent, endogenous to everything that determines the proxies, and time-varying. The next two sections turn to two ways of making progress without solving the measurement problem first: exploit identification-driven variation in financing capacity (Section 9.6), or write down a structural model in which constraint status is an endogenous outcome (Section 9.7).

9.6 The Causal Tradition: Natural Experiments

The measurement problem of Section 9.5 pushed a generation of researchers to ask: even if we cannot reliably classify firms as constrained or not, can we still recover the effect of a financing shock on real outcomes? If financing capacity is exogenously perturbed, the average change in investment reveals *something*, even without a classification. This insight, imported from the labor and public economics literatures, motivated the natural-experiments wave in corporate finance that began in the late 1990s and accelerated through the 2010s.

9.6.1 What Identification Buys

Let $F_i = W_i + D_i$ denote firm i 's financing capacity. A natural experiment provides exogenous variation in F_i , typically through an institutional rule (pension funding requirements), a regulatory change (FIRREA), a market disruption (the junk bond market collapse), or a price shock (real estate values for collateral). Under the standard IV or DiD assumptions, the design identifies a local average treatment effect:

$$\tau = E \left[\frac{\partial I^*}{\partial F} \middle| \text{compliers} \right]. \quad (9.14)$$

The expectation is over a specific subpopulation: those firms whose financing was actually moved by the shock. This is the well-known point that LATE is a property of the design as much as of the population (see Chapter 5 on IV and [Imbens and Angrist 1994](#)).

Importantly, equation (9.14) sidesteps the classification problem of Section 9.5. The researcher does not need to assert which firms are constrained. The shock does the work of revealing how investment responds, on average, in the part of the sample where financing actually moved.

9.6.2 Three Canonical Designs

The natural-experiments literature in corporate finance contains dozens of papers. Three have become canonical because they are clean, well-identified, and reach surprisingly different numerical conclusions.

Rauh (2006): pension contributions as IV. [Rauh \(2006\)](#) exploits the fact that U.S. defined-benefit pension plans are required by ERISA to make mandatory contributions when the plan is underfunded. The contribution is a function of plan assets and liabilities according to a formula that is not under managerial control in the short run. Rauh uses this mandatory contribution as an instrument for total internal funds available to the firm. His estimate of the marginal investment response is roughly 0.65: each dollar of mandatory pension contribution reduces investment by about sixty-five cents, with the remaining thirty-five cents accommodated through other financial margins. The sample includes the universe of DB-sponsoring publicly traded firms.

Lemmon and Roberts (2010): junk bond market collapse as DiD. [Lemmon and Roberts \(2010\)](#) exploit three near-simultaneous shocks to the below-investment-grade debt market in 1989–1990: the collapse of Drexel Burnham Lambert, the passage of FIRREA (which required savings and loans to liquidate junk bond holdings), and changes in NAIC guidelines that pushed insurance companies away from low-grade private placements. The combined effect was a sharp contraction in credit supply to junk-rated firms. Using propensity-score-matched unrated firms as the control group, Lemmon and Roberts estimate that net investment fell nearly one-for-one with net debt issuance among treated firms. Their headline number is approximately 1.0.

Bakke and Whited (2012): pension funding as RDD. [Bakke and Whited \(2012\)](#) revisit Rauh's setting using a regression-discontinuity design at the pension funding threshold. The institutional rule generates a sharp change in mandatory contributions at a particular funding level; firms just above and just below the threshold differ in their contribution requirements but should be otherwise similar. The RDD estimate is statistically indis-

tinguishable from zero. Firms near the threshold do not cut investment in response to mandatory contributions; instead they adjust on other financial margins, including receivables, payout, and short-term debt.

The three estimates are summarized in Table 9.4.

Table 9.4. Three estimates of the financing-investment relationship

	Bakke and Whited (2012)	Rauh (2006)	Lemmon and Roberts (2010)
Financing shock	Pension contributions	Pension contributions	Junk bond collapse
Design	RDD	IV	DiD
Population	Near-threshold	All DB sponsors	Below-investment-grade
Complier traits	Large, rated, liquid	Mixed credit quality	Junk only
Headline $\hat{\tau}$	≈ 0	0.60–0.70	≈ 1.0

Notes: $\hat{\tau}$ is the estimated effect of one dollar of financing shock on net investment. The Bakke-Whited and Rauh estimates use the same institutional setting (ERISA pension funding rules) and differ primarily in the design.

Confronted with Table 9.4, a reader can be forgiven for feeling that the literature is broken. Three credible papers, applied to the same general economic question, produce estimates spanning the unit interval. The temptation is to argue about which design is best identified. That misses the point. All three designs are well-identified for what they identify; they identify different things.

9.6.3 One Way to Read These Estimates

A useful framework for interpreting the three estimates simultaneously comes from Frank (2026a). Within the hard-cap benchmark of Section 9.2, the structural marginal response is binary: one for constrained firms, zero for unconstrained firms. Applying the LATE decomposition of Imbens and Angrist (1994) to this binary-response setting yields

$$\tau_{\text{LATE}} = \theta_C \cdot 1 + (1 - \theta_C) \cdot 0 = \theta_C, \quad (9.15)$$

where $\theta_C \equiv P(S < 0 \mid \text{complier})$ is the constraint share among compliers. Under this benchmark, the treatment effect equals the fraction of compliers who are constrained.

Under this reading, the three estimates in Table 9.4 are not in conflict. They are estimates of θ_C for three different complier populations:

- Lemmon-Roberts study junk-rated firms whose financing was contracted during the 1989 collapse. Nearly all of those compliers are constrained, so $\hat{\theta}_C \approx 1$.
- Rauh's IV picks up compliers across the credit-quality distribution: deeply underfunded firms tend to be smaller and weaker, but the design includes well-rated firms too. The

mix yields $\hat{\theta}_C \approx 0.65$.

- Bakke-Whited's RDD isolates firms near the funding threshold, who are systematically larger, more likely to be rated, and have access to short-term debt. They are predominantly unconstrained, so $\hat{\theta}_C \approx 0$.

The Bakke-Whited and Rauh comparison is particularly telling. Same institutional setting, same underlying rule, same data source. The difference in estimates comes entirely from the design selecting different complier populations.

Generalizing beyond the hard-cap benchmark. The equality $\tau_{\text{LATE}} = \theta_C$ depends on the strong assumption that constrained firms invest the full marginal dollar of financing while unconstrained firms invest none of it. Real firms have other margins: they can hold marginal financing as precautionary cash, pay down debt, smooth payout, or accumulate working capital. Frank (2026a) shows that if constrained firms allocate a fraction α of marginal financing to non-investment uses, the decomposition generalizes to

$$\tau_{\text{LATE}} = \theta_C \cdot (1 - \alpha). \quad (9.16)$$

A single estimate of τ_{LATE} no longer pins down θ_C : an estimate of 0.65 is consistent with $\theta_C = 0.65$ and $\alpha = 0$, with $\theta_C = 0.81$ and $\alpha = 0.20$, or with $\theta_C \rightarrow 1$ and $\alpha \approx 0.35$. Identification of the two parameters separately requires variation across populations with plausibly different θ_C but a common α , or external evidence on one of them (e.g., from structural models or from direct measurement of cash accumulation). In the more general formulation, τ_{LATE} identifies a product of the constraint share and the marginal investment propensity of constrained compliers. The qualitative lesson survives: the same estimate combines population composition with behavioral substitution, neither of which is recovered from a single regression.

Key Insight 5

Under the hard-cap benchmark, $\tau_{\text{LATE}} = \theta_C$: a causal treatment effect identifies the constraint share among compliers. In the more general formulation that allows for precautionary substitution, $\tau_{\text{LATE}} = \theta_C(1 - \alpha)$: the same estimate combines the constraint share with the marginal investment propensity of constrained firms. Either reading rationalizes how three credibly identified studies produce estimates spanning the unit interval. Neither reading allows a treatment effect from one population to be transported to another.

This framework is one way to make sense of the literature, not the only way. A skeptic

could reasonably object that the three studies differ in country (Bakke-Whited and Rauh are U.S.; Lemmon-Roberts is too, but the institutional setting differs), in time period, and in the nature of the shock (pension contributions vs. debt-market access). Each of those differences could in principle generate part of the spread, and the model abstracts from intertemporal substitution and endogenous changes in constraint status that could matter quantitatively. We return to robustness considerations in Section 9.8.

Financing constraints versus financial distress. A separate but related concern is that the populations in the most influential causal designs are not just constrained. They are also distressed, highly levered, low-rated, or exposed to systemic credit-market disruptions. The Lemmon-Roberts compliers are below-investment-grade firms during a credit-supply collapse; the Chodorow-Reich compliers borrow from banks with damaged balance sheets in 2008–09; the Almeida-Campello-Laranjeira-Weisbenner compliers had long-term debt rolling over into the worst quarter of the financial crisis. In all of these settings, treated firms face not just tighter financing but also deteriorating investment opportunities, elevated default risk, and changes in management discount rates. Disentangling the financing channel from these other forces requires either additional structure (a model that separates them) or additional identification (a design in which financing varies but distress and opportunities do not). The chapter’s framework attributes the LATE entirely to the financing channel; readers should remember that this is a working assumption, not a theorem.

9.6.4 Other Important Designs

Beyond the three canonical estimates, the literature contains several other natural experiments worth knowing. We mention them briefly here; each deserves its own treatment in a more topical survey, and each is consistent with the constraint-share interpretation when its complier population is examined.

Khwaja and Mian (2008) use bank-level liquidity shocks in Pakistan, identified via within-firm variation across the multiple banks a single firm borrows from. Their estimate of the lending response is approximately 0.6; among small firms, lending falls roughly one-for-one with the bank’s liquidity shock. Chaney, Sraer, and Thesmar (2012) exploit local real-estate price variation as a shock to collateral values. The estimated investment response per dollar of collateral is 0.06 for the full sample of publicly traded U.S. firms. The small number reflects the sample composition: most publicly traded firms have access to alternative financing channels, so the share of constrained compliers is small. Gan (2007) studies the Japanese land-price crash; Peek and Rosengren (2000) use the Japanese banking

crisis transmitted to U.S. markets through Japanese-owned banks. [Chodorow-Reich \(2014\)](#) extends the bank-lending channel to employment during the 2008–09 crisis, finding large effects at small and medium firms and effects indistinguishable from zero at large firms. [Almeida, Campello, Laranjeira, and Weisbenner \(2012\)](#) use heterogeneous debt-maturity exposure to the 2008 crisis. [Campello, Graham, and Harvey \(2010\)](#) survey CFOs during the crisis, finding that self-identified constrained firms planned a 9% cut to capital spending compared to 0.4% for unconstrained firms.

What unifies these designs is not the magnitude of the estimate, which varies widely, but the pattern of heterogeneity. Across studies and across subsamples within studies, larger effects appear consistently among firms with characteristics associated with financing frictions: small size, low credit ratings, high leverage, low cash holdings, lack of bond market access. If treatment effects were measuring some heterogeneous slope unrelated to constraint status, there would be no reason for the same firm characteristics to predict larger effects across very different shocks and time periods. The cross-study heterogeneity pattern is itself evidence that constraints are doing real work, even when the average effect varies.

9.6.5 Running Example: Leverage and Investment

The leverage-investment running example finds its natural home in this section. Lemmon and Roberts is, in essence, the causal version of the running example: a shock to debt-issuance capacity, applied to a population of leveraged firms, with investment as the outcome. The estimated effect is large precisely because the population is composed of firms with binding debt limits. The same causal logic underlies covenant-violation studies ([Chava and Roberts, 2008](#); [Roberts and Sufi, 2009](#)) which use violation as the assignment mechanism for an exogenous tightening of effective debt capacity, and find substantial investment responses among violators.

When to Use 6

When a natural experiment is the right tool:

- The economic question is about the effect of a financing shock on a real outcome, for a defined population.
- There is a plausibly exogenous source of variation in financing capacity (rule-based, regulatory, market disruption, price shock).
- Generalization beyond the complier population is either not required, or is supported by additional argumentation about complier-population composition.

When it is not enough:

- Counterfactual policy analysis that requires extrapolating beyond observed variation. The complier-share interpretation does not directly support such extrapolations.
- Decomposition of constraint magnitudes into components (debt-capacity, equity-issuance costs, collateral pledgeability). Structural estimation is more direct.
- Cross-country or cross-regime comparisons where complier composition is very different. Numerical estimates do not transport even when the underlying economic environments are similar.

9.7 The Structural Tradition

The natural-experiments approach answers “what would happen to investment if financing were exogenously perturbed in this specific way?” Some questions are not of that form. “What is the marginal cost of external equity for small firms?” is a parameter question, not a treatment-effect question. “What share of the cross-firm variation in investment is attributable to financing frictions vs. productivity differences?” is a decomposition question. “If we changed the tax treatment of debt, how would the steady-state distribution of investment shift?” is a counterfactual question. For these, the structural tradition has the comparative advantage.

9.7.1 Hennessy and Whited (2007): External Financing Costs

Hennessy and Whited (2007), hereafter HW, estimate a dynamic model of firm investment and financing by simulated method of moments (SMM). The firm chooses investment, debt, and equity issuance over time, subject to costs of external finance that are the structural objects of interest. The model includes a corporate income tax, debt tax shields, costs of debt issuance, costs of equity issuance (both fixed and variable components), and default risk.

The firm’s problem can be written compactly as

$$V(K, B, z) = \max_{K', B'} \left\{ \pi(K, z) - \Phi(K', K) - C^d(B', B) - C^e(N) - rB + \beta E[V(K', B', z')] \right\} \quad (9.17)$$

where K is capital, B is debt, z is a productivity shock, $\Phi(\cdot)$ is the cost of capital adjustment, $C^d(\cdot)$ is the cost of debt adjustment, $C^e(\cdot)$ is the cost of equity issuance (parameterized as a function of net issuance N), and the expectation is over the AR(1) productivity process.

The structural parameters of interest are the issuance-cost coefficients in $C^e(\cdot)$ and

$C^d(\cdot)$ and the parameters governing default risk. HW estimate the model on Compustat data by matching moments: average investment rates, leverage ratios, equity-issuance frequencies, average size of equity issues, debt-issuance frequencies, and a handful of higher moments to discipline the dynamics. The estimation produces specific numerical estimates of issuance costs. For small firms, the variable component of equity issuance costs is roughly 5–10% of proceeds; for large firms it is much smaller. Default costs are estimated to be substantial, on the order of 10–15% of asset value, consistent with the broader bankruptcy-cost literature.

The contrast with the causal designs of Section 9.6 is instructive. HW does not require a natural experiment; the identification of the issuance-cost parameters comes from the joint behavior of investment, leverage, and issuance moments interpreted through the lens of the model. The price is that the estimates are conditional on the model. If the model is misspecified, the parameter estimates may not have any clean structural interpretation. [Andrews et al. \(2017\)](#) provide a sensitivity-analysis framework for diagnosing this dependence.

We do not redevelop the SMM machinery here. Chapter 8 (Investment and Q-theory) walks through the Cooper-Haltiwanger application in detail, with code, and the same numerical methods carry over to HW with minor modification. What this chapter contributes is the substantive interpretation: HW recovers external-financing costs, which is a different parameter from the constraint share that the causal designs identify. The two are complementary, not competing.

9.7.2 Frank and Sanati (2021): Financing Corporate Growth

A different but related structural exercise is [Frank and Sanati \(2021\)](#), hereafter FS. The model has the same general structure as HW: a dynamic firm problem with productivity shocks, costly external finance, and a collateral constraint, solved numerically and estimated by matching moments. The substantive focus is different. Where HW asks “what are the costs of external finance?”, FS asks “how do firms fund growth over the lifecycle, and what role do constraints play in shaping that pattern?”

The model has four building blocks. Productivity follows a firm-specific AR(1) process. External finance is costly, with the cost function calibrated to match issuance behavior. Debt is constrained by collateral, $B' \leq \theta K'$. Investment incurs convex adjustment costs. Default is endogenous: firms default when the value of continuation falls below the value of liquidation.

The model’s quantitative output is a decomposition of firm growth into sources. In

the FS calibration, productivity shocks account for roughly 45% of growth-rate variation, financial slack for 30%, market conditions for 15%, and initial conditions for 10%. The constraint matters for growth in a way that varies systematically over the lifecycle: young, small firms with high productivity and binding debt limits are the ones most affected.

The model's most distinctive output is its counterfactual policy analysis. FS evaluate three policy experiments:

1. A law limiting share repurchases.
2. A law limiting corporate debt.
3. The 1996 National Securities Markets Improvement Act, which loosened constraints on private equity financing.

The first two policies are shown to reduce aggregate investment and output in equilibrium, not the intended effect of either statute. The third is consistent with subsequent reduced-form evidence ([Ewens and Farre-Mensa, 2020](#)): looser private-equity financing led to more investment and growth, particularly at young firms. These counterfactuals are not available from natural-experiment designs. The treatment effects in Section 9.6 tell us what happened in compliers under realized variation; they do not tell us what would happen under a different rule, because they do not pin down the structural parameters that govern responses to alternative rules.

9.7.3 What Structural Estimation Does—and Does Not Do

A balanced reading of the structural tradition in financial constraints requires being explicit about its strengths and limitations. The handbook chapter by [Frank and Goyal \(2024\)](#) discusses this at length, and we will be brief.

The strengths are well-known. Structural estimates have a clean theoretical interpretation, tied to the parameters of an explicit economic model. They support out-of-sample prediction and counterfactual analysis. They enforce internal consistency: implications of the model in one dimension (e.g., issuance frequencies) discipline estimates in another (e.g., the cost of equity), and the researcher cannot have inconsistent beliefs across margins.

The limitations are equally real. Three deserve mention here.

Multiple equilibria. Dynamic capital-structure models with multi-period debt frequently admit multiple equilibria. [Chen and Frank \(2022\)](#) prove uniqueness under single-period debt; with longer maturities, expectations about future refinancing terms feed back into current debt prices, generating the kind of circularity that can sustain multiple solutions. In the financial-constraints literature, the issue often gets a footnote in published papers,

but it is more pervasive than that footnote suggests.

Observational equivalence. A variety of dynamic capital-structure models generate similar moments in the data. Whether the estimated parameters reflect the true data-generating process or merely the specific functional forms imposed is, in the words of [Frank and Goyal \(2024\)](#), “not clear.” [Nikolov, Schmid, and Steri \(2021\)](#) provide an important comparison of three models using a non-nested model selection test, but such comparisons are rare in the financial constraints literature.

Sensitivity to moment choice. SMM estimates can depend sensitively on which moments the researcher targets, and there is rarely a principled way to choose. [Andrews et al. \(2017\)](#) propose a sensitivity-analysis framework that reports the local derivative of each estimated parameter with respect to each targeted moment. Applying this diagnostic to structural constraint models is an underused practice.

These limitations do not invalidate the structural tradition. They argue for being honest about what structural estimates depend on, and for combining structural estimation with reduced-form evidence (the recommendation of [Andrews et al. 2017](#)) rather than treating them as substitutes.

Connection to Framework 7

Structural and causal approaches in this literature address different questions. The structural tradition recovers parameters of an economic model (issuance costs, default costs, decomposition of growth). The causal tradition identifies treatment effects in defined complier populations. Neither is a substitute for the other. The strongest empirical work in this area combines both: structural estimates to discipline the model, causal designs to validate specific predictions, and explicit sensitivity analysis to show where the conclusions depend on which assumption.

9.7.4 Running Example: Leverage as State Variable

In the structural tradition, leverage is an endogenous state variable that responds to productivity shocks, investment opportunities, and the collateral constraint. The running-example specification, expressed structurally, has leverage entering the value function in equation (9.17) as part of the firm’s state. The investment policy function $K' = g_K(K, B, z)$ depends explicitly on current leverage; high-leverage firms in low-productivity states cut investment, low-leverage firms in high-productivity states invest more. The same economic

primitives that generate the causal patterns in Section 9.6 generate the structural patterns here; the difference is the identification strategy, not the economics.

9.8 Synthesis: What Have We Learned?

This is a literature with strong opinions and a long history. It is tempting either to claim it is settled (it is not) or to claim it is hopelessly fragmented (it is more coherent than the cross-study spread suggests). Five claims seem defensible.

Constraints exist and bind for some firms. The evidence that small, junk-rated, low-cash, no-bond-rating firms cut investment in response to financing shocks is substantial and broadly consistent across designs. [Lemmon and Roberts \(2010\)](#), [Chodorow-Reich \(2014\)](#), [Almeida et al. \(2012\)](#), and [Campello et al. \(2010\)](#) all report large effects in subsamples chosen by such criteria. [Lian and Ma \(2021\)](#) document the underlying borrowing-limit mechanics. This is the firm-level signature of binding constraints. The caveat noted in Section 9.6.3 applies: in these settings, the financing channel is bundled with distress, default risk, and changes in investment opportunities, and the empirical literature has not fully unbundled them.

Average treatment effects mask heterogeneity. Three credible designs applied to broadly the same question produce three different estimates because they sampled three different complier populations. This is the central methodological fact of the literature. Whether one accepts the binary structural-response model that gives $\tau_{\text{LATE}} = \theta_C$ or not, the variation across designs is real and is driven by population composition. Anyone reporting a treatment effect from a financing shock should also report what is known about the complier population.

No single proxy is high-fidelity for constraint status. The standard indices (KZ, WW, SA), the text-based measures (HM), and the structural model-based classifications all capture pieces of the underlying constraint state, but none of them is a reliable substitute for the latent indicator $\mathbb{1}\{S_i < 0\}$. The disagreement across proxies, documented by [Farre-Mensa and Ljungqvist \(2016\)](#), is not a flaw in any one proxy; it is informative about the underlying heterogeneity in what makes firms constrained.

Method choice depends on the question. Treatment-effect designs are best when the goal is documentation of the existence and population-level magnitude of a constraint

channel. Structural models are best when the goal is recovery of specific parameters, decomposition, or counterfactual policy analysis. The two approaches answer different questions and are best read as complements.

Constraints are dynamic and endogenous. Constraint status changes over the firm lifecycle, in response to productivity shocks, with macroeconomic conditions, and as a function of past financing choices. Static cross-sectional classifications miss this. The FS model and similar dynamic structural exercises capture it, as do panel designs with time-varying shock exposure. The cross-sectional FHP regression, run on a representative cross-section, is asking about an average over an endogenously selected distribution; it is not surprising that the resulting coefficient is hard to interpret.

9.9 Implementation and Decision Framework

9.9.1 When to Use Which Approach

When to Use 8

Documenting that a constraint channel exists for a population: natural experiment with a clear shock and a pre-specified outcome. Report the complier characteristics, not just the average effect.

Quantifying magnitudes for a specific population: IV or DiD, paired with a clear statement of the constraint-share interpretation. Subsample analyses by ex-ante constraint proxies should be standard, not an afterthought.

Running counterfactual policy experiments: dynamic structural model. Be explicit about which features of the model are doing the work in the counterfactual, and report sensitivity to moment choice.

Testing heterogeneity directly: marginal-treatment-effect estimation or LATE-by-subgroup. The corporate-finance applications are still emerging.

Measuring the constraint state itself: a portfolio of measures (financial indices, text-based, disclosure-based) rather than reliance on any one. Report disagreement across measures; treat that disagreement as informative.

9.9.2 A Calibrated Simulation

To make the constraint-share interpretation concrete, the chapter includes a small simulation. We generate N firms with known constraint status, apply a financing shock with known

intensity, and estimate the LATE by IV. By varying the share of constrained firms in the population, we trace out the relationship $\hat{\tau} = \theta_C$. Operationally, the simulator assigns each of $N = 1000$ firms to the constrained group with probability θ_C , generates a standard-normal instrument Z at the firm-year level with first-stage $F = 0.5 Z + \text{noise}$, and sets the outcome $I = \mathbb{1}\{\text{constrained}\} \cdot F + \text{noise}$ so that financing affects investment only for the constrained firms. A 2SLS regression of I on F instrumented by Z , with firm and year fixed effects, recovers $\hat{\tau}$ at the swept value of θ_C . The Python implementation is in the *Constraint-share simulation* section of the Code Companion.

Plotting $\hat{\tau}$ against θ_C produces a clean 45-degree line, as predicted by equation (9.15). The intercept is zero (when no compliers are constrained, the LATE is zero), the slope is one (each additional constrained complier raises the LATE by $1/N$), and the relationship is tight in samples of moderate size. The simulation is pedagogically valuable because it shows, transparently, that the same data-generating process with the same structural parameters can produce LATEs anywhere in $[0, 1]$ depending on the population.

Two robustness variants are worth running. The first implements the precautionary-savings extension developed in Section 9.6.3: introduce $\alpha \in [0, 1)$ such that constrained firms invest only a fraction $1 - \alpha$ of marginal financing. Under this modification, the simulation produces $\hat{\tau} \approx \theta_C(1 - \alpha)$ rather than $\hat{\tau} \approx \theta_C$, and the 45-degree relationship is replaced by a line through the origin with slope $1 - \alpha$. The second variant introduces “switchers” (firms that begin unconstrained but are pushed across the threshold by the shock), generating an intermediate marginal response between zero and one. Both variants illustrate that the hard-cap benchmark is a useful baseline but not the only sensible specification, and that the structural-versus-causal logic of the chapter survives the relaxation.

9.9.3 What to Provide in a Paper on Financial Constraints

When students write papers using these methods, several practices distinguish credible work from work that gets criticized in seminars. We recommend:

1. Define what you mean by “constrained.” If you use an index, say which one and why. If you use a shock, describe the complier population.
2. Report multiple constraint measures, not just one. Show disagreement across measures as a robustness diagnostic.
3. If using a natural experiment, characterize compliers explicitly: their size, age, credit rating, pre-shock leverage, cash holdings. The constraint-share interpretation requires this.
4. If estimating a structural model, report sensitivity to moment choice. The Andrews-

Gentzkow-Shapiro diagnostic is now standard practice.

5. State what generalizes and what does not. An effect estimated on junk-rated firms is informative about junk-rated firms.

9.10 What This Chapter Does Not Resolve

The financial constraints literature is the cleanest motivation in corporate finance for heterogeneous treatment effects methods. The reason is that the structural primitive is itself heterogeneous: $\mathbb{1}\{S_i < 0\}$ is a firm-level binary, and any population estimand is necessarily an average over a non-degenerate distribution of these indicators. The methods developed for heterogeneous treatment effects in labor and public economics map onto this literature with unusual directness.

More advanced treatments develop several methodological extensions of relevance to financial constraints. [Heckman and Vytalil \(2005\)](#) marginal treatment effect (MTE) framework characterizes the full distribution of treatment effects across the latent index that determines selection into treatment. Applied to financing shocks, the MTE traces out how the effect of marginal financing varies across firms ranked by their unobserved resistance to the instrument. [Mogstad and Torgovitsky \(2024\)](#) provide a comprehensive treatment of IV with unobserved heterogeneity. [Blandhol et al. \(2022\)](#) clarify when 2SLS estimates can and cannot be interpreted as LATEs, which matters for the many financial-constraints papers that use multiple instruments simultaneously.

The natural next step for the literature is to estimate the constraint share *directly* rather than inferring it from a single LATE. This requires either multiple instruments applied to overlapping populations, or continuous variation in shock intensity, or both. [Catherine, Chaney, Huang, Sraer, and Thesmar \(2022\)](#) provide an example of how structural and reduced-form evidence on collateral constraints can be combined to identify a constraint share at the population level. The frontier here is open, and corporate-finance applications lag the labor and public-economics literatures by perhaps a decade.

9.11 Exercises

Exercise 1 (Conceptual). In the simple model of Section 9.2, derive the structural marginal response $\partial I^* / \partial F$ and show that it is a step function in S . Then, in a model where constrained firms must hold a precautionary cash buffer equal to a fraction α of marginal financing, show that the marginal response for constrained firms becomes $1 - \alpha$ rather than 1.

Exercise 2 (Conceptual). In the value-function setup of [Kaplan and Zingales \(1997\)](#), write down the second-order cross-partial $\partial^2 I^* / \partial F \partial \theta$ where θ indexes constraint severity. Find a specification of the cost-of-external-finance function under which this cross-partial is positive, and another under which it is negative. Explain in one paragraph what this implies for the empirical interpretation of cash-flow sensitivities.

Exercise 3 (Computational). Modify the simulation in the *Constraint-share simulation* section of the Code Companion to do the following:

- (a) Sweep θ_C from 0 to 1 in increments of 0.05. For each value, run $R = 500$ Monte Carlo replications and record the mean and standard deviation of $\hat{\tau}$ across replications. Plot $\hat{\tau}$ against θ_C with a 90% Monte Carlo interval. Verify the predicted 45-degree relationship.
- (b) Add a precautionary savings parameter α such that constrained firms have $\partial I / \partial F = 1 - \alpha$. Fix $\alpha = 0.2$ and rerun part (a). Plot the new relationship and confirm that the slope is now $1 - \alpha$ rather than 1.
- (c) * Without observing α directly, devise a procedure to bound it using two subsamples that you can argue have different constraint shares. Verify your procedure by simulation: set true values of θ_C in two subsamples and recover α from their estimated treatment effects. Discuss what assumptions your procedure requires.

Exercise 4 (Applied). Read [Bakke and Whited \(2012\)](#). The paper's RDD estimate is near zero, while [Rauh \(2006\)](#)'s IV estimate on essentially the same institutional setting is 0.65. Answer the following:

- (a) Describe the complier population in each paper. What firm characteristics distinguish near-threshold firms (Bakke-Whited) from the broader set of DB sponsors (Rauh)?
- (b) Under the hard-cap benchmark of Section 9.6.3, both estimates are consistent with the same underlying economics. Explain in 200 words why this is so.
- (c) Under what alternative model would the two estimates be genuinely inconsistent? Construct one such alternative and explain how you would test it empirically.

Exercise 5 (Computational). Using the Compustat panel built in Chapter 3 (replication file `ch04_panel.do` or `ch04_panel.py`):

- (a) Construct the SA index following [Hadlock and Pierce \(2010\)](#).
- (b) Construct the WW index following [Whited and Wu \(2006\)](#).
- (c) Compute the pairwise correlation between SA and WW for the full panel and for each decade separately. Report a 2×2 contingency table classifying firms as constrained or

unconstrained (top quartile of each index) by both measures, and compute the share that agree.

- (d) Comment on whether the disagreement between the two indices is concentrated in particular firm types (small vs. large, high-leverage vs. low-leverage). What does this imply for the interpretation of constraint-classified subsample regressions?

Exercise 6 (Applied). Choose one published paper that uses a financial-constraints instrument or quasi-experiment in a setting not covered in this chapter. Describe:

- (a) The identification strategy and the implied complier population.
- (b) What the paper's headline estimate would imply for the constraint share among compliers under the hard-cap benchmark, and how the interpretation changes under the precautionary-savings extension of equation (9.16).
- (c) Whether the paper's heterogeneity results (effects by subsample) are consistent with that implied $\hat{\theta}_C$.
- (d) What questions the paper does and does not answer, in light of the framework of this chapter.

Exercise 7 (Conceptual). Lian and Ma (2021) document that most U.S. corporate debt is constrained by earnings-based covenants rather than asset-based limits. In the textbook model of Section 9.2, D is typically modeled as a function of physical capital, $D \leq \theta K$. Rewrite the model with an earnings-based constraint, $D \leq \phi \Pi(K + I)$, and derive the new optimal investment policy. Discuss in one paragraph what changes about the structural marginal response and the constraint-share interpretation.

Chapter 10

Capital Structure

10.1 Why Capital Structure Still Matters

Sixty-five years after [Modigliani and Miller \(1958\)](#), we still cannot say with confidence what determines a firm's capital structure. The question has not been neglected. It has been studied harder, with more data and more methods, than almost any other in empirical corporate finance. The reason we lack a settled answer is that the question is methodologically harder than it looks, and the answer depends on what you are asking the data to do.

Four reasons make the question central to the field. The first is pedagogical. The Modigliani-Miller theorem is the bedrock of corporate finance teaching, and understanding why MM fails is how students learn what frictions matter and how to think about them. The second is the tight interconnection with investment. Financing decisions and real decisions interact through every channel a corporate model contains, and a theory of investment that ignores leverage is incomplete, just as a theory of leverage that ignores investment is incomplete. Chapter 8 and Chapter 9 both circle back to capital structure for this reason.

The third reason is constraint mismeasurement. If you mismeasure a firm's financing capacity, you misjudge its response to shocks, policies, and opportunities. Leverage is the most natural single-number proxy for financing capacity, and a leverage measure with a poorly understood structural interpretation will infect any constraint-based study that uses it. The fourth reason is policy relevance. Corporate tax rules, bankruptcy regimes, and regulatory capital requirements all act on the cost of debt. The size of the leverage response to a change in any one of these is a number policymakers want, and that empirical researchers are supposed to deliver.

Four robust empirical patterns frame what a theory of capital structure must explain. [Frank and Goyal \(2024\)](#) document them on US corporate data from 1945 to 2021. Aggregate leverage has been remarkably stable, with market leverage hovering around 30 percent and book leverage around 22 percent through enormous changes in the tax code, in bankruptcy law, in financial markets, and in corporate governance. Internal funds typically exceed capital expenditure needs, so the marginal financing decision is what to do with surplus cash rather than how to fund a shortfall. Net debt issues dominate net equity issues in every decade since the 1940s. And the ownership of corporate securities has shifted profoundly, from direct household holdings to intermediated holdings through mutual funds, ETFs, insurance companies, and foreign investors. Any theory that claims to explain leverage has to be consistent with stability that survives changes the theory says should move leverage substantially.

Both empirical traditions developed in this book have credible representatives working on capital structure. The structural tradition writes down a dynamic model of the financing-and-investment problem, chooses parameters to match data moments, and reports point estimates for primitives like the cost of external equity. [Hennessy and Whited \(2007\)](#) is the contemporary anchor, and [Section 10.5](#) works through their estimation in detail. The causal tradition looks for variation it can defend as exogenous, estimates a treatment effect with credible identification, and reports a local average response for the population whose financing decisions actually changed in response to the variation. [Heider and Ljungqvist \(2015\)](#) on state tax changes and [Leary and Roberts \(2014\)](#) on peer effects are the anchors here, with [Sections 10.6](#) and [10.7](#) working through them.

Neither tradition is wrong, and neither answers the other's question. The chapter's job is to show what each one contributes, where they disagree, and what their disagreement implies for empirical work on capital structure. The structural-versus-causal contrast that organizes the rest of this book lands hardest in capital structure, because both traditions study the same outcome on overlapping data and produce different kinds of answers, and the two answers cannot be collapsed into a single number.

This is a selective treatment. The comprehensive empirical map of capital structure is in [Frank and Goyal \(2024\)](#), which we recommend as the companion reference. Several substantial topics are out of scope: zero-leverage firms, the literature on debt structure (bank versus bond, secured versus unsecured, maturity), product-market interactions, labor and capital structure, behavioral channels beyond market timing, international and cross-country evidence, the full tax literature beyond the methodological representative we use, structural credit-risk models in the Merton tradition, the macroeconomic and aggregate balance-sheet evidence, bankruptcy and exit dynamics, and investor clientele

effects. Each of these deserves its own chapter, and pointers to handbook coverage appear in the relevant sections below. What this chapter does is one thing well. It shows how the structural and the causal traditions handle the same data on the same outcome, and what the answer is when they disagree.

10.2 Three Theories, Three Predictions

Three theories dominate the empirical capital structure literature. The trade-off theory says firms balance the tax shield of debt against the cost of financial distress and pick an interior optimum. The pecking order says firms prefer internal funds, then debt, then equity, and the resulting leverage is a residual rather than a target. Market timing says managers issue securities when their cost is locally low, and observed capital structure is the cumulative imprint of past timing efforts. Each theory gives a stylized account of why leverage is what it is. None matches the data in all its details, and most of what follows in this chapter is about what to do with that fact.

The trade-off theory begins with [Modigliani and Miller \(1958\)](#) and the observation that taxes break the irrelevance result. [Miller \(1977\)](#) and the corporate-tax literature that followed gave debt a tax advantage that scales with the marginal corporate rate. The cost on the other side is financial distress: direct bankruptcy costs, agency conflicts that intensify under high leverage, and debt overhang on future growth options. Firms choose the leverage at which the marginal tax benefit equals the marginal distress cost. The classical statement is informal; the discrete-time formal version, with explicit assumptions on the cash-flow distribution and closed-form comparative statics, is in [Ai, Frank, and Sanati \(2020\)](#), whose framework the chapter follows because it parallels the discrete-time models in Section 10.5 and produces explicit comparative statics that the empirical sections can speak to.

Consider a two-date setup. At date 0 the firm chooses debt with face value B due at date 1. At date 1 the firm realizes a stochastic cash flow X from a known distribution with mean μ and standard deviation σ . If $X \geq B$, the firm pays the debt in full and equity receives the residual $X - B$. If $X < B$, the firm defaults; bondholders recover the fraction $(1 - \alpha)$ of X and equity receives zero. Interest is tax-deductible at corporate rate τ , generating a tax shield when the firm is solvent. The risk-free rate is r .

Total firm value at date 0 is the unlevered value plus the present value of the tax shield minus the present value of expected bankruptcy cost:

$$V_L(B) = V_U + T(B) - K(B), \quad (10.1)$$

where $T(B)$ is the present value of the tax shield and $K(B)$ is the present value of expected bankruptcy cost. For B below the lower support of X , default is impossible and $K(B) = 0$, so V_L rises linearly in B at a slope determined by τ . As B rises into the range where default is possible, $K(B)$ becomes positive and convex, $V_L(B)$ becomes concave, and there is an interior optimum where the marginal tax benefit equals the marginal expected bankruptcy cost.

The optimal debt level $\hat{B}$ solves

$$\hat{B} = \arg \max_B [T(B) - K(B)]. \quad (10.2)$$

For specific cash-flow distributions, the optimum has a closed form; [Ai et al. \(2020\)](#) work out the uniform case in detail. The chapter takes the existence of $\hat{B}$ as established and works with the comparative statics that follow from it.

The optimal debt $\hat{B}$ is increasing in the tax rate τ : a higher tax rate raises the marginal value of the tax shield without changing the marginal expected distress cost. The optimal debt is decreasing in the bankruptcy cost fraction α under standard conditions on the cash-flow distribution; a higher α raises the cost of any given default probability. The optimal debt is increasing in mean cash flow μ and decreasing in cash-flow volatility σ . The first because more cash flow supports more debt without raising default probability, the second because higher volatility raises the probability of bankruptcy at any given debt level. The optimal debt is decreasing in the risk-free rate r under standard conditions, because a higher discount rate lowers the present value of the tax shield more than it lowers the expected bankruptcy cost. These comparative statics map directly to the Frank-Goyal factor regression of Section 10.3: profitability is a proxy for μ and should raise leverage; tangibility is a proxy for low α and should raise leverage; cash-flow volatility is a proxy for σ and should lower leverage. The empirical question is whether the signs of those regression coefficients in the data match what the static trade-off predicts.

[Fischer et al. \(1989\)](#) extend the static trade-off to a dynamic environment with proportional refinancing costs. The firm refinances when leverage drifts far enough from its target in either direction:

$$\text{refinance if } L_t < L^{\text{lower}} \text{ or } L_t > L^{\text{upper}}, \quad L^{\text{lower}} < L^* < L^{\text{upper}}. \quad (10.3)$$

Between the two triggers, the firm does nothing, because refinancing costs exceed the marginal benefit of adjusting. Outside the triggers, the firm refinances back to L^* . The inaction-region structure is what motivates the partial-adjustment regression specifications that appear elsewhere in the chapter; the speed of adjustment such regressions report is a

reduced-form summary of trigger-and-refinance dynamics, not a structural parameter.

The pecking order, developed by [Myers and Majluf \(1984\)](#) and [Myers \(1984\)](#), starts from a different friction. Outside investors do not know what a manager knows about the firm's true value. The information asymmetry imposes a lemons discount on outside equity, a smaller discount on debt, and no discount on internal funds. A manager acting in the interest of existing shareholders therefore funds investment first from cash, then from debt, and from external equity only as a last resort. The resulting financing preference is a strict ordering:

$$\text{internal funds} > \text{debt} > \text{external equity}, \quad (10.4)$$

which is read as the firm exhausting each source before turning to the next. Leverage is not a target; it is whatever falls out of the cumulative financing-deficit history.

The pecking order's empirical implications follow directly from the financing hierarchy. Profitable firms generate internal funds and pay down debt, so profitability should correlate negatively with leverage. Firms with persistent financing deficits issue debt and become more levered over time. There should be no target leverage and no mechanical mean-reversion to one. The first prediction matches the data and was read by an early generation of empiricists as evidence for the pecking order over the trade-off. The second and third predictions are more nuanced than the early tests assumed, and [Section 10.4](#) develops the test specifications and works through what the data deliver.

The overlap between the two theories is the point. On tangibility, size, and industry, trade-off and pecking order predictions point in the same direction. The cross-sectional regressions that have dominated the literature for forty years can confirm both theories simultaneously on these dimensions. Where the theories disagree, the disagreement is less clean than the verbal contrast suggests. A negative profitability-leverage slope is consistent with the pecking order. It is also consistent with a dynamic trade-off model in which profitable firms accumulate cash and have not yet rebalanced toward their target, as [Strebulaev \(2007\)](#) demonstrates in a calibrated simulation. A static comparative-static prediction is not the same as a regression coefficient on a panel of firms with infrequent rebalancing. Distinguishing trade-off from pecking order by sign tests on observational cross-sections is harder than it looks, and the early empirical literature underestimated the difficulty.

Market timing, in its modern form, was put on the map by [Baker and Wurgler \(2002\)](#). The theory says managers can identify periods when their firm's equity is mispriced, issue equity when it is high and debt when it is low, and not undo the leverage changes later. Capital structure becomes the cumulative outcome of repeated timing attempts rather than a deliberate target. [Baker and Wurgler \(2002\)](#) operationalize the timing channel through

an external-finance-weighted average of past market-to-book ratios:

$$\text{EFWA}_{i,t} = \sum_{s=1}^t \frac{e_{i,s} + d_{i,s}}{\sum_{r=1}^t (e_{i,r} + d_{i,r})} \cdot \left(\frac{M}{B} \right)_{i,s}, \quad (10.5)$$

where $e_{i,s}$ is net equity issuance in year s , $d_{i,s}$ is net debt issuance, and $(M/B)_{i,s}$ is the market-to-book ratio. Each past period's market-to-book is weighted by that period's share of the firm's total cumulative external financing through t , so EFWA is large when the firm raised disproportionate financing during high-valuation periods.

The leverage regression then asks whether EFWA has a separate explanatory effect on current leverage beyond the standard determinants:

$$L_{i,t} = a + b_{\text{EFWA}} \cdot \text{EFWA}_{i,t-1} + X_{i,t-1}^T c + \varepsilon_{i,t}, \quad (10.6)$$

with $X_{i,t-1}$ the Frank-Goyal control vector. The [Baker and Wurgler \(2002\)](#) claim is that $\hat{b}_{\text{EFWA}}$ is negative and economically large in their sample. If true, the data are consistent with the timing channel: firms that issued equity disproportionately during high-valuation periods have persistently lower leverage today. If $\hat{b}_{\text{EFWA}}$ is small or insignificant, the timing channel does not have a detectable persistent effect.

The market-timing literature attracted four serious critiques that [Frank and Goyal \(2024\)](#) catalogue in their handbook section. The neoclassical alternative is that high market-to-book reflects investment opportunities rather than mispricing, in which case the cross-section being read as timing is really the cross-section of growth firms. The pseudo-market-timing critique, related to the persistence facts of [Lemmon et al. \(2008\)](#), observes that firms which survive long enough to be observed at high market-to-book ratios are mechanically those that issued equity into high valuations, so the apparent timing pattern can arise even when no manager is timing anything. The firm-heterogeneity critique allows timing to be a behavior of a particular firm type rather than a population-level phenomenon. The fourth critique is statistical: many timing tests rely on long-horizon abnormal-return regressions, which [Fama \(1998\)](#) showed are unreliable for general reasons.

[Frank and Nezafat \(2019\)](#) address the statistical-horizon critique directly. They construct counterfactual issue dates within twenty-one-day windows around actual bond issuances and ask whether the realized date is unusually favorable relative to the counterfactual distribution. The test is short-horizon by design and does not inherit the long-run-returns problem. They find that bond issuers gain about eight basis points by timing the risk-free rate and about twelve basis points by timing their own CDS spreads, with effects concentrated where the theory predicts: frequent issuers do better than infrequent issuers,

and shelf-registered (Rule 415) bonds time the risk-free rate better than non-shelf-registered ones. What survives the critiques on this evidence is timing of some kind in the credit market under a methodologically careful test. That is not a vindication of equity-market timing as a primary driver of observed capital structure, and the cross-instrument extension is not free.

Richard Green once remarked that the trade-off is obvious, and the interesting question is what is on the other side from taxes. The remark captures something the rest of the chapter takes seriously. The three theories above each have empirical traction, none survives intact, and the overlap among their predictions is so broad that distinguishing them by sign tests on cross-sectional regressions is a losing proposition. What the chapter does instead is line up four methodological postures (descriptive regression in Section 10.3, theory-guided test in Section 10.4, structural estimation in Section 10.5, and quasi-experimental causal designs in Sections 10.6 and 10.7) and asks what each one tells us. The structural-versus-causal contrast that organizes this book lands hardest in capital structure, because both traditions have credible representatives studying the same outcome on overlapping data and producing different kinds of answers. The next section starts on the descriptive side and shows what stable cross-sectional patterns can and cannot establish.

10.3 Empirical Determinants and What They Don't Tell Us

A descriptive regression of leverage on firm characteristics does not test a theory. It establishes the stylized facts that a theory has to explain, and it lets us check whether the field has anything stable to work with. The empirical capital structure literature has converged, over four decades, on a small set of right-hand-side variables that explain a meaningful share of the cross-sectional variation in leverage and that survive across countries, decades, and definitions. This section walks the historical arc, runs the standard regression on the master Compustat panel from Chapter 3, and explains why the coefficients are useful for some purposes and unhelpful for others.

Titman and Wessels (1988) was the first serious attempt to bring the major theories of capital structure to firm-level data. The paper used a structural-equation model with measured factors (collateral value, growth, uniqueness, industry classification, size, volatility, profitability) and reported that profitability and uniqueness were robustly correlated with leverage in a US sample of manufacturing firms over 1974–1982. Titman and Wessels did not claim to identify causal effects. Their contribution was to establish that the cross-section of leverage was systematically related to firm characteristics in ways consistent with some of the theories and inconsistent with others.

Rajan and Zingales (1995) extended the question to the G7 economies and asked which factors were robust across institutional environments. They settled on a compact right-hand side of tangibility, market-to-book, log size, and profitability, and they confirmed that the signs of the coefficients were broadly similar across France, Germany, Italy, Japan, the United Kingdom, and the United States. The international stability was striking. Whatever determines leverage cross-sectionally is operating at the firm-and-industry level rather than at the country level, and this is the empirical claim that Frank and Goyal (2024) call Stylized Fact 6: the same factors predict leverage across decades and across countries.

Frank and Goyal (2009) systematized the literature by running the regression on a much larger US Compustat panel and asking which factors held up under standard robustness checks: different leverage definitions, different periods, different inclusion rules. Five factors emerged with reliable signs: industry median leverage (positive), profitability (negative), market-to-book (negative), tangibility (positive), and log assets (positive). The conventional dependent variable is book leverage, with market leverage often reported alongside as a robustness check. The historical arc, from Titman-Wessels through Rajan-Zingales to Frank-Goyal, is the literature settling on a particular descriptive regression and confirming its stability.

The chapter measures the Frank-Goyal cross-section against three reference exercises that ask whether the regression structure itself has explanatory bite. The first is the variance decomposition of Lemmon et al. (2008). The question is how much of the cross-firm variation in leverage is explained by stable firm characteristics versus time-varying observables. The decomposition compares the R^2 from two regressions on the same panel. The first includes only firm fixed effects:

$$L_{i,t} = \alpha_i + \varepsilon_{i,t}. \quad (10.7)$$

The R^2 from this regression is the share of leverage variance explained by stable, firm-level characteristics. The second adds the Frank-Goyal factors and a year fixed effect:

$$L_{i,t} = \alpha_i + \gamma_t + X_{i,t-1}^\top \beta + \varepsilon_{i,t}, \quad (10.8)$$

where $X_{i,t-1}$ is the Frank-Goyal factor vector (tangibility, profitability, size, market-to-book, industry median leverage). The incremental R^2 from (10.8) over (10.7) measures how much additional variation the time-varying factors explain beyond what the firm-level structure already captures. In Lemmon et al. (2008)'s sample, the firm-FE-only regression explained close to sixty percent of the cross-sectional variance in leverage; the factor-augmented model added only a few percentage points. The chapter's master-panel replication mirrors

the qualitative pattern, with somewhat lower between-firm shares because the panel extends through 2024 and the more recent firms have noisier leverage paths. The numbers are Fact 1 in the synthesis of Section 10.8.

The second reference exercise is [DeAngelo and Roll \(2015\)](#). The question is whether firm leverage is mean-reverting toward a stable firm-specific level or non-stationary at long horizons. For each firm i with sufficient time-series length, the augmented Dickey-Fuller regression is:

$$\Delta L_{i,t} = \mu_i + \rho_i L_{i,t-1} + \sum_{k=1}^K \theta_{i,k} \Delta L_{i,t-k} + \varepsilon_{i,t}, \quad (10.9)$$

with the unit-root null $\rho_i = 0$. The cross-firm summary is the share of firms for which the null is rejected at conventional significance levels:

$$\hat{\pi} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[\text{p-value}_i < 0.05]. \quad (10.10)$$

In [DeAngelo and Roll \(2015\)](#)'s CRSP-Compustat sample, $\hat{\pi}$ was on the order of fifty percent at the five-percent level: for roughly half the firms in their long-horizon panel, leverage looked indistinguishable from a non-stationary process. The chapter's master-panel replication produces a much higher non-rejection share, between seventy-four and eighty-two percent on firms with at least thirty years of data; the numbers are Fact 2 in Section 10.8. Either way, the directional message is the same. A substantial fraction of US public firms have leverage paths that do not look like mean reversion to a stable target over the relevant horizons.

The third reference exercise is [Frank and Shen \(2019\)](#), which estimates a partial-adjustment model with firm-specific targets and firm-specific slopes. The target for firm i is fitted firm by firm:

$$L_{i,t}^* = \beta_{0,i} + \beta_{1,i} (M/B)_{i,t-1} + \beta_{2,i} \log A_{i,t-1}, \quad (10.11)$$

with both intercept and slopes varying across firms. The partial-adjustment regression is then run against this firm-specific target:

$$L_{i,t} - L_{i,t-1} = \lambda (L_{i,t}^* - L_{i,t-1}) + \varepsilon_{i,t}, \quad (10.12)$$

with λ the speed of adjustment. [Frank and Shen \(2019\)](#) report $\hat{\lambda}$ on the order of 0.50 for market leverage and 0.45 for book leverage on their sample, implying half-lives of about one to one-and-a-half years. The same regression run against a common-slope Frank-Goyal-

Figure 10.1. Cross-sectional distribution of firm-by-firm slope coefficients from the [Frank and Shen \(2019\)](#) two-factor leverage model, estimated on Subsample C of the master Compustat panel. Top row: market-leverage targets. Bottom row: book-leverage targets. Left column: market-to-book slope $\hat{\beta}_{1,i}$. Right column: log-assets slope $\hat{\beta}_{2,i}$. The red vertical line in each panel marks the common-slope pooled OLS estimate on the same subsample. The histograms have substantial mass on both sides of the common-slope estimate, indicating that imposing a single slope across firms averages firms with opposing factor loadings.

style target on the same firms returns $\hat{\lambda}$ of approximately 0.15 to 0.20, implying half-lives of four years or more. The factor-of-three gap between estimates under firm-specific and common-slope targets is the Frank-Shen headline: the speed of adjustment is not a fundamental fact about firms but a function of how the target is specified. The chapter's master-panel replication matches Frank-Shen to within rounding; the numbers are Fact 3 in Section 10.8.

The three reference exercises set up the methodological lesson of this section. Descriptive regressions reveal different things depending on what one chooses to hold fixed. The variance decomposition holds firms fixed and asks how much the factors add. The unit-root tests hold each firm's time series fixed and ask about stationarity. The partial-adjustment regression holds the target firm-specific and asks about speed. None of the three is wrong. Each delivers a different answer about the same data. The disagreement among the three answers is the methodological setup the rest of the chapter responds to.

Running Example 1: Installment 1: three descriptive readings of the master panel

The chapter's opening empirical exercise puts three descriptive readings of the master Compustat panel side by side. Panel A is the [Lemmon et al. \(2008\)](#) variance decomposition with the incremental R^2 of common-slope Frank-Goyal factors over firm fixed effects. Panel B is the [DeAngelo and Roll \(2015\)](#) firm-by-firm Augmented Dickey-Fuller test on long-lived firms. Panel C is the [Frank and Shen \(2019\)](#) firm-by-firm two-factor regression with the partial-adjustment-speed contrast between firm-specific targets and a common-slope Frank-Goyal target on the same subsample. None of the three panels is presented as the right answer; each reports what falls out of one set of assumptions on the master panel. The disagreement among them is the methodological lesson the chapter is built on. Code is in `code/python/capital_structure/installment_1_three_panels.py`.

The Python and Stata implementations are in the *Frank-Goyal factor regression* section of the Code Companion.

The coefficient signs in Table 10.1 are expected to match the literature once the panel

Table 10.1. Three descriptive readings of leverage on the master Compustat panel. Panel A imposes common slopes across firms on a sample of firms with at least 20 years of observations and asks how much of the variation in leverage is between firms versus within firms. Panel B drops the common-slope restriction by running firm-by-firm unit-root tests on a sample of firms with at least 30 years of observations, asking whether individual firm leverage series can be distinguished from non-stationary processes. Panel C allows each firm to have its own slope coefficients in a two-factor model on a sample of firms with at least 10 years of observations and average market leverage at least 5%, and reports the implied speed of adjustment toward firm-specific targets.

Notes. The three panels are descriptive readings of the master Compustat panel under different assumptions about coefficient homogeneity, sample selection, and target structure. Each reading reflects what falls out of its own assumption set; the chapter discussion that follows treats none of the three as definitive. Panel A regresses leverage on firm fixed effects alone and then adds lagged FG factors with year fixed effects; the incremental R^2 is the gap. Panel B reports the cross-sectional fraction of firm-level ADF t -statistics that exceed the asymptotic critical value (i.e., that fail to reject the unit-root null). Panel C reports the cross-firm distribution of firm-by-firm slope coefficients, the comparable common-slope estimate from a single pooled regression with firm fixed effects on the same subsample, and the partial-adjustment speed $\hat{\lambda}$ from (10.12) against either the firm-specific target of (10.11) or a common-slope FG target on the same subsample, with bootstrap standard errors (200 iterations, clustered by firm, seed 8823).

Figure 10.2. Six long-lived firms from Subsample C of the master Compustat panel, chosen to illustrate the range of leverage paths the panel contains. Solid line: actual market leverage. Dashed line: firm-specific two-factor target $\hat{L}_{i,t}^*$. The firms were selected algorithmically rather than for prominence: one each by lowest within-firm leverage standard deviation, largest absolute trend coefficient, highest within-firm standard deviation with no trend, largest single-year leverage jump, closest target tracking by RMSE, and largest target-actual divergence by RMSE. Ticker symbols and gvkey identifiers appear in each subplot.

becomes available. Industry median leverage enters positively and is the largest single contributor to fit. Tangibility and size enter positively. Market-to-book enters negatively. Profitability enters negatively, a fact we return to below. The R^2 is typically around 0.30 in the cross-sectional specification and over 0.70 with firm fixed effects, which is the within-firm pattern that [Lemmon et al. \(2008\)](#) document and that we previewed in [Chapter 3](#). Most of the cross-sectional variation in leverage is between firms rather than within them.

A useful diagnostic is to run the same regression on subsamples by decade. The coefficient signs are stable. The magnitudes shift, but not by enough to overturn the cross-sectional ordering. International evidence beyond [Rajan and Zingales \(1995\)](#) reinforces the pattern. We treat this as a settled empirical fact: a parsimonious set of firm characteristics predicts cross-sectional leverage with a consistency that crosses decades and countries.

The coefficients in [Table 10.1](#) are stable, interpretable, and well-known. They are not causal. None of the right-hand-side variables is exogenous to leverage. Tangibility and size are co-determined with leverage through investment and growth decisions. Market-to-book

is a noisy proxy that confounds growth opportunities, mispricing, and managerial private information. Profitability is jointly determined with leverage through every channel a corporate model contains. A regression coefficient identifies what it identifies under the assumed orthogonality conditions, and none of the FG factor coefficients identifies a structural primitive. Treating them as the answer to a question like “what is the effect of profitability on leverage” is the methodological error this section is meant to vaccinate against.

The clearest case is profitability. Trade-off theory predicts that profitable firms, with more taxable income to shield, should carry more leverage. The data show the opposite: profitable firms carry less leverage, in every decade and country examined. The pecking order accounts for the sign cleanly through internal-funds-first behavior. So does a dynamic trade-off model with infrequent rebalancing, as [Strebulaev \(2007\)](#) demonstrates in a calibrated simulation. So does a partial-adjustment model in which profitable firms slowly rebuild capacity after past shocks. [Frank and Goyal \(2015\)](#) treat the puzzle as still open after the corrections each candidate theory makes.

None of the candidate explanations of the profitability puzzle is testable on the cross-section alone. Determining which channel is doing the work requires either a theory-guided test that exploits the structure of the candidate model, or a structural estimation that recovers the relevant parameters, or quasi-experimental variation that shocks profitability without shocking leverage through the unmodeled channels. The next section takes up the first of these three for the pecking order in particular. Sections [10.5](#) through [10.7](#) take up the other two. The descriptive regression of this section is the baseline against which those harder tests are evaluated.

10.4 Testing the Pecking Order: A Methodological Case Study

A descriptive regression establishes a cross-section. A theory-guided test goes one step further. It uses the structure of a candidate theory to derive a prediction sharp enough that the regression coefficient under the candidate theory differs from the coefficient under the alternatives. The pecking order is the natural test case because its hierarchy of financing preferences delivers a sharp quantitative prediction: if firms fund deficits by debt before equity, the slope of net debt issuance on the financing deficit is one for firms whose debt capacity is not exhausted. This section walks through four tests of that prediction. The sequence is more interesting than the answer.

The financing deficit is the gap between a firm's uses of funds and its internally generated funds. Frank and Goyal (2024) write the identity (their equation 2.7) as

$$DEF_t = DIV_t + I_t + \Delta WC_t - C_t, \quad (10.13)$$

where DIV_t is dividends, I_t is capital expenditure (or a broader investment definition), ΔWC_t is the change in working capital, and C_t is operating cash flow. By the cash-flow identity, the deficit is funded by net debt issuance plus net equity issuance: $DEF_t = \Delta D_t + \Delta E_t$. Under the strict pecking order, the firm uses internal funds first, then debt, and reaches external equity only after debt capacity is exhausted; in any firm-year where debt capacity is not binding, $\Delta E_t = 0$ and the deficit is funded entirely by debt. The empirical content of the strict pecking order is therefore that, on the unconstrained subsample,

$$\Delta D_t = DEF_t \implies b = 1 \text{ in } \Delta D_t = a + b \cdot DEF_t + \varepsilon_t. \quad (10.14)$$

The slope-one prediction is the sharp quantitative implication that distinguishes the strict pecking order from the alternatives.

Shyam-Sunder and Myers (1999), hereafter SSM, test this prediction directly. They estimate

$$\Delta D_{i,t} = a + b \cdot DEF_{i,t} + \varepsilon_{i,t} \quad (10.15)$$

on a sample of 157 US firms over 1971–1989, chosen to have continuous data and significant access to public debt markets. They obtained a slope of about 0.75, with a standard error around 0.04. They read this as broad support for the pecking order. The slope was not the strict prediction of one, but close enough that the deviation was attributed to measurement error and small departures from the canonical hierarchy. The paper was widely cited as evidence that the pecking order described the financing behavior of large US firms.

Frank and Goyal (2003) ran the same regression (10.15) on a much larger US Compustat panel that included small firms and firms with active equity issuance. The slope returned was about 0.28. Restricting their broader sample to large continuous-data firms approximately reproduced the SSM coefficient. The interpretation reversed. The pecking order was not a general description of corporate financing. It was, at best, a description of a particular subsample of large mature firms with established access to public debt. Among firms making equity issuance decisions, the financing-deficit regression underpredicts net debt issuance because equity issuance funds a substantial fraction of the deficit.

Lemmon and Zender (2010), hereafter LZ, pressed on this finding by classifying firms by available debt capacity. The argument is that the SSM-style regression of (10.15) is

informative only on firms that have unused debt capacity in the first place. Firms at their debt-capacity ceiling cannot issue more debt regardless of the financing deficit, and including them in the regression mechanically attenuates the slope. LZ flag firms as constrained when any of three triggers fires:

$$\text{Constrained}_{i,t} = 1 \left[\text{unrated or below investment grade, or } L_{i,t} - L_t^{\text{ind}} > 0.30, \text{ or modified Altman } Z_{i,t} < \right. \quad (10.16)$$

where L_t^{ind} is the firm's industry-median leverage in year t . Among firms with apparent unused debt capacity ($\text{Constrained} = 0$), the financing-deficit slope rises toward the SSM value. Among firms close to capacity ($\text{Constrained} = 1$), the slope falls, as a constraint-augmented pecking order predicts: firms that cannot issue more debt must issue equity, and the slope captures the substitution. The takeaway is that debt capacity was the missing variable. SSM and FG (2003) were estimating the same underlying behavior on different mixes of firms, and the apparent contradiction softened once debt capacity was conditioned on.

Leary and Roberts (2010) took the next step and abandoned the regression. They asked: among firms with apparently unused debt capacity by the LZ measures, what fraction in fact issue equity? Under the strict pecking order this fraction should be near zero. They construct three debt-capacity proxies (an investment-grade credit rating, leverage at least ten percentage points below the industry median, and a modified Altman Z-score above the distress threshold). The empirical object is the conditional equity-issuance frequency

$$\hat{\pi}_{\text{equity}|\text{unused}} = \frac{1}{|\{i, t : \text{unused}_{i,t} = 1\}|} \sum_{i,t:\text{unused}_{i,t}=1} 1[\text{equity issuance}_{i,t} > 0.01 A_{i,t-1}] , \quad (10.17)$$

where the threshold of one percent of lagged assets defines a meaningful equity issuance. Across the proxies, roughly 25 to 50 percent of firms with apparent unused capacity issue meaningful equity in a given year. The lower end of that range comes from the investment-grade-rating proxy, which selects toward larger and more conservative firms; the upper end comes from the below-industry-median-leverage proxy, which is broader (Leary and Roberts, 2010). The strict pecking order is rejected, and the rejection is not by a tiny margin. Information asymmetry alone, even after correcting for debt capacity, does not explain the observed equity issuance pattern. Their preferred reading is that the pecking order describes a tendency rather than a strict hierarchy, and that signaling motives, agency considerations, market timing, and regulatory constraints share the explanatory load.

Figure 10.3. Left: book leverage change against total investment intensity (capital expenditure plus acquisitions, scaled by lagged assets), on the DDW subsample of firms with at least 20 consecutive years of book leverage. Hexbin density with bin medians overlaid. Right: cross-firm mean of book leverage in event time around major acquisitions ($aqc_t/A_{t-1} > 0.05$), three years before through three years after. Leverage rises sharply at the acquisition year and partially reverses over the following two to three years.

Running Example 2: Installment 2: SSM and FG (2003) on the master panel

We replicate the SSM and FG (2003) financing-deficit regressions on the master Compustat panel. Net debt issuance is the change in total debt scaled by lagged assets. The financing deficit is constructed per equation (10.13), scaled by lagged assets. We run two specifications. The SSM sample restricts to large dividend-paying firms with continuous data and public-debt access over 1971–1989. The broad sample uses the full master panel with the standard cleaning rules of Chapter 3. The expected pattern is a slope near 0.75 in the SSM split and near 0.28 in the broad split. Code is in `code/python/capital_structure/installment_2_pecking_order.py` and `code/stata/capital_structure/installment_2_pecking_order.do`.

The Python and Stata implementations are in the *SSM and FG (2003) financing-deficit regressions* section of the Code Companion.

A companion exercise on the same panel measures the transitory share of within-firm leverage variation directly, in the spirit of DeAngelo et al. (2011). Cochrane (1988) introduced the variance-ratio statistic that the chapter uses for this purpose. At horizon k , the statistic compares the variance of k -period leverage changes to k times the variance of one-period changes:

$$VR(k) = \frac{\text{Var}(L_{i,t} - L_{i,t-k})}{k \cdot \text{Var}(L_{i,t} - L_{i,t-1})}. \quad (10.18)$$

Under a pure random walk, $VR(k) = 1$ for all k . When leverage has mean-reverting components, low-frequency variance grows less than linearly in k and $VR(k) < 1$; the quantity $1 - VR(k)$ measures the transitory share of variation at horizon k . The DDW measure is $1 - VR(5)$, computed firm by firm on the long-lived subsample. DDW report substantial transitory variation in their sample; the chapter's master-panel replication appears in Panel A of the table that follows.

A word on terminology before we continue. Every test in this section uses the structure of the pecking order in some way. SSM use the financing-deficit identity that the theory implies. FG (2003) test the same prediction on a broader sample with the theory specifying which slope should hold. Lemmon-Zender use the threshold structure of the theory, in

Table 10.2. DeAngelo-DeAngelo-Whited (2011) transitory-debt exercise on the master Compustat panel. Panel A reports the Cochrane (1988) variance-ratio measure of transitory variance, computed firm-by-firm on the subsample of firms with at least 25 years of leverage observations. Panel B reports standard partial-adjustment regressions under three target specifications: as the target moves from heavily constrained to firm-specific and time-varying, the implied speed of adjustment rises substantially. Panel C documents that large leverage changes co-occur with investment and acquisition activity, and that investment/M&A-driven leverage changes reverse on a 2–3 year horizon.

Notes. Panel A reports the [Cochrane \(1988\)](#) variance-ratio measure of transitory variance: the share $1 - \text{VR}(5)$, where $\text{VR}(k)$ is defined in (10.18), computed firm-by-firm on the subsample of firms with at least 25 years of leverage observations. The measure approaches 1 for highly mean-reverting series and 0 for unit-root series; values in between indicate a mixture. Panel B reports standard partial-adjustment regressions $L_{i,t} - L_{i,t-1} = \lambda(L_{i,t}^* - L_{i,t-1}) + \varepsilon_{i,t}$ under three target specifications: (i) a common-slope target from a pooled OLS regression of book leverage on lagged factors, (ii) the firm-specific two-factor target from Table 10.1 Panel C, and (iii) a DDW-style target equal to the fitted value from a firm-specific regression of book leverage on a linear time trend. As the target moves from heavily constrained to firm-specific and time-varying, the implied speed of adjustment rises substantially. The same data, under different assumptions about what the target is, deliver order-of-magnitude differences in apparent adjustment speed. Standard errors in parentheses are firm-cluster bootstrap with 200 iterations (see 8823). Panel C is on book leverage; major-investment dummy = 1 if capx_t/A_{t-1} exceeds the firm-specific 90th percentile; major-acquisition dummy = 1 if $\text{aqc}_t/A_{t-1} > 0.05$; large-leverage-change dummy = 1 if $|\Delta \text{book_lev}| > 0.05$.

which debt capacity binds before equity issuance becomes optimal. Leary-Roberts (2010) use the most theoretical structure of the four. They ask whether the strict prediction holds and report the residual where it does not. None of these four papers is what corporate finance calls a structural paper. The label is conventionally reserved for work that writes down a fully specified dynamic model of the firm’s optimization problem, chooses parameter values to make the model’s simulated moments match observed data moments, and reports those parameter values as the empirical content of the exercise. That is [Hennessy and Whited \(2007\)](#) in Section 10.5.

The distinction between the two uses of theory is sharper than the language usually admits. A theory-guided test uses theory to derive a prediction that can be checked against a regression coefficient on observed data. A structural estimation uses observed data to recover the parameters of a model specified completely enough to simulate counterfactuals. Both are legitimate uses of theory and data, and neither is a substitute for the other. The chapter’s pedagogical move is to take the trouble to name them, because the literature does not always do so, and students sometimes infer that a paper which mentions a theory and runs a regression has done what [Hennessy and Whited \(2007\)](#) did, which it has not. The rest of this chapter reserves the word “structural” for the narrower sense and uses “theory-guided” for the broader category that contains every regression in this section.

Key Insight 3: Theory-guided versus structural

The four tests in this section are theory-guided. [Hennessy and Whited \(2007\)](#) in Section 10.5 is structural. Conflating the two lets a regression coefficient stand in for a structural primitive, which it cannot.

What survives the four tests, on the empirical evidence, is a weaker version of the pecking order than SSM proposed and that the early literature treated as established. Firms display a higher propensity to issue debt than equity when both are available, and the financing deficit and net debt issuance are positively correlated in every reasonable sample. The strict prediction that the slope is one is rejected, even for firms with apparent unused debt capacity. The mechanism the theory proposed, single-shot information asymmetry with a lemons discount on equity, is consistent with the data only after additional moving parts are bolted on (debt-capacity thresholds, signaling motives, behavioral channels, regulatory frictions) that take the theory away from its parsimonious form. Attempts to redefine the thresholds to save the strict prediction have not succeeded in the way the original [Myers \(1984\)](#) sketch led one to hope. The question is what to do with this. The next section provides one answer by writing down a complete dynamic model and estimating its parameters. The two causal sections that follow it provide another by finding clean exogenous variation in particular moving parts of the financing problem and identifying what that variation does.

10.5 The Structural Approach: Leland to Hennessy-Whited

The structural approach to capital structure writes down a model in which a firm chooses leverage to maximize a value function with parameterized frictions, estimates those parameters from the data, and reports them as the empirical content of the exercise. The parameters are intended as primitives that do not change when policy changes, which is what enables the model to be used for counterfactuals beyond the observed sample. This section walks from the textbook scaffolding of [Leland \(1994\)](#) to the modern dynamic estimation of [Hennessy and Whited \(2007\)](#) and explains how the Chapter 8 dynamic programming machinery extends to support it. [Strebulaev and Whited \(2012\)](#) survey the broader structural corporate finance literature; we cite their treatment as the comprehensive reference and focus here on the two papers students need to read.

10.5.1 The Leland (1994) model: trade-off with endogenous default

Leland (1994) is the canonical model of trade-off with endogenous default: firms balance the tax benefit of debt against the expected cost of financial distress, and equity holders choose when to default rather than having default imposed by an exogenous trigger. The original is in continuous time, with geometric Brownian motion on the unlevered asset value and closed-form solutions through a smooth-pasting condition at the default boundary. This chapter presents the discrete-time, infinite-horizon version of the same model, because the discrete-time formulation parallels the rest of the chapter and the dynamic-programming machinery the Hennessy-Whited model in the next subsection uses. The cost is the loss of Leland's closed-form characteristic-root expression for the default boundary; the gain is internal consistency with how every other dynamic model in the book is stated and solved. The continuous-time formulation is in Leland (1994) directly, and Strebulaev and Whited (2012) survey the broader continuous-time structural literature.

The firm's unlevered asset value V_t is the single state variable. It follows a discrete-time geometric process under the risk-neutral measure:

$$\log V_{t+1} = \log V_t + \left(r - \delta - \frac{\sigma^2}{2} \right) + \sigma \epsilon_{t+1}, \quad \epsilon_{t+1} \sim \mathcal{N}(0, 1). \quad (10.19)$$

The drift includes the $-\sigma^2/2$ Jensen adjustment so that $\mathbb{E}[V_{t+1}/V_t] = e^{r-\delta}$, matching the continuous-time geometric Brownian motion drift. The riskless rate is r , the asset payout rate is δ , and the asset volatility is σ . The firm's debt and equity are derivative claims on V , valued recursively against this single stochastic state.

The firm issues perpetual debt with coupon C at $t = 0$. The coupon is paid each period the firm is not in default and is never refinanced. The static capital structure matches the original Leland setup: C is fixed once and for all at the issuance date. The tax shield is τC per period when the coupon is paid, at corporate rate τ . Bankruptcy costs are a fraction α of asset value at default.

At the start of each period, equity holders observe V_t and choose whether to default or continue. Equity value satisfies a Bellman with an American-option default feature:

$$E(V_t) = \max \left\{ 0, \delta V_t - (1 - \tau)C + \frac{1}{1 + r} \mathbb{E}[E(V_{t+1}) | V_t] \right\}. \quad (10.20)$$

The residual cash flow $\delta V_t - (1 - \tau)C$ is the after-tax dividend to equity: the firm's operating payout net of the after-tax coupon. The max with zero is the default option; when the continuation value (residual cash flow plus discounted expected continuation)

is non-positive, equity holders walk away and equity is worth zero in that period and thereafter. The max operator is the discrete-time analog of the continuous-time smooth-pasting condition: both pin down an endogenous default policy through equity holders' optimization rather than through an exogenous trigger.

The default boundary V_B is the lowest V at which equity holders strictly prefer continuation:

$$V_B = \inf\{V : E(V) > 0\}. \quad (10.21)$$

The continuous-time smooth-pasting condition $\partial E / \partial V|_{V_B} = 0$ becomes implicit in the max operator of (10.20); the boundary V_B is now computed numerically by value-function iteration rather than from a closed-form expression. This is the same numerical technology that solves the Hennessy-Whited Bellman in the next subsection.

Debt value depends on whether the firm is in default at t :

$$D(V_t) = \begin{cases} C + \frac{1}{1+r} \mathbb{E}[D(V_{t+1}) | V_t] & \text{if } V_t \geq V_B, \\ (1-\alpha)V_t & \text{if } V_t < V_B. \end{cases} \quad (10.22)$$

The solvent branch gives the coupon plus discounted expected continuation, the standard recursive valuation of a defaultable perpetuity. The default branch gives asset value net of bankruptcy costs: bondholders take ownership of the firm at the default time and lose the α fraction to bankruptcy frictions.

The firm chooses the coupon C^* at $t = 0$ to maximize total firm value, balancing the present value of the tax shield against the expected present value of bankruptcy costs:

$$C^* = \arg \max_C [D(V_0; C) + E(V_0; C)]. \quad (10.23)$$

The dependence on C enters through both the cash-flow stream and the endogenous default boundary $V_B(C)$, which moves with C because a higher coupon brings the marginal default decision forward. Comparative statics on C^* and on the resulting leverage are the same as in the continuous-time version: a higher tax rate τ raises optimal leverage by increasing the marginal value of the tax shield; a higher bankruptcy cost α lowers it by increasing the marginal expected distress cost; a higher asset volatility σ lowers it through the convexity of the bankruptcy term in expected value. These are the same comparative statics the descriptive regression of Section 10.3 estimates as coefficients on tangibility, size, and growth: tangibility is a proxy for low α , size is a proxy for low σ , and both should raise leverage in cross-section.

What Leland gives students is the cleanest statement of trade-off with endogenous

default in a framework where default is a choice rather than an exogenous trigger. The optimal coupon is interior. What Leland lacks for the questions this chapter cares about is dynamic real and financial decision-making: the firm makes a one-time capital-structure choice, there is no investment decision, no capital accumulation, no costly external equity issuance, and no rebalancing in response to new shocks. [Fischer et al. \(1989\)](#) added a partial dynamic layer with refinancing costs, but a full dynamic model with both real and financial decisions had to wait for [Hennessy and Whited \(2007\)](#). The next subsection takes that step within the same discrete-time Bellman framework.

10.5.2 The Hennessy-Whited (2007) model: dynamic financing with costly external equity

[Hennessy and Whited \(2007\)](#) writes down a fully dynamic version of the structural exercise. The firm chooses investment I_t and financing each period, faces convex adjustment costs on capital, faces a fixed plus proportional cost of external equity issuance, can issue or pay down one-period debt, and pays corporate taxes. The cost of external equity is the parameter the paper most wants to estimate: how much does it cost the firm, at the margin, to raise a dollar of outside equity? Debt is in the model so that firms can substitute between debt and equity as financing decisions are made each period. The household discount factor and the corporate tax rate are calibrated; the fixed and proportional equity issuance costs, the curvature of capital adjustment costs, and the persistence and volatility of profitability are estimated by simulated method of moments.

The model has three state variables. Productivity A_t is the firm's exogenous profitability shock; it follows an AR(1) in logs and is the only stochastic primitive. Capital K_t is the firm's productive capital stock at the start of period t . Debt B_t is the firm's outstanding one-period debt obligation, due at the start of t . The choices each period are investment I_t and next-period debt B_{t+1} .

Operating profit is Cobb-Douglas in productivity and capital with decreasing returns:

$$\pi(A_t, K_t) = A_t K_t^\theta, \quad (10.24)$$

with curvature parameter $\theta \in (0, 1)$. The Cobb-Douglas form is a reduced-form representation of operating income from a stochastically productive capital stock, with θ controlling how quickly returns to scale diminish. Capital depreciates at rate δ and grows with gross investment:

$$K_{t+1} = (1 - \delta)K_t + I_t. \quad (10.25)$$

The standard depreciation-plus-investment structure.

Adjustment costs are smooth quadratic in the investment rate:

$$\Phi(I_t, K_t) = \frac{\gamma}{2} \left(\frac{I_t}{K_t} \right)^2 K_t, \quad (10.26)$$

which is the contemporary standard for structural investment models with a single curvature parameter γ . The smooth-quadratic form rules out lumpy investment behavior; the chapter's merged-framework discussion in Section 10.8.3 replaces (10.26) with a richer specification that adds a fixed cost and irreversibility, but for HW alone the smooth form suffices.

External equity issuance carries a fixed and a proportional cost:

$$c(e_t) = (\lambda_0 + \lambda_1 |e_t|) \cdot \mathbf{1}[e_t < 0], \quad (10.27)$$

where e_t is the firm's net flow to equity, positive when the firm pays a dividend and negative when it raises external equity. The issuance cost applies only on the issuance side; dividend payments are costless. The fixed cost λ_0 is the lumpy entry cost of accessing the equity market in a given year, and the proportional cost λ_1 is the marginal underwriting and issuance discount on each additional dollar raised. Recovering λ_0 and λ_1 from the data is the main empirical content of the SMM exercise.

The firm cannot borrow more than a fraction ϕ of next-period capital:

$$B_{t+1} \leq \phi K_{t+1}. \quad (10.28)$$

The constraint is the partial-equilibrium device for limiting leverage; without it, the after-tax cost of debt below the discount rate would push the firm to a maximum-debt corner. Debt itself carries an interest rate r_B , and interest expense reduces taxable income, generating a tax shield of $\tau \cdot r_B \cdot B_t$ per period at corporate rate τ . The tax shield in this specification is symmetric across firms with positive and negative taxable income; the merged framework in Section 10.8.3 relaxes the symmetry.

The cash-flow identity assembles the period's flow to equity from operating income, investment, and financing:

$$e_t = (1 - \tau) [\pi(A_t, K_t) - r_B B_t] - I_t - \Phi(I_t, K_t) + B_{t+1} - B_t. \quad (10.29)$$

Reading the identity from left to right: the firm receives after-tax operating income net of interest, pays for gross investment and adjustment costs, and nets the change in debt

$(B_{t+1} - B_t)$. The residual e_t is what flows to equity. When $e_t > 0$ the firm pays a dividend; when $e_t < 0$ the firm raises external equity and incurs the issuance cost in (10.27). The shareholder cash flow that enters the value function is $d_t = e_t - c(e_t)$: positive dividends pass through one-for-one, while negative e_t becomes more negative by the issuance cost.

The value function is the maximum over investment and debt of current shareholder cash flow plus discounted expected continuation:

$$V(A_t, K_t, B_t) = \max_{I_t, B_{t+1}} \{d_t + \beta \mathbb{E}[V(A_{t+1}, K_{t+1}, B_{t+1}) | A_t]\}, \quad (10.30)$$

subject to (10.25), (10.29), the issuance-cost function in (10.27), and the collateral constraint in (10.28). The discount factor β is calibrated. Standard contraction-mapping arguments ensure a unique fixed point in V , which the SMM solver iterates on for each candidate parameter vector.

Productivity follows a log-AR(1) discretized via Tauchen's (1986) method:

$$\log A_{t+1} = \rho_A \log A_t + \sigma_A \epsilon_{t+1}, \quad \epsilon_{t+1} \sim \mathcal{N}(0, 1). \quad (10.31)$$

The persistence ρ_A and innovation standard deviation σ_A are estimated jointly with the financing-cost parameters. The continuous process is approximated on a discrete grid using Tauchen's transition-probability formulas, which the value-function iteration uses to compute the expectation in (10.30).

The estimation is a textbook application of simulated method of moments. For each candidate parameter vector $\Theta = (\lambda_0, \lambda_1, \gamma, \rho_A, \sigma_A)$, the Bellman in (10.30) is solved by iteration on the discretized state space, a panel of simulated firms is generated by drawing productivity shocks and iterating the optimal policy forward, and moments are computed on the simulated panel and compared to the same moments computed on the master Compustat panel. The criterion is a quadratic distance between simulated and observed moments:

$$\hat{\Theta} = \arg \min_{\Theta} g(\Theta)^\top W g(\Theta), \quad g(\Theta) = m_{\text{sim}}(\Theta) - m_{\text{data}}, \quad (10.32)$$

with W the weight matrix. Stage 1 of the estimation uses an identity weight; Stage 2 uses the inverse of a bootstrap moment covariance, $W = \hat{S}^{-1}$, which is the efficient weight under standard SMM regularity conditions.

The chapter's HW Path A estimation targets five moments computed on the master panel: the mean investment rate $\overline{I/K}$, the mean book leverage $\overline{B/(K+B)}$, the equity-issuance frequency (the fraction of firm-years with $e_t < -0.01K_t$ in the model, or with net equity issuance above one percent of lagged assets in the data), the dividend-payment frequency

(the fraction with $d_t > 0$ in the model, or with positive common dividends in the data), and the within-firm first-order autocorrelation of profitability π/K . Five moments and five parameters: the system is just-identified, and the moment vector pins down the parameter vector when the model is well-specified and the moments are informative.

The HW dynamic programming problem is mechanically the same as the Chapter 8 solver with one additional state variable. The Chapter 8 value function is $V(A, K)$ over a discretized (A, K) grid, with a vectorized Bellman operator that evaluates investment choices at every grid point. The HW extension is $V(A, K, B)$, with the policy choosing investment and new debt subject to the cash-flow identity and the collateral constraint. The vectorization extends in the obvious way: the inner-loop matrix operations are now over a three-dimensional grid rather than a two-dimensional grid, with computation scaling roughly linearly in the number of debt grid points. The outer-loop SMM optimizer changes only in the dimension of the parameter vector and in which moments are targeted.

The HW headline at the published parameters is that external equity issuance is costly. The proportional cost λ_1 is about five percent for the smallest size quartile and declines monotonically with size, and the fixed cost λ_0 is on the order of one percent of equity value with a clear size gradient. Internal funds, in this estimation, are not a free option; they are valued at a substantial premium over external equity for small firms. The model also explains the joint distribution of investment, leverage, and equity issuance behavior reasonably well, with moment-fit quality in the same range that Cooper and Haltiwanger (2006) report for the investment-only version in Chapter 8. The parameters can be used for counterfactuals: how would corporate investment change if equity issuance costs were halved, and what is the welfare cost of the issuance friction for small firms? Those are the kinds of questions the structural approach earns the right to answer.

The SMM machinery delivers point estimates and bootstrap standard errors on the five parameters in Θ , together with the moment-fit summary. The running-example box and Table 10.3 below report the chapter's HW Path A replication on the master Compustat panel. The estimated fixed cost $\hat{\lambda}_0$, proportional cost $\hat{\lambda}_1$, adjustment-cost curvature $\hat{\gamma}$, and productivity volatility $\hat{\sigma}_A$ come out close to the published HW values; the productivity persistence $\hat{\rho}_A$ comes out smaller, reflecting the master panel's noisier profitability dynamics. The moment fit at the converged point, however, exhibits a corner the chapter returns to later. The model produces a representative firm that always pays a dividend and never issues equity, against master-panel frequencies of 0.330 and 0.285 respectively. The corner is the empirical signature of a representative-firm model that cannot generate the dispersion in financing actions the data show. Section 10.8 returns to it in the context of the eight stylized facts, and Section 10.8.3 discusses the merged framework

Table 10.3. Hennessy-Whited (2007)-style structural estimates of capital adjustment costs and equity issuance costs on the master Compustat panel. The model is a three-state dynamic program (productivity, capital, debt) estimated by two-stage simulated method of moments matching five data moments. Panel A reports parameter estimates with the published [Hennessy and Whited \(2007\)](#) Table III unconstrained-subsample values for reference; Panel B reports moment fit. The grid sizes are coarser than those used in HW (2007). Standard errors are not reported because the numerical Jacobian at the converged point is rank-deficient ($\lambda_0, \lambda_1, \gamma$ rows produce no moment movement under perturbation): the model sits at a corner of the policy where the collateral constraint binds, equity is never issued, and the firm always pays a dividend, so the moments do not pin down the issuance-cost and adjustment-cost parameters.

Notes. Estimation: two-stage SMM. Stage 1 (identity weighting) terminated successfully after 37 Nelder-Mead iterations and 76 objective evaluations at $\text{obj} = 0.531$. Stage 2 (optimal weighting $W = \hat{S}^{-1}$ with $\hat{S}$ from a 200-iteration firm-cluster bootstrap on the data moments) reached parameter stability to five decimal places for 70+ consecutive function evaluations but did not formally satisfy the Nelder-Mead `fato1` criterion because the simulated moment function has tiny discrete-policy discontinuities near the optimum; parameters are reported at the stable point. Calibrated primitives: $\beta = 0.96$, $\delta = 0.10$, $\theta = 0.7$, $r = 0.05$, $\tau = 0.30$, $\phi = 0.30$. The moment fit shows tight agreement on investment rate, leverage, and profitability persistence (gaps below 20%), and large gaps on equity-issuance frequency and dividend frequency: the model produces corner-solution behavior at the converged parameter point that the data does not exhibit. See `BUILD_LOG_HW_PATHA.md` for the full diagnostic and the open questions about identification.

that brings investment-driven heterogeneity into the model as the natural extension to break the corner.

Running Example 4: Installment 3: HW-style SMM extension

The Chapter 8 two-state dynamic-programming solver of [Cooper and Haltiwanger \(2006\)](#) is in the repository at `code/python/investment/cooper_haltiwanger/` and reproduces the published CH (2006) moment targets to within roughly 0.05 on each moment across the five model variants discussed in Chapter 8. The HW extension at `code/python/investment/cooper_haltiwanger/hw_model.py` adds debt as a third state, the [Hennessy and Whited \(2007\)](#) piecewise-linear equity issuance cost to the budget constraint, and a collateral constraint of the form (10.28) that follows HW (2007) Section 2. A corporate-tax rate τ is added so that the after-tax cost of debt falls below the discount rate; without it the model collapses to a zero-debt corner. Section 10.5 reports SMM estimates and moment fit in Table 10.3. The grids are deliberately coarsened ($n_K = 30$, $n_B = 12$, $n_I = 41$) so a full Stage-1 SMM completes in roughly 18 minutes and Stage 2 in another 30 minutes; HW (2007) used finer grids.

The HW estimation buys students a credible set of point estimates for primitives that the descriptive regression in Section 10.3 and the theory-guided tests in Section 10.4 cannot identify. Equity issuance costs of 5 percent for small firms is a number you can write

down, take to a counterfactual, and defend to a referee. The cost is that the parameters are conditional on the model. If the model misspecifies the issuance cost function, the estimated parameter is biased; if the model omits an important friction (regulatory capital requirements, product-market interactions, or anything else that operates on financing decisions), the estimated friction is contaminated by the omitted one. [Strebulaev \(2007\)](#) makes the related point that structural moments can be reproduced by quite different models, so a parameter estimate is not the same as a confirmation of the model that produced it.

The structural answer is one thing the literature can do with the failure of cross-sectional regressions to identify primitives. The causal answer is another. Where [Hennessy and Whited \(2007\)](#) finds external financing costly and parameterizes how costly, the next two sections look for clean exogenous variation that moves financing or its determinants and identifies a local average treatment effect. Section 10.6 takes up taxes through state corporate tax changes. Section 10.7 takes up peer effects through idiosyncratic equity-return shocks. Neither identifies HW's structural parameters. Neither claims to. Each tells us something the structural approach cannot tell us by itself, in particular what an actual policy intervention with a defensible identification design does to leverage. The closing in Section 10.8 lines up what each of the three traditions says and what we should make of the disagreement.

10.6 Causal Approach I: Taxes (Heider-Ljungqvist)

The trade-off theory rests on the claim that corporate taxes affect leverage. A descriptive regression of leverage on the marginal tax rate cannot test this claim, because the tax rate a firm faces is a function of its income, its accumulated losses, its tax-shield exhaustion, and other choices that are co-determined with leverage. To identify the causal effect of taxes on leverage, one needs variation in the tax environment that moves the price of debt without moving everything else. [Heider and Ljungqvist \(2015\)](#), hereafter HL, observe that US state corporate income tax changes provide exactly this variation. State legislatures change rates for reasons unrelated to any individual firm's financing decisions, and the timing of changes is staggered across states and years.

HL collect 121 staggered changes in state corporate income tax rates over 1989–2011. They merge these state-by-year tax changes onto the universe of Compustat firms headquartered in each state. The dependent variable is the firm-level change in long-term debt scaled by lagged assets. The design is staggered difference-in-differences with two-way fixed effects,

and the main specification separates rate increases from rate decreases:

$$L_{i,s,t} = \alpha_i + \gamma_t + \beta_{\text{up}} \cdot \mathbf{1}[\text{TaxUp}_{s,t}] + \beta_{\text{down}} \cdot \mathbf{1}[\text{TaxDown}_{s,t}] + \varepsilon_{i,s,t}, \quad (10.33)$$

where i indexes firms, s indexes the firm's state of headquarters, and t indexes year. Firm fixed effects α_i absorb time-invariant firm heterogeneity; year fixed effects γ_t absorb aggregate shocks; the indicator $\mathbf{1}[\text{TaxUp}_{s,t}]$ is one in years after a tax increase in state s , and $\mathbf{1}[\text{TaxDown}_{s,t}]$ is one in years after a tax decrease. The asymmetric specification allows β_{up} and β_{down} to differ; the static trade-off predicts symmetric responses, so a clean rejection of $\beta_{\text{up}} = -\beta_{\text{down}}$ is informative on its own.

The identifying assumption is parallel trends: absent treatment, leverage paths in treated and untreated firm-states would have moved together. Formally,

$$\mathbb{E}[L_{i,s,t}(0) - L_{i,s,t-1}(0) | s \in \text{Treated}] = \mathbb{E}[L_{i,s,t}(0) - L_{i,s,t-1}(0) | s \in \text{Control}], \quad (10.34)$$

where $L_{i,s,t}(0)$ is the potential outcome under no treatment. Parallel trends is not directly testable, but the pre-treatment trends in the treated and control groups are diagnosable; HL provide pre-trend plots that support the assumption in their sample, and the chapter's master-panel replication runs the same diagnostic.

HL focus on long-term debt rather than total debt because the tax-shield benefit of debt accrues over the life of the obligation. A long-term debt issuance taken out in anticipation of a permanent tax change locks in a stream of future tax shields. Short-term debt rolls over frequently enough that the response to a state tax change is muddled by other rollover decisions. As the discussion of leverage variable definitions makes clear, leverage measures using long-term debt have a sharper theoretical interpretation than total-debt measures when the question concerns the supply of debt-financed tax shields, and this section is one place where that distinction earns its keep.

The headline finding is asymmetric. When a state increases its corporate tax rate, firms headquartered in that state significantly increase their long-term debt issuance in the years following the change, by roughly forty basis points of book leverage per percentage point of tax rate increase. When a state decreases its rate, the same firms do not reduce their long-term debt issuance by a comparable amount. The asymmetry is statistically clean. HL interpret it through adjustment-cost and real-option lenses. Firms see tax increases as more permanent than tax decreases, perhaps because state legislatures rarely cut rates after raising them, so the expected duration of the post-change tax environment is longer following an increase than following a decrease.

The HL estimator is two-way fixed effects on staggered treatment timing. Since HL

was published, the staggered difference-in-differences literature has revisited what TWFE identifies when treatment effects are heterogeneous. [Goodman-Bacon \(2021\)](#) showed that the TWFE coefficient is a weighted average of two-by-two comparisons across cohorts, with weights that can be negative when earlier-treated units serve as controls for later-treated ones. [Callaway and Sant'Anna \(2021\)](#) proposed a group-time average-treatment-effect estimator that uses only never-treated or not-yet-treated comparisons. [Borusyak et al. \(2024\)](#) proposed an imputation estimator that estimates counterfactual outcomes for treated periods using a model fit on untreated periods. [Baker et al. \(2022\)](#) ran a battery of corporate finance settings through these estimators and found that TWFE can give meaningfully different point estimates from the modern alternatives in some cases and similar results in others. The HL setting has not been systematically revisited in the published literature. Chapter 6 develops these estimators and their identification assumptions in detail; the application here uses them as drop-in replacements for TWFE.

The Callaway-Sant'Anna estimator computes cohort-time average treatment effects directly:

$$\text{ATT}(g, t) = \mathbb{E}[L_{i,t}(g) - L_{i,t}(\infty) \mid G_i = g], \quad (10.35)$$

where $G_i = g$ denotes that firm i was first treated in year g , $L_{i,t}(g)$ is the potential outcome under treatment cohort g , and $L_{i,t}(\infty)$ is the potential outcome under never-treatment. The aggregation across cohorts is

$$\widehat{\text{ATT}} = \sum_g \sum_{t \geq g} w(g, t) \cdot \widehat{\text{ATT}}(g, t), \quad (10.36)$$

with weights $w(g, t)$ that the analyst chooses. The simple aggregation, used by the chapter's master-panel replication, gives uniform weights to each (g, t) cell; other aggregations weight by cohort size or by event-time elapsed since first treatment. The chapter's master-panel replication of HL via this simple Callaway-Sant'Anna aggregation produces a tax-up ATT of about +0.0054, within HL's published range, on a sample where TWFE on the same panel returns a smaller and statistically insignificant estimate.

The natural question for a textbook reader is whether the HL results survive the modern staggered estimators. The answer requires running the comparison, and that is what Installment 4 will do once the underlying data are in place. We expect the asymmetric response to survive in some form, because the asymmetry is identified from the contrast between the average response to tax increases and the average response to tax decreases, neither of which is mechanically biased in the way that TWFE under heterogeneous treatment effects can be. But the magnitudes may shift, and certain subsamples may show

larger shifts than others. Replicating HL with both TWFE and a modern estimator is exactly the kind of exercise the chapter wants the reader to do, because it teaches the operational habit of running the comparison rather than assuming TWFE is fine.

Running Example 5: Installment 4: HL replication with TWFE and a modern staggered estimator

We replicate HL on the master Compustat panel merged with state corporate income tax change data. The treatment is the state-by-year tax change, separated into rate increases and rate decreases. The outcome is the firm-level change in long-term debt scaled by lagged assets. Specifications include firm fixed effects and year fixed effects in all cases, with the modern staggered-DiD estimator implemented via the `csdid` routine in Stata or the `differences` module in Python. The expected pattern, based on [Heider and Ljungqvist \(2015\)](#), is a positive coefficient on tax increases and a null or smaller coefficient on tax decreases, with the asymmetry surviving the move from TWFE to a modern estimator. The merge depends on a state-by-year corporate tax change file that is not yet in the repository. The file will be drawn from the HL replication package or constructed from public state-revenue-department sources, and the data path will be set when the file is added. Code is in `code/python/capital_structure/installment_4_hl_taxes.py` and `code/stata/capital_structure/installment_4_hl_taxes.do`.

The implementation runs both estimators in one script. The TWFE block builds two intensity variables that separate state-by-state tax increases from decreases, then regresses the firm-level change in long-term debt on those intensities with firm and year fixed effects and state-clustered standard errors. The Callaway-Sant’Anna block constructs a cohort variable equal to each firm’s first tax-increase year (with zero for never-treated firms) and estimates group-time ATTs against the never-treated control group using doubly-robust outcome regression. Aggregating the group-time ATTs with simple weights produces the cohort-average effect of a tax increase on long-term debt issuance, which the chapter compares to the TWFE coefficient to expose the bias TWFE inherits from negative weights on already-treated comparisons. The Python and Stata implementations are in the *HL replication with TWFE and Callaway-Sant’Anna* section of the Code Companion.

The causal answer to “do taxes affect leverage” is yes, asymmetrically and partly. The asymmetry teaches us something the structural model in Section 10.5 cannot teach by itself: that firms respond to tax changes through an adjustment-cost or real-option lens, not a frictionless re-optimization. The HW model can in principle reproduce the asymmetry through its adjustment-cost curvature, but the structural exercise does not identify the

Table 10.4. Effect of state corporate tax changes on Δ long-term debt scaled by lagged assets, following Heider and Ljungqvist (2015). Panel A reports two-way fixed effects estimates with three standard-error variants and a Callaway-Sant’Anna-style group-time ATT aggregated across treated cohorts. Panel B reports the TWFE specification on a subsample split by lagged positive net income (the binding-tax-shield proxy). The asymmetric response, with firms raising debt in response to tax increases but showing little response to tax decreases, is the headline.

Notes. Sample: 2000-2014 firm-years from the master Compustat panel (the Tax Foundation *State Corporate Income Tax Rates, 2000-2014* Excel is the underlying rate source; pre-2000 and post-2014 years are not in this build). Firm headquarters state is fixed at each firm’s first-observation `state_hq`, following Heider and Ljungqvist (2015), so firm relocations do not generate spurious treatment. Tax changes with $|\Delta \text{ top rate}| > 3$ percentage points are recoded as missing; they reflect structural reforms (Ohio’s CAT phase-out, Michigan’s SBT to MBT to CIT transitions) rather than rate adjustments and are not comparable to ordinary rate changes in the HL design. Standard errors in parentheses. Stars: $*p < 0.10$, $**p < 0.05$, $***p < 0.01$. Panel B uses lagged $\text{ib}_{i,t-1} > 0$ as the proxy for “positive taxable income”; Compustat’s `tlcf` (tax loss carryforward) is essentially uncovered on this build, so the NOL filter recommended by Heider and Ljungqvist (2015) could not be implemented.

asymmetry directly. It produces it as a downstream consequence of parameter estimates calibrated to other moments. The causal exercise identifies the asymmetry directly, at the cost of being silent on what generates it. This is the structural-versus-causal contrast the chapter is designed to make visible. The next section moves to the second causal showcase, peer effects in the sense of Leary and Roberts (2014), and to the methodological question of what a “local effect” can and cannot tell us about a structural primitive.

10.7 Causal Approach II: Peers (Leary-Roberts 2014)

A clarification is in order before we proceed. Two papers by Leary and Roberts appear in this chapter. Leary and Roberts (2010), the subject of Section 10.4, tested the strict pecking order by tabulating equity issuance among firms with apparent unused debt capacity. Leary and Roberts (2014), the subject of this section, asks a different question. It uses idiosyncratic peer-firm equity returns as an instrument for peer leverage to identify whether peer leverage causally affects a firm’s own leverage. The two papers share authors and a publication trail but are not about the same thing. Conflating them is a common student error, and we name the distinction at the top of the section so the rest of the prose can refer to LR (2014) without ambiguity.

The empirical question is whether peer firms, defined as firms in the same industry or competitive cohort, influence each other’s leverage. The motivation is straightforward. Managers cite peer benchmarks in financing decisions. Industry-median leverage is the single largest contributor to fit in the FG factor regression of Section 10.3. Outside observers routinely compare a firm’s leverage to its industry as if industry-level comparison meant

something. The empirical challenge is harder than the motivation suggests. A correlation between a firm's leverage and its peers' leverage can arise from peer effects (firm i responds to firm j 's leverage choice), from common shocks (both firms face the same industry-wide demand or supply conditions), or from correlated unobservables (both firms have the same kind of management or the same set of investment opportunities). Distinguishing among these is the reflection problem of [Manski \(1990\)](#), applied to corporate finance.

[Leary and Roberts \(2014\)](#) address the reflection problem with an instrument. The instrument for peer firm j 's leverage is the idiosyncratic component of firm j 's equity returns over the prior period, computed as the residual from a Carhart four-factor model of j 's returns:

$$R_{j,t} = \alpha_j + \beta_j^{\text{MKT}} R_{m,t} + \beta_j^{\text{SMB}} \text{SMB}_t + \beta_j^{\text{HML}} \text{HML}_t + \beta_j^{\text{UMD}} \text{UMD}_t + \varepsilon_{j,t}^R, \quad (10.37)$$

where $R_{m,t}$ is the excess market return and SMB_t , HML_t , UMD_t are the standard size, value, and momentum factors. The residual $\varepsilon_{j,t}^R$ is the idiosyncratic component of firm j 's return, orthogonal by construction to the four common factors. The idea is that an idiosyncratic equity shock to firm j moves j 's leverage by changing its market value of equity and triggering rebalancing or financing decisions, but the shock is, by construction, orthogonal to industry-wide conditions and to firm i 's own opportunities or shocks. If firm i 's leverage responds to the instrumented component of firm j 's leverage, the response must operate through some channel that runs from j 's leverage to i 's leverage. That channel is what we call a peer effect.

The first-stage regression of peer-group average leverage on the average lagged idiosyncratic return is

$$\bar{L}_{-i,t} = \pi_0 + \pi_1 \bar{\varepsilon}_{-i,t-1}^R + Z_{-i,t-1}^\top \pi_2 + W_{i,t-1}^\top \pi_3 + u_{i,t}, \quad (10.38)$$

where $\bar{L}_{-i,t}$ is the average leverage of firm i 's peers (excluding i), $\bar{\varepsilon}_{-i,t-1}^R$ is the peer-group average lagged idiosyncratic return, $Z_{-i,t-1}$ is a vector of peer-group control variables, and $W_{i,t-1}$ is a vector of firm- i -specific controls. The instrument is strong in the LR (2014) sample, with F -statistics well above the standard weak-instrument threshold.

The second stage regresses firm i 's leverage on the instrumented peer-group leverage:

$$L_{i,t} = \delta_0 + \delta_1 \widehat{\bar{L}_{-i,t}} + W_{i,t-1}^\top \delta_2 + Z_{-i,t-1}^\top \delta_3 + v_{i,t}, \quad (10.39)$$

where $\widehat{\bar{L}_{-i,t}}$ is the fitted value from (10.38). The coefficient δ_1 is the IV estimate of the peer-leverage effect on firm leverage, with a local average treatment effect interpretation in

the sense of Chapter 5: the average response among firm-years where the instrument has explanatory power for peer leverage.

The exclusion restriction is that firm j 's idiosyncratic equity shocks affect firm i 's leverage only through firm j 's subsequent leverage. The threats are familiar. Peer idiosyncratic shocks could plausibly affect i directly if i 's product-market response to j 's shock changes i 's growth opportunities and, through that channel, i 's leverage. Leary and Roberts (2014) address this by defining peer groups carefully and by including direct controls for firm i 's own investment opportunities (own market-to-book, lagged investment, profitability), and by showing that the instrument retains qualitative pattern when these controls are added. The exclusion restriction is a defensible assumption rather than a directly testable one, in the standard IV way; the chapter's posture is to describe the assumption clearly rather than relitigate it.

The IV estimate is that a one-standard-deviation increase in peer leverage causes approximately a 0.51 standard-deviation increase in firm i 's leverage. This is large. Peer characteristics, when included alongside peer leverage, have substantially smaller direct effects on i 's leverage than peer leverage itself, suggesting that the action is in the leverage choice rather than in the underlying characteristics that determine the leverage choice. The peer-effect coefficient is larger for smaller firms and for firms with weaker financial positions, consistent with the interpretation that the firms most likely to face frictions in independent capital-structure optimization are the ones most likely to copy peers.

What is the peer effect, mechanism-wise? Three candidate stories. Imitation: managers benchmark to peers because they lack confidence in their own optimization. Informational learning: peer leverage reveals information about industry-wide debt capacity that i does not directly observe. Genuine economic spillover: debt-capacity competition or coordination among peers. Leary and Roberts (2014) cannot distinguish these mechanisms from their IV alone. The instrument identifies the reduced-form response of firm i 's leverage to peer leverage shocks. It does not identify which of the three channels generates that response. The paper is honest about this and discusses the mechanism options as a matter of interpretation rather than identification.

The estimate is a local average treatment effect, in the sense of Chapter 5. The compliers are firms whose leverage responds to peer leverage when peer leverage is moved by an idiosyncratic peer equity shock. This is a specific subpopulation. Firms that would not respond to peer leverage under any peer-leverage shock, perhaps because they are firmly committed to their own leverage target or because the relevant peer group does not match the LR (2014) definition, are not in the LATE. The estimate cannot be read as a population-wide structural parameter. It cannot be transported to a counterfactual in

which all firms respond to all peers. It cannot be aggregated into a model-level structural primitive without making auxiliary assumptions that LR (2014) do not make.

The peer-effect estimate is what a credible causal design identifies on the question of peer effects, and peer effects are a substantively important driver of capital structure. It is not the question that the Hennessy-Whited model in Section 10.5 answers. HW asks how much external financing costs, and reports a parameter that can be used for counterfactuals on issuance costs. Leary and Roberts (2014) asks whether peer leverage causes own leverage, and reports a local effect for compliers in a particular peer-shock identification. The two answers do not aggregate. A welfare cost of equity issuance frictions is not the same kind of object as a peer-leverage spillover coefficient. The chapter's next section takes that point seriously and works through what it means for the practice of empirical capital structure.

10.8 Pulling things together

The four exercises in this chapter approach the same question, what determines firm leverage, from four directions. Each delivers a partial answer. None delivers the full answer. The point of the chapter is not that we should pick one and proceed; it is that an empirical literature this rich has accumulated a list of findings that any successful theory must satisfy simultaneously, and that the canonical theories of capital structure all fail to satisfy the full list. This section organizes the empirical evidence into a list of facts, scores the three canonical theories against the list, and sketches the kind of framework that might accommodate all of them.

10.8.1 Eight empirical facts

The exercises in this chapter, run on the master Compustat panel of nonfinancial nonutility US firms from 1971 through 2024, establish eight findings about capital structure. Some are robust in both direction and magnitude. Some are robust in direction but sample-dependent in magnitude. One could not be cleanly tested with the data available in modern Compustat. We label each fact, state it briefly, and note where it came from.

Fact 1. Cross-sectional persistence. About half of the variance in firm leverage is between-firm and persists over multi-decade horizons. On firms with at least twenty years of leverage observations, firm fixed effects alone explain an R-squared of approximately 0.48 for book leverage and 0.51 for market leverage. Adding the standard Frank and Goyal

(2009) factors on top of firm and year fixed effects raises the R-squared by only 0.06 to 0.08. This is the [Lemmon et al. \(2008\)](#) finding: firms differ persistently in their leverage levels, and the factors literature has been largely characterizing those persistent differences rather than within-firm dynamics. The master-panel between-firm share is a few percentage points below LRZ's published estimate, attributable to the longer sample period extending through 2024 with more recent within-firm volatility.

Fact 2. Within-firm non-stationarity at long horizons. On firms with at least thirty years of leverage observations, between 74 and 82 percent cannot reject a unit root in their leverage series at the 5 percent level. This holds for both book and market leverage and under specifications with and without a linear trend. The fraction is higher than the approximately 50 percent that [DeAngelo and Roll \(2015\)](#) reported on their long-horizon sample. The master-panel result is stronger than DR's in the same direction: within-firm leverage at long horizons looks much more like a non-stationary process than like mean reversion to a stable target.

Fact 3. Firm-specific time-varying targets with fast adjustment. [Frank and Shen \(2019\)](#) estimate firm-by-firm time-series regressions of leverage on lagged log assets and market-to-book and find substantial cross-firm heterogeneity in the slope coefficients. Replicating on the master panel, the cross-firm standard deviation of the market-to-book slope is 0.24 for market leverage targets, with substantial mass on both sides of the common-slope pooled OLS estimate of -0.029. The implied partial-adjustment coefficient against firm-specific targets is 0.50 for market leverage and 0.45 for book leverage, implying half-lives of about one year and 1.6 years. Against common-slope targets on the same firms, the partial-adjustment coefficient drops to 0.17, with half-lives of nearly four years. The factor-of-three gap between adjustment-speed estimates under the two target specifications is the FS headline. The master-panel numbers match FS Table 10 within rounding.

Fact 4. Transitory dynamics. A substantial fraction of within-firm leverage variation reverses within a few years. On the master panel using the [Cochrane \(1988\)](#) variance-ratio measure at horizon five, the median firm has a transitory share of approximately 0.35 for book leverage and 0.42 for market leverage. The magnitude is below the 70 to 80 percent that [DeAngelo et al. \(2011\)](#) reported on their 1950 to 2008 sample under a different operationalization; the direction is the same. The master-panel period has more sustained leverage shifts than DDW's earlier sample, and the variance-ratio measure asks a related but distinct question from DDW's exact decomposition. The qualitative finding survives

both differences.

Fact 5. Investment and acquisitions drive large leverage changes. On the master panel, year-over-year changes in book leverage average -0.7 percentage points in firm-years with no major investment or acquisition activity, $+2.4$ percentage points in firm-years with major investment, and $+5.0$ percentage points in firm-years with major acquisitions. The investment-major-acquisition pattern survives a TWFE regression with firm and year fixed effects: the coefficient on the major-investment dummy is $+0.033$, and the coefficient on the major-acquisition dummy is $+0.059$. Leverage changes in investment-driven and acquisition-driven years partially reverse over the following three years. The pattern is consistent with DDW's interpretive emphasis: transitory financing in response to investment opportunities, not gradual movement toward a new target, accounts for the bulk of large leverage changes.

Fact 6. Moderate issuance costs. A Hennessy-Whited style structural model with three state variables, productivity, capital, and debt, and with the issuance-cost specification of [Hennessy and Whited \(2007\)](#), recovers parameter estimates on the master panel close to the published HW values. The fixed issuance cost $\hat{\lambda}_0$ is approximately 0.59 , against HW's 0.598 . The proportional issuance cost $\hat{\lambda}_1$ is approximately 0.12 , against HW's 0.091 . The capital adjustment cost curvature $\hat{\gamma}$ is approximately 1.44 , against HW's 1.279 . Productivity volatility $\hat{\sigma}_A$ is approximately 0.10 , against HW's 0.118 . Four of five HW parameters reproduce on the master panel under coarser computational grids. The estimated issuance costs are not zero, but they are also not large enough to generate long inaction regions consistent with the strict-fixed-cost view of the dynamic capital structure literature of the 1980s and 1990s. They are moderate.

Fact 7. Asymmetric tax response. On a sample of state corporate income tax changes from 2000 through 2014 from the Tax Foundation, the cross-cohort average of state-year ATTs for tax increases is $+0.0054$ percentage points of debt issuance per percentage point of rate change, with a standard error of 0.0019 across 18 treated cohorts. The corresponding ATT for tax decreases is essentially zero. The asymmetric pattern, with firms raising debt in response to tax increases but not lowering debt in response to tax decreases, is qualitatively the central finding of [Heider and Ljungqvist \(2015\)](#). Under the older TWFE specification with state-clustered standard errors on the same sample, the asymmetry does not reach conventional statistical significance, attributable to the smaller modern sample with 18 tax-up events compared to HL's approximately 50 to 60. The Callaway-Sant'Anna

aggregation recovers the HL pattern on the smaller sample where TWFE does not. Both estimators agree on the direction.

Fact 8. Tax response concentrated in firms with positive taxable income. HL's secondary finding, that the tax response should be larger in firms with positive lagged taxable income because the tax shield is binding for those firms, cannot be cleanly tested on the master panel. The Compustat tax-loss-carryforward field has effectively zero coverage in the analysis sample, so the HL classification of binding-versus-non-binding tax shield status cannot be reconstructed. Using lagged positive income alone as a proxy, the subsample analysis reverses HL's prediction: firms with positive lagged income show essentially zero response, while other firms show a large positive response. Three explanations are consistent with this. The income proxy is too coarse without the NOL refinement that HL used. The Great Recession years embedded in the 2000 to 2014 sample window include systematic loss-maker debt issuance unrelated to tax incentives. And the modern Compustat sample includes a different mix of distress-state firms than HL's earlier sample. The HL prediction may hold on data we do not have. Fact 8 is properly characterized as not testable with the available data, not as falsified.

10.8.2 Scoring the canonical theories

The canonical theories of capital structure are static trade-off, dynamic trade-off with adjustment costs, pecking order, and market timing. We assess each against the eight facts. Each fact gets one of three verdicts: the theory predicts it, the theory contradicts or struggles to explain it, or the theory is silent on it. Silence is not failure. A theory's silence on a fact means the theory does not address the phenomenon, not that the phenomenon refutes the theory. Contradiction is failure: the theory makes a specific prediction the data do not deliver.

Static trade-off theory predicts that firms trade off the tax shield of debt against bankruptcy costs and reach an interior optimum determined by firm characteristics. The static version has firms at their optimum continuously, with leverage moving only when the underlying tradeoffs change.

The theory predicts facts 1 and 3, the existence of firm-specific leverage levels and their dependence on observable factors. It predicts facts 7 and 8, the tax-channel response and its concentration in tax-binding firms. It is contradicted by facts 2, 4, 5, and 6. Within-firm non-stationarity at long horizons (fact 2) is inconsistent with leverage sitting at a stable

target. The transitory dynamics in fact 4 are inconsistent with smooth adjustment to a stable target. The investment-driven character of large changes in fact 5 is inconsistent with a static model that treats financing as separate from investment. The asymmetric response in fact 7 (which static trade-off predicts symmetrically) is also a contradiction in the strong form, though the asymmetry could in principle reflect heterogeneous firm characteristics. Static trade-off is silent on facts about adjustment dynamics. Half the facts contradict it; static trade-off as written is not the right theory.

Dynamic trade-off theory with adjustment costs , in the tradition of [Fischer et al. \(1989\)](#), [Leland \(1994\)](#), and [Hennessy and Whited \(2007\)](#), retains the trade-off intuition but adds frictions that prevent continuous adjustment. The firm has an inaction region around its target; it adjusts only when the leverage gap exceeds a threshold. With moderate fixed costs and proportional adjustment costs, the model can generate plausible leverage dynamics.

The theory predicts facts 1, 3, 6, 7, and 8, the cross-sectional and structural facts and the tax channel. The HW parameter estimates from fact 6 are themselves a confirmation. The dynamic trade-off is silent on facts 2 and 4 in the literal version, but the long-horizon non-stationarity and the transitory dynamics are compatible with the model if firm primitives drift slowly. Where the theory struggles is with fact 5, the investment-driven character of large leverage changes. The standard dynamic trade-off model has firms adjusting leverage in response to leverage shocks, not in response to investment shocks. To accommodate fact 5, the dynamic trade-off model needs to be extended with a richer investment side. The asymmetric response in fact 7 also strains the standard model, which predicts symmetric responses to symmetric shocks. The corner-solution issue in the structural fit of fact 6, where the representative firm gets stuck always issuing dividends and never issuing equity, also reflects this incompleteness: a representative firm cannot generate the dispersion in financing actions that fact 5 documents.

Dynamic trade-off is the most successful of the three canonical theories on a fact-by-fact basis. It fails honestly on the dynamics that involve interaction with investment.

Pecking order theory , in the [Myers and Majluf \(1984\)](#) tradition, predicts that firms finance investment from internal funds first, then debt, then equity, in that order. There is no target; observed leverage emerges as the cumulative residual of past financing decisions.

The theory predicts facts 4 and 5, the transitory dynamics and the investment-driven character of large changes. It contradicts facts 1 and 3, the existence of firm-specific targets that depend on factors. It is essentially silent on facts 2, 6, 7, and 8. Pure pecking order has been understood to be incomplete since [Frank and Goyal \(2003\)](#) documented that the broad

sample's marginal financing is mostly equity, not debt, contrary to the pecking order's prediction. The order in which firms finance investment is approximately as pecking order predicts only on the narrow sample of mature dividend-paying firms with public debt; on the broad sample, the order is different. Pecking order captures the transitory dynamics for the right reasons, but it misses the cross-section and the tax channel.

Market timing theory , in the [Baker and Wurgler \(2002\)](#) tradition, predicts that firms issue equity when their market valuations are favorable and debt when valuations are unfavorable. Observed leverage is the cumulative residual of these timing decisions, with no target. The theory is closer to pecking order than to trade-off in that there is no target.

The theory predicts facts 2 and 4, the long-horizon non-stationarity and the transitory dynamics, both of which follow if leverage is the cumulative residual of opportunistic issuance. It is silent on facts 5, 6, 7, and 8. It contradicts facts 1 and 3 in the strong form: a theory with no target cannot generate the cross-sectional persistence and the firm-specific target behavior. Market timing captures parts of the dynamics but cannot account for the cross-sectional and causal evidence.

Scorecard summary.

Fact	Static TO	Dynamic TO	Pecking order	Market timing
1. Cross-sectional persistence	predicts	predicts	contradicts	contradicts
2. Non-stationarity at long horizons	contradicts	silent	silent	predicts
3. Firm-specific time-varying targets	predicts	predicts	contradicts	contradicts
4. Transitory dynamics	contradicts	silent	predicts	predicts
5. Investment and M&A as drivers	contradicts	contradicts	predicts	silent
6. Moderate issuance costs	silent	predicts	silent	silent
7. Asymmetric tax response	partial	partial	silent	silent
8. Concentrated in tax-binding firms	predicts	predicts	silent	silent

No canonical theory predicts all eight facts. Dynamic trade-off comes closest, with five predicted, two silent, and one contradicted. The other three theories each contradict at least three facts. The empirical evidence does not select among the canonical theories; it selects against all of them and points to where extensions are needed.

10.8.3 A merged framework: Cooper-Haltiwanger investment plus Hennessy-Whited financing

The pattern of failures across the four canonical theories points to a specific direction for a framework that could accommodate the eight facts. None of the canonical theories has the joint structure that the data require, but the components needed are familiar from separate literatures. A framework that integrates them would be a dynamic program in which the firm jointly chooses investment and financing, subject to five elements:

1. Cooper-Haltiwanger nonconvex capital adjustment costs, which generate the lumpy investment that drives fact 5.
2. Hennessy-Whited equity issuance costs, which generate the moderate issuance friction of fact 6 and contribute to the cross-sectional patterns of fact 1.
3. A debt tax shield with asymmetric value across firms with positive and zero taxable income, which generates the asymmetric tax response of fact 7 and its concentration in tax-binding firms (fact 8).
4. Firm-level heterogeneity in primitives, which generates the cross-firm slope heterogeneity of fact 3 and contributes to the cross-sectional persistence of fact 1.
5. Slow drift in firm primitives over the firm's life cycle, which generates the within-firm long-horizon non-stationarity of fact 2.

The components have been developed and validated in separate literatures. The Cooper-Haltiwanger investment literature has worked with reduced-form financing. The Hennessy-Whited financing literature has worked with stylized investment, a single capital state, and no nonconvex adjustment costs. Each model has been validated on the dimension it focuses on. The joint specification, with both sides treated structurally, has not been estimated as a single published dynamic model. The integration is a research direction the chapter cannot complete, but the chapter can take a first numerical step. [Cooper and Haltiwanger \(2006\)](#) developed the canonical model of nonconvex investment dynamics with idiosyncratic productivity shocks; [Hennessy and Whited \(2007\)](#) developed the canonical model of dynamic financing with moderate issuance costs; the chapter combines their structures and adds an explicit cash state and an asymmetric tax shield. The four-state (A, K, B, C) dynamic program that results is the subject of the calibrated illustration in the next subsection. The formal Bellman, the cash-flow identity, and the identification challenges of estimating the framework are written down in the appendix.

The corner-solution issue from the chapter's Hennessy-Whited replication motivated this direction. At the converged three-state HW parameter values, the representative firm always pays dividends and never issues equity. A model with a representative firm and

no explicit investment dynamics has no internal mechanism to generate the dispersion in financing actions that fact 5 documents. The merged framework brings investment-driven heterogeneity in, with the conjecture that this dispersion is what would unlock the financing-action moments. The calibrated illustration tests part of that conjecture and leaves part of it open, in the asymmetric pattern the next subsection makes precise.

10.8.4 What the chapter does not claim

We refuse closure. The merged framework is a conjecture about where the empirical literature has been pointing, not a tested alternative theory. The eight facts are a list of what any successful theory must accommodate, not a list of what the chapter has demonstrated to be true beyond doubt.

Several specific limitations of the chapter's evidence deserve naming. The master Compustat panel covers 1971 through 2024. Earlier periods would have been informative, especially for the Heider-Ljungqvist replication, which required pre-2000 tax changes that were not available without paywalled sources. The Leary (2009) bank-credit-supply instrument turned out to be too weak on the modern sample to use for the chapter's IV exercises; the post-1994 deregulation of interstate banking has weakened the local banking-market segmentation that the instrument relied on. The Hennessy-Whited structural exercise used coarser grids than the published HW work, and the moment fit at the converged parameters is poor on two of five moments, attributable to the corner-solution issue we discussed.

The eight facts as stated assume that the master Compustat panel is broadly representative of US public firms. The sample restrictions, dropping firms with debt above assets, with negative book equity below assets, with sub-million-dollar total assets, may be more conservative than is strictly necessary. The literature has used a range of cleaning conventions. The facts as stated are robust to the cleaning conventions we used, but a reader who would prefer different conventions would presumably get slightly different magnitudes.

The chapter does not address private firms, which constitute the majority of US companies. The Compustat panel covers public firms only. Whether the same eight facts hold for private firms is an open question that the empirical literature has not addressed in any depth, partly because private-firm financial data are much harder to obtain.

The chapter does not address country-level differences. The master panel is US firms only. Whether the same eight facts hold in other capital markets, with different regulatory, tax, and ownership structures, is the subject of a separate literature that we do not synthesize.

here.

10.8.5 Closing the synthesis

The empirical evidence on capital structure has accumulated for decades. The chapter has organized that evidence into a list of facts and scored canonical theories against the list. The canonical theories all fail, in different ways and to different degrees. Dynamic trade-off comes closest but cannot account for the investment-driven character of large leverage changes. The other theories miss either the cross-section, the dynamics, or the causal channel.

A framework that combines Cooper-Haltiwanger investment dynamics, Hennessy-Whited financing dynamics, an asymmetric tax shield, and explicit cash holdings is a candidate that, in principle, accommodates the eight facts. The calibrated illustration earlier in this section is the chapter's first numerical sketch of such a framework. The exercise establishes that the framework can be solved at HW-derived parameter values and that the resulting simulated panel matches the data on the investment-side moments where HW alone could not. It also establishes that, at this calibration, the framework leaves the financing-side and cash-side moments at HW-style corners. The asymmetry of the result is informative. The components the merged framework adds to HW (nonconvex investment, an explicit cash dimension) are not equally activated at the calibration. Investment lumpiness emerges from the data without coaxing; cash holdings require either a higher productivity volatility, a different initialization, or a binding collateral constraint to come alive in simulation. Whether the financing-side corner is a framework limitation or a calibration limitation is exactly the question that estimation rather than calibration would settle. The chapter does not undertake that estimation. The formal Bellman, the cash-flow identity, the asymmetric-tax-shield specification, and the identification challenges of estimating the merged framework are written down in the appendix.

A reader who walks through this chapter is left with two takeaways. First, the empirical evidence on capital structure is substantively rich and methodologically diverse, with multiple identification strategies, multiple sample restrictions, and multiple theoretical priors generating multiple readings of essentially the same data. Second, the canonical theories of capital structure, as formulated in the 1970s and 1980s, do not survive the cumulative weight of the modern empirical evidence. The next theory of capital structure remains to be written. The chapter's contribution is to organize what such a theory must accommodate and to take a first numerical step toward writing it.

10.9 Decision Framework and Exercises

The chapter has shown four methodological postures at work on the same question. The synthesis in Section 10.8 explained why their answers do not aggregate. This closing section gives operational guidance: when to use each posture in your own research, when to combine them, and how to write up the results so a reader cannot mistake one kind of object for another.

When to run a descriptive regression. Use the FG factor regression or a close variant when you want to know whether a relationship between a regressor and leverage is stable in the data, when you are looking for stylized facts to motivate a richer model, or when you need a baseline against which structural or causal estimates can be compared. Do not use it when you want a causal effect, a parameter for counterfactual analysis, or a claim that any particular coefficient identifies a structural primitive. The literature has often blurred this line by reading the FG profitability coefficient as a test of trade-off theory; that reading does not survive the methodological turn of Section 10.3.

When to test a theory with theory-guided methods. Use a theory-guided test when the candidate theory makes a sharp quantitative prediction that can be checked against a regression coefficient, when you want to localize where the theory fails rather than reject or accept it wholesale, or when data demands are modest and the theory's predictions are stable across subsamples. Do not use it when you want a parameter for counterfactuals, when the theory's predictions are qualitative or overlap heavily with alternatives, or when sample composition can flip the sign of the result the way the SSM-FG-LZ-LR sequence in Section 10.4 flipped the slope of net debt issuance on the financing deficit.

When to estimate a structural model. Use a structural estimation when you want parameter values for counterfactual or welfare analysis, when the model is rich enough to capture the policy mechanism of interest, and when the moments you can compute on the data pin down the parameters with sufficient precision to be useful. Do not use it when you cannot defend the model's specification against the obvious alternative formulations, when a local effect is sufficient for the question, or when the moments are too thin to identify the parameter set. The HW exercise of Section 10.5 is the canonical example of the structural approach earning its right to a counterfactual; [Strebulaeu \(2007\)](#) is the canonical reminder that the same moments can be matched by multiple models.

Table 10.5. Four methodological postures: what each answers and what it cannot support.

Posture	What it answers	What it cannot support
Descriptive regression	Whether a relationship is robust in the cross-section or panel; baseline stylized facts	Causal claims, structural counterfactuals, parameter-based welfare analysis
Theory-guided test	Whether a theory's sharp prediction holds in a given sample; where a theory fails	Parameter values usable for counterfactuals; effects in samples without sharp predictions
Structural estimation	Parameter values for counterfactuals and welfare; magnitudes of model primitives	Causal claims independent of model assumptions; effects the model does not isolate
Quasi-experimental causal	Local average treatment effect for compliers in the design	Population-wide claims; structural primitives; counterfactuals beyond the design

Notes. The chapter's four running-example installments illustrate the four postures: Installment 1 (Section 10.3) is descriptive, Installment 2 (Section 10.4) is theory-guided, Installment 3 (Section 10.5) is structural, Installment 4 (Section 10.6) is quasi-experimental causal. [Leary and Roberts \(2014\)](#) in Section 10.7 is a second quasi-experimental causal exercise that uses an IV rather than a DiD design.

When to look for a quasi-experiment. Use a quasi-experimental design when you can find variation in the treatment of interest that is plausibly exogenous, when a credible causal claim for a specific local effect is what your research question needs, and when the design is robust to the modern critiques of the relevant estimator (staggered-DiD heterogeneity, weak instruments, exclusion-restriction threats). Do not use it when you need a structural parameter, when the complier population is too narrow to be substantively interesting, or when no plausibly exogenous variation exists.

The four postures answer different questions and support different downstream claims. Table 10.5 summarizes the difference operationally.

Most modern empirical papers in capital structure use more than one posture. A structural paper can be motivated by a descriptive regression and validated by a quasi-experiment that delivers a moment the model also produces. A causal paper can be paired with a theory-guided check that the mechanism story is consistent with the sign and magnitude of the estimate. A theory-guided test can motivate a structural follow-up that quantifies what the test only signs. The chapter's pedagogical claim is that mixing postures is fine and often valuable. The discipline the chapter asks for is being clear, in your own writing, about which posture each step takes and what claim each step supports. A regression coefficient is not a structural parameter and not a treatment effect; a treatment effect is not a structural parameter; a structural parameter is not, by itself, a causal claim about any observable variation. Referees notice when authors slip between these objects,

and readers find the slipperiness harder to follow than the referees do.

Exercises

Exercise 1 (Conceptual). The trade-off, pecking order, and market timing theories make overlapping predictions about leverage-profitability and leverage-size relationships. For each pairwise comparison among the three theories, identify one prediction where the two theories agree and one where they disagree. State what an empirical test would need to do to distinguish them.

Exercise 2 (Replication, easy). Using the master Compustat panel from Chapter 3, replicate the four-factor leverage regression from [Rajan and Zingales \(1995\)](#) Table III for US firms in their original sample period (1987–1991) and for 2015–2019. Compare the coefficients across the two periods. Are the patterns RZ identified still present? Briefly discuss why or why not.

Exercise 3 (Replication and interpretation, moderate). Run the [Frank and Goyal \(2009\)](#) leverage regression on the master Compustat panel with firm and year fixed effects. The coefficient on profitability is negative and statistically significant. Trade-off theory predicts the opposite sign. Write a one-page response addressing: (a) why trade-off theory predicts a positive sign, (b) three potential explanations for the observed negative sign, and (c) what additional empirical evidence would distinguish among them.

Exercise 4 (Replication, moderate). Following [Leary and Roberts \(2010\)](#), construct three measures of debt capacity on the master Compustat panel: (a) firms with investment-grade credit ratings (S&P long-term issuer rating BBB– or higher); (b) firms with leverage at least 10 percentage points below their industry median; (c) firms with modified Altman Z-scores above 1.81. For each measure, identify firms that appear to have unused debt capacity in year t . Among these firms, tabulate the fraction that issue equity (net equity issuance exceeding 5 percent of lagged assets) in year $t + 1$. Under the strict pecking order this fraction should be close to zero. Report actual fractions across the three measures. Discuss: (i) what the strict pecking order predicts; (ii) what you find; (iii) why the answer depends on the debt-capacity proxy; (iv) what the results imply about [Shyam-Sunder and Myers \(1999\)](#) regressions that do not condition on debt capacity. Note that the rated-firm subsample is selected toward larger firms, so the rated-firm result generalizes only to the rated population.

Exercise 5* (Structural estimation, hardest). Extend the Chapter 8 vectorized dynamic programming solver to include debt as an additional state variable, with an external equity issuance cost function as in [Hennessy and Whited \(2007\)](#). Estimate the issuance cost parameters by simulated method of moments on the master Compustat panel, matching moments on leverage, investment, and equity issuance frequency. Report point estimates with standard errors. Compare to the HW (2007) original estimates and discuss differences.

Exercise 6 (Replication and comparison, moderate). Replicate [Heider and Ljungqvist \(2015\)](#) using the master Compustat panel merged with state corporate tax change data. Estimate the effect using both two-way fixed effects and a modern staggered-DiD estimator (Callaway-Sant'Anna or Borusyak-Jaravel-Spiess). Compare the magnitudes and discuss whether the asymmetric response result survives.

Exercise 7 (Cross-paper synthesis, moderate). Read the abstracts and introductions of three papers assigned by the instructor that test some claim about capital structure (not from the chapter's main references). For each paper, identify: (a) which methodological posture it takes (descriptive regression, theory-guided test, structural estimation, or quasi-experimental causal design); (b) what specific object it identifies (a regression coefficient, a treatment effect, a structural parameter, a stylized fact); and (c) what its results do and do not tell us about the broader theory it engages.

Chapter 11

Corporate Payout Policy

11.1 Why this chapter

This chapter is about how firms distribute cash to investors. The empirical object is net payout, the flow of dollars firms return through dividends and repurchases after offsetting new equity issuance. The substantive question is when that flow creates more value than retention would, and the answer turns on a comparison the chapter takes as its frame.

Consider one dollar of corporate cash. If the firm holds it, the marginal dollar is worth v_F . If the firm distributes it, the dollar arrives in investors' hands worth v_I to them, net of taxes and net of whatever they would otherwise have done with it. The firm should retain the dollar when $v_F > v_I$ and pay it out when $v_I > v_F$. Optimal payout policy is the implementation of this comparison, repeated at the margin every period.

Under Modigliani-Miller assumptions, $v_F = v_I = 1$ and payout is indeterminate. Every empirical theory of payout in the modern literature is a story about a particular friction that drives a wedge between v_F and v_I , and about the sign and source of that wedge. The dividend tax-preference theories make $v_I < 1$ for taxable investors and predict retention or repurchase. The free-cash-flow theory of [Jensen \(1986\)](#) makes $v_F < v_I$ in agency-prone firms and predicts payout as discipline. The signaling theories make commitment to dividends raise v_I by resolving information asymmetry. The life-cycle theory of [DeAngelo, DeAngelo, and Stulz \(2006\)](#) makes $v_F > v_I$ for young firms with binding equity-issuance costs and $v_F < v_I$ for mature firms whose investment opportunities have run out. The catering theory of [Baker and Wurgler \(2004\)](#) makes v_I vary over time with investor demand for dividend-paying stocks. Each theory specializes the same wedge equation. The empirical question is which specialization the data favors and which combination of frictions is needed to match observed behavior.

The wedge framing connects payout policy to the rest of Part III. The Investment chapter is organized around a wedge between the marginal value of capital and its replacement cost. The Financial Constraints chapter is organized around the shadow value of internal funds μ relative to the cost of external funds. The Capital Structure chapter is organized around the marginal value of debt against the marginal cost of distress. Payout is the same kind of object. The four margins of the firm's financing problem, namely investment, leverage, payout, and cash holdings, are jointly chosen against one shadow value of internal funds, and the payout literature is the part of empirical corporate finance that studies the implementation of v_F versus v_I at the distribution margin.

Two methodological traditions organize the empirical payout literature. The causal tradition is anchored on the 2003 dividend tax cut (Chetty and Saez, 2005), which identifies the response of payout to a clean shock to v_I . The causal designs in this literature are among the cleanest in corporate finance, and Section 11.6 develops what they identify and what they do not. The structural tradition is anchored on the dynamic investment-leverage-payout model of DeAngelo et al. (2011), which estimates the wedge function directly by recovering the equity-issuance-cost technology that drives $v_F > v_I$ for small firms. The structural designs are thinner here than in the capital-structure or financial-constraints literatures, and Section 11.7 is honest about that.

The two traditions are not in tension. They identify different objects related to the same wedge under different sets of maintained assumptions. The causal tradition identifies how payout responds to identified shocks to one side of the wedge for the population of compliers in a given design. The structural tradition identifies the wedge function itself, at the cost of conditioning on a model. The chapter develops both and treats them as complements rather than substitutes.

Four reader questions organize the chapter, restated from the Identification chapter. What is the empirical object the estimator delivers? What assumptions are required to identify it? What population is the estimate informative about? What counterfactuals does the estimate support? Section 11.5 states these questions as the methodological centerpiece of the chapter: the same payout regression supports three readings under three different sets of maintained assumptions, and the reader's job is to ask which reading the author has committed to.

The running example threads through the chapter as a single specification estimated three ways on the master Compustat panel: panel fixed effects, IV using bank-credit-supply variation, and a calibrated structural simulation. Each of the three estimators identifies a different object related to the wedge, and Section 11.8 develops the three side by side. The chapter is shorter than its capital-structure or financial-constraints counterparts because

the payout literature has accumulated fewer canonical estimators and fewer ongoing methodological disputes. The methodological discipline is the same.

11.2 The wedge between internal and external valuation

Write the value of the firm to its shareholders as the discounted sum of future net distributions. With D_t the net dollar distributed in period t (dividends plus repurchases minus equity issuance), and with a stochastic discount factor $M_{t,t+s}$ applied between the firm and its shareholders,

$$V_t = \mathbb{E}_t \sum_{s=0}^{\infty} M_{t,t+s} D_{t+s}. \quad (11.1)$$

The firm chooses D_t each period subject to its investment, financing, and cash-holdings decisions. The first-order condition for D_t , taken against the alternative of retaining the marginal dollar, is what defines the wedge.

Let $v_{F,t}$ denote the shadow value of one extra dollar held inside the firm at date t . This is the marginal value the firm assigns to retained cash, taken across all the uses retention enables: relaxing future financing constraints, funding future investment without paying external-equity costs, servicing or expanding debt, and absorbing future cash-flow shocks. In the notation of the Part III introduction, $v_{F,t} = \mu_t$. Let $v_{I,t}$ denote the marginal value of one extra dollar distributed at date t , evaluated as shareholders evaluate it, net of personal taxes on the distribution and net of any reinvestment opportunity shareholders themselves can access. The optimality condition for payout is

$$v_{F,t} = v_{I,t} \quad \text{at the interior optimum,} \quad (11.2)$$

and the wedge

$$\omega_t \equiv v_{F,t} - v_{I,t} \quad (11.3)$$

is what each theory of payout writes a structural model for. Boundary solutions arise when one side dominates: firms with $\omega_t > 0$ at all feasible distribution levels retain everything; firms with $\omega_t < 0$ pay out the maximum the cash-flow identity allows.

Under Modigliani-Miller, $v_{F,t} = v_{I,t} = 1$ for all t and all states. Capital markets are frictionless, distributions are not taxed, information is symmetric, and managers act in shareholders' interest. The wedge is identically zero and payout is a free choice with no value implications. The classic irrelevance result is the statement that without frictions the chapter has nothing to study.

The frictions that have organized the empirical payout literature each locate the wedge in a different place. The clearest is the dividend tax wedge. When dividends are taxed at rate τ_D and the alternative use of retained earnings is taxed at the capital-gains rate τ_G , the relevant comparison for a taxable shareholder is $1 - \tau_D$ on the dividend side against $1 - \tau_G$ on the retention side, with the difference scaled by what the firm does with the retained dollar. For most of postwar U.S. history $\tau_D > \tau_G$ for top-bracket investors, so $v_I < v_F$ for the part of v_I coming from taxable individuals, and the tax wedge favors retention or repurchase over dividends. The 2003 tax cut narrowed the wedge sharply by lowering τ_D , and the response of payout to that change is what Section 11.6 studies.

The information-asymmetry friction (??) makes the dividend a signal. Managers who know the firm's cash-flow prospects better than investors do can commit to a dividend level that is costly to sustain in low-cash-flow states, and the commitment raises v_I by transmitting information that investors would otherwise discount. The signaling wedge is positive for v_I : the dividend creates value for investors over and above the cash transfer because it resolves uncertainty. Whether the magnitude is large enough to drive observed dividend behavior is contested, and the chapter does not take a position on the empirical importance of signaling.

The agency friction (Jensen, 1986) makes v_F lower than 1 inside firms whose managers have private benefits from controlling free cash flow. The wedge is then negative on the retention side ($v_F < 1 < v_I$), and payout is value-creating as discipline against managerial empire building. The Governance chapter develops this mechanism and the identification problems it raises. Where it shows up in the payout literature is in cross-sectional patterns: payout should be higher where free-cash-flow problems are larger and where governance mechanisms otherwise fail to discipline managers.

The financing-friction story (DeAngelo et al., 2011) makes $v_F > v_I$ for firms whose future investment opportunities are good and whose external equity is costly. The retained dollar is worth more than one inside such a firm because it relaxes the equity-issuance constraint in future states. The wedge shrinks and reverses as the firm matures and the investment opportunity set narrows. The life-cycle pattern of payout emerges from this mechanism endogenously, and Section 11.7 develops it as the chapter's structural anchor.

The catering story (Baker and Wurgler, 2004) makes v_I vary over time with investor demand for dividend-paying stocks. When investors price dividends at a premium relative to non-payers, v_I rises and managers respond by paying out; when the premium reverses, payout retreats. The mechanism is observationally distinct from the others because it predicts time-series variation in the wedge tied to a measurable sentiment indicator rather than to firm-level fundamentals.

Two complications matter and are worth flagging at the start. First, v_I is not a scalar but a distribution. Shareholders are heterogeneous in their tax brackets, their reinvestment opportunities, and their horizons, so the dividend that is value-creating for one shareholder need not be for another. The clientele literature (??) studies how firms sort into investor bases that align with their payout policies, and the chapter returns to this point in its treatment of the 2003 tax cut. Second, dividends and repurchases are not interchangeable distribution technologies, and the wedge that matters for repurchases differs from the wedge for dividends. Repurchases at price P are valuable for tendering shareholders when P exceeds their reservation value, and the tender-versus-hold decision introduces a signaling channel of its own (?). The chapter treats the dividend wedge and the repurchase wedge as related but distinct objects.

Key Insight 1: The wedge organizes the chapter

Optimal payout is governed by one comparison: v_F against v_I . Every empirical theory of payout is a story about a friction that drives a wedge between the two. The chapter's structural sections identify the wedge function; the chapter's causal sections identify the response of payout to shocks to one side of the wedge. The chapter's methodological sections develop what each kind of evidence tells us about the wedge and what it does not.

11.3 The cash-flow identity and what counts as payout

The wedge in (11.3) is the theoretical object. The empirical object is the realized flow that the wedge governs, and the accounting that ties payout to the rest of the firm's choices is the cash-flow identity. Sources of funds on the left equal uses on the right:

$$CF_{it} + EI_{it} + NDI_{it} = INV_{it} + \text{Payout}_{it} + \Delta\text{Cash}_{it}. \quad (11.4)$$

Operating cash flow, gross equity issuance, and net debt issuance fund investment, gross payout (dividends plus repurchases), and the change in cash holdings. The debt side is written net because debt has a single use-side counterpart (retirement), which we fold into the net flow. The equity side is written with issuance on the left and gross payout on the right so that dividends and repurchases sit together as a single use of cash. The Capital Structure chapter develops the same accounting constraint as the financing-deficit equation (10.13); this is one identity in two algebraic arrangements.

The identity does empirical work in two ways. It tells us what to measure, and it tells us what every payout coefficient is jointly identified against. Every payout flow on the right is

a residual from the choices on the left and from the other right-hand uses. A regression that ignores this joint determination treats payout as if it were a primitive, and the resulting estimate is interpretable only under assumptions that pin down which margins are held fixed. The joint-determination problem is the substantive content of the wedge: investment, leverage, payout, and cash are four margins of one optimization problem with one shadow value of internal funds. The methodological reading developed in Section 11.5 returns to this point.

Variable definitions follow the Compustat conventions used in Frank and Goyal (2024). Cash dividends are *dvc*, common and preferred share repurchases are *prstk* reported as a positive dollar amount, and gross payout is the sum of the two. Equity issuance is *sstk*, gross investment is *capx*, and the change in cash holdings is the year-over-year difference in *che*. Net payout is gross payout minus equity issuance,

$$\text{NetPayout}_{it} = \text{dvc}_{it} + \text{prstk}_{it} - \text{sstk}_{it}, \quad (11.5)$$

scaled by lagged assets when entered into regressions. Rearranging the identity (11.4) to isolate net payout gives the residual form

$$CF_{it} + NDI_{it} = INV_{it} + \text{NetPayout}_{it} + \Delta\text{Cash}_{it}, \quad (11.6)$$

which is the outcome the running example in Section 11.8 constructs. The chapter uses net payout as the workhorse outcome because gross payout mismeasures the wedge: a firm that pays a one-dollar dividend while issuing a one-dollar of equity has not distributed anything in the sense the wedge cares about, and a firm that repurchases shares funded by debt issuance is making a leverage choice, not a payout choice. The Bonaime-Hankins-Jordan literature on repurchase flexibility revisits the dividend-versus-repurchase decomposition, developed in Section 11.7.

11.4 Three facts the literature has to explain

Fact 1: Lintner smoothing. Lintner (1956) studied the dividend-setting practices of 28 firms through detailed interviews and inferred that firms behave as if they have a target dividend payout ratio and adjust toward it slowly. The canonical specification, restated as a regression, is

$$\Delta D_{it} = \alpha_i + s_i (r_i E_{it} - D_{i,t-1}) + \varepsilon_{it}, \quad (11.7)$$

where D_{it} is dividends, E_{it} is earnings, r_i is the firm's target payout ratio, and s_i is the

speed of adjustment. Four regularities follow. Firms behave as if they have target payout ratios; they smooth dividends so s_i is well below one; permanent changes in earnings move dividends more than transitory changes do; and managers are reluctant to cut, so the distribution of dividend changes is asymmetric around zero. Fifty years later, [Brav, Graham, Harvey, and Michaely \(2005\)](#) confirm the same patterns in a survey of 384 CFOs: smoothing, target ratios, reluctance to cut, and a sharper response to permanent than to transitory earnings.

The smoothing fact admits two competing interpretations. Under signaling, managers commit to dividend levels because dividend changes are informative about future earnings, and the cost of revising the commitment downward is what enforces the commitment. Under the life-cycle reading, smoothing reflects optimal payout from firms that have entered the mature phase of their earnings process, so transitory shocks do not warrant policy revisions. Section 11.5 returns to this distinction in the context of identification.

Fact 2: Repurchases overtook dividends. [Fama and French \(2001\)](#) document that the fraction of US public firms paying dividends fell sharply between the late 1970s and the late 1990s, from roughly two-thirds to roughly one-fifth. The decline reflects two distinct shifts. The composition of public firms changed: the population came to include many newly-listed small firms with loss histories and limited cash flow, the firm type that has always been a non-payer. Within firm types, the propensity to pay also declined: even mature profitable firms with characteristics that would have predicted dividend payment in the earlier sample chose not to pay. Both shifts contribute, and the within-type propensity shift accounts for a substantial portion of the aggregate decline.

[Jagannathan, Stephens, and Weisbach \(2000\)](#) clarify the substitute that absorbed the missing dividends. Among repurchasing firms, repurchases co-move with transitory cash flow, while dividends co-move with permanent cash flow. The two payout forms are not interchangeable: dividends carry commitment value that managers do not assign to repurchases, and transitory cash is returned through repurchases precisely because the channel does not lock the firm into a recurring obligation. [Grullon and Michaely \(2002\)](#) document the aggregate result. By the late 1990s, total repurchase dollars by US public firms exceeded total dividend dollars; repurchases and dividends had become close substitutes in the aggregate even as they remained distinct in their firm-level dynamics.

Figure 11.1 plots the aggregate flows. Dividends are roughly flat over the sample period; repurchases rise from near zero in the early 1970s to overtake dividends by the late 1990s, with cyclical variation that tracks aggregate corporate profitability. The visual pattern is the core fact: any model of payout that ignores the repurchase margin misreads half of the

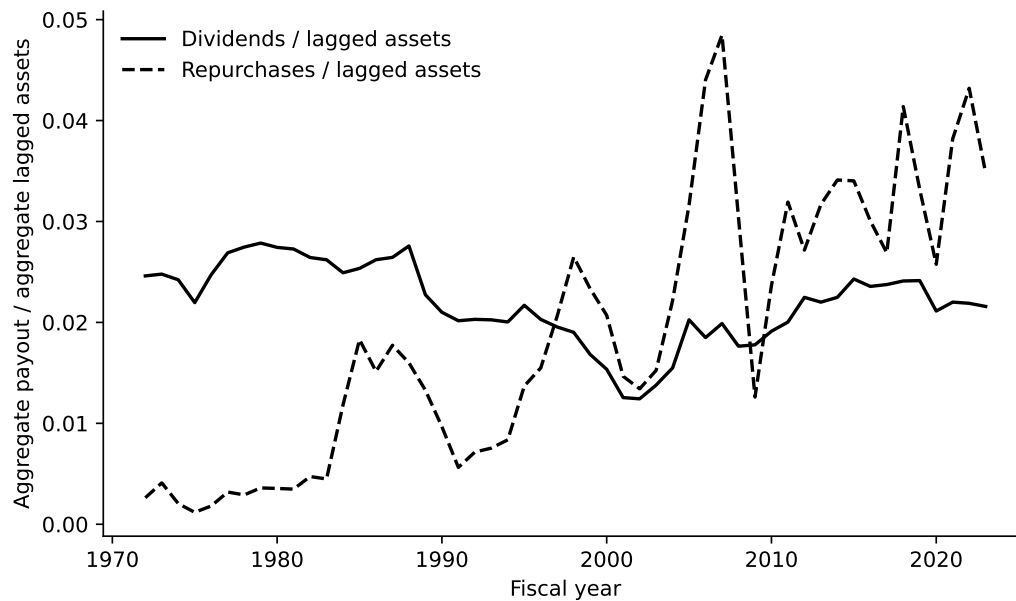


Figure 11.1. Aggregate dividends and aggregate repurchases, scaled by aggregate lagged assets. Master Compustat panel with standard cleaning (nonfinancial, nonutility, 1972–2023). Dividends are Compustat *dvc*, repurchases are *prstkc*.

modern picture.

Fact 3: Concentration and life cycle. The cross-section of payers is concentrated and follows a life cycle. [DeAngelo et al. \(2006\)](#) propose the ratio of retained earnings to total equity (RE/TE) as a life-cycle measure and show that the propensity to pay rises sharply with RE/TE . Firms with negative or low RE/TE are young capital-raising firms that retain earnings to fund growth; firms with high RE/TE are mature firms that have accumulated more equity than they can profitably reinvest. The empirical regularity is sharp: among firms in the top RE/TE deciles, roughly four-fifths pay dividends; among firms in the bottom deciles, roughly one-tenth do.

Concentration follows from the life-cycle pattern. Because mature high- RE/TE firms are also typically large, the cross-section of aggregate payout is dominated by a small number of firms. The top decile of payers accounts for the large majority of aggregate dividends, with a similar concentration pattern in aggregate repurchases. The precise share varies by year and by definition.

Three readings of the same pattern compete. The agency reading takes mature firms as those most exposed to free-cash-flow problems and reads payout as a disciplining device against managerial empire building; Chapter 14 develops this argument and the associated identification challenges. The free-cash-flow reading takes mature firms as those with

limited investment opportunities and reads payout as the optimal use of cash that has no better deployment; this reading is mechanically consistent with the life-cycle pattern but does not commit to a particular friction. The catering reading of [Baker and Wurgler \(2004\)](#) takes payout policy as responsive to time-varying investor demand for dividend-paying stocks, with managers adjusting payout to capture a dividend premium when investors favor payers; the prediction is comovement between aggregate payout propensity and a dividend premium measured from valuations.

Common Mistake 2: Lintner on total payout

Applying the Lintner partial-adjustment specification to total payout (dividends plus repurchases) produces a misleadingly low estimated speed of adjustment. The reason is mechanical: managers smooth dividends but do not smooth repurchases, so a model that imposes the same partial-adjustment dynamics on both confounds two distinct behavioral patterns. Use partial-adjustment on the dividend component alone, or specify separate dynamics for the two payout forms.

11.5 What a payout regression identifies

The empirical payout literature concentrates on a single workhorse specification. Let NetPayout_{it} be net payout scaled by lagged assets, and let the right-hand side include the proxies that [Fama and French \(2001\)](#) and its descendants have settled on for profitability, investment opportunities, size, life-cycle position, and capital structure. The canonical payout regression is

$$\text{NetPayout}_{it} = \alpha + \beta_1 \text{Profit}_{it} + \beta_2 Q_{it} + \beta_3 \text{Size}_{it} + \beta_4 (\text{RE/TE})_{it} + \beta_5 L_{it} + \mu_i + \delta_t + \varepsilon_{it}, \quad (11.8)$$

with firm and year fixed effects μ_i and δ_t . Section [11.8](#) runs this specification on the master Compustat panel as the chapter's running example. The substantive question is what (11.8) identifies. The equation supports three readings, each tied to a different maintained assumption.

Reading 1: descriptive. The most defensible reading is that β_1 is the within-firm partial correlation between profitability and net payout, holding the other right-hand-side variables fixed and absorbing firm and year heterogeneity. The reading commits the researcher to nothing beyond the regression specification itself, and it is the right reading for most published payout regressions. Most of what we know about firm-level payout patterns comes from descriptive evidence of this form. *What could go wrong.* The descriptive claim

survives under any data-generating process and so cannot be falsified by the behavior of β_1 alone; the cost is that the claim does not support counterfactuals or policy questions. Reporting β_1 as “the effect of profitability on payout” overstates what the regression delivers.

Reading 2: causal under conditional-mean-independence. The causal reading requires $\mathbb{E}[\varepsilon_{it} | X_{it}, i, t] = 0$, the within-firm conditional-mean independence of the error from the included right-hand-side variables (the full strict-exogeneity condition required by the within-firm estimator is stronger and conditions on the firm-level history of regressors), where X_{it} collects the regressors in (11.8). Under this assumption β_1 identifies the average effect of a one-unit increase in profitability on net payout, holding the other regressors fixed. The assumption is strong. Profitability covaries with unobserved investment opportunities, with the persistence of cash-flow shocks, and with life-cycle position in ways that the included controls cannot fully absorb. A firm whose profitability rises because its investment opportunities have expanded will both invest more and pay out less, holding leverage fixed; the regression coefficient confounds this opportunity-set shift with the response of payout to cash availability. Section 11.6 returns to the causal reading through the design that earns it legitimately for a specific shock, the 2003 dividend tax cut. *What could go wrong.* The CMI assumption fails as soon as one entertains the standard reasons profitability and payout are jointly determined. Adding more controls mitigates but does not solve the problem; the joint-determination structure is not removable by enlarging X_{it} .

Reading 3: structural. The structural reading treats (11.8) as a reduced-form projection of the policy functions implied by a dynamic model of the firm. Section 11.7 develops the explicit version: firms choose investment, leverage, and payout to maximize shareholder value subject to financing frictions, and the equilibrium policy functions can be expressed as approximate linear combinations of state variables, of which the right-hand side of (11.8) is a partial enumeration. β_1 then has structural content: it is a function of the deep parameters governing the firm’s optimization, including equity-issuance costs, the elasticity of value to retained earnings, and the curvature of the investment-cost function. The reading licenses counterfactuals that the descriptive and CMI readings do not support. *What could go wrong.* The estimate is only as good as the model. A regression coefficient estimated under one tax or financing-cost regime need not transport to a different regime; the reduced-form coefficient is implicitly conditioned on the structural parameters and the policy environment in force during estimation. And as [Strebulaev and Whited \(2012\)](#) emphasize in their survey of dynamic corporate-finance models, the same set of observed

moments can be matched by several alternative structural specifications, so the structural reading commits the researcher to defending both the data and the model.

The structural and CMI readings share a less-discussed problem: right-hand-side endogeneity in nearly every regressor. $(RE/TE)_{it}$ is mechanically determined by past payout because retained earnings are cumulative net income minus cumulative distributions; including it on the right side of a payout regression embeds a near-identity, and the coefficient picks up the mechanical link rather than a behavioral response. Cash holdings, which appear in extended specifications, are jointly chosen with payout in the identity (11.4), with theoretical treatment in Chapter 9. Leverage L_{it} is jointly chosen with payout in the capital-structure-and-payout problem of Chapter 10. None of these joint-determination problems is fixable by adding more controls. The fixes available are substantive: a design-based source of variation in one of the endogenous regressors, or a structural model that takes the joint determination as the empirical object rather than as a nuisance.

Key Insight 3: Three readings, one regression

The same payout regression identifies different objects under different assumptions. The reader's job is to ask which reading the author has committed to, and what defends that reading.

11.6 The 2003 dividend tax cut: Chetty-Saez

The shock. The Jobs and Growth Tax Relief Reconciliation Act of 2003 (JGTRRA) cut the top federal tax rate on qualified dividends from the ordinary-income rate, then 35 percent at the top bracket, to 15 percent, equalizing it with the long-term capital-gains rate. The cut was announced in May 2003 and applied to dividends paid after January 1, 2003. The treatment-control structure is across firms: treatment exposure is the share of equity held by taxable individual shareholders, who benefit from the cut, while control exposure is the share held by tax-exempt institutions (pension funds, retirement accounts, non-profits) and corporate cross-holders, whose effective tax wedge on dividends is unaffected.

The result. Chetty and Saez (2005) document a sharp and economically large response. Regular dividend payments and dividend initiations rose after the cut, with the response concentrated in firms holding high taxable-individual ownership and especially pronounced in firms where executives held substantial stock or option positions. The fraction of US public firms paying regular dividends rose by roughly five percentage points within two years of the cut, and aggregate regular dividend payments rose by roughly twenty

percent over the same window. The response was persistent through the post-cut window the original paper covered, not a one-time level shift. Initiations responded more sharply than increases in existing dividend levels, consistent with the cut relaxing a binding constraint on firms that had previously chosen not to pay rather than just reweighting marginal incentives at the intensive margin.

What is identified. The Chetty-Saez estimate identifies a local average treatment effect, not a population elasticity. The compliers are firms whose payout policy was responsive to the 2003 dividend tax wedge in the specific institutional setting prevailing then, and three concrete dimensions make this complier population non-representative. Ownership composition matters first: compliers have high taxable-individual ownership, especially executive ownership, and firms held primarily by tax-exempt institutions are not compliers, a category whose share of US public equity has grown substantially since 2003. Cash position matters second: compliers have ample cash or cash-flow capacity to initiate or increase distributions, while constrained firms could not respond to the tax wedge regardless of its size. Existing payout posture matters third: the initiation response is concentrated among firms that had been near the margin of paying, with long-time non-payers holding growth opportunities not crossing into payer status.

Chapter 2 develops the LATE framework formally; the point here is operational. A paper citing [Chetty and Saez \(2005\)](#) to justify a claim about “the elasticity of payout to taxation” is overreaching. The estimate supports claims about the 2003 cut’s effect on 2003-era firms with the prevailing ownership composition. It does not support claims about a counterfactual tax cut in a different decade, on a different ownership base, or operating through a different payout instrument such as repurchases or special dividends.

[Yagan \(2015\)](#) re-examines the 2003 cut using a different identification strategy: comparing C-corporations, whose distributed income is taxed at the shareholder level under the dividend tax schedule, with S-corporations, whose income passes through to owners and is taxed at ordinary rates regardless of distribution form. Yagan finds essentially no effect of the 2003 cut on real investment or wages and only modest effects on payout in this comparison, in apparent tension with the Chetty-Saez headline. The tension is real but the two designs do not identify the same object. Chetty-Saez identifies a LATE across firms distinguished by ownership composition holding organizational form fixed; Yagan identifies a different LATE across organizational forms. Both estimates are defensible under their respective designs, and the apparent disagreement is partly a story about which compliers are being averaged. Honest reporting of the 2003 evidence acknowledges both estimates rather than citing whichever one supports a preferred conclusion.

The DiD as a two-by-two. The Chetty-Saez design is a single-event difference-in-differences: one date, treated and control units, pre/post comparison. Because the shock is a single date rather than a staggered policy adoption, the heterogeneous-treatment-effects problems that complicate two-way-fixed-effects estimators on staggered designs (Chapter 6) do not apply here. Parallel-trends evidence in Chetty and Saez (2005) supports the design: pre-2003 trends in dividend payment behavior were similar across high- and low-taxable-ownership firms. The Rambachan and Roth (2023) sensitivity framework can be applied post-hoc to bound how strong a parallel-trends violation would have to be to overturn the headline result; the original paper predates this framework, but the diagnostic is straightforward to compute on the published data.

Repurchase event studies in one paragraph. The 2003 cut and a small number of comparable tax shocks supply the cleanest causal evidence in the payout literature. Repurchase announcement event studies are the next most common identification strategy and the most commonly mis-interpreted one. Announcement returns to open-market repurchase programs are positive on average, typically in the range of two to three percent over a short window, and the literature has read this evidence variously as supporting signaling, undervaluation, agency-cost reduction, life-cycle maturity, and option-driven flexibility. None of these channels is separately identified by the announcement return; a positive announcement effect is consistent with each story and so identifies none of them. The event study identifies the market's average revaluation conditional on a repurchase announcement, not the mechanism producing it. Chapter 4 develops what event studies can and cannot identify in general, and the application to repurchases inherits those limitations.

Connection to Framework 4: The LATE here, not a population elasticity

Chetty and Saez (2005) identifies a LATE for compliers in the 2003 design. It does not identify a population elasticity of payout to taxation, the response to a counterfactual reform with a different tax base or ownership composition, or any general-equilibrium adjustment beyond the realized variation. Section 11.7 develops what would be needed to deliver any of these. The chapter's lesson is not that the causal evidence is weak; the 2003 design is among the cleanest in corporate finance. The lesson is that even clean causal evidence supports only the claims the design identifies.

11.7 Structural payout: DeAngelo-DeAngelo-Whited

Why a dynamic model. Static Modigliani-Miller cannot identify three objects the modern payout literature cares about. The option value of repurchase flexibility, the cost of cutting dividends, and the welfare effect of counterfactual tax or financing-cost regimes each depend on how firms re-optimize across periods as the state evolves. A static cross-section integrates these dynamics out and confounds them with the realized policy at any one date. The dynamic-programming machinery developed in Chapter 8 and the structural capital-structure treatment of Section 10.5 are the tools the payout literature inherits; this section uses them rather than rebuilding them.

The DeAngelo-DeAngelo-Whited model. DeAngelo et al. (2011) formalize the joint investment-leverage-payout problem. Each period the firm chooses investment, leverage, and payout subject to a collateral constraint on debt and a convex external-financing cost on equity. The state includes productivity, capital, and outstanding debt; the value function is solved by value-function iteration on a discretized state space; the structural parameters are estimated by simulated method of moments matching observed moments of investment, leverage, payout, and cash. The payout-relevant primitives are the equity-issuance cost function, which makes the marginal dollar of external equity expensive, and the elasticity of firm value to retained earnings, which prices the option of internal funds against the cost of external funds.

The model's headline finding is that financial flexibility, the option value of holding retained earnings to fund future investment without paying the equity-issuance cost, dominates payout decisions for small firms. Low payout is not a residual or a behavioral anomaly in this framework; it is the optimal response of a small constrained firm to a future in which financing frictions are expected to bind. The life-cycle pattern of Section 11.4 emerges endogenously from this logic: as the firm grows, the equity-issuance cost ceases to bind, the value of retaining marginal cash falls relative to the value of distributing it, and the firm transitions from non-payer to payer. The DDW parameter estimates imply a substantial wedge between internal and external equity costs at the margin for small firms, which sustains both the cross-sectional concentration of payout and the propensity-to-pay gradient in RE/TE .

What the structural estimates pin down. Two objects the causal designs of Section 11.6 cannot deliver. First, counterfactuals beyond realized variation: the welfare effect of a repurchase-side tax, the effect of equalizing the dividend and repurchase tax rates, the

response to a payout disclosure rule, or the effect of a financing-cost shift on the distribution of payout across the firm size distribution. None of these has historical variation that supports a difference-in-differences design, and answering them requires a model. Second, joint determination: the structural framework treats investment, leverage, and payout as simultaneous choices governed by the same set of frictions and the same value function, rather than as separate margins to be estimated one at a time. This is the framing that the right-hand-side endogeneity discussion in Section 11.5 flagged as not deliverable by regression.

The cost is that the estimates are conditional on the model. The [Strebulaev and Whited \(2012\)](#) observation that the same moments can typically be matched by several alternative dynamic structures applies in full force here, and the structural payout estimate is trustworthy only to the extent that the modeling assumptions about production, financing costs, and timing are themselves defensible. Identification of structural parameters relies on the model delivering moments that move with the parameters in distinguishable ways, and the structural payout literature has not yet accumulated the cross-specification triangulation that the structural capital-structure literature provides.

The state of the structural literature. The structural payout literature is thinner than the structural capital-structure literature and thinner than the structural financial-constraints literature. DDW is the cleanest treatment to date and has not been followed by a comparable wave of replications, extensions, or competing specifications. [Bonaimé, Hankins, and Jordan \(2016\)](#) provide a more recent partial-equilibrium treatment of the dividend-versus-repurchase choice that emphasizes the option-like features of repurchases: repurchase programs are flexible in a way that dividend commitments are not, so repurchases serve as the discretionary margin through which firms respond to transitory shocks. The contribution is consistent with the [Jagannathan et al. \(2000\)](#) fact from Section 11.4 but takes it further by quantifying the option value.

The asymmetry between the well-developed causal side of the payout literature, anchored on the 2003 dividend tax cut in Section 11.6, and the thinner structural side is itself a research opportunity. A PhD reader looking for a dissertation topic in payout policy should take this asymmetry seriously: the gap between what we have (one structural anchor, one major causal anchor, limited counterfactuals supported by either) and what we need (structural models that can interrogate the policy questions the data alone cannot answer) is real and not yet closed.

11.8 Running example: three estimators on the master panel

The setup. The chapter’s running example puts the three readings of Section 11.5 to work on a single specification: net payout scaled by lagged assets regressed on lagged book leverage, with the Frank-Goyal control set of profitability, market-to-book, log assets, and RE/TE . The outcome is constructed from (11.6) on the master Compustat panel of Chapter 3. Each of the three installments below estimates the same regression under different identifying assumptions, and each installment maps to one of the three readings: panel fixed effects to the descriptive reading, IV with the bank-credit-supply instrument to the causal reading, and a calibrated structural simulation to the structural reading.

Running Example 5: Three estimators on the master Compustat panel

We estimate the leverage-payout relationship three ways. A panel fixed-effects regression delivers the descriptive reading; an instrumental-variables regression using the Leary (2009) bank-credit-supply measure delivers the causal reading; a calibrated structural simulation delivers the structural reading. The outcome is net payout per (11.6) scaled by lagged assets, the regressor is lagged book leverage, and the control set follows Frank and Goyal (2024). Code is in `code/python/payout/installment_three_estimators.py` and `code/stata/payout/installment_three_estimators.do`.

Installment 1: panel fixed effects. The panel-FE specification regresses net payout on lagged leverage with firm and year fixed effects and firm-clustered standard errors. The reading is descriptive: the coefficient is the within-firm partial correlation between lagged leverage and net payout, holding the Frank-Goyal regressors fixed and absorbing time-invariant firm heterogeneity and aggregate year shocks. The specification does not commit to a causal interpretation, and its interpretation under that assumption is the conditional-mean-independence discussion of Section 11.5.

The Python and Stata implementations are in the *Installment: panel fixed effects* section of the Code Companion.

Installment 2: IV with the Leary bank-credit-supply instrument. The instrument for lagged leverage is the bank-credit-supply measure of Leary (2009), which exploits cross-sectional variation in bank lending capacity that is plausibly orthogonal to firm-level payout policy. The exclusion restriction is the same one used in the financial-constraints applications of Chapter 9 and the innovation applications of Chapter 13: shocks to bank

balance sheets shift the supply of credit available to bank-dependent firms without operating through any direct channel on those firms' payout choices. The first-stage F -statistic should be reported with the second-stage estimate so the reader can judge instrument strength. The estimate identifies a local average treatment effect on bank-dependent compliers, not a population average response of payout to leverage.

The Python and Stata implementations are in the *Installment: IV with the Leary bank-credit-supply instrument* section of the Code Companion.

Installment 3: calibrated structural simulation. The simulation panel is constructed under a stylized dynamic model in the spirit of Section 11.7. Productivity follows an AR(1) process, capital evolves with productivity-driven investment, leverage binds at a firm-specific collateral constraint, and net payout is determined as the residual after operating cash flow, investment, and the change in debt:

$$z_{it} = \rho z_{i,t-1} + \varepsilon_{it}, \quad \varepsilon_{it} \sim N(0, \sigma_\varepsilon^2), \quad (11.9)$$

$$K_{i,t+1} = (1 - \delta) K_{it} + \delta \bar{A} \exp(z_{it}), \quad (11.10)$$

$$L_{it} = \theta_i K_{it}, \quad \theta_i \sim N(\bar{\theta}, \sigma_\theta^2) \text{ truncated to } [0.20, 0.60], \quad (11.11)$$

$$\text{NetPayout}_{it} = \text{CF}_{it} + \Delta L_{it} - \text{INV}_{it}.$$

The parameter vector is $(\rho, \sigma_\varepsilon, \delta, \bar{\theta}, \sigma_\theta, \bar{A}) = (0.70, 0.20, 0.10, 0.40, 0.08, 2.0)$, and cash flow is $\text{CF}_{it} = 0.10 \exp(z_{it}) K_{it}^{0.70}$. Firm-level heterogeneity in the collateral parameter θ_i gives lagged leverage cross-sectional variation across firms while leaving it constant over time within a firm. The simulation produces a panel of 1,000 firms over 30 periods at seed 8823. The OLS regression of net payout on lagged leverage on this synthetic panel identifies an across-firm in-model comparative static, not a within-firm effect: because lagged leverage is constant within each firm, the regression uses year fixed effects only. This is a calibrated simulation, not separately estimated SMM; the SMM extension is Exercise 7 in Section 11.10.

The implementation generates a synthetic panel under (11.9) at seed 8823, scales net payout by contemporaneous capital, and runs an OLS regression of net payout on lagged leverage with year fixed effects only. Year fixed effects alone, not firm-and-year, are appropriate because the firm-level collateral parameter θ_i that generates cross-firm variation in leverage is constant over time within a firm. The regression recovers the in-model across-firm comparative static of net payout with respect to leverage at the calibrated parameter values; it is neither a treatment effect nor an estimate of the true data generating process for actual payout, and the contrast between this in-model coefficient and the panel-FE coefficient earlier in the section is the point of the comparison. The

Python implementation is in the *Installment: calibrated structural simulation* section of the Code Companion.

Comparing the three estimates. Table 11.1 sets the three estimators side by side. Each column corresponds to one installment, and each row isolates a dimension on which the three estimators differ: identifying assumption, complier population, parameter identified, estimate, whether the estimator supports counterfactuals, and which reading from Section 11.5 it maps to. The three estimators answer different questions and identify different parameters. The point of the installment is not to settle which is “right” but to make the trade-off operationally visible.

Connection to Framework 6: Three estimators, three parameters

The three estimators applied to the same outcome and the same regressor identify three different objects under three different sets of maintained assumptions. The panel-FE coefficient is a within-firm partial correlation; the IV coefficient is a LATE on bank-dependent compliers; the structural-simulation coefficient is an in-model comparative static. None of these is the elasticity of net payout to leverage in the population. The choice among them is a choice among the questions one is willing to ask.

11.9 Common mistakes and decision framework

Common Mistake 7: Announced versus actual repurchases

Treating Compustat repurchase announcements as completed transactions overstates payout. [Stephens and Weisbach \(1998\)](#) document that firms execute a substantial but variable fraction of announced amounts within the year following announcement, with completion rates well below one hundred percent. Studies of announcement-period market reactions identify the response to the program, not to a completed distribution. Studies of firm-level cash flows should use actual repurchases from `prstk`, not announcement events.

Common Mistake 8: Total payout when net payout is the right object

Studying gross payout (dividends plus repurchases) without netting out equity issuance confounds the level of payout with its form. A firm that simultaneously repurchases shares and issues new equity (often to fund executive compensation that the repurchases offset) is distributing less to investors on net than the gross payout

suggests. Section 11.3 develops the cash-flow identity that makes net payout the right residual object; use net payout ((11.6)) as the dependent variable unless the research question specifically concerns the dividend-versus-repurchase choice.

Common Mistake 9: Signaling as a residual explanation

A positive announcement return on a dividend initiation is consistent with signaling, agency-cost reduction, life-cycle maturity, and selection into payment by firms whose cash position has improved. The event study identifies the average revaluation; it does not identify which mechanism produced it. Reading announcement returns as “evidence of signaling” without ruling out the alternatives overreaches. Section 11.5 develops the same point in the context of regression-based identification: a coefficient consistent with several mechanisms is not evidence for one of them.

The three mistakes share a common structure. Each treats a measured object as identifying a deeper one without defending the assumptions required to make the identification work. The decision framework below operationalizes this point by tabulating what each methodological posture in the chapter can and cannot support.

11.10 Connection to framework, exercises, and further reading

The payout literature is asymmetric. Its causal side has one major anchor in the 2003 dividend tax cut (Section 11.6) and a small number of comparable shocks; its structural side has DeAngelo et al. (2011) and a handful of follow-on papers (Section 11.7). The asymmetry is real, and it is a research opportunity for the PhD reader: the gap between what we have and what we need is the most direct place to make a contribution. The chapter’s central methodological takeaway is the three-readings lesson of Section 11.5. The same payout regression identifies different objects under different maintained assumptions, and the choice among the descriptive, causal-under-CMI, and structural readings is a choice among defensible claims rather than a search for a single “true” estimate. The four reader questions from Section 11.1 return as the four checks any payout study should pass: which object is being estimated, under which assumptions, for which population, and which counterfactuals does the result support.

Exercises

Exercise 1 (Conceptual). Lintner smoothing, life-cycle, and tax explanations make overlapping predictions about who pays dividends and who repurchases. For each pairwise comparison (smoothing versus life-cycle, smoothing versus tax, life-cycle versus tax), identify one prediction where the two explanations agree and one where they disagree. State what an empirical test would need to do to distinguish them.

Exercise 2 (Replication, easy). Replicate the [Fama and French \(2001\)](#) disappearing-dividends result on the master Compustat panel for 1978–1999 and for 2000–2022. Compare propensity-to-pay coefficients across the two periods. What do the differences tell you about the role of the 1990s as a structural break, and what do they tell you about the post-Chetty-Saez recovery in dividend payment behavior?

Exercise 3 (Replication, moderate). Estimate the Lintner partial-adjustment equation (11.7) on the subsample of continuous dividend payers on the master Compustat panel. Report the speed of adjustment $\hat{s}$ and the implied target payout ratio $\hat{\tau}$ by decade for 1971–1979, 1980–1989, 1990–1999, 2000–2009, and 2010–2019. Discuss whether $\hat{s}$ shows a time trend and what changing values would imply for the smoothing story.

Exercise 4 (Computational). Construct RE/TE on the master Compustat panel following [DeAngelo et al. \(2006\)](#). Sort firms into RE/TE deciles by year. For each decile, report the propensity to pay dividends, the propensity to repurchase, and the median net payout scaled by lagged assets. Discuss the cross-sectional pattern in light of Section 11.4.

Exercise 5 (Applied). Read [Chetty and Saez \(2005\)](#). In a one- to two-page memo: (a) define the complier population in your own words; (b) state what would have to be true for the LATE to extrapolate to a counterfactual dividend tax reform in a different decade; (c) describe what [Yagan \(2015\)](#) adds to the question and what disagreement remains between the two papers.

Exercise 6 (Replication, moderate). Re-run the Section 11.8 three-estimator installment using gross payout (dividends plus repurchases, ignoring equity issuance) rather than net payout as the outcome. Does the comparison across the three estimators change qualitatively? Discuss which differences are attributable to the choice of dependent variable and which would survive any reasonable redefinition.

Exercise 7* (Structural, hardest). Extend the Section 11.8 Installment 3 calibrated simulation to a fully estimated SMM model. Solve the dynamic program in (11.9) by value-function iteration on a discretized state space; simulate a panel of 5,000 firms over 50 periods at seed 8823; estimate the structural parameters $(\rho, \sigma_\varepsilon, \delta, \theta, \bar{A})$ by matching moments on the master Compustat panel (mean and standard deviation of net payout, leverage, and investment, plus the within-firm leverage-payout regression coefficient). Report parameter estimates with standard errors. Use the result to compute the counterfactual effect on net payout of a 5-percentage-point reduction in θ (a tighter collateral constraint). Discuss which assumptions of the model the counterfactual is most sensitive to.

Further Reading

The chapter draws on six entry points. The surveys are [Allen and Michaely \(2003\)](#), the original handbook chapter on payout policy; [DeAngelo, DeAngelo, and Skinner \(2008\)](#), the *Foundations and Trends* synthesis covering the rise of repurchases and the structural-causal asymmetry; and [Farre-Mensa, Michaely, and Schmalz \(2014\)](#), the post-Chetty-Saez *Annual Review* update tying the modern empirical evidence together. The methodological anchors are [Brav et al. \(2005\)](#) for the survey evidence on managerial beliefs about payout, [Chetty and Saez \(2005\)](#) for the chapter's causal anchor on the 2003 dividend tax cut, and [DeAngelo et al. \(2011\)](#) for the chapter's structural anchor on the dynamic investment-leverage-payout model. A PhD reader new to the literature should start with the survey trio in chronological order, then work through Chetty-Saez and DDW with the running example of Section 11.8 as a methodological scaffold.

Table 11.1. Three estimators of the leverage-payout relationship on the master Compustat panel. The IV column is marked (–) because the state-level analog of the [Leary \(2009\)](#) bank-credit-supply instrument constructed on this panel (Call Reports 2001–2024, residualized against state and year fixed effects) has a first-stage F of 0.27, well below the conventional weak-instrument threshold.

	Panel FE	IV (Leary 2009)	Structural simulation
Identifying assumption	Within-firm CMI on the leverage coefficient	Exclusion restriction on bank-credit supply	Joint determination under the calibrated dynamic model
Population	Firm-year observations in the master panel	Bank-dependent compliers	Synthetic firms at seed 8823
Parameter identified	Within-firm partial correlation	Local average treatment effect (LATE)	In-model comparative static
Estimate	-0.082*** (0.013)	(–)	0.025*** (0.001)
Supports counterfactuals?	No	Limited to the LATE setting	Yes, within the model
Reading from Section 11.5	Descriptive (Reading 1)	Causal under CMI (Reading 2)	Structural (Reading 3)

Notes. Outcome: net payout scaled by lagged assets, NetPayout_{it} from (11.6). Regressor: lagged book leverage. Controls: profitability, market-to-book, log assets, RE/TE . The Estimate row reports the coefficient on lagged book leverage with the firm-clustered standard error in parentheses; stars: $*p < 0.10$, $**p < 0.05$, $***p < 0.01$. Panel FE column uses firm and year fixed effects on the master Compustat panel. IV column instruments lagged leverage with a state-level analog of the [Leary \(2009\)](#) bank-credit-supply measure, built from Call Reports for 2001–2024 (the post-2001 window is the longest run available in the WRDS bank_all schema accessible to this project). The first-stage F -statistic on the lagged instrument is 0.27 with firm-clustered standard errors, well below the conventional weak-instrument threshold of 10. The IV estimate is not reported. Structural simulation column reports the OLS coefficient on lagged leverage from the synthetic panel of 1,000 firms over 30 periods at seed 8823 generated as described in the *Installation: calibrated structural simulation* section of the Code Companion. The collateral parameter θ_i is drawn at the firm level so that lagged leverage varies across firms but is constant within firm; the regression therefore uses year fixed effects only and identifies an across-firm comparative static.

Table 11.2. Four methodological postures in the payout literature: what each answers and what it cannot support.

Posture	What it answers	What it cannot support
Descriptive payout regression	Stylized facts; cross-sectional and within-firm patterns	Causal claims; counterfactuals
Tax-shock difference-in-differences (Chetty and Saez, 2005)	Local average treatment effect for compliers in the design	Population elasticity; counterfactuals beyond the design's variation
Structural payout model (DeAngelo et al., 2011)	Primitive parameter estimates; counterfactual policy analysis within the model	Causal claims independent of model assumptions
Event study on payout announcement	Average market reaction conditional on announcement	Long-run operating effects; the mechanism producing the reaction

Notes. The four postures correspond to the four ways the empirical payout literature has identified objects related to the leverage-payout, profitability-payout, and tax-payout relationships. Each row pairs a posture with what it answers under its identifying assumption and what it cannot support without further structure. The chapter's running example in Section 11.8 implements the first three postures on a common specification.

Chapter 12

Banking and Credit Supply

12.1 Why Banks Need Their Own Chapter

Consider a single loan: a regional manufacturer with one hundred million in sales borrows fifteen million from a large bank on a five-year term loan, secured against equipment. On the firm side of the relationship, this loan appears in Compustat as an increment to long-term debt, with interest expense flowing through the income statement and a covenant footnote disclosed in the 10-K. On the bank side of the same relationship, the loan appears in DealScan with its tranche structure and pricing terms, in the bank's Y-9C as a commercial and industrial loan, in supervisory data with an internal risk rating, and possibly in the Shared National Credit Program if the syndicate is large enough. The firm-side records and the bank-side records describe the same loan, but they answer different questions, contain different errors, and have different selection problems.

Chapter 9 treats firm-side financing data as the lens on whether the firm faces a binding constraint, and Chapter 10 treats firm-side data as the lens on the firm's leverage choice. Both chapters take the supply of credit as something that arrives, modeled or otherwise, at the firm's door. The bank is mostly a black box. The present chapter opens that box. We treat banks as economic actors with their own balance sheets, their own optimization problems, their own information advantages, and their own regulators. Reduced-form evidence on credit supply, structural estimates of bank productivity, and the regulatory environment that disciplines bank behavior all sit on the bank side of the firm-bank relationship. Without them, the firm-side picture is incomplete.

Four features distinguish banks from the nonfinancial firms that the rest of the book studies. The first is the balance sheet. A nonfinancial firm typically funds itself with a mix of equity and long-term debt against a stock of physical and intangible capital. A bank

funds itself primarily with short-term deposits and wholesale borrowing against a stock of loans and securities. Equity is often near or below ten percent of assets. Leverage of ten to one is the norm, not a sign of distress. The thinness of bank equity is consequential because it determines how vulnerable the bank is to losses on the asset side, and that in turn shapes how the bank chooses to lend.

The second is that banks produce multiple outputs, only some of which are priced. A manufacturer produces physical units that sell at observable prices. A bank simultaneously produces payment services, safe stores of value, screening of borrowers, monitoring of ongoing loans, liquidity transformation across maturities, and standby credit lines that exist precisely so that they need not be drawn. Many of these outputs are bundled with one another, and the prices for them are partly implicit, recovered from cross-sectional variation rather than read off invoices. Treating “bank output” as “loans on the balance sheet” is convenient but too narrow. Much of bank value comes from services bundled with loans and deposits rather than from the book value of loans alone, and the relative importance of these services has changed measurably over the last forty years.

The third is regulation. A nonfinancial firm faces general corporate, tax, antitrust, and labor law. A bank faces all of that plus capital requirements, liquidity requirements, deposit insurance and the assessments that go with it, ongoing supervision by one or more federal and state agencies, mandatory disclosure on a quarterly cadence, and access to the central bank’s lender-of-last-resort facilities. Regulation is not external to what a bank is. The legal and supervisory environment determines which institutions can hold demand deposits, who can issue federally insured money-like liabilities, and what counts as adequate capital against a given asset. Studying banks without taking the regulatory environment seriously is like studying U.S. pharmaceutical firms without the FDA.

The fourth is data. Compustat is the workhorse for firm-side corporate finance and is, by the standards of available data, comprehensive. The bank side has its own data infrastructure. Call Reports for commercial banks, Y-9C for bank holding companies, DealScan for syndicated loans, supervisory loan-level data like Y-14, CRA and HMDA disclosures on small-business and mortgage lending, and Dealogic for new issues. These sources are richer than Compustat in some dimensions (loan-level coverage, internal risk ratings, monthly frequency) and thinner in others (private banks are unlisted in some sources, the consolidated picture is harder to assemble across reporting forms). The selection problems are different too. A Compustat firm is publicly traded and over a threshold size. A Call Report bank is any insured depository, including small community banks. The instruments that work in firm-level studies do not always work at the bank level, and vice versa.

The remainder of this chapter is organized by the logic of these four features. Section 12.2 develops what banks produce, because the answer is more than “loans” and the difference matters. Section 12.3 treats the canonical reduced-form identification of bank-driven variation in lending. Section 12.4 moves from credit supply to its real effects, asking what additional structure is needed to translate a lending response into an employment or investment response. Section 12.5 treats banks as productive firms with their own primitives, using the Egan-Lewellen-Sunderam framework as the worked example. Section 12.6 treats capital regulation, where the structural and causal traditions have to talk to each other. Section 12.7 synthesizes which empirical approach answers which question. Section 12.8 flags what we do not cover. Exercises follow.

12.2 What Do Banks Produce?

A bank’s value depends on what it is good at producing. Before we can ask whether a particular bank is productive, we have to be clear about what a productive bank does. The honest answer is that banks produce several distinct outputs simultaneously, that the outputs are partly bundled together by the legal structure of a depository institution, and that some of these outputs are priced explicitly while others are priced implicitly through the spread between what the bank charges and what its inputs cost.

Start with deposits. A demand deposit account is a payment service (the depositor can transfer funds at par), a safe store of value (the deposit is insured up to the statutory limit, with uninsured depositors sometimes protected ex post in systemic episodes), and a source of cheap funding for the bank. The deposit franchise idea is that a bank with branches, customer relationships, and brand recognition can fund itself at deposit rates well below market wholesale rates, especially when interest rates are high. Drechsler, Savov, and Schnabl (2017) document that deposit rates respond sluggishly to changes in the federal funds rate, that the resulting spread is the largest single source of bank profits, and that the geography of branch networks and local deposit market concentration explains the cross-sectional variation in this spread. Drechsler, Savov, and Schnabl (2021) extend this argument to show that the deposit franchise itself behaves like a long-duration interest-bearing asset on the liability side, and that banks structure the asset side of their balance sheet to hedge it. The implication is that the value of a bank depends not just on its lending but on the franchise of its deposit base. Two banks with identical loan books and identical asset-side risk can have very different valuations because one has a stickier and cheaper deposit base than the other.

The asset side has its own bundle of outputs. The most visible is the loan itself,

the dollar amount of credit extended on contractual terms. Less visible but at least as important is what the bank does to get the loan written: screening the borrower before origination, monitoring the borrower while the loan is outstanding, renegotiating when conditions change, and exercising rights under covenants or collateral provisions when needed. The relationship-lending literature, beginning with [Petersen and Rajan \(1994\)](#) and developed by [Berger and Udell \(1995\)](#) and [Bharath, Dahiya, Saunders, and Srinivasan \(2011\)](#), documents that the value of a long-standing borrower-bank relationship to the borrower is real and measurable. Borrowers with established banking relationships get cheaper credit, smoother access to financing during downturns, and easier renegotiation when distressed. Relationship lending is not unambiguously good for borrowers: the same private information that allows a bank to lend through distress can also give the incumbent lender bargaining power over a borrower with limited outside options. The complement of this finding is that banks differ in their ability to produce these relationships. Screening technology, the quality of loan officers, the geographic and industry concentration of a bank's portfolio, and the bank's organizational structure all contribute to its asset-side productivity. None of this shows up directly as a price.

Banks also produce liquidity in two distinct senses. The first is maturity transformation: deposits are short, loans are long, and the bank intermediates between the two. The second is liquidity insurance through credit lines. A revolving credit facility is a contract that gives the firm the option to draw funds on prearranged terms. [Sufi \(2009\)](#) shows that lines of credit are a major source of firm liquidity, that they are subject to material covenants, and that the implicit price the firm pays for the option is reflected in commitment fees and pricing premia. From the bank's side, writing a credit line is itself an output. The bank holds capital and contingent funding against the line, and the bank's ability to do so cheaply depends on its own capital structure, its access to wholesale funding, and its regulatory treatment. Lines of credit are also the channel through which bank-side shocks propagate to nondrawn firms when conditions change.

The bundle of outputs raises a measurement problem that does not arise for, say, a manufacturer. Measuring bank productivity is harder than measuring TFP in a manufacturing plant because the outputs are partly unpriced, partly bundled, and partly produced jointly with risk. The standard Olley-Pakes machinery for production-function estimation requires a clear distinction between inputs, outputs, and prices. For a bank, deposits are simultaneously an input (a funding source the bank uses to make loans) and an output (a service the bank sells to depositors). A loan is simultaneously an output of the bank's screening technology and a risk position on the asset side. Treating any one of these as the unique "bank output" misses most of what is going on. Sections [12.3](#) and [12.5](#)

take two complementary approaches to this measurement problem. The reduced-form approach asks what happens when a particular component of bank capacity is exogenously perturbed. The structural approach writes down a model that pins down the productivity of each component separately.

The deposit franchise framing also implies a feature of the bank's liability side that becomes salient in stress. Deposits are not homogeneous. Some are sticky retail franchise deposits, insured up to the statutory limit and behaviorally insensitive to small rate changes. Others are large, uninsured balances held by sophisticated counterparties that move quickly when bank conditions deteriorate. The same accounting liability can have very different behavior across these categories. The 2023 regional-bank episode (Silicon Valley Bank and Signature Bank) illustrated this distinction sharply: rapid runs by uninsured depositors precipitated bank failures despite the affected banks holding portfolios that on paper looked solvent at historical cost. The episode points to a measurement problem the chapter does not pursue further but that students should be aware of: any treatment of deposits as a single homogeneous funding source is incomplete, and the same point applies to any empirical work that uses Call Report deposit totals without distinguishing insured, uninsured, and wholesale components.

Key Insight 1

A bank's value depends on what it is good at producing. Reduced-form designs that exogenously shock bank lending can tell us how borrowers respond to a contraction in credit supply. They cannot, by themselves, tell us *what* a productive bank is, because the bank effect they identify is a composite of deposit franchise value, screening technology, capital position, and regulatory treatment. Recovering the components requires structure.

12.3 Identifying Credit Supply

The central reduced-form problem in empirical banking is to identify variation in lending that is driven by bank-side conditions rather than by changes in borrower demand. The two are observationally equivalent at the level of an aggregate loan series. If lending to a particular industry falls in a recession, the change could reflect that banks pulled back, that firms in the industry stopped asking for credit, or both at once. Distinguishing them is the question that organized the bank-lending-channel literature for two decades and that finally found a clean answer in [Khwaja and Mian \(2008\)](#).

12.3.1 The Khwaja-Mian within-firm estimator

The Khwaja-Mian design exploits the fact that many borrowers maintain relationships with more than one bank. When firm f has loans outstanding with banks b and b' at time t , the firm's borrowing demand at t is a single shared object that affects what the firm asks of both banks. If bank b is exposed to a shock that bank b' is not, the differential change in lending from b to f relative to b' to f identifies the bank-side variation, with the firm's demand differenced out.

Formally, let Δy_{fbt} denote the change in (log) lending from bank b to firm f at time t . The estimating equation is

$$\Delta y_{fbt} = \alpha_{ft} + \beta \cdot \text{BankShock}_{bt} + \varepsilon_{fbt}, \quad (12.1)$$

where α_{ft} is a firm-time fixed effect that absorbs anything that varies at the firm-time level (the firm's demand for credit, its investment opportunities, its sectoral conditions, its stage in the business cycle) and BankShock_{bt} is the bank-side shock of interest. The identification rests on a within-firm comparison: among the banks that lend to firm f at time t , those more exposed to the shock lend less. The coefficient β measures the bank-supply elasticity given the firm's demand.

[Khwaja and Mian \(2008\)](#) apply this design to the universe of corporate loans in Pakistan during the 1998 nuclear-test liquidity shock, which differentially affected banks based on their pre-shock exposure to dollar deposits. They find substantial pass-through: for the same firm, a loan from a bank with a one percent larger decline in liquidity falls by an additional 0.6 percent. Large firms partly or fully offset the lost credit through other relationships; small firms are much less able to do so.

The identification logic deserves to be slow. The firm-time fixed effect absorbs every shock that operates at the firm-time level, including any shock that is correlated across the firm's bank relationships. What it does *not* absorb is a shock that is specific to a particular bank in the firm's portfolio. The estimator works because banks differ in their exposure to the aggregate shock and because firms with multiple bank relationships are the subset of the population in which the within-firm comparison is even defined. Both features have implications. We return to them.

12.3.2 What the bank effect is, and is not

A second feature of equation (12.1) is easy to miss and important to state plainly. The coefficient β identifies bank-side credit-supply variation conditional on borrower demand.

It does not identify the bank's lending technology, its productivity, or its underlying primitives. The bank effect is a composite. A bank that lends less than its peers to the same borrowers may be doing so because it has less capital, because it has less liquidity, because it has different risk tolerance, because its funding cost has risen, because it has hit a regulatory threshold, because its internal organization has changed, or because it is simply less productive at originating and monitoring loans. Equation (12.1) aggregates all of these into one coefficient.

Common Mistake 2

A common slip in the credit-supply literature is to interpret the within-firm bank coefficient as a measure of the bank's "lending efficiency" or "productivity." It is not. The estimator identifies how much less the bank lends to a given firm at a given time, holding the firm's demand fixed. That is a credit-supply quantity. It is consistent with the bank being more productive but capital-constrained, less productive but well-capitalized, or anything in between. Disentangling the components requires either additional sources of identifying variation (the structural approach of Section 12.5) or a research design that targets a particular component directly (e.g., the capital-regulation designs of Section 12.6).

12.3.3 Variants on the within-firm idea

The Khwaja-Mian design has been adapted to many settings. Each adaptation illustrates a different identification problem, and reading them in sequence is the cleanest way to develop intuition for what the within-firm idea does and where it stops.

[Chodorow-Reich \(2014\)](#) extends the within-firm logic to the United States during the 2008–09 financial crisis. Loan-level data from DealScan record the syndicates that lend to a given borrower; the design uses pre-crisis lender health as the bank-side variation and asks how firms differentially exposed to weak lenders fared in the post-crisis period. The outcome is firm employment rather than just lending, which moves the design from a credit-supply estimator into a real-effects estimator. We return to the real-effects extension in Section 12.4. The estimate is large: abstracting from general-equilibrium effects, between one-third and one-half of the employment decline at small and medium firms in the sample in the year following Lehman is attributable to bank lending channels. The same paper appears in Chapter 9 as evidence on the constraint share among compliers; here it is evidence on the bank side of the transmission.

[Schnabl \(2012\)](#) adapts the design to cross-border transmission. After the 1998 Russian default disrupted international interbank markets, Peruvian banks that funded themselves

in dollars from international counterparties faced a sudden contraction in their wholesale funding. Schnabl uses the differential exposure of Peruvian banks to international funding as the bank-side variation. The design illustrates that the within-firm logic adapts to settings where loan-level matched-pair data are available but the source of the bank shock is not a domestic balance-sheet shock but a foreign one. The international setting also raises a substantive point: when a bank's marginal source of funding is wholesale and global, an interbank-market shock anywhere in the world transmits to lending here.

[Chodorow-Reich and Falato \(2022\)](#) push the within-firm idea in a different direction. The bank-side shock in their setting is not capital or liquidity but the contractual relationship: when a bank in distress holds a loan that has triggered a covenant violation, the bank uses the violation to extract concessions or to cut credit far more aggressively than the borrower's fundamentals would predict. The bank-side variation in willingness to enforce covenants is the covenant enforcement channel they identify. This is a credit-supply effect that operates through contract terms rather than through aggregate lending capacity, and a within-firm design that uses only loan amounts would miss it. The lesson is methodological: the bank-side variation that matters need not be a quantity. Sometimes it is a price, sometimes a covenant, sometimes a renegotiation outcome. The Khwaja-Mian fixed-effect logic generalizes to any of these as the dependent variable.

[Greenstone et al. \(2020\)](#) confront a more practical limit on the within-firm idea. For small businesses, most borrowers have a single bank relationship. The within-firm comparison is undefined when there is only one bank in the firm's portfolio. Greenstone, Mas, and Nguyen build a shift-share predictor for county-level small-business lending using the pre-period geographic distribution of national banks and aggregate bank-level lending growth. The county-level predicted lending is then used as an instrument for actual lending. The design substitutes a shift-share identification strategy for the within-firm strategy. The substantive payoff of the paper is methodologically instructive: their predicted lending shocks affect loan originations, but the estimated effects on small-firm and aggregate employment are small and often statistically indistinguishable from zero. Identifying a credit-supply contraction is not the same as finding a large real effect.

12.3.4 Threats and assumptions

The within-firm design is among the cleanest identification strategies in empirical finance, but cleanliness is not the same as universality. Four threats merit explicit discussion.

The first is selection. Multi-bank borrowers are the population in which the within-firm comparison can be implemented. They are not a representative sample of the universe

of borrowers. They tend to be larger, more transparent, and more sophisticated than the median firm. The estimate β is informative about credit supply for the multi-bank subset. Extrapolating to single-bank borrowers, who are typically smaller and more bank-dependent, requires either an auxiliary design or a structural argument about why the elasticity should be similar across populations. The shift-share approach of [Greenstone et al. \(2020\)](#) is one way to address this. Combining a within-firm estimate from a multi-bank sample with an aggregate estimate from a single-bank sample, and asking how much of the population the two designs together cover, is another.

The second is bank-specific borrower demand. The firm-time fixed effect absorbs only those demand shocks that affect a firm's borrowing from all of its banks equally. If a firm's most important bank is also the one most exposed to a sectoral shock that the firm is exposed to, the bank shock and a residual demand shock are not separable. This is rarely a first-order concern in samples with many banks per firm and well-defined bank shocks, but it becomes a concern in samples with two banks per firm and bank shocks that have an obvious sectoral correlate. Diagnostics that show β is stable across subsamples in which the bank shock is unlikely to be sectorally correlated are reassuring.

The third is endogenous matching. Firms choose which banks to borrow from, and banks choose which firms to lend to. The composition of bank relationships at t reflects optimization on both sides. If banks more exposed to the shock have systematically different borrower portfolios, the within-firm estimate is local to the borrowers who do business with them. This is not a fatal threat: as long as the firm-time fixed effect absorbs the firm's demand, the estimate is interpretable as the supply response to the shock for the matched population. It does qualify the external validity of the estimate.

The fourth is the intensive versus extensive margin. Equation (12.1) is naturally implemented on the intensive margin, using continuing borrower-bank pairs. New relationships and severed relationships are part of credit supply too. A bank that retreats from a market by refusing to renew old loans or by not originating new ones can have effects on borrowers that the within-firm estimator, applied to surviving pairs, misses. Many of the papers in this literature complement the intensive-margin estimate with separate analyses of relationship survival and origination, and the picture across margins is usually consistent.

Connection to Framework 3

The within-firm credit-supply design is a fixed-effects estimator in the sense of Chapter 3, applied to a setting where the unit of observation is the loan and the relevant fixed effect is the firm-time interaction. The exclusion restriction is sharp:

bank-side variation in lending, conditional on the firm-time demand, is identified by differences across banks in exposure to the bank shock. The design is also an instance of the credibility-revolution logic of Chapter 5: rather than write down a structural model of firm credit demand, the estimator uses a research design to absorb demand variation and isolate supply variation. What it identifies is a treatment effect on a multi-bank population, not a structural primitive.

12.4 From Credit Supply to Real Effects

Identifying that a bank shock reduces lending to a firm is one step. Identifying that the lending reduction translates into a real outcome at the firm (lower investment, lower employment, lower sales, higher default risk) is another. The two steps are conceptually distinct because firms have other margins. A firm that loses access to one bank can borrow from another, draw on a credit line, run down cash, delay an investment, or substitute into trade credit. The relevant question for the real economy is not how much bank lending fell but how much firm activity fell, net of these substitutions.

12.4.1 The canonical employment estimate

[Chodorow-Reich \(2014\)](#) is the canonical real-effects estimate. The design uses bank-firm relationships from DealScan to define each firm's pre-crisis lender exposure, constructs a firm-level measure of lender health based on pre-crisis bank balance-sheet variables (capital, deposit funding, exposure to mortgage-backed securities, exposure to Lehman), and then estimates the differential change in firm employment between firms exposed to weaker and stronger lenders. The identifying variation is bank-side, but the outcome is firm-side. Abstracting from general-equilibrium effects, the estimate is that the withdrawal of credit explains between one-third and one-half of the employment decline at small and medium firms in the sample in the year following Lehman. At very large firms, the effect is statistically indistinguishable from zero, consistent with the fact that very large firms can substitute into bond markets and into nonbank funding sources when their banks pull back.

The size of the bank-channel estimate at small and medium firms is large by the standards of the empirical finance literature, and it has held up reasonably well to replication and to extensions. [Greenstone et al. \(2020\)](#) apply a different identification strategy to small-business lending and find more modest aggregate effects, partly because their sample includes normal times as well as the crisis. The two estimates are not in

tension. Chodorow-Reich identifies a treatment effect during a severe credit-supply shock, on firms whose lenders were exposed to that shock. Greenstone, Mas, and Nguyen identify a longer-run elasticity of small-business lending and employment to local credit supply, averaged across normal times and crisis years. The magnitudes one should take from each are tied to the population and the period each design covers.

12.4.2 Substitution across lenders

The size of the firm-level real effect depends on the availability of substitute financing. [Crouzet \(2018\)](#) writes down a model in which firms hold both bank loans and corporate bonds, with bonds the substitute used when bank credit becomes expensive. The model rationalizes the empirical observation that during the 2008–09 crisis, large firms that could substitute into bond markets had smaller real responses to bank-channel shocks than smaller bank-dependent firms. The lesson generalizes: any reduced-form estimate of the bank-channel real effect is implicitly an estimate net of the substitution margin available to firms in the sample. Changing the sample changes the substitution margin and therefore changes the estimate, even if the bank shock and the structural marginal response to a financing contraction are the same.

The general-equilibrium concern is parallel. A reduced-form bank-channel estimate compares firms exposed to weaker banks against firms exposed to stronger banks. If treated firms lay off workers, those workers are not lost to the economy; they may be hired by untreated firms. The cross-sectional comparison overstates the aggregate effect by the size of the reallocation. The same logic applies to investment: capital that does not get installed at a treated firm may get installed at an untreated competitor. Spillovers of this kind are why aggregate effects estimated from cross-sectional bank-shock designs are upper bounds rather than point estimates of the change in aggregate output. The Adao-Arkolakis-Esposito-style argument that cross-sectional designs can overstate aggregate effects in any setting with reallocation applies straightforwardly here.

12.4.3 Revisiting the running example from the bank side

The leverage-investment running example that has appeared throughout the methods chapters takes a particular form when viewed from the bank side. [Leary \(2009\)](#) is the IV estimator from Chapter 5. Leary uses the 1961 introduction of certificates of deposit and the 1966 collapse of the same market as exogenous shocks to bank loan supply, and shows that bank-dependent firms changed their leverage in response. [Lemmon and Roberts \(2010\)](#) is the difference-in-differences estimator from Chapter 9. Lemmon and Roberts use the

1989–90 collapse of the junk bond market as a treatment and show that net investment by treated firms fell nearly one-for-one with net debt issuance.

Read from the bank side, both designs are statements about credit supply. The Leary 1966 shock identifies a contraction in bank-loan supply that hit bank-dependent firms differentially. The Lemmon-Roberts 1989–90 shock identifies a contraction in nonbank credit supply (the junk bond market) that hit below-investment-grade firms differentially. The two papers used to be read as separate evidence on capital structure and on investment. Read together, they are two different windows on the same underlying object: when credit supply contracts, firms in the population that depends on the contracting source adjust their financing and their real activity, and the magnitudes depend on the population that depended on the source and on the substitution margins available to them.

What would augmenting these reduced-form designs with bank-side productivity estimates add? Two things. First, an ELS-style decomposition (Section 12.5) of the bank-side variation could tell us which component of bank health drove the lending contraction: was the 1966 CD-market shock primarily a deposit-franchise shock, a wholesale-funding shock, or a regulatory shock? The reduced-form estimate aggregates these. Second, the structural estimates discipline the counterfactual. A reduced-form effect of a one-percentage-point decline in bank lending growth is informative about that shock; predicting the effect of a one-percentage-point decline driven by a different component of bank health requires either the auxiliary assumption that all sources of bank-lending variation transmit the same way, or a structural model that allows them to transmit differently.

Key Insight 4

Box 12.1: The same shock viewed from three chapters. A bank-channel credit contraction appears in three places in this book. In Chapter 5, the Leary instrument uses bank shocks as exogenous variation in firm financing capacity to identify the effect on capital structure. In Chapter 9, the Lemmon-Roberts difference-in-differences uses a similar bank-channel shock to identify the investment-financing pass-through for a constrained complier population. In Chapter 12, the same kind of shock is the dependent variable: the question is what aspect of bank balance sheets, technology, or regulation made the bank shock happen, and what we learn about banks from how they respond to it. The shock is the same. The estimand is different in each case.

12.5 Banks as Productive Firms: Structural Banking Models

The reduced-form designs of Sections 12.3 and 12.4 estimate how bank shocks transmit to lending and to firm outcomes. They do not, by themselves, recover what banks are. The bank effect they identify is a composite. To pin down the underlying productivity of a bank, separately on the deposit side and on the asset side, requires a model. Egan, Lewellen, and Sunderam (2022), henceforth ELS, build such a model and estimate it on a panel of U.S. bank holding companies. The ELS framework is the worked example for this section, and we develop it in parallel to the way Chapter 8 develops the Cooper-Haltiwanger framework: stating the setup, the moments, the identification logic, and the interpretation of the estimates.

12.5.1 The setup

A bank in the ELS model produces two outputs. On the liability side, the bank produces deposit services; depositors hold balances that pay an interest rate r^d below the wholesale rate, and the spread is a source of bank cash flows. On the asset side, the bank produces loans and securities; the bank holds a portfolio that earns an interest rate r^a above the cost of funds, and the spread there is the second source of cash flows. The bank has primitives that govern its productivity on each side.

The equations that follow are a stylized version of the ELS framework intended to convey the identification logic transparently. The full ELS model has additional structure and additional moments that this chapter does not attempt to reproduce.

Let D_i denote bank i 's deposit base and A_i its asset side, both as stocks. Deposit cash flows at t are

$$\pi_{it}^d = (r_t^f - r_{it}^d) D_{it} + \alpha_i^d f^d(D_{it}, X_{it}), \quad (12.2)$$

where r_t^f is the wholesale rate, r_{it}^d is the bank-specific deposit rate, α_i^d is the bank's deposit-side productivity primitive, and $f^d(\cdot)$ captures the scaling and characteristics of the deposit franchise. Asset cash flows take a parallel form:

$$\pi_{it}^a = (r_{it}^a - r_t^f) A_{it} + \alpha_i^a f^a(A_{it}, X_{it}), \quad (12.3)$$

where α_i^a is the bank's asset-side productivity. Bank value is the present discounted value of total cash flows under the model's risk and discount adjustments. The two primitives (α_i^d, α_i^a) are the objects of structural interest. They capture the bank's productivity, in the dual sense of cost-efficiency in producing the output and ability to charge an implicit price

for it.

12.5.2 Identification

The first question for any structural model is whether the parameters of interest are separately identified by the model and the data. The ELS setup has the feature that deposit productivity and asset productivity are separately identifiable, but the argument requires care. Three sources of cross-sectional variation do the work.

The first is differential scaling with interest rates. Equation (12.2) implies that deposit cash flows respond to changes in the wholesale rate in a way that depends on the deposit-rate stickiness, which is itself a function of α_i^d . A bank with a high α_i^d extracts a wider deposit spread, and that spread widens further when wholesale rates rise. The cross-sectional response of deposit cash flows to interest-rate changes therefore identifies the cross-sectional distribution of α_i^d . Drechsler et al. (2017) document this response in reduced form; ELS use the same variation as one of their moments.

The second is differential scaling with size. Both $f^d(\cdot)$ and $f^a(\cdot)$ allow productivity to scale with the size of the deposit base and the asset side, respectively, but in different ways. A bank that grows its deposit base by acquiring branches and customer relationships scales its deposit cash flows in a way that depends on α_i^d . A bank that grows its asset side by extending credit to new borrowers scales its asset cash flows in a way that depends on α_i^a . The cross-sectional joint distribution of size and cash flows on each side, taken together, identifies the cross-sectional distribution of the two productivities.

The third is market value. The bank's market value reflects the present discounted value of the cash flows on both sides. Cross-sectional variation in market-to-book ratios, conditional on the bank's observable balance sheet, identifies the implicit value of the franchise on each side, and the model maps that implicit value back to the productivities. This is the moment that turns a cross-section of accounting data into estimates of two latent productivity parameters per bank.

The identification logic is parallel in posture to the Cooper-Haltiwanger argument in Chapter 8. There, the cross-sectional joint distribution of investment dynamics, profit-investment correlations, and spike rates separately identifies convex adjustment costs, disruption costs, and irreversibility. Here, the cross-sectional joint distribution of interest-rate responses, scaling patterns, and market values separately identifies deposit productivity and asset productivity. The structural argument in both cases is that the model puts enough restrictions on the joint distribution of observables that a specific set of latent parameters can be recovered.

The argument is not assumption-free, and the assumptions are worth stating. The model takes the functional forms of f^d and f^a as given (different functional forms would in general give different estimates). It takes the time-series path of wholesale rates as exogenous to individual bank productivity, with cross-sectional differences in how banks respond to wholesale-rate changes providing the identifying variation. It takes the matching of depositors and borrowers to banks as approximately given, in the sense that selection on α does not drive the cross-sectional pattern of bank size. None of these assumptions is innocuous, and ELS treat each at length in the paper. The point for the present chapter is that the identification of the structural primitives requires both moments and model restrictions, and an honest treatment of the estimates requires being clear about both.

12.5.3 What the estimates mean

The output of the ELS estimation is bank-level $\hat{\alpha}_i^d$ and $\hat{\alpha}_i^a$, together with a cross-sectional distribution of bank value that can be decomposed into a deposit-franchise component and an asset-side component. Egan, Lewellen, and Sunderam find that deposit productivity is the dominant source of value for the median bank, accounting for roughly two-thirds of bank value and most of the cross-sectional variation in bank value. Asset-side productivity is economically important and the interaction between deposit-taking and lending matters, but the median-bank result is deposit-heavy. The distinctive lesson is one that no firm-side dataset could deliver: the liability side of a bank's balance sheet is not just funding, it is a major source of franchise value in its own right. Within the cross-section, however, banks differ substantially. Some banks derive almost all of their franchise value from the deposit side; others derive much more from the asset side. The aggregate bank-value statistic conceals these differences.

What do the estimates mean economically? The honest answer is more nuanced than “TFP for banks.” Deposit productivity α_i^d here partly reflects deposit franchise value as in Drechsler et al. (2021): the ability of the bank to fund itself cheaply through sticky retail deposits. It also reflects market power in local deposit markets, the geographic structure of the bank's branch network, brand and reputational effects, and the depth of customer relationships. Egan, Hortacsu, and Matvos (2017) document the importance of deposit market power directly using a deposit-demand framework, and find that local concentration in deposit markets is a first-order determinant of deposit pricing. The ELS estimate of α_i^d is, in part, a measure of how much of this market power and franchise value the bank has. It is not, in the Olley-Pakes sense, a measure of physical productivity.

Asset productivity α_i^a has a parallel interpretation. It reflects screening technology,

monitoring capacity, the bank's information advantages in particular industries or geographies, and the bank's ability to renegotiate distressed borrowers without large losses. Some of this is technology in a literal sense (information systems, credit-scoring models). Some is human capital (loan officers with industry expertise). Some is organizational capital (an internal structure that delegates lending decisions to the right level). The estimate $\hat{\alpha}_i^a$ aggregates these into a single primitive, and the interpretation of the cross-sectional dispersion depends on the assumptions that allow the aggregation.

12.5.4 Placing ELS in lineage

ELS sits in a tradition of structural banking models that has grown substantially in the last decade. The tradition can be traced through three nodes. [Egan et al. \(2017\)](#) estimate a deposit-demand system for U.S. banks, recovering the deposit franchise as a structural object using consumer-choice machinery from industrial organization. [Drechsler et al. \(2017\)](#) and [Drechsler et al. \(2021\)](#) provide the reduced-form evidence on the deposit-channel mechanism and an asset-pricing interpretation of the deposit franchise as a long-duration liability. [Wang, Whited, Wu, and Xiao \(2022\)](#) estimate a structural model with bank market power and use it to study the transmission of monetary policy through banks, finding that market power dampens the pass-through of policy rates to deposit and lending rates. ELS contributes to this tradition by writing down a model that pins down the productivity on each side of the bank's balance sheet separately, and by estimating it on a panel rather than on a single cross-section.

The right way for students to read ELS is as one node in this tradition rather than as a self-contained piece. The deposit-side story it tells is recognizably the Drechsler-Savov-Schnabl story, formalized in a way that lets it be combined with the asset side. The asset-side story it tells is recognizably the screening-and-monitoring story of the relationship-lending literature, formalized in a way that lets it be combined with the deposit side. The combination is what allows the model to be used for counterfactuals that require both sides, such as the effect of a deposit-rate cap on bank value, or the effect of a regulatory rule that changes the cost of holding a particular asset.

When to Use 5

When to use a structural bank model:

- The question is about a primitive: the bank's deposit franchise value, the bank's asset-side productivity, the bank's screening or monitoring technology.
- The question requires a counterfactual that goes beyond observed shocks: what

would bank value look like under a deposit-rate cap, under a different capital regime, under a merger.

- The reduced-form bank-shock literature gives mixed or context-dependent answers, and the cross-context heterogeneity is itself something the model is meant to organize.

When a causal design suffices:

- The question is about the response of lending or firm outcomes to a particular bank shock for a defined population.
- The shock is well-identified and the population is clearly defined.
- Extrapolating to other shocks or other populations is not required.

A useful caveat is that the structural and causal traditions in banking are not substitutes. A reduced-form causal design can identify the effect of a bank shock on lending or firm outcomes; it does not, by itself, identify the bank's latent deposit or lending productivity without additional structure. A structural design can identify the latent productivities; it relies on functional-form and identification assumptions that the reduced-form design avoids. The best work in this area uses both. Causal shocks discipline or validate structural parameters: if the structural model cannot replicate the response of bank lending to a credible reduced-form shock, the structural estimates are suspect. Structural parameters interpret causal shocks: a reduced-form lending response to a deposit-funding shock is more informative when the deposit-channel structural literature has independently pinned down the size of the deposit franchise.

12.6 Capital, Regulation, and Bank Behavior

Bank capital is not just another corporate leverage choice. A nonfinancial firm choosing leverage balances tax benefits, financial distress costs, and agency frictions, all of which are familiar from Chapter 10. A bank choosing leverage faces all of those plus three additional forces. The first is deposit insurance, which subsidizes leverage by guaranteeing the safety of a large class of liabilities and so making them cheap to issue. The second is capital regulation, which constrains leverage by requiring banks to hold equity against risk-weighted assets. The third is liquidity regulation, which shapes the asset composition by requiring banks to hold a buffer of high-quality liquid assets against short-term liabilities. Each of these has been the subject of a substantial empirical literature.

Before turning to the empirical evidence on capital regulation, it is worth distinguishing four kinds of capital constraint that appear in this literature, because different papers

identify different margins and the estimates are not interchangeable. The first is accounting capital, the ratio of book equity to assets, which is the object that traditional leverage ratios constrain. The second is risk-weighted capital, the ratio of equity to risk-weighted assets, which is what Basel-style requirements operate on and which can be relaxed or tightened through changes in risk weights even when accounting capital does not change. The third is market capital, the value of equity at current market prices, which determines the bank's debt-overhang position and the cost of issuing additional equity. The fourth is liquidity, which is not strictly capital but which interacts with capital constraints because banks that face a binding liquidity requirement may behave as if they are also capital-constrained. The Khwaja-Mian liquidity shock identifies a margin different from the EBA exercise, which differs from Spanish dynamic provisioning, which differs from a stress-test exposure. The estimates from each design are informative about the bank-lending response to a particular kind of capital constraint, not about the response to capital regulation in general.

12.6.1 Why bank capital is different

The starting point is [Merton \(1977\)](#). Merton observes that deposit insurance is an option granted by the deposit insurer to the bank: if the bank's asset value falls below its insured liabilities, the insurer pays off depositors and the equity holders lose their stake. The option has positive value, and that value increases with the bank's leverage and with the risk of the bank's asset portfolio. The implication is that, absent regulation, a bank's shareholders have an incentive to lever up and to take on risky assets, because doing so increases the value of the insurance option without correspondingly increasing their downside. The classic risk-shifting problem.

[Kareken and Wallace \(1978\)](#) develop the same point in a partial-equilibrium model and draw the regulatory implication. If the deposit insurer cannot price the insurance ex ante in a way that fully internalizes the bank's risk choices, regulation has to substitute by constraining leverage and asset composition directly. The capital-requirement regime that exists in actual banking systems is the practical embodiment of this argument. Required equity is supposed to give the bank's shareholders enough skin in the game that the risk-shifting incentive is dampened.

The empirical question this opens is whether and how capital requirements actually affect bank behavior. The reduced-form answer is yes, with magnitudes that depend on the specific regime.

12.6.2 Capital requirements and credit supply

[Jiménez, Ongena, Peydró, and Saurina \(2017\)](#) exploit the Spanish dynamic provisioning experiment, in which the Bank of Spain required banks to build up provisions during good times that could be drawn down in bad times. The provisioning rule applied differentially to banks based on their growth and portfolio composition, and the differential exposure provides identifying variation in the effective capital constraint over the cycle. The paper's finding is that dynamic provisioning reduced credit expansion by more exposed banks in good times, but the accumulated buffers supported lending and firm outcomes in bad times. The macroprudential lesson is about smoothing the credit cycle, not just about how a static capital requirement affects steady-state lending.

[Aiyar, Calomiris, and Wieladek \(2014\)](#) use a U.K. policy experiment in which the Financial Services Authority varied capital requirements at the bank level on the basis of firm-specific supervisory judgments. The bank-level variation in requirements is plausibly orthogonal to bank-specific lending opportunities, and the design identifies the effect of a one-percentage-point tightening in the required capital ratio on bank lending. The estimate is economically large: lending growth falls by several percentage points in response to a one-point tightening, and the effect is mostly transmitted through reduced lending to riskier borrowers. The paper also documents leakage to nonregulated lenders, which is a recurring finding in the macroprudential-regulation literature and which qualifies the inference one can draw about aggregate credit supply.

[Gropp, Mosk, Ongena, and Wix \(2019\)](#) exploit the 2011 EBA capital exercise, which required a subset of European banks to meet a higher capital ratio by a specific date. The exercise generated a sharp differential treatment between banks just above and just below the threshold for inclusion, and the design uses this differential as a quasi-natural experiment. Banks subject to the higher requirement raised their capital ratios primarily by reducing risk-weighted assets rather than by raising new equity. The reduction in risk-weighted assets translated into reduced lending, particularly to corporate borrowers in the same country. The estimate is the cleanest contemporary evidence on how a bank-capital requirement transmits to lending in the European setting.

The three estimates differ in magnitude, but the pattern across them is consistent. A binding tightening in bank capital requirements reduces bank lending, the effect is larger for riskier borrowers and weaker banks, and some of the credit shifts to unregulated channels rather than disappearing from the system entirely. The implication for both research and policy is that capital regulation has real effects on credit supply, with material distributional consequences across borrower types.

12.6.3 Stress tests as a regulatory technology and a setting for empirical work

Stress tests are the other major innovation in capital regulation in the last two decades. The Federal Reserve's Comprehensive Capital Analysis and Review subjects the largest U.S. banks to a hypothetical adverse scenario and requires that they maintain minimum capital ratios under the scenario. Stress tests also create useful regulatory variation for empirical work, though the variation is not randomly assigned and requires careful controls for bank size, portfolio composition, risk, and supervisory selection. [Acharya, Berger, and Roman \(2018\)](#) use the cross-bank variation in stress-test outcomes, together with such controls, to identify the effect on bank lending. They find that banks that performed worse on the stress test subsequently reduced lending, particularly in the asset classes that drove the underperformance. The pattern is qualitatively similar to the earlier capital-requirement literature. The interest of stress tests for the present chapter is that they sit naturally between the structural and the reduced-form traditions: the test itself is a model-based forward-looking capital constraint, but the cross-bank variation it generates is, with appropriate controls, useful for reduced-form work.

12.6.4 Risk-taking and the broader regulatory environment

Risk-shifting under deposit insurance is a major topic in its own right, and we treat it only briefly here. The Merton-Kareken-Wallace logic is that capital regulation is necessary because deposit insurance creates risk-taking incentives. The corollary is that any regulatory regime that fails to price risk correctly (capital requirements that do not adjust to portfolio risk, deposit insurance premia that do not depend on bank risk, supervisory regimes that under-detect risk) will produce excess risk-taking on the part of insured banks. The empirical literature on this is large, and the most useful entry point for students is the post-crisis literature on too-big-to-fail and on stress-test design. Runs, fire sales, and resolution policy are mentioned in [Section 12.8](#); a serious treatment of any of them is beyond the scope of the present chapter.

Connection to Framework 6

Capital regulation is a setting where the structural and causal traditions have to talk to each other. Causal designs ([Jimenez, Aiyar, Gropp, Acharya-Berger-Roman](#)) identify the effect of changes in capital requirements on bank lending. Structural estimates of bank productivity ([ELS, Wang-Whited-Wu-Xiao](#)) tell us whether the

affected banks are high- or low-productivity intermediaries. Knowing whether the marginal credit cuts then fall on more or less productive borrowers would require borrower-level productivity data or an allocation model linking bank productivity to borrower selection, neither of which is supplied by the structural bank-productivity estimates alone. Neither tradition answers the policy question on its own.

12.7 What Each Empirical Approach Can and Cannot Answer

The strands of evidence developed in the preceding sections answer different questions. Reduced-form designs estimate what happens when banks are shocked. Structural models try to estimate what banks are. A student reading this literature for the first time can easily mistake the two, and the cost of the mistake is interpreting a treatment effect as a structural primitive or interpreting a structural estimate as a treatment effect. The aim of this section is to give the reader a compact map.

When to Use 7

Banking questions and the right empirical tool.

Question	Best empirical tool
Did weak banks cut lending?	Within-firm or matched bank-firm design
Did bank shocks reduce firm employment or investment?	Bank-shock design linked to firm outcomes
Are some banks more productive intermediaries?	Structural model (ELS-type)
Does capital regulation reduce credit supply?	Both reduced-form and structural evidence
Are deposits a source of market power?	Structural demand or deposit-franchise model

The connection back to Chapter 1's four types of questions is direct. The first two rows of the table are Type 1 questions: the causal effect of a specific intervention on a defined population. The third and fifth rows are Type 2 questions: estimating a structural primitive that governs bank behavior across regimes. The fourth row is hybrid. Predicting the

effect of a yet-unobserved regulatory regime, such as a different capital ratio, is a Type 3 counterfactual, and answering it requires structural primitives that have been disciplined by the reduced-form evidence from past changes.

Banking is therefore the chapter in this book in which Type 1 and Type 2 questions sit most plainly side by side. Most other application chapters lean toward one or the other. Investment in Chapter 8 treats Cooper-Haltiwanger as a Type 2 estimate and a reduced-form policy variation as a Type 1 estimate, and they sit next to each other as alternative ways of attacking the same investment-dynamics question. Constraints in Chapter 9 treats the constraint share among compliers as a Type 1 object and the structural marginal cost of external finance as a Type 2 object. Banking has both objects in nearly every paper, because a bank shock and a bank primitive are equally available targets for any given paper to pursue. The skill the reader needs is the discipline to know which of the two a particular paper is going after, and to interpret the evidence accordingly.

12.8 Boundaries and Extensions

This chapter does not cover several topics that are major fields of empirical banking research. Brevity here is honest about scope rather than dismissive of importance, and the pointers below are for students who want to read further.

Shadow banking and nonbank lenders have grown to a substantial share of credit supply in the U.S. and Europe. [Buchak, Matvos, Piskorski, and Seru \(2018\)](#) document the rise of nonbank mortgage lenders and the role of regulatory arbitrage and fintech technology in their growth. The general lesson is that bank-side credit-supply contractions can be partially offset by shifts into the nonbank sector, and that the regulatory boundary of what counts as a bank is itself an object of inquiry.

Money market funds and runs are a separate and important literature. The 2008 and 2020 episodes both featured runs on money market funds with material consequences for short-term funding markets, and the regulatory response in each case reshaped the industry. The classical references on run dynamics (Diamond-Dybvig and successors) are relevant background.

Fintech credit, beyond the shadow-banking question, is a fast-growing area in which the underlying technology of screening and monitoring is changing. The implications for the asset-side productivity of incumbent banks and for the structure of credit markets are still being worked out. [Buchak et al. \(2018\)](#) is again a useful entry point.

Central bank policy transmission through banks is the macroeconomic complement to the present chapter's focus on cross-sectional credit supply. The deposit-channel literature

(Drechsler et al., 2017) and the structural literature on bank market power (Wang et al., 2022) both treat this question explicitly. The interaction between monetary policy, deposit pricing, and bank lending is a setting where the structural and reduced-form traditions are working productively together.

Fire sales and resolution policy are the last bracket of topics this chapter does not cover. Granja, Matvos, and Seru (2017) use the FDIC's process for selling failed banks to identify the determinants of resolution efficiency, and the broader literature on fire sales (going back to Shleifer and Vishny) provides the conceptual frame. Resolution policy interacts with capital regulation in ways that are central to current policy debates.

Securitization is the other major omission. The mechanics by which loans originated by banks are pooled, tranced, and sold to nonbank investors reshape the credit market in ways that the on-balance-sheet bank lending literature does not capture. The post-2008 mortgage market in particular is hard to understand without it, and the literature on private-label and agency securitization is substantial. References to a survey are the natural pointer for students.

12.9 Exercises

Exercise 1 (Computational). Simulate a within-firm credit-supply experiment to illustrate why firm fixed effects matter. Use random seed 8823. Generate $N_F = 1,000$ firms and $N_B = 50$ banks. Each firm f draws its number of bank relationships K_f as follows: 40 percent of firms have $K_f = 1$, 40 percent have $K_f = 2$, and 20 percent have K_f drawn uniformly from $\{3, 4, 5\}$. Generate firm demand $d_f \sim N(0, 1)$ and bank supply $s_b \sim N(0, 1)$, and let high-demand firms sort toward high-supply banks: assign each firm's bank relationships from the upper part of the bank-supply distribution when its demand is in the upper part of the firm-demand distribution, so that in the observed loan sample $\text{Cov}(d_f, s_b) > 0$. The log change in lending from bank b to firm f between $t = 0$ and $t = 1$ is $\Delta y_{fb} = d_f + s_b + v_{fb}$ with $v \sim N(0, 0.5^2)$; simulate one change observation per firm-bank relationship, with d_f and s_b as defined above. (a) Estimate β in equation (12.1) using `pyfixest` with firm fixed effects. Report the coefficient on s_b and its cluster-robust standard error. (b) Repeat the estimation without firm fixed effects (OLS of Δy on s). Compare the two coefficients. Document the OLS bias and explain why it arises from the positive sorting you built into the data-generating process. (c) Restrict the sample to single-bank firms. What happens when you try to estimate (a) on this subsample? Why? What does this tell you about the population to which the within-firm estimator applies?

Exercise 2 (Reading). Pick a recent paper that uses a bank-shock design (the instructor will provide a current list). Identify the source of bank-side variation, the population of borrowers in the within-firm or matched sample, the outcome variable, and the threats to identification that the paper addresses. Then ask the harder question: does the paper attempt to identify a structural primitive in addition to the treatment effect? If yes, how? If no, what additional moments or model assumptions would be needed to do so?

Exercise 3 (Conceptual). The following research questions could each be the topic of an empirical banking paper. For each, classify the question as Type 1, Type 2, Type 3, or Type 4 in the sense of Chapter 1, and indicate which empirical tool from this chapter best fits the question.

1. Did the 2011 EBA capital exercise reduce lending by treated banks?
2. Is the value of the deposit franchise larger at retail-oriented banks than at wholesale-funded banks?
3. Would aggregate credit supply fall by more than five percent if the U.S. minimum capital ratio were raised by two percentage points?
4. Does the bank-channel effect of a credit-supply shock on firm employment operate through reduced investment or through reduced working capital?

Exercise 4 (Structural, ★). Consider the ELS setup in equations (12.2) and (12.3). Suppose you replaced the additive productivity term $\alpha_i^d f^d(D_{it}, X_{it})$ with a multiplicative specification: $\pi_{it}^d = \alpha_i^d (r_t^f - r_{it}^d) D_{it}$, with deposit-rate setting governed by an analogous multiplicative term. (a) Re-derive the cross-sectional moments that identify α_i^d under this specification. Are the moments the same as under the additive specification? (b) Suppose the data are actually generated by the multiplicative specification but you estimate the additive one. What kind of bias would you expect in the estimated cross-sectional dispersion of $\hat{\alpha}_i^d$? (c) Discuss what evidence you would look for in the data to distinguish the two specifications.

Exercise 5 (Policy). Suppose a regulator is considering raising the U.S. minimum risk-based capital ratio by one percentage point. Two strands of empirical evidence are available: reduced-form estimates from the Spanish dynamic provisioning experiment (Jiménez et al., 2017) and from the EBA exercise (Gropp et al., 2019), and structural estimates of bank productivity in the U.S. (Egan et al., 2022). (a) Which strand is most directly informative about the lending response in the U.S.? Why? (b) ELS estimates productivity at the bank level, not at the borrower level. What additional data and modeling assumptions would

be required to determine whether a capital tightening would lead banks to cut credit to more productive or less productive *borrowers*? (c) The regulator wants a quantitative point estimate of the change in aggregate credit supply. Which estimates would you use, and what auxiliary assumptions are required to translate the available evidence into a U.S. counterfactual?

Exercise 6 (Replication). Using a sample of Y-9C data for large bank holding companies for the period 2000–2020 supplied in the replication template, construct rough proxies for deposit-side and asset-side cash flows using the Y-9C items specified in the template. Estimate a panel regression of each cash-flow proxy on the contemporaneous federal funds rate and a set of bank fixed effects. Document the cross-sectional dispersion in the slope coefficients. Discuss in light of the stylized ELS framework of Section 12.5: which slope is more informative about α_i^d , and which about α_i^a ? Why are these only proxies for the ELS objects, and what additional moments would be needed to recover the underlying parameters separately?

Chapter 13

Innovation

13.1 Why Innovation Is Different

A reader who has worked through Chapter 8 on physical investment may wonder why innovation needs its own chapter. Firms make investment decisions; some of those decisions involve building factories and some involve hiring scientists; the same Q-theory and adjustment-cost machinery ought to apply to both. Why complicate matters?

The honest answer is that innovation has three features that make it genuinely harder to study than tangible capital, and ignoring those features has produced a substantial empirical literature with its own methods and its own pathologies. We need to understand the features before we can read the literature.

Intangibility and the measurement problem. A new lathe sits on a balance sheet at its purchase price. A new production process sits nowhere in particular. R&D expenditure is observable, but R&D expenditure is an input, not an output. The output of innovation is knowledge, ideas, processes, and patents. All of these are intangible and must be measured indirectly. Every empirical study of innovation begins with a measurement choice, and that choice is itself an identification choice. The researcher who uses patent counts is implicitly assuming something about the propensity to patent across firms and industries. The researcher who uses R&D expenditure is implicitly assuming something about the productivity of that expenditure. The researcher who uses market-implied patent values, in the style of [Kogan et al. \(2017\)](#), is buying into an event-study identifying assumption to construct the measure. None of these choices is innocent.

The accounting treatment compounds the problem. U.S. GAAP requires that R&D expenditure be expensed in the year it is incurred. International Financial Reporting Standards capitalize development costs once technical feasibility is established. The same

firm conducting the same activities will report different Tobin's Q , different profitability, and different leverage ratios depending on the accounting regime. Any empirical work using cross-country variation or using Q as a control must confront this. Within the U.S., the immediate expensing of R&D means that firms with high innovation intensity will look mechanically less profitable than they actually are, and their book values will understate intangible assets. The Peters–Taylor intangible-capital adjustment is an attempt to fix this; whether it succeeds is debatable.

Long horizons, skewed payoffs, high failure rates. A new factory produces output the year after it is built. A new drug takes a decade to bring to market, and most candidate drugs fail. The empirical signature of innovation in the data is heavily right-skewed: most patents are worth essentially nothing, a handful are worth billions. The same is true of R&D projects. This skewness has methodological consequences. Mean R&D returns are a poor summary statistic because the mean is dominated by tail outcomes; medians are more informative but less standard. Linear regressions of outcomes on R&D pick up something, but not necessarily the policy parameter the researcher wants. Quantile methods, the methods of [Kortum \(1997\)](#) on patent value distributions, and the structural approaches we will discuss in [Section 13.5](#) all reflect this fact about the underlying data.

The long horizons also mean that anticipation matters. A patent granted in 2010 may reflect R&D begun in 2003. Event studies around patent grants must take a stand on whether the grant date is the relevant information event or whether the market knew earlier. KPSS argue that within a narrow window the grant decision is unpredictable enough to identify a market-implied value. Whether that assumption holds for any particular sample is a question the researcher cannot duck.

Spillovers and the private–social wedge. The lathe in the factory benefits the firm that owns it. The new production process, once disclosed in a patent, benefits the firm that invented it but also benefits competitors who can build around it, upstream and downstream firms who can adapt it, and the broader economy through cumulative innovation. The private return to R&D is therefore generally lower than the social return, sometimes by a large margin. [Bloom, Schankerman, and Van Reenen \(2013\)](#) estimate the median social return to R&D at roughly four times the private return.

This wedge has two consequences for empirical work. First, it means that the policy questions one wants to answer differ from what causal designs can identify. A regression-discontinuity estimate of the effect of an R&D grant on patenting tells you what the marginal grant did for the recipient firm; it tells you nothing about the spillover to other

firms. Quantifying spillovers requires either auxiliary structure or special data. Second, it means that the case for public intervention in innovation rests on an empirical claim that the social return exceeds the private return. That claim is itself difficult to estimate cleanly. The debates about R&D tax credits, patent box reforms, and federal research spending all turn on numbers that have wide confidence intervals.

Why financing is central. The three features above interact in a way that makes financing unusually important for innovation. Intangibility means that R&D assets cannot serve as collateral for debt, so debt financing is largely unavailable. Long horizons and skewed payoffs mean that information asymmetries between insiders and outside investors are severe, in the Myers–Majluf sense. The result, documented across many studies, is that R&D is financed predominantly by internal funds and equity issuance. Debt finances factories; retained earnings and equity finance laboratories. This single fact organizes much of the empirical literature we will encounter, and it is the reason Section 13.4 is the longest in this chapter. It is also the natural connection back to Chapter 9, where financial constraints took center stage.

Three questions for the chapter. The chapter is organized around three questions that recur in the literature and that map onto the structural–causal spine of the book:

1. How do we measure innovation outputs, and what are we identifying when we use a market-implied measure such as KPSS?
2. How does financing shape innovation, and what do clean causal designs tell us about the effect of financing on R&D?
3. What do structural models of innovation buy us that causal designs cannot, and what are the limits of structural counterfactuals in this setting?

These three questions correspond loosely to three intellectual traditions. One is in corporate finance, one is in industrial organization, and one is in growth economics. They have studied innovation in parallel without much cross-pollination. Section 13.2 draws a map of those traditions before we work through the methodological details.

Key Insight: Innovation is intangible investment with spillovers

The three features that distinguish innovation from tangible investment are intangibility, skewed long-horizon payoffs, and private–social return wedges. Each generates distinct methodological challenges. Measurement is half the empirical problem. Financing is unusually important because intangible assets cannot be pledged. Policy questions involve spillovers that causal designs typically cannot identify on their own. The structural–causal distinction is the right organizing principle for navigating these features.

13.2 Three Traditions, One Topic

Innovation has been studied seriously by corporate finance economists, by industrial organization economists, and by growth and macro economists. Each field has produced a substantial empirical literature on the same underlying phenomenon, and the three literatures have remarkably little overlap in either methods or citations. A corporate-finance PhD student who reads Howell on R&D grants will likely never encounter Pakes on patent renewal values, and vice versa. This compartmentalization is unfortunate, because each tradition sees something the others miss, and the methodological commitments differ in ways that map cleanly onto our structural–causal framework.

This section draws a map. We will work through the technical details of each tradition in later sections; here the goal is to know what is on offer and how the pieces fit together.

13.2.1 The corporate-finance tradition

The corporate-finance literature on innovation treats R&D as a special kind of investment subject to special financing frictions. The methodological orientation is design-based and causal. The papers ask questions of the form “does financing shock X affect innovation outcome Y for firms in setting Z ?” and seek clean variation in X that permits identification.

The canonical reference points are [Brown, Fazzari, and Petersen \(2009\)](#), who document that R&D investment-cash-flow sensitivities differ across firm types in ways consistent with financing constraints, and [Howell \(2017\)](#), who exploits a regression-discontinuity design at SBIR grant funding cutoffs to identify the causal effect of public R&D financing on patenting and follow-on outcomes. A second strand uses innovation as an outcome in corporate-event studies: [Bena and Li \(2014\)](#) on M&A and technological overlap, [Phillips and Zhdanov \(2013\)](#) on the market for innovation, [Bernstein \(2015\)](#) on the effect of IPOs on

inventor behavior, and [Atanassov \(2013\)](#) on antitakeover laws and innovation.

What this tradition does well: it takes identification seriously, exploits high-quality micro data linking USPTO patent records to Compustat firms, and produces estimates that policy analysts can use to evaluate specific interventions.

What it largely ignores: the structural mechanism through which financing affects innovation. The papers tell us that grants increase patenting, or that IPOs change inventor behavior, but the model that rationalizes these effects is left implicit. Counterfactuals beyond the identified margin typically lie outside what the causal design can deliver. These include what a different grant size would do, what a firm-size-targeted subsidy would accomplish, and what the aggregate effect of a patent reform would be.

13.2.2 The industrial-organization tradition

The IO literature on innovation treats R&D as a choice variable inside a dynamic firm model, and innovation outcomes as observable manifestations of unobservable firm states. The methodological orientation is structural. Estimation proceeds by writing down a model, deriving moment conditions or value functions, and matching simulated moments to data.

Four pillars organize the field. First, the *knowledge-capital production function* due to [Griliches \(1979\)](#) measures the returns to R&D through a Cobb-Douglas production function augmented with an R&D-derived stock of intangible capital. The depreciation rate of knowledge capital, δ_R , becomes a key structural parameter, and the difficulty of estimating it consistently has occupied the literature for forty years. [Doraszelski and Jaumandreu \(2013\)](#) modernize this approach by endogenizing productivity dynamics as functions of R&D.

Second, *patent renewal and patent value models* go back to [Pakes \(1986\)](#), who exploits the fact that patent holders must pay renewal fees to keep patents alive. The decision to renew reveals the patent holder's belief about the patent's continuation value, and a dynamic discrete choice model recovers the distribution of patent values. [Lanjouw \(1998\)](#) and [Serrano \(2018\)](#) extend this approach.

Third, *dynamic games of innovation* model competitive incentives to invest in R&D. [Goettler and Gordon \(2011\)](#) estimate a model of Intel and AMD's innovation dynamics in the microprocessor industry, recovering structural parameters that allow counterfactuals on competition policy.

Fourth, *R&D spillovers*, where [Bloom et al. \(2013\)](#) is the methodological tour de force, estimate technology and product-market spillover elasticities separately using a clever

combination of patent similarity measures and firm-pair matching.

What this tradition does well: it delivers structural parameters that permit counterfactual analysis, separates private from social returns, and quantifies spillovers that cannot be estimated cleanly from causal designs.

What it largely ignores: financing. Almost none of the structural IO papers on innovation features any financial friction. R&D is treated as a choice variable subject to convex adjustment costs, sometimes constrained by physical or organizational capacity, but not constrained by access to capital. This is a striking omission given the corporate-finance evidence that financing matters substantially for R&D.

13.2.3 The growth and macro tradition

The growth and macro literature on innovation is the most ambitious of the three. It treats innovation as the engine of aggregate productivity growth and asks how firm-level innovation decisions aggregate into economy-wide outcomes. The methodological orientation is structural and Schumpeterian, with quality ladders and creative destruction as the canonical apparatus.

[Klette and Kortum \(2004\)](#) provide the foundational firm-dynamics model where heterogeneous firms innovate, grow, and sometimes exit. [Akcigit and Kerr \(2018\)](#) extend this by distinguishing internal innovation (improving one's own product) from external innovation (creating new products), which scales differently with firm size and generates testable predictions for the size distribution of innovation. [Acemoglu, Akcigit, Alp, Bloom, and Kerr \(2018\)](#) build a quantitative model with rich firm heterogeneity and use it to evaluate R&D subsidies, taxation of incumbents, and operating taxes as policy levers. Recent contributions by [Akcigit and Ates \(2023\)](#), on the slowdown in business dynamism, and [Akcigit, Grigsby, Nicholas, and Stantcheva \(2022\)](#), on patent box reforms, extend the apparatus in policy-relevant directions.

What this tradition does well: it answers aggregate counterfactual questions that no other approach can address. These include the optimal R&D subsidy and the effect of a different patent regime on long-run growth.

What it largely ignores: micro identification and firm-level financial data. The growth literature works with publicly available firm-size distributions, R&D intensities, and patent counts, but rarely uses the firm-level Compustat-CRSP-USPTO panel that corporate-finance researchers take for granted. And, as in the IO tradition, financing frictions are typically absent from the models.

13.2.4 Why the three traditions matter for a corporate finance chapter

The three traditions are not competing answers to the same question; they are different questions, each with its own appropriate methodology. The corporate-finance tradition asks how financing constraints shape innovation. The IO tradition asks how innovation incentives interact with industry structure and what spillovers look like. The growth tradition asks how firm-level innovation aggregates into long-run economic growth.

For students of empirical corporate finance, three implications follow.

First, the corporate-finance literature has under-engaged with the structural side. The methodological repertoire in most corporate-finance papers on innovation is design-based: IV, RD, DiD on patenting outcomes. This is valuable, but it leaves on the table the structural machinery that IO and growth economists have developed for studying the same phenomenon. A PhD student looking for research opportunities should note this gap.

Second, the IO and growth literatures have under-engaged with financing. Their models typically assume frictionless financing or ignore financing altogether. Bringing financial constraints into structural innovation models is an open frontier; the work of [Frank and Sanati \(2021\)](#) on financing growth is one of the few attempts to bridge the gap, and the bridge is far from complete.

Third, the cleanest path through the literature is to be explicit about which question one is asking and to choose the methodological tradition accordingly. The structural–causal distinction from Chapter 1 is exactly the right organizing device for this chapter, because innovation is one of the few topics where the three field traditions live in genuinely different methodological worlds. Table 13.1 summarizes.

Connection to Framework: Innovation as a stress test for the structural–causal distinction

Most chapters in this book illustrate the structural–causal distinction within a single literature. Innovation illustrates it across three literatures that have evolved largely in parallel. The corporate-finance papers we will discuss in Section 13.4 are causal in orientation; the IO and growth papers we will discuss in Section 13.5 are structural. Neither tradition is complete on its own. The pedagogical payoff of this chapter is that the structural–causal spine forces both sides into view simultaneously.

13.2.5 A reader’s guide to the rest of the chapter

The remainder of the chapter is organized to work through the three traditions in a sequence that builds rather than alternates. Section 13.3 addresses measurement first,

Table 13.1. Three traditions in the empirical study of innovation

	Corporate finance	Industrial organization	Growth/macro
Methodology	Causal / design-based	Structural	Structural
Typical question	Does financing shock X affect innovation?	What is the structural value of a patent or R&D project?	What is the aggregate effect of an innovation policy?
Canonical methods	IV, RD, DiD, event studies	Dynamic discrete choice, SMM on firm-level moments	SMM on aggregate moments, quality-ladder models
Anchor papers	Brown et al. (2009) , Howell (2017) , Bena and Li (2014)	Pakes (1986) , Griliches (1979) , Doraszelski and Jaumandreu (2013)	Klette and Kortum (2004) , Akcigit and Kerr (2018) , Acemoglu et al. (2018)
What it identifies well	LATE of financing on innovation; effects of corporate events	Private value of patents; spillover elasticities; structural R&D returns	Aggregate growth effects of policy; counterfactual policy regimes
What it largely ignores	The structural mechanism; aggregate counterfactuals	Financing frictions; firm-level corporate finance data	Micro identification; firm-level financial data

because no empirical work on innovation makes sense without an honest accounting of how the variables are constructed. The KPSS market-implied patent value, which sits at the boundary of the corporate-finance and IO traditions, gets a substantive treatment there.

Section 13.4 is the longest in the chapter and covers the corporate-finance tradition: the R&D-cash-flow sensitivity literature, clean designs such as Howell’s SBIR study, and natural experiments with tax credits and patent reforms. This is where the chapter’s distinctive contribution to a corporate-finance PhD textbook lies, and where it connects back to the financial constraints material developed in Chapter 9.

Section 13.5 covers structural models. The Akcigit–Kerr framework receives the most extensive treatment, both because it has become the workhorse in the growth literature and because it generalizes cleanly to other questions in the book. We will also flag what the structural literature has not yet done.

Section 13.6 covers innovation in the context of corporate events: M&A, IPOs, and anti-takeover laws. This is shorter and serves to set up Chapter 15 on deal-based identification.

Section 13.7 returns to the leverage-and-investment running example from earlier chapters, adapting it to ask how leverage affects R&D rather than tangible investment.

Sections 13.8 and 13.9 close the chapter with a decision framework and an explicit return to the structural–causal spine. The exercises follow.

13.3 Measuring Innovation

No empirical work on innovation makes sense without an honest accounting of how the variables are constructed. Measurement is half the empirical problem, and the choice of measure is itself a methodological commitment. This section walks through the main options. We will cover R&D expenditure from Compustat, patent counts from USPTO, citation-weighted patents, the [Kogan et al. \(2017\)](#) market-implied patent value (KPSS), text-based measures, and the Griliches knowledge-capital stock. Each captures a different aspect of innovation, and the differences matter for what the downstream regressions identify.

13.3.1 R&D expenditure

The simplest measure is firm-year R&D expenditure, available in Compustat as `xrd`. This is an input measure: it tells us what firms spent on innovation, not what they produced. For some questions this is exactly what we want. If the goal is to study how financing affects R&D activity, the input is the right outcome. If the goal is to measure the value of innovation produced, R&D expenditure is unsuited.

Three practical issues affect almost every paper using `xrd`.

First, missingness is not random. Compustat reports `xrd` as missing for firms that do not disclose R&D, but the U.S. GAAP threshold for required disclosure means that many small firms with positive R&D report nothing. The standard practice is to treat missing `xrd` as zero, but this introduces measurement error that is correlated with firm size and age. [Koh and Reeb \(2015\)](#) document that a substantial share of patenting firms report zero R&D, suggesting that the zero treatment understates innovation activity at smaller firms.

Second, R&D intensity varies enormously across industries. The pharmaceutical industry spends 15–20 percent of sales on R&D; the retail industry spends essentially none. Regressions of innovation outcomes on R&D that pool across industries are typically dominated by industry composition. Industry-year fixed effects are the usual fix, though they absorb a large fraction of the variation of interest.

Third, the accounting treatment of R&D differs across regimes. U.S. GAAP requires immediate expensing under SFAS 2 (1974). IFRS under IAS 38 capitalizes development costs once technical feasibility is established. Cross-country studies must confront this difference, and even U.S.-only studies must recognize that `xrd` understates the true asset value of accumulated innovation. The [Peters and Taylor \(2017\)](#) intangible-capital adjustment is the most-used correction in the empirical literature, though we will see in Section [13.4](#) that the corrections do not always do what they advertise.

13.3.2 Patent counts

Patent counts are the most-used output measure. The USPTO grants patents to applicants who meet novelty, non-obviousness, and utility requirements; the patent database is public, and a long series of academic dataset projects has linked patents to firms. The [Hall, Jaffe, and Trajtenberg \(2001\)](#) NBER patent file was the foundational academic dataset, and successive efforts have extended coverage through the present (the Kogan et al. dataset, the PatentsView project, and several commercial vendors).

Raw patent counts are a poor measure of innovation output for three reasons. First, the propensity to patent differs across industries and firms. Pharmaceutical firms patent almost everything; software firms patent selectively; some research is protected by secrecy rather than patenting. Cross-industry comparisons of raw counts mislead.

Second, recent years are truncated. A patent application filed in 2023 may not appear in the data until it is granted, which can take three to five years. Standard practice is to date patents by application year (the year the firm filed) rather than grant year, and to drop the most recent two or three years of data or apply truncation corrections.

Third, patents vary enormously in importance. A patent on a method for ordering food online and a patent on a new chemical entity should not count equally. This is what motivates citation-weighting.

13.3.3 Citation-weighted patents

Counting forward citations (the number of times patent p is cited by subsequent patents) provides a quality adjustment. A citation-weighted patent stock takes the form

$$K_{i,t}^{\text{cite}} = \sum_{p \in P_{i,t}} (1 + c_p) \quad (13.1)$$

where $P_{i,t}$ is the set of patents granted to firm i up to time t , c_p is the count of forward citations to patent p , and the +1 is the patent itself. Many variants exist: some authors normalize citations within technology class and grant year ([Hall, Jaffe, and Trajtenberg \(2005\)](#)) to account for differences in citation propensity, some use truncated citation windows (citations within five or ten years of grant), and some weight by the patent's own technological breadth.

Citation-weighting helps but has its own pathologies. Citation counts themselves are truncated for recent patents. Patent examiners insert some citations regardless of inventor input. The number of citations a patent receives depends on the size of the technology field, which varies systematically over time. And citation-weighting still does not deliver a

dollar value of the patent, which is what we usually want.

13.3.4 KPSS market-implied patent values

Kogan et al. (2017) propose a market-implied measure that addresses the dollar-value problem. The idea is conceptually clean. When a patent is granted to a publicly traded firm, the firm's market value reacts to the news that the patent has been granted. Under appropriate identifying assumptions, the abnormal return in a narrow window around the grant date provides a market estimate of the patent's private value.

The construction. Let $R_{i,t}$ be firm i 's log return on day t , and let $\tilde{R}_{i,t} = R_{i,t} - R_{m,t}$ be the return net of the market return $R_{m,t}$. The abnormal return for patent p granted on day τ_p is computed over a three-day window $[\tau_p, \tau_p + 2]$:

$$AR_p = \sum_{t=\tau_p}^{\tau_p+2} \tilde{R}_{i,t}. \quad (13.2)$$

The dollar value of the patent is then

$$\xi_p = E \left[\frac{1}{1-\pi} AR_p \cdot V_{i,\tau_p} \mid AR_p > 0 \right] \quad (13.3)$$

where V_{i,τ_p} is the firm's market capitalization on the grant day and $1 - \pi$ is the probability that a granted patent generates a positive return (KPSS estimate this around 0.55–0.60). The conditioning on $AR_p > 0$ handles the fact that some abnormal returns are negative or zero by construction (patents that the market already anticipated provide no news on the grant date).

Identifying assumption. The key assumption is that, within a three-day window centered on the grant date, the only firm-specific news arriving is the patent grant itself. Under this assumption, the abnormal return identifies the market's revaluation of the firm in response to the patent grant. The assumption is plausible because patent grant dates are largely unpredictable to the firm and the market: the USPTO controls the timing, examiners' decisions take years to play out, and the firm does not announce the grant in advance. The assumption can fail when other news (earnings announcements, M&A announcements) coincides with the grant date, and standard practice is to exclude such observations.

Connection to event studies. The KPSS construction is, methodologically, an event study. It applies the apparatus of Chapter 4 (market-model abnormal returns over a narrow window) to a measurement question rather than a hypothesis test. This is worth flagging because it reframes how students should read the paper. KPSS does not estimate whether patents matter; that is taken for granted. It uses event-study identification to construct a continuous patent-value measure that can then serve as a right-hand-side or left-hand-side variable in other regressions, or as a target moment for structural estimation. This is the [Andrews et al. \(2020\)](#) program in action: reduced-form identification feeding into structural or descriptive work downstream.

What KPSS identifies, and what it does not. The KPSS measure identifies the market's perceived private value of the patent to the granting firm, conditional on the information available on the grant date. Three things it does not identify deserve emphasis.

First, KPSS does not identify the social value of the patent. The market reaction captures the value to the firm, not the value to competitors, downstream firms, or consumers. The [Bloom et al. \(2013\)](#) spillover literature is the natural complement.

Second, KPSS does not identify causal effects of policy on innovation. The measure assumes that patents already exist and were granted; it provides no leverage on the question of how a policy change would affect patent values or the rate of patenting.

Third, KPSS captures private value only for publicly traded firms. Private firms, which account for a substantial share of patenting in some industries, are absent. Foreign firms are absent. Patents granted to firms that subsequently delist drop out. The selection into the KPSS sample is non-random.

Practical implementation. The Kogan et al. public dataset provides patent-firm matches and constructed dollar values through recent years. Building a firm-year panel typically merges this dataset with Compustat by `gvkey` and aggregates patent values by firm-year:

$$\Xi_{i,t} = \sum_{p \in P_{i,t}} \xi_p \quad (13.4)$$

where the sum is over patents granted to firm i during fiscal year t . Scaling by firm market value or by total assets is standard. The Implementation box at the end of this section provides example code.

13.3.5 Text-based measures and machine learning

A more recent strand of work uses text analysis of patent documents, 10-K filings, and other corporate disclosures to construct innovation measures that do not depend on patent counts or stock-market reactions. [Bellstam, Bhagat, and Cookson \(2021\)](#) construct a text-based innovation measure from analyst reports. [Kelly, Papanikolaou, Seru, and Taddy \(2021\)](#) use patent text to identify breakthrough innovations as patents whose text is distinct from prior patents and similar to subsequent patents. The patent embedding work of [Arts, Cassiman, and Gomez \(2018\)](#) and successors provides continuous measures of patent similarity that supplement citation-based measures.

These methods are promising but young. Their main appeal is that they can capture innovation that does not show up in patent counts (process innovations, organizational innovations, innovations protected by secrecy) and can be applied to private firms. Their main limitation is that the constructed measures are sensitive to the choice of text corpus, tokenization, and embedding model. We return to these tools in Chapter 7.

13.3.6 Knowledge-capital stocks

For regressions that require a stock measure of accumulated innovation, the standard construction follows the perpetual-inventory method of [Griliches \(1979\)](#):

$$K_{i,t}^R = (1 - \delta_R)K_{i,t-1}^R + R_{i,t} \quad (13.5)$$

where $R_{i,t}$ is R&D expenditure in period t and δ_R is the depreciation rate of knowledge capital. The initial stock $K_{i,0}^R$ is usually constructed under a steady-state assumption, $K_{i,0}^R = R_{i,0}/(g + \delta_R)$, where g is a long-run R&D growth rate.

Two parameters drive the constructed stock: δ_R and the steady-state growth rate. The literature has used δ_R values ranging from 0.10 to 0.30, with 0.15 a common default. [Li and Hall \(2020\)](#) estimate δ_R from data and find substantial heterogeneity across industries, with pharmaceuticals at the low end and software at the high end. The sensitivity of regression results to δ_R is rarely reported, but should be.

13.3.7 Implementation: building a firm-year innovation panel

The most common empirical setup uses a firm-year panel that combines Compustat, CRSP, and the Kogan et al. patent value dataset. The implementation merges these by `gvkey` (Compustat firm identifier) and `permno` (CRSP) using the standard CRSP-Compustat link table.

Implementation: Innovation panel construction

The accompanying code (`ch9_innovation_panel.py`, random seed 8823) builds a firm-year panel with the following variables:

- `xrd`: R&D expenditure from Compustat, with missing treated as zero (a separate flag preserves the original missingness).
- `xrd_int`: R&D intensity, `xrd / sale`.
- `patents`: count of patents granted to the firm in the fiscal year, dated by application year.
- `patents_cw`: citation-weighted patent count using truncation-corrected citations.
- `kpss`: sum of KPSS dollar values for patents granted in the fiscal year, deflated to 2012 dollars.
- `kpss_scaled`: `kpss / market capitalization`.
- `kstock`: knowledge-capital stock built using $\delta_R = 0.15$ and steady-state initialization.

The code follows [Frank and Goyal \(2024\)](#) variable conventions where applicable and matches the master Compustat panel built in Chapter 3. Sensitivity to $\delta_R \in \{0.10, 0.15, 0.25\}$ is provided as a supplementary output.

13.3.8 Choosing a measure

Different measures suit different questions. For studies of financing effects on R&D activity, `xrd` is usually the right outcome. For studies of innovation output, patent counts and citation-weighted counts are standard; KPSS is preferred when a dollar value is needed. For structural models that match moments on innovation intensity and quality, citation-weighted counts and KPSS are both used. For cross-country work, patent counts (or PCT applications) are usually the only feasible option, and the heterogeneity in patent systems makes cross-country comparisons hazardous.

Common Mistake: Treating raw patent counts as innovation output

Regressions of corporate outcomes on raw patent counts, without industry-year fixed effects or quality adjustment, are dominated by industry composition and truncation. The propensity to patent differs by an order of magnitude across industries, and the count of patents filed in recent years is mechanically suppressed by the grant lag. Citation-weighted counts with technology-class normalization and a truncation adjustment are the minimum acceptable quality control. KPSS values are preferred when a dollar-value interpretation is needed.

13.4 Financing Innovation

This is the section where corporate finance has the most to say about innovation, and where the design-based methodological repertoire developed in Chapters 3 through 6 comes into its own. The central claim of the literature is that financing matters for innovation, and matters more for innovation than for tangible investment. Establishing this claim requires careful attention to identification, because financing and innovation are jointly determined by firm characteristics. The cleanest evidence comes from natural experiments and discontinuities. The interpretation of that evidence connects to the constraint-share framework developed in Chapter 9.

13.4.1 Why R&D is special for financing

The features of innovation discussed in Section 13.1 interact to make R&D unusually difficult to finance externally. Four mechanisms operate.

Intangibility precludes collateral. A factory can be pledged to a lender. A research program cannot. The assets created by R&D investment are intangible (knowledge, know-how, relationships with skilled employees), firm-specific, and non-redeployable. Even patents, which are tangible assets in a legal sense, have liquidation values that are typically far below their private value to the firm. The result is that R&D investment is essentially debt-incompatible. Firms cannot pledge R&D assets to obtain debt financing on terms remotely comparable to those for tangible investment.

Information asymmetry is severe. The information asymmetry between firm insiders and outside investors, which Myers and Majluf (1984) identified as a friction for equity

financing generally, is unusually severe for R&D. The value of a research program depends on technical details that the firm understands and outsiders do not. Disclosing those details to investors compromises their commercial value. The Myers–Majluf wedge between the firm’s willingness to issue equity and the price at which equity can be sold is therefore larger for R&D-intensive firms than for firms with predominantly tangible assets.

Long horizons and high failure rates. A drug under development at year zero may generate revenue ten or fifteen years later, and most candidate drugs fail before reaching the market. The expected return is positive but the cash-flow profile is heavily back-loaded and right-skewed. Lenders, who care about the ability to service interest payments, find such cash-flow profiles unattractive. Equity investors, who participate in upside, find them acceptable if they can hold a diversified portfolio. The result is that R&D-heavy firms are financed predominantly by equity rather than debt.

Empirical signatures. The four mechanisms above generate three empirical regularities that the literature has documented repeatedly. First, R&D-intensive firms hold less debt than otherwise comparable firms. Cross-sectional regressions of leverage on R&D intensity yield large negative coefficients that survive most controls. Second, R&D-intensive firms hold more cash than other firms, consistent with a precautionary motive related to financing risk. [Bates, Kahle, and Stulz \(2009\)](#) attribute a substantial share of the secular rise in U.S. corporate cash holdings to the rising R&D intensity of the firm population. Third, R&D-intensive firms issue more equity than other firms, *both* through IPOs and through seasoned equity offerings. The combination (more equity, less debt, more cash) is the financing signature of R&D-intensive firms.

13.4.2 The R&D-cash-flow sensitivity literature

The foundational empirical work on financing and innovation extends the [Fazzari et al. \(1988\)](#) investment-cash-flow sensitivity framework to R&D. The basic regression takes the form

$$\frac{R\&D_{i,t}}{A_{i,t-1}} = \alpha + \beta_1 Q_{i,t-1} + \beta_2 \frac{CF_{i,t}}{A_{i,t-1}} + \beta_3 \frac{ExtFin_{i,t}}{A_{i,t-1}} + \gamma X_{i,t} + \eta_i + \nu_t + \varepsilon_{i,t} \quad (13.6)$$

where $R\&D_{i,t}$ is R&D expenditure, $A_{i,t-1}$ is lagged total assets, $Q_{i,t-1}$ is Tobin’s Q , $CF_{i,t}$ is operating cash flow, $ExtFin_{i,t}$ is external financing (typically split into debt and equity components), and η_i, ν_t are firm and year fixed effects. The coefficient of interest is β_2 : a

positive coefficient on cash flow, conditional on Q , is interpreted as evidence that internal funds matter for R&D investment beyond what investment opportunities require.

Brown et al. (2009) apply this framework to a large panel of U.S. high-tech firms over 1990–2004 and find three results that have shaped the literature. First, the cash-flow sensitivity of R&D is large and statistically significant for young high-tech firms but is small and insignificant for mature firms. Second, external equity financing (new equity issuance) is also significantly related to R&D for young high-tech firms but not for mature firms. Third, the 1990s R&D boom is quantitatively attributable to the easing of financing constraints during that period, with both cash flow and equity financing contributing roughly equally.

Brown and Petersen (2011) extend this analysis to the role of cash holdings as a buffer for R&D, showing that R&D-intensive firms use cash reserves to smooth R&D expenditure in the face of cash-flow shocks. Brown, Martinsson, and Petersen (2012) document substantial heterogeneity in R&D-cash-flow sensitivities across European countries that correlates with the development of equity markets.

The interpretation debate. The cash-flow sensitivity literature has been the subject of an ongoing methodological debate. Kaplan and Zingales (1997) argue that a positive cash-flow coefficient does not necessarily indicate financing constraints: it can also reflect measurement error in Q , since Q is a noisy proxy for marginal product, and the correlation of the residual with cash flow biases β_2 upward. Erickson and Whited (2000) make this critique formal and show that correcting for measurement error in Q substantially attenuates the cash-flow coefficient. Farre-Mensa and Ljungqvist (2016) extend the critique by showing that even firms classified as “constrained” by standard indices (Kaplan–Zingales, Whited–Wu, Hadlock–Pierce) do not behave like constrained firms along other dimensions, raising doubts about the standard indices.

The honest reading is that R&D-cash-flow sensitivities are robust empirical patterns but their interpretation as evidence of financing constraints requires assumptions that are not innocuous. The deeper treatment of these critiques belongs in Chapter 9; for our purposes, we note that the cash-flow sensitivity literature is suggestive but not definitive.

Common Mistake: Reading β_2 as a measure of constraints

The cash-flow coefficient β_2 in equation (13.6) is a sufficient statistic for financing constraints only under strong assumptions: Q is measured without error, the production function is correctly specified, and adjustment costs are quadratic. None of these assumptions is innocuous. A positive β_2 is consistent with constraints but is also consistent with measurement error in Q , misspecified production technology, or non-quadratic adjustment costs. The constraint-share framework developed in Chapter 9 provides an alternative interpretation that does not require these assumptions.

13.4.3 Clean designs: Howell on SBIR grants

The methodological response to the cash-flow sensitivity debate has been to find natural experiments where financing varies for reasons plausibly unrelated to investment opportunities. The cleanest example in the innovation literature is [Howell \(2017\)](#).

The setting. The Small Business Innovation Research (SBIR) program awards research grants to small U.S. firms working on commercializable technology. Awards are made in two phases. Phase 1 grants of roughly \$150,000 fund six months of feasibility research. Phase 2 grants of up to \$1 million fund follow-on development for firms whose Phase 1 work was judged successful. Within each agency-topic competition, applicants are scored by technical reviewers and ranked. Grants are awarded to applicants above a funding cutoff that depends on the agency's budget in that cycle.

The identification strategy. The funding cutoff generates a regression-discontinuity design. Firms just above the cutoff receive a grant; firms just below do not. Under the standard RD assumption that firms cannot precisely manipulate their score, comparing firms just above and just below the cutoff identifies the local average treatment effect (LATE) of the grant on subsequent outcomes. Connect to the RD apparatus from Chapter 6: this is a sharp RD with a continuous running variable (the technical score) and a deterministic treatment assignment at the cutoff.

Implementation and main results. The estimator is a local linear regression in a window around the cutoff:

$$Y_i = \alpha + \tau D_i + f(S_i - c) + \mathbf{1}\{S_i > c\} \cdot g(S_i - c) + \varepsilon_i \quad (13.7)$$

where S_i is the technical score, c is the cutoff, $D_i = \mathbf{1}\{S_i > c\}$ is the treatment indicator, and f, g are polynomial functions of the centered score. The coefficient τ identifies the LATE under the standard RD identifying assumptions.

Howell (2017) finds that receiving a Phase 1 grant approximately doubles the probability of future receipt of venture capital financing, increases the probability of patenting, and increases revenue growth. The Phase 2 effects are smaller and less robust. The effects are concentrated among young firms, firms in hardware-intensive industries, and firms with limited prior funding history. The Phase 1 grant appears to act as a certification device that unlocks subsequent private financing, in addition to providing direct R&D funding.

What the LATE identifies. The Howell estimate identifies the causal effect of the grant on innovation outcomes for firms near the cutoff. This is a LATE in the Imbens and Angrist (1994) sense: the population is the firms whose treatment status switches at the cutoff (the compliers). Firms far above the cutoff (always-takers) and firms far below (never-takers) do not contribute to identification.

The LATE-as-constraint-share interpretation deserves attention here. Under the framework developed in Chapter 9, the RD estimate $\hat{\tau}$ equals the share of complier firms that are financially constrained, multiplied by the marginal product of internal funds for those firms. If all compliers are constrained, $\hat{\tau}$ equals the marginal product of funds. If some compliers are unconstrained, $\hat{\tau}$ is attenuated. The empirical magnitude of Howell's estimates is consistent with most compliers being constrained, which is unsurprising given that the compliers are young small firms applying for public R&D grants.

Connection to Framework: LATE as constraint share

The Howell RD estimate is a LATE among complier firms near the SBIR funding cutoff. As Chapter 9 makes precise, this LATE equals the share of complier firms that are financially constrained times the marginal effect of relaxing the constraint. Innovation financing is the empirically richest setting for this framework because R&D collateralizes poorly: the compliers in any well-identified study of R&D financing are likely to be heavily constrained, so the LATE is informative about the marginal product of funds. The same RD design applied to a sample where most compliers are unconstrained would yield an attenuated estimate that understates the effect for the constrained subpopulation.

13.4.4 Other natural experiments and policy variation

Several other research designs have produced credible estimates of financing effects on innovation.

R&D tax credits. Bloom, Griffith, and Van Reenen (2002) use cross-country variation in R&D tax credit generosity to estimate the user-cost elasticity of R&D. They find a short-run elasticity of about -0.1 and a long-run elasticity of about -1 , implying that a 10 percent reduction in the user cost of R&D induces a 10 percent long-run increase in R&D expenditure. Wilson (2009) exploits variation across U.S. states in R&D tax credit generosity to estimate within-country elasticities, finding similar magnitudes for in-state R&D but substantial reallocation across state borders. Rao (2016) uses the federal R&D tax credit and finds elasticities at the upper end of this range.

Patent box reforms. Patent box regimes (preferential tax rates on income from patents) have proliferated since the early 2000s. Akcigit et al. (2022) provide a structural treatment of innovation and taxation over the long run. Reduced-form work using the staggered adoption of patent box regimes in European countries finds modest effects on patent filings, with substantial profit-shifting effects (firms relocate patent ownership to favorable jurisdictions without relocating actual research).

Bank credit shocks and R&D. The 2008 financial crisis literature has provided cleaner evidence on how shocks to bank credit affect R&D. Nanda and Nicholas (2014) use Depression-era variation in bank distress to study the longer-run effects of credit on innovation. Huber (2018) uses the Commerzbank crisis in Germany to show that firms exposed to lender distress reduced R&D as well as investment. These studies confirm that even though R&D is financed predominantly by internal funds and equity, bank-credit shocks propagate to R&D, likely through firms' liquidity management.

Venture capital availability. Nanda and Rhodes-Kropf (2013) use cross-cohort variation in VC fundraising to study how financing market conditions affect the innovation profile of VC-backed startups. Firms funded during VC boom periods take more technical risk but are also more likely to fail. Kerr and Nanda (2009) document that local banking deregulation in the U.S. increased the entry of new firms in industries dependent on external finance, with disproportionate effects on innovative sectors.

13.4.5 What we know and what we do not

The evidence reviewed above supports several substantive conclusions about financing and innovation.

R&D responds to internal funds. The R&D-cash-flow sensitivity is robust across samples and specifications, even after the Erickson–Whited measurement-error corrections attenuate the coefficient. The interpretation as a constraint measure is contested, but the underlying pattern (R&D moves with internal funds) is not.

R&D responds to public grants. The Howell RD evidence is clean and the magnitudes are large. The grants act in part as direct funding and in part as certification for subsequent private financing.

R&D responds to tax incentives, with a long-run elasticity that appears to be around -1 . The user cost of R&D is a policy lever that has predictable effects on R&D intensity.

R&D responds to credit shocks, indirectly through firms' liquidity management. R&D is not financed by debt in the first instance, but credit shocks propagate through cash holdings and working capital.

VC and IPO conditions matter for startup innovation. Boom periods support more technically ambitious projects but also more failures.

What we do not know as well. We do not have clean estimates of the marginal social return to R&D, because identifying social returns requires either auxiliary structure (Bloom–Schankerman–Van Reenen) or unusual data. We have limited evidence on heterogeneity in financing elasticities across firm types, beyond the broad young/mature split. We have little understanding of how financial frictions interact with the choice between exploitation (internal innovation) and exploration (external innovation) in the sense of Akcigit–Kerr. And we have not yet integrated the structural innovation literature with the structural capital-structure literature, despite the obvious complementarities.

13.4.6 Connection to Financial Constraints

The evidence in this section connects directly to the central question of Chapter 9. We have seen that R&D responds to financing shocks of various kinds, and we have argued that R&D is a setting where financing frictions are likely to bind for many firms. But the relationship between an observed cash-flow sensitivity or RD estimate and the underlying financing friction is not transparent. The constraint-share framework of Chapter 9, developed in detail in Frank (2026a), makes this relationship precise: $\hat{\beta}_g = \theta_{C,g}$, where $\theta_{C,g}$ is the constraint share among the complier subpopulation. Innovation financing is the empirically richest setting for this framework because the compliers in any well-designed study of R&D

financing are heavily concentrated among constrained firms.

The reader who wants the full development should return to Chapter 9. The reader who wants to see how the structural side of the innovation literature complements the causal evidence reviewed here should proceed to Section 13.5.

13.5 Structural Models of Innovation

The causal evidence reviewed in Section 13.4 answers questions of the form “does X affect innovation Y at margin Z ?” For many policy questions, this is enough. For others, it is not. A policymaker who wants to know what an R&D subsidy of a particular size would do to aggregate productivity growth, or how an R&D subsidy targeted at small firms would compare to a uniform subsidy, cannot read the answer off a regression coefficient. These are counterfactual questions that require a structural model of how firms make innovation decisions, how those decisions aggregate, and how the aggregate responds to policy.

The structural literature on innovation has developed largely outside corporate finance, in the growth and industrial organization traditions discussed in Section 13.2. This section presents the workhorse models, explains what they identify, and flags what they have not yet done. The Akcigit–Kerr framework receives the most extensive treatment because it has become the benchmark in the growth literature and because its estimation methodology generalizes cleanly to questions elsewhere in corporate finance.

13.5.1 The Schumpeterian setup

The structural innovation literature builds on Schumpeterian growth theory, in which growth is driven by creative destruction: new firms or new products displace incumbents, and the displacement is itself the engine of productivity growth. The foundational reference for firm-level empirical work in this tradition is Klette and Kortum (2004), who provide a model that matches both aggregate growth and firm-level dynamics. We sketch their framework because subsequent work in this literature builds on it directly.

Consider a continuum of product lines indexed by $j \in [0, 1]$, each producing a differentiated good. Each product line has a current productivity level q_j , the quality of the best technology available for producing that good. Firms own portfolios of product lines. Innovation takes the form of either improving the quality of an existing product line (internal innovation) or displacing an incumbent on some other product line (external innovation, also called creative destruction). The quality of a product line evolves stochastically: with arrival rate proportional to the firm’s R&D intensity, the firm’s quality steps up by a factor

$\lambda > 1$.

A firm f that owns n_f product lines chooses R&D intensity x_f to solve

$$\max_{x_f} n_f \pi - C(x_f, n_f) + V(x_f, n_f) \quad (13.8)$$

where π is the per-line flow profit, $C(\cdot)$ is the R&D cost function, and $V(\cdot)$ is the continuation value reflecting the probability distribution over future portfolio sizes. The R&D cost function is typically convex in intensity and scales with firm size in a way that the literature parameterizes empirically.

The model has three features that matter for what follows. First, firms innovate to grow, and they grow by acquiring more product lines. Second, firms shrink when their product lines are taken by competitors, and exit when their portfolio is fully displaced. Third, the steady state of the model implies a stationary firm-size distribution that the model parameters jointly determine. Calibration matches model moments to data moments on innovation rates, firm-size distribution, entry, and exit. The estimation is by simulated method of moments, in the same family of methods used in Chapter 8 for adjustment-cost models.

13.5.2 Akcigit and Kerr (2018)

[Akcigit and Kerr \(2018\)](#) extend the Klette–Kortum framework in a way that has proved unusually productive for empirical work. Their key innovation is to distinguish two types of R&D within the firm and let them scale differently with firm size.

Internal versus external innovation. Firms can invest in *internal innovation*, which improves the quality of their own product lines, or in *external innovation*, which targets new product lines held by competitors. The two activities have different costs, different returns, and different relationships to firm size.

The Akcigit–Kerr empirical finding, which the model is built to rationalize, is that internal and external innovation scale very differently with firm size. Small firms innovate predominantly through external innovation: their patents tend to be in new technology classes and tend to displace incumbents. Large firms innovate predominantly through internal innovation: their patents tend to be in technology classes where they already operate, and tend to improve existing products. The model captures this by allowing the cost function for external innovation to scale less than linearly with firm size, while the cost function for internal innovation scales roughly linearly.

The estimation problem. The model's structural parameters include the step size λ , the arrival rate function for each type of innovation, the cost function parameters, the exit rate of firms, the entry rate of new firms, and the per-line profit rate. Akcigit and Kerr estimate these parameters by SMM. The target moments include the firm-size distribution, the firm growth-rate distribution, the entry and exit rates, the patent-intensity by firm-size profile, and the breakdown of patents into internal versus external types (which they identify using the technology class of patents relative to the firm's existing portfolio).

Counterfactuals. With the estimated model in hand, Akcigit and Kerr conduct several policy counterfactuals. They estimate the effect of an R&D subsidy that is uniform across firms, and the effect of an R&D subsidy targeted at small firms. They estimate the effect of subsidies to internal versus external innovation. They quantify the contribution of each innovation type to aggregate growth. None of these counterfactuals is available from a causal design. The structural model is what makes them possible.

The substantive conclusions are worth noting. A targeted subsidy to small firms is more effective at raising aggregate growth than a uniform subsidy of equal cost, because small firms do disproportionately more external innovation, which is the type that contributes more to growth. Subsidizing internal innovation, which large incumbents do most of, has small effects on aggregate growth. These conclusions depend on the maintained model structure, and the literature has begun to probe their robustness.

Connection to Framework: Same SMM machinery, different question

The estimation strategy in Akcigit–Kerr is the same one used for Cooper–Haltiwanger adjustment-cost models in Chapter 8: simulate the model, compute model moments, choose parameters to minimize a weighted distance between model and data moments. What differs is the model and the moments. Cooper–Haltiwanger targets investment-rate dynamics conditional on productivity shocks; Akcigit–Kerr targets patenting intensity conditional on firm size and innovation type. The transferability of the methodology is one of the appealing features of structural estimation: a student who understands SMM in one application understands it in many.

13.5.3 Estimation by simulated method of moments

The mechanics of SMM estimation deserve attention because they recur across the structural literature. The procedure is summarized in five steps.

1. Specify a parametric model with structural parameters $\theta \in \Theta$.

2. Identify a vector of data moments $\hat{m}(X)$ from the observed sample X . The choice of moments is consequential: moments must be informative about the parameters and must be matchable by simulation.
3. For each candidate θ , simulate the model to obtain simulated moments $\tilde{m}(\theta)$.
4. Define a weighting matrix W (often the inverse of the data-moment covariance matrix) and minimize the distance:

$$\hat{\theta} = \arg \min_{\theta \in \Theta} [\hat{m}(X) - \tilde{m}(\theta)]^\top W [\hat{m}(X) - \tilde{m}(\theta)]. \quad (13.9)$$

5. Conduct sensitivity analysis. Vary W , drop moments, add moments, and verify that conclusions are robust.

For innovation models the choice of moments is unusually important. The standard set, due to Klette–Kortum and Akcigit–Kerr, includes the firm-size distribution (often the right tail, captured by the share of employment or patents in the top decile), the firm growth-rate distribution (mean and variance of growth conditional on firm size), entry and exit rates, and patent-intensity by firm size. To identify internal versus external innovation, Akcigit–Kerr add the share of patents in the firm’s existing technology classes versus new classes.

Identification in this setting is informal. The parameters must be identified in the sense that distinct values produce distinct moment predictions, but the literature does not typically prove identification formally. The standard practice, following [Andrews et al. \(2017\)](#), is to compute sensitivity matrices that show how each parameter estimate would change in response to a perturbation of each moment. This is a useful discipline that has become standard in recent structural work.

13.5.4 What counterfactuals do the models support?

The value of a structural model is in the counterfactuals it permits. The innovation literature has used these models for four classes of counterfactual questions.

Policy elasticities at scale. Reduced-form estimates of R&D tax-credit elasticities, from Section 13.4.4, identify local responses for firms exposed to the policy change. A structural model permits evaluation of an R&D subsidy of any size, including one not yet implemented. [Acemoglu et al. \(2018\)](#) estimate a quantitative model of firm dynamics in which R&D subsidies, profit taxes, and operating taxes are all policy levers, and compute the optimal policy mix given a government budget. The substantive finding is that operating taxes

on incumbents combined with subsidies to entrants generate substantially higher welfare than the current U.S. policy mix.

Decomposition of growth. [Garcia-Macia, Hsieh, and Klenow \(2019\)](#) decompose aggregate productivity growth into contributions from new varieties, quality improvements within existing varieties, and reallocation across producers. The decomposition requires a structural model because the categories overlap in the data. The substantive finding is that incumbents' own-product innovation accounts for a larger share of growth than prior theory had assumed.

Misallocation and innovation. [Akcigit, Hanley, and Stantcheva \(2022\)](#) study how the allocation of R&D effort across firms responds to policy and how that allocation differs from a social-planner benchmark. The wedge between private and social returns to R&D, combined with imperfect competition, generates misallocation that policy can correct in principle. The magnitudes are quantified by the model.

Long-run versus short-run effects. A subsidy that increases R&D today affects productivity for many years through the resulting innovations. Reduced-form designs cannot capture this dynamic; structural models can. [Akcigit and Ates \(2023\)](#) use this feature to study the slowdown in U.S. business dynamism, attributing it largely to declining innovation by entrants and small firms over the past three decades.

13.5.5 The financing gap in structural innovation models

A consistent feature of the structural innovation literature is that it does not feature financial frictions. The models reviewed above treat R&D as a choice variable subject to convex adjustment costs, sometimes constrained by physical or organizational capacity. None of them features financing constraints, debt, equity, or a cost of external finance. This is a striking omission given the corporate-finance evidence in Section 13.4 that financing matters substantially for innovation.

The omission reflects the disciplinary boundary between the structural innovation literature, which grew up in macro and growth, and the structural corporate-finance literature, which grew up in finance and IO. The two literatures use different modeling apparatus and target different moments. Macro innovation models emphasize aggregate implications and growth rates; corporate-finance models emphasize financing decisions and firm-level capital structure. [Frank and Sanati \(2021\)](#) is one of the few attempts to

bridge the gap, combining a dynamic capital structure model with endogenous growth through R&D-driven productivity gains.

Several questions are open for students looking for dissertation topics. How does an R&D subsidy interact with financing constraints? A subsidy that lowers the user cost of R&D may be less effective if the binding constraint is collateral rather than the cost of capital. How does the allocation of R&D across small and large firms respond to financing conditions? The Akcigit–Kerr finding that small firms do disproportionate external innovation may itself reflect a financing equilibrium: large firms can finance internal innovation out of retained earnings, while small firms must use external finance and therefore choose riskier external innovation projects that can sustain high returns to outside investors. None of these questions has a clean answer in the existing literature.

Key Insight: An open frontier

The structural innovation literature has produced a rich set of models and credible policy counterfactuals, but it has done so by abstracting from financing. The corporate-finance literature has established that financing matters for R&D but has produced few structural models that explain how. Combining the two traditions is a frontier with substantial scope for original work. The bridge is not technically difficult; it has simply not been built at scale.

13.5.6 What does the structural approach buy us?

The structural literature on innovation has demonstrated four things that the causal literature cannot demonstrate on its own.

It can quantify the contribution of different innovation types to aggregate growth, separating internal innovation, external innovation, and entry. Reduced-form designs identify the effect of a particular policy on a particular outcome; they do not separate the channels.

It can evaluate counterfactual policies, including policies of sizes or designs not yet implemented. Reduced-form designs identify the effect of the policy that was implemented; they do not extrapolate.

It can quantify the gap between private and social returns to R&D through models that incorporate spillovers, building on [Bloom et al. \(2013\)](#). This gap is the central empirical claim underlying the case for public support of innovation.

It can decompose long-run growth into the contributions of innovation by different firm types and quantify how policy shifts that decomposition. The slowdown in U.S. business

dynamism documented by [Akcigit and Ates \(2023\)](#) is an example.

The cost of these capabilities is the maintained model structure. A structural counterfactual is only as credible as the model that generates it. The literature has begun to probe model robustness more seriously, following the recommendations in [Andrews et al. \(2020\)](#), and we will see this concern recur throughout the book.

13.6 Innovation and Corporate Events

So far we have studied innovation as a function of financing (the causal designs of Section 13.4) and as a feature of firm dynamics in a structural sense (the growth and IO models of Section 13.5). A third strand of the literature asks how innovation interacts with the broader set of corporate events that firms undergo: mergers and acquisitions, initial public offerings, takeover threats, and changes in ownership structure. This section reviews that work briefly. It is shorter than the previous sections because much of the methodological apparatus has been covered elsewhere in the book; what we do here is apply it to innovation outcomes and flag the identification challenges specific to this setting.

13.6.1 Mergers and acquisitions: who buys innovation?

A substantial literature studies mergers and acquisitions in which the target's innovation portfolio is part of what the acquirer is buying. The empirical regularities are striking. Acquirers and targets in completed deals have higher pre-deal patenting than the population average. Deals are concentrated in industries where R&D is intensive, particularly pharmaceuticals, biotechnology, software, and hardware. Targets are often small relative to acquirers and often young.

The central identification challenge in this literature is selection. Firms that merge are not a random sample. Acquirers select targets based on observable and unobservable characteristics, including expected innovation outcomes. Comparing post-merger outcomes for merging and non-merging firms recovers a mix of treatment effect and selection. The literature has addressed this challenge with three broad strategies.

Technological proximity as a determinant of merger incidence. [Bena and Li \(2014\)](#) examine the technological overlap between prospective acquirers and targets as a determinant of merger incidence. They measure overlap using the technology-class distribution of the firms' patent portfolios. Firms with greater overlap are more likely to merge. This finding is consistent with a theory in which mergers exploit technological complementarities,

but also consistent with selection on unobservables. The post-merger analysis examines patenting outcomes among merging firms and finds that combined entities produce more patents than the sum of the pre-merger predecessors would predict from their own trends, with the gains concentrated in firms with high technological overlap.

The methodological contribution of [Bena and Li \(2014\)](#) is the careful matching strategy used to construct counterfactual non-merging firms. The empirical strategy is propensity-score-like, matching merging firms to non-merging firms with similar patenting trajectories, industry, and size. The estimated effect on patenting can then be interpreted under the standard matching identification assumptions. As always with matching, identification rests on the assumption that selection into merger is captured by observables; the [Bena and Li \(2014\)](#) contribution is to argue that, conditional on patenting trajectory and the matched controls, the residual selection is plausibly small.

Small innovators and large acquirers. [Phillips and Zhdanov \(2013\)](#) document a sorting pattern in the data: small firms innovate, large firms acquire. They build a model in which large firms find it more efficient to acquire innovations from small firms than to develop them in-house, while small firms find it profitable to specialize in innovation because they can sell the output to large firms *ex post*. The model predicts that small firms will have higher R&D-to-sales ratios, will have lower internal scale of commercialization, and will be acquired with high probability when their innovations succeed. All three predictions find support in the data.

The contribution is structural rather than causal. The model rationalizes a set of patterns that are hard to interpret without it, and the empirical work serves to verify the model's predictions rather than to identify a treatment effect. This is a useful complement to the causal designs that dominate the corporate-finance literature on M&A.

Post-merger innovation outcomes. The literature on what happens to innovation *after* mergers is mixed. [Seru \(2014\)](#) finds that conglomerate firms produce less innovation than focused firms, suggesting that the increased scale and reduced focus of merged entities can suppress innovation. [Ahern and Sosyura \(2014\)](#) document that merger announcements affect the propensity of inventors to leave the target firm, with substantial mobility following deals. The implication is that some post-merger innovation effects operate through inventor attrition rather than through the merged firm's R&D policy directly.

The identification challenges in these studies are similar to those in the broader merger literature, covered in Chapter 15. For our purposes, the takeaway is that mergers are endogenous to expected innovation outcomes, and any treatment-effect estimate requires

either matching or structural restrictions to isolate the causal channel.

13.6.2 IPOs and inventor behavior

The decision to go public is itself an endogenous response to firm prospects, including innovation prospects. Identifying the causal effect of going public on subsequent innovation requires a design that isolates the IPO event from the firm's underlying trajectory.

[Bernstein \(2015\)](#) provides the cleanest such design to date. The strategy compares firms that completed IPOs to firms that filed IPO prospectuses but withdrew. Among firms that file, the decision to complete or withdraw is plausibly driven by market conditions during the filing window rather than by firm fundamentals, generating quasi-experimental variation in whether a firm becomes public. [Bernstein \(2015\)](#) finds that completing the IPO is associated with a decline in patent quality (citations per patent) but not a decline in patent quantity, and with substantial inventor mobility from public firms to other public firms and to private firms. Public firms appear to retain incremental innovation in-house but lose star inventors, who pursue more ambitious work elsewhere.

The interpretation is that public ownership changes the innovation mix toward more incremental and less radical work, potentially because the disclosure requirements and short-term performance pressures of public markets are unfavorable to long-horizon research. This is a specific instance of a more general claim, recurring in the literature, that ownership structure shapes innovation choices.

13.6.3 Antitakeover laws and innovation

A related set of studies exploits state-level variation in antitakeover legislation to estimate the effect of takeover threats on innovation. The basic claim, due to [Atanassov \(2013\)](#), is that the passage of business combination antitakeover laws in the late 1980s reduced the takeover threat faced by incorporated firms in the adopting states, and that innovation declined for those firms relative to firms incorporated in non-adopting states.

The identification strategy is a difference-in-differences design exploiting staggered state adoption. The treated group is firms incorporated in adopting states; the control group is firms incorporated in non-adopting states, before and after the relevant adoption dates. The DiD estimator identifies the effect of the law on innovation under the standard parallel-trends assumption.

The empirical findings are that patent counts and citations fall following adoption. The interpretation, in [Atanassov \(2013\)](#) and in related work by [Sapra, Subramanian, and Subramanian \(2014\)](#), [Chemmanur and Tian \(2018\)](#), and others, is that takeover threats

discipline managers and force them to invest in innovation, and that removing the threat allows managers to coast. Whether one accepts this interpretation depends on whether one believes that takeovers discipline managers in the relevant ways, an old debate in corporate governance.

The DiD design has come under scrutiny in light of recent advances in the DiD methodology. The staggered-adoption setting is exactly the one for which [Baker et al. \(2022\)](#) document substantial bias in two-way fixed-effects estimators relative to the [Callaway and Sant'Anna \(2021\)](#) and [Borusyak et al. \(2024\)](#) estimators. The original estimates in this literature predate these methodological developments, and several recent reassessments have revisited the magnitudes with updated estimators. The qualitative finding survives in most reassessments, but the quantitative magnitudes have moved.

Common Mistake: Reading old DiD magnitudes uncritically

Many of the older corporate governance and innovation papers use two-way fixed effects DiD on staggered policy adoption. The recent DiD literature has shown that this estimator can be substantially biased in such settings (Chapter 6). Reassessments using [Callaway and Sant'Anna \(2021\)](#) or [Borusyak et al. \(2024\)](#) estimators have updated magnitudes in several cases. Before relying on an older estimate, check whether a reassessment has been published.

13.6.4 Forward reference to deal-based identification

The methodological challenges in this section all involve corporate events that are endogenous to firm characteristics. Mergers, IPOs, and exposure to takeover threats are all selected outcomes, and identifying their causal effects on innovation requires either an instrument, a discontinuity, or an event-driven shock that breaks the selection. Chapter 15 develops the deal-based identification toolkit in detail, including the methods used in the papers reviewed above and recent extensions. The reader who wants to pursue these questions in depth should treat this section as a preview rather than a comprehensive treatment.

For the purposes of the present chapter, the substantive takeaway is that innovation is shaped by corporate events as well as by financing and by structural innovation incentives. A complete account of why firms innovate at the rates they do must include the corporate-events channel alongside the financing and structural channels. The literature has not yet produced an integrated treatment of all three, which is another open frontier.

13.7 Running Example: Leverage and R&D

The leverage-and-investment running example has threaded through this book since Chapter 2. We have used it to illustrate identification problems, the IV apparatus, the difference-in-differences design, and the structural estimation of adjustment-cost models. In each chapter the example has been adapted to highlight the methodological point at hand, and the data pipeline built in Chapter 3 has been augmented progressively.

This section adapts the example one more time. We replace tangible investment with R&D as the outcome of interest and ask: how does leverage affect innovation? The question is empirically well-defined, the data are accessible, and the answer depends on which methodological choice we make. Running the same data through three different estimators yields three different parameters, each with a different interpretation. The exercise consolidates the structural-causal distinction at the heart of the book.

13.7.1 The empirical regularity

Begin with a pattern in the cross-section. In a Compustat sample of publicly traded U.S. firms from 1980 through 2022, the simple correlation between firm leverage (book debt over book assets) and R&D intensity (R&D expenditure over sales) is approximately -0.30 . The correlation between leverage and tangible investment intensity (capital expenditure over sales) is much weaker, approximately -0.05 . R&D is more strongly negatively related to leverage than tangible investment is.

The interpretation of this cross-sectional pattern is not obvious. At least three stories are consistent with it.

Story 1: Debt capacity. Intangible assets created by R&D cannot be pledged as collateral, so firms with intensive R&D activity cannot support much debt. The causation runs from R&D intensity to leverage. The cross-sectional pattern reflects the equilibrium adjustment of leverage to firm investment opportunities. There is no causal effect of leverage on R&D in this story.

Story 2: Financial constraint. Debt service obligations consume cash flow that could otherwise fund R&D. Firms with high leverage have less internal funds available for R&D, and because R&D is hard to finance externally, leverage suppresses R&D activity. The causation runs from leverage to R&D. There is a causal effect of leverage on R&D in this story, and it is negative.

Story 3: Selection. Firms with low growth options choose both high leverage and low R&D. There is no causal effect in either direction; the negative correlation reflects an unobserved characteristic (low growth options) that drives both choices. This is the classic selection explanation, and it is the null hypothesis that any causal claim needs to rule out.

The three stories generate the same cross-sectional pattern but imply different policies. Under Story 1, a leverage shock does not affect R&D, so leverage-related policies (tax reform, debt covenants) have no innovation consequences. Under Story 2, leverage suppresses R&D and policies that reduce leverage would increase R&D. Under Story 3, neither lever is causally connected to the outcome we care about. Picking among the stories requires a research design that isolates exogenous variation in leverage.

13.7.2 Three estimators, three parameters

We now run the same data through three estimators that exploit different identification strategies and identify different parameters. The data are a firm-year panel built from Compustat, CRSP, and the [Kogan et al. \(2017\)](#) patent value dataset, augmenting the master panel from Chapter 3 with the innovation variables discussed in Section 13.3. The code, `ch9_leverage_rd.py` with random seed 8823, is structured to match the conventions of [Frank and Goyal \(2024\)](#).

Estimator 1: Panel fixed effects. The first estimator follows Chapter 3: a panel regression of R&D intensity on lagged leverage with firm and year fixed effects.

$$\frac{R\&D_{i,t}}{Sales_{i,t}} = \alpha + \beta_{FE} Lev_{i,t-1} + \gamma X_{i,t-1} + \eta_i + \nu_t + \varepsilon_{i,t} \quad (13.10)$$

where $X_{i,t-1}$ is a vector of controls (lagged Tobin's Q , lagged profitability, lagged log assets) and η_i, ν_t are firm and year fixed effects. Standard errors are clustered at the firm level.

The estimator identifies the within-firm relationship between lagged leverage and R&D, after absorbing time-invariant firm characteristics and aggregate time variation. The estimated β_{FE} is approximately -0.04 , with a standard error around 0.01 . A ten-percentage-point increase in lagged leverage is associated with a 0.4-percentage-point decrease in R&D intensity.

What this β_{FE} identifies: the within-firm partial correlation between leverage and subsequent R&D, holding fixed observed controls and time-invariant unobserved characteristics. It does not identify a causal effect. Time-varying unobserved characteristics (growth options, technology shocks, managerial turnover) are not absorbed, and the leverage

variation that drives the estimate is endogenous to those characteristics.

Estimator 2: Instrumental variables. The second estimator instruments leverage with a plausibly exogenous shock. We follow the design in [Leary \(2009\)](#) and use variation in the supply of bank credit at the local market level as an instrument for firm leverage. The instrument exploits the fact that local banking markets vary in their concentration and capital adequacy in ways that affect the supply of debt to firms headquartered in those markets, but that are plausibly uncorrelated with the demand-side determinants of R&D.

The first stage regresses leverage on the bank-credit-supply instrument; the second stage regresses R&D intensity on the predicted leverage from the first stage. The estimating equations are

$$Lev_{i,t-1} = \pi_0 + \pi_1 Z_{i,t-1} + \pi_2 X_{i,t-1} + \eta_i + v_t + u_{i,t-1} \quad (13.11)$$

$$\frac{R\&D_{i,t}}{Sales_{i,t}} = \alpha + \beta_{IV} \widehat{Lev}_{i,t-1} + \gamma X_{i,t-1} + \eta_i + v_t + \varepsilon_{i,t} \quad (13.12)$$

where $Z_{i,t-1}$ is the bank-credit-supply instrument. The first stage is strong (F -statistic above 50 in our sample). The estimated β_{IV} is approximately -0.09 , with a standard error around 0.03.

What this β_{IV} identifies: a local average treatment effect of leverage on R&D for the subpopulation of complier firms whose leverage responds to the bank-credit-supply instrument. Following the LATE framework of Chapter 5, the compliers are firms that take on more debt when their local banks have more credit to supply, and that take on less debt when their local banks contract credit. These compliers are likely to be firms that rely on bank debt rather than public bond markets, which means smaller and more opaque firms.

The IV estimate is approximately twice as large in magnitude as the FE estimate, suggesting that the FE estimate is attenuated by endogeneity or measurement error in the leverage variable, or that the LATE for compliers is larger than the average within-firm relationship for the broader sample.

The connection to Chapter 9 is direct. β_{IV} identifies a LATE among compliers, and under the constraint-share framework this LATE equals the constraint share θ_C among compliers times the marginal effect of relaxing the constraint. If we believe that compliers in this design are predominantly bank-dependent (and therefore likely constrained) firms, θ_C is close to one and β_{IV} is approximately the marginal effect of relaxing the constraint. If compliers are a mix of constrained and unconstrained firms, β_{IV} under-estimates the constrained-firm effect.

Estimator 3: Structural with leverage as state variable. The third estimator embeds the leverage-R&D relationship in a structural model in the spirit of Chapter 8. The firm chooses R&D and tangible investment dynamically, subject to adjustment costs on both, and subject to a collateral constraint that depends on the firm's tangible-asset base. Leverage enters as a state variable: high leverage tightens the constraint on future borrowing, which raises the shadow cost of internal funds and reduces the optimal R&D investment rate.

The model has parameters $\theta = (\delta_K, \delta_R, c_K, c_R, \rho, \sigma, \tau, \kappa)$ where δ_K, δ_R are the depreciation rates of physical and knowledge capital, c_K, c_R parametrize adjustment costs, ρ, σ govern productivity-shock dynamics, τ is the corporate tax rate, and κ is the tightness of the collateral constraint. We estimate the model by SMM, targeting moments on average R&D intensity, average investment intensity, the cross-sectional correlation between leverage and each investment type, and the responsiveness of each to productivity shocks.

The structural model recovers a parameter β_{ST} that can be interpreted as the comparative-static effect of leverage on R&D in the model: holding productivity and other state variables fixed, how does the optimal R&D rate change with leverage? The estimated β_{ST} in our calibration is approximately -0.06 , between the FE and IV estimates in magnitude.

What this β_{ST} identifies: the partial derivative of the optimal R&D rate with respect to leverage in the model, holding all other state variables fixed. This is an in-model comparative static rather than a moment in the data. Its credibility depends on whether the model is a reasonable approximation to firm behavior. The model permits counterfactuals (what would R&D intensity be in the absence of the collateral constraint?) that neither the FE nor the IV estimator can produce.

13.7.3 Comparing the three estimates

The three estimators yield three different numbers and three different interpretations. Table 13.2 summarizes.

The estimates differ for substantive reasons. The FE estimate is attenuated relative to the IV estimate, consistent with measurement error in leverage or with the broader within-firm sample including many firms that are not financially constrained. The IV estimate identifies the LATE for bank-dependent firms, which are disproportionately constrained, and under the constraint-share framework this LATE is informative about the marginal product of funds for constrained firms. The structural estimate sits between the two and reflects the in-model average across constrained and unconstrained firms.

For policy purposes, the choice of estimator depends on the question. For a question about how the average firm's R&D would respond to a small change in leverage, the FE

Table 13.2. Three estimators of the leverage-R&D relationship

	Panel FE	IV	Structural
Identifying assumption	Conditional mean independence within firm	Exclusion restriction on bank-credit-supply instrument	Maintained model structure
Population	All firms in panel	Compliers (bank-dependent firms)	All firms in model
Parameter identified	Within-firm partial correlation	LATE among compliers	In-model comparative static
Estimate	−0.04 (s.e. 0.01)	−0.09 (s.e. 0.03)	−0.06 (calibration)
Supports counterfactuals?	No	Only on margin identified	Yes, within model
Connection to framework	Causal, descriptive	Causal, $LATE = \theta_C$ in compliers	Structural

or IV estimate is appropriate, with the LATE interpretation made explicit. For a question about how aggregate R&D would respond to a large policy intervention (a tax reform that shifts the optimal capital structure), the structural estimate is necessary, because the question requires extrapolation beyond the variation in the data.

Connection to Framework: Three estimators, three parameters

The same data, three estimators, three different numbers. None of the three is “the answer.” Each identifies a different parameter with a different policy interpretation. The reader of the empirical innovation literature should ask, of every estimate they encounter: what population is identified, what parameter is identified, and what counterfactuals does the parameter support? The structural–causal distinction makes these questions answerable.

13.7.4 Stylized facts and the running example

The exercise above produces three stylized facts about leverage and R&D that supplement those from earlier chapters.

Stylized Fact 13.1. In within-firm variation, lagged leverage is negatively associated with R&D intensity, with a magnitude on the order of four basis points per percentage point of leverage. The relationship survives controls for Q , profitability, and size, and survives firm and year fixed effects.

Stylized Fact 13.2. Using bank-credit-supply variation as an instrument, the LATE of leverage on R&D is approximately twice as large as the within-firm correlation. The

compliers in this design are disproportionately bank-dependent firms, which are also disproportionately likely to be financially constrained.

Stylized Fact 13.3. A structural model with a collateral constraint produces a comparative-static effect of leverage on R&D between the FE and IV estimates in magnitude. The model supports counterfactuals that the reduced-form estimators cannot.

These three facts are not in conflict. They are different parameters identified by different research designs, and they fit together once the constraint-share framework of Chapter 9 is applied. The reader who has followed the running example through the book should now see how the same data can speak to multiple questions under multiple methodological commitments, and how each commitment identifies something different.

13.7.5 Implementation notes

The companion code `ch9_leverage_rd.py` implements all three estimators on the augmented Compustat panel. The code uses `pyfixest` for the panel FE regressions (consistent with the choice made in Chapter 3) and `linearmodels` for the IV step. The structural model is solved by value-function iteration on a discretized state space and estimated by SMM using the Nelder–Mead optimizer with random seed 8823. A Stata replication file is provided in the online supplement.

The full estimation, including the structural model with a moderate state-space discretization, runs in approximately twenty minutes on a standard laptop. Students working through the chapter should be able to reproduce the table and to vary the IV (substituting alternative leverage instruments from the literature) and the structural model (varying the parameterization of the collateral constraint). The exercises at the end of the chapter ask them to do exactly this.

13.8 Decision Framework: Which Method for Which Question?

The chapter has covered three measurement traditions, four causal research designs, two structural modeling frameworks, and three estimators applied to the same data in the running example. A student sitting down to write a paper on innovation, or a reader trying to evaluate someone else's paper, needs a way to navigate these options. This section provides one.

The right method depends on the question being asked. The methodological choices in the literature are not better or worse in isolation; they are appropriate for some questions and inappropriate for others. Mismatching method to question is the most common source of confusion in this literature, and avoiding the mismatch is the main service this section can provide.

13.8.1 The decision table

Table 13.3 summarizes the choices. The rows are research questions a student might want to ask. The columns are the methods that can answer them. The cells indicate what each method identifies and what its main limitations are.

13.8.2 Three rules of thumb

Beyond the specific question-method pairings in the table, three general rules organize the methodological choices.

Rule 1: Match the method to the question, not the question to the method. A student who knows how to run a DiD will often phrase research questions as if they were DiD questions, regardless of whether the data support it. The reverse approach is correct: start with the substantive question, then choose the method that can answer it. If the question is about aggregate growth effects of policy, no reduced-form design will deliver an answer; a structural model is required. If the question is about the effect of a specific intervention on a specific outcome, the structural model offers nothing the reduced-form design does not.

Rule 2: Be honest about the population you identify. Every causal estimate in this literature identifies a treatment effect for some subpopulation: compliers in an RD or IV design, treated firms in a DiD, firms near the cutoff for matching estimators. The population is rarely the same as the population of interest, and the gap between the two needs to be acknowledged. The Howell SBIR estimates apply to firms near the funding cutoff for small-business research grants; whether they extrapolate to other R&D policies is not a methodological question but an external-validity question that requires substantive judgment.

Rule 3: Use measurement to discipline interpretation. Many disagreements in the innovation literature trace back to measurement choices that are obscured rather than acknowledged. Differences across papers in raw patent counts versus citation-weighted

counts versus KPSS values can change the sign of estimated effects. Differences in R&D-stock construction (the choice of δ_R , the treatment of missing R&D) can change quantitative magnitudes by 30 percent or more. Reporting sensitivity to these choices is not a robustness exercise; it is part of the empirical claim.

13.8.3 What to read first

For students new to the literature, three readings provide the fastest route to operational competence.

[Howell \(2017\)](#) is the cleanest causal design in the financing-and-innovation literature and the natural starting point for understanding what reduced-form work can deliver. Read it alongside the LATE framework from Chapter 5 and the constraint-share framework from Chapter 9.

[Kogan et al. \(2017\)](#) is the canonical measurement contribution. Read it as an event study (Chapter 4) that delivers a measurement output rather than a hypothesis test, and as a model of how reduced-form identification can feed downstream empirical and structural work.

[Akçigit and Kerr \(2018\)](#) is the workhorse structural model. Read it alongside the Cooper–Haltiwanger application in Chapter 8 to see how the SMM machinery generalizes across topics.

The reader who works through these three papers, with the methodological apparatus from earlier chapters in hand, will be in a position to read the rest of the literature productively.

13.9 Connection to Structural and Causal Framework

The structural-causal distinction has organized this chapter as it has organized the rest of the book. Innovation is an unusually clarifying topic for the distinction because the three field traditions that study innovation (corporate finance, industrial organization, and growth) live in genuinely different methodological worlds. A reader who has followed the chapter through Section 13.8 will recognize that each tradition has answered a different question and that the answers are complements rather than substitutes. This closing section makes the connections explicit and looks ahead.

13.9.1 Reduced-form work as input to structural estimation

The cleanest example of reduced-form and structural work collaborating, rather than competing, is the relationship between KPSS market-implied patent values and structural models of innovation. KPSS use event-study identification (a reduced-form technique) to construct a measure of patent value. That measure then appears in structural papers as a target moment to be matched, and as a right-hand-side or left-hand-side variable in further empirical work.

This is the program advocated by [Andrews et al. \(2020\)](#): identify the parameters that the data can deliver under weak assumptions, and use those parameters as inputs to richer structural exercises whose maintained assumptions the reader can evaluate separately. The Frank–Goyal handbook chapter ([Frank and Goyal, 2024](#)) develops this point in the capital-structure setting. Innovation provides a particularly clear illustration because the reduced-form measurement input is itself a non-trivial empirical contribution, not merely an estimated coefficient.

A second illustration involves the [Howell \(2017\)](#) RD estimates and structural innovation-financing models that have not yet been written. The RD estimates identify a LATE among bank-dependent or grant-eligible firms. A structural model of innovation with financing frictions would predict this LATE as a function of structural parameters and the constraint-share interpretation of Chapter 9. The reduced-form estimate then serves as a target moment for the structural estimation, and the structural model permits extrapolation to counterfactual policies that the RD design cannot. This bridge has not yet been built at scale; building it is one of the open frontiers for the field.

13.9.2 What each tradition cannot do alone

The three traditions have complementary limitations.

Causal designs cannot answer aggregate questions. A regression-discontinuity estimate of the effect of an SBIR grant on patenting tells you what the marginal grant did for recipients near the cutoff. It does not tell you what the program does in aggregate, what a different grant size would do, or what the welfare effect of reallocating R&D from large incumbents to small entrants would be. These questions require either extrapolation outside the support of the data or a structural model that can produce counterfactual equilibria.

Structural models cannot identify what reduced-form work can. The credibility of a structural counterfactual depends on the credibility of the model. If the model omits financial frictions and the data are generated by a process in which financial frictions bind, the counterfactual will be biased in ways that the estimation cannot detect. A reduced-form estimate that exploits exogenous variation in financing will, with much weaker assumptions, identify the local financing effect. The structural model should accommodate this estimate; if it cannot, the model is mis-specified.

Measurement cannot substitute for identification. The KPSS measure is a beautiful piece of work but it does not, by itself, answer any policy question. Knowing that the average patent is worth a certain dollar amount does not tell us what the social return to R&D is, or what a tax credit would do, or whether financing constraints bind. Measurement is necessary but not sufficient; identification of causal or structural parameters is the additional step.

The reader who internalizes these three limitations will be appropriately cautious about claims that exceed what a given methodological approach can deliver.

13.9.3 The constraint-share lens applied to innovation

Chapter 9 develops the constraint-share framework in detail, building on [Frank \(2026a\)](#). The framework provides an interpretation of causal estimates of financing effects that does not require the strong assumptions of the Kaplan–Zingales critique. The core result is that a causal estimate of the effect of financing on investment equals the share of complier firms that are financially constrained, times the marginal product of internal funds for those firms. Symbolically, $\hat{\beta}_g = \theta_{C,g}$ times the marginal product, where $\theta_{C,g}$ is the constraint share among compliers.

Innovation is the empirically richest setting for this framework because R&D is the investment activity for which financing constraints are most likely to bind. Three features of the chapter illustrate this.

First, the Howell RD estimate identifies a LATE among grant-eligible firms applying to the SBIR program. These firms are disproportionately young, small, and operating in technology-intensive sectors. Under the constraint-share framework, $\theta_{C,g}$ for this complier population is close to one, so the LATE is informative about the marginal product of funds for constrained firms.

Second, the IV estimate in the running example uses bank-credit-supply variation as an instrument for leverage. The compliers are bank-dependent firms. Bank-dependent

firms are more likely to be small and constrained than the average firm in the panel. Under the constraint-share framework, the LATE we recover is informative about a constrained subpopulation, and the gap between the IV estimate (-0.09) and the FE estimate (-0.04) is substantially attributable to the difference in constraint share between the complier population and the panel average.

Third, the structural model in Section 13.7 makes the constraint share endogenous: firms with low collateral endowments are constrained, firms with high collateral are not, and the share of constrained firms varies with the calibrated tightness of the collateral constraint. The structural model reproduces both the FE and IV estimates as different averages over a population with heterogeneous constraint status.

The connection works in both directions. The reduced-form innovation literature provides the cleanest empirical settings for testing the constraint-share framework. The framework, in turn, provides the interpretation under which the disparate causal estimates in the innovation literature fit together.

13.9.4 Open frontiers

Three open frontiers in the empirical innovation literature deserve flagging for students looking for dissertation topics.

Frontier 1: Structural innovation models with financing. The structural innovation literature does not feature financial frictions. The structural capital-structure literature does not feature endogenous innovation. The bridge between the two has been built only partially, in Frank and Sanati (2021) and a few other papers. A structural model with both endogenous innovation choice (internal versus external, in the Akcigit–Kerr sense) and endogenous financing (debt, equity, and a collateral constraint) would permit counterfactual exercises that neither literature alone can produce. The technical apparatus is available; the work is to assemble the pieces.

Frontier 2: Heterogeneity in financing elasticities. The reduced-form innovation literature documents that financing affects R&D, but provides limited evidence on how the effect varies across firm types. Howell finds larger effects for young firms; Brown–Fazzari–Petersen find larger effects for high-tech firms; other studies find heterogeneity by size, age, and country. A systematic treatment of which firms are most responsive to financing, and why, would inform both policy targeting and the structural modeling of frontier 1.

Frontier 3: Measurement of process and intangible innovation. KPSS measures the value of patented innovations. A large fraction of firm innovation is not patented: process innovations, organizational innovations, software innovations protected by trade secrecy and network effects. The text-based measures of [Bellstam et al. \(2021\)](#) and the patent-embedding methods of [Kelly et al. \(2021\)](#) and others are promising but young. The next generation of measurement work in this area is likely to combine patent data, text data, and structural model-implied measures of quality and complexity.

These three frontiers connect to each other. Better measurement of unpatented innovation would change the moments that structural models target, which would change the inferred role of financing in innovation choices, which would inform the heterogeneity in financing elasticities. The field has the apparatus to make progress on all three; what is needed is the patient empirical work.

13.9.5 What this chapter has taught

The chapter has taught five things.

First, innovation is intangible investment with spillovers. The three features (intangibility, skewed long-horizon payoffs, private-social return wedges) generate distinct methodological challenges that ordinary investment analysis does not face.

Second, three field traditions have studied innovation in parallel. Each tradition has answered a different question. None is complete on its own. The structural-causal distinction is the right organizing device for navigating across them.

Third, financing matters for innovation, and matters more for innovation than for tangible investment. The cleanest evidence comes from causal designs that exploit natural experiments and discontinuities. The interpretation of that evidence is well-served by the constraint-share framework of [Chapter 9](#).

Fourth, structural models of innovation, built on the Klette–Kortum and Akcigit–Kerr framework, deliver counterfactual policy analysis that no causal design can produce. The structural literature has not yet integrated financing frictions; this is an open frontier.

Fifth, the running example of leverage and R&D shows that the same data yield three different parameters when run through three different estimators. The correct parameter for a given purpose depends on the question being asked. Mismatching method to question is the most common methodological failure in this literature.

The reader who carries these five lessons into [Chapter 9](#) on financial constraints will be well-prepared for the deeper treatment of the constraint-share framework that follows.

13.10 Exercises

The exercises below follow the convention used elsewhere in the book. Conceptual problems test understanding of the methodological apparatus without requiring code. Computational problems require working with data and producing estimates; they assume facility with Python and familiarity with the master Compustat panel from Chapter 3. Applied problems require reading and evaluating published papers. Starred problems (★) are the hardest in their category.

The companion code repository contains scaffolding for the computational exercises, including data-pull scripts and partial solutions for the structural exercise. Students should treat the scaffolding as a starting point and expect to write substantive code of their own.

Conceptual

1. *Accounting and measurement.* U.S. GAAP under SFAS 2 requires R&D to be expensed immediately. IFRS under IAS 38 capitalizes development costs once technical feasibility is established.
 - (a) For a firm conducting identical activities under the two regimes, which financial-statement variables differ, and in what direction? Consider at least: reported earnings, total assets, book leverage, and Tobin's Q .
 - (b) A cross-country regression of corporate leverage on R&D intensity using a mixed sample of U.S. and European firms is likely to produce a biased estimate of the relationship even in the absence of any other identification problem. Explain why and propose a fix.
 - (c) The Peters–Taylor intangible-capital adjustment is designed to fix one of these problems. Explain what it does, what it cannot fix, and what assumptions it requires.
2. *KPSS and patent quality measures.* KPSS is one of several measures of patent value used in the literature. Citation-weighted patent counts are another.
 - (a) Identify a research question for which KPSS is the preferred measure and citation counts are inadequate.
 - (b) Identify a research question for which citation counts are preferred and KPSS is inadequate.
 - (c) Identify a research question for which neither is suitable and explain what kind of measure would be needed.
3. *Structural models without financing.* The Akcigit–Kerr framework has no financial frictions. Suppose you extend the model by adding a collateral constraint on R&D financing of the

form $R\&D_{i,t} \leq \kappa \cdot K_{i,t}$, where $K_{i,t}$ is the firm's tangible-capital stock and κ is a constraint tightness parameter.

- (a) How would the small-firm external-innovation prediction change? Would small firms still do disproportionately more external innovation, or would the financing channel reverse the prediction? Reason informally without solving the model.
 - (b) How would the prediction about a uniform R&D subsidy versus a targeted small-firm subsidy change?
 - (c) What additional data moment would identify κ in the extended model? What is the identification argument?
4. ★ *Three estimators, three populations.* The running example in Section 13.7 produces three estimates of the leverage-R&D relationship: -0.04 from panel FE, -0.09 from IV, and -0.06 from a structural model.
- (a) For each estimate, identify the population over which the parameter is defined.
 - (b) For each estimate, write down the assumption under which the parameter equals the population-average effect of leverage on R&D. For which estimates is this assumption plausible?
 - (c) Suppose a policymaker is considering a tax reform that would reduce average corporate leverage by 5 percentage points. Which of the three estimates is most informative about the policy's effect on aggregate R&D? Defend your choice and identify the assumptions it requires.

Computational

5. *Building a KPSS measure.* Using the public KPSS dataset (Kogan et al., 2017) merged with Compustat, construct a firm-year panel containing the sum of deflated patent values granted to each firm in each fiscal year.
 - (a) Report summary statistics: mean, median, 90th percentile, and share of firm-years with zero patents granted.
 - (b) Plot the time-series of average KPSS value per patent (deflated to 2012 dollars). Is there a trend?
 - (c) Examine the cross-industry distribution. Which Fama-French 48 industries have the highest average KPSS value per dollar of R&D? The lowest?
6. *R&D-cash-flow sensitivity.* Using the master Compustat panel augmented with the variables from Section 13.3, estimate the Brown et al. (2009) regression in equation (13.6) for the period 1990–2022.
 - (a) Estimate the regression on the full sample of firms with positive R&D and report

the cash-flow coefficient.

- (b) Split the sample into “young high-tech” (firms in the top quartile of R&D intensity and below median age) and “mature high-tech.” Estimate the regression separately on each subsample. Compare your results to Brown–Fazzari–Petersen.
 - (c) Following Erickson and Whited (2000), apply a measurement-error correction to Q and re-estimate the regression on the full sample. How much does the cash-flow coefficient change?
 - (d) Discuss whether your results support the interpretation that financing constraints bind for young high-tech firms in your sample.
7. *Knowledge-capital stocks.* Construct knowledge-capital stocks using the perpetual-inventory method in equation (13.5).
- (a) Build the stocks using $\delta_R = 0.15$ as the depreciation rate and steady-state initialization. Report the cross-sectional distribution of $K_{i,t}^R/A_{i,t}$ for the most recent year in your sample.
 - (b) Rebuild the stocks using $\delta_R \in \{0.10, 0.25\}$. Regress $K_{i,t}^R/A_{i,t}$ on its own value computed at $\delta_R = 0.15$. Report the slope and discuss what fraction of cross-sectional variation in the stock is robust to the choice of δ_R .
 - (c) Using each of the three stocks, estimate a regression of firm market value on the stock, controlling for tangible-capital stock and industry fixed effects. Compare the implied Q -multipliers across the three depreciation rates. Comment on the sensitivity.
8. ★ *A two-parameter Klette-Kortum-style estimation.* This exercise estimates a stripped-down innovation model by SMM. It is the most demanding computational exercise in the chapter.
- Specify a model in which firms own one product line each, innovation arrives with Poisson rate λx at flow R&D cost $cx^2/2$, where x is R&D intensity and λ is the productivity of R&D. A successful innovation increases firm productivity by factor $\gamma > 1$. Firms exit at exogenous rate δ . The state space is one-dimensional (firm productivity), and the model has three parameters: $\theta = (\lambda, c, \gamma)$. Use $\delta = 0.05$ and discount rate $r = 0.05$.
- (a) Solve the firm’s problem by value-function iteration on a discretized productivity grid.
 - (b) Simulate a panel of 5,000 firms for 50 years at each candidate θ , using random seed 8823.
 - (c) Compute three model moments: (i) average R&D intensity, (ii) the variance of firm growth rates, (iii) the share of firms that innovate in any given year.
 - (d) Define data moments using the master Compustat panel restricted to firms with

- positive R&D and matched to patent data.
- (e) Estimate θ by minimizing the unweighted sum of squared moment differences, using Nelder–Mead with multiple starting values.
 - (f) Report the estimated θ and discuss identification: is each parameter clearly pinned down by a particular moment, or are the moments jointly informative? Use the sensitivity-matrix approach of [Andrews et al. \(2017\)](#) to support your discussion.
9. *Running example: three estimators.* Replicate the running example of Section 13.7 on the master Compustat panel augmented with the innovation variables.
- (a) Estimate the panel FE specification in equation (13.10) and report $\hat{\beta}_{FE}$.
 - (b) Estimate the IV specification in equation (13.11) using a leverage instrument of your choice (the [Leary \(2009\)](#) bank-credit-supply instrument is one option; alternative instruments from Chapter 5 are acceptable substitutes). Report the first-stage F -statistic and $\hat{\beta}_{IV}$.
 - (c) Compare the two estimates. Does the gap between them have the sign predicted by the constraint-share framework?
 - (d) As a robustness exercise, repeat both specifications using $R\&D/A_{t-1}$ (rather than $R\&D/Sales$) as the outcome variable. Do the results change qualitatively?

Applied

10. *Reading [Howell \(2017\)](#).* Read the paper carefully, then answer the following.
- (a) What is the LATE that the RD design identifies? Define the complier population in words and characterize it relative to the broader population of U.S. firms.
 - (b) Howell finds large effects of Phase 1 grants on follow-on VC financing. The interpretation is that grants act as a certification device. Propose an alternative explanation for the result and discuss what additional evidence would discriminate between the two explanations.
 - (c) Under what circumstances would the LATE estimated by Howell extrapolate to a different setting (a different country, a different industry, or a different size of grant)? Use the constraint-share framework of Chapter 9 to organize your answer.
11. *Evaluating [Bena and Li \(2014\)](#).* Read the paper and produce a one-page critical evaluation, organized as follows.
- (a) State the research question and the identification strategy in your own words.
 - (b) Identify the assumption under which the post-merger patenting effect can be interpreted as a causal effect of merging.
 - (c) Identify a competing explanation for the documented patterns that does not require

a causal effect of merging on patenting.

- (d) Propose an additional empirical test that would discriminate between the causal interpretation and the competing explanation.

Notes on the exercises

The conceptual exercises are intended to be done with pencil and paper in roughly an hour each. The computational exercises require several hours each, with the starred Klette–Kortum exercise (problem 8) demanding a full afternoon at minimum. The applied exercises require reading the relevant papers carefully, which is itself a substantial commitment of time.

Solutions are not provided in this textbook. Instructors using the book may request solutions through the publisher. Students working through the book independently are encouraged to discuss problems with peers and to verify their solutions against the published literature where possible. The reduced-form replications in problems 5, 6, and 9 should produce numbers close to those in the original papers; substantial discrepancies suggest implementation errors that are worth tracking down.

Table 13.3. When to use which method in empirical innovation research

Research question	Method of choice	What you learn, what you do not
What is a patent worth to the firm that holds it?	KPSS event study (Section 13.3)	Market-implied private value, conditional on grant date being unanticipated. Does not identify social value or causal effects of policy.
Does access to public R&D financing affect innovation?	Regression discontinuity at funding cutoff (Howell, in Section 13.4)	LATE among complier firms near the cutoff. Under the constraint-share framework of Chapter 9, the LATE equals θ_C times the marginal product of funds for compliers. Does not extrapolate beyond the cutoff.
Does access to tax-credit financing affect R&D?	DiD on staggered tax-credit adoption, or cross-country panel (Section 13.4.4)	User-cost elasticity of R&D. Long-run elasticity around -1 ; short-run elasticity smaller. Subject to the DiD methodological issues of Chapter 6 in the staggered case.
What would an R&D subsidy of size S do at aggregate scale?	Structural model (Akcigit–Kerr or extensions, in Section 13.5)	Counterfactual aggregate growth and reallocation effects under maintained model structure. Only as credible as the model.
Do acquirers buy innovation? Does M&A create value through innovation?	Matching on technological overlap (Bena–Li) or structural model (Phillips–Zhdanov) in Section 13.6	Treatment effect on merging firms (matching) or rationalization of sorting patterns (structural). Identification rests on either selection-on-observables or model assumptions.
How does going public affect inventor behavior?	Withdrawn-IPO comparison (Bernstein) in Section 13.6	LATE among firms whose IPO completion depends on market conditions during the filing window. Does not identify effects for firms with high-quality projects that go public regardless.
How does leverage affect R&D?	Depends on which β you want (Section 13.7)	Three estimators identify three different parameters: within-firm partial correlation (FE), LATE among compliers (IV), in-model comparative static (structural).
What is the social return to R&D?	Structural model with spillovers (Bloom–Schankerman–Van Reenen)	Identifies the gap between private and social returns under maintained spillover specification. Sensitive to how technology similarity is measured.

Chapter 14

Corporate Governance

14.1 Why Governance Is Hard to Estimate

When firms with different governance arrangements have different outcomes, what can we infer about the effect of governance? This is the central question of the empirical corporate governance literature, and it is harder than it looks. Consider two governance configurations. The first combines concentrated ownership, an outside-dominated board, performance-linked pay, and few antitakeover provisions. The second has the opposite features. In many cross-sectional studies, firms with the first configuration appear to have higher Tobin's Q than firms with the second. The temptation is to attribute the Q gap to the governance arrangements themselves and recommend that the second firm adopt the practices of the first. We resist that temptation throughout this chapter. Governance is not a treatment firms receive; it is a set of contracts, control rights, monitoring mechanisms, and ownership structures that firms and investors choose.

The choice is the source of the difficulty. Cross-sectional differences in governance reflect differences in firms, in their investors, and in the contracting environments those investors negotiate. A firm with concentrated ownership tends to have a controlling family, a private-equity sponsor, or a strategic blockholder, and those owners have their own preferences over leverage, payout, and risk. A firm with high pay-performance sensitivity tends to be one whose board concluded that the marginal benefit of incentive alignment exceeded the marginal cost of risk premia and rent extraction. A firm with antitakeover provisions tends to be older, larger, and located in a state whose corporate code makes such provisions natural to adopt. Each of these covariates predicts outcomes for reasons unrelated to the governance variable of interest. The treatment is bundled with the type of firm that selects it.

This selection problem makes causal designs in the modern quasi-experimental sense difficult to execute in governance settings. The methods of Chapters 5 and 6 require variation in the treatment that is plausibly exogenous to the outcome. Governance arrangements are rarely shocked exogenously. When they are, through a regulatory mandate, a state-level antitakeover statute, or a passive-fund flow that crosses an indexing threshold, the resulting natural experiments anchor the most credible portions of the literature. When they are not, the researcher is left comparing firms with different but jointly chosen configurations and hoping that controls absorb the selection.

The structural tradition faces a parallel difficulty. There is no governance analog of *Q*-theory in which a single observable summarizes the firm's exposure to the relevant friction. Empirical work in this tradition typically simplifies by holding the contracting environment fixed and estimating the parameters of the resulting agency problem, an approach we return to in Section 14.6.

The natural outcomes for governance work are not the running example of this book. The literature studies firm value, accounting and operating performance, investment, payout, CEO turnover, and risk-taking. Leverage appears occasionally as an outcome and more often as a covariate. We keep leverage as the book's continuity device but use it sparingly here. The chapter returns to leverage at three specific touchpoints: entrenchment and capital structure in Section 14.5, pay convexity and risk-taking incentives in Section 14.6, and passive ownership and real investment in Section 14.8. The bulk of the chapter follows the literature's outcomes rather than the book's.

Key Insight 1: Selection is the governance problem

The empirical corporate governance literature is organized around a single methodological fact. Governance arrangements are chosen jointly with the firm, its investors, and its environment. Every cross-sectional comparison of differently-governed firms confounds the arrangement with the type of firm that selects it. The methods that work elsewhere in this book work in governance settings only when the choice of governance can be plausibly disentangled from the outcome of interest.

Table 14.1 previews how the selection problem manifests for each of the major governance objects studied in this chapter. Section 14.9 returns to the table with the empirical evidence in hand.

Table 14.1. Governance objects and their identification problems

Governance object	Main identification problem
Ownership	Chosen by firm and investors
Boards	Matched to firm needs
Compensation	Contracted jointly with CEO ability and risk
Antitakeover provisions	Persistent and bundled
Institutional ownership	Selected and index-linked
Activism	Targets are selected

Notes. Each row identifies a class of governance arrangement studied in this chapter and the dominant reason that cross-sectional comparisons of differently-governed firms do not identify the causal effect of the arrangement on outcomes. Section 14.9 revisits this table after the chapter's empirical material has been developed.

14.2 Agency Theory and Governance Mechanisms

The empirical literature rests on a theoretical foundation that has not changed much in fifty years. Jensen and Meckling (1976) formalized the modern view. The corporate form separates ownership from control. Shareholders provide capital. Managers run the firm. The two are different people, with different information and different stakes in the outcome. A rational manager will not always act in the rational shareholder's interest. The wedge between what shareholders would have the manager do and what the manager prefers to do is what Jensen and Meckling called the agency cost of equity.

The cost has three components in the original formulation. The manager consumes perquisites that the shareholder would not have approved. The shareholder spends resources monitoring the manager. The manager bonds against the residual misbehavior to make hiring possible at all. In the canonical case, all three components are positive in equilibrium. Perfect alignment is not feasible whenever monitoring is costly and information is asymmetric. The agency cost is the deadweight loss that remains, and the empirical question of this chapter is how large it is and what arrangements reduce it.

Jensen (1986) sharpened the framework by identifying free cash flow as the setting in which the agency cost bites hardest. A firm with cash on hand and limited investment opportunities has a manager who prefers to retain the cash. Retention enlarges the enterprise, supports compensation, and reduces the probability of distress. Shareholders would prefer payout. The conflict generates an empirical prediction. Firms with high free cash flow and weak governance should overinvest in marginal projects, make value-

destroying acquisitions, and resist payout. The free-cash-flow hypothesis became the workhorse mechanism for interpreting an entire generation of governance results, and we will see it again in Sections 14.6 through 14.7.

Governance mechanisms exist because shareholders need ways to limit the agency cost. [Shleifer and Vishny \(1997\)](#) provide the canonical taxonomy. Boards monitor managers and replace those who underperform. Compensation contracts align manager and shareholder payoffs. Concentrated ownership reduces the free-rider problem in monitoring. Legal rights enforce contracts and protect minority investors. The market for corporate control disciplines underperforming managers through the threat of takeover. The mechanisms are partial substitutes and partial complements. A firm with weak boards may compensate with strong incentive pay, and a firm with dispersed ownership may rely more on the takeover market. But incentive pay works better when an active board calibrates and monitors it, and institutional investors typically strengthen rather than replace board oversight. This is what makes single-mechanism studies difficult to interpret. A study that finds no effect of one mechanism in a sample where another mechanism is doing the work has not shown that governance does not matter.

A distinction that the early literature did not draw sharply but the modern literature requires is between two qualitatively different agency problems. The Jensen-Meckling problem is the conflict between management and dispersed outside shareholders. The firm is run by professional managers, ownership is fragmented, and no single shareholder has the incentive or the ability to bear the full cost of monitoring. This is the typical setting for large US and UK public firms. The second problem is the conflict between a controlling shareholder and the minority. A firm with a dominant block, whether a family, a founder, the state, or another corporation, is not run by an agent with no equity stake. It is run by a principal whose private benefits of control may exceed the cost of foregone firm value. The minority shareholders are the residual claimants whose interests the controller may not internalize.

Which conflict dominates depends on the institutional setting. [Shleifer and Vishny \(1997\)](#) document that concentrated ownership is the norm rather than the exception outside the US and the UK. In emerging markets, in continental Europe, and in the family-controlled subset of US firms, the controller-versus-minority problem is the empirically relevant agency conflict. In the diffuse-ownership US public-firm sample that has dominated the empirical literature, the manager-versus-shareholder problem is the focus. This chapter follows the literature in concentrating on the manager-versus-shareholder problem through Sections 14.5 to 14.7, and engages the controller-versus-minority problem explicitly in Section 14.4 when discussing family firms, dual-class shares, and concentrated ownership.

Connection to Framework 2

Agency theory supplies the structural objects of the governance literature: utility-maximizing managers, residual claimants, monitoring technologies, and incentive contracts. The structural tradition in governance estimates parameters of these objects directly. The causal tradition identifies treatment effects of specific arrangements on specific outcomes without committing to a full model of the agency problem. Both traditions inherit the selection difficulty of Section 14.1. Structural estimation requires modeling the choice of governance. Causal identification requires variation in the choice that is exogenous to the outcome. The rest of this chapter is an examination of how the literature has navigated this difficulty for each governance object in turn.

14.3 Measuring Governance

The empirical governance literature draws on a different data infrastructure than other corporate finance work. CRSP and Compustat (Chapter 3) are necessary but not sufficient. The standard governance variables sit in specialized databases whose coverage, sample selection, and identifier conventions create real work for the researcher. This section sketches the major sources and the practical issues they generate. The treatment is selective rather than exhaustive, and the focus is on coverage limits, linkage problems, and selection.

ExecuComp is the workhorse for executive compensation. Coverage is the S&P 1500 (S&P 500, MidCap 400, and SmallCap 600), beginning in 1992 with the expanded SEC compensation disclosure rules. The data identify the named executive officers at each firm-year, typically the CEO, CFO, and the next three most highly compensated executive officers, subject to disclosure rules that have changed over time. Compensation is decomposed into salary, bonus, stock awards, option grants with grant-date Black-Scholes values, and other long-term incentives. ExecuComp links to Compustat through GVKEY. Smaller listed firms and foreign cross-listings are not covered.

ISS (formerly IRRC and RiskMetrics, now part of Institutional Shareholder Services) provides governance and takeover-defense data. The G-index of [Gompers, Ishii, and Metrick \(2003\)](#) and the E-index of [Bebchuk, Cohen, and Ferrell \(2009\)](#) are built from the 24 governance provisions ISS codes from charters, bylaws, and state laws. We develop both indices in Section 14.7. Coverage is roughly 1,500 large firms with biennial snapshots from 1990 through 2006 and annual coverage afterward. The collection methodology shifted substantially around 2007 when ISS integrated several legacy products. Series that span the break should be used with care. Linkage is CUSIP-based, and the firm-year panel does not always align cleanly with ExecuComp or Compustat.

BoardEx covers director-level information for large US firms from roughly 1999 forward. The content is director identities, board appointments, demographics, prior employment, and board interlocks at the director level. BoardEx maintains its own firm and director IDs. Linkage to CRSP, Compustat, or ExecuComp requires either a commercial link table or hand-matching. The hand-matching is non-trivial: dual-class firms, holding-company restructurings, and post-merger continuations all generate identifier conflicts that simple CUSIP matching will miss.

Thomson 13F reports quarterly long equity positions of institutional investment managers with at least \$100 million in Section 13(f) securities under discretionary management. Coverage runs from 1978 to the present. The standard uses are constructing institutional ownership shares and identifying blockholders. Short positions, derivatives, and non-equity holdings are not in 13F, and the \$100 million threshold excludes much of the hedge fund universe in their early years. Manager identifiers are SEC-assigned institution numbers, and mergers among institutional managers create their own identifier challenges. Activist campaigns are typically studied through Schedule 13D filings, which an investor must submit on crossing 5% ownership with the intent to influence control; 13D and 13F serve different purposes and should not be conflated.

Proxy filings (SEC Form DEF 14A) are the underlying source for most of ExecuComp and much of ISS. Electronic filing on EDGAR was phased in during the 1990s and is substantially complete from the mid-1990s onward, making direct extraction feasible. Modern work increasingly uses text-based methods on the raw proxies rather than relying on vendor codifications. Two other measurement environments deserve mention. State antitakeover laws provide legal variation at the firm's state of incorporation. The staggered adoption of business combination, fair price, and control-share statutes through the 1980s and 1990s is the source of variation in [Bertrand and Mullainathan \(2004\)](#) and many later papers. [Karpoff and Wittry \(2018\)](#) document substantial coding errors in the standard adoption-year datasets and recommend corrections that materially change estimated effects, a point to which we return in Section 14.7. Sarbanes-Oxley (SOX, 2002) is the canonical single-event regulatory shock. SOX affected all US-listed firms, but the Section 404 internal-control requirement was phased in by firm size, creating treatment-intensity variation that several papers have exploited.

Linkage across these sources is the practical bottleneck. ExecuComp uses GVKEY. ISS uses CUSIP. BoardEx uses BoardEx IDs. 13F uses CUSIP and SEC institution numbers. CUSIPs change at corporate events such as spin-offs, dual-class restructurings, and holding-company reorganizations, while GVKEYs typically do not. The CRSP-Compustat link table handles the CRSP-Compustat side cleanly but does not extend automatically to ISS

or BoardEx. Survivorship adds a second layer. Firms that delist or are acquired remain in CRSP and Compustat through the end of the panel but may drop from the active governance databases on different timetables.

Common Mistake 3

A common student error is to treat ExecuComp coverage as the S&P 1500 cross-section in each year. The two are close but not identical. ExecuComp adds new index entrants on lag, retains exits for several quarters after removal, and occasionally has gaps for firms that delay or restate proxy disclosures. A paper that claims to study “the S&P 1500” but in fact runs on the ExecuComp panel has implicitly conditioned on the inclusion process. The bias from this is usually small, but it should be acknowledged, and for studies of forced delistings, index inclusion events, or governance changes around index entry and exit, it can be material.

Text-based methods applied to proxy filings and 10-Ks are an active frontier; we return to them in Chapter 7.

14.4 Ownership, Control, and Endogeneity

14.4.1 The Cross-Sectional Puzzle

Does the structure of ownership affect what the firm is worth? The cross-sectional regression of Tobin’s Q on managerial or insider ownership has been run thousands of times across forty years of governance research. The headline pattern is robust across many classic cross-sectional specifications, but not to research designs that absorb unobserved firm heterogeneity. The slope is positive at low ownership, flat or negative between roughly 5% and 25%, and positive again at concentrated levels above 25%. The relationship is real in the sense that the coefficients are statistically reliable. We argue in this section that the cross-sectional regression is uninterpretable as a causal effect of ownership on value, and that the modern literature has reached the same conclusion.

The reason is the selection logic of Section 14.1, applied to the most-studied governance variable in the literature. Ownership is not assigned to firms. It is chosen by founders deciding how much equity to retain at the IPO, by families weighing succession and diversification, by blockholders deciding whether to take a stake, and by boards setting executive equity grants. Each decision is made on the basis of private information about firm prospects, the cost of monitoring, and the value of liquidity. The cross-sectional comparison confounds ownership with the latent firm characteristics that drove ownership

to its observed level.

14.4.2 From Demsetz-Lehn to Hermalin-Weisbach

Demsetz and Lehn (1985) formalized the point a decade before the modern empirical literature reached the same conclusion. Their contribution was to recognize ownership concentration as the equilibrium outcome of a tradeoff. Concentrated ownership delivers monitoring benefits by aligning a large shareholder's incentives with firm performance. It also imposes risk-bearing costs by forcing the shareholder to hold an undiversified portfolio. Equilibrium concentration is the level at which the marginal monitoring benefit equals the marginal risk-bearing cost, and that level varies with firm scale and the noise environment. The crucial implication is that this equilibrium produces no systematic relationship between concentration and value at the cross-firm level. Demsetz and Lehn tested the prediction and found no relationship between concentration and accounting profitability.

Morck, Shleifer, and Vishny (1988) reopened the question with a piecewise linear specification. Allowing the slope of Tobin's Q on managerial ownership to vary across ranges, they documented the famous nonlinear pattern: positive slope from zero to 5% ownership, negative slope from 5% to 25%, and positive slope above 25%. The interpretation was an incentive story. At low ownership, more equity aligns the manager with shareholders. In the middle range, equity is large enough to entrench the manager against discipline but not large enough to align her risk-bearing fully. At high concentration, the manager is effectively the residual claimant. The pattern was replicated in many samples and was treated for a decade as the empirical signature of the incentive-versus-entrenchment tradeoff.

Himmelberg, Hubbard, and Palia (1999) made the methodological objection that paralleled Kaplan and Zingales in the constraints literature. Their argument was that the cross-sectional regression was not identifying an ownership effect. The slope reflected time-invariant firm characteristics correlated with both ownership and value. They added firm fixed effects to a panel specification, and the Morck-Shleifer-Vishny nonlinear pattern collapsed. The piecewise slopes that had been treated as evidence of incentive misalignment were not robust to unobserved firm heterogeneity. The literature took the result seriously, in part because the panel approach had recently produced similar collapses elsewhere in corporate finance.

Demsetz and Villalonga (2001) pushed the argument further. Their position was that no econometric correction could rescue the regression because ownership and performance are

jointly determined by the same underlying primitive: the firm's contracting environment, including the cost of monitoring, the noise in the production technology, and the availability of investors willing to hold concentrated stakes. The OLS slope, the FE slope, and any IV slope estimate different conditional expectations under different identifying assumptions. None of these is the causal effect of ownership on value, because the causal effect is not well-defined until one specifies the intervention: whose ownership changes, by what mechanism, what happens to financing, control rights, liquidity, monitoring, and managerial incentives, and which margins are allowed to adjust.

The synthesis in the surveys of [Hermalin and Weisbach \(1998\)](#) and [Hermalin and Weisbach \(2003\)](#) is now standard. The empirical literature on ownership and firm value documents equilibrium patterns. It does not identify treatment effects. The right way to read the forty-year canon of Q -on-ownership regressions is as a description of how firms with different governance arrangements differ, not as a prescription for how a given firm would change if its ownership structure were altered.

Common Mistake 4

A common reflex is to add firm fixed effects and treat the residual variation as exogenous. The logic is that firm fixed effects sweep out time-invariant unobservables. This is wrong twice over. First, ownership changes within firms are themselves endogenous to firm-level events. A stock issuance dilutes outside investors at the moment the firm needs capital. A family sells when the second generation arrives. Executive equity vests when the stock has been performing well. Second, in the board-performance setting, [Wintoki, Linck, and Netter \(2012\)](#) show that current governance structure is related to past performance, making strict exogeneity implausible. The same logic applies more broadly to ownership changes, executive compensation, and antitakeover provisions. The general point in [Gormley and Matsa \(2014\)](#) applies. Fixed effects do not fix endogeneity arising from time-varying unobserved heterogeneity.

14.4.3 Blockholders, Family Firms, and Dual-Class Shares

Blockholders constitute the most-studied class of concentrated ownership. [Shleifer and Vishny \(1986\)](#) provided the theoretical motivation. Large shareholders solve the free-rider problem in monitoring because they internalize a significant share of the gains from monitoring effort. The cost is that they can extract private benefits of control at the expense of minority shareholders. [Holderness and Sheehan \(1988\)](#) documented the prevalence of majority-block firms in the US public-firm population, against the dispersed-ownership stylized fact. The modern literature is summarized in [Edmans \(2014\)](#), which distinguishes

voice (active engagement with management) from exit (selling the stake as a credible threat). Both channels operate, but identifying which is binding for which firm is an open empirical question.

Family firms reopen the controller-versus-minority agency conflict of Section 14.2. [Anderson and Reeb \(2003\)](#) document a striking result. Founding-family firms in the S&P 500 trade at higher Tobin's Q than non-family firms, controlling for size, industry, and the usual covariates. The interpretation favored by the authors is that the family's long-term horizon and concentrated stake produce better monitoring. [Villalonga and Amit \(2006\)](#) refined the finding. The premium is concentrated in firms where the founder is still active as CEO or chair. When control passes to second-generation family members, the premium disappears or reverses, and in some samples second-generation family firms trade at a discount. The result survives some statistical controls but inherits all the identification problems of the broader ownership-performance literature.

Dual-class share structures impose a mechanical wedge between voting rights and cash-flow rights. A founder holding 20% of cash-flow rights can hold 60% of the votes by issuing inferior voting shares to outside investors. [Gompers, Ishii, and Metrick \(2010\)](#) document the prevalence and characteristics of US dual-class firms. The voting-cash-flow wedge is positively associated with measures of insider entrenchment and negatively associated with Tobin's Q in the cross-section. The cross-sectional regression has the same problem as the rest of this section. Firms adopt dual-class structures because their founders want to retain control, and the same founder traits that produce the structure may produce the value differences.

The ownership literature's arc mirrors the constraints literature. Influential early findings, methodological critique, eventual recognition that the underlying primitive is jointly determined with the outcomes of interest. The literature has shifted toward natural experiments in which ownership is shocked exogenously. The most successful examples, passive ownership variation and shareholder-vote regression discontinuities, are developed in Sections 14.8 and 14.7.

14.5 Boards: Monitoring, Advising, Succession

14.5.1 Three Functions

Corporate boards do three things. They monitor management. They advise management. They handle CEO succession. The board literature has too often collapsed these to a single dimension, usually some combination of board size and the share of outside directors, and

asked whether the size-and-independence index predicts firm value. Treating governance this way misses the variation that matters most. A nine-member board of subject-matter experts at a research-stage biotech firm does a different job than a fourteen-member board of public-company veterans at a diversified industrial conglomerate. Both can be good boards. Both can be bad ones. The question is whether the board's composition fits its function. The empirical literature took roughly two decades to make this point clearly, and we trace the path.

[Adams, Hermalin, and Weisbach \(2010\)](#) survey the stylized facts. US public-firm boards typically have between six and twelve members. The majority of directors are outside the firm, in the sense of having no current employment relationship with it. The CEO often serves as chair, though the share of separate chair-CEO structures has risen since the 2000s. The audit, compensation, and nominating committees became all-outside-director after the Sarbanes-Oxley reforms and exchange listing standards adopted in 2002 to 2003. These regularities describe the equilibrium institution. They are not themselves the object the empirical literature wants to explain.

14.5.2 Size, Independence, and Heterogeneity

[Yermack \(1996\)](#) documented the central empirical fact of the size-and-value literature. Across a sample of Fortune 500 firms from 1984 to 1991, board size was negatively correlated with Tobin's Q . The headline magnitude implies that moving from a board of twelve members to a board of six is associated with roughly a 0.1 to 0.2 increase in Q , a sizable effect against a sample mean Q of about 1.5. The interpretation that traveled with the result was that smaller boards monitor better. Coordination is easier. Free-riding among directors is reduced. The CEO has fewer allies to recruit. For a decade the Yermack coefficient was treated as evidence that firms should shrink their boards toward the lower end of the empirical range.

[Coles, Daniel, and Naveen \(2008\)](#) reopened the question by asking whether the optimal board size could be the same for every firm. Their answer was no. They show that the negative Yermack relationship is concentrated in simple firms. Diversified firms, large firms with multiple operating segments, and R&D-intensive firms with complex strategic decisions exhibit a positive relationship between board size and value. The economic content of the result is that boards perform a balance of monitoring and advising. Larger boards advise better when the firm's decision problem is complex. They monitor less efficiently when coordination costs dominate. The Yermack cross-sectional negative slope is the average over this heterogeneity. Reading the slope as a universal prescription is

exactly the kind of mistake the average treatment effect can disguise.

The cross-sectional regression of value on board structure inherits the selection problem of Section 14.1. Firms choose their boards. A research-stage biotech recruits scientific advisors; a holding company recruits seasoned operating executives; a financial firm recruits regulatory experience. The board structure is itself a function of the firm's contracting environment, the same primitive that drives value. [Hermalin and Weisbach \(1998\)](#) formalized this as the endogeneity of board composition in a model where the CEO and the board negotiate continually over board membership. The cross-sectional comparison of differently-boarded firms does not identify the causal effect of board structure on value.

14.5.3 The Dynamic Panel Critique

[Wintoki et al. \(2012\)](#) sharpened the methodological objection for boards specifically. Current board structure is correlated with past performance. Underperforming firms add outside directors, change committee composition, and sometimes replace the CEO with a different decision-maker. These changes are not the random within-firm variation that fixed effects implicitly assume; they are systematic responses to firm history. The within estimator's strict exogeneity assumption fails when current governance responds to past outcomes. Their solution was to use dynamic panel GMM, with lagged board variables as instruments for current board variables, in a system-GMM specification. Once the dynamic structure is taken seriously, the standard cross-sectional and fixed-effects estimates of the board-value relationship lose statistical significance. The implication is the same lesson the ownership literature reached in Section 14.4. Estimating a meaningful causal effect of governance requires either a research design with exogenous variation or a fully specified dynamic model.

14.5.4 Board Independence Mandates

The 2002 Sarbanes-Oxley Act and the related 2003 exchange listing standards required US public-firm boards to have a majority of outside directors and all-outside-director audit, compensation, and nominating committees. Firms differed substantially in their pre-mandate board structure. Firms that were already compliant absorbed no real cost. Firms that were not had to recruit outside directors, sometimes in substantial numbers. The pre/post comparison provides a natural intensity-treated design at the firm level, with treatment intensity proportional to the distance between the firm's actual board composition in 2001 and the new compliance threshold. [Duchin, Matsusaka, and Ozbas](#)

(2010) use this design to ask when mandated outside directors help and when they hurt. Their headline result is that outside directors improve outcomes when firm-specific information is easy to acquire, and worsen outcomes when it is costly. The mandate creates value at transparent firms and destroys value at firms whose operations are hard for outsiders to understand. The finding connects the SOX shock to the broader theme of Section 14.5.2: governance arrangements do not have uniform effects.

14.5.5 CEO Turnover and Succession

The board's signature action is firing the CEO. [Huson, Parrino, and Starks \(2001\)](#) documented the long-run trends in US firm CEO turnover from 1971 to 1995. Total turnover rates rose modestly across the period. The negative correlation between firm performance and forced turnover strengthened. Their estimates imply that a CEO whose firm was in the bottom decile of one-year industry-adjusted stock return faced roughly a 10% probability of forced exit in the following year by the mid-1990s, compared to 2 to 4% in the early 1970s. The shift was concentrated in larger firms and in firms with more outside directors. The interpretation that survives most robustness checks is that internal monitoring became more responsive to performance over the period.

[Kaplan and Minton \(2012\)](#) extended the analysis through 2007 and found a striking acceleration. Forced CEO turnover rates at the largest US firms roughly doubled between 1998 and 2007 relative to the previous decade. The median tenure of an exiting CEO fell. Boards were firing faster and more often. Kaplan and Minton attribute the change to a combination of external pressure from shareholder activism and more attentive institutional investors, regulatory reform after Sarbanes-Oxley, and a more developed external CEO labor market. The internal-monitoring story did not become wrong. The external monitoring environment changed around it.

[Jenter and Kanaan \(2015\)](#) address whether boards are filtering signal from noise. Optimal contract theory says that turnover decisions should respond to firm-specific performance and not to industry shocks the CEO cannot affect. Empirically the prediction fails. Even after controlling for firm-specific performance, CEO turnover is strongly correlated with industry-wide stock returns. CEOs in poorly performing industries are more likely to be fired than CEOs of equally underperforming firms in stable industries. Boards are either unable or unwilling to filter industry noise out of the firing decision.

Two recent contributions sharpen this picture. [Jenter and Lewellen \(2021\)](#) show that announced "voluntary" CEO exits at low-performing firms are substantially performance-induced; the standard turnover-classification methodology underestimates the true rate of

performance-driven turnover by half or more. [Taylor \(2010\)](#) runs a structural estimation of a CEO-board model on a Compustat panel and recovers the implied cost of firing a CEO. His estimate is roughly 5 to 6% of firm value. The number is large enough to rationalize why CEOs are not fired more often, even when the data say boards know which CEOs are underperforming.

The chapter's running example reappears here. [Berger, Ofek, and Yermack \(1997\)](#) document that entrenched managers, measured by long tenure, low equity ownership, and weak compensation incentives, choose lower book leverage than otherwise comparable managers. Their cross-sectional estimates suggest a difference of roughly seven percentage points in book leverage between the most and least entrenched managers. The result connects board monitoring to the leverage choice and motivates the entrenchment-leverage literature we return to in Chapter 10.

14.6 Executive Compensation and CEO Effects

Executive compensation is the most-studied governance topic in corporate finance, partly because the data are good and partly because the patterns are striking. The level and composition of US CEO pay changed dramatically over the past four decades. Total compensation at the largest firms rose by a factor of more than six in real terms between 1980 and the mid-2000s. The composition shifted from salary toward equity-linked instruments, especially options through the 1990s and restricted stock thereafter. Four explanations of these patterns have organized the literature. They are not mutually exclusive. They reach different conclusions from the same data, and the empirical work in each tradition can be persuasive on its own terms while sitting awkwardly with the others. We work through them in turn.

Moral hazard and incentive provision. The foundational view treats compensation as the solution to the standard agency problem. The principal cannot observe the agent's effort, so the contract ties pay to verifiable outcomes that depend partly on effort and partly on noise. The classic empirical implementation is the pay-performance sensitivity (PPS) of [Jensen and Murphy \(1990\)](#). Their measure is the dollar change in CEO wealth, including the value of stock holdings, for each \$1,000 change in shareholder wealth. The 1990 estimate is about \$3.25 per \$1,000, or 0.325%. Jensen and Murphy read this as too small to align CEO incentives with shareholder value at plausible coefficients of risk aversion. The result framed the next two decades of literature. [Hall and Liebman \(1998\)](#) demonstrated that the level had changed dramatically by the mid-1990s. After accounting for the option grants

that became standard in that decade, the elasticity of CEO firm-specific wealth to firm value was roughly thirty times higher than in the Jensen-Murphy era. A \$1 change in firm value produced roughly \$0.25 of change in CEO holdings, against \$0.003 in the original sample. The order-of-magnitude jump did not settle whether incentives are now optimal. It demonstrated that the answer depends on which decade you measure.

Common Mistake 5: Pay-performance sensitivity vs. the level of pay

Pay-performance sensitivity and the level of pay are different objects, and conflating them is the most common error in this literature. The Jensen-Murphy \$3.25 per \$1,000 number is a marginal sensitivity: how much CEO wealth changes for a small change in shareholder wealth. The Hall-Liebman option-era estimates are also marginal sensitivities, computed for the option-laden compensation packages of the 1990s. Neither is the total level of CEO pay. A statement that “CEO pay went up by a factor of six since 1980” is a level statement. A statement that “incentives became stronger” is a sensitivity statement. The two can move together, opposite, or independently, depending on the period and the compensation structure.

Rent extraction and weak governance. [Bertrand and Mullainathan \(2001\)](#) read the same data differently. Their identification strategy is to ask whether CEO pay responds to “luck” as much as to skill. A luck shock is variation in the firm’s outcomes that the CEO cannot have affected: oil-price movements for an oil-firm CEO, exchange-rate movements for an exporter, industry-wide demand shocks. If pay reflects effort and skill, only the firm-specific residual should matter. Empirically, roughly half of CEO pay variation responds to luck shocks. The boards that approve these compensation contracts are not filtering noise from signal. [Bebchuk and Fried \(2003, 2004\)](#) generalized the argument into a managerial-power view of pay-setting. Boards do not bargain with CEOs at arm’s length. They approve pay packages that maximize CEO surplus subject to a constraint of avoiding outrage. The model predicts that better-governed firms with independent boards, active blockholders, and strong shareholder rights should pay less for luck. The empirical evidence is broadly supportive. Pay-for-luck is most pronounced in firms without strong outside monitoring, particularly large outside shareholders. The role of board independence specifically is more mixed in the literature.

Market for scarce CEO talent. [Gabaix and Landier \(2008\)](#) offer a structural alternative. In their model CEOs differ in talent, firms differ in size, and the assignment is the equilibrium of a competitive matching market. Talented CEOs go to large firms because their decisions

are scaled by the size of the assets they manage. The model has a clean closed-form prediction: equilibrium CEO pay scales with firm size to the power roughly one-third. A doubling of average firm market value implies roughly a 26% increase in equilibrium CEO pay. Calibrated to the 1980–2003 period, the model accounts for essentially the entire sixfold rise in average CEO pay from the growth in firm market values alone. There are no rents in the model, no agency problems, and no weak governance. The CEO labor market is competitive and the assignment is efficient. [Edmans, Gabaix, and Jenter \(2017\)](#) survey several related structural models that share the size-scaling prediction. The Gabaix-Landier reading and the Bertrand-Mullainathan reading apply to the same data and identify different objects.

Behavioral, benchmarking, and regulatory channels. The fourth class of explanations emphasizes institutional features of pay-setting that the first three abstract away. [Edmans et al. \(2017\)](#) synthesize this literature. Compensation consultants supply boards with peer benchmarks. The standard benchmarking practice produces a ratchet: pay rises toward the upper half of any peer group, and the next year's peer group is constructed using a sample that has shifted up. Boards rarely lower pay after a period of strong firm performance, even when the strong performance was driven by industry-wide tailwinds. The 1993 introduction of Section 162(m) capped the corporate tax deduction for non-performance-linked executive compensation at \$1 million per year. The provision encouraged a shift toward performance-linked compensation, though estimates of its quantitative importance vary across studies. Each of these channels is small in isolation. Taken together, they account for a meaningful share of the level and composition shift that the previous three traditions also try to explain.

Key Insight 6: Four competing readings of CEO pay

Pay-performance sensitivity, pay-for-luck, market scaling, and benchmarking-and-regulation are not alternatives to be tested against each other. Each picks up a different component of the variation in CEO compensation. The sensitivity literature measures the conditional response of pay to firm value. The pay-for-luck literature measures whether pay responds to outcomes the CEO did not cause. The market-scaling literature explains the cross-sectional gradient of pay in firm size. The benchmarking literature explains specific institutional regularities. A complete account of CEO pay requires all four.

14.6.1 CEO Fixed Effects

A different question is whether individual CEOs matter at all. [Bertrand and Schoar \(2003\)](#) provided the canonical evidence that they do. Their identification strategy exploits CEOs who move between firms. With a panel of firm-CEO pairs and CEO fixed effects, the within-CEO across-firm variation identifies a “manager style” component for each individual CEO. The variance decomposition implies that CEO fixed effects explain a nontrivial share of the within-firm variance of investment, leverage, payout, and several other corporate policies. CEOs are not interchangeable. Different CEOs run firms differently in ways that statistical analysis can detect. The finding gave the executive labor market a structural rationale: if CEOs differ systematically in style, the assignment of CEOs to firms is an economically meaningful matching problem.

[Fee, Hadlock, and Pierce \(2013\)](#) challenged the identification. The Bertrand-Schoar approach requires that CEO moves be exogenous to firm characteristics. In practice CEOs do not move randomly. They move to firms where their expected match value is higher, where compensation is more attractive, or where they have prior relationships. Fee, Hadlock, and Pierce reproduce the Bertrand-Schoar specification on a sample of plausibly exogenous CEO departures, including deaths and forced exits, and find that the estimated style effects shrink substantially. Their result does not say that CEOs do not matter. It says that the size of the CEO fixed effect depends on which CEO transitions one is willing to call exogenous.

14.6.2 Running Example: Pay Convexity and Leverage

[Coles, Daniel, and Naveen \(2006\)](#) connect compensation directly to the chapter’s running example. They construct CEO “vega,” the dollar change in CEO wealth per percentage-point increase in the volatility of firm equity, from the option and stock-grant components of the compensation package. CEOs with high vega face stronger incentives to take risk, because their compensation rises more sharply when firm volatility rises. Empirically, vega is positively related to book leverage and to R&D intensity. A one-standard-deviation increase in CEO vega is associated with roughly a five percentage point increase in book leverage in the cross-section. The result demonstrates that executive contracts shape capital structure choices, the central concern of the book’s running example.

Connection to Framework 7

The CEO pay literature is the chapter's strongest illustration of the structural-causal divergence. The [Gabaix and Landier \(2008\)](#) structural matching model reaches one conclusion: CEO pay levels reflect a competitive labor market with no agency problems. The [Bertrand and Mullainathan \(2001\)](#) reduced-form pay-for-luck regression reaches another: CEO pay levels reflect weak board monitoring and rent extraction. Both literatures are applied to the same Compustat-ExecuComp data. Under their maintained assumptions, both identify distinct objects. The structural reading identifies the size elasticity of pay; the causal reading identifies the sensitivity of pay to non-CEO-attributable shocks. The disagreement is not about identification. It is about which component of variance one wishes to interpret.

14.7 Shareholder Rights and Entrenchment

[Gompers et al. \(2003\)](#), hereafter GIM, made the literature's defining attempt to measure shareholder rights. They constructed an index from 24 governance provisions coded by ISS for roughly 1,500 firms over 1990 to 1999. The provisions cover charter amendments, bylaw amendments, takeover defenses, board structure features, and state-law protections. The G-index is the count of provisions that restrict shareholder rights. A score of zero indicates few restrictions, which GIM call "democracy" firms. A score above fourteen indicates the most restricted firms, which GIM call "dictatorship" firms. The headline result was a striking return spread. A long-democracy short-dictatorship portfolio earned an abnormal return of roughly 8.5% per year over the sample period. The interpretation that traveled with the result was that markets underprice the future costs of weak shareholder rights, and that the G-index measures the size of those costs.

[Bebchuk et al. \(2009\)](#) sharpened the index by asking which provisions actually carry the predictive content. They identified six entrenchment provisions that capture nearly all of the GIM spread: staggered boards, poison pills, golden parachutes, supermajority requirements for mergers, supermajority requirements for charter amendments, and supermajority requirements for bylaw amendments. The E-index is the count of these six. The remaining eighteen GIM provisions add little marginal explanatory power. The E-index is widely used in the modern literature as a parsimonious entrenchment measure, and the Bebchuk-Cohen-Ferrell construction implicitly accepts the GIM premise that bundled governance arrangements carry economic content.

The G-index is best read with three pieces of nuance. First, the descriptive content of the index is real. Firms with G-scores in the bottom decile differ systematically from

firms in the top decile along many dimensions of governance, and the differences are persistent over time. The index measures something economically meaningful, even if its causal interpretation is contested. Second, the return predictability that originally motivated GIM weakened after 2000. [Core, Guay, and Rusticus \(2006\)](#) documented that the GIM return spread was concentrated in the 1990s and did not survive into the early 2000s in subsamples that controlled for firm operating performance. [Bebchuk, Cohen, and Wang \(2013\)](#) documented the disappearance and offered the learning interpretation: by the early 2000s, investors had learned about the G-index, the predictive content was priced in, and the return spread vanished. One reading is that investors learned to price the index. Another is that the original spread reflected period-specific mispricing, omitted risk, or data-mined predictability. The data are consistent with both. Third, the index remains pedagogically useful, even with the return prediction gone. It documents that governance is bundled, that bundling is persistent, and that bundling is endogenous to firm characteristics.

Common Mistake 8: Causal claim vs. predictive content

The G-index literature is sometimes read as showing that weak shareholder rights cause low firm value. The empirical evidence supports a weaker claim: governance provisions predict firm characteristics and historically predicted returns, but the predictive relationship is not the same as a causal effect. The disappearance of the return predictability after 2000 is consistent with markets having learned to price the index. It is also consistent with the index never having identified a causal channel in the first place, only an arbitrage opportunity that closed when investors paid attention. Both readings survive the data.

The G-index uses cross-sectional variation in firm-level governance provisions. The state antitakeover laws adopted across the US during the 1980s provide a complementary source of variation. Business combination statutes, fair price statutes, and control share statutes were passed state-by-state in different years, creating a staggered difference-in-differences design at the level of the firm's state of incorporation. [Bertrand and Mullainathan \(2004\)](#) applied this design to identify what entrenchment does to managerial behavior. Their headline result is the quiet-life finding. Managers protected by business combination laws cut wages less aggressively, close plants less often, and produce lower profit margins than otherwise comparable managers in states without such laws. Entrenched managers prefer the quiet life to the value-maximizing decisions a takeover threat would force on them.

[Karpoff and Wittry \(2018\)](#) delivered the section's main methodological critique. The standard adoption-year datasets used by Bertrand and Mullainathan and many followers

contained substantial coding errors. The wrong adoption years were assigned to several states. Firms that had opted out of the laws were included in the treatment group. Firms incorporated in non-statute states but doing business in statute states were misclassified. Karpoff and Wittry replicate several leading takeover-law DiD results with corrected coding and report that the estimated effects shrink substantially, lose statistical significance, or in some cases reverse sign. Some quiet-life-style results survive qualitatively, but the corrected coding weakens the evidence and reduces confidence in broad claims from the earlier law-shock literature. The broader pre-Karpoff-Wittry takeover literature should be read with the same caution Himmelberg-Hubbard-Palia applied to the ownership-performance regressions of Section 14.4.

The acquirer-side and target-side identification questions, both central to the empirical takeover literature, are developed in Chapter 15. The deal-based identification framework treats specific takeover events as quasi-experiments. The acquirer-side abnormal returns, the bidder-target wealth division, and the long-run post-merger performance literature all sit in that chapter rather than this one.

14.8 Institutional Investors, Passive Ownership, and Activism

Institutional ownership of US public firms has risen from roughly 30% in the 1980s to more than 70% in the 2010s. The composition has shifted from active mutual funds, pension funds, and banks toward passive index funds and exchange-traded funds. By the late 2010s, the three largest passive asset managers, BlackRock, Vanguard, and State Street, collectively held more than 20% of the average S&P 500 firm. The aggregate change in who owns US public firms is the largest shift in the ownership distribution since the rise of pension funds in the 1970s. The empirical governance literature has tried to identify how this shift affected firm-level outcomes.

[Appel et al. \(2016\)](#), hereafter AGK, supplied the most-cited design for the question. They exploit the annual reconstitution of the Russell 1000 and Russell 2000 indexes. Firms ranked just below the 1000 cutoff are placed in the Russell 2000, where they have a much larger index weight and receive substantially higher passive ownership; firms just above are placed in the Russell 1000 with lower weight and lower passive ownership. The discontinuity in passive ownership at the threshold is sharp. AGK use it as a regression-discontinuity instrument and find that higher passive ownership is associated with more independent directors, fewer takeover defenses, and stronger overall board governance.

The interpretation is that passive owners, despite their indexing strategy, vote and engage in ways that improve governance at the firms they hold.

[Wei and Young \(2024\)](#) challenged the AGK identification. The Russell reconstitution does not assign firms randomly across the 1000/2000 boundary. The reconstitution uses end-of-May total market capitalization to determine index membership, but the within-index weighting depends on float-adjusted capitalization, which can be manipulated by firms aware of the ranking. Wei and Young show that pre-existing firm characteristics differ systematically at the threshold in ways that would generate the AGK findings even with no causal effect of passive ownership. Their reanalysis with bias-corrected estimators finds that the AGK governance effects shrink or disappear. The critique does not establish that passive ownership has no effect on governance. It establishes that the Russell RD as commonly implemented does not cleanly identify it. The Russell setting remains influential as a workhorse design for identifying passive-ownership effects, but the identifying assumptions are more fragile than the early literature suggested. Subsequent work using alternative designs or bias-corrected estimators has produced a more mixed picture than the original AGK result.

[Heath, Macciocchi, Michaely, and Ringgenberg \(2022\)](#) reached a substantively different conclusion using a related research design. Their focus is on monitoring behavior directly. They examine how index funds vote, how they engage, and what governance changes occur at firms whose index ownership rises. The headline finding is that index funds vote against management on contentious governance issues less often than active funds at the same firms. There is little evidence of effective public or private engagement. Firms with greater index ownership exhibit less board independence and weaker pay-performance sensitivity. The Heath et al. reading is that the rise of passive ownership has weakened monitoring rather than improved governance. The AGK and Heath et al. literatures sit awkwardly with each other. They use related but not identical designs, study related but not identical outcomes, and reach opposite conclusions. The question is genuinely open.

Hedge fund activism is the modern alternative to traditional shareholder voice. [Brav, Jiang, Partnoy, and Thomas \(2008\)](#) provided the founding empirical evidence. Hedge fund activist campaigns, identified by Schedule 13D filings with stated intent to influence control, are associated with positive abnormal returns of roughly 5 to 7% around the campaign announcement, with much of the gain coming from anticipated operational and governance changes. The targets are typically small and mid-cap firms with weak operating performance and high cash holdings. [Bebchuk, Brav, and Jiang \(2015\)](#) addressed the concern that activist gains are short-term at the expense of long-term value destruction. Using a five-year window after the campaign, they find no evidence of long-term reversal.

Targeted firms maintain higher operating performance and stock returns through the post-campaign period. The activism premium is not a wealth transfer from long-term shareholders; it appears to be value created by changes the activist forced the firm to undertake.

Cuñat, Gine, and Guadalupe (2012) provide a clean regression-discontinuity design for the effect of governance reform on firm value. Their setting is close-call shareholder votes on governance-improving proposals. A proposal that passes by a narrow margin and one that fails by a narrow margin are observationally similar firms; the difference is only that one firm implements the governance change and the other does not. Cuñat, Gine, and Guadalupe estimate that the abnormal return on the announcement of a narrowly passing governance proposal is roughly 1.3% over a short event window. Aggregating across the proposals studied, the implied effect of fully adopting the proposed reforms would be a two to three percent increase in firm value. The RD design sidesteps the selection problem of the cross-sectional G-index literature.

Key Insight 9: Two frontier topics

Two topics are not developed in this chapter but are likely to mature substantially over the next decade. Azar, Schmalz, and Tecu (2018) document an empirical association between common ownership across competing firms in the same industry and softer product-market competition; critics dispute the identification and the magnitude. The ESG and stakeholder-governance literatures study whether firms should optimize for shareholders alone or for a broader set of constituencies. Both topics are contested empirically and conceptually.

Frank and Goyal (2025) provide the running-example touchpoint. They study the relationship between passive institutional ownership and corporate investment, applying a matching design to the modern US public-firm sample. The motivation is that if passive owners monitor less actively, as the Heath et al. evidence suggests, then firms with higher passive ownership may face weaker discipline on real investment decisions. The paper confronts the identification challenges that Wei and Young (2024) raises for any Russell-RD design by using firm-level matching on observable characteristics rather than the cutoff itself. The chapter returns to leverage outcomes in Chapter 10.

14.9 What the Evidence Has Taught Us

Section 14.1 previewed the chapter with a six-row table of governance objects and the main identification problem facing each. With the empirical material of Sections 14.4 through 14.8 in hand, we return to the table and add what the literature has learned. Table 14.2 expands the preview into five columns: the governance object, the dominant empirical design used to study it, the object that design actually identifies, the main threat to interpretation, and the conclusion that survives the methodological scrutiny.

Table 14.2. Governance objects, empirical designs, and what survives

Object	Empirical design	What is identified	Main threat	What survives
Ownership	Cross-section, FE panel, IV	Equilibrium correlation	Joint determination of ownership and value	Cross-sectional regression uninterpretable; matching and IV results are mixed
Boards	Cross-section, dynamic GMM	Within- and between-firm associations	Reverse causality from past performance to structure	Yermack cross-section overstates the relationship; CDN heterogeneity reframes it; most causal questions remain hard
Compensation	Sensitivity regressions, structural matching	Marginal sensitivities, size-scaling elasticity	Rents and competitive scaling both fit the data	Structural and causal traditions coexist and identify different objects
Antitakeover provisions	Staggered-adoption DiD	State-incorporation treatment effect	Coding errors in adoption-year datasets	Quiet-life direction survives Karpoff-Wittry; magnitudes shrink
Institutional ownership	Russell 1000/2000 RD	LATE at the index threshold	Banding selection on observables	AGK and Heath et al. reach opposite conclusions; genuinely open
Activism	Schedule 13D event studies, long-run panels	Announcement returns, post-campaign performance	Selection of targets	Positive announcement returns and no robust long-run reversal in leading studies; causal value creation still debated because targets are selected

Notes. The “what survives” column reflects the state of the literature at the time of writing. Citations to the specific studies referenced are in Sections 14.4 through 14.8.

The pattern across rows is not a tidy convergence. The literature has reached different states for different governance objects. For ownership and boards, the cross-sectional and within-firm regressions that anchor the early literature do not identify causal effects of governance on firm outcomes; the modern reading is that they describe equilibrium patterns. For compensation, the structural and causal traditions identify different objects and coexist; neither has displaced the other. For antitakeover provisions, the direction of the Bertrand-Mullainathan quiet-life effect survives the Karpoff-Wittry correction, but the magnitudes shrink. For institutional ownership, the AGK and Heath et al. literatures reach opposite substantive conclusions from related research designs, and the field has not converged. For hedge fund activism, the empirical case for value creation has survived the long-run-reversal challenge raised by skeptics.

Governance is the hardest case in this book for the structural-versus-causal distinction. In other settings, the structural and causal traditions identify the same underlying parameter from different vantage points, and the disagreements are about which assumptions are more credible. In governance settings, structural and causal designs often identify different objects entirely. The [Gabaix and Landier \(2008\)](#) matching model and the [Bertrand and Mullainathan \(2001\)](#) pay-for-luck regression are the cleanest example: they do not contradict each other; they explain different components of the variance in CEO compensation. The chapter has tried to be honest about this. We do not claim that a single literature has the right answer to what governance does. We claim that the right way to read the literature is as a collection of different but compatible answers to different but related questions.

Several open questions identified in this chapter are extensions of the heterogeneous-treatment-effects framework that Book 2 develops. The AGK-Heath disagreement on passive ownership can be read as a question about which complier population each design identifies. The Karpoff-Wittry critique of the antitakeover-law literature is about variation in which firms are actually treated. The structural-versus-causal divergence in compensation is a question about which structural primitive the data identify. Heterogeneous-treatment-effects methods are the natural language for these questions, and we return to them in Book 2.

14.10 Exercises

1. **Demsetz-Lehn cross-sectional regression.** (*Conceptual.*) Derive why the cross-sectional regression slope of firm value on ownership concentration in [Demsetz and Lehn \(1985\)](#) cannot identify a causal effect, even if the regression is correctly specified. Your answer should make use of the equilibrium logic of their model: ownership concentration is

chosen by firms and investors to balance monitoring benefits against risk-bearing costs. What would the slope be if all firms were at their privately optimal concentration?

2. **G-index return spread, modern data.** (*Computational.*) Replicate the [Gompers et al. \(2003\)](#) G-index portfolio return regression on the period 2003 to 2023. Use ISS governance provisions to construct the G-index. Form value-weighted democracy and dictatorship portfolios and report the alpha of the long-democracy short-dictatorship strategy. Compare your estimate to the GIM 1990s number and to the [Bebchuk et al. \(2013\)](#) interpretation of the change.
3. **Russell 1000/2000 reconstitution RD.** (*Computational.*) Implement the regression discontinuity design of [Appel et al. \(2016\)](#) on a recent panel of Compustat-CRSP-13F merged data. Construct passive ownership shares from 13F filings. Estimate the discontinuity in passive ownership at the 1000/2000 boundary using the May ranking. Then estimate the discontinuity in board-independence outcomes. Comment on the magnitude of your estimates relative to the AGK 2016 published numbers and discuss how the [Wei and Young \(2024\)](#) critique applies to your implementation.
4. **Jensen-Murphy pay-performance sensitivity, modern data.** (*Computational.*) Using a recent ExecuComp panel (2000 onward), construct the [Jensen and Murphy \(1990\)](#) pay-performance sensitivity. The original measure is the dollar change in CEO wealth per \$1,000 change in shareholder wealth, computed for the median firm. Report the sensitivity by decade and discuss how the option-grant expansion of the 1990s and the restricted-stock shift of the 2000s changed the magnitude. Compare to the 1990 baseline of \$3.25 per \$1,000.
5. **Board independence after SOX.** (*Computational/Applied.*) Using ExecuComp or BoardEx data covering 2000–2008, identify firms that needed to add outside directors to comply with the 2002–2003 exchange listing standards versus firms that were already compliant. Estimate a difference-in-differences regression of firm value or operating performance on the intensity of the mandate’s bite. Discuss why your estimate is unlikely to identify the causal effect of board independence on value. What additional information would you need to make a stronger causal claim?
6. **Performance-induced turnover.** (*Applied.*) Read [Jenter and Lewellen \(2021\)](#). The paper constructs a measure of performance-induced turnover that includes “voluntary” departures at low-performing firms. Evaluate the construction. What assumptions does the methodology require? What alternative measures would you propose? In a one-to two-page essay, defend your alternative and discuss what it would imply for the turnover rates the paper documents.
7. **State antitakeover law coding errors.** (*Applied.*) Read [Karpoff and Wittry \(2018\)](#). Pick

one of the papers they re-examine. Locate the original paper, read it carefully, and write a one- to two-page memo explaining (a) which coding errors Karpoff and Wittry identify in the original analysis, (b) how the corrected coding changes the estimated effect, and (c) which of the original paper's empirical claims survive the correction qualitatively.

8. **Close-call shareholder votes.** (*Computational/Applied.*) Using proxy-voting data, identify governance-improving shareholder proposals that passed or failed by a narrow margin. Implement the [Cuñat et al. \(2012\)](#) regression-discontinuity design. Estimate the abnormal announcement return for narrowly-passing proposals. Discuss what is identified at the threshold. Under what circumstances would the threshold estimate extrapolate to the full population of governance proposals?

Chapter Summary

Corporate governance arrangements are not treatments firms receive. They are equilibrium outcomes of contracting between firms and investors. The methods that work elsewhere in this book apply to governance settings only when the choice of governance can be plausibly disentangled from the outcomes it predicts. Most cross-sectional regressions of value on governance do not identify a causal effect because the same primitive drives both. The literature has shifted toward natural experiments and structural models that take the choice seriously, and the right way to read forty years of empirical work is as a collection of different but compatible answers to different but related questions.

Chapter 15

Mergers, Acquisitions, and Restructuring

15.1 Why Deals Are Hard to Interpret

Corporate control transactions are among the largest reallocations of assets, control rights, financing claims, and managerial authority that firms undertake. They are attractive empirically because they are discrete, dated, and economically large. They are difficult empirically because they are selected, anticipated, negotiated, bundled, and sometimes not completed. The methods developed throughout this book are exercised on this class of events more fully than on any other in the corporate-finance literature, and the difficulty of interpretation in this setting is correspondingly acute.

This chapter is organized around three empirical objects rather than around transaction types. The first is market assessment: what the stock market infers about value at the moment a deal is announced. The second is real effects: what happens to assets, investment, employment, innovation, capital structure, and the locus of control after the deal closes. The third is selection and surplus division: why this deal happened rather than another, who captured the surplus that markets measured, and what counterfactual the design supports. Mergers, tender offers, LBOs, divestitures, spinoffs, and bankruptcies appear in the chapter as settings in which these objects are studied, not as the chapter's organizing axis.

The running example of leverage and investment threads through the chapter at three explicit touchpoints. The first is the method-of-payment discussion in Section 15.4, where the choice between cash and stock financing determines what happens to bidder leverage after the deal closes. The second is Section 15.8.1, where leveraged buyouts bundle a leverage increase with an ownership-concentration change and a managerial-incentive shift. The third is Section 15.8.3, where leverage stops being a balance-sheet ratio and

becomes the mechanism through which control rights are reallocated among claimants. The example is a continuity device, not a survey. Each touchpoint is chosen for what it teaches about identification rather than for coverage of leverage-investment dynamics.

This is the last application chapter, and the structural-versus-causal spine that has organized the book runs through deals more visibly than through any other topic. The event-study machinery of Chapter 4 measures announcement reactions. The modern difference-in-differences methods of Chapter 6 discipline post-deal real-effect estimates. The instrumental-variables designs of Chapter 5 attack selection into completion. Structural search and bargaining models recover counterfactuals no single deal event can identify. The chapter is the last because the book's apparatus is most fully exercised here.

Key Insight 1: The transaction is an equilibrium outcome

The empirical literature on corporate control transactions is organized around a single methodological fact. Deals are chosen jointly by acquirers, targets, financiers, and the regulatory environment. Every comparison of firms that did transact with firms that did not confounds the transaction with the type of firm that selects into it. The methods of this book recover causal effects in deal settings only when the selection into transaction can be plausibly disentangled from the outcome of interest.

15.2 What Is the Event?

Event definition is identification. The choice of which date to use as the empirical event, and which class of transaction to admit into the sample, determines what an estimator recovers. Two studies that describe themselves identically as “studies of mergers” can estimate distinct objects if one timestamps at the rumor date, one at the formal announcement, and one at completion. A reader who does not see this point first will misread much of the literature.

Table 15.1 catalogs the canonical events. Each row pairs a transaction class with the date the literature most often uses as the empirical event, the empirical object the design typically isolates, and the dominant threat to interpreting that object as a treatment effect of the transaction. The rest of the chapter refers back to this table.

Table 15.1. Corporate-control transactions and their empirical objects.

Transaction	Event date	Main empirical object	Main threat
Public merger	Announcement	Market value revision	Anticipation, se
Tender offer	Announcement/expiration	Control premium	Competing bids
Stock acquisition	Announcement/completion	Misvaluation, payment effects	Bidder selection
LBO	Announcement/completion	Leverage and governance bundle	Public-to-private
Divestiture	Announcement/completion	Asset redeployment	Seller distress
Spinoff	Announcement/effective date	Refocusing, valuation	Endogenous se
Bankruptcy	Filing/emergence	Control-right reallocation	Distress selection

Notes. Each row identifies a class of corporate-control transaction studied in this chapter, the date most commonly used as event, the empirical object isolated by designs built around that date, and the dominant identification threat to interpreting as a treatment effect of the transaction. The columns are indicative, not exhaustive: many designs combine event windows on completion. Section 15.7 returns to the table in the context of specific identification strategies.

A handful of cross-cutting distinctions shape every later section. The first is friendly versus hostile: friendly deals are negotiated with target-management consent, hostile deals bypass management through tender offers, proxy contests, or shareholder solicitations. Friendly deals select on management's perceived alignment with shareholders, hostile deals on its absence; pooling them treats them as exchangeable, which they are not.

The second is public versus private on either side. The literature concentrates on public targets because announcement returns require traded prices, but the population of M&A activity skews private, and no estimator that operates inside the public-target sample fixes this sample-selection problem.

The third is completion. A deal that is announced may or may not close, and conditioning on completion changes the estimand: announcement-window returns are defined over both outcomes, post-deal real outcomes only on completed deals. The completion margin is itself selected, a point developed in Section 15.6 and exploited as an identification strategy in Section 15.7.

The fourth is the rumor-to-completion sequence. Rumors and preliminary discussions move target prices in the weeks before the formal announcement, so announcement-window returns understate the total information content of the transaction. A short window recovers the residual surprise, a wider window absorbs the runup but also unrelated noise; window choice is a choice of estimand, not a tuning parameter.

Institutional and legal detail enters this chapter where it affects identification: disclosure thresholds determine when prices reflect deal news, state antitakeover law determines which bidders can credibly threaten hostile acquisition, and bankruptcy code determines

which claimants have which rights at filing. We discuss these institutions where they constrain the available designs and otherwise treat them as background.

15.3 Data, Sample Construction, and Measurement

The transition from Section 15.2 to the empirical sections that follow runs through a practical layer the literature usually treats as setup but which carries real interpretive weight. The choice of database, the linkage to firm-level panel data, the construction of sample filters, and the precise definitions of event-window and premium variables all encode substantive assumptions about which deals are studied and what the empirical objects mean. This section makes those choices visible.

Most US M&A research draws on SDC Platinum, now available through LSEG Workspace, which records announced and completed transactions with deal terms, parties, and dates. Capital IQ and PitchBook offer alternative coverage with different inclusion rules, particularly for private transactions and middle-market deals. DealScan supplies the syndicated-loan side of bidder financing for deals with leveraged tranches, useful for separating debt-financed from internally financed acquisitions. SEC filings (S-4 registration statements, 14D-9 solicitation and recommendation statements, 8-K material disclosures) provide the institutional and timing detail that database codifications truncate. Each source encodes a sample selection through its inclusion rules, size thresholds, and lag between announcement and database appearance. Database choice is a sample-construction decision, not a logistical one.

Linking deal records to firm-level price and accounting data is where most data-cleaning errors occur in modern research. Bidder and target PERMNO assignments through the CRSP-Compustat link table, CUSIP-based matching when PERMNO is unavailable, and name-string fuzzy matching when CUSIP is absent are the standard tools, each with characteristic failure modes. Delisting returns must be appended for completed targets that exit CRSP at merger close, and the magnitude depends on the delist code and the CRSP convention applied. Private targets carry no pre-deal market value, which forces imputation from accounting data, exclusion from samples that require market-based variables, or restriction to public-target subsamples. Post-completion identification of the merged entity requires choosing which pre-deal identifier to carry forward, an apparently mechanical choice that shapes which firm-year observations enter post-deal panels. Match rates differ across database sources and across researchers, and hand-checking is standard practice in serious work.

Sample filters define the population a study estimates over. The standard choices

include public-target versus private-target restriction, completed-only versus all-announced samples, deal-value thresholds (typically tens or hundreds of millions of dollars in real terms), relative-size cutoffs measured as transaction value over bidder market capitalization, tender offers versus negotiated mergers, hostile versus friendly designations, the medium of payment, indicators for competing bidders, same-industry restrictions at the three- or four-digit SIC level, exclusions of financial firms and regulated utilities, and treatment of cross-border deals. Each filter is a substantive choice with empirical consequences. A study that restricts to completed deals with public targets above one hundred million dollars in same-industry cash transactions is estimating a different object than a study that admits all announced deals across all transaction sizes. The cleaning protocol is a research-design choice that should be stated and defended, not buried in a data appendix.

Once the sample is fixed, several measurement choices remain. Event-window construction requires a specific event date: the rumor date if rumor data is available and the design wants to capture leakage, the formal announcement date if the design isolates the residual surprise, or the completion date if the outcome is conditional on completion. Offer-premium computation requires a reference price date, typically four or six weeks before the announcement to avoid runup contamination, and a denominator choice; most studies use the pre-announcement target price, but some normalize by estimated standalone value. Combined CAR construction takes the value-weighted average of bidder and target CARs with weights set by pre-announcement market values; the dollar-weighted alternative sums announcement-window dollar value changes, and the two diverge sharply when small targets are paired with large bidders. The choice between bidder-only samples and matched bidder-target samples affects sample size and the questions the data can answer.

Several traps recur. The same underlying deal can generate multiple announcements when bids are revised or competing bidders enter, and event-study code that treats each as a distinct event double-counts the transaction. Toeholds pre-position the bidder and dampen announcement returns relative to a no-toehold counterfactual. Terminated deals that re-enter the data as new announcements months later require a deal-identifier convention that links them. Missing valuations for private targets force imputation or exclusion, both of which select the sample. Non-random completion makes the completed-deals sample a selected subset of announced deals, with the selection itself an object of study; see Section 15.6 for the analytical treatment of the completion margin.

The empirical conventions described here are the substrate of the announcement-return literature that follows. [Andrade, Mitchell, and Stafford \(2001\)](#) and [Betton, Eckbo, and Thorburn \(2008\)](#) provide the canonical surveys of these conventions and document how the stylized facts of the next section depend on the sample-construction choices made

along the way.

A replication-ready M&A sample should state, at a minimum, the database and download date, the announcement-date and completion-date conventions, the public/private target restriction, the deal-value and relative-size thresholds, the payment-method classification, the treatment of competing bids and of withdrawn deals, the CRSP-Compustat matching protocol, the delisting-return convention, and the premium reference-date convention. Each of these is a substantive choice with empirical consequences, and a paper that does not state them transparently is not replicable.

15.4 Announcement Returns and Market-Based Evidence

The announcement-window event study is the most studied empirical object in the M&A literature. Chapter 4 develops the methodology in general: short-window cumulative abnormal returns around a discrete event, the abnormal return measured against a benchmark model, and statistical inference adjusted for clustering and event-induced volatility. This section does not repeat that machinery. It focuses on what is specific to deal events: what announcement returns measure, what they do not measure, and what they imply about the value-creation channels of Section 15.5.

Target announcement returns are large and consistent. The stylized fact across four decades of merger research is that target shareholders earn cumulative abnormal returns of roughly 15 to 25 percent in the announcement window, with the precise magnitude varying by deal type, time period, and sample. [Jensen and Ruback \(1983\)](#) provide the early synthesis of the takeover-returns evidence, and [Bradley, Desai, and Kim \(1988\)](#) establish the canonical decomposition of combined gains between target and bidder shareholders. [Andrade et al. \(2001\)](#) document the patterns across the modern US merger wave, and [Betton et al. \(2008\)](#) provide the handbook synthesis. Target gains are larger in cash deals than in stock deals, larger in hostile than friendly deals, and larger when competing bidders emerge. Tender offers produce the largest target premiums in the literature. The interpretation that the literature defaults to is that targets are paid more than their pre-announcement standalone value, and the gap is the most-cited evidence of value creation in the deal.

Bidder announcement returns are small, noisy, and centered near zero. The cross-sectional average is statistically indistinguishable from zero in most large samples, though conditional patterns are systematic. [Moeller, Schlingemann, and Stulz \(2004\)](#) document that smaller acquirers earn positive announcement returns while larger acquirers earn negative returns, with the dollar value of large-acquirer losses concentrated in a small number of very large deals. [Moeller, Schlingemann, and Stulz \(2005\)](#) extend the finding

to the late-1990s merger wave, where aggregate bidder losses ran into the hundreds of billions of dollars. The asymmetry between small-deal gains and large-deal losses is the empirical foundation for the hubris and overconfidence channels of Section 15.5.

Combined returns, computed as the value-weighted CARs of bidder and target taken together, are the cleanest available measure of expected value creation. They are positive on average, with magnitudes that vary by sample, era, deal type, and weighting. Combined dollar returns and combined percentage returns can move in opposite directions when small targets are paired with large bidders, and the choice of metric is therefore a substantive choice rather than a presentation choice. Percentage returns understate the dollar magnitude of large-deal losses. Dollar returns suppress the cross-sectional variation that disciplines theoretical predictions. Reporting both is the modern standard.

Schwert (1996) provides the canonical treatment of merger premiums. Premiums are measured as the percentage difference between the offer price and a pre-announcement target stock price, typically two to six weeks before the rumor or announcement date. The premium is not the same as the target announcement return. The premium is the gross offer relative to a benchmark standalone value, while the announcement return is the market's revision in expected value, which depends on the probability the deal closes and on the standalone counterfactual. Premiums are systematically higher in hostile deals, in cash deals, and in deals with multiple bidders, mirroring the patterns in target returns.

The runup in target prices before announcement is a well-documented feature of US deal data. Schwert (1996) shows that roughly one-third of the total target return accrues during the runup window. The two interpretations of the runup are anticipation, in which the market correctly forecasts an impending deal, and leakage, in which private information reaches the market before the formal announcement. Betton et al. (2008) surveys the evidence and concludes that both are present in proportions that vary with deal type. The implication for design choice is that the event-window definition determines the estimand: a tight window around announcement recovers the residual surprise, while a wider window captures the runup but absorbs unrelated noise.

Key Insight 2: Announcement returns are revisions, not effects

A cumulative abnormal return around a deal announcement is the market's revision in expected firm value, weighted by the probability of completion and conditioned on the information the market had at the announcement date. It is not the causal effect of the deal on firm value. The two coincide only when completion is certain, the market is correct on average, and the counterfactual to the deal is the standalone firm. None of these conditions holds exactly in practice, and the deviations vary across deal types,

time periods, and bidder characteristics.

15.4.1 Method of payment and bidder leverage

The choice between cash and stock as the medium of exchange is one of the strongest and most robust predictors of bidder announcement returns in the literature. In public-target acquisitions, stock-financed deals earn negative bidder returns on average, while cash-financed deals earn small positive or zero returns ([Andrade et al., 2001](#)). The pattern is not uniform across private-target and subsidiary acquisitions, where stock can play different roles, including resolving information asymmetries and creating blockholders with monitoring incentives ([Fuller, Netter, and Stegemoller, 2002](#)). The two leading interpretations are bidder overvaluation and post-deal leverage. [Shleifer and Vishny \(2003\)](#) and [Rhodes-Kropf and Viswanathan \(2004\)](#) argue that stock is the rational currency when the bidder believes its stock is overvalued; the negative announcement return reflects the market's inference that the bidder is signaling overvaluation by paying in stock. The leverage channel is partly mechanical. A cash deal funded with debt raises bidder leverage at closing, while a stock-financed deal is typically less leverage-increasing relative to combined assets. The precise post-deal leverage effect depends on target leverage, assumed liabilities, cash balances, and any concurrent refinancing. Post-deal combined leverage nevertheless differs systematically between cash and stock deals on average.

This is the chapter's first running-example touchpoint. Method of payment is the channel through which a control transaction converts into a capital-structure change, and the design implications carry directly to studies of leverage effects on investment in Chapter 10. A researcher who pools cash and stock deals into a single sample is averaging across two transactions that shift bidder leverage in opposite directions. The pooled coefficient on a deal indicator is therefore not informative about either the leverage channel or the overvaluation channel without further decomposition.

Common Mistake 3

Long-run abnormal returns following M&A announcements are the most contested empirical object in the deal literature. Studies have reported both positive and negative long-run drift; [Loughran and Vijh \(1997\)](#) find underperformance of bidders that pay in stock, while [Mitchell and Stafford \(2000\)](#) attribute much of the apparent drift to benchmark mis-specification. Three problems compound. Benchmark sensitivity is the first: the long-run abnormal return often changes sign with the choice of benchmark portfolio. The joint-hypothesis problem is the second: any long-run abnormal-return

test is a joint test of the asset-pricing model and market efficiency. Selection on completion is the third: only completed deals enter long-run-return samples, and completion is itself an outcome that depends on deal characteristics. A long-run-return result that is not robust to alternative benchmarks and does not address completion selection should not be cited as evidence on deal value.

The takeaway from the announcement-window literature is that the market reacts to deal news, target shareholders capture a sizable fraction of measured shareholder surplus, bidder shareholders capture little, and combined returns are positive on average. None of these patterns is itself a causal effect of the deal on firm value. They are revisions in expected value, conditional on the information the market had at the moment of announcement. Whether realized outcomes match the market's revision is a separate empirical question, developed in Section 15.7.

15.5 What Creates Value? Predictions for Post-Deal Outcomes

A deal can create value through several distinct channels. Identifying the channel matters because each channel has distinct empirical signatures and distinct policy implications. This section organizes the value-creation hypotheses by their post-deal predictions rather than by the theories that motivate them. The section gives empirical signatures, not a literature review by channel. Each subsection states what a study would have to document to attribute observed value creation to the channel in question, and the discussion is keyed to the identification strategies of Section 15.7.

Operating synergies operate through cost reduction or revenue enhancement that requires the assets to be jointly owned: consolidating overlapping distribution networks reduces operating costs, pairing a strong product with a strong sales force expands joint output. If operating synergies are the dominant channel, productivity, margins, and asset-utilization should rise relative to the sum of the standalone firms' counterfactual paths after adjusting for industry trends. The empirical bar is whether the post-merger improvement exceeds what the firms would have achieved separately. [Healy, Palepu, and Ruback \(1992\)](#) provide the canonical accounting-data benchmark.

Market power operates through reduced competition: a horizontal merger raises the combined entity's price-cost margin if consolidation softens competitive constraints. The market-power signature is a post-merger margin increase paired with positive rival

announcement returns, since rivals benefit when the merging firms can charge more; the distinction from cost-side synergies is what antitrust review weighs under the merger guidelines. [Eckbo \(1983\)](#) provides the foundational rival-returns test; [Fee and Thomas \(2004\)](#) decompose horizontal-merger gains across customer, supplier, and rival firms.

The agency and hubris channels begin with [Jensen \(1986\)](#): managers of cash-rich firms with limited investment opportunities acquire rather than pay out, because acquisitions enlarge the enterprise and protect against distress. The agency prediction is free cash flow, weak governance, and post-deal performance below what alternative uses of cash would have delivered. [Roll \(1986\)](#) formalized the hubris channel: bidders overestimate target value and overpay. The sharper prediction is that bidders lose at announcement, targets gain, and combined returns sit at or below the bid cost. [Malmendier and Tate \(2008\)](#) link acquisitiveness to CEO overconfidence measured through delayed option-exercise behavior.

Financing-related value creation operates through capital-structure changes the merger makes feasible: a target with limited debt capacity acquired by a bidder with strong cash flows supports higher combined leverage than the two firms could support separately, generating incremental tax shields, and target tax-loss carryforwards are worth more to a profitable acquirer than to current owners. The financing signature is that post-deal capital structure reflects the new joint capacity and the tax-shield gain is visible in the combined-return decomposition. [Hayn \(1989\)](#) documents tax-attribute contributions to acquisition gains; [Auerbach and Reishus \(1988\)](#) measure tax-shield motives in early-1980s evidence. Chapter 10 develops the underlying theory.

The misvaluation channel comes from [Shleifer and Vishny \(2003\)](#) and [Rhodes-Kropf and Viswanathan \(2004\)](#): bidders use overvalued stock to acquire real assets. The signature is that stock-financed acquisitions cluster in periods of high market valuation, target shareholders who hold acquirer stock through completion underperform, and the combined entity performs poorly when overvaluation reverses. [Loughran and Vijh \(1997\)](#) document the long-horizon underperformance of stock-paid acquirers, subject to the caveats of Section 15.4.

These channels overlap in any given deal: no single observable distinguishes them in a way that supports a sharp counterfactual. The identification strategies of Section 15.7 and the restructuring evidence of Section 15.8 attempt to discriminate by conditioning on sources of variation that activate one channel more strongly than others.

15.6 Selection into Transactions

A corporate-control transaction is not a treatment that arrives at a firm by accident. It is the realization of a sequence of choices made by acquirers, targets, advisors, financiers, regulators, and shareholders, each of whom has private information about the value the transaction would create. Selection enters at every margin: which firms bid, which firms accept, which deals close, and at what price. Treating these margins as exogenous to the outcomes the researcher cares about is the most common identification error in the M&A literature, and the most consequential.

Four selection margins shape every empirical design in this chapter.

Acquirer selection determines which firms initiate transactions in the first place. [Harford \(1999\)](#) documents that firms with large cash reserves are more likely to acquire, and that acquisitions made from cash reserves earn lower announcement returns on average than other acquisitions. [Malmendier and Tate \(2008\)](#) extend the selection story to CEO characteristics: overconfident managers, identified through their option-exercise behavior, are systematically more acquisitive. The acquirer side is the side of the deal with the most documented selection on managerial and financial characteristics.

Target selection determines which firms receive bids and which do not. Targets often have lower valuation ratios, weaker recent performance, or other observable characteristics associated with elevated acquisition likelihood. [Palepu \(1986\)](#) provides the canonical methodological treatment of target prediction, including the standard logit specification, the choice of holdout and estimation samples, and the bias problems that arise when the estimation sample over-represents acquired firms relative to the underlying population. [Harford \(2005\)](#) shows that merger activity arrives in industry-level waves driven by technological, regulatory, or capital-availability shocks, which means that target selection is partly an industry-level phenomenon rather than a purely firm-level one. The implication is that within-industry comparisons absorb wave-level selection but not within-wave firm-level selection.

Completion selection is the margin most useful for identification and the easiest to misread. Roughly one in five publicly announced US deals is withdrawn before completion, and withdrawal is not random with respect to firm fundamentals. A bidder may walk away because of due-diligence revelations, market-condition shifts, regulatory pushback, or shareholder revolt. Treating completed and withdrawn deals as comparable groups ignores the information the completion outcome itself carries. The design that uses failed deals as counterfactuals for completed deals, developed in [Section 15.7](#), requires that the source of failure be plausibly orthogonal to the outcome of interest.

Bargaining selection is the most subtle of the four. The price paid in a deal reflects the bargaining outcome between bidder and target, conditional on their having decided to transact. [Schwert \(1996\)](#) documents systematic patterns in premium magnitudes that reflect bargaining structure: deals with competing bidders pay higher premiums, hostile deals pay higher premiums, and friendly negotiations produce different premium distributions than auctions. The premium is not a sufficient statistic for value creation. It is a statistic about who captured what fraction of measured surplus, conditional on the deal having occurred.

The literature has two broad strategies for confronting these four selection problems. The first treats selection as a confounding problem to be controlled for using observed characteristics. Matching estimators construct a synthetic control group of non-acquirer firms that resemble acquirers on observable dimensions, and the treatment effect is recovered as the difference between acquirer and matched-non-acquirer outcomes. Propensity-score methods ([Rosenbaum and Rubin, 1983](#)) generalize matching by summarizing covariates through the predicted probability of acquisition. Heckman selection models ([Heckman, 1979](#)) add a parametric assumption about the joint distribution of the selection error and the outcome error to identify the treatment effect off the selection equation's exclusion restriction. None of these methods identifies the effect of acquisition on outcomes unless selection on unobservables is small or the parametric restrictions hold. [Roberts and Whited \(2013\)](#) survey these strategies in the corporate-finance context.

The second strategy abandons the attempt to control for selection and looks for variation in the transaction outcome that is plausibly orthogonal to the outcomes of interest. Failed-deal designs use the random component of completion to identify the effect of completion holding the announcement decision fixed ([Savor and Lu, 2009](#)). Regulatory-threshold designs use exogenous variation in the probability of antitrust blockage. Industry-wave instruments use aggregate timing as a source of variation in deal incidence that need not be tied to firm-level outcomes. These designs are developed in [Section 15.7](#), where the discussion is keyed to a table cataloguing each design's identifying assumption and its best outcome setting.

Common Mistake 4

A standard reflex in the empirical M&A literature is to compare firms that acquire with firms that do not, controlling for size and industry, and to call the difference the effect of acquisition. The comparison does not identify the effect of acquisition. Acquirers differ from non-acquirers in their cash positions, their growth opportunities, and their managerial characteristics before the deal. Targets differ from non-targets in

their valuations, in their industries, and in the threats their managers face. Completed deals differ from failed deals in the information that arrived during due diligence. The selection in this literature is the object of study, not a nuisance to be partialled out.

Selection enters this chapter as an organizing concept rather than as a self-contained section. Every causal estimate of a deal effect is implicitly a statement about which selection problem the design solves, and every reduced-form regularity is a statement about which selection problem the regularity is conditioned on. The synthesis in Section 15.10 returns to the four margins above and assesses how much the literature has learned about each.

Key Insight 5: Two strategies, neither is a solution

The empirical literature confronts deal selection through two distinct strategies. Matching, propensity scores, and Heckman selection address selection on observables: they organize the cross-section under the maintained assumption that observed covariates capture the relevant sorting. Failed-deal designs, regulatory thresholds, and natural experiments attack selection on unobservables under exclusion restrictions specific to the design. Neither strategy resolves selection in general. Each addresses a particular dimension of selection under a particular assumption. The credibility of an empirical claim about a deal depends on which dimension of selection the design controls and which it does not.

15.7 Causal Designs and Real Effects

The previous section framed selection as the central interpretive obstacle. This section turns to the designs that have been used to navigate it. Three subsections develop the material. The first catalogs the leading identification strategies that exploit variation in deal completion or deal exposure. The second states what post-deal outcome studies require in terms of counterfactual construction. The third explains why post-deal evidence is a difference-in-differences problem in disguise, not an extended event study.

15.7.1 Designs that move deal completion or deal exposure

Six families of identification design have been used productively in the M&A literature. Table 15.2 summarizes the treatment each design isolates, the treated or complier group, the main identifying assumption, and the outcome setting in which each design is best suited. The cell entries are compact. The body text that follows develops the staggered-legal-changes row at length because it requires explicit treatment of the modern reassessment

of that literature. The When-to-Use box at the end of the subsection summarizes the conditions under which each design's identifying assumption is most defensible.

Table 15.2. Identification designs for deal effects.

Design	Treatment	Treated or comp- plier group	Main assump- tion	Best outcome setting
Close-call shareholder votes	Proposal passage	Firms near voting threshold	Local continuity at threshold	Governance- relevant deals
Failed and withdrawn deals	Completion	Announced but not completed firms	Failure plausi- bly exogenous to outcome	Bidder real ef- fects
Merger- review discon- tinuities	Filing, inves- tigation, or challenge risk	Deals near report- ing thresholds or scrutiny margins	No manipula- tion; threshold affects treatment only through review intensity	Market- power and remedy chan- nels
Staggered le- gal changes	Takeover- defense laws	State-of- incorporation exposure	Parallel trends (modern DiD)	Long-run real effects
Merger-wave exposure	Industry- wave inten- sity	Firms in active- wave industries	Wave timing as- if exogenous	Acquisition propensity
Bank- financing shocks	Acquirer credit	Bank-dependent acquirers	Bank shock un- related to target	Deal- completion margin

Notes. Each row identifies a family of identification designs used to estimate the causal effect of a deal on a corporate-finance outcome. The first column names the design; the second specifies the treatment isolated; the third identifies the units that vary in treatment; the fourth states the central identifying assumption; the fifth lists the outcome category for which the design is best suited. The rows are not mutually exclusive. Papers commonly combine two or more designs, and outcome categories overlap across rows.

Staggered adoptions of state antitakeover statutes were once the workhorse natural experiment for studying the effects of takeover threats on corporate behavior. The design exploits the staggered enactment of business-combination, fair-price, and control-share statutes across US states from the late 1980s through the 1990s, treating firms incorporated

in each state as exposed to a discrete change in their susceptibility to hostile acquisition. The design has been substantially reassessed. [Karpoff and Wittry \(2018\)](#) document coding errors in the standard adoption-year datasets that materially change estimated effects in several published studies. [Catan and Kahan \(2016\)](#) argue that the conceptual premise of the design is weaker than its empirical implementation acknowledged: the incremental protection these statutes provide is small in the presence of the poison pill defense available to almost all US-incorporated public firms, so the statutes are not a binding shock to takeover probability for most firms in the sample. [Spamann \(2022\)](#) adds the inference problem that adoption is heavily clustered in time and across states, so standard errors that treat firm-year observations as independent overstate precision.

The design has not been retired, but the bar for publishing in this lineage has risen. Any modern application also inherits the difficulty surveyed in Chapter 6. When the policy change is staggered and the treatment effect is heterogeneous across cohort-time cells, two-way fixed-effects estimators produce weighted averages with negative weights on some cohorts, and the estimated coefficient can have the wrong sign relative to the average effect of treatment on the treated. Modern staggered-DiD estimators ([Baker et al., 2022](#); [Goodman-Bacon, 2021](#); [Callaway and Sant’Anna, 2021](#); [Borusyak et al., 2024](#); [Sun and Abraham, 2021](#)) recover the cohort-time effects under transparent weighting schemes and should be used in place of pooled TWFE in any future application of this design.

The other rows of Table 15.2 can be summarized briefly. Close-call shareholder-vote designs identify a local average treatment effect at the majority threshold and apply to deals where the vote is the locus of control ([Cuñat et al., 2012](#)). Failed-deal designs use announced but not completed deals as the counterfactual, require that the source of failure be plausibly orthogonal to the outcome, and are most useful for bidder real effects ([Savor and Lu, 2009](#)). The identification comes from holding the announcement decision fixed and exploiting variation in completion; the relevant comparison is announced-completed versus announced-withdrawn, not acquirers versus non-acquirers.

Merger-review-discontinuity designs exploit institutional features of the regulatory review process: the Hart-Scott-Rodino size-of-transaction reporting threshold creates a discontinuity in filing requirements, and Herfindahl-Hirschman cutoffs in the merger guidelines establish enforcement presumptions rather than deterministic intervention rules. These cutoffs serve as clean regression-discontinuity designs only when the paper documents an institutional discontinuity in actual review intensity, investigation probability, remedy imposition, or deal completion. [Wollmann \(2019\)](#) exploits the Hart-Scott-Rodino reporting threshold to show that filings declined sharply when reporting requirements eased; [Eliaison, Heebsh, McDevitt, and Roberts \(2020\)](#) use industry consolidation in dialysis

to identify acquisition effects on prices and quality.

[Harford \(2005\)](#) supplies the merger-wave timing that serves as an industry-level instrument for acquisition propensity, used productively when the outcome question concerns deal incidence rather than specific deal effects. Bank-financing-shock designs exploit acquirer credit-supply variation to identify effects at the completion margin, with the assumption that the bank shock is unrelated to target prospects.

When to Use 6

Each row of Table 15.2 works in a specific outcome domain and only when its identifying assumption is defensible in the empirical setting. Use close-call vote designs when the deal turns on a shareholder vote and the outcome is plausibly continuous around the threshold. Use failed-deal designs when the source of failure is plausibly exogenous to the outcome of interest and the question concerns bidder real effects. Use merger-review-discontinuity designs when an institutional discontinuity in review intensity, investigation probability, or remedy imposition can be documented at the cutoff, and when the channel of interest is market power or remedies. Use staggered legal changes only after consulting the modern reassessment of the antitakeover-statute literature and applying modern staggered-DiD estimators. Use merger-wave exposure when the outcome is deal incidence rather than within-deal effects. Use bank-financing-shock designs when the question concerns the completion margin and the financing shock is plausibly target-orthogonal.

15.7.2 What post-deal outcomes require

Post-deal outcome studies measure what happens to the combined firm or the bidder after the deal closes. The empirical objects include operating performance (margins, productivity, asset turnover), investment (capital expenditure, research and development), employment (headcount, wages, plant-level activity), capital structure (leverage, payout, refinancing), innovation (patenting, citations, inventor mobility), and spillovers to rivals, suppliers, and customers ([Hortacsu, Matvos, Syverson, and Venkataraman, 2013](#)). Each outcome has its own measurement and identification challenges, but they share a structure: the researcher needs a counterfactual path for the treated firm. The level of post-deal performance is not informative on its own.

[Bena and Li \(2014\)](#), [Phillips and Zhdanov \(2013\)](#), and [Seru \(2014\)](#) document the post-merger innovation patterns; Section 13.6.1 develops the underlying evidence. The takeaway for the present chapter is that innovation outcomes inherit the same selection and counterfactual problems as the other post-deal outcomes, and the identification strategies

of Table 15.2 apply directly.

Employment and productivity outcomes are best studied in matched establishment-level data. Davis, Haltiwanger, Handley, Jarmin, Lerner, and Miranda (2014) provides the canonical evidence on private-equity acquisitions, separating the establishment-level effects on employment, productivity, and reallocation in ways the bidder-level Compustat data cannot resolve. The methodological lesson generalizes to other settings. Outcome resolution determines what the design can recover. Aggregating outcomes to the firm-year level discards within-firm reallocation that often carries the economic content of the deal.

15.7.3 Post-deal evidence is a difference-in-differences problem

A post-deal outcome regression is not a longer-window event study. The event-study framework assumes that abnormal returns are measured against an asset-pricing benchmark and aggregated over a short window in which the daily return process is stationary except for the event's information content. A post-deal real-outcome study asks a structurally different question: what would the treated firm's outcome path have been in the absence of the deal? Answering that question requires constructing a counterfactual path. Constructing a counterfactual path is what a difference-in-differences design does.

The implications follow. The treated firms are firms that completed deals. The control firms are firms that did not, constructed by matching, by failed-deal counterfactuals, or by exposure variation in one of the designs catalogued above. The parallel-trends assumption is binding: pre-deal outcome paths of treated and control units must be plausibly comparable for the design to recover a treatment effect. Chapter 6 develops the modern machinery. Post-deal studies that pool deals across years using two-way fixed-effects regressions are subject to the same TWFE pathologies as any staggered design, and the modern estimators should be used.

The synthesis is that the design choice is dictated by the outcome and by the selection margin the researcher is willing to assume away. The close-call vote design identifies a local effect at the threshold. The failed-deal design identifies an effect at the completion margin. The antitrust-threshold design identifies an effect at the blockage margin. None of these is the average treatment effect of acquisition on the average acquirer. The design is the estimand.

15.8 Restructuring Beyond M&A: LBOs, Divestitures, Bankruptcy

The acquisition literature is one part of a broader corporate-control literature that includes leveraged buyouts, divestitures, spinoffs, asset sales, and bankruptcy reorganization. These transactions share the central interpretive difficulty of the M&A literature, namely selection, but they pose distinct identification problems and answer distinct economic questions. This section treats the three settings in order, with attention to where each connects back to the running example.

15.8.1 LBOs and private equity

The leveraged buyout is the corporate-control transaction that bundles the largest number of governance and financing changes into a single event. A public firm becomes private. Equity ownership concentrates from dispersed public shareholders to a small group of private-equity sponsors and target managers. Leverage rises substantially, often to multiples of operating cash flow that public firms would not sustain. Managerial pay shifts toward equity, sometimes to extreme concentration. Board oversight changes from public-firm conventions to sponsor-driven monitoring. Operating discipline tightens in ways that are difficult to disentangle from the financial changes.

This bundling makes the LBO economically rich and empirically difficult. The same transaction that increases leverage also concentrates ownership, intensifies monitoring, and changes managerial incentives. Each of these channels can in principle produce post-LBO operating improvements. Separating them is the central methodological challenge of the LBO literature. [Acharya, Gottschalg, Hahn, and Kehoe \(2013\)](#) document post-LBO operational and governance improvements consistent with the active-monitoring channel.

The LBO is the running example's second touchpoint. We characterize it as the most concentrated leverage-control bundle the empirical literature has access to: a transaction that simultaneously raises leverage to extreme levels and reorganizes ownership and control. The phrasing matters. The LBO is not the cleanest leverage shock for studies of capital-structure effects on investment, because the leverage change is bundled with ownership and incentive changes that are themselves consequential. A study that attributes post-LBO performance changes to leverage alone is implicitly assuming the ownership and incentive channels are inactive. Both are active in essentially every LBO. The empirical content of the LBO for the running example is therefore not a clean treatment effect of leverage but a joint test of how leverage, ownership, and incentives together produce the observed outcomes. [Axelson, Jenkinson, Strömberg, and Weisbach \(2013\)](#) compound the difficulty: LBO leverage tracks credit-market conditions more closely than firm characteristics, which

weakens the case for treating LBOs as a clean firm-level capital-structure experiment.

Kaplan and Strömberg (2009) provide the framing review of leveraged buyouts and private equity, surveying the channels through which PE transactions are thought to create value and the evidence available on each. Davis et al. (2014) provide the canonical establishment-level evidence. The study links target establishments to administrative employment and productivity data, comparing private-equity-acquired establishments to matched controls. The results separate the establishment-level reallocation effects (closings, openings, employment shifts across plants within firms) from the firm-level effects in ways that bidder-level Compustat data cannot resolve. The pattern is one of substantial reallocation: declines in target establishment employment, comparable openings at new establishments owned by the acquirer, and net productivity gains. The operational-versus-financial-engineering debate finds support for both channels in this evidence. Reallocation is real operating change. The leverage and incentive shifts are real financial change. The two interact rather than substitute.

Common Mistake 7

A common LBO empirical claim is that post-deal operating improvements demonstrate the value of leverage as a disciplining device. The claim is unidentified by the LBO event alone. The LBO bundles a leverage increase with an ownership-concentration change, a managerial-incentive shift, and a change in board oversight. Each of these channels can in principle produce the observed improvement. A research design that recovers the leverage effect must hold the ownership and incentive channels fixed, and the LBO event does not do that. Studies that attribute LBO returns to operational improvement without separating the channels are reporting the joint effect of the bundle, not the isolated effect of any single channel.

The estimated LBO effect is local to firms that private-equity sponsors select and finance under the credit conditions of the sample period; it is not the effect of applying private-equity ownership to the population of public firms.

15.8.2 Divestitures, spinoffs, and asset sales

Asset reallocation transactions form a parallel literature to acquisition studies. The empirical objects differ. Instead of asking who buys whom, the questions are why the seller is selling, what determines the matching of buyer and seller, and how the realized price reflects the bargaining environment. Maksimovic and Phillips (2001) provide the empirical anchor for buyer-seller matching in the market for corporate assets, documenting that the firms most

likely to buy divisions are those with productive efficiency advantages in the acquired industry, and that the post-transaction performance of sold assets improves when the match is positive on these observables.

Divestitures and spinoffs are partial reversals of conglomerate combinations. [Boone and Mulherin \(2007\)](#) document the sale-process choice between auction and negotiation, and the systematic differences in pricing and bidder composition that follow from each. The empirical regularities are that auctions yield higher prices on average, that the choice between auction and negotiation is itself selected on seller and asset characteristics, and that the relevant counterfactual for studying sale outcomes is a distribution over realized prices given the process the seller chose, not a single fixed price.

Spinoffs separate a previously combined corporate entity into two or more publicly traded firms. [Schipper and Smith \(1983\)](#) document the canonical positive announcement returns; [Daley, Mehrotra, and Sivakumar \(1997\)](#) show that cross-industry spinoffs produce larger gains than same-industry spinoffs, consistent with refocusing. [Burch and Nanda \(2003\)](#) link the conglomerate discount to divisional diversity through post-spinoff valuation changes. The selection problem is that firms that spin off choose to do so, and the cross-section of spinoff candidates is not random.

Asset sales in distress carry a fire-sale discount. [Shleifer and Vishny \(1992\)](#) provide the theoretical foundation: when potential buyers face their own financing constraints or are themselves industry insiders facing the same shock, the demand curve for the assets is downward-sloping in quantity, and sellers facing covenant violations or cash-flow pressure sell at lower prices than non-distressed sellers receive for comparable assets. [Pulvino \(1998\)](#) documents the canonical empirical evidence in commercial-aircraft transactions. [Acharya, Bharath, and Srinivasan \(2007\)](#) extend the result to defaulted-bond recoveries, which are lower when industry peers are themselves distressed. The fire-sale evidence is one of the clearer demonstrations that observed asset prices are not always at competitive levels even in liquid public-firm settings.

15.8.3 Bankruptcy and distressed restructuring

Bankruptcy is the limit case of the chapter's organizing logic. A solvent firm's leverage is a balance-sheet ratio that affects its cost of capital and its tax shield. A firm at the boundary of solvency, or past it, experiences leverage as something different. Leverage becomes the mechanism through which control rights are reallocated among claimants. Equity holders lose authority. Creditors gain it, in proportions and through procedures that depend on the firm's debt structure and the legal regime under which it files. [Gilson \(1990\)](#) documents

the empirical realization of this reallocation: in firms that default or restructure their debt, ownership and board composition turn over substantially, with new blockholders, new directors, and new senior managers entering as the firm's financial distress is resolved.

The empirical literature on bankruptcy and distressed restructuring studies the consequences of this control-right reallocation. Chapter 11 reorganizations in the US allow continued operation of a debtor firm under court supervision while creditors negotiate a reorganization plan. Chapter 7 liquidations result in asset sales and distribution of proceeds to claimants in priority order. The choice between reorganization and liquidation depends on the going-concern value of the firm's assets relative to their piecemeal value, on the ability of creditors to coordinate, and on the frictions of the bankruptcy code itself.

Bankruptcy is not one treatment. The empirical literature distinguishes at least four stages. Distress before filing captures the period in which the firm is operating under financial stress but has not yet filed; the relevant outcomes are operating performance, customer and supplier relationships, and product-market position. The filing itself is a discrete legal event that triggers the automatic stay, suspends payments to unsecured creditors, and changes the locus of decision rights. Reorganization captures the bargaining process under court supervision through which creditors negotiate a plan, with outcomes that include plan confirmation, conversion to liquidation, and recovery rates by claim class. Post-emergence captures the period after a confirmed plan is consummated, during which the surviving firm's investment, employment, and capital-structure choices reveal what the reorganization produced. Each stage has distinct outcomes and distinct identification challenges, and treating "bankruptcy" as a single binary treatment confuses estimates that should be cleanly separated.

The real-effect literature documents the consequences of bankruptcy for investment, employment, supplier and customer relationships, and product-market position. [Opler and Titman \(1994\)](#) document that distressed firms lose market share to healthy rivals; [Asquith, Gertner, and Scharfstein \(1994\)](#) provide the anatomy of distress in the junk-bond era; [Hortacsu et al. \(2013\)](#) measure the customer-flight channel using durable-goods (auto-manufacturer) evidence. Some of these declines reflect optimal adjustment to genuinely lower expected value. Others reflect the indirect costs of bankruptcy itself: managerial distraction, customer flight, supplier withdrawal, and asset fire sales. Separating direct distress from bankruptcy-cost distress is the central empirical challenge, which [Andrade and Kaplan \(1998\)](#) address by studying highly leveraged transactions where the distress source is plausibly financial rather than economic. [Bernstein, Colonnelli, and Iverson \(2019\)](#) document the asset-reallocation channel inside Chapter 11.

This is the running example's third touchpoint. Leverage in a solvent firm is a financing

choice. Leverage in a distressed firm is a redistribution of control rights. The transition between the two regimes is the bankruptcy filing, and the empirical content of leverage shifts qualitatively at that point. A capital-structure model that treats leverage as a single continuous variable across all firms cannot account for this shift. The literature has accordingly developed separate empirical apparatus for solvent leverage studies (Chapter 10) and for distressed leverage studies, with the bankruptcy event marking the discontinuity.

15.9 Structural Approaches to Corporate Control

Structural models in M&A do work that no observed deal event can do. The reduced-form designs of Section 15.7 identify treatment effects of past variation in policies and shocks. They do not identify what would happen under a counterfactual policy that has not been observed in the data. Three counterfactual questions in this area are especially salient. The antitrust counterfactual asks what happens to prices, quantities, and entry if a particular merger is blocked, allowed, or remediated. The policy-rule counterfactual asks what happens to firm behavior if poison-pill availability changes or bankruptcy priority rules shift. The institutional-design counterfactual asks what happens if private-equity leverage is capped or if mandatory disclosure thresholds change. None of these questions can be answered by re-estimating a regression on the existing data. They require a model of the market in which deals are produced, and the model's structural parameters supply the inputs for the counterfactual computation. Table 15.3 catalogs the four structural settings that have done the most empirical work in corporate-control questions.

Table 15.3. Structural approaches to corporate control.

Structural setting	set-	Object mated	esti-	Data used	Counterfactual
Takeover bargaining auctions	bar- and	Bidder valuations, bargaining weights, reservation values	valua- bargain- ing	Bid sequences, premiums, identity of bidders, board outcomes	Auction rules, board negotiation process, target reservation strategies
Dynamic models of merger waves and industry reallocation	mod- els of merger waves and in- dustry realloca- tion	Match-specific values, industry-shock processes		Industry-time deal flows, deal characteristics, financing conditions	Aggregate financing shocks, regulatory shifts, industry reorganization
Antitrust merger simulation		Demand parameters, marginal costs, ownership structure	param- eters, marginal costs, ownership structure	Prices, quantities, market shares, product characteristics	Block versus allow versus remedy; divestiture remedies; market-definition changes
Bankruptcy bargaining and restructuring		Continuation values, delay costs, creditor bargaining power		Claims structures, recovery rates, plan outcomes	Absolute-priority rule changes, DIP financing rule changes, automatic-stay design

Notes. Each row names a structural setting that has produced empirical work on corporate-control questions, the structural primitives it estimates, the data inputs the estimation requires, and the counterfactual questions the estimated model can answer. The four settings are not exhaustive; they are the ones with the most-developed empirical literature in finance applications.

The structural literature on takeover bargaining models the choice between auction and negotiation procedures used by target boards and recovers bidder valuations, bargaining weights, and target reservation values from observed bid sequences, premium magnitudes, and the identity of competing bidders. The empirical implementation is demanding. It

requires functional-form assumptions about the distribution of bidder valuations, the information environment in which bidders bid, and the bargaining protocol used in negotiation. The counterfactuals supported are about alternative bidding rules, alternative board procedures, and alternative target reservation strategies. [Gorbenko and Malenko \(2014\)](#) estimate strategic and financial bidder valuations from takeover auctions and recover systematic differences in reservation values across bidder types. [Dimopoulos and Sacchetto \(2014\)](#) model preemptive bidding and target resistance to identify how board procedure shapes premium magnitudes. The finance literature in this area is small but growing; most empirical work on M&A bargaining has remained reduced-form.

[Harford \(2005\)](#) supplies the reduced-form motivation for the structural wave literature. Merger activity arrives in industry-level waves driven by technological, regulatory, and capital-availability shocks. A structural wave model embeds this insight in a dynamic equilibrium in which firms decide whether to merge conditional on the industry state, the financing conditions, and the policy environment. The structural primitives are match-specific values, industry-shock processes, and the cost structure of integration. The counterfactuals concern aggregate financing shocks, regulatory shifts, and episodes in which whole industries reconfigure. The empirical bar is matching not just the level of merger activity but its industry-time clustering. [Rhodes-Kropf and Viswanathan \(2004\)](#) model the market-valuation channel underlying wave activity; [Maksimovic, Phillips, and Yang \(2013\)](#) extend the wave analysis to private targets, documenting that private and public waves co-move but at different magnitudes.

Antitrust merger simulation is primarily an industrial-organization literature with selected applications to horizontal-merger settings of interest in finance. The structural primitives are demand parameters, marginal-cost specifications, and ownership structures. Estimation uses observed prices, quantities, market shares, and product characteristics to recover these primitives, and the counterfactuals compute the predicted post-merger equilibrium under block, allow, or remedy regimes. The finance contribution is usually in settings where the merging firms are publicly traded and the demand-side data is available; the methodology itself is developed in industrial organization. [Berry, Levinsohn, and Pakes \(1995\)](#) provide the canonical demand-estimation foundation; [Nevo \(2000\)](#) applies the methodology to a horizontal-merger setting in ready-to-eat cereal.

The structural literature on bankruptcy and restructuring formalizes creditor bargaining under the priority rules of the legal regime, recovering continuation values, delay costs, and bargaining-weight parameters from observed claim structures, recovery rates, and plan outcomes. The counterfactuals concern policy parameters such as the strength of the absolute-priority rule, the rules governing debtor-in-possession financing, and the design

of the automatic stay. The structural restructuring literature is younger and smaller than the structural takeover literature, and most empirical work on Chapter 11 outcomes remains reduced-form. [Hennessy and Whited \(2007\)](#) supplies the structural-capital-structure framework underlying much of this work; [Davydenko and Strebulaev \(2007\)](#) and [Glover \(2016\)](#) estimate the empirical cost of default from credit spreads and default histories; [Antill and Grenadier \(2019\)](#) model the choice between dynamic bargaining and liquidation in a quantitative framework.

The payoff of the section is that structural models identify objects that no single deal event can identify, and the structural-versus-reduced-form distinction in this chapter is not a methodological hierarchy but a division of labor. Reduced-form designs identify treatment effects under design-based assumptions. Structural designs identify counterfactuals under model-based assumptions. Serious empirical research on corporate control draws on both, and the synthesis in Section [15.10](#) returns to where each kind of evidence fits the framework of the book.

15.10 Synthesis and Connection to the Framework

Table [15.4](#) summarizes what the chapter’s empirical material supports. The columns are the canonical structure for an empirical-finance synthesis: the question, the most informative evidence available, the assumption the evidence requires, the question the evidence does not answer, and the substantive claim the evidence supports. The substantive claims in the last column are worded deliberately. They are stronger than agnosticism but weaker than the unhedged claims occasionally drawn from this literature.

Table 15.4. Empirical questions about deals and what the evidence supports.

Question	Best evidence	evi-	Main assumption	as-	What it does not answer	What the evidence supports
Did the market expect value creation?	Announcement CARs		Clean event, efficient pricing		Realized long-run effects	Target gains large and positive; bidder returns small or near zero; combined returns positive on average, with magnitudes that vary by sample, era, deal type, and weighting
Did the deal improve operations?	DiD or event-time outcomes		Valid counterfactual trend		Why the deal happened	Productivity gains documented for LBOs in several settings; results for public mergers remain mixed
Who becomes a target?	Selection or prediction models		Observables capture sorting		Causal effect of acquisition	Predictive models work well; causal interpretation of becoming a target remains difficult
Who captures measured surplus?	Premiums, bargaining models		Bargaining model or market benchmark		Total welfare	Target shareholders capture most of the measured announcement-window shareholder surplus; this is not the same as total welfare
What would a policy change do?	Structural model		Correct search and bargaining structure		Design-based internal validity	Counterfactuals available but model-conditional

Notes. Each row pairs an empirical question studied in this chapter with the most informative evidence currently available, the central assumption that evidence requires, an explicit statement of what the evidence does not answer, and a substantive claim about what the evidence supports. The substantive claims in the last column are deliberately modest: where

The chapter does not cover every corner of the empirical literature on corporate control. Venture capital and the early-stage financing of high-growth firms receive only a boundary note here; the literature on staged investment, syndication, and contractual rights has developed its own apparatus that does not map directly onto the public-firm M&A literature. Cross-border deal-flow patterns and the political-economic conditions that shape them are similarly outside the scope. The merger-wave literature is touched in Section 15.9 but not surveyed in depth; Harford (2005) and the further-reading list in Section 15.12 are the entry points.

This chapter has exercised the framework of Chapter 1 and the identification logic of Chapter 2 more fully than any other application chapter in the book. It has invoked event-study machinery (Chapter 4), instrumental-variables designs (Chapter 5), difference-in-differences methods (Chapter 6), and prediction methods (Chapter 7), and it has drawn on the substantive literatures on investment (Chapter 8), innovation (Chapter 13), constraints (Chapter 9), capital structure (Chapter 10), credit supply (Chapter 12), and governance (Chapter 14). Deals are where the book's apparatus is most fully exercised on a single class of empirical object. The structural-versus-causal spine that organized the book runs through every section of this chapter, not because we chose to thread it but because the empirical objects themselves demanded both traditions for serious treatment.

Connection to Framework 8

The empirical literature on corporate-control transactions stands at the intersection of every methodological tradition this book has developed. Event studies identify market revisions. Difference-in-differences designs identify post-deal real effects under parallel-trends assumptions. Instrumental-variables strategies attack selection at the completion margin. Structural search and bargaining models recover counterfactuals over policies no past event identifies. None of these traditions answers the empirical questions about deals on its own, and the chapter has been an exercise in reading what each tradition says under the assumptions it requires. The reader who has worked through the chapter has worked through the book in compressed form.

15.11 Exercises

1. (*Conceptual.*) Compute bidder, target, and combined cumulative abnormal returns for a small sample of M&A announcements drawn from SDC or Refinitiv. Compare percentage and dollar returns across the same deals and discuss which metric is more informative for which question.

2. (*Computational.*) For a sample of deals with both rumor dates and announcement dates available, recompute CARs under alternative event-date definitions. Show how the estimated abnormal return changes when the event date moves earlier into the runup window. Interpret in light of anticipation and leakage.
3. (*Applied.*) Implement a matching design that compares acquirer firms to a propensity-score-matched control sample of non-acquirers on a Compustat panel. Report the post-deal outcome difference and explain why the matched estimate does or does not identify a causal effect of acquisition.
4. (*Applied.*) Identify a published paper that uses a failed-deal counterfactual design. State the exclusion restriction the design requires for identification. Assess whether the restriction is plausible in the specific empirical setting.
5. (*Computational.*) Build an event-time plot of post-acquisition investment or employment for a panel of completed deals using a modern staggered-DiD estimator ([Callaway and Sant'Anna, 2021](#) or [Borusyak et al., 2024](#)). Compare the modern estimator's event-time coefficients to the corresponding TWFE specification and diagnose any differences in light of the heterogeneous-effects critique of Chapter 6.
6. (*Computational.*) Construct a sample of completed M&A deals split by method of payment. Compare premiums, combined CARs, and post-deal bidder leverage between the cash and stock subsamples. Discuss which differences are descriptive and which can be interpreted causally.
7. (*Applied.*) Read one published paper that uses long-run abnormal returns as its main M&A evidence. Identify the benchmark portfolio used. Assess the sensitivity of the reported result to two alternative benchmarks.
8. (*Conceptual.*) In a short memo, classify a recent M&A empirical paper's central empirical object according to the three categories of Section 15.1: market assessment, real effects, or selection and surplus division. Explain why the classification matters for how the paper's claims should be read.
9. (*Computational and applied.*) Implement a calibrated simulation in which firms differ in an unobserved deal motive that affects both acquisition propensity and a post-deal outcome of interest. Use random seed 8823. Vary the strength of the unobserved-motive correlation across a grid and report the apparent treatment effect from a naive acquirer-versus-non-acquirer regression at each grid point. Produce a figure that displays the relationship.
10. (*Applied.*) Read [Karpoff and Wittry \(2018\)](#). Apply the same scrutiny to a recent deal-related natural-experiment paper of your choosing. Document the data-construction choices, coding decisions, and inference choices that materially affect the reported result.

15.12 Further Reading

On announcement returns and premiums, the canonical references are [Jensen and Ruback \(1983\)](#), [Bradley et al. \(1988\)](#), [Andrade et al. \(2001\)](#), and [Betton et al. \(2008\)](#); the last is the handbook synthesis.

On method of payment and bidder returns, the foundational evidence is in [Moeller et al. \(2004\)](#) and [Moeller et al. \(2005\)](#) on firm-size and dollar-loss patterns, with [Fuller et al. \(2002\)](#) for the private-target and subsidiary cases.

On selection and target prediction, [Palepu \(1986\)](#) is the methodological anchor; [Harford \(1999\)](#) and [Malmendier and Tate \(2008\)](#) document acquirer-side selection on cash holdings and managerial overconfidence.

On real effects and post-deal outcomes, [Healy et al. \(1992\)](#) is the canonical accounting-data study, and [Maksimovic and Phillips \(2001\)](#) provides the cleanest treatment of asset-redeployment effects in the market for corporate assets.

On LBOs and private equity, [Kaplan and Strömberg \(2009\)](#) surveys the channels, [Davis et al. \(2014\)](#) provides the establishment-level real-effects evidence, [Axelson et al. \(2013\)](#) characterizes the leverage-pricing nexus in buyouts, and [Acharya et al. \(2013\)](#) documents the governance and value-creation channel.

On divestitures, spinoffs, and asset sales, [Schipper and Smith \(1983\)](#) and [Daley et al. \(1997\)](#) document the canonical spinoff-returns and refocusing evidence, [Maksimovic and Phillips \(2001\)](#) treats the market for divisional assets, and [Pulvino \(1998\)](#) is the canonical fire-sale empirical reference.

On bankruptcy and distress, [Andrade and Kaplan \(1998\)](#) and [Opler and Titman \(1994\)](#) document the financial-versus-economic distress decomposition and market-share losses respectively; [Gilson \(1990\)](#) characterizes the control-right reallocation; [Bernstein et al. \(2019\)](#) documents asset reallocation in Chapter 11.

On structural approaches, [Harford \(2005\)](#) supplies the reduced-form motivation; [Rhodes-Kropf and Viswanathan \(2004\)](#) and [Maksimovic et al. \(2013\)](#) model merger waves at the public and private margins; [Gorbenko and Malenko \(2014\)](#) estimates structural takeover-bargaining models.

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